// ZPWR-PATCH-CORE //

ZPWR Patch-Core - Block Catalog

v0.3.0

MODULE REFERENCE

the complete block catalog of the modular patch graph

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MenkeTechnologies

Contents

Catalog at a glance

4218 blocks across **25 categories** and the consuming plugins, including 194 component-level analog-circuit models and 441 control / modulation (CV-badge) sources.

In plugin zpwr-fx 3346 zpwr-synth 3655 zpwr-midi-fx 563

Registry Audio 3346 Synth 309 MIDI 563

Output jack Audio 3462 MIDI 563 CV 193

By category (block count, largest first):

Oscillator	2104
Modulation	450
Sequencing	249
Dynamics	225
Probability	180
Distortion	173
Filter	168
Analog	155
Utility	151
Effect	57
Delay	53
Harmony	51
Reverb	43
Routing	30
Pitch	29
Amp	24
Physical	20
Stereo	18
Spectral	13
Envelope	7
Control	5
Voicing	5
Expression	3
Tuning	3
Timing	2

Architecture

Every block in this catalog runs through one shared patch-graph engine. The full signal and control path - from the host buffer, through the layered engine and the per-voice patch graph, onto the shared bus, and back out - is below. Signal flows top to bottom; control and modulation enter from the right; the engine's CPU optimizations are called out where they apply.

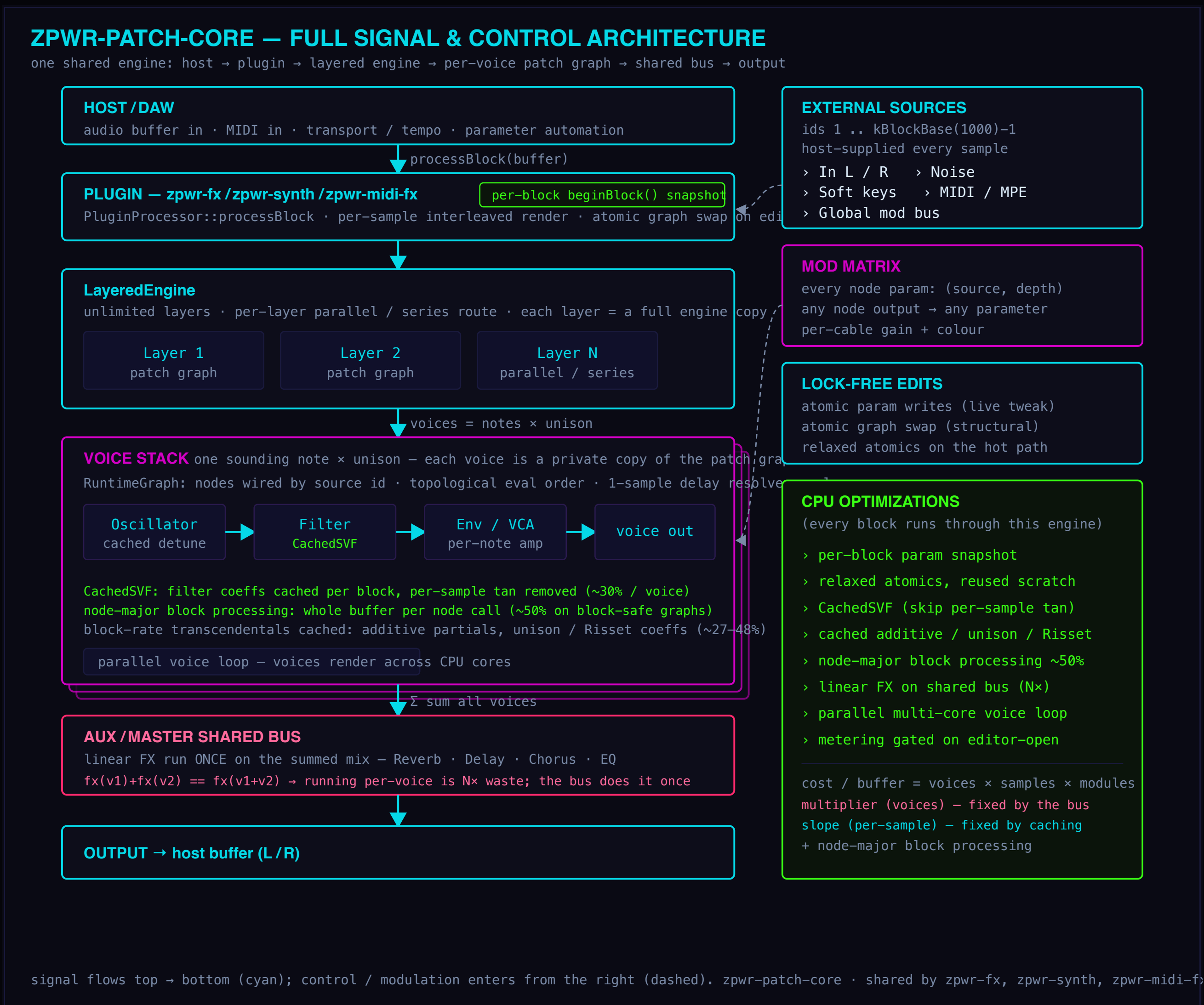

Figure 1: Full signal & control architecture of zpwr-patch-core

Performance

Every block in this catalog runs through one shared patch-graph engine, so every optimization in that engine speeds up every patch. The full list of CPU work shipped to date is below, grouped by what it attacks. The cost of one audio buffer is voices x samples x modules: the multiplier (voices x per-voice work) is what pins a chord; the slope (per-sample dispatch and coefficient recompute) is the constant overhead. The phases below attack each in turn. Measured figures are arm64, -O3, from interleaved before/after benchmark runs.

1. **Engine hot path** - shaves the per-sample slope; together ~18% off per-sample DSP:

    > **Relaxed atomic ordering** on every hot-path load (params, sync-divisor, gain, bypass, bus-send, modulation), dropping the arm64 acquire barrier on the inner loop.
    > **Reused compute-context scratch** - a single member buffer reused each sample instead of constructing one per sample.
    > **Tempo-sync scan skipped** for nodes that have no syncable parameter.
    > **Per-block parameter snapshot** - on the host's `beginBlock` call, unmodulated nodes do zero atomic parameter loads per sample; a safe per-sample fallback runs when a host never calls it.

2. **Coefficient & transcendental caching** - block-rate values that used to be recomputed every sample are now computed once per block and reused, recomputed only when an input parameter changes (all bit-identical):

    > **CachedSVF** caches the state-variable (TPT) filter coefficients, skipping a per-sample `tan`; one subclass fixes every SVF-based block with no call-site edits. ChoirPad 3-voice: 106 to 76 ns/sample (~30%).
    > **Additive partial-amplitude cache** - a dense additive voice was running up to 70 `pow` per sample (14 partials x 5 internal voices); recomputed only on a Partials / Rolloff / Odd-Even change. 1140 to 600 ns/sample (~48%).
    > **Unison detune-ratio cache** - a `pow` plus a `sqrt` per voice per sample, cached once; benefits every unison oscillator. Osc 5-voice: 246 to 180 ns/sample (~27%).
    > **Per-voice detune-ratio cache** for Screech, Hoover, Reese, Wt, SynthFM and SynthSync (the per-voice `pow(2, spread x detune)` ratio).
    > **Risset decay coefficients** - the seven-partial `exp` decay loop cached.
    > **Additator amplitude-table cache** - up to 32 `pow` per sample on block-rate Partials / Tilt / Balance, cached like the additive table.

3. **Voice multiplier** - the real fix for chord cost; turns N copies of an effect back into one:

    > **Linear FX kept on the shared bus.** Reverb and delay are linear (`fx(v1)+fx(v2) == fx(v1+v2)`), so running them per voice is N-times waste for a bit-identical result. They live on the shared aux / master bus; only synthesis and per-note timbre stay in the voice patch.
    > **Every factory preset audited clean** - no linear FX sits in any voice layer.
    > **Regression gate** - the preset validator rejects any reverb placed in a voice layer, so the multiplier cannot creep back in.

4. **Host & parallelism** - removes wasted per-voice work in the synth host:- **Metering gated on editor-open** - the per-sample O(nodes) metering scan is skipped entirely while the editor is closed. - **Parallel voice loop** - voices render across CPU cores through a persistent worker pool, bit-identical to serial, opt-in.

5. **Block (node-major) processing** - each node processes a whole buffer in one call instead of one sample at a time, amortizing the per-sample cable-gather, context-build and dispatch overhead (~50% on block-safe graphs; a per-sample-capture fallback covers graphs with buses or modulation):

    > **11 FX blocks converted** to whole-buffer processing - the medium FX. Each mirrors its per-sample body over the buffer with block-rate coefficients hoisted out of the loop, and is verified bit-identical to the per-sample path. Converted so far: Bitcrush, Chorus, Comb, Drive, EQ, Filter, Flanger, Gate, Lofi, Tremolo, Waveshaper.

Reading this catalog (badge key)

An icon glyph precedes each block name; then its badges, then in: its input jacks and out: its output-jack type, then the description:

- > [CAT] **category** - **AMP** Amp, **ANA** Analog, **CTL** Control, **DLY** Delay, **DST** Distortion, **DYN** Dynamics, **ENV** Envelope, **FLT** Filter, **FX** Effect, **HRM** Harmony, **MOD** Modulation, **OSC** Oscillator, **PIT** Pitch, **PRB** Probability, **REV** Reverb, **RTE** Routing, **SEQ** Sequencing, **SPC** Spectral, **STR** Stereo, **TUN** Tuning, **UTL** Utility, **VOI** Voicing.
- > [CV] a **control / modulation source** (outputs CV, not audio).
- > [CIRCUIT] a **component-level analog-circuit model** (per-sample nodal / Newton solve).
- > [PHY] a **physical-model instrument** (waveguide / modal / friction synthesis of strings, membranes and resonant bodies).
- > [FX] [SYNTH] [MIDI] which **plugin(s)** ship the block - zpwr-fx, zpwr-synth (gets the whole audio pack) and/or zpwr-midi-fx.
- > **in:** the block's input jacks, and **out:** its output-jack signal type - out: Audio, out: MIDI (note stream) or out: CV (control / modulation).
- > **icon:** the glyph before each block name shows its kind at a glance:

Oscillator	Filter	Delay
Reverb	LFO / mod	Envelope
Dynamics	Stereo	Pitch
Spectral	Harmony	Sequencer
Routing	Voicing	Effect
Control	Probability	Tuning
Utility	Guitar amp	Pedal / stompbox
Speaker cabinet	Analog circuit	Other

Audio (zpwr-fx / shared pack) (3346 blocks)

Amp (24)

AC30 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Vox AC30 Top Boost-style amp: two cascaded Koren 12AX7 stages into a chimey top-boost tone stack (flat ~800 Hz mid, bright treble) and a class-A EL84 push-pull power stage that compresses and sags early - the jangly British chime. EL84 (lower max current) + class-A bias distinguish it from the EL34 Marshalls.

Bandmaster [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Bandmaster-style amp: two Koren 12AX7 stages into a deeply scooped blackface tone stack and a class-AB 6L6 push-pull power stage - a piano-clean ~40 W blackface head with big headroom and a glassy, scooped top, breaking up only when pushed hard.

Bass410 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Ampeg-style 4x10 bass cabinet: a deep extended low end, a gentle low-mid bump and an early treble roll-off for a round, punchy bass tone.

Bassman [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender 5F6-A tweed Bassman / JTM45-lineage tube amp: two cascaded 12AX7 gain stages modeled with the Koren triode plate-current equation (real self-bias, grid-conduction asymmetry - not a tanh) into a Fender-style tone stack (wide ~650 Hz mid), then a class-AB EL34 push-pull power stage whose plate currents subtract through the output transformer (even harmonics cancel, odd-harmonic clip + supply sag emerge from the physics), 4x oversampled.

Blue212 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Vox AC30-style 2x12 with Celestion Blue alnico voicing: an open low end, an extended top and a sweet chimey presence - the jangly British combo cab.

Bluesbreaker [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall 1962 Bluesbreaker-style amp: two Koren 12AX7 stages into a Marshall tone stack and a class-AB KT66 push-pull power stage - the JTM45-lineage combo behind the classic British blues-rock ‘Beano’ tone, with a warm, bloomy, KT66 break-up and more low-end body than the later EL34 Plexi.

Bogner [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Bogner Ecstasy-style amp: three cascaded Koren 12AX7 stages into a tight, focused tone stack and a class-AB EL34 push-pull power stage - smooth, singing, high-gain modern British-voiced lead with a refined, liquid sustain.

Brit412 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall 1960A-style 4x12 with Celestion Greenback voicing: a tight low-cut, a roll-off around 4.5 kHz and a forward upper-mid presence - the definitive British rock guitar cabinet.

DeluxeRev [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Deluxe Reverb-style amp: two Koren 12AX7 stages into a blackface tone stack and a class-AB 6V6 push-pull power stage - the warm, touch-sensitive American combo that breaks into a sweet, compressed overdrive when cranked. Lower headroom than the Twin.

EVH5150 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EVH 5150III-style amp: three cascaded Koren 12AX7 stages into a scooped, aggressive tone stack and a class-AB EL34 push-pull power stage driven hard - the tight, saturated, percussive modern metal/shred standard with massive gain and a fast attack.

JCM800 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall JCM800 2203-style high-gain amp: three cascaded Koren 12AX7 stages (the extra gain stage = hot-rodded crunch) into a scooped Marshall tone stack and a class-AB EL34 push-pull power stage. The 80s rock/metal standard - tight, aggressive, harmonically rich.

JTM45 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall JTM45-style amp: the original Marshall, two cascaded Koren 12AX7 stages into a Marshall tone stack and a class-AB KT66 push-pull power stage (more headroom and a rounder break-up than the later EL34 Plexi). Bluesy British crunch with bloom.

Jubilee [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall Silver Jubilee 2555-style amp: three cascaded Koren 12AX7 stages into a scooped Marshall tone stack and a class-AB EL34 push-pull power stage - a hot-rodded JCM800 with a tighter, more compressed high-gain crunch. The 80s/90s rock-lead Marshall.

MarkV [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mesa/Boogie Mark V-style amp: three cascaded Koren 12AX7 stages into a deeply scooped Mesa tone stack and a class-AB power stage - the smooth, compressed, mid-scooped American high-gain lead (the 'Santana/Metallica' voice) with singing sustain.

Matchless [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Matchless DC30-style boutique amp: two Koren 12AX7 stages into a chimey top-boost-style tone stack and a class-A EL84 push-pull power stage - the AC30 chime with tighter low end and a more focused, hi-fi breakup. Boutique British class-A.

Orange412 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Orange-style 4x12: a thick, dark, low-mid-forward voicing with an earlier treble roll-off - the heavy, woody stoner/doom cabinet.

Plexi [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall Plexi 1959 Super Lead-style amp: two cascaded Koren 12AX7 stages into a scooped Marshall tone stack (~500 Hz mid) and a class-AB EL34 push-pull power stage - the bright, crunchy British roar, with real push-pull even-harmonic cancellation and supply sag. Same physics engine as the Bassman, Marshall voicing.

Princeton [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Princeton Reverb-style amp: two Koren 12AX7 stages into a blackface tone stack and a small class-AB 6V6 push-pull power stage - the low-power (~12 W) blackface combo that breaks into a sweet, touch-sensitive overdrive earlier than the Deluxe, the studio recording favourite. Lower power-tube headroom = earlier sag and bloom.

Recto412 [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mesa/Boogie Rectifier-style 4x12 (V30 loaded): a big low end, a slightly later treble roll-off and an aggressive 3-4 kHz bite - the modern high-gain metal cab.

Rectoverb [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mesa/Boogie Rectoverb-style amp: three cascaded Koren 12AX7 stages into a loose, scooped Recto-voiced tone stack and a class-AB EL34 push-pull power stage - thick, saturated, sagging modern high-gain with a huge low end. The nu-metal/modern-rock rhythm crush.

Showman [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Dual Showman-style amp: two Koren 12AX7 stages into a scooped blackface tone stack and a big, stiff class-AB 6L6 push-pull power stage - the loudest, cleanest blackface (~85 W), enormous headroom and a tight, sparkling, scooped clean that barely breaks up. Stiffer power supply (less sag) than the Twin.

SuperReverb [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Super Reverb-style amp: two Koren 12AX7 stages into a blackface tone stack and a class-AB 6L6 push-pull power stage voiced bright and chimey through its 4x10 cab - a livelier, more touch-responsive blackface clean than the Twin, singing with pick dynamics and breaking up sweetly when cranked.

Twin [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Twin Reverb-style amp: two Koren 12AX7 stages into a deeply scooped blackface tone stack and a big class-AB 6L6 push-pull power stage with huge clean headroom - the loud, sparkling, scooped American clean. Breaks up only when pushed hard.

Vibrolux [AMP] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Vibrolux Reverb-style amp: two Koren 12AX7 stages into a blackface tone stack and a class-AB 6L6 push-pull power stage voiced bright and chimey - a smaller, livelier blackface clean that sings with pick dynamics.

Analog (155)

Acido [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

TB-303-style 4-pole diode ladder: per-stage diode soft-clip and a diode-shaped feedback stage for high-resonance acid squelch.

Ampeg [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Ampeg SVT-style bass amp: two triode stages, thick low-mid focus, power-supply sag and an output-transformer stage with a dark cab - built for bass.

Ampex [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Ampex ATR-102-style mastering tape: gentle magnetic hysteresis (B/H + remanence), a soft wide head bump and smooth head-gap HF roll-off - subtle mix-bus polish (pro deck: flutter negligible by design).

Aphex [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Aphex Aural Exciter-style enhancer: a tuned high-pass side chain into an asymmetric class-A harmonic generator (2nd+3rd) and a phase-shift network, blended back for air and presence.

API2500 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

API 2500-style VCA bus compressor: a Thrust sidechain tilt with Old (feedback) or New (feedforward) detection, through a dB-linear VCA cell with control-port slew and gain-cell THD.

API550 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

API 550-style 3-band EQ with proportional-Q (bands widen at low boost, tighten at high boost) and op-amp soft-clip bite.

API560 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

API 560-style graphic EQ: fixed octave bands with proportional-Q so adjacent bands sum smoothly (4-band).

ApiPre [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

API 312-style preamp: a discrete 2520 op-amp (level-dependent knee) into an API output transformer with LF-flux iron, plus a bright top; the harmonic tilt shifts upward as level rises.

ARP2600 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ARP 2600-style 4072 four-pole ladder: aggressive, clean self-oscillating resonance with a brighter, wetter bite than a Moog ladder.

Audimax [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

CBS Audimax-style broadcast AGC: a slow, bidirectional gain-riding automatic level control that holds a target output loudness - boosting quiet material UP and pulling loud material DOWN over seconds, unlike a compressor (downward-only, fast). Its 'platform' gate FREEZES the gain when the program falls silent so it never rides room noise up in the pauses. Target sets the held level, Speed the gain-ride time, Range the maximum boost, Gate the platform threshold.

Avalon [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Avalon VT-737-style tube channel EQ: clean, musical bell and shelf bands with gentle tube warmth.

Baxandall [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Baxandall negative-feedback tone network: the classic hi-fi Bass/Treble - a gentle low shelf (~100 Hz) and high shelf (~10 kHz) that boost or cut smoothly and symmetrically, with none of the midrange scoop of a passive stack. Bass and Treble in dB.

BBE [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

BBE Sonic Maximizer-style processor: a phase-alignment network advances the highs ahead of the (allpass-delayed) lows and adds exciter-style HF harmonics for clarity and punch.

BiasTrem [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tube bias tremolo: the LFO shifts the tube grid bias (operating point), so the throb modulates both amplitude and 2nd-harmonic content - a richer, harmonically-moving wobble than plain amplitude tremolo.

BigMuff [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Big Muff Pi-style fuzz: two cascaded silicon diode clipping stages (Newton-solved Shockley pairs, 4x oversampled) that square up into a wall of sustain, with a scooped-mid tone.

BiPhase [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mu-Tron Bi-Phase dual phaser: TWO complete 6-stage OTA phasers with independent LFO rates (RateA / RateB), whose two moving notch combs beat against each other for a swirling, three-dimensional sweep no single-sweep phaser (Phase 90, Small Stone) can reach. Depth sets the sweep width, Feedback the resonant emphasis of the notches, Mix the wet blend. Six 1st-order all-pass stages per side.

Blackface [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Blackface-style amp: a triode stage, scooped-mid clean headroom (~500 Hz notch), a stiff (low-sag) power section and an output-transformer stage with a bright, glassy top.

BlueBox [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Blue Box-style octave fuzz: cascaded flip-flops divide the pitch down two octaves into a gritty square sub-octave, blended with a fuzzed dry signal - the synthetic, glitchy -2-octave fuzz.

BluesDriver [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss BD-2 Blues Driver-style overdrive: a JFET-buffered asymmetric soft-clipper that stays bright, dynamic and amp-like - cleans up with pick attack, breaks up when pushed. Gain sets the drive, Tone the top-end tilt, Level the output.

BodeRing [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Bode / Moog 6401-style ring modulator: the input multiplied by an internal tunable sine carrier through a diode-ring nonlinearity, giving the metallic sum-and-difference sidebands of the classic sci-fi/bell sound. Freq tunes the carrier, Drive sets the diode-bridge bite, Mix blends modulated against dry.

BusSat [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

SSL bus-style VCA saturation into an output transformer: a thin 'glue' veneer for the mix bus - subtle thickening, not obvious distortion.

BX20 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

AKG BX-20-style torsional-spring reverb: a dispersive modulated all-pass tank models the spring chirp smeared smooth - long, dark, no boing.

Cassette [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tascam Portastudio-style 4-track cassette: the lo-fi tape end - a slow narrow capstan gives heavy wow (slow pitch drift) plus flutter (fast wobble) through a modulated delay, a restricted bandwidth (HF gap loss + LF rolloff) and early-saturating tape hysteresis, the bedroom-demo cassette character versus the clean Ampex/Studer mastering decks. Wow sets the pitch instability, Sat the tape drive, Tone the HF rolloff, Mix the blend.

CE1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss CE-1-style BBD chorus/vibrato: a triangle-LFO chorus or a wet-only vibrato through a companded bucket-brigade line, with preamp coloration.

CEM3320 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

Curtis CEM3320-style 4-pole OTA filter (Prophet-5 rev3 / SH-101 / Pro-One): four OTA integrator poles with per-stage soft clip and a bass-compensated resonance path, so it stays fat where a Moog ladder thins out - cleaner, brighter, more aggressive into self-oscillation. Cutoff, Reso to self-osc, Mod-CV depth (in 3), Drive pushes the cells.

Chamber [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Chamber reverb: higher modal density than a plate with a diffuse onset and warm, non-metallic tail - between a plate and a room.

Chandler [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Chandler Curve Bender-style EQ (EMI/Abbey Road): musical stepped bands with transformer/valve harmonic colour.

ChandlerPre [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Chandler TG2-style preamp (EMI/Abbey Road): an input stage into an iron-core transformer (LF flux saturation) - punchy with a distinctive forward presence.

CryBaby [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Dunlop Cry Baby / Vox inductor wah: a resonant inductor-cap band-pass swept by Pedal over the throaty 350 Hz - 2.2 kHz range, with the Q peaking through mid-sweep the way a real wah inductor does and falling at the heel/toe. Pedal sets the position (modulate it for auto-wah), Q the bite, Mix the blend.

CS80 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Yamaha CS-80 dual resonant filter: a 2-pole resonant high-pass in series with a 2-pole resonant low-pass, each swept independently for the lush, vocal, brass-and-strings voice (Vangelis / Blade Runner) a single multimode filter cannot make. HP Cutoff lifts the bottom, LP Cutoff closes the top, Reso peaks both corners (soft-saturated, musical - not a screaming self-oscillator), Drive pushes the input.

CultureVulture [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Thermionic Culture Vulture valve distortion: Triode mode is an asymmetric, even-harmonic soft saturation (warm, musical thickening); Pentode mode is a harder odd-harmonic clip with a push-pull crossover dip near zero for the snarling, aggressive ‘P’ destruction. Bias shifts the valve operating point (asymmetry / gate), Drive pushes it from fattening to full disintegration, Tone tilts the output, Level trims it. A valve character/destruction unit, not a pedal or amp.

Dbx160 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

dbx 160-style VCA compressor: true-RMS detection and an ‘over-easy’ soft knee through a dB-linear VCA cell with control-port slew and gain-cell THD; punchy and musical.

Dbx165 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

dbx 165-style VCA compressor: true-RMS detection into a dB-linear VCA with the over-easy soft knee, faster and more aggressive than the 160, plus the PeakStop clamp - a click-free output brickwall that stops transients dead. Threshold/Ratio/Knee shape the compression, Makeup the makeup gain.

dbxNR [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

dbx Type II-style companding noise reduction: a 2:1 RMS-compress encode and complementary 1:2 expand decode with HF pre-emphasis - run as one box the two halves track at slightly different speeds, so steady tone passes clean but transients pump and HF content (cymbals) modulates the floor: the signature dbx ‘breathing’. Amount sets the companding depth, Emphasis the HF-triggered breathing, Mix the blend.

Diezel [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Diezel VH4-style amp: three cascaded triode stages, ultra-tight aggressive high-gain with a scooped low end, reservoir sag and an output-transformer stage.

DimensionD [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Roland Dimension D-style BBD chorus: two anti-phase bucket-brigade lines (compander + clock-tracked reconstruction filter) for a wide, ‘motionless’ thickening with no audible wobble.

Distressor [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Distressor-style aggressive compressor: a FET gain cell whose bite grows with gain reduction, switchable harmonic distortion (Dist2 = genuine 2nd, Dist3 = 3rd) and a British-mode all-buttons slam.

Drawmer [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Drawmer DS201-style noise gate: opens fast above Threshold and closes after Release, with hysteresis so it doesn't chatter on the edge. Range sets how far it ducks when closed (hard gate vs gentle expander), Attack the open speed - the studio-standard gate for tightening drums and killing hum between phrases.

DS1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss DS-1-style distortion: a Newton-solved symmetric silicon diode pair (4x oversampled - the real shunt clipper, not a hard jlimit) with an active tilt-tone control - brash and cutting.

Dumble [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Dumble Overdrive Special amp (full circuit): smooth, singing high-gain with a spongy supply for bloom and sustain - the holy-grail boutique lead voice. Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for Dumble's compressed sustain.

DynaComp [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Dyna Comp-style OTA pedal compressor (CA3080): a full-wave-rectified envelope drives a high-ratio OTA gain cell for the squishy, sustaining 'country squeeze' - fast clamp, slow bloom, and the signature high-end softening. Sensitivity sets how hard it squeezes, Attack the detector speed, Output the makeup.

Echoplex [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Maestro Echoplex EP-34-style tape echo: warm, wow+flutter-modulated repeats with magnetic hysteresis (asymmetric saturation + remanence) and head gap-loss HF roll-off.

Echorec [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Binson Echorec-style magnetic drum delay: several fixed playback heads give rhythmic multi-tap echoes with rotating-drum wobble and magnetic hysteresis in the feedback.

EF86 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EF86 pentode preamp (Vox AC15/AC30, Matchless front end): a pentode gain stage modeled by the 3/2-power Child-Langmuir plate law with a sharp cutoff (ckt::pentodeIp) - far more gain and a harder, faster-compressing grind than the triode stages in the amp models, with the asymmetric even-harmonic colour of a single tube stage. Drive sets the grid swing, Bias the operating point (edge of breakup / asymmetry), Tone a post treble tilt, Level the output.

ElectricMistress [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EHX Electric Mistress-style flanger: a swept companded bucket-brigade line forms moving comb notches; Filter Matrix freezes the sweep for a fixed metallic tone.

EMS [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EMS VCS3-style 3-pole (18 dB) diode ladder: asymmetric diode shaping gives a sharp, acidic resonance.

EMT140 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EMT 140-style plate reverb: a dispersive modulated all-pass tank (the plate shimmer where HF and LF propagate at different speeds), instant dense diffusion, bright sheen, HF decaying faster than the low tail.

EMT250 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EMT 250-style early digital reverb: a dense tank with first-generation converter quantization grain - bright and unmistakably early-digital.

ENGL [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ENGL-style amp: three cascaded triode stages, tight scooped modern-metal high-gain, reservoir sag and an output-transformer stage with a dark cab.

EP3 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Echoplex EP-3 preamp: the Echoplex's FET preamp section, famous as an always-on tone boost (Page, Van Halen). A touch of FET saturation plus a presence bump and slightly tightened lows make whatever runs through it fuller, clearer and more present - a buffer/boost with character, not a distortion or amp. Gain sets the boost, Bright the presence bump, Level the output.

Fairchild670 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fairchild 670-style vari-mu tube limiter: a remote-cutoff 6386 twin-triode is the gain cell, its mu falling as control bias rises so the ratio is soft and program-dependent, and the 6-position time-constant switch sets attack/release from snappy to molasses. Threshold leans it in, TimeConst picks the program-dependent ballistics, the vari-mu adds gentle glue and tube harmonics, Output is makeup.

FATSO [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Empirical Labs FATSO-style warmth/comp: a soft-knee compressor into a tape-style saturation stage (a tilting HF soft-clip) that rounds transients and adds harmonic 'warmth', the way the hardware glues and thickens a mix. Comp sets the squeeze, Warmth the saturation drive, Output the makeup.

FET [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

1176-style FET compressor: ultra-fast attack, fixed high ratios, and an asymmetric FET gain cell whose harmonic bite grows with gain reduction (max ratio = all-buttons slam).

Friedman [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Friedman BE-100-style amp: hot-rodded Marshall - two cascaded triode stages, tight low end, singing mids, reservoir sag and an output-transformer stage.

Fulldrive [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fulltone Fulldrive-style overdrive: a TS-derived circuit opened up with less mid-hump and more gain on tap, asymmetric soft clipping that stays articulate when pushed, plus a cleaner Boost stage. Gain sets the drive, Tone the brightness, Boost stacks gain, Level the output.

FuncGen [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

H3000-style function generator: a modulation source with 19 selectable waveshapes plus random sample-and-hold, smooth random and noise (no audio input).

FuzzFace [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fuzz Face-style germanium fuzz: two cascaded germanium common-emitter transistor stages (Ebers-Moll, 4x oversampled) whose asymmetric cutoff/saturation gives the gated, spitty fuzz; Bias starves the second stage toward the splatty/gated voicing, and it cleans up as you lower Input - just like rolling back the guitar volume.

FuzzFactory [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Z.Vex Fuzz Factory-style two-transistor fuzz: massive asymmetric gain into a hard clip, with Gate choking the decay into a sputtery rip and Stab starving the bias toward a self-oscillating squeal that ducks under the note - the pinched, gated, velcro fuzz. 4x-oversampled.

FZ1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Maestro FZ-1 Fuzz-Tone: the first fuzz pedal (the ‘Satisfaction’ buzz). Three germanium stages give a thin, splatty, ripped-speaker square-wave fuzz that GATES hard as the note decays into the famous sputtering decay. Thinner, harder and more gated than the thick Tone Bender or the FuzzFace. Attack sets the gain, Bias starves the supply (more gate/sputter), Level the output.

GML [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

GML 8200-style parametric EQ: ultra-clean, transparent constant-Q bands with no saturation - the surgical reference where boost and cut cancel exactly.

Guvnor [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Marshall Guv’nor-style distortion: hard symmetric diode clipping after a Marshall-voiced mid push with an active treble tilt - the ‘amp in a box’ crunch-to-roar. Gain sets the clipping, Tone the brightness, Body the low-mid push, Level the output.

H3000 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Eventide H3000-style micro-pitch doubler: two slightly detuned, short-delayed voices thicken the signal into a wide, lush double-track.

Helios [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Helios Type 69-style console EQ: a low bell, a switchable peak-or-trough mid at stepped frequencies, and a 10 kHz high shelf - punchy and musical.

HeliosPre [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Helios console-style preamp: punchy class-A transistor saturation into a console transformer (LF-flux iron) with an asymmetric Bias and a forward ~1.5 kHz midrange push. Drive sets the grit, Tone rolls the top, Mix blends dry/wet - aggressive British-console colour.

Hiwatt [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Hiwatt DR103 amp (full circuit): three could-be-cleaner cascaded triode stages with a stiff power supply and a bright, high-headroom voice - the clean-but-LOUD British military-grade tone (Townshend / Gilmour) that stays articulate where a Marshall breaks up. Gain drives the preamp, Tone tilts the stack, Master pushes the power stage, Level trims. Cascaded triode transfer + voiced tone stack + supply sag, not a single tanh + fixed EQ.

HM2 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss HM-2 Heavy Metal ('Swedish chainsaw'): a high-gain hard clipper into the HM-2's scooped dual-band Color EQ. Turn Color up and it guts the mids while pushing a resonant low end and a fizzy top into the unmistakable buzzsaw wall of Swedish death metal (Entombed). Dist sets the gain, Color sweeps flat -> full chainsaw scoop, Level the output. Distinct from the mid-scoop MetalZone by the dual-band buzzsaw voicing.

JC120 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Roland JC-120 Jazz Chorus (solid-state clean): the transistor clean platform - a stiff power supply (no sag), big bright headroom that stays glassy and articulate, and a harder, more brittle clip when finally pushed, unlike the soft, compressing tube amps. Drive sets the gain, Tone the brightness, Level the output. Pair with a Chorus block for the full JC stereo shimmer.

JfetStage [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

JFET common-source overdrive (true square-law device model, ckt::jfetId): the signal swings the gate of a J201-style JFET and its drain current Id = Idss(1-Vgs/Vp)^2 forms the output, which clips ASYMMETRICALLY - compressing toward zero-bias, cutting off at pinch-off - for the even-harmonic, dynamic, 'amp-like' warmth of a FET preamp stage. Drive sets the gate swing, Bias the operating point (asymmetry / edge of breakup), Tone a post treble tilt, Level the output.

Jup-8 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Jupiter-8-style 4-pole OTA ladder: per-stage OTA transconductance saturation with smooth resonance that keeps the low end (no Moog bass loss).

Klon [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Klon Centaur-style overdrive: a Newton-solved germanium diode pair (4x oversampled) summed back with the clean signal - 'transparent' via the dual-path blend, not soft clipping.

Korg35 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Korg MS-10-style Korg35 Sallen-Key low-pass: hard diode-clipped resonance that screams and squares up at high settings - more aggressive than the later MS-20 OTA revision.

LA2A [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Teletronix LA-2A-style opto leveling amplifier: a T4 electro-optical cell compresses slowly and smoothly with the cell's two-stage, program-dependent release - fast for the first part of recovery, then a long gentle tail - so it sits musically without surgical control. Peak Reduction sets how hard it leans, Gain the makeup, Emphasis tilts the detector toward highs (de-essing).

LA3A [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Teletronix LA-3A-style optical compressor: the LA-2A photocell gain law but a fast-attack, brighter solid-state amp - punchier optical leveling with program-dependent release.

Lex224 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Lexicon 224-style digital hall: a long, lush, slowly-modulated reverb tail (chorusing avoids metallic ringing) with bright, dense diffusion.

Lex480 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Lexicon 480L-style digital reverb: a brighter, denser random-hall/plate with a modulated tail and fast build-up.

Lockhart [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Lockhart-style transistor wavefolder (west-coast): the signal reflects off the rails through several folding stages, multiplying harmonics as Fold rises - the Buchla/Serge timbre. 4x-oversampled.

Maag [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Maag EQ4-style EQ: musical low and mid bands plus the famous Air Band - a very high shelf for open, silky top.

Manley [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Manley Massive Passive-style EQ: wide bandwidth limits the available gain (opposite of proportional-Q), passive LC curves into an output transformer with LF-flux iron harmonics.

MarkIIC [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mesa/Boogie Mark IIC+ amp (full circuit): tight, scooped, high-gain lead with a stiff supply for fast attack and a singing sustain (the 80s metal lead voice). Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for the Mark's tight low end and scooped mids - distinct from the looser, fatter Rectifier.

MassivePassive [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Manley Massive Passive-style EQ: broad, gentle passive-network shelf and bell curves with wide, interactive Q - boosts feel airy and musical rather than surgical, the reason it's a mastering staple. Low shelf, a wide high bell at HiFreq, and an Air shelf on top, each in dB.

Matrix12 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Oberheim Xpander / Matrix-12 multimode VCF (CEM3372): a 4-pole OTA ladder whose Mode taps a binomial mix of the four pole outputs to produce low-pass, high-pass and band-pass responses at 1-, 2- and 4-pole slopes from a single filter

– the signature Oberheim pole-mixing the SEM's 2-pole SVF cannot do. Per-stage tanh models the OTA soft clip and Reso feeds the 4-pole output back for resonance up to self-oscillation. Cutoff sweeps it, Drive pushes the stages.

MemoryMan [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EHX Memory Man-style analog BBD delay: a companded bucket-brigade line whose clock-tracked reconstruction filter darkens each repeat with delay length, gently chorused, with line-level regeneration.

MEQ5 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Pultec MEQ-5-style midrange passive EQ: a low-mid boost, a mid dip and a high-mid boost for vocal/guitar presence.

MetalZone [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss MT-2 Metal Zone-style high-gain distortion: two 4x-oversampled clip stages into an active dual-peak mid scoop with a sweepable parametric mid.

Mini [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Minimoog-style 4-pole transistor ladder: per-stage saturation thins the bass as resonance climbs to self-oscillation.

MS-20 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MS-20-style 2-pole Sallen-Key low-pass: input op-amp saturation and a saturated resonance integrator plus diode clipping in the resonance path - screams and distorts at high resonance.

MuTron3 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mu-Tron III-style envelope auto-wah: an OTA state-variable filter whose cutoff is swept by an envelope follower of the playing dynamics. Sens sets the sweep depth, Direction flips up/down, Peak sets resonance, Mode picks LP/BP/HP.

MXRDist [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Distortion+-style overdrive: a Newton-solved ASYMMETRIC germanium diode pair (4x oversampled) - mismatched diodes give the even-harmonic, woolier-than-a-DS-1 crunch - with an output low-pass.

MXRFlanger [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Flanger-style BBD flanger: a companded bucket-brigade line swept for a deep, metallic comb with line-level regeneration for jet-plane sweeps.

N1073 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Neve 1073-style EQ: fixed 12 kHz HF shelf, sweepable mid bell and LF shelf, into an iron-core transformer whose flux saturates the low frequencies first for genuine LF harmonic warmth.

Neve1081 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Neve 1081-style 4-band EQ: the 1073's bigger brother with a fixed HF and LF shelf PLUS two independently sweepable mid bells (the 1073 has only one), into an iron-core transformer whose LF flux saturates first for genuine low-end warmth. High/Low set the shelves, LoMid/HiMid the two bell gains, LoMidFreq/HiMidFreq their centres.

Neve33609 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Neve 33609-style diode-bridge bus compressor: smooth glue through a diode-bridge gain cell with control-port slew, a dual-time-constant release and transformer saturation.

NevePre [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Neve 1073-style preamp: a class-A triode gain stage (grid conduction) into an iron-core output transformer whose LF flux saturates first - the bass 'sings' with iron warmth.

OCD [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fulltone OCD-style overdrive: 4x-oversampled asymmetric MOSFET hard clipping (smooth-edged) with a bright/dark voicing switch.

Octavia [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Octavia-style octave-up fuzz: 4x-oversampled full-wave rectification doubles the pitch, then heavy fuzz - the ring-mod-like upper-octave scream.

OD1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss OD-1 OverDrive-style pedal: the original asymmetric soft-clipper (two diodes one way, one the other) for a smooth, warm, even+odd-harmonic break-up - lower gain and rounder than a Tube Screamer, with no mid hump. Drive sets the overdrive, Level the output (the real OD-1 has no tone knob).

ODR1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Nobels ODR-1-style natural overdrive: an op-amp soft-clipper with the signature Spectrum control that scoops the mids while bumping bass and treble together for a transparent, open, amp-like break-up - the Nashville session staple. Drive sets the gain, Spectrum the bass+treble vs mid balance, Level the output.

Odyssey [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ARP Odyssey 2-pole filter (4023): brighter and reedier than the 4-pole ladders, with a loud, aggressive resonance that bites and sings - the thin, vocal ARP lead voice. A soft-saturated 2-pole state-variable filter tapping LP / BP / HP via Mode. Cutoff sweeps it, Reso peaks the corner toward self-oscillation, Drive pushes the input.

Opto [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

LA-2A-style optical compressor: a T4 photocell whose photoresistor gain law gives a soft knee and a ratio that rises with level, plus program-dependent release (recovers slower after sustained compression).

OptoTrem [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender optical tremolo: a lamp drives a photocell whose asymmetric lag (cools slower than it warms) and sublinear I-V curve round the throb; Shape sets the cell response speed (smooth to choppy).

Orange [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Orange OR-series amp (full circuit): mid-forward, thick and woody with more preamp gain and a spongier supply than the Hiwatt - the chunky British crunch. Gain, Tone, Master, Level. Shares the cascaded-triode + voiced-tone-stack + supply-sag circuit model, voiced for Orange's mid push.

Peavey5150 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Peavey 5150 / 6505 amp (full circuit): tight, aggressive, high-gain crunch with a stiff supply for percussive attack - the metalcore / Van Halen rhythm and lead standard. Gain, Tone, Master, Level. Shares the cascaded-triode + voiced-tone-stack + supply-sag model, voiced for the 5150's tight low end and biting upper mids.

Percolator [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Interfax Harmonic Percolator-style fuzz: mixed germanium/silicon asymmetric clipping generates strong even-order harmonics for a wooly, octave-tinged thickness unlike symmetric fuzzes. Balance skews the asymmetry from even-rich to odd-rich. 4x-oversampled.

Phase100 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Phase 100-style 6-stage phaser: six JFET all-pass stages give three evenly-tracking notches - deeper and more resonant than the 4-stage Phase 90 - with a 4-position Intensity that adds feedback resonance for the thick, swirling, peakier sweep. Rate sets the LFO speed, Intensity the notch depth/feedback, Mix the blend.

Phase90 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

MXR Phase 90-style 4-stage phaser: four JFET all-pass stages (bias-dependent frequency shift + transfer-curve saturation) give two evenly-tracking notches; Script (clean) or Block (resonant feedback).

Polivoks [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Soviet Polivoks 2-pole OTA filter: its resonance path clips like a comparator, so at high Reso it does not ring sweetly - it slams into a hard square-wave self-oscillation and screams. A 2x-oversampled two-integrator SVF whose band-pass feedback is hard-clipped (not soft tanh) gives that brutal, buzzy Eastern-bloc character no Moog/SEM filter has. Cutoff, Reso, Drive into the core, and Mode taps LP / BP / HP.

Polysix [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Korg Polysix-style SSM2044 four-pole low-pass: gentle OTA saturation gives a smooth, creamy resonance that stays musical right up to self-oscillation.

Pultec [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Pultec EQP-1A-style passive EQ: low boost and cut use offset corners so they do not cancel (the boost-with-dip trick), plus gentle tube warmth.

Rangemaster [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Dallas Rangemaster-style treble booster: a high-pass into a real germanium common-emitter transistor stage (Ebers-Moll, Newton-free closed form, 4x oversampled) - its asymmetric cutoff/saturation adds the bright, singing even+odd harmonics on top of the dry signal.

RAT [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ProCo RAT-style distortion: an LM308-slewed gain stage into a Newton-solved silicon diode pair clamping to ground (the real shunt clipper, 4x oversampled - not a hard jlimit), tamed by a single low-pass Filter knob.

RectifierSag [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tube-rectifier power-supply sag: a model of the B+ supply drooping under load and recovering with the reservoir cap's RC time. Loud transients momentarily collapse the headroom (duck/compress) then BLOOM back as the supply recovers - the springy, touch-sensitive 'sag' of a tube amp with a 5U4/GZ34 rectifier, distinct from a feed-forward compressor. Drive sets the stage gain, Sag the droop depth, Recover the supply recovery time, Level the output.

Recto [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Mesa Rectifier-style high-gain amp: three cascaded asymmetric triode stages with tight inter-stage high-passing, a deep mid scoop, selectable reservoir sag and an output-transformer stage.

RMX16 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

AMS RMX16-style digital reverb (Nonlin program): a dense, gated tail that holds then cuts off instead of decaying naturally - the classic 80s gated sound.

RossComp [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Ross Compressor-style OTA pedal (CA3080): the smoother, longer-sustaining cousin of the Dyna Comp - a softer knee and slower release give singing sustain with less squash and a more open top end. Sustain sets how hard it squeezes, Attack the detector speed, Level the makeup.

SansAmp [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tech 21 SansAmp-style amp/cab DI: a Character tilt into cascaded asymmetric soft-clipping (4x oversampled) and a speaker-cabinet low-pass, giving the recorded amp tone direct with no mic. Drive sets the gain into the stages, Character tilts presence vs warmth, Level the output.

SCComp [ANA] [CIRCUIT] [FX] [SYNTH] in: In, Sidechain out: Audio

Sidechain compressor: compresses the signal (in 0) by the level of a key input (in 1) through a dB-linear VCA cell with control-port slew and gain-cell THD - cable the host sidechain (SC L/R) to the key for ducking/pumping.

SD1 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Boss SD-1 Super Overdrive: an op-amp gain stage into an ASYMMETRIC soft clipper (two diodes one way, one the other) for a more open, even-harmonic crunch than the symmetric Tube Screamer, with the familiar mid-hump tone that cuts through a band. Drive sets the gain, Tone the mid/treble balance, Level the output.

SDD3000 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Korg SDD-3000-style delay: a characterful input preamp into a modeled converter (ADC anti-alias + 13-bit quantization) - clean and bright with subtle digital grain.

SEM [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Oberheim SEM-style 2-pole state-variable filter: VCA-transistor saturation in the resonance integrator, continuously morphing low-pass through notch to high-pass.

ShadowHills [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Shadow Hills Mastering-style compressor: a photocell opto stage (soft-knee, rising-ratio law, program-dependent release) into a discrete fixed-ratio VCA stage, with a transformer-flavour select (nickel/iron/steel).

SmallStone [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EHX Small Stone-style 4-stage OTA phaser: nonlinear OTA all-pass stages; the Color switch adds feedback, drives the OTAs harder and widens the notch sweep for a more vocal, resonant swirl.

Soldano [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Soldano SLO-100-style amp: three cascaded triode stages for a smooth, saturated lead with singing sustain, reservoir sag and an output-transformer stage.

Solina [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Solina/string-machine-style ensemble: three BBD taps on 120-degree-phased dual LFOs (slow + fast) through one companded bucket-brigade line for a lush, moving multi-voice swirl.

Sontec [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Sontec MEP-250-style mastering parametric EQ: clean Class-A bell bands with a musical, selectable Q - precise without sounding clinical, the mastering standard. Two parametric bells (Lo/Hi freq + gain) sharing the Q control.

SpaceEcho [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Roland Space Echo-style tape delay: wow+flutter-modulated tape transport with magnetic hysteresis (asymmetric saturation + remanence) and tone-shaped head gap loss in the feedback repeats.

SSLBus [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

SSL bus-style VCA glue compressor: a dual-time-constant program-dependent auto-release for pumping-free glue, through a dB-linear VCA cell with control-port slew and gain-cell THD that grows with gain reduction.

SSLE [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

SSL E-series-style 4-band parametric EQ: constant-Q (bandwidth stays put as you boost or cut) for a clean, surgical response.

SSLG [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

SSL G-series-style 4-band parametric EQ: proportional-Q (the more you boost or cut, the narrower the band) for a musical, self-limiting curve.

StaLevel [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Gates Sta-Level-style tube vari-mu limiter: a very slow program-dependent auto-release and gentle ~3:1 leveling through a remote-cutoff tube cell whose 2nd-harmonic warmth (Warmth) grows with gain reduction.

Studer [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Studer A800-style tape: magnetic hysteresis (B/H saturation + remanence), a firm low-frequency head bump and gentle head-gap HF loss - punchy track-level glue (pro deck: flutter negligible by design).

Summit [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Summit TLA-100-style tube/opto leveler: photocell gain law (soft knee, rising ratio) with program-dependent release, transformer warmth and tube makeup.

SunnModelT [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Sunn Model T amp (full circuit): fat, dark, high-headroom and loose - the wall-of-low-end doom / sludge / stoner voice, the opposite of a tight bright amp. Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for huge bass, dark top and a spongy, blooming low end.

SuperFuzz [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Univox Super-Fuzz-style octave fuzz: 4x-oversampled full-wave rectification for the octave-up into hard germanium square clipping, with a switchable 1 kHz mid-scoop notch.

Supro [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Supro 1624T-style octal amp: 6973 power tubes on a tube-rectifier supply with no negative feedback, so loud transients sag the rails - the amp ducks then blooms, breaking up asymmetrically and early. Sag sets how hard the supply droops.

Synthacon [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Steiner-Parker Synthacon-style diode multimode filter: an asymmetric diode-clipped resonance path gives a snarly, vocal bite, and Mode sweeps the output continuously LP -> BP -> HP - rougher than a clean SVF.

TimeCube [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Cooper Time Cube-style delay: a short, dark acoustic-tube delay (sound piped through a garden hose between transducers) for tight doubling and slapback. Time sets the short delay, Feedback the repeats and Mix the blend. Naturally band-limited and woody - characterful on vocals and snare, not a clean digital echo.

Timmy [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Paul C Timmy-style transparent overdrive: low-gain op-amp soft clipping with cut-only Bass and Treble shaped BEFORE the gain (so it tightens and de-fizzes without adding colour) - clean, dynamic, amp-like. Gain sets the drive, Bass/Treble cut, Level the output.

ToneBender [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tone Bender MkII germanium fuzz: three cascaded germanium transistor stages, each asymmetric (germanium clips harder on one rail) with interstage coupling-cap high-pass between them, for the thick, sustaining, slightly-broken vintage fuzz. Bias starves the supply so quiet/decaying notes GATE into the famous spitty sputter; Drive sets the gain, Tone the output roll-off, Level the volume. Distinct from FuzzFace (2 transistors) and Big Muff (4 silicon stages + diodes).

Trident [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Trident A-Range-style console EQ: four fixed musical bands (LF shelf, two mid bells, HF shelf) with an aggressive, forward voicing.

TriStereo [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Dytronics Tri-Stereo Chorus-style ensemble: three bucket-brigade voices (compander + clock-tracked reconstruction) at staggered rates for a lush, wide three-dimensional chorus.

TubePre [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Vintage tube console-style preamp: two cascaded triode stages with an interstage coupling cap (2nd-harmonic-dominant asymmetric clipping) and a warm low shelf.

TubeScreamer [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Ibanez Tube Screamer-style overdrive: a real op-amp feedback diode clipper - the mid-focused clean gain is clamped by an antiparallel Shockley diode pair, Newton-solved and 4x oversampled (not a tanh) - with the famous ~720 Hz mid hump. Asym mismatches the two diodes for the asymmetric-clip mod, adding the even harmonics a symmetric clipper can't.

TubeTech [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Tube-Tech CL 1B-style opto compressor: a photocell gain law (soft knee, rising ratio) with program-dependent release and tube makeup warmth.

Tweed [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fender Tweed 5E3-style amp: two asymmetric triode stages into a physical B+ reservoir sag (droops and blooms under pick attack) and an output-transformer stage - warm, loose, early breakup.

UA610 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Universal Audio 610 tube preamp (Bill Putnam): a 12AX7 valve gain stage between input and output transformers. Push Drive to overdrive the tube into warm asymmetric saturation while Level pads the output – the '610 move' of cranking the valve and taming the volume – with transformer low-end thickening and the 610's Low/High program EQ. Distinct from the generic tube preamp by the transformer saturation and the gain/level interaction.

UniVibe [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Uni-Vibe-style 4-stage phaser: staggered all-pass frequencies driven by an asymmetric lamp-filament LFO through a power-law photocell I-V curve, with nonlinear stages - the watery, throbbing swirl. Chorus or Vibrato.

Urei1176 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

UREI 1176LN-style FET peak limiter: a field-effect transistor as the variable-gain cell gives the near-instant attack (20 us-800 us) no opto or vari-mu matches, with fixed program ratios and the all-buttons-in mode that slams every ratio at once for the aggressive, distorted British crush. Input drives it into reduction, Attack/Release are the (reversed on the real unit) time constants, Ratio selects 4/8/12/20:1 or All, Output is makeup.

V76 [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Telefunken V76-style tube preamp: two cascaded tube stages with interstage coupling into an output transformer - rich, warm 2nd-harmonic saturation with a low-mid body.

VactrolLPG [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Buchla 292-style low-pass gate: a vactrol (LED + photocell) opens a 2-pole low-pass and a VCA together, and the photocell's slow nonlinear light-decay gives the plucky 'bongo' ring as transients fade. Mode picks gate (VCF+VCA), filter-only, or amp-only.

VaractorFilter [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Varactor-tuned filter: a varactor (voltage-controlled capacitance) sets the cutoff, so the cutoff RISES with the instantaneous signal voltage - modulating at audio rate, not via an LFO or envelope. The filter intermodulates the signal with itself, adding gritty, phase-coherent partials no static filter or envelope-wah makes (a clean linear SVF whose tuning is the only nonlinearity). Cutoff sets the base, Depth the varactor sensitivity (self-modulation amount), Reso the resonance, Drive the level into the varactor, Level the output.

VariMu [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Fairchild/Manley-style tube vari-mu compressor: program-dependent ratio rises with level, and a remote-cutoff tube transfer whose genuine 2nd-harmonic warmth grows as gain reduction biases the tube toward cutoff.

Wasp [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

EDP Wasp-style 12 dB multimode filter: the CMOS-inverter gain-stage outputs hard-clip at the rails for a fizzy, gritty bite. Mode 0 LP / 1 BP / 2 HP.

WoolyMammoth [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ZVex Wooly Mammoth-style gated fuzz: a huge, square-wave synth-bass fuzz with the signature Pinch control - a starve/gate that chokes the low-level tail into a spluttery, ripped-speaker gate while keeping a massive low end. Wool sets the fuzz, Pinch the gate/starve, EQ the tone, Level the output.

Zendrive [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Hermida Zendrive-style overdrive: a smooth, mid-focused 'Dumble-in-a-box' with singing sustain and a touch-sensitive, vocal break-up. Gain sets the drive, Tone the top end, Voice the upper-mid focus (the Dumble cut), Level the output.

ZenerClip [ANA] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Back-to-back Zener shunt clipper (Newton-solved, 4x oversampled): two Zeners anode-to-anode clip symmetrically at about +/-3.3 V - far more headroom than a silicon diode pair - so the signal passes clean until it slams into a hard knee. Louder and punchier than soft diode overdrive, the 'stays clean then cracks' character of high-headroom Zener/LED clipping stages. Drive sets the gain into the clipper, Tone a tilt around 1.2 kHz, Level the output.

Delay (53)

AlignDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Alignment delay (Ableton Align Delay-style): a clean dry delay used to time-align tracks or push a signal later by a precise amount - no feedback, no filtering, no colour. Time sets the delay in milliseconds; leave Mix fully wet for pure alignment, or blend for a simple slap double.

BBDDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Bucket-brigade style echo whose delayed tap is passed through a one-pole low-pass and a tanh saturation in the feedback path, so each repeat grows darker and grittier. Time sets the delay length (1-8000 ms), Feedback sets regeneration up to 0.9, Tone sets the high-frequency rolloff of the repeats, and Mix blends the filtered echo with the dry signal. Lower Tone yields the murky, decaying character of analog BBD delays.

BBDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Bucket-brigade delay: emulates an analog BBD chip with input companding, a darkening low-pass on each repeat and gentle saturation, for the warm, lo-fi, increasingly-murky echoes of a vintage analog delay.

Beatmasher [DLY] [FX] [SYNTH] in: Audio, Gate out: Audio

Beat-repeat: while the gate on in1 is held it records a slice of audio on the first pass then loops that captured slice; releasing the gate passes the dry signal through. Length sets the size of the captured/looped slice (20..1000 ms) and Mix blends the looped slice against the dry input. Hold the gate to freeze and stutter a fragment, release to drop back to live audio.

BeatSlicer [DLY] [FX] [SYNTH] in: Audio, Gate out: Audio

Captures a buffer while the gate on in1 is held, divides it into 8 slices, and replays them in one of several reorder patterns; releasing the gate passes dry audio through. Length sets the buffer/loop size (100..2000 ms), Pattern selects the slice reordering (0..4, e.g. straight, swapped pairs, doubled, reversed, interleaved), and Mix blends the sliced output against dry. Hold the gate to scramble a phrase into a rhythmic rearrangement.

Bouncer [DLY] [FX] [SYNTH] in: Audio, Gate out: Audio

A feedback delay whose delay time continually shrinks, so the echoes speed up toward a stutter like a bouncing ball; a gate on in1 re-arms the delay back to its start time. Time sets the initial delay (20..600 ms), Accel sets how fast the time shrinks, Feedback sets the echo regeneration (up to 0.95), and Mix blends wet against dry. Higher Accel collapses the repeats into a rapid buzz, while a gate retrigger resets the bounce.

Chorus [DLY] [FX] [SYNTH] in: Audio out: Audio

Single-voice chorus: an LFO (Rate, Depth) modulates a short base Delay (1-30 ms) and the detuned copy is blended back via Mix to thicken and widen the source. Short delays with light depth give subtle doubling; longer and deeper settings approach vibrato and flange. See Ensemble for a three-voice string-machine variant.

▶▷▷ **Comb** [DLY] [FX] [SYNTH] in: Audio out: Audio

Tuned comb resonator: a short feedback delay set by Freq (20-2000 Hz) with a damping low-pass, which doubles as a Karplus-Strong plucked string when excited by a transient. Feedback sets resonance and sustain while Damp darkens the decay. High feedback rings at the tuned pitch; pair it with noise or an impulse for string and membrane tones.

▶▷▷ **CrossFeedbackDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Cross-coupled delay: two delay lines at incommensurate times each feed the OTHER line's input through a damping filter, so echoes bounce between the two streams and interleave into a denser, evolving pattern than a single tap can make.

▶▷▷ **Delay** [DLY] [FX] [SYNTH] in: Audio, Feedback Mod, Time Mod out: Audio

Feedback delay line (1-8000 ms) with a damping low-pass in the feedback path that progressively darkens each repeat. Time is modulated by in 3 (scaled by Mod) for chorus and flange-style movement, Feedback sets regeneration up to 0.99, and Mix sets dry/wet. Damp at 0 gives bright digital echoes; higher values emulate tape/BBD high-frequency loss.

▶▷▷ **DiffuseDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Feedback delay whose tap is smeared through a 4-stage all-pass diffusion chain, pushing echoes toward reverb. Time sets the delay (1-1000 ms), Feedback sets regeneration up to 0.9, and Diffuse sets the all-pass coefficient that controls how much each repeat is blurred. Mix sets dry/wet; low Diffuse gives clean echoes, high Diffuse dissolves them into a diffuse wash.

▶▷▷ **DubDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Dub-techno echo: a feedback delay whose repeats are progressively band-limited and tape-saturated inside the loop, so each echo gets darker, warmer and more degraded - the endless, dubby Basic-Channel chord-stab echo of dub techno and dub house. Time sets the delay, Feedback how many repeats (high = near-infinite), Tone darkens each repeat (low-pass in the loop), Drive the in-loop tape saturation, and Mix the wet/dry blend. Long Time + high Feedback + low Tone gives the classic sinking dub tail. Distinct from the clean Delay and the wow/flutter TapeDelay - the feedback path is filtered and saturated.

▶▷▷ **DubEcho** [DLY] [FX] [SYNTH] in: Audio out: Audio

Dub echo: each repeat passes through a low-pass and soft saturation in the feedback loop, so echoes get progressively darker and warmer as they decay - the classic dub/reggae spring-into-the-mix delay.

▶▷▷ **DuckDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Feedback delay line whose wet output is attenuated by an envelope follower tracking the dry signal, so repeats drop out while you play and swell up in the gaps. Time sets the delay length (1-8000 ms), Feedback sets regeneration up to 0.9, Duck sets how hard the dry level suppresses the echoes, and Mix blends wet and dry. Keeps echoes from cluttering busy passages while letting them bloom in the silences.

▶▷▷ **DuckingDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Ducking delay: a feedback echo whose wet output is pulled down by an envelope of the dry input, so the repeats stay out of the way while you play and swell up in the gaps - the classic technique for delays that add space without cluttering the mix. Time sets the delay, Feedback the repeats, Depth how hard the dry ducks the wet.

Ensemble [DLY] [FX] [SYNTH] in: Audio out: Audio

Three-voice string-machine chorus: one LFO modulates three delay taps at evenly spread phases, summing into a lush, wide thickening richer than the single-voice Chorus. Rate and Depth set the swirl and Mix the blend. Classic for pads, strings and synth-brass.

FilterDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Filtered multi-tap delay (Ableton Filter Delay-style): three delay taps at spreading times, each band-passed at Color before feedback, so the echoes are tuned and coloured rather than full-range. Time sets the first tap, Spread the spacing of the other two, Color the band-pass centre, Feedback the regeneration, and Mix the wet blend. (Mono sum of the three taps; the device's L/C/R panning needs a stereo rack.)

Flanger [DLY] [FX] [SYNTH] in: Audio out: Audio

Swept-comb flanger: an LFO sweeps a very short delay (0.5-10 ms) mixed with the dry signal to create moving comb notches, with bipolar Feedback resonating the peaks (negative values invert for a hollow tone). Depth sets the sweep width and Mix the blend. Short delays with high feedback give the classic metallic jet-whoosh.

FoldDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Folding echo: a long delay whose feedback path is wavefolded each pass, so every repeat grows brighter and more harmonically dense instead of darker - an inverted, ever-spikier echo tail that a damped delay never produces.

FractalEcho [DLY] [FX] [SYNTH] in: Audio out: Audio

Fractal echo: five taps spaced at successive golden-ratio fractions of the base time, each quieter than the last, so one hit scatters into a self-similar cluster of echoes that bunch up as they fade - a fractal stutter, not an even repeat.

Freeze [DLY] [FX] [SYNTH] in: Audio, Freeze Gate out: Audio

Audio freeze / looper: while in1 is high it captures the recent input and loops a window of it indefinitely (an instant sustain/pad-freeze); while in1 is low it passes the dry signal and keeps recording. Size sets the loop window length, Mix the wet level.

FreqEcho [DLY] [FX] [SYNTH] in: Audio out: Audio

Frequency-shifted echo: a delay line whose feedback is single-sideband frequency-shifted via a polyphase Hilbert pair, so every repeat slides up or down in Hz rather than in pitch. Time sets the delay (1 to 1000 ms), Shift sets the linear frequency offset (-500 to +500 Hz), Feedback (0 to 0.95) sets repeat count, and Mix blends dry/wet. Because the shift is inharmonic, the tail detunes and smears with each pass for metallic, ever-evolving echoes.

Glitch [DLY] [FX] [SYNTH] in: Audio out: Audio

Probabilistic buffer glitcher that records the input and, at each slice boundary, randomly jumps to a captured slice and may play it forward or reversed. Slice sets the segment length in milliseconds and Chance sets the probability that a given slice is repeated or reversed rather than passed through cleanly. Mix blends the glitched playback against the dry signal for stutter, repeat and reverse artifacts.

GrainDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Granular feedback delay that reads overlapping windowed grains from the delay line, pitch-shifting them and scattering their read positions. Time sets the base tap (10-8000 ms), Pitch transposes the grains +/-12 semitones, and Scatter randomizes each grain's offset for a smeared, diffuse cloud. Mix sets dry/wet; the shifted grains are fed back at reduced gain, so repeats drift in pitch and position.

GrainReverse [DLY] [FX] [SYNTH] in: Audio out: Audio

Reverse-grain smear: continuously records the input and reads each short grain backwards through a Hann window, so the sound is chopped into little reversed snippets - a stuttering, sucking, backwards texture from a live signal, no buffer-capture needed.

LagrangeDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Lagrange fractional delay: reads the delay line at a sub-sample position using 4-point (cubic) Lagrange interpolation rather than the usual linear interpolation, so sub-sample delays keep their high-frequency content instead of being dulled - the interpolation used in high-quality tuning, chorus and physical-modeling delay lines. Delay in ms, wet/dry Mix.

Looper [DLY] [FX] [SYNTH] in: Audio out: Audio

Circulating-buffer looper that re-reads a fixed-length loop and either overdubs new input onto it or recirculates it unchanged. Length sets the loop time (100-2000 ms) and Record switches between overdub (input mixed with the looped buffer at half gain) and pure recirculation when off. Mix sets dry/wet; leave Record on to layer takes, turn it off to freeze and play back the captured loop.

Morphagene [DLY] [FX] [SYNTH] in: Audio out: Audio

Tape-splice granular (Make Noise / Soundhack Morphagene): continuously records the in0 input onto a virtual tape, then plays back a Gene (grain) from it. Slide scans the playback position along the reel, Vari sets the varispeed (pitch + direction, reverse below centre), Gene the grain length, Morph the grain overlap/smear, and Mix the blend. Scrub, stretch, reverse and granulate live audio - distinct from the fixed-loop Looper and the cloud-style granular blocks.

Multitap [DLY] [FX] [SYNTH] in: Audio out: Audio

Multi-tap delay: the input is written into a circular buffer and read back at several time-spaced taps that are summed, averaged, and fed back into the buffer. Time sets the base delay in ms, Taps chooses 1 to 6 read points, Feedback recirculates the averaged taps for repeats, Spread pulls the taps toward the longer end for uneven echo timing, and Mix crossfades dry input against the wet sum. Low Spread gives evenly stacked echoes while high Spread spaces them out, and higher Feedback builds longer rhythmic delay trails.

MultiTapDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Multi-tap delay: reads four taps at 1/4, 1/2, 3/4 and 1x of the base time off a single delay line with decreasing levels, building a rhythmic pattern of echoes from one knob.

NoiseModDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Random-warble delay: the delay time wanders on a smoothed random-step modulator that picks a new target at the set rate, giving an unpredictable detuned warble between a chorus and a broken-tape pitch drift.

▶▷▷ **PeakFlanger** [DLY] [FX] [SYNTH] in: Audio out: Audio

A comb flanger whose delay time is swept by a peak envelope follower on the input, so transients drive the flange motion (delay ranges ~0.5..9.5 ms). Soften sets the envelope release time, Feedback sets the comb regeneration (+/-0.95, negative inverts the comb), Amount scales how far each peak sweeps the delay, and Mix blends wet against dry. The Traktor Flanger Pulse character: the jet sweep follows the music's own dynamics rather than a fixed LFO.

▶▷▷ **PingPong** [DLY] [FX] [SYNTH] in: Audio out: Audio

Ping-pong delay for a mono source: a single line is read at time T or 2T so the two outputs hear staggered taps that appear to bounce between speakers, with regeneration fed back through the longer tap. Time sets the base delay, Feedback the number of repeats up to 0.9, and Mix dry/wet. Out picks L or R, where R hears T and L hears 2T to create the alternating bounce.

▶▷▷ **PitchEcho** [DLY] [FX] [SYNTH] in: Audio out: Audio

Pitch-locked echo: the delay time is set to a whole number of the input's detected pitch periods, so the repeats land exactly in phase and reinforce the tone instead of smearing it.

▶▷▷ **Rainmaker** [DLY] [FX] [SYNTH] in: Audio out: Audio

Rhythmic tapped delay (Intellijel / Cylonix Rainmaker): spreads up to Taps echoes across the delay Time in a decaying rhythm, then runs the result through a tuned Comb resonator for ringing, pitched repeats - rhythmic delays, resonant reverbs and string-like tones in one. Time sets the span, Taps the number of echoes, Feedback the regeneration, Comb the resonator pitch/amount, and Mix the wet blend. A 16-tap rhythm engine, distinct from the fixed-tap Multitap.

▶▷▷ **RectDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Octave-rising echo: the feedback is full-wave rectified (DC-removed) each pass, so every repeat jumps an octave and the tail climbs into a shimmering high cloud - a pitched, ascending echo unlike any flat-pitch delay.

▶▷▷ **Reverse** [DLY] [FX] [SYNTH] in: Audio out: Audio

Reverse delay: snapshots a Window-length slice of input and plays it backward on a loop, a glitchy pre-echo/reverse texture, blended via Mix. Short windows give fluttering grains; longer windows reverse recognizable phrases. See ReverseGrain and ReverseVerb for related effects.

▶▷▷ **ReverseDelay** [DLY] [FX] [SYNTH] in: Audio out: Audio

Captures the input into a delay buffer and reads it back newest-to-oldest within a repeating window, so each chunk plays in reverse. Time sets the window length (50-8000 ms) that defines how much audio is flipped per cycle, and Mix blends the reversed playback against the dry signal. Useful for backward-tape swells and disorienting rhythmic stutters.

▶▷▷ **ReverseGrain** [DLY] [FX] [SYNTH] in: Audio, Gate out: Audio

Captures a buffer while the gate on in1 is held, then plays it backward at a pitch-shifted read rate for a reversed, stuttered texture; releasing the gate passes dry audio through. Length sets the captured grain size (50..1500 ms), Pitch transposes the reverse playback rate (+/-12 semitones), and Mix blends the reversed grain against dry. Hold the gate to suck a phrase into a backward swell.

▶▷▷ Rotary [DLY] [FX] [SYNTH] in: Audio out: Audio

Leslie rotary speaker: combines a doppler pitch wobble (modulated short delay) with a phase-offset amplitude tremolo so the tone swirls as the virtual horn spins. Speed sets the rotation rate and Depth its intensity. Iconic on organ, electric piano and guitar - automate Speed for the slow-to-fast chorale ramp.

▶▷▷ Scratch [DLY] [FX] [SYNTH] in: Audio out: Audio

Turntable-scratch effect where a raised-cosine LFO sweeps the delay read head back and forth through recent input. Rate sets the sweep speed (0.1-8 Hz) and Depth sets how far back the head travels (1-200 ms). Mix sets dry/wet; faster Rate and deeper Depth give more pronounced vinyl-scratch wobble.

▶▷▷ Shuffler [DLY] [FX] [SYNTH] in: Audio out: Audio

Beat-shuffler that records the input and, at each segment boundary, may jump its playback to one of a few earlier segment offsets. Segment sets the slice length (50-500 ms) and Chance sets the probability that a new boundary triggers a jump to a randomly chosen prior offset. Mix sets dry/wet; low Chance plays mostly straight, high Chance stutters and rearranges the beat.

▶▷▷ Slapback [DLY] [FX] [SYNTH] in: Audio out: Audio

Slapback echo: a single short delayed repeat (typically 60-150 ms) layered under the dry signal - the snappy rockabilly/vocal doubling slap, no feedback wash. Time sets the delay, Mix the level of the repeat.

▶▷▷ SlapDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Slapback echo: a single short delayed repeat (with optional feedback for a few taps) - the rockabilly/vocal slap that fattens a sound without turning into a wash.

▶▷▷ SliceShuffler [DLY] [FX] [SYNTH] in: Audio out: Audio

Continuous buffer-shuffler (Max-for-Live Buffer-Shuffler style): records the incoming audio one bar at a time and, while recording the next bar, plays the last bar back as Steps slices re-ordered by a Seed-driven permutation, with a fraction of slices (Reverse) played backwards and a 16-step on/off Gate mask muting chosen slots. Length sets the bar in milliseconds, Steps the slice count (4/8/16), Seed the rearrangement (every value a different reproducible shuffle), and Mix blends against dry. Unlike BeatSlicer's five fixed patterns this is a fully addressable per-slice remap that runs without a trigger.

▶▷▷ SpectralTiltDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Spectral-tilt delay: each pass through the feedback loop tilts the spectrum around a pivot, so successive echoes get progressively darker (or brighter) - tape-like high-end decay or a reverse 'opening up' tail, set by one bipolar Tilt knob.

▶▷▷ Stutter [DLY] [FX] [SYNTH] in: Audio out: Audio

Auto-rhythmic buffer repeat: at each Rate cycle it snapshots a Grab-length window and loops it for the cycle, creating stutters and beat-repeat artefacts, with Mix blending against the live signal. Short grabs at high rates give granular buzz; longer grabs repeat recognizable slices. See Beatmasher and BeatSlicer for tempo-locked glitching.

TapeDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Vintage tape echo: a ~1.5 Hz wow modulates the delay time while a Tone low-pass and tanh tape Saturation in the feedback path darken and thicken each repeat. Feedback sets regeneration and Mix the blend. Unlike the clean Delay, repeats degrade and warble musically - raise Wow and Sat for worn-tape character.

TapeFlutter [DLY] [FX] [SYNTH] in: Audio out: Audio

Tape flutter: a short fractional delay whose read head wobbles under three stacked LFOs - slow wow, mid flutter and fast scrape - reproducing the drifting pitch waver of worn tape; no echo, just the wobble.

TapeStop [DLY] [FX] [SYNTH] in: Audio, Stop Gate out: Audio

Tape-stop effect: while in1 is held the playback slows from full speed to a halt over Time, dropping the pitch and smearing into silence - the classic turntable/tape brake. Releasing in1 returns to normal speed. Mix blends against the dry. A performance effect, not a fixed pitch shift.

ThiranAllpass [DLY] [FX] [SYNTH] in: Audio out: Audio

Thiran fractional delay: an integer delay line followed by a first-order Thiran all-pass that adds the sub-sample remainder with a maximally-flat group delay, resolving the delay to a fraction of a sample without the high-frequency loss of linear interpolation. A recursive all-pass method distinct from the FIR Lagrange delay - flat magnitude at the cost of frequency-dependent group delay. Feedback recirculates, Mix blends.

Vibrato [DLY] [FX] [SYNTH] in: Audio out: Audio

Pitch vibrato: an LFO modulates a short delay's read position to bend pitch, output fully wet (no dry blend) for true vibrato rather than chorus. Rate sets the wobble speed and Depth its pitch range. Gentle settings add expression; deep settings give a seasick warble.

WowDelay [DLY] [FX] [SYNTH] in: Audio out: Audio

Wow echo: the delay time drifts under a slow sub-1 Hz wow LFO, so the repeats sag and rise in pitch like a worn tape echo - the warbling, detuning character of an old Echoplex, distinct from the fast flutter of TapeFlutter.

Distortion (173)

AbsShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Odd/even harmonic blender: crossfades between a tanh stage (odd harmonics, symmetric) and a rectifying stage (even harmonics, octave-up), letting you dial the exact odd-vs-even harmonic balance of the distortion.

AdaaCubic [DST] [FX] [SYNTH] in: Audio out: Audio

Anti-aliased cubic clip: the classic cubic soft-clip (1.5x - 0.5x^3, flat past unity) with 1st-order antiderivative anti-aliasing, giving warm odd-harmonic overdrive without the aliasing of the naive cubic.

AdaaFold [DST] [FX] [SYNTH] in: Audio out: Audio

Anti-aliased wavefolder: a sinusoidal folder with 1st-order antiderivative anti-aliasing, so hard driving into many folds generates far less of the harsh aliasing that plagues naive digital folders.

AdaaHardClip [DST] [FX] [SYNTH] in: Audio out: Audio

Anti-aliased hard clip: applies 1st-order antiderivative anti-aliasing to a brick-wall clipper, so the brutal clipping edges generate far less aliasing than a naive hard clip - aggressive yet clean digital clipping.

ADAAsym [DST] [FX] [SYNTH] in: Audio out: Audio

Anti-aliased asymmetric drive: a diode-like curve that saturates the two polarities differently (even + odd harmonics) with 1st-order antiderivative anti-aliasing, so the asymmetry's bright harmonics do not fold back as aliasing.

AdaaTanh [DST] [FX] [SYNTH] in: Audio out: Audio

Anti-aliased tanh: applies 1st-order antiderivative anti-aliasing (ADAA) to a tanh saturator, evaluating the average of the nonlinearity's integral across each sample step - hot drive without the harsh aliasing naive tanh produces.

ALawShaper [DST] [FX] [SYNTH] in: Audio out: Audio

A-law companding: the telecom A-law transfer (linear below 1/A, logarithmic above), which heavily compresses level and lifts quiet detail - a gritty, lo-fi logarithmic saturation distinct from mu-law, with a hard knee at the 1/A breakpoint.

AlgSigmoidShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Algebraic-sigmoid waveshaper: the rational soft-clip x/sqrt(1+(Drive*x)^2), which saturates faster than softsign but more gradually than a hard clip and is cheap to compute (no transcendentals) - a clean, neutral overdrive with a smooth rail approach.

AmpScan [DST] [FX] [SYNTH] in: Audio out: Audio

Amplitude wave-scanner: the instantaneous level indexes a position in a single-cycle waveform, so louder samples read further through the cycle - a 1-D wave-terrain shaper with offset scanning, distinct from a direct sine folder.

AmpSim [DST] [FX] [SYNTH] in: Audio out: Audio

Guitar amp model chaining an asymmetric tanh preamp, a tone-tilt filter and a speaker-cabinet low-pass with output compensation. Drive sets preamp gain (0-48 dB) into the asymmetric clipper, Tone tilts between darker and brighter response, and Cab sets the cabinet cutoff (brighter at higher values). Mix blends the modeled tone against the dry; ranges from edge-of-breakup grit to saturated lead tones.

AnalogClip [DST] [FX] [SYNTH] in: Audio out: Audio

Variable clipper: one knob morphs the clipping knee from a smooth soft-clip to a hard brick wall, with an asymmetry/bias control for even harmonics - a general-purpose analog-flavored clipper covering the whole soft-to-hard range.

Array [DST] [FX] [SYNTH] in: Audio out: Audio

Drawable waveshaper: maps the input at in0 (folded from bipolar into a 0 to 1 index) through a transfer curve drawn in node config via linear interpolation, then back to bipolar, blended by Mix against the dry input. The drawn breakpoints define the input-to-output mapping and Mix sets the wet/dry balance. A diagonal curve is a clean pass-through while steeper or folded curves act as custom distortion and waveshaping.

AsinShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Arcsine waveshaper: maps the clamped input through asin(x)/(pi/2), the inverse of a sine - so it steepens toward the rails and expands the mid-level (the opposite curvature of a sine soft-clip), brightening and adding odd harmonics as Drive pushes into the clamp.

AsymFold [DST] [FX] [SYNTH] in: Audio out: Audio

Asymmetric triangle wavefolder: positive and negative excursions fold against a biased threshold for even-harmonic content, with a leaky DC servo that recenters the folded output.

AsymFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Asymmetric fuzz: a bias offset pushes the signal into a hard clip that ceilings the two polarities at different levels, the gated, sputtery, even-harmonic snarl of a starved-bias fuzz pedal.

AsymSat [DST] [FX] [SYNTH] in: Audio out: Audio

Asymmetric saturator: positive and negative half-cycles are tanh-driven by independent amounts, producing even-harmonic, tube-like asymmetry you dial per polarity.

AsymWrap [DST] [FX] [SYNTH] in: Audio out: Audio

Asymmetric wrap/fold: positive swings WRAP around a ceiling (sawtooth wraparound -> a brash, octave-rich buzz) while negative swings soft-clip with tanh. The deliberate lopsidedness throws strong even harmonics and a glitchy upper octave that symmetric folders never produce. Drive pushes it, Ceiling sets the positive wrap point, Mix blends.

AuralExciter [DST] [FX] [SYNTH] in: Audio out: Audio

Aural exciter: high-passes the signal, generates new upper harmonics through a soft nonlinearity, high-passes again and adds them back - restoring air and presence without an EQ boost.

BentIdentityShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Bent-identity shaper: passes the signal through (sqrt(x^2+1)-1)/2 + x, a curve that is almost a straight line but bends slightly and asymmetrically around zero - so it stays near-transparent while adding a touch of low-order (mostly even) harmonic warmth. A subtle console-style colour rather than an overt distortion. Drive sets the bend amount.

BEOD [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Friedman BE-OD-style high-gain overdrive: a hot op-amp gain stage into a tight asymmetric silicon diode clipper, voiced like the BE-100 amp's lead channel - saturated, focused, amp-in-a-box drive.

BiasBreath [DST] [FX] [SYNTH] in: Audio out: Audio

Breathing bias saturator: the signal envelope drives a DC offset into the input before a tanh stage (and removes it after), so the harmonic asymmetry swells with playing dynamics - quiet stays clean, loud goes asymmetric.

BiasStarve [DST] [FX] [SYNTH] in: Audio out: Audio

Bias starve: a starved-plate, dying-battery fuzz - low supply voltage chokes the gain, so notes splat, sputter, gate and sag with a sick, broken-up 'velcro' decay that collapses as it starves. Starve sets how low the supply runs, Drive the gain into it, Mix the blend.

Bitcrush [DST] [FX] [SYNTH] in: Audio out: Audio

Digital degradation: Bits quantizes the amplitude to 2^Bits levels (quantization noise) while Downsample holds each sample for N frames (sample-rate reduction and aliasing). Mix blends against the dry signal. Low Bits gives gritty lo-fi crunch; high Downsample adds metallic aliasing for retro and 8-bit textures.

BitLogic [DST] [FX] [SYNTH] in: Audio out: Audio

Bitwise logic shaper: quantizes the sample to an integer and combines it with a bit-shifted copy of itself via AND/OR/XOR (the bytebeat trick), producing harsh digital staircase artifacts no analog circuit makes.

BitParity [DST] [FX] [SYNTH] in: Audio out: Audio

Parity fuzz: quantizes the sample and inverts its polarity whenever the parity (XOR of all its bits) is odd - so the waveform is flipped on a fine, signal-dependent bit pattern, generating a dense, dirty, ring-mod-like fuzz that tracks the amplitude in a chaotic, digital way. Bits sets the resolution, Mix the blend.

BitReverse [DST] [FX] [SYNTH] in: Audio out: Audio

Bit-reversal scrambler: quantizes the sample and reverses the order of its bits, so the most and least significant bits swap roles - a violent, structured digital distortion with no analog analogue.

BitRotate [DST] [FX] [SYNTH] in: Audio out: Audio

Bit rotate: quantizes to a byte then circularly ROTATES the bits, so the most-significant (loudest) bits wrap down into the least-significant places - a violent, non-monotonic remapping that turns smooth ramps into buzzing digital staircases. Distinct from bit-crushing: no information is dropped, it's scrambled. Bits sets the rotation amount, Mix the blend.

BitStick [DST] [FX] [SYNTH] in: Audio out: Audio

Stuck-bit glitch: instead of quantizing every sample, the output randomly STICKS at its previous quantized value and only refreshes with probability (1 - Stick) - like a failing converter with stuck pixels, freezing little plateaus of the waveform into a jittery, broken digital texture. Steps sets the resolution, Stick the freeze probability, Mix the blend.

Bitwise [DST] [FX] [SYNTH] in: Audio out: Audio

Bitwise mangler / databender (a first as a synth block): converts each sample to a 16-bit integer and applies a raw bit operation - XOR, AND, OR or bit-rotate - against Mask, then back to audio. Where Bitcrush only quantises the amplitude, this rips into the actual bit pattern: flipping a high bit folds the waveform, toggling low bits adds fizzing digital grit. Op picks the operation, Mask the bits to act on, Mix the blend. Pure circuit-bent digital destruction with no analogue equivalent.

CabSim [DST] [FX] [SYNTH] in: Audio out: Audio

Generic speaker-cabinet voicing: a 90 Hz high-pass trims sub rumble, a high-frequency rolloff tames fizz, and a peak around 1.5 kHz adds cabinet body, all blended by Mix. Body sets how much of the resonant midrange peak is added and Bright raises the rolloff corner for a more open top end. A lightweight tone-shaping voicing for amp signals, not an impulse-response capture of a real cabinet.

CapLeak [DST] [FX] [SYNTH] in: Audio out: Audio

Capacitor leak: a slow DC bias leaks in proportional to the recent signal and bleeds away in silence, drifting the operating point so the clipping turns lopsided and wanders - the bias-creep of a leaky coupling capacitor that never quite settles. Amount sets the leak, Mix the blend.

ChaosClip [DST] [FX] [SYNTH] in: Audio out: Audio

Chaos clipper: a hard clipper whose ceiling is jittered every sample by a logistic chaos map, so the clip level dances unpredictably between tight and loose - the distortion shimmers and crackles with a living, never-repeating edge instead of a static wall. Drive sets the push, Chaos the ceiling jitter, Mix the blend.

ChaosCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Chaotic bitcrusher: a logistic-map chaos generator wobbles the quantizer's bit depth on every decimated sample, so the crush noise floor wanders unpredictably instead of sitting at a fixed grit - a living, never-repeating lo-fi.

ChaosDrive [DST] [FX] [SYNTH] in: Audio out: Audio

Chaos-modulated drive: a logistic-map iterator nudges the saturation gain every sample, so the distortion shimmers and never sits perfectly still - controlled instability between clean and crushed.

ChaosFold [DST] [FX] [SYNTH] in: Audio out: Audio

Chaotic wavefolder: the fold density is steered every sample by a logistic chaos map running in its turbulent regime, so the harmonics shimmer and scramble unpredictably - a folder that never repeats, between a controllable warble (low Chaos) and a fully scrambled metallic roar (high Chaos). Drive sets the input gain, Chaos the strangeness of the modulation, Mix the blend.

Character [DST] [FX] [SYNTH] in: Audio out: Audio

Character drive with five voicings (Mode): Push (soft clip), Heat (S-clip), Soar (sine fold), Howl (triangle fold) and Shred (wrap fold), each driven by Drive and blended via Mix. One module spanning warm saturation to wild folding distortion. Modulate Mode or Drive for evolving textures.

ChebyBank [DST] [FX] [SYNTH] in: Audio out: Audio

Chebyshev harmonic bank: generates harmonics 2..5 simultaneously via Chebyshev polynomials and mixes them by a spread control, sculpting a precise additive harmonic spectrum from a single waveshaper.

ChebyShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Chebyshev harmonic shaper: a Chebyshev polynomial of order N maps a sine input to its Nth harmonic exactly, letting you dial a specific overtone (2nd..6th) instead of a generic distortion smear.

Chebyshev [DST] [FX] [SYNTH] in: Audio out: Audio

Chebyshev-polynomial shaper that synthesizes a single chosen Harmonic (2nd-8th) from the input via the T_n recurrence, blended by Amount and Mix. Unlike broadband clipping it adds one targeted overtone - the 2nd for octave thickness, higher orders for metallic ring. Pairs well with a clean sine input for precise harmonic generation.

Clip [DST] [FX] [SYNTH] in: Audio out: Audio

Clipper that morphs from soft (tanh) to hard clipping at the Threshold via Hardness, with Output trim and dry/wet Mix. Soft adds rounded saturation harmonics; hard adds bright odd harmonics and caps peaks dead. Use as a tone shaper or a transparent peak ceiling ahead of a limiter.

Corrode [DST] [FX] [SYNTH] in: Audio out: Audio

Corrode: a bit-depth that DROPS the longer the signal is sustained loud, so a held note progressively decays into a coarse, quantized, corroded version of itself, then recovers resolution in the gaps. Rate sets how fast it corrodes, Mix the blend.

CoshShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Cosh expander: sign(x)(cosh(|x|drive)-1), normalized, which grows faster than a square so loud signal is pushed up hard while quiet signal is suppressed - an aggressive upward-expansion shaper that adds odd harmonics and exaggerates dynamics, the opposite of a soft clip.

CrossDist [DST] [FX] [SYNTH] in: Audio out: Audio

Multiband distortion: two crossovers split the signal into low, mid and high bands that are each saturated independently before recombining, so you can grind one band hard while keeping the others clean.

CrossFold [DST] [FX] [SYNTH] in: In A, In B out: Audio

Cross-modulated wavefolder: the fold threshold is set live by a second input, so one signal carves the folding ceiling of another - audio-rate fold-threshold modulation for spectra that move with the modulator.

CrossoverDist [DST] [FX] [SYNTH] in: Audio out: Audio

Crossover distortion: introduces a dead-band notch around zero like an underbiased class-B push-pull amplifier, adding the gritty, buzzy artefact that appears on low-level signal near the zero crossing.

CrushFold [DST] [FX] [SYNTH] in: Audio out: Audio

Crush-then-fold: the signal is bit- and sample-rate-crushed, then the stepped result is driven into a triangle wavefolder, so the quantization steps become hard fold edges - a harsh, aliased, digital-destruction timbre neither stage makes alone.

Decimator [DST] [FX] [SYNTH] in: Audio out: Audio

Bit-crusher + sample-rate reducer: quantises the in0 signal to Bits resolution and holds each sample for Downsample steps, for everything from gentle lo-fi grit to full 8-bit / telephone destruction. Mix blends against the clean signal. Distinct from Bitcrush in adding the sample-rate-reduction (aliasing) stage.

DeltaSigma [DST] [FX] [SYNTH] in: Audio out: Audio

Delta-sigma crush: re-quantizes the audio to a single bit with first-order noise shaping (an error-feedback modulator), turning the signal into a dense +/-1 pulse-density stream like a 1-bit DSD converter - harsh, grainy, with the quantization noise pushed up high. Then a low-pass recovers a gritty version. Tone sets the recovery filter, Mix the blend.

Diode [DST] [FX] [SYNTH] in: Audio out: Audio

Asymmetric diode clipper that drives the input into an exponential soft-knee saturation, shaping the positive and negative halves differently. Drive sets the input gain in dB into the nonlinearity, while Bias increases the curvature of the negative half for stronger asymmetry and added even harmonics. Mix blends the clipped signal against the dry input for germanium/silicon-style warmth and grit.

DiodeClip [DST] [FX] [SYNTH] in: Audio out: Audio

Antiparallel-diode clipper: the exponential Shockley curve (1 - e^-|x|) soft-clips with the smooth, rounded knee of a real diode pair rather than a hard or tanh limiter.

DomainFlip [DST] [FX] [SYNTH] in: Audio out: Audio

Domain flip: a hysteretic staircase quantizer modelling magnetic domains that only flip once the input pushes far enough past the current level - so the output sticks on a step then jumps, tracing the jagged Barkhausen staircase rather than the smooth signal. Step sets the flip threshold, Mix the blend.

Drive [DST] [FX] [SYNTH] in: Audio out: Audio

Waveshaping overdrive with three curves (Type: tanh soft-clip, sine fold, or hard clip) driven by up to 48 dB of gain plus a Bias offset for even-harmonic asymmetry. A post Tone low-pass tames fizz, Mix blends dry/wet, and Output trims the final level. Spans subtle warmth to aggressive distortion depending on Drive and curve.

DrumBuss [DST] [FX] [SYNTH] in: Audio out: Audio

Drum-bus processor (Ableton Drum Buss-style): a one-stop chain for drum groups - Transient sharpens or softens the attack, Drive adds tube grit, Crunch a harder harmonic edge, and Boom adds a tuned resonant low-end thump at BoomFreq. Mix blends it back. Glues and beefs up a drum bus in one block.

DualBandDrive [DST] [FX] [SYNTH] in: Audio out: Audio

Two-band saturator: a single crossover splits low and high bands that are tanh-driven independently before recombining, so you can grit the highs without muddying the lows (or the reverse).

DynSat [DST] [FX] [SYNTH] in: Audio out: Audio

Dynamic saturation: the drive amount tracks the input envelope, so louder passages distort more and quiet ones stay clean (Sens sets how strongly). Adds harmonics on transients and peaks without muddying sustained low-level material. Good for program-dependent grit on drums and bus.

EluShaper [DST] [FX] [SYNTH] in: Audio out: Audio

ELU waveshaper: the exponential-linear-unit transfer passes positive signal linearly but bends negative excursions through alpha*(e^x-1) toward a soft floor at -Alpha - a strongly asymmetric saturation that adds even harmonics and a tube-like one-sided compression.

EnvBitcrush [DST] [FX] [SYNTH] in: Audio out: Audio

Dynamic bit-crusher: the quantizer's bit depth rides the input envelope, so loud passages stay clean and quiet tails crumble into lo-fi grit (or the reverse) - level-dependent digital decay.

EnvCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Envelope-driven bit-crusher: the bit depth tracks the input LEVEL, so loud passages stay clean and quiet passages dissolve into gritty quantization - the opposite of how digital normally behaves, like a dying battery or a sample-player running out of headroom on the tails. Floor/Ceil set the bit-depth range, Sens how fast the level pulls it, Mix the blend.

EnvFold [DST] [FX] [SYNTH] in: Audio out: Audio

Envelope-driven wavefolder: the drive into a triangle folder rises with the input envelope, so quiet passages stay clean and loud ones fold into ever-brighter harmonics - dynamics-dependent folding with a DC servo.

Erosion [DST] [FX] [SYNTH] in: Audio out: Audio

Noise-modulation distortion: sprays band-limited noise (Freq sets its colour) onto the signal by multiplying it with the input, eroding the tone into gritty granular texture. Amount sets intensity and Mix dry/wet. From subtle dirt to harsh destruction - a creative degrade rather than a clean overdrive.

Exciter [DST] [FX] [SYNTH] in: Audio out: Audio

Harmonic exciter: splits off the band above Freq, generates new upper harmonics from it with a tanh Drive, and blends them back via Amount for added air and presence. Subtle settings add sheen and intelligibility without raising broadband level. Most effective on dull vocals, drums and full mixes.

ExpShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Exponential waveshaper: maps the signal through an exponential transfer curve so small signals stay small and large ones swell sharply, an aggressive convex distortion unlike the concave tanh.

FeedbackSaturator [DST] [FX] [SYNTH] in: Audio out: Audio

Feedback saturation: the saturated output is low-passed and fed back into the saturator's input, so the distortion compounds on itself and the harmonics bloom and self-bias the way a cranked amp with a feedback loop does - richer and more compressed than a single tanh stage.

FeedbackShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Self-modulating waveshaper: the block's OWN previous output is fed back into the shaping curve - y = tanh(Drivein + Feedbacky_prev) - so the distortion bends itself every sample. From warm saturation up into chaotic screaming as Feedback rises; a living, unstable fuzz no static waveshaper can make. Drive sets gain, Feedback the self-modulation, Mix the blend.

FloatCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Floating-point crusher: quantizes the MANTISSA at the signal's own exponent, so the quantization step scales WITH the level (like low-bit floating-point audio) - the noise floor rides up and down with the signal instead of sitting at a fixed level as in a normal (linear) bit-crusher. A subtler, more 'alive' digital grit. Mantissa sets the kept bits, Mix the blend.

FoldCascade [DST] [FX] [SYNTH] in: Audio out: Audio

Cascaded wavefolder: drives the signal through several fold stages at progressively tighter thresholds, multiplying the fold density into a dense, West-Coast complex-oscillator timbre.

FoldSine [DST] [FX] [SYNTH] in: Audio out: Audio

Phase-warped sine folder: folds through sin() like a sinusoidal folder but pre-warps the phase with a second-harmonic term, breaking the pure odd-harmonic symmetry to give a brighter, asymmetric fold spectrum.

FractalFold [DST] [FX] [SYNTH] in: Audio out: Audio

Recursive folder: the folded output is fed back into the folder's input, so each sample folds against a memory of the last fold - the harmonics build into a self-similar, evolving, fractal-like spectrum that a single-pass folder cannot make.

Fuzz [DST] [FX] [SYNTH] in: Audio out: Audio

High-gain fuzz: heavy drive soft-saturates then hard-clips into a square wave for a thick, sustaining buzz, with a Tone low-pass to tame the fizz and Level to set output. Mix blends dry/wet. Aggressive and compressed - see the BigMuff and FuzzFace analog models for circuit-specific fuzz voicings.

GatedFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Gated fuzz: a hard fuzz clip followed by a fast noise gate, so the fuzz collapses into stuttering, velcro-ripping decay as notes die - the dying-battery / starved fuzz sound.

GeluShaper [DST] [FX] [SYNTH] in: Audio out: Audio

GELU waveshaper: the Gaussian-error-linear-unit transfer 0.5x(1+tanh(sqrt(2/pi)(x+0.0447x^3))), which gates the signal by an approximate Gaussian CDF - a soft, slightly dipping asymmetric saturation that fades small signals and passes large ones, a gentler-than-Mish self-gate.

GeoFold [DST] [FX] [SYNTH] in: Audio out: Audio

Geometric folder: a wavefolder whose fold spacing is logarithmic rather than linear, so the harmonics it adds are spread geometrically (octave-like) instead of evenly - a brighter, bell-ier, more musical fold than the usual triangle/sine wrap. Drive sets how deep it folds, Mix the blend.

GompertzShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Gompertz waveshaper: shapes the signal through the asymmetric double-exponential sigmoid $2e^{(-e}(-Drivex))-1$, which approaches its lower and upper rails at different rates - unlike the symmetric tanh, this lopsidedness injects strong even harmonics and a tube-like asymmetry. A distinctly coloured saturation; a transfer-curve use of the Gompertz growth function.

GravityQuant [DST] [FX] [SYNTH] in: Audio out: Audio

Magnetic quantizer: instead of hard-snapping to a grid, the output is PULLED toward the nearest grid level by a continuous spring (Strength). At low Strength it just nudges the signal toward the steps for a subtle pitch/level magnetism; at full Strength it becomes a hard quantizer. The in-between gives a soft, gravitational stair-step. Levels sets the grid, Strength the pull.

GrayFold [DST] [FX] [SYNTH] in: Audio out: Audio

Gray-code folder: quantizes the sample and XORs it with its own Gray code (v ^ (v»1)), refracting the waveform into a self-similar staircase of digital harmonics.

GridConduction [DST] [FX] [SYNTH] in: Audio out: Audio

Grid conduction: when the drive swings positive past the bias the tube's grid starts drawing current and clamps that half hard, while the negative half stays soft - the asymmetric blocking distortion of an overdriven valve grid that gives tube amps their compressed, sagging bite. Drive sets the push, Bias where conduction starts, Mix the blend.

GrowlBox [DST] [FX] [SYNTH] in: Audio out: Audio

Growl box: at random zero crossings the polarity of an added sub-octave FLIPS, so the period randomly doubles and the tone breaks into a gnarly, sputtering sub-harmonic growl/multiphonic - the controlled chaos of a fuzz on the edge of tracking. Amount sets the sub level, Chaos how often it flips (smooth octave-down at 0, chaotic growl high), Mix the blend.

GudermannianShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Gudermannian waveshaper: maps the input through the Gudermannian gd(x)=2*atan(e^x)-pi/2, a transcendental soft-clip with a gentler knee than tanh and a softer approach to the rails - a smooth, transparent saturation that adds subtle odd harmonics without harsh edges.

HarmonicMul [DST] [FX] [SYNTH] in: Audio out: Audio

Harmonic multiplier: passes the signal through the Nth Chebyshev polynomial, which maps a sine to its exact Nth harmonic - a precise single-harmonic generator (octave at N=2, fifth-of-octave at N=3, etc.) for additive coloring.

HermiteShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Hermite waveshaper: maps the input through a probabilist's Hermite polynomial He_n of selectable order, generating a specific harmonic (like Chebyshev shaping but on the Hermite basis), then tanh-limited - a precise, order-selectable harmonic adder.

HysteresisQuant [DST] [FX] [SYNTH] in: Audio out: Audio

Hysteresis quantizer: a stepped quantizer with a DEADBAND - it holds its current step and only jumps to a new one once the input moves past the deadband, so dithery, wandering signals stop chattering between adjacent steps. Glitch-free stair-stepping with a sticky, lagged digital character. Steps sets the resolution, Deadband the stickiness, Mix the blend.

HystShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Backlash distortion: the output only moves once the driven input pulls beyond a deadband around its last position, modelling mechanical play / relay hysteresis - it sticks then jumps, adding memory-dependent grit unlike any clip or fold.

LangevinShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Langevin waveshaper: shapes the signal through the Langevin function L(x)=coth(x)-1/x, the saturation curve of paramagnetic and ferrofluid magnetization - a sigmoid that approaches the rails more gently than tanh and with a softer knee, giving a smooth, magnetic-style saturation. Normalized so unity input maps to full scale at the chosen Drive.

LockhartFold [DST] [FX] [SYNTH] in: Audio out: Audio

Lockhart wavefolder: the closed-form transfer curve of the Lockhart transistor folding cell (the heart of many West-Coast folders), giving smooth, analog-accurate fold harmonics rather than a piecewise reflection.

Lofi [DST] [FX] [SYNTH] in: Audio out: Audio

Lo-fi degrader: sample-rate reduction (holds each sample for Rate frames) plus a Tone low-pass and added tape Hiss, blended via Mix. Unlike Bitcrush it quantizes in time rather than bit-depth, layering filtering and noise for a warmer vintage-sampler grime. Push Rate for aliasing; lower Tone and raise Hiss for cassette/SP-style character.

LogShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Logarithmic shaper: a log transfer curve lifts quiet detail steeply and compresses loud peaks, the opposite curvature of the exponential shaper - a bright, dense, low-level-emphasizing distortion.

LorenzDrive [DST] [FX] [SYNTH] in: Audio out: Audio

Lorenz-modulated drive: the Lorenz attractor's wandering x coordinate continuously bends the gain of a tanh saturator, so the distortion character drifts chaotically and organically as the signal passes through.

LPCWhiten [DST] [FX] [SYNTH] in: Audio out: Audio

Linear-prediction whitener: a one-tap adaptive predictor (NLMS) continuously guesses the next sample from the last one and outputs only the prediction ERROR, so the predictable, resonant part is subtracted away and what's left is the flat, whitened, edgy residual - instant presence and de-resonance. Amount sets how much residual replaces the signal, Mix the blend.

Magnetize [DST] [FX] [SYNTH] in: Audio out: Audio

Magnetize: a saturator with magnetic HYSTERESIS - the transfer curve lags differently on the rising and falling edges, tracing a loop like tape magnetization, which adds warmth, a soft memory smear and even-order colour the way a real magnetic medium does. Drive sets the level into the loop, Mix the blend.

MishShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Mish waveshaper: the self-gated neural activation x*tanh(ln(1+e^x)) used as a transfer curve - nearly linear for loud positive signal but with a smooth non-monotonic dip just below zero, adding a soft asymmetric saturation with subtle even harmonics distinct from tanh.

MorningGlory [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

JHS Morning Glory-style transparent overdrive: a low-gain symmetric LED-clipper drive that stays open, dynamic and amp-like, cleaning up with the guitar volume - a pedalboard always-on.

MuLaw [DST] [FX] [SYNTH] in: Audio out: Audio

Mu-law compander: the logarithmic mu-law curve from telephone codecs squashes the dynamics nonlinearly, adding the gritty, level-dependent quantization grain of an 8-bit phone line.

Mulholland [DST] [FX] [SYNTH] in: Audio out: Audio

Tanh overdrive feeding a resonant Chamberlin state-variable lowpass, with a slow internal LFO wobbling the cutoff. Drive sets the input gain into the saturator (1..30), Cutoff sets the filter frequency (~80 Hz..14 kHz), Reso sets the filter Q, Mod sets how much the LFO modulates cutoff, and Mix blends wet against dry. Goes from gentle filtered grit to a screaming self-resonant squeal as Drive and Reso climb.

MultibandFold [DST] [FX] [SYNTH] in: Audio out: Audio

Two-band wavefolder: a crossover splits the signal into low and high bands that are wavefolded with separate drive amounts, so the bass can stay clean while the highs shred (or vice versa) - frequency-selective folding that a full-band folder cannot do.

MultiDist [DST] [FX] [SYNTH] in: Audio out: Audio

Three-band distortion: one-pole filters at 300 Hz and 3 kHz split the input into low, mid and high bands, each saturated through its own tanh stage before being summed. Low, Mid and High set the drive into each band's saturator and Mix blends the result against dry. Per-band drive lets you grind the highs without muddying the lows, or add weight to bass alone.

Munge [DST] [FX] [SYNTH] in: Audio out: Audio

Munge: a digital mangler that flips between a small bank of per-sample operations (clean, invert, full-wave rectify, sine fold) on a random clock, so the waveform is mauled into a glitchy, ever-shifting destruction. No two passes mangle the same way. Rate sets how fast the operation switches, Amount the wet level, Mix the blend.

NoiseShapeQuant [DST] [FX] [SYNTH] in: Audio out: Audio

Noise-shaped quantizer: crushes the sample value to N levels but feeds the quantization error forward into the next sample, pushing the crush noise up out of the band a plain bit-crusher leaves it in.

Nroot [DST] [FX] [SYNTH] in: Audio out: Audio

Signed nth root: sign(x)*|x|^(1/N), which lifts low-level signal toward unity while leaving the sign intact - so quiet detail is brought up (the opposite of squaring), brightening and thickening as N rises, a continuously-tunable root-law shaper.

OctaveDivider [DST] [FX] [SYNTH] in: Audio out: Audio

Analog-style sub-octave: a flip-flop toggles on every rising zero crossing of the input, producing a square one octave below that tracks the pitch; scaled by the input's own envelope so it follows dynamics, then blended under the dry signal for a thick sub.

OctaveFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Octave fuzz: blends a fuzz-distorted signal with a DC-corrected full-wave-rectified upper octave, the classic Hendrix/Octavia scream. Drive sets gain, Octave the upper-octave blend and Mix dry/wet. Tracks best on clean single notes near the 12th fret; chords intermodulate, by design.

OctFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Octave fuzz: blends a straight hard fuzz with a full-wave-rectified fuzz (an octave above), the classic Octavia/octave-up fuzz voice that screams an octave over the note when pushed.

OverdriveOS [DST] [FX] [SYNTH] in: Audio out: Audio

Oversampled overdrive: linearly upsamples 2x, applies tanh saturation at the higher rate and averages back down, so the alias products generated by the distortion fall mostly above the audible band - cleaner high-gain tone.

ParabolicSatShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Parabolic-saturation waveshaper: passes the signal through x*(2-|x|) up to unity then hard-limits, so the approach to the rails is a smooth parabola rather than a tanh curve - a punchy soft-clip with a distinct second-harmonic-tinged knee, then a clean ceiling.

PhaseBurn [DST] [FX] [SYNTH] in: Audio out: Audio

Phase burn: a first-order all-pass whose coefficient is driven by the SIGNAL ITSELF (a shaped, level-dependent phase shift), so the waveform's phase smears and self-warps more as it gets louder - a frequency-dependent, dynamic phase distortion that adds a hollow, vowel-like movement and grit on transients without changing the amplitude curve. Drive scales the modulation, Amount the burn depth, Mix the blend.

Polarize [DST] [FX] [SYNTH] in: Audio out: Audio

Polarize: the asymmetry of a soft clipper is slowly swept by an LFO, so the bias drifts and the generated even harmonics wax and wane over time - the tone breathing between symmetric (odd-only) and lopsided (even-rich) saturation. Rate sets the sweep, Depth the bias swing, Mix the blend.

PowerShape [DST] [FX] [SYNTH] in: Audio out: Audio

Power-law shaper: raises the signal to a variable exponent (preserving sign), so below 1 it expands quiet detail and rounds the peaks while above 1 it contracts, hollowing the body and sharpening the spikes - a continuous control over the whole transfer curve's shape from soft-and-fat to hard-and-thin. Exp sets the exponent, Mix the blend.

PowShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Power-law shaper: raises the signal's magnitude to a variable exponent (preserving sign), sweeping the transfer curve from expanding/concave below 1 to compressing/convex above - continuous control of harmonic content.

Pulverize [DST] [FX] [SYNTH] in: Audio out: Audio

Pulverize: the signal is run through a stack of sine wavefolders in series, each one pulverizing the last, so a clean tone is ground into a dense, buzzing cloud of high harmonics. The more stages, the finer the powder. Drive sets the input level, Stages the grind depth, Mix the blend.

RectFold [DST] [FX] [SYNTH] in: Audio out: Audio

Rectify-then-fold: full-wave rectification followed by triangle folding stacks octave-up rectifier content with fold harmonics for a dense, aggressive shaper unlike either stage alone.

RectFuzzGate [DST] [FX] [SYNTH] in: Audio out: Audio

Gated rectifier fuzz: full-wave rectification feeds a hard tanh fuzz (octave-up, even harmonics) that only opens when the envelope crosses a threshold, with a touch of crackle on the gate - a spitty, gated octave-fuzz.

Rectify [DST] [FX] [SYNTH] in: Audio out: Audio

Rectifier waveshaper: full-wave (Mode 0) folds the negative half upward to add strong octave-up and even harmonics, while half-wave (Mode 1) keeps only the positive half. Mix blends with the dry signal. The classic octave-fuzz colour - aggressive alone, useful in small amounts for harmonic lift.

RectMorph [DST] [FX] [SYNTH] in: Audio out: Audio

Rectifier morph: sweeps continuously from half-wave to full-wave rectification, generating octave-up and even-harmonic content; dry blend keeps the fundamental.

ResonantFold [DST] [FX] [SYNTH] in: Audio out: Audio

Resonant folder: a state-variable low-pass with adjustable resonance feeds a triangle wavefolder, so the resonant peak is driven into folding and rings into a screaming formant of fold harmonics - a vocal, whistling distortion that tracks the cutoff.

RingFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Ring fuzz: ring-modulates the input against an internal sine, then slams the result into a hard fuzz clip, stacking clangorous inharmonic sidebands with gated fuzz spit for a destroyed, robotic tone.

Riot [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Suhr Riot-style distortion: a high-gain asymmetric-clipper drive that nails a tight, saturated, amp-like high-gain rhythm and lead - JCM800-into-a-box aggression with a smooth top.

Roar [DST] [FX] [SYNTH] in: Audio out: Audio

Feedback waveshaper (Ableton Roar-style): drives the signal through a selectable saturation curve, but wraps it in a resonant feedback loop - the distorted output is band-passed at Color and fed back into the input, so it rings, blooms and self-oscillates into the gnarly 'roar' that plain saturation can't make. Drive sets the push into the Shape (soft sine / tube / soft clip / diode / fold), Tone tilts the spectrum before the shaper, Feedback how much output returns, Color the feedback resonance frequency, and Mix the dry/wet. (Single-shaper core; the full device's serial/parallel/multiband/mid-side routings need a rack.)

Rumble [DST] [FX] [SYNTH] in: Audio out: Audio

Techno rumble bass: the input on in0 (a kick) drives an envelope follower that swells two slightly detuned sub-sines running underneath it, so a low rolling sub locks to the kick pattern and the slow Release lets the rumble roll on between hits - the Drumcode-style rumble built by sub-under-kick rather than a reverb send. Tune sets the sub pitch, Detune beats the two oscillators against each other for movement, Release sets how long the rumble tail rolls, Drive saturates it for weight, and Mix blends the rumble under the dry input.

SagClip [DST] [FX] [SYNTH] in: Audio out: Audio

Sag clipper: the clipping ceiling DROPS as the running level rises, modelling a power supply that sags under load - so sustained loud passages compress and clip harder while transients punch through before the rail collapses. A living, dynamic distortion that breathes, unlike a fixed clipper. Drive pushes it, Sag how far the rail collapses, Mix the blend.

SampleRot [DST] [FX] [SYNTH] in: Audio out: Audio

Sample rot: the effective sample rate decays the longer a note sustains - the decimation deepens over a held tone so it grows progressively more aliased and gritty, then refreshes in the gaps. A time-axis companion to bit-rot, on the rate axis instead of the depth axis. Rate sets the rot speed, Mix the blend.

Saturator [DST] [FX] [SYNTH] in: Audio out: Audio

Multi-mode saturator: drives the in0 signal into one of four waveshaping curves - tanh, soft tube, sigmoid or hard clip - with Tone tilting the result darker/brighter, output-level compensation, and a dry/wet Mix. A clean, controllable distortion stage distinct from the modeled fuzz/amp circuits.

Schmitt [DST] [FX] [SYNTH] in: Audio out: Audio

Schmitt-trigger fuzz (a first as a synth block): a comparator with memory - the output snaps fully high once the driven input rises past +Hyst and stays high until it falls past -Hyst, squaring the signal into a hard two-level wave that resists chattering near zero. The hysteresis band makes it 'stick', cleaning the buzz of a raw square-up while adding a gated, brutal edge. Drive sets the gain into the trigger, Hyst the dead-band width, Level the output, Mix the blend. A hysteretic square fuzz no clipper or fold reproduces.

SergeFold [DST] [FX] [SYNTH] in: Audio out: Audio

Serge wave multiplier: six folding cells at staggered offsets sum into the dense, bright, vocal-formant-like spectrum of a Serge VCM, a richer fold than a single threshold reflection.

Shaper [DST] [FX] [SYNTH] in: Audio out: Audio

SynthMaster-style 12-mode waveshaper: Shape selects soft/hard clip, tube, diode, sine and triangle fold, full/half rectify, sine, crush, sync-wrap and Chebyshev modes, each driven by Drive with Bias for symmetry. Mix and Output blend and trim. Modulate Shape to morph between distortion characters - a one-stop tone-shaping distortion.

ShotNoise [DST] [FX] [SYNTH] in: Audio out: Audio

Shot noise: adds grit whose loudness follows the PHYSICAL shot-noise law - its amplitude scales with the square root of the instantaneous signal level (sqrt(|x|)), the way photon/electron shot noise grows with current. So the noise tracks the signal's contour instead of sitting flat: a self-masking, signal-dependent fizz that disappears in the gaps. Amount sets the noise gain, Mix the blend.

SignedSquare [DST] [FX] [SYNTH] in: Audio out: Audio

Signed square: sign(x)*x^2, which squares the magnitude while keeping the sign - so loud parts are pushed louder and quiet parts quieter (a downward expander curve) without the octave-up of a plain square, adding odd-harmonic edge as it expands.

SinhExciter [DST] [FX] [SYNTH] in: Audio out: Audio

Sinh exciter: shapes the signal through sinh(Drive*x)/sinh(Drive), an expanding transfer curve that is the mirror image of tanh saturation - instead of squashing peaks it pushes mid and high levels outward, exaggerating dynamics and adding bright odd harmonics. A de-compressing exciter rather than a soft-clipper. Normalized so unity input stays unity.

Sizzle [DST] [FX] [SYNTH] in: Audio out: Audio

Sizzle: an HF exciter that generates fizzy new top end from the signal's OWN high frequencies (it shapes the highpassed content, so it only sizzles where there is already brightness, never on bass). Adds air, presence and a metallic crackle. Drive sets the harmonic heat, Amount the added sizzle, Mix the blend.

SlewChew [DST] [FX] [SYNTH] in: Audio out: Audio

Slew chew: a slew limiter with DIFFERENT rise and fall rate caps, so the waveform is chewed asymmetrically - sharp on one edge, rounded on the other - bending the shape and pumping in even harmonics, the lopsided cousin of a symmetric slew. Rise sets the up-rate, Fall the down-rate, Mix the blend.

SlewClip [DST] [FX] [SYNTH] in: Audio out: Audio

Slew-rate distortion (a first as a synth block): limits how fast the output can move between samples, so when Drive pushes the input past that maximum slope the block cannot keep up - fast transients are sheared into ramps and the waveform is dragged toward a buzzing triangle. Where a clipper flattens the peaks, SlewClip mangles the slopes: a harsh, gritty, distinctly digital lo-fi grind that also drops level as it limits. Rate sets the maximum step per sample, Drive the input gain, Mix the blend.

SlewCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Slew + crush: a slew-rate limiter rounds the fast edges, then a noise-shaped quantizer drops the bit depth - the combined softening and grit of a cheap converter feeding a slow line.

SlewFold [DST] [FX] [SYNTH] in: Audio out: Audio

Tracking wavefolder: the fold threshold slews toward the signal level, so the fold density pumps and breathes with dynamics rather than staying fixed - a self-adjusting fold.

SlewFuzz [DST] [FX] [SYNTH] in: Audio out: Audio

Slew-then-fuzz: a slew-rate limiter rounds the fast edges (program-dependent triangular softening) before a hot tanh fuzz, so the fuzz bites on sustained energy but the rounded transients keep it from fizzing - a thick, controlled gnarl.

SlewShape [DST] [FX] [SYNTH] in: Audio out: Audio

Slew-rate distortion: separate rise and fall step limits soften fast edges asymmetrically (TIM-style), generating triangular, program-dependent harmonics; mix blends the slewed and dry signals.

SlewStep [DST] [FX] [SYNTH] in: Audio out: Audio

Slew step: quantizes to a coarse staircase but then GLIDES toward each new step at a limited rate instead of jumping, so you get the lo-fi stepping of a bit-crusher with smooth ramps between treads - a soft, gooey digital quantization rather than a hard one. Steps sets the resolution, Slew the glide rate, Mix the blend.

SlopeOverload [DST] [FX] [SYNTH] in: Audio out: Audio

Slope overload: a slew-rate-limited follower that can only change so fast, so when the input moves quicker than the limit it lags behind and overshoots - the slope-overload and granular-noise distortion of a delta / CVSD modulator. Slow signals stay clean, fast ones tear. Slew sets the rate limit, Mix the blend.

SmoothClamp [DST] [FX] [SYNTH] in: Audio out: Audio

Smooth clamp: passes the signal linearly until it reaches the Knee, then smoothly tanh-saturates the excess toward the +/-1 rails - a soft-knee limiter with a genuinely flat (undistorted) centre, distinct from a plain tanh that curves everywhere.

Snarl [DST] [FX] [SYNTH] in: Audio out: Audio

Snarl: a high-resonance band-pass parked in the low-mids feeds a saturator, so the signal grows a vicious vocal-formant growl - a snarling, throaty bite midway between a wah stuck in place and a fuzz. Freq sets where it snarls, Drive the aggression, Mix the blend.

SoftplusShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Softplus waveshaper: the centred softplus ln(1+e^x)-ln2 as a transfer curve - it passes positive signal almost linearly but smoothly compresses negative excursions toward a floor, an asymmetric soft half-wave shaper rich in even harmonics (octave-up colour).

SoftsignShaper [DST] [FX] [SYNTH] in: Audio out: Audio

Softsign waveshaper: the rational sigmoid x/(1+|x|), a soft clipper that saturates far more gradually than tanh and never quite reaches the rails - a smooth, gentle overdrive with a softer knee and longer transition than transcendental saturators.

SovtekMuff [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Sovtek (Green Russian) Big Muff-style fuzz: cascaded transistor gain into two symmetric silicon-diode clipping stages for the thick, smooth, mid-scooped wall of sustain - darker and bassier than the US Muff.

SpectralClip [DST] [FX] [SYNTH] in: Audio out: Audio

Band clipper: splits the signal into low and high bands and clips ONLY the highs, leaving the lows clean - so you get crunchy, distorted top end riding over an undistorted, solid bottom, the inverse of the usual full-range fuzz that turns bass to mush. Drive sets the high-band clip, Mix the blend.

StochasticResonance [DST] [FX] [SYNTH] in: Audio out: Audio

Stochastic resonance: the counter-intuitive effect where adding NOISE to a weak, sub-threshold signal HELPS it through a nonlinearity instead of burying it. The signal drives a bistable double-well system (x' = Barrierx - x^3 + Drivein + noise) that hops between its two states; a tuned amount of noise makes those hops follow the weak input, recovering and amplifying detail a gate or expander would simply cut. Too little noise and nothing crosses, too much and it is random - the sweet spot in between is the resonance, so SWEEP Noise to find it. A gritty, alive texture/recovery tool with no equivalent in the catalog. Drive scales the input into the wells, Noise the assist level, Barrier the well depth/threshold, Mix the blend.

SubEnhance [DST] [FX] [SYNTH] in: Audio out: Audio

Sub-bass enhancer: a flip-flop toggled per rising zero crossing generates a half-frequency square tracked by the input envelope and low-passed, added under the dry signal at Sub. Reinforces weak low end on kicks, bass and 808s. Monophonic by nature - best on clean low-register sources.

Subharmonic [DST] [FX] [SYNTH] in: Audio out: Audio

Subharmonic generator: zero-crossing detection on the input drives a divide-by-four flip-flop, multiplying the input's rectified amplitude to synthesize a tone two octaves below, then smooths it with a one-pole low-pass. Sub sets the level of the generated sub, Tone opens the low-pass cutoff (darker to brighter sub), and Mix sums the sub onto the dry signal. Adds weight and low-end body to bass, kicks and leads.

TanhLadder [DST] [FX] [SYNTH] in: Audio out: Audio

Cascaded tanh stages: several tanh gain stages in series compound their soft saturation, building the progressive, harmonically dense overdrive of a multi-stage tube preamp rather than a single clip.

TapeMachine [DST] [FX] [SYNTH] in: Audio out: Audio

Tape machine: combines soft tape saturation, a slow wow pitch wobble via a modulated delay, and a hint of hiss into one block - the glued, warm, slightly-unstable character of analog tape.

TapeSat [DST] [FX] [SYNTH] in: Audio out: Audio

Tape-style saturation: a soft tanh stage with a touch of linear blend emulates magnetic-tape compression, with Warmth rolling off the highs for a rounder tone and Mix blending dry/wet. Gentle glue and harmonic thickening rather than overt distortion. See the Studer and Ampex analog models for machine-specific tape.

TiltDistort [DST] [FX] [SYNTH] in: Audio out: Audio

Tilt-weighted saturation: the spectrum is tilted before a tanh stage and tilted back after, so the distortion bites hardest on the boosted band - dial Tilt positive to fizz only the highs, negative to grind only the lows, while the overall balance is restored.

TransformerSat [DST] [FX] [SYNTH] in: Audio out: Audio

Transformer saturation: a hysteresis loop (the output lags and remembers, like a magnetized iron core) plus soft clipping models an audio transformer's low-end thickening and gentle history-dependent distortion.

TransientExciter [DST] [FX] [SYNTH] in: Audio out: Audio

Transient exciter: harmonics are added ONLY during attacks - a fast/slow envelope split detects each onset and injects bright odd harmonics that fade as the note settles, so picks, hits and consonants get crisp bite while sustains stay clean. The opposite of smearing. Amount sets the bite, Sharp the harmonic edge, Mix the blend.

TransientFold [DST] [FX] [SYNTH] in: Audio out: Audio

Transient-keyed folder: a fast-minus-slow envelope detector measures the attack of each event and opens a wavefolder only on those transients, so the body of a sound stays clean while every hit cracks with a burst of fold harmonics - percussive brightening, not sustained grind.

TriState [DST] [FX] [SYNTH] in: Audio out: Audio

Tri-state quantizer: collapses the signal to just three levels - full positive, full negative, or zero inside a central dead band - so it squares up like an infinite clipper but with a silent gap around zero, giving a gated, telegraph-like buzz. Dead sets the size of the zero zone, Mix the blend.

TriToSine [DST] [FX] [SYNTH] in: Audio out: Audio

Triangle-to-sine shaper: the soft-clipping transfer that classic analog VCOs use to round a triangle into a near-perfect sine, useful for warming any signal or deriving a sine from a triangle core.

Tube [DST] [FX] [SYNTH] in: Audio out: Audio

Tube-style saturation: an asymmetric transfer curve (offset by Bias) adds the even and odd harmonics of a valve stage, tamed by a Tone low-pass and blended via Mix. Warmer and rounder than the harder Drive/Waveshaper - subtle settings thicken and glue, higher Drive gives singing tube grit. See the TubePre and VariMu analog models for circuit-specific versions.

Tumnus [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Wampler Tumnus-style transparent boost/overdrive (Klon-lineage): a clean op-amp boost summed with a mild germanium-diode clip stage for a touch of grit while keeping the dry signal intact - the famous transparent mid-push.

TurboRAT [DST] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

ProCo Turbo RAT-style distortion: a high-gain op-amp stage into hard LED clipping for a sharper, louder, more aggressive RAT with a fierce midrange bite.

VHS [DST] [FX] [SYNTH] in: Audio out: Audio

Cassette/VHS tape emulation: the input runs through a short modulated delay whose time wobbles via an internal 0.7 Hz LFO, then a one-pole low-pass rolls off highs and random pseudo-noise hiss is summed in. Wow sets the LFO delay-modulation depth (pitch wobble), Hiss scales the added noise level, and Dropout raises the probability of brief signal dropouts driven by a per-sample random envelope. Mix blends the degraded signal against the dry input for subtle aging or full lo-fi destruction.

Vinyl [DST] [FX] [SYNTH] in: Audio out: Audio

Vinyl character that reads the signal through a slowly modulated delay for wow pitch-drift, injects random impulses for surface crackle, and pushes the result through a tanh saturator for groove distortion. Wow scales the pitch-drift depth, Crackle sets how often surface pops appear, Drive sets the saturation amount, and Mix sets dry/wet. Use it to age a clean source with turntable wobble, dust noise and warm grit.

VirtualBass [DST] [FX] [SYNTH] in: Audio out: Audio

Virtual bass / missing-fundamental enhancer: isolates the sub-bass band, rectifies it to synthesize a harmonic series (2f, 3f, 4f…) of the fundamental, then mixes those upper-bass harmonics back with the dry signal. The ear fuses the harmonics into the residue pitch of the absent fundamental, so a small speaker that cannot reproduce 50 Hz still seems to - the MaxxBass/psychoacoustic-bass principle. Distinct from a sub-octave generator, which adds a lower octave the speaker still cannot play.

WaveCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Crush-then-fold: a low-bit quantizer's staircase is fed straight into a triangle folder, so the quantization steps themselves get folded - a layered digital-plus-fold timbre neither stage makes alone.

Wavefolder [DST] [FX] [SYNTH] in: Audio out: Audio

West-coast wavefolder that multiplies the signal by Fold and reflects it through triangle folding boundaries, adding bright harmonics that bloom as level rises. Bias shifts the fold symmetry for asymmetric timbres, Mix blends dry/wet and Output trims gain. Unlike clipping, folding keeps energy moving, so the tone stays lively even at extreme settings.

WaveMirror [DST] [FX] [SYNTH] in: Audio out: Audio

Reflecting waveshaper: excursions past a slowly DC-tracked threshold are mirrored back down by an adjustable amount, folding ring-like overtones that recenter as the signal's bias drifts.

WaveMix [DST] [FX] [SYNTH] in: Audio out: Audio

Blended waveshaper: one knob crossfades the transfer curve from soft tanh saturation to sinusoidal folding, sweeping the harmonic character from warm clip to bright fold in a single stage.

Wavemult [DST] [FX] [SYNTH] in: Audio out: Audio

West-coast wave multiplier (Serge/Buchla-style): cascades several triangle wavefolders so the signal is folded again and again as Drive rises, multiplying the harmonics far beyond a single Wavefolder for ringing, metallic, formant-rich timbres. Drive sets the fold depth, Stages the number of cascaded folders (more = denser), Bias the fold asymmetry, and Mix the dry/wet blend.

Waveset [DST] [FX] [SYNTH] in: Audio In out: Audio

Waveset distortion (Trevor Wishart, 'Audible Design'): slices the in0 signal into wavesets - the segments between upward zero-crossings - and plays each one Repeat times before moving on, dropping the input in between. Repeating every waveset time-stretches and drops the pitch in gritty, vowel-like steps that track the signal's own period, not a fixed grid - the classic granular-but-pitch-synchronous Wishart sound. Repeat sets how many times each waveset loops, Mix blends dry/wet. in0 = Audio In.

WaveshapeMorph [DST] [FX] [SYNTH] in: Audio out: Audio

Morphing waveshaper: one knob crossfades the drive curve from a tanh saturator (odd harmonics, soft clip) to a sine wavefolder (dense fold harmonics), sweeping continuously from warm overdrive to bright metallic folding on the same input.

Waveshaper [DST] [FX] [SYNTH] in: Audio out: Audio

Memoryless waveshaper with four transfer curves (Shape: tanh, atan, cubic soft-clip, sine fold) driven by Drive and blended via Mix. Each curve adds a different harmonic signature, from smooth saturation (tanh/atan) to a foldback character (sine). Lighter-weight than Drive when you just want one fixed shaping curve.

WaveWrap [DST] [FX] [SYNTH] in: Audio out: Audio

Sinusoidal wavefolder: drives the signal through sin() so that increasing index folds it through multiple sine lobes (Serge-style), staying perfectly bounded while adding rich symmetric harmonics.

XformerSat [DST] [FX] [SYNTH] in: Audio out: Audio

Transformer saturation: asymmetric soft-clipping knees (different for positive and negative swings) add even-harmonic colour, plus a low-passed component for iron-core low-end thickening. Subtle weight and warmth on bass, drums and the mix bus. See the NevePre analog model for a transformer-preamp variant.

XorFold [DST] [FX] [SYNTH] in: Audio out: Audio

XOR fold: bit-wise exclusive-ORs the current sample against a delayed copy of itself, a digital comb that carves jagged, metallic, glitch-ridden notches no analog filter can make - the harmonics interfere at the BIT level rather than the amplitude level. Time sets the delay, Mix the blend.

ZenerClamp [DST] [FX] [SYNTH] in: Audio out: Audio

Zener clamp: a voltage-reference clamp that limits asymmetrically - a soft forward knee one way, a harder reverse-breakdown wall the other - so the waveform is squashed lopsidedly the way a Zener-diode protection circuit clips, adding strong even harmonics. Forward sets the soft side, Reverse the breakdown wall, Mix the blend.

ZeroCrossGlitch [DST] [FX] [SYNTH] in: Audio out: Audio

Zero-cross sample-hold: tracks the signed peak within each half-cycle and freezes the output to it until the next zero crossing, rebuilding the waveform as flat plateaus that snap on every crossing - a PWM-like, octave-rich square reconstruction.

ZeroCrush [DST] [FX] [SYNTH] in: Audio out: Audio

Zero-cross sample-hold: latches the input value at each rising zero crossing and holds it until the next, reducing the waveform to one stair-step per pitch period - a pitch-synced lo-fi staircase.

Dynamics (63)

AmpHold [DYN] [FX] [SYNTH] in: Audio out: Audio

Peak-hold envelope: captures the loudest recent peak and holds it, then decays slowly - a control-rate 'freeze the loudness' source for driving other parameters, distinct from a plain release-only follower.

AsymComp [DYN] [FX] [SYNTH] in: Audio out: Audio

Asymmetric compressor: independent envelope followers level the positive and negative half-waves with separate ratios, so unequal polarity gains introduce even harmonics and a tube-like asymmetry on top of the leveling - part compressor, part saturator.

AutoGain [DYN] [FX] [SYNTH] in: Audio out: Audio

Automatic gain control: tracks the signal's average level and slowly scales it toward a target, evening out long-term loudness swings (quiet passages come up, loud ones come down) the way a broadcast leveler does.

Choke [DYN] [FX] [SYNTH] in: Audio out: Audio

Choke gate: every new onset slams the gain shut for a brief window, killing whatever tail was still ringing - the hi-hat choke, applied to anything. Successive hits clamp each other off into a tight, staccato, gasping rhythm. Time sets the choke window, Mix the blend.

CompProg [DYN] [FX] [SYNTH] in: Audio out: Audio

Program-dependent compressor: blends a fast and a slow release envelope so brief peaks recover quickly while sustained loud passages release slowly, the auto-release behaviour of classic bus compressors - transparent glue.

Compressor [DYN] [FX] [SYNTH] in: Audio out: Audio

Peak-following compressor: tracks the input envelope with Attack/Release ballistics and, above Threshold, reduces gain by the set Ratio, with Makeup to restore level. Fast attack tames transients while slow release glues sustained material. A clean general-purpose dynamics stage - see the analog pack (FET, Opto, VariMu) for coloured compression.

CompUpward [DYN] [FX] [SYNTH] in: Audio out: Audio

Upward compressor: instead of pulling loud parts down, it pushes signal below the threshold up, raising low-level detail and ambience - density and presence without limiting the peaks.

ContrastEnv [DYN] [FX] [SYNTH] in: Audio out: Audio

Amplitude contrast: a power curve applied to the signal's envelope around a reference level pushes loud parts louder and quiet parts quieter (positive) or evens them out (negative), preserving waveform shape.

CrestFactor [DYN] [FX] [SYNTH] in: Audio out: Audio

Crest-factor follower: tracks the running peak-to-RMS ratio of the input (about 1.4 for a steady sine, ~1 for a square, high for spiky/transient or sparse material) - a control signal measuring how 'peaky' the dynamics are, for transient-aware modulation or auto-dynamics. Time sets the averaging window.

DeEsser [DYN] [FX] [SYNTH] in: Audio out: Audio

Tames sibilance by splitting off a high band with a one-pole low-pass, tracking its level with a fast-attack envelope, and applying gain reduction to that band only when it crosses the threshold before recombining it with the lows. Freq sets the sibilance band, Threshold sets where reduction begins, Amount scales how hard the high band is pulled down, and Mix sets dry/wet. Because only the highs are compressed, it removes harsh esses without dulling the rest of the signal.

Deflate [DYN] [FX] [SYNTH] in: Audio out: Audio

Deflate: the opposite of an inflator - a cubic curve that sucks the body out of low-level signal so quiet detail shrinks back and the sound feels smaller, thinner and pulled-in. Use it to deflate an over-full source or for a claustrophobic effect. Amount sets the deflation, Mix the blend.

DeSibilance [DYN] [FX] [SYNTH] in: Audio out: Audio

De-esser: detects energy in a high band (the 's' and 't' sounds) and ducks only that band when it exceeds a threshold, taming harsh sibilance on vocals without dulling the whole signal.

Ducker [DYN] [FX] [SYNTH] in: Audio, Sidechain out: Audio

Sidechain ducker that follows the level of the key input (in 2) with a release-smoothed envelope and pulls down the main signal whenever that key exceeds the threshold. Threshold sets the key level that triggers ducking, Amount sets how far the main signal drops, and Release sets how quickly it recovers once the key falls back. Use it for music-under-voice ducking or rhythmic pumping driven by a separate trigger.

DynEQ [DYN] [FX] [SYNTH] in: Audio out: Audio

Dynamic EQ band: an SVF band-pass at Freq/Q tracks an envelope follower, and when that band's energy crosses Threshold the band is subtracted from the signal in proportion to Amount. Freq and Q place and narrow the band, Threshold sets where reduction begins, and Mix blends the result against dry. It only acts when the band gets loud, so it tames resonances and harshness without static EQ cuts.

EntropyGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Entropy gate: measures zero-crossing density (how noisy vs tonal the signal is) and gates on it, so you can pass tone and mute hiss/noise, or the reverse - a timbre-keyed gate, not a level gate.

EnvClip [DYN] [FX] [SYNTH] in: Audio out: Audio

Adaptive clipper: the soft-clip ceiling follows the signal envelope, so the amount of clipping stays proportional as levels change - consistent grit from quiet to loud passages instead of a fixed wall.

Envelope [DYN] [FX] [SYNTH] in: Audio out: Audio

Envelope follower that rectifies the input and smooths it with independent Attack and Release times into a control signal. Gain scales the output for driving a filter cutoff, VCA or any mod input (auto-wah, ducking, dynamics). Fast attack tracks transients; a longer release gives a smoother contour.

EnvFollow [DYN] [FX] [SYNTH] in: Audio out: Audio

Envelope follower: tracks the rectified amplitude of the input with separate Attack and Release times, outputting a smooth control signal that rises quickly on transients and falls slowly back - the standard way to turn audio into a modulation source. Patch its output into any parameter's mod input so a filter, VCA or effect responds to playing dynamics; Gain scales the result. Unlike the duckers (which bury the follower inside one effect) this exposes the envelope itself.

EnvGateAR [DYN] [FX] [SYNTH] in: In A, In B out: Audio

Gated AR envelope VCA: a second-input gate opens an attack/release envelope that shapes the audio input's level - a self-contained note-on amplitude envelope without a full synth voice.

EnvSlope [DYN] [FX] [SYNTH] in: Audio out: Audio

Envelope slope: outputs the per-sample rate of change of the amplitude envelope - strongly positive on attacks, negative on decays, near zero on steady sustain - a bipolar onset/transient detector and dynamics-derived modulation source. Gain scales the (small) derivative to a usable range.

EnvSwell [DYN] [FX] [SYNTH] in: Audio out: Audio

Auto volume swell: a slow-attack envelope suppresses each note's initial transient and fades the body in (the violin bow-swell / no-pick effect), with no pedal or LFO.

Expander [DYN] [FX] [SYNTH] in: Audio out: Audio

Downward expander: attenuates signal that falls below Threshold by the set Ratio (the inverse of a compressor), tightening quiet passages and reducing bleed and hiss between notes. Attack/Release set the ballistics. Gentler than a gate - it fades low-level material rather than hard-muting it.

FeedbackCompressor [DYN] [FX] [SYNTH] in: Audio out: Audio

Feedback-topology compressor: the level detector watches the already-compressed output instead of the input (the classic 1176/opto behavior), so the gain reduction is self-referential - this softens the knee, makes the effective ratio program-dependent, and gives the smoother, 'rounder' character feedback designs are prized for. Threshold, Ratio, Attack/Release in ms, Makeup.

Gate [DYN] [FX] [SYNTH] in: Audio out: Audio

Noise gate: when the detected level falls below Threshold it attenuates the signal toward Range (the floor), opening and closing with Attack/Release ballistics. Range at -80 dB fully mutes between events; smaller values leave a residual for gentler ducking. Use it to silence hiss and bleed between phrases or for rhythmic gating.

GateBounce [DYN] [FX] [SYNTH] in: Audio out: Audio

Contact bounce: when the gate opens it chatters - a few fast random on/off bounces before it settles, the way a mechanical relay or switch contact bounces on closing. Adds a glitchy, machine-gun stutter to every note onset. Bounce sets how much it chatters, Mix the blend.

HystGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Hysteresis gate: separate open and (lower) close thresholds give Schmitt-trigger immunity to chatter near the threshold, with a smoothed gain ramp so the gate fades rather than clicks.

Inflate [DYN] [FX] [SYNTH] in: Audio out: Audio

Inflate: a psychoacoustic loudness exciter - a gentle cubic curve that lifts the body and harmonics of the signal so it sounds bigger and louder without raising the peak level. The 'inflator' move for making a mix feel fuller. Amount sets the inflation, Mix the blend.

Limiter [DYN] [FX] [SYNTH] in: Audio out: Audio

Brick-wall peak limiter: instantly clamps gain so peaks never exceed Threshold, then releases smoothly back to unity over Release, with Output trimming the post-limit level. Set Threshold as the ceiling and drive the input for transparent loudness. See Maximizer for loudness-optimized limiting.

LookaheadLimit [DYN] [FX] [SYNTH] in: Audio out: Audio

Lookahead limiter: the signal is delayed through a short buffer while the gain is computed from the (still-undelayed) incoming peak, so the gain reduction is already in place when the peak reaches the output - catching transients a feed-forward limiter would overshoot. Release sets the recovery, Thresh the ceiling.

MaskGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Temporal masking gate: a loud event briefly DEAFENS the block to what follows, modelling forward auditory masking - quiet detail in the wake of a transient is pushed down (the ear can't hear it anyway), tightening busy material and emphasising the hits. Distinct from a normal gate: the threshold is set by the recent loudest peak, not a fixed level. Amount sets the masking depth, Hold the masking time, Mix the blend.

Maximizer [DYN] [FX] [SYNTH] in: Audio out: Audio

Loudness maximizer that applies a gain boost, tracks the peak envelope, and divides back any sample exceeding the ceiling before hard-clamping to it, acting as a brick-wall limiter. Boost sets how hard the input is driven up and Ceiling sets the absolute output peak in dB. Push Boost while keeping Ceiling near 0 dB to raise apparent loudness without overshooting the output.

MBDynamics [DYN] [FX] [SYNTH] in: Audio out: Audio

Multiband dynamics processor that splits the input into low, mid and high bands (crossovers near 250 Hz and 2.5 kHz) and applies independent compression and gating to each. Threshold sets the level above which a band is compressed at Ratio, while Gate sets the level below which a band is attenuated downward. Mix blends the processed sum against the dry signal, useful for tightening or de-noising specific frequency regions at once.

MultiComp [DYN] [FX] [SYNTH] in: Audio out: Audio

Three-band compressor: crossovers split the signal into low, mid and high bands (Low/Mid and Mid/High), each compressed independently against a shared Threshold and Ratio with fast-attack/slow-release ballistics, then summed with Makeup. Tame a busy mix's bands separately - control boomy lows without dulling the highs. See the SSL/dbx analog models for coloured bus compression.

MultiCompress [DYN] [FX] [SYNTH] in: Audio out: Audio

Two-band compressor: splits the signal at a crossover and compresses the low and high bands independently before recombining, so a loud low end stops pumping the highs (and vice versa) - the core of multiband dynamics.

NoiseGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Noise gate: passes the in0 audio only while its level is above Threshold, otherwise silences it - cleans up hiss, bleed and tails between notes. A peak follower with a Release time smoothly opens and closes the gate so it doesn't chatter.

NoiseGateChop [DYN] [FX] [SYNTH] in: Audio out: Audio

Chopping gate: while the input sits above threshold it passes clean, but as it falls below the gate chops it with a fast square on/off instead of fading - so decaying tails break into a rhythmic glitchy stutter rather than smoothly muting. Rate sets the chop speed.

NoiseGateHF [DYN] [FX] [SYNTH] in: Audio out: Audio

High-frequency-keyed gate: triggered by high-band energy (a one-pole high-pass detector), it clamps hiss and cymbal bleed without choking sustained low-frequency tone.

Omnipressor [DYN] [FX] [SYNTH] in: Audio out: Audio

Extreme dynamics on one knob: Function sweeps from gating expansion through limiting compression to full dynamic reversal, where loud passages turn quiet and quiet turns loud.

OTT [DYN] [FX] [SYNTH] in: Audio out: Audio

Over-the-top multiband compressor (Xfer OTT-style): splits the signal into low / mid / high bands and applies BOTH downward and upward compression to each at once - loud parts are pulled down and quiet detail is pushed up - for the dense, controlled, minimal-dynamic-range sound on modern EDM / dubstep / future-bass synths and drums. Depth sets how hard the up + down compression bites (the OTT % knob), Time the attack/release across all bands (fast = tight, slow = more transient), Low / Mid / High trim each band's output in dB, and Mix blends against dry. Unlike MultiComp / MBDynamics this also compresses upward (raises the quiet parts), which is what gives OTT its signature.

ParallelComp [DYN] [FX] [SYNTH] in: Audio out: Audio

Parallel (New York) compression: a heavily-compressed copy is blended back UNDER the dry signal, so the transients and peaks stay intact while the body and low-level detail are lifted - punch and density without squashing the dynamics (the drum-bus / mix-bus trick). Distinct from a normal compressor, which replaces the signal: here the crush sits in parallel. Threshold/Ratio set the parallel crush, Attack/Release its ballistics, Makeup the parallel gain, Blend how much of the compressed copy is mixed under the dry.

PeakLimiter [DYN] [FX] [SYNTH] in: Audio out: Audio

Peak limiter: gain drops instantly whenever the signal would exceed the ceiling and recovers gradually, holding output below a hard threshold without the slow ratio behaviour of a compressor - transparent loudness control.

PercentileGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Percentile gate: the threshold is not fixed - it slowly tracks the signal's own loud peaks, so the gate passes only the top fraction of dynamics and adapts as the material gets louder or softer over time. Self-calibrating, unlike a fixed-dB gate. Open sets how much of the top passes, Mix the blend.

Pump [DYN] [FX] [SYNTH] in: Audio out: Audio

Tempo-synced sidechain-style volume pump that ducks the input each cycle and recovers using a sample-clock phase for coherent timing. Rate sets the duck frequency in Hz, Depth sets how far the level drops, and Curve shapes the recovery ramp from snappy to gradual. Mix blends the pumped signal against the dry input for the classic EDM ducking effect without an external sidechain.

RateGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Slew gate: opens on MOTION, not loudness - it passes signal only while its rate of change (slew) is high, so fast transients and bright content get through while slow, steady tones are gated out. A movement-sensitive gate that does the opposite of a sustain pedal: keeps the attacks, drops the drones. Threshold sets the motion needed, Mix the blend.

RmsComp [DYN] [FX] [SYNTH] in: Audio out: Audio

RMS compressor: detects level with a root-mean-square window instead of peaks, so gain reduction follows perceived loudness smoothly and ignores brief spikes - the gentle, musical compression of classic RMS-detector units.

Sidechain [DYN] [FX] [SYNTH] in: In A, In B out: Audio

Sidechain ducker: the audio level is pulled down in proportion to a second input's envelope, the pumping kick-ducks-bass / broadcast voiceover effect, driven by an external key.

SlewGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Self-adjusting gate: the threshold slews toward a slow running average of the level, so the gate auto-tracks a drifting noise floor and opens only on signal that rises above its own recent average.

SoftExpand [DYN] [FX] [SYNTH] in: Audio out: Audio

Soft downward expander: below the threshold the gain falls off by the set ratio (smoothly, not a hard gate), widening dynamic range and pushing low-level noise/bleed down without chopping.

SoftKnee [DYN] [FX] [SYNTH] in: Audio out: Audio

Soft-knee compressor: a smooth knee region around the threshold eases gain reduction from 1:1 into the set ratio, so compression blends in gradually instead of switching hard at the threshold.

SoftThreshold [DYN] [FX] [SYNTH] in: Audio out: Audio

Soft thresholding: the wavelet-denoising shrinkage operator sign(x)*max(|x|-T,0), which subtracts the threshold from the magnitude rather than hard-gating - so signal below T is removed and signal above is passed but pulled down by T, a continuous deadband distinct from a hard noise gate.

Splat [DYN] [FX] [SYNTH] in: Audio out: Audio

Splat: a transient that crosses the threshold briefly blows the gain into hard saturation, then recovers - so big hits ‘splat’ with a momentary overdriven smear before settling clean, the overshoot artifact of a slamming converter or a maxed preamp. Thresh sets the trigger, Amount the splat heat, Mix the blend.

Squash [DYN] [FX] [SYNTH] in: Audio out: Audio

Aggressive parallel smash: a fast peak-following envelope drives heavy gain reduction above a fixed threshold, and the gain-reduced signal is pushed through a tanh saturator for crushed, distorted dynamics. Amount steepens the gain-reduction curve (harder squash), Tone scales the saturated output level, and Mix blends the smashed signal in parallel with the dry input. Useful for adding punch and grit to drums and bus material.

SubBoost [DYN] [FX] [SYNTH] in: Audio out: Audio

Sub-bass synthesizer: isolates the low band and drives it through a soft nonlinearity to regenerate and reinforce the fundamental, adding felt low end on small speakers without just boosting EQ.

ThyratronZap [DYN] [FX] [SYNTH] in: Audio out: Audio

Thyratron latch: a gas-tube trigger that FIRES when the input crosses a strike level and stays latched on - passing full signal - until the level falls below a lower hold point, then snaps off. A latching, hysteretic on/off zap that holds a note open after a hit. Strike sets the fire level, Hold the drop-out level, Mix the blend.

Transient [DYN] [FX] [SYNTH] in: Audio out: Audio

Transient shaper: a fast vs slow envelope follower separates the attack (fast > slow) from the sustain (fast < slow), letting Attack and Sustain each be boosted or cut independently, with Output trimming the result. Add punch to drums, or tighten/lengthen sustain - all independent of input level, unlike a compressor.

TransientDesign [DYN] [FX] [SYNTH] in: Audio out: Audio

Transient designer: a fast and a slow envelope track the signal; their difference isolates the attack while the slow part is the sustain, so you can punch up or soften each independently - add snap or glue without a threshold.

TransientGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Transient-only gate: passes the signal while the fast envelope outruns the slow one (the attack) and mutes the sustain - isolates clicks / picks / hits, the inverse of a sustain ducker.

TransientSculpt [DYN] [FX] [SYNTH] in: Audio out: Audio

Transient designer: a fast and a slow envelope follower track the signal; their difference is the attack and the slow one is the body, letting you boost or cut the punch and the sustain of a sound independently.

TransientSplit [DYN] [FX] [SYNTH] in: Audio out: Audio

Transient designer: a fast minus slow envelope isolates attack vs sustain energy, scaled independently so you can sharpen or soften hits and swell or dampen tails without a threshold.

TruePeakLimit [DYN] [FX] [SYNTH] in: Audio out: Audio

True-peak limiter: estimates the inter-sample peak with a 4-point band-limited reconstruction of the midpoint between samples and limits against that, catching the over-0 dBFS overshoots a sample-peak limiter misses and that clip a downstream converter (the ITU-R BS.1770 true-peak concern). Output is one sample delayed so the gain aligns with the reconstructed peak - an inter-sample-aware ceiling, distinct from a plain lookahead limiter. Ceiling sets the ISP ceiling, Release the recovery.

UpComp [DYN] [FX] [SYNTH] in: Audio out: Audio

Upward compressor that tracks the input envelope and, when it sits below the threshold, applies makeup gain proportional to how far below it is and to the ratio, lifting quiet passages toward the threshold. Threshold sets the level below which boosting begins, Ratio sets how aggressively the deficit is filled, Makeup adds a fixed output gain, and Mix sets dry/wet. Use it to raise low-level detail and tighten dynamic range from the bottom up rather than taming peaks.

VarianceGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Variance gate: opens not on loudness but on COMPLEXITY - it tracks the short-term variance (the energy of the fluctuation around the local mean) and passes only when the signal is busy/noisy, muting steady tones and DC-like calm. A texture-sensitive gate. Threshold sets the activity needed, Mix the blend.

ZeroCrossGate [DYN] [FX] [SYNTH] in: Audio out: Audio

Zero-cross gate: a noise gate that only ever opens or closes AT a zero crossing of the signal, so it never produces the click a normal gate makes when it slams shut mid-waveform. The gate decision is latched at each crossing from the envelope vs Threshold. Click-free gating for harsh, percussive, or already-distorted material. Threshold sets the open level, Mix the blend.

Effect (51)

AMRadio [FX] [FX] [SYNTH] in: Audio out: Audio

AM-radio lo-fi: band-limits in0 to a tinny midrange, adds a faint carrier hum and a wash of static, the sound of a distant transistor radio. Band sets the midrange tightness, Static the noise level, Mix the blend. in0 = Audio.

Avalanche [FX] [FX] [SYNTH] in: Audio out: Audio

Avalanche: a feedback line whose gain briefly exceeds unity after a loud trigger, so the sound builds on itself into a swelling roar before a tanh ceiling and a decay tame it back down - a self-amplifying cascade that snowballs and settles. Trigger sets the trip level, Decay the avalanche length, Mix the blend.

BarkhausenBuzz [FX] [FX] [SYNTH] in: Audio out: Audio

Barkhausen buzz: magnetic-domain jumps emit a crackle whose intensity tracks the RATE of magnetization change (dB/dt) - so fast-moving signal hisses and buzzes with domain noise while steady levels stay quiet. The audible signature of a magnetizing core. Amount sets the buzz level, Mix the blend.

Boomerang [FX] [FX] [SYNTH] in: Audio out: Audio

Boomerang: a reverse delay that catches the signal and throws it back REVERSED through a sliding window, with feedback so each throw comes back again - swelling, backwards-shimmer echoes that arc out and return. Time sets the window length, Feedback how many returns, Mix the blend.

ClothRustle [FX] [FX] [SYNTH] in: Audio out: Audio

Cloth rustle: soft high-frequency noise bursts triggered by how fast the signal MOVES, like fabric brushing past a microphone - it whispers and rustles on motion and falls silent when the signal holds still. Amount sets the rustle level, Mix the blend.

Creak [FX] [FX] [SYNTH] in: Audio out: Audio

Creak: a stress-driven stick-slip squeak - the input level winds up tension that periodically slips, sweeping a thin resonant tone upward in little jerks, like a creaking door, rope or floorboard. The louder you push, the faster and higher it creaks. Tune sets the squeak pitch, Mix the blend.

Crumble [FX] [FX] [SYNTH] in: Audio out: Audio

Crumble: the longer a note sustains the more of its samples drop out and hold, so a held tone progressively crumbles into a stuttering, gap-riddled fragment - then knits back together in the silences. Rate sets the crumble onset, Mix the blend.

Crystallize [FX] [FX] [SYNTH] in: Audio out: Audio

Crystallize: each transient seeds a high shimmering partial locked to a tuned pitch grid (so the sparkle is always musical, never random), ringing out and accreting into a crystalline upper layer. Tune sets the grid root, Amount the shimmer level, Mix the blend.

DiodeClipper [FX] [FX] [SYNTH] in: Audio out: Audio

Diode clipper: a symmetric pair of clipping diodes giving a soft-knee crunch that rounds into hard saturation as it's pushed - the core of countless distortion pedals. Drive sets the gain, Bias skews the symmetry toward even harmonics, Level the output. in0 = Audio.

Dropout2 [FX] [FX] [SYNTH] in: Audio out: Audio

Tape dropout: random brief mutes, as if the oxide flaked off the tape - the signal cuts out for a few milliseconds at random, with a quick fade so it ticks rather than clicks. Sparse and subtle for vintage wear, dense for a glitchy, failing-media stutter. Rate sets how often dropouts happen, Length their duration, Mix the blend.

Foldback [FX] [FX] [SYNTH] in: Audio out: Audio

Foldback distortion: instead of clipping, any signal past the threshold is reflected back on itself, generating bright, ring-mod-like upper harmonics that grow chaotically with drive - harsher and more digital than soft clipping. Drive sets the level into the fold, Thresh the fold point, Level the output. in0 = Audio.

FreqShifter [FX] [FX] [SYNTH] in: Audio out: Audio

Frequency shifter: builds an approximate quadrature (Hilbert) pair of the input through two allpass paths and ring-modulates them with a complex exponential, shifting every partial by a fixed Hz amount (not a pitch ratio) - the inharmonic, Bode-style shifter.

GeigerNoise [FX] [FX] [SYNTH] in: Audio out: Audio

Geiger counter: emits sharp random CLICKS whose density tracks the input level - loud input is a frantic crackle of clicks, quiet input an occasional tick, like a Geiger counter responding to radiation. Each click is a fast-decaying impulse fired by a Poisson process whose rate is driven by |signal|. A novel level-to-texture sonifier. Sensitivity sets the click rate, Decay the click length, Mix the blend.

GravelCrunch [FX] [FX] [SYNTH] in: Audio out: Audio

Gravel crunch: dense bursts of low-passed noise grains gated by the signal level, like boots grinding on gravel - the more signal, the more it crunches underneath. A grounded, gritty texture rather than a hiss. Density sets the crunch amount, Mix the blend.

Groan [FX] [FX] [SYNTH] in: Audio out: Audio

Groan: the input level wakes a deep, slowly wavering sub-tone that moans underneath the signal - a wobbling low resonance whose pitch drifts on a slow random LFO, adding a creaking, living, beast-like undertone to drones and pads. Tune sets the groan pitch, Mix the blend.

IntegrateDump [FX] [FX] [SYNTH] in: Audio out: Audio

Integrate-and-dump: the input is accumulated into a bucket and, the moment the bucket overflows past Threshold, it DUMPS its whole contents as one spike and resets to zero. Loud or sustained input fills it fast (rapid spikes), quiet input fills it slowly (sparse spikes) - a charge-pump rhythm generator whose pulse rate tracks the signal energy. Rate sets the fill speed, Threshold the dump level, Mix the blend.

Ionize [FX] [FX] [SYNTH] in: Audio out: Audio

Ionize: transients ignite bursts of high, crackling electric sparks - bandpassed noise gated into random arcs, like a plasma discharge or a Tesla coil snapping. Distinct from a tonal sparkle: this is pure electric noise. Amount sets the arc level, Mix the blend.

JitterClock [FX] [FX] [SYNTH] in: Audio out: Audio

Clock jitter: the sampling instant is dithered by a tiny random delay each sample, so the signal acquires the smeared, noisy edge of a jittery sample clock - subtle phase-noise grime that frosts transients and high frequencies. Amount sets the jitter, Mix the blend.

MagnetostrictRing [FX] [FX] [SYNTH] in: Audio out: Audio

Magnetostrictive ring: a magnetic core changes shape with the SQUARE of the drive (magnetostriction), exciting a high metallic resonance an octave up - so the signal grows a singing, ferromagnetic ring that tracks its envelope, the hum you hear from transformers and tape heads. Freq sets the ring pitch, Amount the ring level, Mix the blend.

Megaphone [FX] [FX] [SYNTH] in: Audio out: Audio

Megaphone / bullhorn: forces in0 through a harsh, peaky midrange band and clips it, the distorted, honking voice-of-authority sound. Honk sets the resonant peak, Drive the clipping, Mix the blend. in0 = Audio.

MetalStress [FX] [FX] [SYNTH] in: Audio out: Audio

Metal fatigue: the input's energy accumulates as 'stress' in a metal part; when the stress crosses a building threshold the metal CRACKS - a bright inharmonic ping - and partially relieves, so sustained loud input produces irregular, escalating metallic snaps and creaks, like a stressed sheet or cooling engine. Buildup sets how fast stress accrues, Threshold the crack point, Mix the blend.

Overdrive [FX] [FX] [SYNTH] in: Audio out: Audio

Tube-style overdrive: a smooth tanh soft-clip with a midrange-focused drive, warming and compressing the signal the way a pushed valve preamp does - the Tube Screamer voicing. Drive sets the gain into the clip, Tone the post brightness, Level the output. in0 = Audio.

PhaserStages [FX] [FX] [SYNTH] in: Audio out: Audio

Multi-stage phaser: a cascade of all-pass sections swept by a built-in LFO, with feedback for deeper notches - from a subtle shimmer to a vocal, resonant sweep. Rate sets the LFO, Stages the notch count, Feedback the resonance, Mix the depth. in0 = Audio.

PlosivePop [FX] [FX] [SYNTH] in: Audio out: Audio

Plosive pop: each transient triggers a short low-frequency thump under the signal - the 'p'/'b' plosive blast or a mic-pop, a dull pressure pulse that fattens hits with sub-bass weight. Tune sets the pop pitch, Amount the thump level, Mix the blend.

PrintThrough [FX] [FX] [SYNTH] in: Audio out: Audio

Tape print-through: on a wound reel the magnetic field of a loud passage slowly imprints a faint GHOST of itself onto the tape layers pressed against it - so on playback a dull copy appears both one layer BEFORE the sound (the eerie pre-echo that creeps in just ahead of a loud transient on old tapes) and one layer AFTER it (the post-echo). A plain delay cannot make a pre-echo; this does, at the cost of one print-spacing of latency. Spacing sets the reel-layer time between the ghosts and the original, Amount the print strength, Tone how dull the ghosts are, Mix the print intensity. A vintage-tape character and a uniquely 'impossible' pre-echo effect.

RatDist [FX] [FX] [SYNTH] in: Audio out: Audio

RAT-style distortion: a hard op-amp clip into a sweepable lowpass 'filter' control, giving the aggressive, fuzzy-yet-focused ProCo RAT grind. Drive sets the gain into the clip, Filter rolls off the top (counter-clockwise = darker), Level the output. in0 = Audio.

Rattle [FX] [FX] [SYNTH] in: Audio out: Audio

Rattle: while the signal is loud enough a loose part buzzes against it - fast random noise bursts gated by the input envelope, like a snare wire, a blown speaker or a rattling panel that only sounds when driven hard. Goes quiet the instant the signal drops. Thresh sets the drive needed, Amount the rattle level, Mix the blend.

RectifierHum [FX] [FX] [SYNTH] in: Audio out: Audio

Power-supply hum injector: adds mains-frequency hum (and its octave, the way a full-wave rectifier ripples at 2x mains) whose level GROWS with the signal - modelling a sagging power supply that hums harder when the amp is pushed. Distinct from a static hum: it ducks away in silence and blooms on transients. Freq sets the mains frequency, Amount the worst-case hum, Mix the blend.

✧ Robotize [FX] [FX] [SYNTH] in: Audio out: Audio

Robot voice: ring-modulates the input against a fixed carrier and imposes a vowel formant, turning speech or any signal into the classic talking-robot timbre. Pitch sets the carrier, Vowel the formant, Mix the wet. in0 = Audio.

✧ RotarySpeaker [FX] [FX] [SYNTH] in: Audio out: Audio

Rotary speaker (Leslie): a spinning-horn cabinet that adds pitch-doppler vibrato and amplitude tremolo, sweeping between slow (chorale) and fast (tremolo) - the swirling organ and guitar cabinet. Speed morphs slow to fast, Depth the swirl, Mix the wet. in0 = Audio.

✧ Rust [FX] [FX] [SYNTH] in: Audio out: Audio

Rust: a corrosion that ACCUMULATES the longer the signal is present - a crackle/grit bed that grows as the contact oxidizes, then slowly cleans up in silence. Unlike a static crackle, it gets worse over a held passage. Speed sets the rusting rate, Mix the blend.

✧ ScrapeFlutter [FX] [FX] [SYNTH] in: Audio out: Audio

Scrape flutter: the fast, nervous high-frequency wobble of tape scraping past a fixed guide or head - a rapid (tens of Hz) random micro-pitch and amplitude jitter plus a faint signal-gated scrape hiss, the gritty motion noise distinct from the slow musical wow. Rate sets the flutter speed, Depth the jitter intensity, Mix the blend.

✧ Shatter [FX] [FX] [SYNTH] in: Audio out: Audio

Shatter: the signal is smashed into tiny grains that play back from random points in a short buffer at random gains (some silent), so the sound fractures into a jagged spray of shards - a stochastic granular shatter, not a tidy looped stutter. Size sets the shard length, Chaos how often and how violently it re-shards, Mix the blend.

✧ Sigh [FX] [FX] [SYNTH] in: Audio out: Audio

Sigh: when a note releases (the level falls away) it triggers a downward filter sweep, so phrase endings trail off into a soft darkening sigh. A release-keyed motion you can't get from a static EQ. Fall sets how long the sigh takes, Mix the blend.

✧ SignFlip [FX] [FX] [SYNTH] in: Audio out: Audio

Sign flip: at a steady clock it randomly inverts the polarity of the whole signal for that window, scrambling the phase into a harsh, fizzing, broken texture - a pure waveform-polarity destroyer with no level change, just sign chaos. Rate sets the flip clock, Mix the blend.

✧ Sparkle [FX] [FX] [SYNTH] in: Audio out: Audio

Sparkle: every transient triggers a brief random high bell 'ting', so hits and picks throw off glittering, fairy-dust overtones at random pitches above the note - an onset-driven shimmer quite unlike a reverb or a fixed octaver. Amount sets the sparkle level, Decay the ting length, Mix the blend.

Sproing [FX] [FX] [SYNTH] in: Audio out: Audio

Sproing: each transient launches a cartoon spring ‘boing’ - a sine whose pitch DROPS rapidly after the hit (the opposite of WaterDrop) with a dispersive, comedic wobble. Turns drum hits and clicks into bouncy spring-toy sounds. Tune sets the start pitch, Drop how far it falls, Decay the bounce length, Mix the blend.

Sputter [FX] [FX] [SYNTH] in: Audio out: Audio

Sputter: the signal cuts in and out at irregular intervals with little bursts of crackle at each break - a failing cable, a dying tube, a loose connection. The randomness of the timing is the point: it never settles into a rhythm. Rate sets the average break rate, Mix the blend.

Stammer [FX] [FX] [SYNTH] in: Audio out: Audio

Stammer: an onset captures a tiny grain and replays it a handful of times with falling level before the sound continues - a quick involuntary repeat that trips over itself, the verbal stammer made audible. Rate sets the repeat speed, Decay how fast the stammer dies, Mix the blend.

SteamHiss [FX] [FX] [SYNTH] in: Audio out: Audio

Steam hiss: a high, airy band of noise that swells and breathes like escaping steam or a hi-hat wash, with the input level pushing the pressure up. Tone sets how bright/sharp the jet is, Amount the pressure, Mix the blend.

SubBass [FX] [FX] [SYNTH] in: Audio out: Audio

Sub-octave generator: tracks the in0 signal and synthesises a clean sine/square one octave below, fattening basslines and kicks with weight a normal octaver can’t. Tone shapes the sub from pure sine to growl, Mix blends it under the dry signal.

TapeHiss [FX] [FX] [SYNTH] in: Audio out: Audio

Tape noise floor: adds a band-limited hiss that behaves like real tape - it is loudest in the SILENCES and ducks down under the signal (self-masking), instead of the constant flat hiss of just summing in noise. Tone shapes the hiss colour, Amount the floor level, Mix the blend. Pairs with the tape/print-through blocks for an authentic worn-tape bed.

Tinnitus [FX] [FX] [SYNTH] in: Audio out: Audio

Tinnitus: a thin high sine that you only hear in the GAPS - it ducks away while the signal plays and rings out in the silences, building up after loud passages and slowly fading, exactly like the ear-ringing after a loud show. An eerie, fatiguing ambience or a sound-design unease tool. Freq sets the ring pitch, Amount the loudest ring, Mix the blend.

ToneStack [FX] [FX] [SYNTH] in: Audio out: Audio

Passive guitar-amp tone stack: the interacting Bass / Mid / Treble shelving network of a Fender/Marshall preamp, with the characteristic mid scoop when the controls interact. Bass boosts the lows, Treble the highs, Mid cuts or boosts the middle. in0 = Audio.

Tumble [FX] [FX] [SYNTH] in: Audio out: Audio

Tumble: a feedback echo whose delay time slowly GROWS after each onset, so the repeats droop downward in pitch and spread apart - the sound of something tumbling away down a flight of stairs. Time sets the starting gap, Feedback the tumble length, Mix the blend.

Twang [FX] [FX] [SYNTH] in: Audio out: Audio

Twang: each transient excites a tuned resonator whose pitch droops as it decays, so picks and hits bloom into a springy, boinging twang - the wobble-board / saw-blade / loose-spring voice. Tune sets the resonant pitch, Decay the twang length, Mix the blend.

Vaporize [FX] [FX] [SYNTH] in: Audio out: Audio

Vaporize: the attack passes clean but the sustain dissolves into a cloud of filtered noise driven by the signal's own envelope, so notes evaporate from solid tone into airy vapor as they hold. Amount sets how much vaporizes, Mix the blend.

VinylSim [FX] [FX] [SYNTH] in: Audio out: Audio

Vinyl record emulation: passes in0 through a wow/flutter pitch wobble (a slowly-modulated delay), a band-limiting filter and added surface crackle - the warm, worn character of a played record. Wow sets the pitch wobble, Crackle the surface noise, Mix the blend.

Warps [FX] [FX] [SYNTH] in: Carrier, Modulator out: Audio

Cross-modulator (meta): blends two inputs through one of four cross-modulation algorithms - crossfade, amplitude mod, ring mod, or sine wavefold of in0 by in1 - with Amount morphing from the clean carrier into the modulated signal. One knob from subtle movement to clangorous metallic timbres.

WaterDrop [FX] [FX] [SYNTH] in: Audio out: Audio

Water drop: each transient in the input strikes a resonator whose pitch RISES as it rings out - the unmistakable 'bloop' of a drip into water (the cavity shrinks as the bubble collapses, so the frequency sweeps UP). A percussion-to-droplet converter. Tune sets the base pitch, Rise how far the pitch sweeps up, Decay the ring, Mix the blend.

WaxCylinder [FX] [FX] [SYNTH] in: Audio out: Audio

Wax cylinder: the 1900s phonograph sound - a steep midrange band (everything below ~250 Hz and above ~3 kHz gone), heavy slow wow from the hand-cranked mandrel, constant surface crackle, and soft horn-driven saturation. The most extreme lo-fi in the box, older than tape. Age scales the bandlimiting + saturation, Wow the pitch drift, Surface the crackle, Mix the blend.

Envelope (6)

AD [ENV] [CV] [FX] [SYNTH] in: Gate out: CV

Attack-Decay envelope: a rising edge on the in0 gate fires a one-shot ramp up over Attack then back down over Decay, ignoring how long the gate is held - the classic percussive/pluck envelope. Patch its CV output into a VCA, filter cutoff or any param.

ADSR [ENV] [CV] [FX] [SYNTH] in: Gate out: CV

Full ADSR envelope: a held in0 gate ramps up over Attack, falls to the Sustain level over Decay and holds there; releasing the gate fades to zero over Release - the standard four-stage envelope for any voice or modulation.

AR [ENV] [CV] [FX] [SYNTH] in: Gate out: CV

Attack-Release (ASR) envelope: while the in0 gate is high the output ramps up to full over Attack and holds; when the gate falls it ramps back to zero over Release - a smooth sustaining envelope for pads and swells.

DADSR [ENV] [CV] [FX] [SYNTH] in: Gate out: CV

Delayed ADSR: like ADSR but with a Delay stage before the attack - the gate opens, nothing happens for Delay, then the four-stage envelope runs. Useful for staggered swells and per-voice timing offsets.

EnvADSR [ENV] [CV] [FX] [SYNTH] in: In A, In B out: CV

ADSR envelope: a gate drives the classic attack/decay/sustain/release contour, applied as a VCA to the audio input - the fundamental synth envelope as a self-contained block.

HoldDecay [ENV] [CV] [FX] [SYNTH] in: Audio out: CV

Peak-hold envelope with linear decay: latches the loudest recent peak, holds it for a set time, then ramps down at a constant slope (not exponentially) - a control source with a straight-line fall a plain release follower cannot make.

Filter (166)

Acid303 [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

TB-303-style acid filter: a smooth, squelchy 18 dB resonant lowpass that screams and 'wows' as Reso climbs - the liquid, rubbery acid-bassline character. Drive overdrives the input for grit, Reso self-oscillates near max. in1 modulates cutoff for that classic envelope-swept squelch.

AdaptiveNotch [FLT] [FX] [SYNTH] in: Audio out: Audio

Adaptive notch: a constrained second-order IIR notch whose centre frequency is steered by normalized-gradient descent on the output power, so it hunts down and removes a single steady whistle, hum or feedback tone wherever it sits - no reference input or manual tuning. Distinct from the reference-based LMS/NLMS cancellers (which need a copy of the noise) and from a fixed notch (which can't track). InitFreq seeds the search, Width the notch sharpness, Adapt the tracking speed.

AirAbsorb [FLT] [FX] [SYNTH] in: Audio out: Audio

Air absorption: models how the atmosphere swallows high frequencies over DISTANCE - a gentle, physically-shaped low-pass that gets progressively darker the farther away the source is, the cue your ear uses to hear depth. Stack it for distance/perspective without reverb. Distance sets how far, Mix the blend.

AllpassFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Cascaded allpass: passes all frequencies at unity gain but shifts their phase, building the swirling notches of a phaser when mixed with the dry signal or smearing transients on its own. Freq sets the centre, Stages the number of allpass sections (steeper phase), Mix the dry/wet for phasing.

AlphaTrimFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Alpha-trimmed-mean filter: sorts a 7-sample window, discards the Trim lowest and Trim highest values, and averages what remains - Trim=0 is a plain moving average, Trim=3 is the median, and values in between blend Gaussian-noise rejection with impulse rejection. A tunable bridge between linear smoothing and rank filtering.

AutoWah [FLT] [FX] [SYNTH] in: Audio out: Audio

Envelope-following resonant band-pass (auto-wah): the input level sweeps a band-pass from Base upward by Range, with Sens setting how strongly playing dynamics open it and Attack the follow speed. Reso sets the peak sharpness and Mix the blend. Funky on guitar, clav and bass; raise Reso for a vocal quack.

AWeighting [FLT] [FX] [SYNTH] in: Audio out: Audio

A-weighting filter: the standardized IEC 61672 / ANSI S1.4 frequency response that approximates the ear's sensitivity at moderate levels - a steep low-cut, a gentle presence rise around 2-4 kHz and a high roll-off - built as the exact 6th-order analog prototype (poles at 20.6, 107.7, 737.9, 12194 Hz) bilinear-transformed and normalized to 0 dB at 1 kHz. Thins a signal to its 'perceptually loud' band; the classic SPL-meter weighting as an audio filter.

BarReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Bar resonator (a first as a synth block): rings the input through the modes of a free-free metal or wooden BAR - the widely-stretched overtones (1, 2.76, 5.40, 8.93…) of a marimba, glockenspiel or music-box tine. Because the partials leap apart so fast it imposes a clear, bright, struck-mallet pitch on whatever passes through, turning drums or noise into tuned-percussion tones. Freq sets the bar's pitch, Sharpness the ring time, Mix the wet/dry blend.

BellReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Bell resonator (a first as a synth block): rings the input through the partials of a cast BELL - the hum, prime, tierce, quint and nominal tones (0.5, 1, 1.2, 1.5, 2, 2.5, 3) whose deliberately-tuned MINOR THIRD gives a bell its dark, mournful clang. Feeding a source through it casts that bronze, struck-bell colour over it. Freq sets the strike tone, Sharpness the ring time, Mix the wet/dry blend. A tuned-bell spectrum distinct from the bar and plate resonators.

BesselLowpass [FLT] [FX] [SYNTH] in: Audio out: Audio

Bessel lowpass: a 2nd-order IIR filter tuned to the Bessel family Q of 1/sqrt(3), which gives maximally flat group delay - so transients pass through with the least possible phase smearing, at the cost of a gentler magnitude rolloff than Butterworth. The filter family of choice when waveform shape matters more than steep cutoff.

BilateralSmoother [FLT] [FX] [SYNTH] in: Audio out: Audio

Bilateral smoother: averages a 7-sample window but weights each neighbour by how close its amplitude is to the centre sample (a Gaussian range kernel of width Range), so smooth regions are denoised while sharp transients - whose neighbours differ a lot - are left intact. A transient-preserving lowpass, unlike a plain moving average. Adds 3 samples latency.

Biquad [FLT] [FX] [SYNTH] in: Audio out: Audio

RBJ biquad: the textbook second-order filter with a type selector (low-pass, high-pass, band-pass, notch) and a resonance/Q control - the general-purpose EQ/filter primitive computed from the standard cookbook coefficients.

BodyResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Acoustic body resonator: three high-Q bandpass modes scaled by a Size knob stamp the boxy, woody resonant signature of an instrument body onto any input, like playing it through a guitar/violin shell.

Buchla292 [FLT] [FX] [SYNTH] in: Audio, Gate CV out: Audio

Buchla 292-style vactrol lowpass gate: a combined lowpass filter + VCA where a control voltage on in1 opens both together through a slow, organic vactrol response - the soft, bongo-like 'bonk' and gentle plucks of West Coast synthesis. Response sets the vactrol attack/decay sluggishness.

BuchlaLPG [FLT] [FX] [SYNTH] in: In A, In B out: Audio

Low-pass gate: a gate input charges a simulated vactrol whose slow, asymmetric response simultaneously opens an amplitude VCA and a low-pass filter, the plucky, organic 'bongo' dynamics of a Buchla LPG.

ButterworthLowpass [FLT] [FX] [SYNTH] in: Audio out: Audio

Butterworth lowpass: a true 4th-order maximally-flat-magnitude lowpass built from two cascaded biquads at the Butterworth Q pair (0.541, 1.307) - the flattest possible passband with a clean 24 dB/oct rolloff and no peak at cutoff. Completes the IIR filter family alongside the Bessel (flat group delay) and Chebyshev (rippled, steeper) lowpasses, for tone shaping where any resonant bump would colour the sound.

CapSag [FLT] [FX] [SYNTH] in: Audio out: Audio

Coupling sag: an AC-coupling high-pass whose corner frequency rises with level, so loud low-end momentarily loses its bottom and then 'sags' back as a thump when the signal drops - the bass pumping of an under-sized coupling capacitor. Amount sets the sag, Mix the blend.

ChaosLfoFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Chaos-swept filter: a logistic-map chaos generator (updated at the set rate) drives the lowpass cutoff, so the tone lurches through deterministic-yet-unpredictable jumps - more structured than random noise modulation, more alive than an LFO sweep.

ChebyshevLowpass [FLT] [FX] [SYNTH] in: Audio out: Audio

Chebyshev type-I lowpass: a 2nd-order IIR whose pole Q is derived from the chosen passband Ripple (in dB), trading a flat passband for a steeper rolloff than Butterworth - more ripple buys a sharper knee and a more resonant edge. The Q comes from the real Chebyshev pole formula, so it is a true filter-family response, not a waveshaper.

ChromaticReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Chromatic resonator (a first as a synth block): rings the input through resonators tuned to all twelve equal-tempered SEMITONES of the octave - a comb of every chromatic pitch, so it imposes a fixed twelve-tone pitch grid on whatever passes through, quantising noise or percussion toward an equal-tempered cluster. Root sets the lowest note, Sharpness the ring time, Mix the wet/dry blend. The densest scale resonator, a chromatic pitch-grid distinct from the consonant major/minor/pentatonic banks.

CicFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

CIC filter: two cascaded boxcar averages of length Taps, giving the sinc^2 magnitude response of a 2-stage cascaded-integrator-comb - a steep, multiply-free lowpass with deep spectral nulls at every multiple of the sample rate over Taps. Sharper rolloff and stronger notches than a single moving average; the workhorse anti-alias filter of decimators.

Cluster [FLT] [FX] [SYNTH] in: Audio out: Audio

Cluster filter (Pigments-style): places up to five resonant band-pass filters around a single Cutoff, fanned outward by Spread, and sums them into one animated, vowel-like comb of peaks. Sweeping Cutoff glides the whole cluster; widening Spread pulls the peaks apart from a single resonance into a chord of formants; Reso sharpens each peak toward self-oscillation; Mix blends against dry. Distinct from the harmonic SMR bank (whose bands track a fundamental's overtones) - here the peaks are symmetric octave-spread satellites of one cutoff.

CombBank [FLT] [FX] [SYNTH] in: Audio out: Audio

Tuned comb filter: feeds the input through a short delay with feedback so it resonates at a pitch (and its harmonics), turning noise into tuned tones and adding a metallic, flange-like comb colour to audio. Freq sets the pitch, Feedback the resonance/ring length, Mix the wet. Distinct from the single Comb in its long-feedback resonant tuning.

CombChord [FLT] [FX] [SYNTH] in: Audio out: Audio

Chordal comb: three feed-forward comb taps tuned to a triad of pitches place harmonically-spaced notch sets at once, coloring the input with a chord's worth of comb resonance.

CombFlutter [FLT] [FX] [SYNTH] in: Audio out: Audio

Fluttering comb: a tuned feedback comb whose delay length wobbles a few percent under an LFO, detuning the resonant peaks continuously so the metallic comb tone shimmers and choruses instead of sitting at a fixed pitch.

CombKeytrack [FLT] [FX] [SYNTH] in: Audio out: Audio

Pitch-tracking comb: detects the input period and tunes a comb filter's notches to a chosen harmonic of it in real time, so the comb coloration locks to the note being played rather than a fixed frequency.

CombMorph [FLT] [FX] [SYNTH] in: Audio out: Audio

Morphing comb filter: one control sweeps continuously from a feed-forward (FIR, notch) comb to a feedback (IIR, resonant) comb at the same tuned delay, with fractional-delay tuning in Hz.

CombResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Tuned comb resonator: a damped feedback comb makes any input (noise, clicks, drums) ring at a chosen pitch, the Karplus-Strong damping filter applied as an effect to turn percussive sources into tones.

CombSweep [FLT] [FX] [SYNTH] in: Audio out: Audio

Envelope-swept comb: the comb's tuned frequency rides the input envelope between two limits, so transients sweep the notches automatically - auto-flange driven by dynamics rather than an LFO.

ComplexResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Complex resonator: a one-pole filter with a complex coefficient Re^(jomega), iterating z = p*z + x so the state spirals as a phasor - the real part is a clean resonance and the two state variables are inherently in quadrature. The building block of the sliding DFT and Mathews' phasor filter, distinct from real-coefficient biquads. Resonance sets the pole radius.

ContactMic [FLT] [FX] [SYNTH] in: Audio out: Audio

Contact mic: the boxy, midrange-honk colour of a piezo disc stuck to a surface - a narrow band-pass plus structure-borne rumble that swells with how fast the signal moves (handling noise), so everything sounds like it was picked up through the body of an object rather than the air. Body sets the rumble, Mix the blend.

ConvolveCab [FLT] [FX] [SYNTH] in: Audio out: Audio

Short convolution: convolves the input with a fixed 16-tap impulse response (a tiny cabinet/early-reflection signature), the genuine FIR-convolution method real cabinet/IR loaders use, in miniature.

CrossComb [FLT] [FX] [SYNTH] in: In A, In B out: Audio

Cross-fed comb: the comb taps a delayed copy of a second input rather than itself, so one signal carves tuned notches into another (cross-synthesis), with the delay set in Hz.

Crossover [FLT] [FX] [SYNTH] in: Audio out: Audio

Two-band splitter that derives a low band from two cascaded one-pole low-passes (an LR2 response) and forms the high band by subtracting it from the input, outputting one band at a time. Freq sets the split frequency and Band chooses whether the low band or the high band is passed. Use it to route or isolate a frequency region, or to feed separate processing chains per band.

CryWah [FLT] [FX] [SYNTH] in: Audio, Pedal CV out: Audio

Wah pedal: a resonant bandpass swept by the Pedal control, the vocal 'wah' of a guitar pedal - rock the pedal (or in1) for the talking sweep. Pedal sweeps the peak, Reso the sharpness, Mix the wet. in1 = Pedal CV.

CWeighting [FLT] [FX] [SYNTH] in: Audio out: Audio

C-weighting filter: the standardized IEC 61672 / ANSI S1.4 response for peak and high-level sound measurement - nearly flat across the midband with gentle -3 dB roll-offs at 31.5 Hz and 8 kHz (a double pole at 20.6 Hz and 12194 Hz, the same outer corners as A-weighting but WITHOUT its midrange emphasis). Keeps the low end A-weighting throws away, so it reads bass-heavy material far closer to flat - the cinema/peak-level counterpart to the A curve.

DCBlocker [FLT] [FX] [SYNTH] in: Audio out: Audio

DC blocker: the standard one-pole/one-zero high-pass (y = x - x_prev + R*y_prev) that removes any constant offset or sub-sonic drift while leaving the audband untouched - essential after rectifiers and asymmetric distortion.

DericheGaussian [FLT] [FX] [SYNTH] in: Audio out: Audio

Recursive Gaussian filter: the Young-van Vliet 3rd-order IIR whose impulse response approximates a true Gaussian of width Sigma samples, giving a maximally smooth lowpass with no ringing - distinct from one-pole and biquad lowpasses, which have exponential or resonant impulse responses. Sigma sets the smoothing radius. Unity DC gain.

DiodeLadder303 [FLT] [FX] [SYNTH] in: Audio out: Audio

Acid diode ladder: a four-stage filter with the asymmetric diode-ladder feedback and gain compensation of the TB-303, giving the rubbery, squelchy resonance that defines acid bass lines.

DispDelay [FLT] [FX] [SYNTH] in: Audio out: Audio

Dispersive resonator: an all-pass in the feedback loop makes high partials travel at a different speed than lows, so the comb rings with the stretched, inharmonic, metallic timbre of a stiff string or bar.

DJFilter [FLT] [FX] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

One-knob DJ filter: with Cutoff at centre the signal is open, turning down sweeps in a low-pass and turning up sweeps in a high-pass, while Reso adds bite at the corner. The Mod input (in 3) offsets the knob for automation. The classic build-and-drop sweep on a single control.

DrumheadReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Drumhead resonator (a first as a synth block): rings the input through a bank tuned to the modes of a circular DRUM MEMBRANE - the inharmonic Bessel-function overtones (1, 1.59, 2.14, 2.30, 2.65…) that give a struck skin its pitch-ambiguous, tom-like ‘boing’ rather than a clear note. Drop a drum loop, voice or noise through it and the source takes on the body of a tuned membrane. Freq sets the membrane pitch, Sharpness the ring time, Mix the wet/dry blend.

DynamicComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Dynamic comb: a tuned comb whose resonant feedback rises with the input’s loudness, so soft passages pass nearly dry and loud hits ring into a sharp metallic resonance - a level-dependent comb intensity, distinct from a comb whose pitch is swept.

DynamicEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Dynamic EQ band: a parametric bell whose gain is driven by the energy in its own band, so it only cuts (or boosts) when that frequency gets loud - surgical, level-dependent tone control a static EQ can’t do.

EddyBrake [FLT] [FX] [SYNTH] in: Audio out: Audio

Eddy-current brake: damping that grows with the signal’s VELOCITY - the faster the waveform moves, the harder it is braked, like a conductive disc dragged through a magnetic field. Fast transients are slugged while slow motion passes, a velocity-dependent low-pass quite unlike a fixed filter. Brake sets the drag strength, Mix the blend.

EmphasisFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Emphasis filter: in Pre mode it applies the speech-coding pre-emphasis y=x-Coeffx[n-1], a single-zero FIR that boosts highs ~6dB/oct to flatten a falling spectrum; in De mode it applies the exact inverse IIR y=(1-Coeff)x+Coeffy[n-1] to restore it. The classic record/playback tilt pair, distinct from a pole-zero shelf. Coeff sets the corner.

EnvCombFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Envelope-swept comb: the comb's tuned resonance sweeps up with the input's loudness, so each transient drives the metallic peaks higher and they fall back as the note decays - a dynamic, vowel-like comb motion that tracks how hard you play.

EnvWah [FLT] [FX] [SYNTH] in: Audio out: Audio

Envelope auto-wah: the input envelope sweeps a resonant bandpass up and down, so playing dynamics open and close the wah - the funk envelope-filter sound, no pedal needed.

EQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Single parametric peaking EQ band (RBJ biquad): Gain boosts or cuts a bell centred at Freq, with Q setting its width - narrow for surgical notches, wide for broad tone shaping. Chain several for a full parametric EQ. See GraphicEQ for fixed bands and the analog pack (Pultec/Neve/SSL) for coloured curves.

Erode [FLT] [FX] [SYNTH] in: Audio out: Audio

Erode: high frequencies wear away the longer a note is held - a low-pass whose cutoff slides down as the signal sustains, so attacks stay bright but tails erode into darkness, recovering in silence. Rate sets the erosion speed, Mix the blend.

FibResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Fibonacci-tuned resonator bank (a first as a synth block): rings the input through resonators tuned to the Fibonacci harmonics of Base (Base1, 2, 3, 5, 8, 13…), so any source is recoloured into a drone whose overtones widen by the golden ratio - the warm, glassy Fibonacci spectrum imposed on whatever passes through. Sharpness sets the resonance Q and Mix the wet/dry blend. A golden-spaced resonator distinct from the prime-tuned and harmonic resonators.

Filter [FLT] [FX] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

Multimode state-variable (TPT) filter switchable between low-pass, high-pass and band-pass via Mode. Cutoff tracks an exponential mod input (in 3) scaled by Mod in octaves, while Reso sets the Q up to self-oscillation and an optional pre-Drive (tanh) dirties the resonance. Mix blends the filtered result back against the dry signal.

FilterFM [FLT] [FX] [SYNTH] in: Audio, FM CV out: Audio

Audio-rate filter FM: a resonant lowpass whose cutoff is frequency-modulated at audio rate by in1, producing metallic, bell-like and clangorous timbres that slow filter sweeps can't reach - the filter becomes a sound source. Cutoff sets the centre, Depth the FM amount in octaves, Reso the resonance.

FilterLFO [FLT] [FX] [SYNTH] in: Audio out: Audio

A 4-pole resonant ladder lowpass swept by an internal LFO that morphs between triangle and sawtooth shapes. Rate sets the LFO speed (Hz), Range sets the sweep depth around the cutoff (~60 Hz..15.6 kHz), Reso sets the resonant feedback, Shape morphs the LFO from triangle to saw, and Mix blends wet against dry. High Reso adds squelch while Shape changes the sweep from symmetric to ramped.

FixedFilterBank [FLT] [FX] [SYNTH] in: Audio out: Audio

Fixed filter bank (Moog 914): three parallel fixed bandpass bands - low, mid and high - each with its own level, a graphic-EQ-style spectral sculptor for formant and tonal shaping. Low / Mid / High set the band gains. in0 = Audio.

Fizz [FLT] [FX] [SYNTH] in: Audio out: Audio

Resonant TPT band-pass filter whose output is mixed with a rectified copy of itself, adding fizzy upper harmonics that grow with resonance. Cutoff sets the band centre (100-12000 Hz), Reso sets both the filter Q and the amount of rectified harmonic content, and Mix blends the result against the dry signal. Use it to add bright, buzzy edge and emphasis to a narrow frequency band.

FoldbackComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Foldback comb: a tuned feedback comb whose recirculating signal is wavefolded each pass, so as the resonance rings it folds into ever-brighter inharmonic overtones - a self-animating metallic drone that gets spikier the harder it is driven.

Formant [FLT] [FX] [SYNTH] in: Audio out: Audio

Morphing vowel filter: three parallel band-pass resonators tuned to formant frequencies are swept continuously across the A-E-I-O-U vowels by Vowel, with Q setting the formant sharpness and Mix the blend. Produces talking, vocal timbres. Sweep Vowel with an LFO or envelope for a wah-like vowel morph.

FormantBP [FLT] [FX] [SYNTH] in: Audio out: Audio

Dual formant filter: two independent resonant bandpass peaks impose vowel-like formant coloration on any input, the spectral signature of a vocal tract applied as an effect.

FractionalDerivative [FLT] [FX] [SYNTH] in: Audio out: Audio

Fractional-order calculus filter: applies the truncated Grunwald-Letnikov operator (an 8-tap FIR with binomial weights) to compute a derivative of fractional Order - positive values differentiate (a tunable +6dB/oct*Order high-boost), negative values integrate (low-boost), and values in between give spectral tilts no integer-order filter can. Distinct from fractional-delay: this changes magnitude slope, not time.

FreqWarpAP [FLT] [FX] [SYNTH] in: Audio out: Audio

Frequency warper: a chain of tunable first-order all-passes warps the effective frequency axis (the bilinear/Laguerre warp used in warped filtering), shifting where the spectrum's detail concentrates - a phasey, formant-bending coloration.

GeometricMeanFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Geometric-mean filter: averages the magnitudes of a 7-sample window in the log domain (exp of the mean of log|x|) and restores the centre sample's sign - because the log domain weights small values more heavily, it pulls toward the quieter samples and tames bursts differently from an arithmetic average. A log-mean smoother. Adds 3 samples latency.

GlassReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Glass resonator (a first as a synth block): rings the input through the sparse, nearly-pure modes of a rubbed wine GLASS or glass-harmonica bowl (about 1, 2.32, 4.25, 6.63, 9.38) - a few widely-spaced, slow-decaying partials that give glass its clear, singing, almost-sine voice. Passing a source through it lends that fragile crystalline ring. Freq sets the glass pitch, Sharpness the (long) ring time, Mix the wet/dry blend. A glassy sparse spectrum distinct from the dense plate and bell resonators.

GoertzelTone [FLT] [FX] [SYNTH] in: Audio out: Audio

Goertzel detector: a leaky single-bin Goertzel recurrence tracks how much energy the input holds at one target frequency and outputs that magnitude as a CV - the efficient tone/DTMF detector used when you only care about one frequency.

GoldenResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Golden-ratio resonator bank (a first as a synth block): rings the input through resonators spaced by powers of the golden ratio (Base, Base1.618, Base2.618, Base*4.236…) - an INHARMONIC bank, so no resonance is a simple multiple of another and the source is recoloured into a shimmering, metallic, bell-like resonance that hangs between a pitch and a clangour. Sharpness sets the resonance Q and Mix the wet/dry blend. The inharmonic counterpart of the prime and octave resonators, ideal for turning drums or noise into struck-metal textures.

GraphicEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Three-band graphic EQ: one-pole filters at 250 Hz and 4 kHz split the input into low, mid and high bands, each scaled by its own +/-12 dB gain and summed back. Low, Mid and High set the per-band gain in dB and Mix blends against dry. A simple tonal shaping tool for broad low/mid/high balance rather than surgical correction.

HampelDeclicker [FLT] [FX] [SYNTH] in: Audio out: Audio

Hampel declicker: over a 7-sample sliding window it finds the median and the median absolute deviation, and replaces the centre sample with the median only when it lies more than Threshold MADs away - so impulsive clicks and digital spikes are removed while the rest of the waveform passes through untouched. A robust despiker, not a lowpass. Adds 3 samples latency.

HarmonicTrap [FLT] [FX] [SYNTH] in: Audio out: Audio

Pitch-tracking harmonic notch: detects the input period from zero crossings and places a feed-forward comb notch on a chosen harmonic of it, removing (or isolating) that overtone as the pitch moves.

HarpReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Harp-string bank (a first as a synth block): rings the input through eight plucked-string loops tuned to a full DIATONIC scale (a whole octave of harp strings), so any source ripples across the strings and rings back as cascading, scale-tuned harp resonance - feed it transients and it answers in glissando-like washes. Root sets the key, Decay the string sustain, Damp the brightness, Mix the wet/dry blend. A diatonic string bank distinct from the chordal and drone banks.

HeadGapLoss [FLT] [FX] [SYNTH] in: Audio out: Audio

Head-gap loss: a tape/disk playback head can't resolve wavelengths shorter than its gap, so the response has a comb of nulls starting at the gap frequency - this models that with a feed-forward comb whose first null tracks the Gap, progressively hollowing and darkening the top end the way a worn or wide-gap head does. Gap sets the loss frequency, Depth how deep the nulls cut, Mix the blend.

ImpulseInvariantLowpass [FLT] [FX] [SYNTH] in: Audio out: Audio

Impulse-invariant lowpass: the analog 2-pole resonator is discretized so its digital impulse response samples the analog one exactly (partial-fraction residues -> a single feed-forward term, no Nyquist zeros). Unlike the bilinear transform it preserves time-domain shape and group delay but can alias, giving a subtly different resonant colour. Resonance sets the pole Q.

KalmanFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Scalar Kalman filter: a smoother whose gain adapts every sample from an estimate of signal vs measurement variance - when the input moves fast the gain rises and it tracks, when steady the gain falls and it denoises. Process sets how quickly the model expects change, Measurement sets assumed input noise. A self-tuning one-pole, not a fixed cutoff.

KarplusComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Tuned comb resonator: a fractional-delay feedback comb tuned in Hz with a damping low-pass in the loop, so whatever you feed it rings at the set pitch like a struck string or metal bar - excitation in, pitched metallic resonance out.

KeytrackFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Key-tracking filter: detects the input's pitch from its zero crossings and sets a resonant low-pass cutoff to a chosen multiple of it, so the filter follows the note and the timbre stays constant across the range - automatic filter keytracking.

KotoReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Koto-string bank (a first as a synth block): rings the input through nine plucked-string loops tuned to a PENTATONIC scale across two octaves, the open, gap-toothed tuning of a Japanese koto or a guzheng. Because the pentatonic scale has no dissonant intervals, any source - even noise - comes back as a consonant, bright, plucked-string cascade. Root sets the key, Decay the string sustain, Damp the brightness, Mix the wet/dry blend. A pentatonic string bank distinct from the diatonic harp and chordal banks.

KWeighting [FLT] [FX] [SYNTH] in: Audio out: Audio

K-weighting filter: the two-stage pre-filter at the heart of the ITU-R BS.1770 / EBU R128 loudness (LUFS) standard - a +4 dB high-frequency shelf (the 'head' filter modelling a head in the soundfield) followed by a ~38 Hz high-pass (the RLB curve). Sample-rate-correct via RBJ biquads rather than the usual hard-coded 48 kHz coefficients. A distinct broadcast-loudness colour, brighter and bass-light, different from the A-weighting curve.

Ladder [FLT] [CIRCUIT] [FX] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

Moog-style 4-pole transistor-ladder low-pass: four cascaded one-poles with a tanh input Drive and global resonance feedback that self-oscillates near maximum Reso. Cutoff tracks an exponential mod input (in 3) in octaves. Warmer and more characterful than the clean Filter - push Drive into the ladder for growl; see the Mini analog model for a Minimoog-voiced variant.

LatticeFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Lattice (Gray-Markel) filter: a two-stage all-pole lattice parameterized directly by reflection coefficients K1, K2 instead of biquad coefficients - the structure used in speech LPC, where each coefficient stays in (-1,1) for guaranteed stability and maps to a resonant pole pair. A different, numerically robust way to dial in two-pole resonance.

LineEnhancer [FLT] [FX] [SYNTH] in: Audio out: Audio

Adaptive line enhancer (ALE): an NLMS predictor fed a decorrelation-delayed copy of the input learns to predict only the part that stays correlated across the delay - the periodic, narrowband content - while broadband noise (uncorrelated past the delay) cancels out. The output is the cleaned-up tone, the dual of the adaptive notch. No reference input needed: it references a delayed version of itself. Delay sets the decorrelation gap, Taps the predictor length, Adapt the convergence speed.

LinkwitzRiley [FLT] [FX] [SYNTH] in: Audio out: Audio

Linkwitz-Riley crossover: cascaded fourth-order low/high splits at two corner frequencies isolate a low, mid or high band with the flat-summing, phase-coherent response used in real multiband splitters and speakers.

LoFi [FLT] [FX] [SYNTH] in: Audio out: Audio

Lo-fi filter (Pigments-style): degrades the signal by undersampling (Downsample sample-and-holds the input at a reduced rate for aliasing grit), bit-reducing (Bits quantises the amplitude for digital crunch), injecting Noise (a hiss/dirt floor), then smoothing with a Cutoff low-pass - from subtle vintage-sampler warmth to extreme broken-converter mangling. Mix blends against dry. A degradation filter rather than a resonant one: use it to age pads, crush drums or add converter character.

LowPassGate [FLT] [FX] [SYNTH] in: Audio, Ping (CV) out: Audio

Buchla west-coast low-pass gate: a vactrol couples a VCA and VCF so a ping opens both with the characteristic nonlinear ringing decay. Mode picks VCA / VCF / both.

LPG [FLT] [FX] [SYNTH] in: Audio, Gate (CV) out: Audio

Vactrol-style low-pass gate: the control input (in 2) simultaneously opens a low-pass filter and a VCA, so brightness and level rise together as the control climbs. Response curves the control law from snappy to sluggish, emulating the vactrol's lag, and Mix blends against dry. Driven by an envelope or trigger it gives the plucky, organic dynamics of Buchla-style patches.

LpHpMorph [FLT] [FX] [SYNTH] in: Audio out: Audio

Morphing SVF: a single Morph knob crossfades the state-variable filter's outputs continuously from low-pass (0) through band-pass (1) to high-pass (2), so one control sweeps the entire filter character at a shared cutoff and resonance - no discrete mode switch.

MajorReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Major-scale resonator (a first as a synth block): rings the input through a bank of resonators tuned to the seven notes of a MAJOR scale (plus the octave), so any source - a drum loop, noise, a voice - is pulled into a bright, consonant, major-key wash, an instant ambient pad. Root sets the key, Sharpness the ring time (turn it up for a sustaining drone), and Mix the wet/dry blend. A scale-tuned resonator distinct from the harmonic and object resonators.

MatchedZLowpass [FLT] [FX] [SYNTH] in: Audio out: Audio

Matched-Z lowpass: the analog 2-pole prototype is mapped to the digital domain by placing each pole directly at z=e^(s*T) and the zeros at Nyquist (z=-1), instead of the usual bilinear transform - so there is no frequency warping and the resonant peak lands exactly where the analog pole sits. Resonance sets the pole Q.

MedianClean [FLT] [FX] [SYNTH] in: Audio out: Audio

Running-median filter: replaces each sample with the median of the last N (odd) samples, removing impulse clicks and crackle while preserving edges in a way a linear filter cannot.

MetalComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Inharmonic dual-tap comb: two detuned feed-forward taps off one delay line place non-integer-spaced notches, giving a metallic, bell-like coloration distinct from a single tuned comb.

MinorReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Minor-scale resonator (a first as a synth block): rings the input through resonators tuned to the seven notes of a NATURAL MINOR scale (plus the octave), recolouring any source into a darker, melancholic, minor-key drone. Root sets the key, Sharpness the ring time, Mix the wet/dry blend. The minor counterpart of the major resonator, ideal for turning drums or noise into a brooding ambient pad.

MoogLadder [FLT] [FX] [SYNTH] in: Audio out: Audio

Transistor ladder low-pass: four cascaded one-pole stages with a tanh-saturated global feedback path model the Moog ladder's fat, self-oscillating 24 dB/oct resonance.

MorphFilter [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

Continuously-morphing state-variable filter: one Morph knob sweeps smoothly from lowpass through bandpass to highpass on a single resonant Chamberlin core, so the filter type itself can be automated - dark to open to thin - without switching blocks. Cutoff sets the corner, Reso the resonance, and in1 modulates the cutoff.

MS20Filter [FLT] [FX] [SYNTH] in: Audio, Cutoff CV, Reso CV out: Audio

Korg MS-20-style Sallen-Key lowpass: a screaming, aggressive resonant filter. Drive pushes the input into a tanh stage before the filter and Reso climbs into self-oscillation - the gritty, acidic MS-20 squelch. in1 modulates cutoff, in2 modulates resonance.

MudSquelch [FLT] [FX] [SYNTH] in: Audio out: Audio

Mud squelch: a resonant low-pass whose cutoff wanders on a slow random walk, dragging the tone through a wet, gloopy, squelching filter sweep - the sound of something sinking into mud. Cutoff sets the centre of the wander, Mix the blend.

MultiMode [FLT] [FX] [SYNTH] in: Audio out: Audio

Morphing multimode filter: a single state-variable core whose output crossfades continuously from low-pass through band-pass to high-pass with one Mode knob - all three responses from one resonant filter.

NasalFormant [FLT] [FX] [SYNTH] in: Audio out: Audio

Nasal formant: adds a nasal pole-zero pair - a resonant peak with a paired anti-resonance just above it - so the tone takes on the pinched, honking 'talking through the nose' colour that a simple band-pass formant can't produce. Freq sets the nasal centre, Amount the nasality, Mix the blend.

NoiseModFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Random-warble filter: the lowpass cutoff wanders continuously over a smoothed white-noise control, so the tone breathes and shifts unpredictably - an organic, never-repeating filter motion distinct from an LFO-swept or sample-and-hold filter.

Notch [FLT] [FX] [SYNTH] in: Audio out: Audio

Band-reject (notch) filter that removes a narrow band around Freq, with Q setting its width and Mix the blend. Use it to kill hum, a resonance or a feedback frequency, or sweep it for phaser-like effects. Narrow Q for surgical removal, wider for broad scooping.

NotchBank [FLT] [FX] [SYNTH] in: Audio out: Audio

Triple notch bank: three independently-spread band-reject notches carve fixed nulls out of the spectrum (the complement of a formant bank), useful for de-essing, feedback control or hollow, phasey coloration.

NotchFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Notch filter: a state-variable band-reject that removes a narrow band around the cutoff while passing everything else, computed as input minus the band-pass - Reso sets how narrow and deep the notch is. For taming a resonance, removing hum, or as a fixed phaser-style scoop.

OberheimSEM [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

Oberheim SEM-style state-variable filter: a smooth, creamy 12 dB multimode (lowpass / bandpass / highpass via Mode) with gentle resonance and the warm, rounded SEM character - the classic fat, musical filter for pads and leads. in1 modulates cutoff.

OctaveResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Octave-stack resonator bank (a first as a synth block): rings the input through resonators tuned to pure octaves of Base (Base1, 2, 4, 8, *16…), so any source is pulled toward a single strongly-reinforced pitch with the bright, hollow body of an organ stop - every resonance an octave of the last, nothing in between. Sharpness sets the resonance Q and Mix the wet/dry blend. A pure-octave resonator distinct from the dense harmonic and prime banks.

OddResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Odd-harmonic resonator bank (a first as a synth block): rings the input through resonators on the ODD harmonics of Base only (Base1, 3, 5, 7, *9…), the spectrum of a square wave or a stopped pipe, so any source takes on a hollow, woody, clarinet-like resonance with no even overtones. Sharpness sets the resonance Q and Mix the wet/dry blend. An odd-harmonic resonator distinct from the prime and Fibonacci banks.

OnePoleAP [FLT] [FX] [SYNTH] in: Audio out: Audio

First-order all-pass: passes every frequency at full level but rotates phase, crossing 90 degrees at its tuned frequency - a phase-alignment / diffusion building block and the cell phasers are built from.

ParaEQ [FLT] [FX] [SYNTH] in: L, R out: Audio

Fully-parametric EQ band with stereo / mid-side / L / R processing - stack several for an Ableton EQ-Eight-style mid-side parametric EQ. Type selects the RBJ filter shape (low cut, low shelf, bell, high shelf, high cut, notch), Freq the centre/corner, Gain the boost or cut (shelf/bell), Q the width, and Mode whether the band acts on the full

Stereo signal, only the Mid, only the Side, or one channel (L/R). Out picks L or R. Reads in0 = L, in1 = R; place two instances (Out=L and Out=R) for a stereo band.

PeakEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Parametric bell EQ: boosts or cuts a band around a centre frequency with adjustable width by adding a scaled band-pass back to the dry signal - one fully-sweepable parametric EQ band.

PeakFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

A biquad peaking EQ band that boosts a single frequency region, followed by a tanh soft limiter to tame the boosted peaks. Freq sets the center frequency (40..16000 Hz), Gain sets the boost amount (0..18 dB), Edge sets the band Q/sharpness, and Mix blends the filtered signal against dry. Useful for emphasizing a resonant frequency without hard clipping when pushed.

PentatonicReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Pentatonic resonator (a first as a synth block): rings the input through resonators tuned to the five notes of a MAJOR PENTATONIC scale (across two octaves) - the gap-toothed, no-semitone scale of folk and gamelan that has no dissonant intervals, so ANYTHING fed through it comes out consonant and open. Root sets the key, Sharpness the ring time, Mix the wet/dry blend. The most foolproof scale resonator for instant harmonious drones from any source.

Pinch [FLT] [FX] [SYNTH] in: Audio out: Audio

Pinch: a notch at Freq that DEEPENS as the level rises, so loud passages get a hollow scooped pinch in the spectrum while quiet ones pass full - a level-keyed dynamic notch, the inverse of a resonant boost. Freq sets where it pinches, Amount the maximum scoop, Mix the blend.

PingBP [FLT] [FX] [SYNTH] in: Audio out: Audio

Pinged bandpass: a high-resonance state-variable bandpass rings on every transient that hits it, turning clicks and percussive input into tuned, decaying tones (a resonator you excite by playing into it).

PingFilter [FLT] [FX] [SYNTH] in: Ping, Pitch CV out: Audio

Pinged resonant filter: a high-resonance bandpass struck by triggers on in0, ringing out a tuned, decaying tone - the classic ‘pinged filter’ tuned-percussion / pluck trick. Freq sets the pitch, Reso the ring length, Level the output. Trigger on in0, in1 = Pitch CV.

Pipe [FLT] [FX] [SYNTH] in: Audio out: Audio

Waveguide pipe resonator (Absynth Pipe-style): a tuned feedback comb that rings at the Tune frequency, switchable between an Open pipe (full harmonic series, positive feedback) and a Closed pipe (only odd harmonics, inverted feedback) for a hollow, clarinet-like colour. Feedback sets the resonance and ring time, Damp rolls off the highs in the loop for a softer pipe, and Mix blends the resonated signal against the dry input. Excite it with a transient or noise burst for plucked/blown-tube tones.

PlateReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Plate resonator (a first as a synth block): rings the input through the dense, inharmonic modes of a vibrating METAL PLATE (about 1, 2.08, 3.41, 3.89, 5.00, 6.45) - the crowded overtone field behind plate reverbs and the shimmer of a cymbal or gong. Its many close, non-integer partials smear any source into a bright metallic wash. Freq sets the plate's tuning, Sharpness the ring time, Mix the wet/dry blend. A 2-D plate spectrum distinct from the 1-D bar and string resonators.

PluckReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Single-string resonator (a first as a synth block): rings the input through ONE tuned plucked-string loop (Karplus-Strong), turning any source into a sustained, pitched, plucked-string resonance at the Root note - a string reverb. Transients excite the string into a clear plucked ring; sustained input drives it like a bowed or e-bowed string. Root sets the pitch, Decay the sustain (toward infinite at the top), Damp the brightness, Mix the wet/dry blend. The single-string core of the sympathetic banks, a tuned string-reverb distinct from the multi-string drones.

PolivoksFilter [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

Polivoks filter: the harsh, screaming Soviet 2-pole resonant filter (Vladimir Kuzmin's design), whose resonance distorts into a raw, biting howl unlike any Western filter. Cutoff, Reso, Mode picks lowpass or bandpass. in1 modulates cutoff.

PrimeResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Prime-tuned resonator bank (a first as a synth block): runs the input through a bank of sharp resonators tuned to the PRIME harmonics of Base (Base2, 3, 5, 7, *11…), so whatever you feed it - noise, a drum loop, a voice - rings out as a pitched drone whose overtones are the primes, the hollow bell-organ colour of the prime spectrum imposed on any source. Sharpness sets the resonance Q (how long each prime rings), and Mix blends the resonated tone against the dry. A number-tuned resonator distinct from a plain comb or formant filter.

PZFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Drawable magnitude-response filter: samples a frequency-to-gain curve drawn in node config at 16 log-spaced bands and bakes one peaking-EQ biquad per band, passing the input at in0 through the series to reproduce the drawn shape. Amount morphs between the dry input and the fully filtered output. Draw boosts and cuts across the spectrum to build custom EQ curves or resonant filter shapes.

Rasp [FLT] [FX] [SYNTH] in: Audio out: Audio

Resonant TPT band-pass filter followed by an asymmetric tanh waveshaper that drives positive half-cycles harder than negative ones, producing a rasping, lopsided distortion. Cutoff sets the band centre (100-12000 Hz), Reso sets the Q and scales the drive so higher resonance increases the snarl, and Mix blends against the dry signal. Ranges from a soft growl at low settings to an aggressive scream at high resonance.

RegaliaMitraEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Regalia-Mitra shelving EQ: realizes a first-order shelf as 0.5((1+Gain)x + (1-Gain)*A(x)) where A is a tunable first-order allpass - the classic allpass-decomposition topology in which the corner frequency (allpass coefficient) and the boost/cut amount are fully independent. Gain=1 is flat, >1 boosts, <1 cuts.

ResHP [FLT] [FX] [SYNTH] in: Audio out: Audio

Resonant high-pass: the high-pass output of a state-variable filter with a resonant peak at the corner, for thinning out low end while adding an emphasised whistle at the cutoff - the counterpart to a resonant low-pass.

ResonantEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

Resonant EQ band: a single sharp, high-Q peak added to the signal - a surgical resonant filter for ringing out or carving one frequency. Freq sets the band, Gain boosts (+) or cuts (-), Q the sharpness.

Resonator [FLT] [FX] [SYNTH] in: Audio out: Audio

High-Q band-pass tuned to a musical pitch, converting the Pitch parameter in semitones to a centre frequency and ringing the input through it like a struck body. Pitch sets the resonant note, Reso sets the Q (sharpness and ring length), and Mix blends the doubled resonant output against the dry signal. Drive it with transients or noise for tuned percussion, pinged-body and comb-resonance textures.

ResoSweepFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Envelope-swept resonant filter: a state-variable low-pass whose cutoff jumps open on each transient and decays back down, with resonance high enough to self-oscillate - an auto-wah/laser-zap that tracks your dynamics.

RIAAFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

RIAA phono EQ: the standard vinyl playback equalization curve - the inverse of the bass-cut/treble-boost used when cutting a record - realized as a bilinear-transformed biquad from the three RIAA time constants (3180/318/75 us, i.e. poles at 50 Hz & 2122 Hz, a zero at 500 Hz). Boosts the lows and rolls off the highs, normalized to unity at 1 kHz. A genuine ‘make it sound like a record’ tone curve, not a generic shelf.

Rings [FLT] [FX] [SYNTH] in: Audio out: Audio

Modal resonator bank: three TPT band-pass resonators tuned around a base pitch are excited by the input, summed to produce a struck/plucked resonant body. Pitch sets the fundamental in semitones, Structure morphs the partial spacing from harmonic (1, 2, 3) to inharmonic (1, 2.76, 5.40), Decay sets the resonator Q (ring length), and Bright scales the summed output level. Mix blends the resonant tone with the dry signal for physical-modeling and bell/string textures.

Ripple [FLT] [FX] [SYNTH] in: Audio out: Audio

Tuned comb resonator built from a short delay line whose length tracks a musical pitch, with a low-passed feedback path that makes it ring at that frequency. Pitch sets the resonant note as a MIDI value (24-108 st), Reso sets the feedback amount up to 0.98 for longer, sharper ringing, Damp sets the high-frequency damping in the feedback path (a darker, woodier ring as it rises), and Mix blends the resonant tone against the dry signal. Pushes input toward a pitched, metallic ring keyed to the chosen note.

Sallen [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

Sallen-Key 2-pole resonant lowpass: the smooth, gently-saturating 12 dB filter of the Korg MS-10/MS-20 and Wasp lineage, growling as Reso climbs into self-oscillation. Cutoff sets the corner, Reso the resonance, Drive the input saturation. in1 modulates cutoff.

SallenKey [FLT] [CIRCUIT] [FX] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

Resonant low-pass built on a TPT state-variable core taking the low-pass output, with the cutoff frequency exponentially modulated by in 3 scaled by Mod in octaves. Cutoff sets the corner, Reso raises the Q toward self-oscillation, and Mix blends the filtered low-pass back against the dry input. The Sallen-Key topology gives a smooth, gentle resonance suited to warm synth and bass filtering.

SavitzkyGolay [FLT] [FX] [SYNTH] in: Audio out: Audio

Savitzky-Golay smoother: convolves a 5-sample window with the exact least-squares quadratic kernel (-3,12,17,12,-3)/35, which fits a parabola through the window and reads off its centre - so it removes noise while preserving the height and width of peaks far better than a moving average. The classic spectroscopy smoother. Adds 2 samples latency.

Shelf [FLT] [FX] [SYNTH] in: Audio out: Audio

Low- or high-shelf EQ (RBJ shelving biquad; Type selects low or high): boosts or cuts everything below/above Freq by Gain at a fixed slope. Use for broad tonal tilt - warming the lows or adding air to the highs. Pair two for a tilt curve, or see TiltEQ for a single-knob version.

ShelfHi [FLT] [FX] [SYNTH] in: Audio out: Audio

High shelf EQ: splits the signal at a corner frequency and applies gain to everything above it, the treble tilt/air control of a mixing console as a single block.

Sherman [FLT] [FX] [SYNTH] in: Audio, Cutoff Mod out: Audio

Sherman Filterbank (circuit model): a faithful model of the Belgian dual analog filter / distortion box. The input hits a VCA-overdrive stage (asymmetric soft clip -> even-harmonic grind, hotter Drive = more distortion and a stronger envelope), then TWO two-integrator (Chamberlin) state-variable filters whose resonance feedback path is saturated, so high Reso screams and self-oscillates instead of blowing up - the Sherman's signature. A rectifier + RC envelope follower tracks the (hot) input and sweeps the cutoff by Env, Freq sets filter 1's base cutoff, Harmonics offsets filter 2 in semitones, and Mode routes the pair Serial (1->2 LP/BP/HP), Parallel (LP+HP or BP+BP) or Ring (the two band-passes ring-modulated). in1 is the FM input (patch an LFO/envelope/audio to frequency-modulate the cutoff). Mix is the Bypass/Effect blend.

SKFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Sallen-Key low-pass: the 2-pole active-filter topology (as in the MS-20 / Korg35) with its own resonant peak and gentle clip, a 12 dB/oct voice distinct from the ladder.

SlapComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Tuned resonator comb: a damped feedback comb tuned in Hz rings the input like a plucked string (Karplus-style damping in the loop); decay sets sustain, mix blends the resonance with the dry signal.

SmithAngellResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Smith-Angell resonator: the classic two-pole resonator improved with zeros at DC and Nyquist (numerator 1 - z^-2), which flattens the skirts and keeps the peak gain nearly constant as the centre frequency sweeps - unlike a plain two-pole resonator whose peak level drifts with tuning. Resonance sets the pole radius (sharpness).

SMR [FLT] [FX] [SYNTH] in: Audio out: Audio

Spectral multiband resonator (4ms SMR): six resonant band-pass filters tuned in a harmonic series excite the in0 input into a ringing, vocal chord - an oscillator bank, resonator and vocoder-like effect in one. Tune sets the lowest band, Spread the spacing of the six bands (harmonic to stretched), Reso the resonance/ring time of each band, and Mix the wet blend. Feed it drums or noise for struck metallic chords, or audio for formant/vocoder colour. Distinct from the single-band Resonator and the modal Rings.

SpeakerSim [FLT] [FX] [SYNTH] in: Audio out: Audio

Speaker cabinet sim: a low-pass roll-off plus two resonant peaks approximate a guitar speaker's frequency response, taming fizz and adding body so a raw distortion sounds like a miked cabinet.

SpectralSmear [FLT] [FX] [SYNTH] in: Audio out: Audio

Allpass smear: six first-order allpass sections in series, fed back on themselves, rotate phase so hard the transients dissolve into a flat-magnitude metallic wash - diffusion without a delay line.

SpectraTilt [FLT] [FX] [SYNTH] in: Audio out: Audio

One-knob spectral tilt: a single control rotates the whole spectrum around a pivot frequency, boosting lows while cutting highs (or vice versa) from one shared low-pass state.

SpectraTiltDyn [FLT] [FX] [SYNTH] in: Audio out: Audio

Dynamic spectral tilt: the brightness tilt around a pivot scales with the input envelope, so the tone opens up as you play harder and darkens when soft - dynamics-linked spectral balance.

SplitEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

A peaking EQ that knows transients from tone: a fast/slow envelope split weighs the band gain toward the attack or the sustain, so you can brighten transients without harshening the body.

Steiner [FLT] [CIRCUIT] [FX] [SYNTH] in: Audio out: Audio

Steiner-Parker style multimode filter whose TPT core derives low-pass, high-pass, band-pass and notch responses simultaneously, with Mode selecting which one is output. Cutoff sets the corner frequency, Reso controls the Q toward self-oscillation, and Mix crossfades the selected response against the dry signal. Switching modes covers everything from dark sweeps to hollow notch tones for synth and effect duty.

SteinerParker [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

Steiner-Parker / Synthacon-style multimode filter: a snappy, vocal-sounding 12 dB filter with lowpass, bandpass, highpass and notch modes (Mode), diode-asymmetric soft clipping in the path. Distinct, reedy character versus the ladder filters. in1 modulates cutoff.

Sweep [FLT] [FX] [SYNTH] in: Audio out: Audio

Onset-triggered filter sweep: an amplitude envelope detects input onsets and restarts a ramp that drives a TPT low-pass cutoff from 100 Hz upward for riser/faller effects. Time sets the ramp duration, Range scales how far the cutoff travels, Reso sets the filter resonance, and Dir flips the ramp between rising and falling. Mix blends the swept output with the dry signal.

SympatheticStrings [FLT] [FX] [SYNTH] in: Audio out: Audio

Sympathetic-string bank (a first as a synth block): rings the input through a bank of tuned, plucked-string loops (Karplus-Strong) set to a major chord, so any source - a drum hit, a breath, a noise burst - excites the strings into a rich, sustaining sympathetic resonance, the way the undamped tarab strings of a sitar or sarangi shimmer behind every note. Unlike the thin 2-pole resonators each string rings its WHOLE harmonic series, giving a dense, lively, plucked-metal sustain. Root sets the chord, Decay the sustain, Damp the brightness, Mix the wet/dry blend.

SympResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Sympathetic resonators: four tuned feedback combs at chord ratios ring in sympathy with whatever you feed them (like a sitar's drone strings or an undamped piano with the pedal down), adding a shimmering resonant halo.

Talkbox [FLT] [FX] [SYNTH] in: Audio out: Audio

Vowel formant filter (talkbox-style): passes the input through three resonant formant band-passes whose centre frequencies morph across the A-E-I-O-U vowels, imposing a singing/talking vocal colour on any sound so a guitar, synth or drum loop 'speaks' a vowel. Vowel selects and morphs the formant, Res sets the formant sharpness, and Mix blends against the dry. Single-input (unlike the Vocoder, which needs a separate modulator) and it resonates the formants rather than shifting existing ones.

TalkVowel [FLT] [FX] [SYNTH] in: Audio out: Audio

Vowel formant morpher: three resonant bandpass formants interpolate across the A-E-I-O-U vowel set with one knob, making any input speak vowels - a talk-box-style spectral overlay.

Tanpura [FLT] [FX] [SYNTH] in: Audio out: Audio

Tanpura-drone string bank (a first as a synth block): rings the input through four sympathetic strings tuned to the classic Indian tanpura drone (Sa, Sa, Pa, high Sa), so any source sets off the endless, buzzing, overtone-rich Sa-Pa-Sa drone that underpins all Hindustani music. The long feedback loops and gentle damping give the characteristic shimmering, never-quite-decaying jivari sustain. Root sets the tonic, Decay the (long) drone length, Damp the brightness, Mix the wet/dry blend. A specific drone tuning distinct from the chordal sympathetic bank.

Telephone [FLT] [FX] [SYNTH] in: Audio out: Audio

Band-limits the input to roughly 300 Hz-3.4 kHz by cascading a one-pole low-pass at 3.4 kHz into a one-pole high-pass at 300 Hz, leaving only the narrow midrange band that survives a phone line. Mix crossfades between the dry signal and the band-passed result. Use it for lo-fi telephone, intercom and AM-radio voice effects.

ThreeSisters [FLT] [FX] [SYNTH] in: Audio out: Audio

Triple resonant filter (Mannequins Three Sisters): three filters - Low, Centre, High - move together by Freq while Span spreads their cutoffs apart. Quality sweeps from a gentle resonant emphasis up to self-oscillating ringing sine tones (a three-voice resonance choir); Mode switches between the spread filter and a Formant voicing. Feed it audio for sweepable formants and resonant drones. Distinct from the single-band filters and the SMR resonator bank.

TiltEQ [FLT] [FX] [SYNTH] in: Audio out: Audio

One-knob spectral tilt that splits the signal at a pivot frequency with a one-pole low-pass and rocks the low and high halves in opposite directions. Tilt sets the slope from -1 (low band up, high band down) through 0 (flat) to +1 (high band up, low band down), Freq places the pivot the split happens around, and Range scales the maximum boost/cut in dB applied to each half. Useful as a fast tone balancer for brightening or warming a source without reaching for a full EQ.

TrimeanFilter [FLT] [FX] [SYNTH] in: Audio out: Audio

Trimean filter: smooths using Tukey's trimean (Q1 + 2*median + Q3)/4 of a 7-sample window - a robust estimate of the centre that weights the median but still uses the quartiles, so it rejects outliers like a median yet tracks the signal more smoothly and efficiently. A middle ground between the median and the mean. Adds 3 samples latency.

TripleComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Triple comb resonator: three tuned feedback combs in parallel, ringing the input at three pitches at once for a chordal, metallic resonance - feed it noise for a struck chord. Root sets the base pitch, Chord the interval spread, Feedback the ring. in0 = Audio.

TripleResonator [FLT] [FX] [SYNTH] in: Audio out: Audio

Triple resonator (Polymoog/Serge): three sharply-tuned resonant peaks laid over the input, ringing out formant-like tones from any source - feed it noise for vowels, audio for metallic body. F1 / F2 / F3 set the three resonance pitches. in0 = Audio.

UniversalComb [FLT] [FX] [SYNTH] in: Audio out: Audio

Universal comb: one tuned delay with separate feed-forward and feedback gains, so a single block morphs between FIR comb, IIR comb, all-pass and flat - the general comb topology from which the others are special cases.

Vocoder [FLT] [FX] [SYNTH] in: Carrier, Modulator out: Audio

Analysis/synthesis vocoder (Pigments-style): splits the carrier at in0 and modulator at in1 into log-spaced bandpass bands, tracks each modulator band envelope, and multiplies the matching carrier band by it before summing, all blended by Mix against the dry carrier. Bands sets the number of analysis bands (4 to 40 - more bands give more intelligible speech imprinting), Q sets band sharpness, Attack sets the envelope follower attack time in ms (shorter tracks transients tighter), Mode picks the band character (Vintage soft to 6 kHz / Modern bright to 10 kHz / Dirty saturated), and Shift formant-shifts the carrier synthesis bands against the modulator analysis bands for chipmunk or deep-throat colours.

VowelBank [FLT] [FX] [SYNTH] in: Audio out: Audio

Vowel formant bank: three resonant peaks that morph through the A-E-I-O-U vowels as Vowel sweeps, turning any source into a talking, singing voice. Vowel selects the vowel, Q the sharpness, Mix the wet. in0 = Audio.

VowelSweep [FLT] [FX] [SYNTH] in: Audio out: Audio

Auto vowel sweep: an internal LFO slides the formant filter continuously through the A-E-I-O-U vowels, making the input talk/sweep on its own - a hands-free talking filter.

Wah [FLT] [FX] [SYNTH] in: Audio, Reso Mod, Sweep Mod out: Audio

Manually-swept resonant band-pass: Pedal sweeps the peak across Range, with Reso setting the quack and Mix the blend. Drive Pedal from the mod input (in 3) - an LFO, envelope or mod wheel - for auto-wah, talk-box or pedal-wah effects. See AutoWah for a built-in envelope follower.

WaspFilter [FLT] [FX] [SYNTH] in: Audio, Cutoff CV out: Audio

EDP Wasp-style CMOS filter: a digital-gate lowpass that distorts hard and dirty as it is driven - the buzzy, gnarly, lo-fi Wasp grind. Drive sets how viciously the CMOS stage clips; Reso adds a sharp, edgy peak. in1 modulates cutoff.

WholeToneReson [FLT] [FX] [SYNTH] in: Audio out: Audio

Whole-tone resonator (a first as a synth block): rings the input through resonators tuned to the six equal WHOLE-TONE steps (plus the octave) - the symmetrical, rootless scale of Debussy and dream sequences, with no perfect fifths to anchor it. Any source takes on a floating, ambiguous, shimmering quality with no clear home note. Root sets the centre, Sharpness the ring time, Mix the wet/dry blend. A symmetrical scale resonator distinct from the major and minor banks.

WinsorizedMean [FLT] [FX] [SYNTH] in: Audio out: Audio

Winsorized-mean filter: sorts a 7-sample window and pulls the Limit most extreme values on each end inward to the nearest surviving value before averaging - unlike the alpha-trimmed mean, which discards them, so outliers are tamed while still contributing. Limit=0 is a moving average, Limit=3 is the median. A robust smoother with a softer outlier response. Adds 3 samples latency.

Wobble [FLT] [FX] [SYNTH] in: Audio out: Audio

Tempo-coherent resonant low-pass wobble: an LFO phased from c.time sweeps a TPT low-pass cutoff between 150 Hz and roughly 6 kHz for dubstep-style movement. Rate sets the LFO frequency, Depth scales the cutoff sweep range, Reso sets the filter resonance, and Shape selects sine, ramp or square LFO contour. Mix blends the wobbled output with the dry signal.

XMF [FLT] [FX] [SYNTH] in: Audio out: Audio

Cross-modulation filter (u-he Zebra XMF-style): two resonant state-variable filters run in parallel, but each one's cutoff is modulated by the OTHER filter's output, so they chase each other into FM-like sidebands, growl and screaming resonant textures a single filter cannot make. Cutoff sets the base frequency, Spread offsets the two filters, XMod the cross-modulation depth, Reso the resonance, and Mix the wet blend. At high XMod and Reso it self-oscillates into metallic, vocal, clangorous tones.

ZdfLadder [FLT] [FX] [SYNTH] in: Audio out: Audio

TPT ladder filter: four topology-preserving one-pole stages with a tanh-saturated resonance feedback model the Moog ladder with the zero-delay-feedback method, holding tuning and resonance accurate where the plain Euler ladder droops.

ZdfSvf [FLT] [FX] [SYNTH] in: Audio out: Audio

Zero-delay-feedback SVF: Zavalishin's topology-preserving-transform state-variable filter solves the feedback loop implicitly each sample, so cutoff and resonance stay accurate right up to Nyquist and it self-oscillates cleanly - a far more analog-faithful SVF than the naive one.

Modulation (386)

AbundanceSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Abundance sequence: compares the sum of a number's divisors to twice the number, outputting deficient (-), perfect (0) or abundant (+); perfect numbers (6,28,496,…) hit exactly zero, a rare event in an otherwise mostly-negative stream.

Aizawa [MOD] [FX] [SYNTH] in: – out: Audio

Aizawa attractor (a first as a synth block): integrates a six-parameter forced system whose trajectory wraps around a sphere while a hole opens through its top axis, producing a shape unlike any other strange attractor - a knotted ball with a drilled core. It free-runs at Rate, outputting x as a smooth bipolar CV that orbits the sphere and dips through the hole, giving rich circular motion with sudden axial excursions. A visually and sonically distinct chaos from the butterfly and spiral attractors.

∿ AliquotSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Aliquot sequence: repeatedly replace a number by the sum of its proper divisors; it may hit a prime then 1, settle on a perfect number, or enter an amicable/sociable cycle (the open Catalan-Dickson question) - a number-theoretic trajectory, reseeded on collapse.

∿ Allpass [MOD] [FX] [SYNTH] in: Audio out: Audio

All-pass diffuser: a cascade of 1-8 first-order all-pass sections (Stages) that rotate phase while leaving magnitude flat, set by Coef. A building block for reverb diffusion, phasers and stereo de-correlation rather than an audible filter on its own. More stages and higher Coef smear transients harder.

∿ AlphaBetaTrack [MOD] [FX] [SYNTH] in: Audio out: Audio

Alpha-beta tracker: the radar constant-velocity estimator applied to a control signal - it predicts the next value from an internal velocity, then corrects position by Alpha and velocity by Beta of the residual. Because it carries momentum it follows ramps and sweeps with far less lag than an exponential smoother of equal noise rejection, settling without the overshoot of a resonant filter - a predictive smoother distinct from the median/trimmed-mean family.

∿ ArpSeq [MOD] [FX] [SYNTH] in: In A, In B out: Audio

Arpeggiator: samples a base pitch CV, then on each clock pulse steps an output CV through a fixed chord shape (root, third, fifth, octave) above it, turning one held note into a running arpeggio.

∿ Astroid [MOD] [FX] [SYNTH] in: — out: Audio

Astroid LFO (a first as a synth block): traces the four-cusped star curve x = cos-cubed(angle) and outputs it directly - a cosine with FLATTENED peaks and steepened zero-crossings, so it lingers at the extremes and snaps quickly through the middle, the inverse of a sine's rounded shape. A gentle, plateau-topped modulation good for slow holds with fast transitions. Rate sets the speed. A cubed-cosine star curve distinct from the sine and triangle shapes.

∿ AsymTremolo [MOD] [FX] [SYNTH] in: Audio out: Audio

Asymmetric tremolo: the amplitude LFO is a skewable ramp - fast-up/slow-down or the reverse - so the volume pumps lopsidedly instead of the usual symmetric sine, giving a rhythmic, sawtooth-like swell.

∿ AutoRing [MOD] [FX] [SYNTH] in: Audio out: Audio

Self ring-mod: the signal is ring-modulated by a DELAYED copy of ITSELF rather than an external oscillator, so the carrier is always related to the source - producing inharmonic, evolving metallic sidebands that move with the material instead of a fixed clang. Time sets the delay (and so the carrier pitch), Depth the ring amount, Mix the blend.

∿ BarberFlange [MOD] [FX] [SYNTH] in: Audio out: Audio

Barber-pole flanger: two delay taps sweep in the same direction and crossfade as each resets, so the comb notches appear to glide endlessly upward (or downward) without ever wrapping - the auditory barber-pole illusion as a flanger.

Barberpole [MOD] [FX] [SYNTH] in: Audio out: Audio

Infinite (barberpole) phaser: two 4-stage allpass banks a half-cycle apart, each amplitude-windowed so one fades in as the other wraps, giving an endless rising or falling sweep. Rate sets the sweep speed (Hz), Depth scales the allpass coefficient swing, Dir flips between rising and falling, and Mix blends wet against dry. A Shepard-tone style effect that seems to climb or descend forever.

Bass [MOD] [FX] [SYNTH] in: – out: Audio

Bass-diffusion source (a first as a synth block): the marketing model of how a new product spreads - a few INNOVATORS adopt on their own, then a wave of IMITATORS follows as they see others using it, giving the classic S-shaped adoption curve. Here the market saturates and resets, so the output is a repeating slow swell from zero to full adoption: a smooth, asymmetric ramp that starts slow, accelerates through the tipping point, then levels off. Innovate sets the spontaneous-adoption rate, Imitate the word-of-mouth strength. A technology-adoption curve distinct from the oscillating dynamical sources. Needs no input.

BaumSweetSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Baum-Sweet sequence: a 2-automatic binary sequence where term n is 1 only if the binary of n contains no block of consecutive zeros of odd length - a deterministic, self-similar gate pattern from automata theory.

BeattySeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Beatty sequence: the gaps in floor(n*phi) alternate between 1 and 2 in the golden-ratio (Fibonacci-word) pattern, partitioning the integers into the complementary Beatty pair - a Sturmian, maximally-even rhythmic gate.

Bedhead [MOD] [FX] [SYNTH] in: – out: Audio

Bedhead-map source (a first as a synth block): iterates the trigonometric map x <- sin(xy/b)y + cos(ax - y), y <- x + sin(y)/b, one of the 'bedhead' attractors whose orbit tangles into a soft, hair-like snarl of overlapping loops. The product term xy inside the sine couples the two coordinates nonlinearly, giving a dense, organic, woven motion. It free-runs at Rate, outputting x as a bipolar CV. A coupled-trig chaos distinct from the additive De Jong and Clifford maps.

BenfordSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Benford's law: draws leading digits with the logarithmic frequencies P(d)=log10(1+1/d) that real-world data (street addresses, stock prices) obey, by taking the first digit of 10^u; digit 1 appears ~30% of the time, 9 under 5% - a skewed digit CV.

BernoulliNoise [MOD] [FX] [SYNTH] in: Audio out: Audio

Bernoulli noise: a biased coin flipped at the clock rate, outputting +1 with probability p and -1 otherwise, the simplest possible random source and the building block of every binomial process - a probability-tunable random gate.

BiasPump [MOD] [FX] [SYNTH] in: Audio out: Audio

Bias pumping: a level-dependent tremolo modelling the way a leaking bias/supply makes an amp 'breathe' - the amplitude wobble gets DEEPER as the signal gets louder, so quiet parts are steady and loud parts pump and throb. Distinct from a plain tremolo whose depth is fixed. Rate sets the pump speed, Depth the worst-case pumping, Mix the blend.

Billow [MOD] [FX] [SYNTH] in: – out: Audio

Billow-noise source (a first as a synth block): a fractal noise that takes the absolute value of each octave (2|noise| - 1), turning the gentle zero-crossings of fBm into rounded, puffy lobes - the look of cumulus clouds, rolling smoke and bubbling foam in procedural textures. The output swells up in soft billows with creased troughs where it folds back on itself, the rounded opposite of the sharp Ridged. Rate sets the base speed and Octaves the detail. A billowy coherent-noise modulator distinct from fBm and Ridged. Needs no input.

BirthDeath [MOD] [FX] [SYNTH] in: – out: Audio

Birth-death (queue) source (a first as a synth block): a population (or a queue of waiting jobs) where new arrivals appear at a steady Birth rate while each member already present has a Death chance of leaving, so the more there are the faster they go - a self-limiting balance that settles around Birth/Death. The output is the population, an ASYMMETRIC mean-reverting wander: it drifts up on a run of arrivals then thins out faster the higher it climbs, the Poisson-occupancy shape of a busy queue. Distinct from the symmetric Ehrenfest diffusion. Rate sets the event speed. Needs no input.

Bloom [MOD] [FX] [SYNTH] in: – out: Audio

Fractal sequencer (Qu-Bit Bloom): grows a melody from a short Trunk seed sequence by recursively grafting variations onto it. Length sets the trunk length; Branch adds fractal sub-sequences derived from the trunk (0 = play the plain trunk, up = ever more elaborate variations); Path chooses which route through the recursive tree is traversed; Scale sets the output range. Self-clocks at Rate. A generative stepped-CV source - quantize for pitch. Distinct from Turing (shift register) and StepSeq (fixed).

BodeShift [MOD] [FX] [SYNTH] in: Audio out: Audio

Frequency shifter: builds an approximate quadrature (90-degree) pair from the input and heterodynes it with a complex oscillator, shifting every partial by a fixed number of Hz - inharmonic, metallic, NOT pitch shifting (a Bode-style shifter).

Bouali [MOD] [FX] [SYNTH] in: – out: Audio

Bouali attractor (a first as a synth block): integrates a three-variable system devised to model fluctuating economic and predator-prey-like quantities - x' = x(4 - y) + 0.3z, y' = -y(1 - x^2), z' = -x(1.5 - 0.05z) - 0.05z - whose trajectory winds onto a broad, layered, sheet-like strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV with a strong slow sway under faster folds. A sheet-like chaos distinct from the scroll attractors. Needs no input.

BounceBall [MOD] [FX] [SYNTH] in: Audio out: Audio

Bouncing-ball envelope: simulates a ball dropped under gravity that loses energy each bounce, so the control output bounces ever faster and lower then resets - an accelerating rhythmic ramp no LFO shape gives.

Breathe [MOD] [FX] [SYNTH] in: Audio out: Audio

Breathe: a slow asymmetric volume swell - a quick inhale rise and a longer exhale fall - that makes the signal seem to breathe in and out. Gentler and more lifelike than a sine tremolo because the shape itself is lung-like. Rate sets the breaths, Depth how deep the breath, Mix the blend.

BrownDrift [MOD] [FX] [SYNTH] in: Audio out: Audio

Brownian drift: a random walk accumulates small steps and reflects off the rails, producing smooth, momentum-carrying wander (1/f-ish) rather than the memoryless jumps of sample-and-hold random.

∿ BrownianBridge [MOD] [FX] [SYNTH] in: – out: Audio

Brownian-bridge source (a first as a synth block): a random walk that is tied down at both ends - it wanders freely in the middle but is mathematically guaranteed to arrive back at zero exactly at the end of each cycle, then starts a fresh bridge. This is the conditioned Brownian motion used to interpolate between fixed points and to model paths with a known destination. Length sets the cycle in steps. A pinned, self-resetting drift distinct from the open random walk. Needs no input.

∿ Brusselator [MOD] [FX] [SYNTH] in: – out: Audio

Brusselator chemical oscillator (a first as a synth block): the textbook model of an autocatalytic chemical clock, where a reactant feeds a loop that periodically builds up and consumes an intermediate, so the concentration pulses up and down on its own - the kind of self-organising rhythm seen in real oscillating reactions. Above the critical feed it locks into a stable limit cycle. Rate sets the speed and B the feed parameter (raise it past 2 to start the oscillation). The output is the concentration x. A chemistry-born oscillation distinct from the biological and chaotic sources.

∿ BurkeShaw [MOD] [FX] [SYNTH] in: – out: Audio

Burke-Shaw attractor (a first as a synth block): integrates x' = -s(x+y), y' = -y - sxz, z' = sxy + V - a tightly-coiled variant of the Lorenz system that winds into a dense, near-toroidal spiral rather than two separate wings, spinning rapidly around a doughnut of trajectories. It free-runs at Rate, outputting x as a smooth, fast-cycling bipolar CV with a strong underlying rotation. A coiled, rotation-dominated chaos distinct from the wing-flipping Lorenz and Chen.

∿ BurningShip [MOD] [FX] [SYNTH] in: – out: Audio

Burning-Ship source (a first as a synth block): iterates z <- (|Re z| + i|Im z|)^2 + c, the Burning Ship fractal's map, where taking the ABSOLUTE VALUE of each part before squaring breaks the usual symmetry and produces the jagged, flame-and-mast shape it is named for. The folding makes its bounded orbits sharper and more angular than a Julia set; escapes reseed near the centre. CReal and CImag pick the region. It free-runs at Rate, outputting the real part. An absolute-value complex map distinct from the Julia and Tricorn sources.

∿ BurstGen [MOD] [FX] [SYNTH] in: Audio out: Audio

Burst / ratchet generator: each gate fires a finite count of evenly-spaced trigger pulses, turning one hit into a programmable roll or ratchet - drum fills and stutters from a single trigger.

∿ Cardioid [MOD] [FX] [SYNTH] in: – out: Audio

Cardioid LFO (a first as a synth block): traces the heart-shaped curve r = 1 - cos(angle) and outputs its horizontal sweep, a strongly ASYMMETRIC modulation - a slow, lingering dwell near one extreme and a quick sweep through the other, the single-cusped 'heart' contour. Its lopsided shape makes it good for envelope-like swells that rise gently and fall fast (or vice versa). Rate sets the speed. A heart-curve LFO distinct from the symmetric sine and the multi-lobe Rose.

∿ CatalanModSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Catalan mod sequence: the Catalan numbers (1,1,2,5,14,42,…) reduced modulo a small base, a combinatorial sequence counting balanced-bracket and tree structures; the modular fold makes a quirky repeating melodic pattern.

CellAuto [MOD] [FX] [SYNTH] in: Audio out: Audio

Cellular-automaton sequencer: evolves a 32-cell elementary CA (any Wolfram rule, e.g. 30) one generation per clock step and outputs the centre cell - emergent, rule-driven gate/CV patterns from a one-line universe.

ChampernowneSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Champernowne digits: the digits of 0.123456789101112… (formed by concatenating the integers), a normal number whose digit stream is deterministic yet statistically uniform - an endlessly-climbing-then-resetting decimal-digit CV.

Chaos [MOD] [FX] [SYNTH] in: – out: Audio

Chaos modulator driven by the logistic map (x = rx(1-x)): at each Rate tick it iterates the map, and Chaos raises r from the periodic into the chaotic regime for unpredictable-but-deterministic motion. Smooth slews between values for a flowing control source rather than hard steps. Use it for evolving, never-quite-repeating modulation.

Chen [MOD] [FX] [SYNTH] in: – out: Audio

Chen attractor (a first as a synth block): integrates Chen's system x' = a(y-x), y' = (c-a)x - xz + cy, z' = xy - bz - a close relative of the Lorenz butterfly discovered by feedback control, but with a more tightly wound double scroll whose two lobes are pulled closer and spun faster. It free-runs at Rate, outputting x as a smooth bipolar CV that loops within one lobe then flips chaotically to the other. A wound-up cousin of Lorenz distinct from the Rossler and Thomas flows.

ChenLee [MOD] [FX] [SYNTH] in: – out: Audio

Chen-Lee attractor (a first as a synth block): integrates x' = ax - yz, y' = by + xz, z' = cz + xy/3, a model of a rigid body rotating under feedback torque, derived from Euler's equations for a gyroscope. Its trajectory traces a pair of stacked, twisted rings (a double-scroll that loops rather than wings), wandering chaotically between them. It free-runs at Rate, outputting x as a smooth bipolar CV with wide swings. A rotational-dynamics chaos distinct from the Lorenz and Rossler families.

ChordRing [MOD] [FX] [SYNTH] in: Audio out: Audio

Chord ring modulator: rings the input against three internal carriers tuned to a triad ratio at once, scattering harmonically-related sidebands instead of the single clangorous tone of a one-carrier ring mod.

ChorusEns [MOD] [FX] [SYNTH] in: Audio out: Audio

Ensemble chorus: three independently LFO-modulated delay taps detune and spread the signal into a lush, string-machine-style ensemble - richer than a single-voice chorus.

Chua [MOD] [FX] [SYNTH] in: – out: Audio

Chua's-circuit chaos source (a first as a synth block): integrates the dimensionless Chua equations - the simplest electronic circuit that is provably chaotic, a real analog oscillator built from an op-amp, two capacitors, an inductor and the piecewise-linear 'Chua diode'. The trajectory winds around two attracting loops and flips unpredictably between them (the famous double-scroll), giving a smooth bipolar CV that meanders forever without repeating. Rate sets the integration speed. A circuit-born chaos, smoother than a map and distinct from the mechanical DoublePendulum or the Lorenz LFO.

ChuaOsc [MOD] [FX] [SYNTH] in: – out: Audio

Chua's-circuit chaos oscillator: a self-running analog-chaos modulation source - a strange attractor that wanders unpredictably yet stays bounded, never repeating. Rate sets the speed of evolution, Shape the attractor regime. A textured, organic chaos for evolving modulation. Needs no input.

CircleMap [MOD] [FX] [SYNTH] in: – out: Audio

Sine-circle-map source (a first as a synth block): iterates theta <- theta + Omega - (K/2pi) sin(2pi theta) on a circle - the model of an oscillator being kicked once per turn, and the textbook route to MODE-LOCKING. For bands of settings (the Arnold tongues) it locks onto a rational rhythm and repeats exactly; between them it runs quasi-periodically or, at high K, chaotically - so sweeping Omega walks a devil's-staircase of locked and unlocked ratios. It free-runs at Rate, outputting the angle as a 0..1 CV. K sets the kick strength, Omega the bare turn rate. A locking oscillator distinct from the dissipative attractors.

Clifford [MOD] [FX] [SYNTH] in: – out: Audio

Clifford-attractor source (a first as a synth block): iterates the strange-attractor map x <- sin(Ay) + cos(Ax), y <- sin(1.6x) + 0.7cos(1.6*y), whose orbit is forever drawn toward a delicate folded-ribbon fractal. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that traces the attractor's swirls. A sets the warp of the figure - small changes redraw the whole attractor - giving a rich family of organic, never-repeating motions distinct from the scrolls of Chua or the kicks of the standard map.

Clock [MOD] [FX] [SYNTH] in: – out: Audio

Tempo-synced clock generator: emits a gate-pulse train locked to the host tempo (or a free Hz Rate). Sync picks the note division (0=free Hz, 1=1/1 .. 8=1/16T), Rate is the free-running frequency when Sync is 0, and Width sets the pulse length as a fraction of the step. Patch its output into envelopes, sample-and-holds, ClockDiv or any gate input to drive rhythmic movement from the host transport.

ClockMultiply [MOD] [FX] [SYNTH] in: Audio out: Audio

Clock multiplier: measures the period between incoming clock pulses and emits a chosen number of evenly-spaced sub-pulses inside each one, generating faster synced clocks (1/8 from 1/4 notes, etc.) from a single trigger stream.

CML [MOD] [FX] [SYNTH] in: – out: Audio

Coupled-map-lattice source (a first as a synth block): eight logistic maps sit in a ring, each iterating its own chaos while a Coupling term shares a fraction of each cell with its neighbours. The competition between local chaos and diffusive smoothing produces spatiotemporal chaos - travelling fronts, frozen-random patches and turbulent bursts that no single attractor shows. The output is the lattice average, a richly textured wandering CV. R sets each map's chaos and Coupling the diffusion. A whole field of coupled chaos rather than one oscillator. Needs no input.

Cobweb [MOD] [FX] [SYNTH] in: – out: Audio

Cobweb-market source (a first as a synth block): models the boom-bust price cycle of a market where producers decide TODAY'S supply based on YESTERDAY'S price - the cobweb theorem behind hog and crop cycles. High prices trigger overproduction, which crashes prices, which triggers underproduction, and so on. With a realistic exponential (Ricker) supply response the price never settles: it oscillates and, when producers over-react (high R), tips into genuine chaos. The output is the price. R sets how strongly supply reacts. A lagged-market chaos distinct from the smooth predator-prey and the logistic map. Needs no input.

∿ CollatzMaxSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Collatz peak: the highest value the 3n+1 trajectory reaches before falling to 1 (27 climbs all the way to 9232), a wildly-varying integer that dwarfs the starting number; log-scaled it makes a jagged, occasionally-spiking stepped CV.

∿ CollatzParityWord [MOD] [FX] [SYNTH] in: Audio out: Audio

Collatz parity word: as the 3n+1 trajectory of the step index unfolds it emits a bit at each step (0 if even, 1 if odd), the 'parity vector' that uniquely encodes the starting number - a deterministic bit-stream from the Collatz dynamics.

∿ CollatzStopping [MOD] [FX] [SYNTH] in: Audio out: Audio

Collatz stopping time: how many 3n+1/halve steps it takes for the step index to reach 1, a famously erratic integer (27 takes 111 steps) - clocked out, it makes a wildly-jumping stepped pattern from the unsolved Collatz conjecture.

∿ CombFlange [MOD] [FX] [SYNTH] in: Audio out: Audio

Feedback flanger: a slow LFO sweeps a short delay line with regeneration, producing the swooshing swept-comb resonance of a classic flanger.

∿ Combinate [MOD] [FX] [SYNTH] in: A, B out: Audio

Combines two modulation or audio inputs (Pigments-style): Mode selects sum, ring (multiply), max, min, difference or sign-XOR, scaled by Mix. A math/logic node for the mod matrix - build complex modulators from simpler ones, or use it as a flexible ring/AM stage. Complements the Expr script node for inline math.

∿ CombSelf [MOD] [FX] [SYNTH] in: Audio out: Audio

Self-modulating comb: the comb's delay length is bent in real time by its own output, a feedback loop that breeds chaotic, screaming, never-quite-repeating resonances - a comb that modulates itself.

∿ CoupledPendula [MOD] [FX] [SYNTH] in: – out: Audio

Coupled-pendulum-chain source (a first as a synth block): a ring of sixteen pendulums, each linked by a spring to its neighbours - the classic lecture-hall wave machine. Give one a push and the swing travels along the chain as a wave, reflecting and interfering with itself (this is the sine-Gordon system, famous for its solitons). With almost no friction the energy sloshes around the ring forever. The output is one pendulum's swing, rising and falling as the waves pass through it. A mechanical wave source distinct from the electronic and chemical oscillators. Needs no input.

∿ CrushChorus [MOD] [FX] [SYNTH] in: Audio out: Audio

Lo-fi chorus: the modulated chorus voice is bit-crushed before mixing, so the doubled layer is grainy and degraded against the clean original - a digital, vintage-sampler shimmer distinct from a clean analog chorus.

∿ Cubic [MOD] [FX] [SYNTH] in: – out: Audio

Cubic-map source (a first as a synth block): iterates x <- A*x - x^3, an ODD-symmetric map (it treats positive and negative x as mirror images) unlike the lopsided logistic. That symmetry gives it a distinctive bifurcation diagram with two mirrored chaotic bands that can suddenly merge, and orbits that swing through both signs. It free-runs at

Rate, stepping the map and outputting x as a bipolar CV. A sets the gain on the road to chaos. A symmetric cubic chaos distinct from the asymmetric quadratic maps. Needs no input.

Cusp [MOD] [FX] [SYNTH] in: – out: Audio

Cusp-map source (a first as a synth block): iterates x <- 1 - 2*sqrt(|x|), a map with a sharp cusp at the origin instead of a smooth parabola. The square-root gives it an infinitely steep peak, so it stretches and folds the line more violently near zero than the polynomial maps, producing fully-developed chaos with a characteristic spiky density. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that hops sharply across [-1, 1]. A non-smooth chaos distinct from the parabolic logistic and the piecewise tent.

DAHDSR [MOD] [FX] [SYNTH] in: Audio out: Audio

Six-stage modulation envelope driven by the gate at in0: on gate-on it runs Delay, Attack, Hold, Decay then holds at Sustain, and on gate-off it enters Release, outputting a 0 to 1 level. Delay/Attack/Hold/Decay/Release are times in seconds and Sustain is the held level. The extra Delay and Hold stages over a plain ADSR let it shape slow swells or gated bursts as a modulation source.

DCWander [MOD] [FX] [SYNTH] in: Audio out: Audio

Slow DC wander: a gliding random offset drifts the signal's baseline like a thermally-unstable analog circuit, adding subtle low-frequency movement and bias for an alive, imperfect feel.

DeBruijnSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

De Bruijn sequence: a cyclic binary string in which every possible k-bit pattern appears exactly once (generated here by the prime-necklace/Lyndon-word concatenation), the maximally-compressed traversal of all sub-patterns - a dense structured gate.

Deffuant [MOD] [FX] [SYNTH] in: – out: Audio

Deffuant opinion-dynamics source (a first as a synth block): models twelve people each holding an opinion between 0 and 1; at each step two meet at random and, only if their views are already within Tolerance of each other, they compromise and move closer - if they are too far apart they simply ignore one another. This 'bounded confidence' makes the crowd settle into one consensus (narrow Tolerance gives several rival camps), occasionally shaken up when someone changes their mind. The output is the SPREAD of opinion (wide = divided, near zero = agreed), drifting as clusters form and dissolve. A social-dynamics source distinct from the physical models. Needs no input.

DeJong [MOD] [FX] [SYNTH] in: – out: Audio

De-Jong attractor source (a first as a synth block): iterates Peter de Jong's map x <- sin(Ay) - cos(Bx), y <- sin(Cx) - cos(Dy), whose orbit weaves into a dense web of looping filaments - one of the most varied strange-attractor families, redrawing completely with the smallest change of constants. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that wanders the web. A warps the figure. A four-sine attractor distinct from the two-sine Clifford. Needs no input.

Deltoid [MOD] [FX] [SYNTH] in: – out: Audio

Deltoid LFO (a first as a synth block): traces the three-cusped hypocycloid (a circle rolling inside one three times its size), outputting (2cos(angle) + cos(2angle)) / 3 - a smooth modulation with a strong main swell and a smaller secondary bump, the triangular cousin of the astroid. Its three-fold asymmetry gives a lopsided, gently galloping motion. Rate sets the speed. A three-cusp curve distinct from the four-cusp astroid and the rolling-circle epicycloid.

∿ Dequan [MOD] [FX] [SYNTH] in: – out: Audio

Dequan-Li attractor (a first as a synth block): integrates a five-term unified chaotic system x′ = a(y - x) + dxz, y′ = kx + fy - xz, z′ = cz + xy - ex*x with large coefficients that drive very energetic, wide-ranging orbits forming a dense double scroll. It free-runs at Rate, outputting x as a smooth bipolar CV with broad excursions. A high-energy unified chaos distinct from the gentler classical attractors.

∿ DetuneDouble [MOD] [FX] [SYNTH] in: Audio out: Audio

Detune doubler: blends in a single copy whose pitch is shifted by a few cents via a slowly-swept delay, fattening a sound into a tight unison double (artificial double-tracking) without the wide motion of a chorus.

∿ DigitalRoot [MOD] [FX] [SYNTH] in: Audio out: Audio

Digital root: repeatedly sum the decimal digits of the step index until one digit remains (equivalently 1 + (n-1) mod 9), producing a perfectly periodic 1..9 staircase - a nine-step modular ramp from elementary number theory.

∿ DigitProductSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Digit product: multiplies together the base-B digits of the step index (which collapses to zero whenever any digit is zero), giving a spiky pattern that drops to silence on every index containing a zero digit - a base-tunable arithmetic CV.

∿ DigitSumSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Digit-sum sequence: the sum of the base-B digits of the step index, a self-similar arithmetic sequence (sawtooth-of-sawtooths in any base) whose averaged growth is logarithmic - a tunable-base stepped control pattern.

∿ Dimension [MOD] [FX] [SYNTH] in: Audio out: Audio

Dimension-D-style BBD chorus: a slow ~0.5 Hz LFO modulates a short delay around 8 ms, with the right channel running anti-phase, and the wet copy is blended with the dry at a fixed ratio. Depth scales how far the delay time is swept. Out picks L or R for the per-channel modulation phase, giving the lush stereo movement of the classic unit rather than a deep vibrato.

∿ DiodeRectChorus [MOD] [FX] [SYNTH] in: Audio out: Audio

Rectified chorus: the modulated delay voice is passed through an asymmetric diode rectifier (the negative half attenuated) before mixing, adding even-harmonic grit and an octave-up shimmer to the usual chorus widening - a dirtier, more vocal double.

∿ DiodeRing [MOD] [FX] [SYNTH] in: In A, In B out: Audio

Diode ring modulator: passes the product of two inputs through the exponential curve of a four-diode bridge, so the sidebands carry the gritty asymmetry and crossover kink of a real ring-mod transformer - not a clean multiply.

∿ DirichletPick [MOD] [FX] [SYNTH] in: Audio out: Audio

Dirichlet pick: draws a random probability vector over several levels from a symmetric Dirichlet distribution (normalized gamma samples) then selects a level by it; low concentration makes one level dominate, high concentration spreads the choice - a structured random quantizer.

DistinctPrimesSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Little-omega omega(n): counts the distinct prime factors of the step index (so 12 = 2^2*3 gives 2), ignoring multiplicity; it grows like log-log n on average and makes a sparse, slowly-rising stepped CV.

DivisorCountSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Divisor count: the number-of-divisors function d(n) (1 for primes, large for highly-composite numbers), a multiplicatively-structured integer sequence; as a CV it spikes on composite-rich steps and dips on primes.

Dopplerize [MOD] [FX] [SYNTH] in: Audio out: Audio

Doppler: a delay line whose length is swept by a velocity LFO, so the pitch bends up as the virtual source rushes toward you and down as it recedes - the siren-passing-by warp. A continuous, physical pitch-vibrato quite unlike a static chorus. Rate sets the pass-by speed, Depth the pitch swing, Mix the blend.

DoublePendulum [MOD] [FX] [SYNTH] in: — out: Audio

Double-pendulum chaos source (a first as a synth block): integrates the real equations of motion of two jointed pendulums - the textbook deterministic-chaos system - and outputs the lower bob's swing as a smooth, never-repeating CV. Tiny differences in motion explode into wholly different trajectories, so it wanders organically forever without an LFO's repetition or a noise source's harshness. Rate sets the simulation speed; the output stays bounded in +/-1 however wildly the arms tumble. A different, smoother chaos than the logistic Chaos or Lorenz blocks.

Drift [MOD] [FX] [SYNTH] in: — out: Audio

Slow smoothed random-walk modulator centred on 0.5, emulating vintage analog drift. Rate sets how often it picks a new target and Depth the deviation. Patch a little into pitch, cutoff or level to take the static, too-perfect edge off a patch.

DriftDetune [MOD] [FX] [SYNTH] in: Audio out: Audio

Analog drift: a single delayed copy whose pitch wanders on a smoothed random walk - the slow, lazy detuning of a warm analog oscillator that never quite stays in tune. Thickens and animates without the regular swirl of a chorus. Drift sets the wander speed, Depth the detune amount, Mix the blend.

Dropout [MOD] [FX] [SYNTH] in: Audio out: Audio

Random rhythmic muting: a clock at Rate decides per cycle whether to drop the signal to silence (Amount sets the drop probability), smoothed by Smooth to avoid clicks. Emulates a bad cable, dying battery or stuttery transmission. Subtle for movement, extreme for glitch and IDM textures.

DrunkLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Drunk-walk modulation: a random source that takes small random steps up or down at a set rate, wandering smoothly within +/-1 and reflecting off the rails - a bounded random walk (the drunk walk) for organic, slowly-drifting modulation, unlike sample-and-hold's independent jumps. Step sets the stride size, Rate the step interval, Depth the output level. A trigger into in0 restarts the walk at centre; unpatched = free-run.

Duffing [MOD] [FX] [SYNTH] in: – out: Audio

Driven-Duffing chaos source (a first as a synth block): integrates the forced double-well Duffing oscillator x'' + dx' - x + x^3 = Drivecos(w t) - a mass in a twin-valley potential shaken by a periodic drive. At chaotic settings it jumps erratically between the two wells, never locking to the drive's period, yielding a smooth bipolar CV with a rhythmic pulse running under the chaos. Rate sets the speed, Drive the forcing amplitude (raise it to push from periodic motion into chaos). A forced-oscillator chaos distinct from the autonomous Chua or DoublePendulum.

DynamicAllpass [MOD] [FX] [SYNTH] in: Audio out: Audio

Level-dependent allpass: a first-order allpass whose phase-rotation coefficient is pushed by the input envelope, so the phase smear (and the phasey notches when mixed dry) breathes with how hard you play.

Ehrenfest [MOD] [FX] [SYNTH] in: – out: Audio

Ehrenfest-urn source (a first as a synth block): the textbook model of diffusion and the arrow of time - Balls are split between two boxes and at each step ONE random ball is moved to the other box. Whichever box has more balls is more likely to lose one, so the split is gently pulled back toward 50/50 and then jiggles around it forever. The output is the fraction in one box: a mean-reverting random wander that, unlike a free random walk, never strays far - a statistical-mechanics restoring force rather than the spring of Ornstein-Uhlenbeck. Rate sets the step speed. Needs no input.

ElFarol [MOD] [FX] [SYNTH] in: – out: Audio

El-Farol-bar source (a first as a synth block): models Brian Arthur's famous bar problem - twelve regulars each decide whether to go out, wanting to go only if the bar will NOT be too crowded. Since everyone is second-guessing everyone else, no fixed rule works, yet through trial-and-error adjustment the crowd self-organises so attendance hovers right around Capacity, never settling. The output is the attendance, fluctuating around the comfort threshold - the founding model of bounded-rationality economics and inductive reasoning. A self-defeating-prediction source distinct from the consensus-seeking opinion models. Needs no input.

EnvFlanger [MOD] [FX] [SYNTH] in: Audio out: Audio

Auto-flanger: the swept comb delay (0.5-10 ms) is driven by the input envelope rather than an LFO, so the jet-like flange whoosh opens with dynamics - quiet passages sit short and bright, loud hits sweep the comb wide. Feedback sharpens the resonant peaks.

EnvPhaser [MOD] [FX] [SYNTH] in: Audio out: Audio

Auto-phaser: a four-stage allpass chain whose coefficient (notch frequency) is pushed up by the input's loudness instead of an LFO, so the phaser sweeps open on every transient and falls back as the note decays - a touch-sensitive sweep that follows your playing.

EnvPluck [MOD] [FX] [SYNTH] in: Audio out: Audio

Transient-triggered envelope: detects attacks in the input and fires a short attack/decay envelope on each one, turning any rhythmic audio into a clean pluck/percussive CV without an external trigger.

∿ **EnvRingMod** [MOD] [FX] [SYNTH] in: Audio out: Audio

Envelope-tracking ring mod: the sine carrier's frequency rises with the input's loudness, so quiet passages ring low and inharmonic while loud hits shoot the sidebands up the spectrum - a dynamics-driven clangour no static ring mod produces.

∿ **EnvTrace** [MOD] [FX] [SYNTH] in: Audio out: Audio

Blendable envelope follower CV: morph between a peak and an RMS detector with one knob and scale the output - a clean control source for driving mods, distinct from the audio-path dynamics blocks.

∿ **Epicycloid** [MOD] [FX] [SYNTH] in: – out: Audio

Epicycloid LFO (a first as a synth block): traces the curve a point on a small circle draws as it rolls around the OUTSIDE of a larger one - the path of a planet in a geocentric sky - and outputs its sweep. With Cusps points it forms a star-like loop of sharp, evenly-spaced cusps, so the LFO sweeps smoothly then snaps at each cusp, a spiky repeating contour. Rate sets the speed. A rolling-circle modulation distinct from the inner-rolling Spirograph.

∿ **EsaFrequency** [MOD] [FX] [SYNTH] in: Audio out: Audio

Energy-separation frequency: uses the Teager operator on both the signal and its difference to solve for the instantaneous normalized frequency, arccos(1 - psi(dx)/(2*psi(x))) - an FFT-free pitch tracker that follows fast frequency modulation sample by sample. Outputs the normalized frequency (0..1 over 0..Nyquist) and is distinct from Teager energy, which measures amplitude.

∿ **Euclid** [MOD] [FX] [SYNTH] in: – out: Audio

Euclidean rhythm generator: distributes Pulses hits as evenly as possible across Steps positions (the algorithm behind countless drum patterns) and clocks them at the tempo-synced step Rate. Steps is the pattern length, Pulses the number of hits, Rotate shifts the pattern start, Rate the per-step note division (1=1/1 .. 8=1/16T), and Width the gate length. Outputs a gate on each active step - feed a drum/pluck envelope for generative rhythm that locks to the host.

∿ **EuclidRhythm** [MOD] [FX] [SYNTH] in: – out: Audio

Euclidean rhythm: spreads a number of pulses as evenly as possible across a number of steps (the Bjorklund algorithm that underlies most world-music rhythms) and clocks it out as a gate - generative grooves from two numbers.

∿ **Excitable** [MOD] [FX] [SYNTH] in: – out: Audio

Excitable-medium source (a first as a synth block): a ring of sixteen FitzHugh-Nagumo cells coupled by diffusion, the model of an excitable tissue like heart muscle or a nerve sheet. A pulse, once lit, travels around the ring re-igniting each cell as it passes - a self-sustaining wave of excitation, the same mechanism behind a heartbeat and the spiral waves of a fibrillating heart. The output is one cell's voltage, spiking each time the wave sweeps by. Drive sets the excitability. A travelling-wave source distinct from the single-cell oscillators. Needs no input.

∿ **FareySeq** [MOD] [FX] [SYNTH] in: Audio out: Audio

Farey sequence: all reduced fractions in [0,1] with denominator up to a chosen order, listed in increasing size (0/1, 1/n, …, 1/2, …, 1/1); the gaps between successive Farey fractions make a symmetric, self-similar stepped CV.

∿ **fBm** [MOD] [FX] [SYNTH] in: – out: Audio

Fractal-Brownian-motion source (a first as a synth block): stacks several octaves of Perlin noise - each one twice as fast and half as loud as the last - into the rich, self-similar randomness of natural terrain, coastlines and mountain skylines. The low octaves give the slow overall drift while the high ones add fine detail, so the motion is organic at every timescale at once. Rate sets the base speed and Octaves how many layers (more = finer, rougher detail). A multi-scale coherent-noise modulator distinct from the single-octave Perlin. Needs no input.

FibSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Fibonacci sequencer: clocks out a modular Fibonacci recurrence (next = (a+b) mod M), a deterministic sequence that visits its values in a fixed but non-obvious order - structured melodic/CV patterns from number theory.

FibWordSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Fibonacci word: the fixed point of the golden substitution 0->01, 1->0, a Sturmian sequence whose 0s and 1s fall at golden-ratio spacing - the most-ordered aperiodic binary pattern, clocked out as a gate.

FinanceChaos [MOD] [FX] [SYNTH] in: – out: Audio

Financial-chaos attractor (a first as a synth block): integrates a model of a chaotic macroeconomy x' = z + (y - a)x, y' = 1 - by - x^2, z' = -x - cz, where x is the interest rate, y the investment demand and z the price index. It shows how a deterministic economy with no outside shocks can still wander unpredictably between booms and busts. It free-runs at Rate, outputting x as a smooth bipolar CV. An econophysics chaos distinct from the mechanical and climate attractors.

Firefly [MOD] [FX] [SYNTH] in: – out: Audio

Firefly-synchronisation source (a first as a synth block): models a swarm of twelve fireflies, each charging toward its flash; when one fires it nudges all the others a little closer to firing too. Through this pulse coupling they spontaneously fall into rhythm - the real mechanism behind fields of fireflies flashing in unison. The output is the swarm's phase COHERENCE (0 = scattered, 1 = all flashing together), which rises as they sync and breaks up again, breathing in and out. Couple sets how strongly each flash pulls the others. A self-organising sync source distinct from the chaos oscillators. Needs no input.

FitzHughNagumo [MOD] [FX] [SYNTH] in: – out: Audio

FitzHugh-Nagumo neuron (a first as a synth block): integrates the two-variable simplification of the Hodgkin-Huxley nerve-impulse equations, v' = v - v^3/3 - w + Drive, w' = 0.08(v + 0.7 - 0.8w). It is the textbook EXCITABLE system: a fast voltage v that fires and a slow recovery w that resets it. With enough Drive it spikes over and over - a sharp upstroke, a refractory dip, a slow recharge - the canonical shape of a firing neuron. Rate sets the speed, Drive the injected current (turn it up to cross from silent into repetitive spiking). A spiking oscillator distinct from the smooth chaos sources.

FoldTremolo [MOD] [FX] [SYNTH] in: Audio out: Audio

Timbral tremolo: an LFO pumps the drive into a wavefolder rather than the volume, so the brightness/harmonic density pulses in rhythm while the level stays steady - a spectral throb distinct from amplitude tremolo.

ForestFire [MOD] [FX] [SYNTH] in: – out: Audio

Forest-fire CA (a first as a synth block): a 4x4 grid of cells that are empty, growing a tree, or burning. Each sample a tree sprouts on an empty cell with chance Growth, a stray tree catches fire by Lightning, and fire spreads to neighbouring trees then burns out - the Drossel-Schwabl model of self-organised criticality. The output is the burning density: quiet while the forest regrows, then sudden sweeping fire-fronts, with the avalanche statistics of real wildfires. A different criticality than the Sandpile. Needs no input.

FourWing [MOD] [FX] [SYNTH] in: – out: Audio

Four-wing attractor (a first as a synth block): integrates x' = ax + yz, y' = bx + cy - xz, z' = -z - xy, a system whose trajectory visits FOUR separate wings instead of the usual two, wandering chaotically between all four quadrants in a butterfly with double the lobes. It free-runs at Rate, outputting x as a smooth bipolar CV that jumps unpredictably between the four wings. A four-lobed chaos distinct from the two-winged Lorenz, Chen and Lu attractors.

FractalDream [MOD] [FX] [SYNTH] in: – out: Audio

Fractal-dream-map source (a first as a synth block): iterates Clifford Pickover's 'fractal dream' map x <- sin(By) + Csin(Bx), y <- sin(Ax) + Dsin(Ay), a four-term sine system whose orbit settles into soft, dreamlike clouds of nested filaments. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that drifts through the dream. A four-sine attractor distinct from the De Jong and Svensson maps. Needs no input.

Fracture [MOD] [FX] [SYNTH] in: Audio out: Audio

Fracture: an onset splits the signal into two delayed copies whose detune WIDENS over time, so the sound starts unified then cracks open and drifts apart into a widening rift. Spread sets how fast and far it fractures, Mix the blend.

Frames [MOD] [FX] [SYNTH] in: A, B, C, D out: Audio

Keyframe morphing mixer (Mutable Instruments Frames): blends its four inputs (A-D) by sweeping one Frame knob through a chain of keyframes, crossfading from A toward D as Frame moves 0 to 1 - the modular take on animation keyframing. Frame sets the position and Anim, when above zero, animates it automatically at that rate (an evolving 4-source morph). Patch four oscillators or modulators in for smooth timbral or CV morphs. Distinct from the static Mixer and the XY-plane Vector.

FuncMaths [MOD] [FX] [SYNTH] in: Audio out: Audio

Function generator: a continuously-cycling rise/fall ramp with adjustable up and down times and a curve control (log to exp), a Maths-style slope source for modulation and looping envelopes.

Galam [MOD] [FX] [SYNTH] in: – out: Audio

Galam majority-rule source (a first as a synth block): models twelve people on a ring each holding a yes/no opinion; at each step a small group is picked and EVERYONE in it adopts the local MAJORITY view - the mechanism Galam used to model how rumours, fashions and political opinions sweep through a crowd toward unanimity. A trickle of independent mind-changing keeps it from freezing forever. The output is the fraction holding 'yes', which lurches toward total consensus (all-yes or all-no) and then gets knocked back. A majority-driven opinion source distinct from the pairwise Deffuant model. Needs no input.

Gallop [MOD] [FX] [SYNTH] in: Audio out: Audio

Gallop: an amplitude pattern that plucks the signal in a galloping long-short-short triplet figure (hits at the downbeat and the back two thirds), turning a pad or a drone into a rhythmic canter. Rate sets the gallop tempo, Mix the blend.

GaloisSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Galois LFSR sequencer: a maximal-length linear-feedback shift register clocks out a long, deterministic pseudo-random bit sequence as a gate - the structured randomness behind Turing-machine-style sequencers.

GamblersRuin [MOD] [FX] [SYNTH] in: – out: Audio

Gambler's-ruin source (a first as a synth block): a fortune that takes a coin-flip step up or down each turn between two walls - bankruptcy at the bottom, the jackpot at the top. The path is a pure random walk, but the moment it touches either wall the game restarts from the middle, so the output is a stream of random excursions: long meandering climbs and slides that always end in a sudden reset to centre. Rate sets the bet speed and Wall the playing-field size (wider walls = longer, rarer excursions). An absorbing-barrier walk distinct from the mean-reverting urns. Needs no input.

GamblersRuinWalk [MOD] [FX] [SYNTH] in: Audio out: Audio

Gambler's ruin: a +/-1 random walk (optionally biased) between two absorbing barriers of 0 and the stake; it wanders until it hits a wall - bust or broke-the-bank - then resets to the middle, giving runs of varying length to absorption.

Gargle [MOD] [FX] [SYNTH] in: Audio out: Audio

Gargle modulator: chops the amplitude on and off at an audio-rate random clock, the classic 'gargle' robot-voice ring/AM chop - between a coarse buzzy ring-modulation and a sputtering bitlike chatter depending on Rate. Rate sets the chop clock, Mix the blend.

GateClock [MOD] [FX] [SYNTH] in: – out: Audio

Clock gate generator: emits a steady rhythmic gate (rate + duty cycle) as a control source for triggering envelopes, plucks or sample-holds - a self-contained tempo pulse, needs no input.

Gater [MOD] [FX] [SYNTH] in: Audio out: Audio

Step-sequenced tremolo gate that reads a per-step on/off pattern from a 16-bit mask, advancing through the steps on a sample-clock phase for tempo coherence. Rate sets the cycle frequency, Pattern is the bitmask of active steps, Steps sets how many steps are used, and Smooth softens the transitions between open and closed. Mix sets how deeply the gate cuts the signal, from subtle pulsing to full rhythmic chopping.

GateRandom [MOD] [FX] [SYNTH] in: Audio out: Audio

Random rhythmic gate: at each step of an internal clock the gate opens or stays shut by a set probability, chopping the signal into an ever-changing stutter pattern (deterministic RNG).

GateShape [MOD] [FX] [SYNTH] in: Audio out: Audio

Shaped rhythmic gate: an internal rate chops the signal, with one knob morphing the gate envelope from hard square (trance gate) to triangle (tremolo) - rhythmic amplitude shaping without a sequencer.

GaussMap [MOD] [FX] [SYNTH] in: – out: Audio

Gauss-map (mouse-map) source (a first as a synth block): iterates x <- exp(-Alpha x^2) + Beta, a bell-shaped map named for the Gaussian in it. Sweeping Beta marches it through a remarkable cascade - fixed points, period-doubling, and period-ADDING windows where the cycle length grows by one - drawing the 'mouse' shape in its bifurcation diagram. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that ranges from steady to multi-step periodic to chaotic as Alpha and Beta move. A bell-curve map distinct from the parabolic logistic and tent maps.

∿ GeometricNoise [MOD] [FX] [SYNTH] in: Audio out: Audio

Geometric noise: the discrete count of failures before the first success in repeated coin flips (the discrete memoryless distribution), quantized into integer levels; lower success probability stretches the tail toward larger counts - a stepped discrete source.

∿ GilbreathSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Gilbreath sequence: repeatedly take absolute differences of consecutive primes and stack the rows; the leading entry of every row is conjectured to always be 1 (Gilbreath's conjecture). This reads out one diagonal of that triangle as a sparse pattern.

∿ Gingerbread [MOD] [FX] [SYNTH] in: – out: Audio

Gingerbreadman-map source (a first as a synth block): iterates the piecewise-linear chaotic map x <- 1 - y + |x|, y <- x - an area-preserving map whose orbits fill a strange six-sided 'gingerbread man' region of interleaved chaotic seas and stable islands. It free-runs at Rate, stepping the map and emitting x as a bipolar stepped CV that hops around the figure forever without repeating. Built from only an absolute value and a subtraction, it is a different, more angular chaos than the trig-based attractors.

∿ Glide [MOD] [FX] [SYNTH] in: Pitch (CV), Gate out: Audio

Portamento: one-pole smooths the pitch/control signal at in0 toward each new value over Time, gated by in1. Time sets the glide duration in seconds (zero snaps instantly), and Mode chooses Always (glide on every note change) or Legato (snap on an isolated note, glide only between overlapping notes). Drop it between a note source and an oscillator for per-layer portamento.

∿ GoldenSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Golden-ratio sequence: an additive recurrence that adds the golden ratio's fractional part each step and wraps, producing the maximally-equidistributed low-discrepancy sequence - quasi-random values that fill the range more evenly than noise.

∿ GolombSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Golomb sequence: the unique non-decreasing sequence where term n states how many times n appears (1,2,2,3,3,4,4,4,…), a self-describing sequence that grows like n^(phi) - a gently-climbing staircase with repeated plateaus.

∿ Goodwin [MOD] [FX] [SYNTH] in: – out: Audio

Goodwin growth-cycle source (a first as a synth block): Richard Goodwin's model of the capitalist business cycle, which borrows the predator-prey equations from ecology - here EMPLOYMENT plays the prey and the WORKERS' WAGE SHARE the predator. High employment lets workers win higher wages, which squeezes profits and investment, which raises unemployment, which weakens wages, which restores profits, and round it goes. The output is the employment rate, cycling endlessly like a boom-and-bust. A class-struggle oscillation, integrated symplectically so the cycle never winds down, distinct from the biological predator-prey. Needs no input.

∿ GrayCodeSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Gray-code sequence: the reflected binary code (each successive integer differs from the last by exactly one bit), read out as a low-bit gate; its single-bit-change property gives a smoothly-rotating, glitch-minimal pattern.

∿ Grids [MOD] [FX] [SYNTH] in: – out: Audio

Topographic drum trigger generator (Mutable Instruments Grids): walks a 32-step groove map and fires a gate when the selected Voice's pattern density at the current step beats the Density threshold, so turning Density up adds hits and down thins them out. Map morphs the rhythm between two feels, Chaos sprays in random hits/drops, and Voice picks Kick, Snare or Hat (run three Grids off the same patch for a full kit). Self-clocks at Rate. Distinct from Euclid's evenly-spread Euclidean rhythms - these are curated grooves.

∿ GumowskiMira [MOD] [FX] [SYNTH] in: – out: Audio

Gumowski-Mira attractor source (a first as a synth block): iterates a map built around the recursive nonlinearity f(x) = A*x + 2(1-A)$x^{2/(1+x}$2), devised at CERN to model the chaotic drift of particles in an accelerator beam. It is famous for the astonishing variety of shapes it draws - rings, swirls, lattices and creatures - from tiny parameter changes. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A particle-physics attractor distinct from the trigonometric and quadratic maps. Needs no input.

∿ HaltonSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Halton sequence: the base-3 radical inverse (the van-der-Corput construction in base 3), filling the interval by repeated trisection; paired with base-2 it forms the 2-D Halton points used to scatter quasi-random samples evenly.

∿ Halvorsen [MOD] [FX] [SYNTH] in: – out: Audio

Halvorsen attractor (a first as a synth block): integrates the cyclically-symmetric system x' = -a*x - 4y - 4z - y^2 (and its x->y->z rotations), whose quadratic cross-terms wind the trajectory into a tight three-armed spiral knot. It free-runs at Rate, outputting x as a smooth bipolar CV that loops and lurches around the attractor's arms. A denser, more wound-up chaos than the open Lorenz butterfly, distinct from the trig-based Thomas. Internally 3 small Euler sub-steps keep the stiff dynamics stable.

∿ HappySeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Happy numbers: repeatedly replace a number by the sum of the squares of its digits; 'happy' starts reach 1, the rest fall into the cycle 4,16,37,58,89,145,42,20,4… - this traces that digit-square trajectory, reseeding at each resolution.

∿ Harmonograph [MOD] [FX] [SYNTH] in: – out: Audio

Harmonograph LFO (a first as a synth block): models the Victorian drawing machine where two pendulums swinging at slightly-detuned frequencies trace a slowly-evolving figure. It sums two sinusoids at the FreqA and FreqB harmonics with a tiny detuning, so the shape drifts and precesses cycle after cycle, never quite repeating - an organic, breathing modulation rather than a fixed loop. Rate sets the base speed, FreqA and FreqB the two pendulum rates. A quasi-periodic curve distinct from the exactly-repeating Rose and Spirograph.

∿ HarmTremolo [MOD] [FX] [SYNTH] in: Audio out: Audio

Harmonic tremolo that splits the signal into low and high bands with a one-pole low-pass and amplitude-modulates them in opposition with a sine LFO, so as the lows swell the highs dip and back again. Rate sets the LFO speed, Depth sets the modulation intensity, Freq places the band split, and Mix sets dry/wet. The opposing band motion gives a richer, more vocal sweep than amplitude tremolo.

∿ Hawkes [MOD] [FX] [SYNTH] in: – out: Audio

Self-exciting (Hawkes) trigger (a first as a synth block): a point process where every event briefly RAISES the chance of another - so triggers arrive in bursts and clusters rather than evenly, the model for earthquake aftershocks, neuron avalanches and viral cascades. Rate sets the background event rate, Excite how strongly each firing kicks the intensity up, and Decay how fast that excitement fades. With low Excite it behaves like a plain Poisson source; raise it and the output clumps into flurries of activity separated by lulls. The branching is kept sub-critical so it can never run away. Needs no input.

HawkesProcess [MOD] [FX] [SYNTH] in: Audio out: Audio

Hawkes process: a self-exciting point process where each event temporarily raises the intensity of further events, so impulses arrive in bursts and aftershock clusters (used for earthquakes, trades, neural spikes) - a clustering, contagious click train.

Henon [MOD] [FX] [SYNTH] in: Clock, Reset out: Audio

Henon-map chaos sequencer (a first as a synth block): iterates the classic 2-D strange-attractor map x' = 1 - Ax^2 + y, y' = Bx on every rising edge of the in0 clock, outputting x as a stepped CV that hops chaotically across the attractor. At the textbook A=1.4, B=0.3 it traces the famous folded Henon curve - deterministic yet never repeating, a sharper, more angular chaos than the smooth DoublePendulum or the logistic Chaos. A rising edge on in1 reseeds to the origin.

HigherOrderCrossings [MOD] [FX] [SYNTH] in: Audio out: Audio

Higher-order crossings: counts zero-crossings not of the signal itself but of its Order-1 th finite difference over a 16-sample window - Order 1 is the ordinary zero-crossing rate, higher Orders emphasize progressively higher frequencies, so the output is a tunable spectral-tilt sensor (Kedem's HOC). A frequency-content control distinct from a plain zero-crossing counter.

HilbertScan [MOD] [FX] [SYNTH] in: Audio out: Audio

Hilbert curve scan: walks the Hilbert space-filling curve and reads out one coordinate, so a 1-D index sweeps a 2-D square while keeping nearby steps spatially close - a locality-preserving, self-similar stepped control source.

HindmarshRose [MOD] [FX] [SYNTH] in: – out: Audio

Hindmarsh-Rose bursting neuron (a first as a synth block): integrates a three-variable neuron model with a slow adaptation current that makes it BURST - it fires a rapid cluster of spikes, then falls quiet while the slow variable recovers, then bursts again, often chaotically so no two bursts match. This spike-then-rest-then-spike rhythm is exactly how many real neurons fire. The output is the membrane voltage: fast spiking riding on a slow up-and-down envelope. Rate sets the speed. A bursting chaos distinct from the steady spiking of FitzHugh-Nagumo and the smooth attractors.

HjorthMobility [MOD] [FX] [SYNTH] in: Audio out: Audio

Hjorth mobility: sqrt(var(dx)/var(x)) over an 8-sample window - the ratio of the signal's slope variance to its amplitude variance, which is a time-domain estimate of mean frequency (brightness). It rises for fast, bright material and falls for slow, dull material, all without an FFT. A spectral-brightness control signal from EEG analysis.

HofstadterFGMG [MOD] [FX] [SYNTH] in: Audio out: Audio

Hofstadter Female-Male sequences: two intertwined recurrences (F(n)=n-M(F(n-1)), M(n)=n-F(M(n-1))) that each call the other (from Godel, Escher, Bach); their interleaved growth makes a gently-jittering self-referential stepped CV.

HofstadterQSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Hofstadter Q sequence: the meta-recurrence Q(n)=Q(n-Q(n-1))+Q(n-Q(n-2)) where the sequence indexes into itself, producing erratic, chaotic-looking yet deterministic growth (from Godel, Escher, Bach) - a self-referential stepped CV.

Hopalong [MOD] [FX] [SYNTH] in: - out: Audio

Hopalong-map source (a first as a synth block): iterates Barry Martin's 'hopalong' map x <- y - sign(x)*sqrt(|x|), y <- A - x, which hurls the orbit back and forth across the origin, the point hopping along an intricate web of nested arcs and swirls. The square-root fold keeps it from running away while letting it explore a wide, lacy attractor. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that leaps across the figure. An orbital, root-folded chaos distinct from the trigonometric and polynomial maps.

IqrSpread [MOD] [FX] [SYNTH] in: Audio out: Audio

Interquartile-range detector: sorts a 7-sample window and outputs the gap between its upper and lower quartiles - a robust measure of local signal spread that ignores the single most extreme sample on each side, so it tracks sustained activity/roughness without being thrown off by isolated clicks. A click-immune alternative to variance as a modulation source.

Ising [MOD] [FX] [SYNTH] in: - out: Audio

Ising-model magnetisation source (a first as a synth block): a 4x4 lattice of up/down spins evolves by the Metropolis rule - a spin flips if it lowers the energy, or randomly with a chance set by Temperature - the textbook model of a magnet. The output is the net magnetisation: cold, the spins align into one domain and it drifts toward +/-1; hot, they tumble and it jitters around zero; near the critical temperature it swings in large, slow, self-similar fluctuations. A statistical-physics modulation source. Needs no input.

JacobiSymbol [MOD] [FX] [SYNTH] in: Audio out: Audio

Jacobi symbol: the Legendre symbol generalized to any odd modulus via reciprocity, giving a +/-1 (or 0) sequence computed by the fast reciprocity flip-and-reduce algorithm - a number-theoretic character pattern over composite moduli.

JacobsthalLucasSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Jacobsthal-Lucas numbers: the companion to the Jacobsthal sequence, 2,1,5,7,17,31,65,… with JL(n)=JL(n-1)+2JL(n-2), stepped live and reduced modulo a base - a doubling-flavoured tunable pattern with a fully-live modulus knob.

JacobsthalSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Jacobsthal sequence: J(n)=J(n-1)+2J(n-2) (0,1,1,3,5,11,21,43,…), the Fibonacci-like sequence whose ratio tends to 2 and which counts tilings and binary strings without adjacent 1s; modded, a doubling-ish stepped pattern.

Julia [MOD] [FX] [SYNTH] in: - out: Audio

Julia-set source (a first as a synth block): iterates the complex map z <- z*z + c, the equation behind the Julia fractals - a point in the complex plane is squared and shifted again and again. For points inside the set it wanders forever in an intricate bounded orbit; when one escapes to infinity the source reseeds near the centre and sets off again, so the output is a restless, fractal-flavoured CV. CReal and CImag pick which Julia set (its shape and dynamics) to explore. It free-runs at Rate, outputting the real part. A complex-dynamics source unlike any real-valued chaos.

∿ JustFriends [MOD] [FX] [SYNTH] in: – out: Audio

Manifold function generator (Mannequins Just Friends): one shape, six harmonically-related copies summed into a rich evolving wave. Time sets the base rate (slow LFO up to audio pitch); Intone spreads the six harmonics from a tight integer series toward stretched, detuned ratios; Curve bends every slope from a linear ramp toward a smooth sine; Level trims output. A function generator and harmonic oscillator in one - drones, organ-ish tones and undulating modulation. Distinct from the partial-summing Additive (pure sines, spectral tilt).

∿ KaprekarMap [MOD] [FX] [SYNTH] in: Audio out: Audio

Kaprekar routine: take a 4-digit number, subtract its ascending from its descending digit arrangement, and repeat; almost every start converges in at most 7 steps to the fixed point 6174 (Kaprekar's constant), then reseeds - a digit-process attractor.

∿ KatzFD [MOD] [FX] [SYNTH] in: Audio out: Audio

Katz fractal dimension: measures waveform crinkliness as log(n)/(log(n)+log(d/L)) over a 16-sample window, where L is the total Euclidean path length along the curve and d is the farthest reach from the first point - 1 for a straight line, rising as the path folds back on itself. Unlike Petrosian's sign-change form, Katz uses real path geometry, so it responds to amplitude excursions too.

∿ KeyTrack [MOD] [FX] [SYNTH] in: Note out: Audio

Keyboard-tracking mod source: turns the note pitch into a centered, scalable modulation - the classic filter keytrack. Patch the Note source into in0; the output is (note - Center) x Amount around the midpoint, so notes above Center push the modulation up and notes below push it down (Amount 1 = full 1:1 tracking across the keyboard, 0.5 = half, negative inverts). Patch the output into a filter cutoff (or any param) so the timbre follows the keyboard. Center sets the pivot note (0-127), Amount the bipolar tracking depth. Unlike the raw Note source this is centered and scalable, so the modulation is zero at the pivot.

∿ Kirman [MOD] [FX] [SYNTH] in: – out: Audio

Kirman-herding source (a first as a synth block): Alan Kirman's 'ants' model of why crowds and markets swing between extremes - twelve agents each hold one of two views, and at each step one agent either switches by IMITATING a random other (herding) or changes its mind spontaneously. With strong herding the population does not settle in the middle but lurches as a herd, swinging to nearly all-one-way, lingering, then stampeding to the other - the bimodal switching behind fads, bank runs and market bubbles. The output is the fraction holding one view. Herd sets the imitation strength. A herding-instability source distinct from the consensus-seeking opinion models. Needs no input.

∿ KolakoskiSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Kolakoski sequence: the unique 1,2 sequence that is its own run-length encoding (its run lengths spell out the sequence itself), a self-referential aperiodic pattern; precomputed once and clocked out as a hypnotically self-similar stepped CV.

∿ KroneckerSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Kronecker symbol: the Jacobi symbol extended to even and negative arguments via the special rule for 2, completing the quadratic character to a fully-multiplicative function on all integers - a +/-1/0 sequence with a distinct even-index structure.

Kuramoto [MOD] [FX] [SYNTH] in: – out: Audio

Kuramoto synchronisation source (a first as a synth block): eight phase oscillators, each at its own natural frequency, are nudged toward each other by Coupling - the model of how fireflies, neurons and metronomes spontaneously fall into step. The output is their combined mean field: with low Coupling the eight detuned waves smear into a restless wobble, and as Coupling rises they lock into one coherent oscillation that swells in amplitude. Rate sets the centre frequency and Spread the detuning. A self-organising motion no LFO or noise source produces. Needs no input.

KuramotoPair [MOD] [FX] [SYNTH] in: Audio out: Audio

Kuramoto coupled oscillators: two phase oscillators at slightly different rates pull on each other; below a coupling threshold they beat, above it they lock - the canonical model of spontaneous synchronization, as an evolving two-tone source.

KuramotoRing [MOD] [FX] [SYNTH] in: – out: Audio

Kuramoto-ring source (a first as a synth block): sixteen phase oscillators arranged in a circle, each pulled toward the phase of its two NEIGHBOURS only (not the whole crowd as in the classic Kuramoto model). Local coupling lets waves of synchrony travel around the ring - bands of in-step oscillators sweeping past out-of-step ones, the metachronal waves seen in beating cilia and centipede legs. The output is one oscillator’s value, beating as the synchrony waves roll by. Couple sets the pull toward neighbours. A spatially-coupled sync source distinct from the all-to-all chaos. Needs no input.

KurtosisDetector [MOD] [FX] [SYNTH] in: Audio out: Audio

Kurtosis detector: computes the 4th standardized moment of a 7-sample window minus 3 (excess kurtosis), which is near zero for noise, negative for a steady sine, and spikes positive on impulsive clicks and transients - a control signal that measures how ‘peaky’ the local waveform distribution is. A statistical transient/impulsiveness sensor, not an envelope follower.

LambdaOmega [MOD] [FX] [SYNTH] in: – out: Audio

Lambda-omega medium source (a first as a synth block): a ring of sixteen cells each running a little rotational oscillator (the lambda-omega reaction-diffusion system, the normal form for any chemical or biological medium near the onset of oscillation), coupled by diffusion. The cells phase-lock into travelling and spiral waves - the spinning patterns of the Belousov-Zhabotinsky reaction and slime-mould aggregation. The output is one cell’s concentration, oscillating as the spiral arms sweep past. A pattern-forming oscillatory medium distinct from the excitable Excitable and the single-cell oscillators. Needs no input.

LargestPrimeFactor [MOD] [FX] [SYNTH] in: Audio out: Audio

Largest prime factor: P(n), the biggest prime dividing the step index (the Stormer / smoothness measure), which equals n on primes and drops sharply on smooth numbers - a spiky log-scaled arithmetic CV.

LegendreSymbol [MOD] [FX] [SYNTH] in: Audio out: Audio

Legendre symbol: outputs +1 if the step index is a quadratic residue modulo the chosen odd prime, -1 if a non-residue, 0 if divisible; this +/-1 pattern (a multiplicative character) underlies quadratic reciprocity and is conjectured pseudo-random.

Lemniscate [MOD] [FX] [SYNTH] in: – out: Audio

Lemniscate LFO (a first as a synth block): traces the figure-eight curve of Bernoulli (r-squared = cos(2*angle)) and outputs its sweep, which produces TWO smooth pulses per cycle separated by flat dead zones where the curve pinches to nothing at the crossing point. The result is a paired-pulse modulation - two lobes of motion, then rest, then two more - quite unlike a single sine swing. Rate sets the speed. A figure-eight LFO distinct from the single-loop curves.

Lengyel [MOD] [FX] [SYNTH] in: – out: Audio

Lengyel-Epstein chemical oscillator (a first as a synth block): the model of the CIMA reaction (chlorine-dioxide / iodine / malonic-acid), the experiment that first produced stationary Turing patterns in a real chemical and which, well stirred, oscillates in time instead. Two coupled concentrations chase each other into a stable limit cycle. Rate sets the speed and Feed the reactant supply. The output is the activator concentration. A pattern-forming chemistry distinct from the Brusselator and the biological oscillators.

Levy [MOD] [FX] [SYNTH] in: – out: Audio

Levy-flight modulation source (a first as a synth block): a random walk whose step sizes are drawn from a heavy-tailed (Cauchy) distribution, so it mostly creeps in tiny steps but every so often makes a sudden enormous leap across the range - the foraging pattern of albatrosses and the maths of anomalous diffusion. Step sets the typical jump scale and Rate how often it moves; the walk reflects off +/-1 to stay in range. Where a smoothed-noise LFO wanders gently, Levy lurks then pounces - clustered stillness punctuated by leaps. Needs no input.

LevyFlight [MOD] [FX] [SYNTH] in: – out: Audio

Levy-flight source (a first as a synth block): a random walk whose step sizes are drawn from a heavy-tailed Cauchy distribution, so most moves are tiny but every so often it takes a wild, far-reaching leap - the search pattern of foraging animals and the signature of anomalous, super-diffusive transport. The value mostly hovers, then teleports across the range and resumes. Jump scales the leaps; the rails wrap so a big jump reappears on the far side. A heavy-tailed flight distinct from the Gaussian random walk. Needs no input.

LFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Low-frequency control generator with sine, triangle, saw and pulse shapes (Shape), free-running at Rate. Depth scales the swing, Bias offsets the centre, Phase shifts the start point, and Width sets the pulse duty cycle. Patch its output into any parameter's mod input for vibrato, tremolo, filter sweeps and rhythmic movement. A trigger into in0 resets the phase (retrigger); leave it unpatched to free-run.

LiouvilleSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Liouville function: lambda(n) is +1 or -1 by the parity of the total number of prime factors of n (with multiplicity); its sign sequence and partial sums are central to analytic number theory - a deterministic, nearly-balanced +/-1 stream.

Lissajous [MOD] [FX] [SYNTH] in: – out: Audio

Lissajous LFO (a first as a synth block): multiplies two sinusoids whose frequencies are in the ratio RatioX:RatioY - the pair that, plotted against each other, trace the classic Lissajous figures on an oscilloscope. The product is a complex, symmetric modulation that beats and nulls as the two rates cross, its character set entirely by the integer ratio. Simple ratios give clean repeating shapes, larger ones dense interference. Rate sets the speed. A two-tone interference LFO distinct from the rolling-circle curves.

LogisticMap [MOD] [FX] [SYNTH] in: – out: Audio

Logistic-map chaos: the classic population-dynamics iterator x = Rx(1-x), stepped at Rate. As R climbs past ~3.57 the output period-doubles into full chaos, then snaps back to order in the periodic windows - a stepped, glitchy random source you can tune from steady to wild. Needs no input.

LookAndSay [MOD] [FX] [SYNTH] in: Audio out: Audio

Look-and-say: each term describes the previous one aloud (1 -> '11' -> '21' -> '1211' -> …), the self-describing sequence whose length grows by Conway's constant; its digit stream (only 1s, 2s and 3s) makes a chattering stepped CV.

Lorenz84 [MOD] [FX] [SYNTH] in: – out: Audio

Lorenz-84 attractor (a first as a synth block): integrates Edward Lorenz's low-order model of the GLOBAL ATMOSPHERIC circulation, x' = -y^2 - z^2 - ax + aF, y' = xy - bxz - y + G, z' = bxy + xz - z, where x is the strength of the westerly wind current and y, z are a travelling weather wave. It chaotically switches between calm and stormy regimes, a toy climate that never repeats. It free-runs at Rate, outputting x as a smooth bipolar CV. A climate-model chaos distinct from the original Lorenz convection attractor.

LorenzLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Lorenz strange-attractor chaos modulator (Serum-style): integrates the Lorenz system and outputs one Axis (X/Y/Z), normalized and scaled by Depth. Produces smooth, organic, never-repeating motion - more flowing than stepped random. Great for evolving cutoff, pitch or pan drift. A trigger into in0 resets the attractor; unpatched = free-run.

LotkaVolterra [MOD] [FX] [SYNTH] in: – out: Audio

Lotka-Volterra predator-prey oscillator (a first as a synth block): the founding equations of mathematical ecology - prey x breed freely but are eaten by predators y, whose numbers in turn rise and fall on the food supply. The two populations chase each other in a perpetual cycle: prey boom, predators follow and crash them, then starve and let the prey boom again. Rate sets the speed and Growth the prey's breeding rate. The output is the prey population as a smooth 0..1 swell, integrated symplectically so the cycle never winds down. An ecological oscillation distinct from the neuron and chaos sources.

Lozi [MOD] [FX] [SYNTH] in: – out: Audio

Lozi-map source (a first as a synth block): iterates x <- 1 - A|x| + y, y <- Bx - the piecewise-linear cousin of the Henon map, with an absolute value where Henon has a square. That sharp corner gives it a strange attractor made of straight-line segments rather than smooth curves, and one of the few chaotic maps proven rigorously to be chaotic. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A sets the fold sharpness. An angular, provably-chaotic attractor distinct from the smooth Henon. Needs no input.

LSystemSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

L-system sequencer: expands a fractal rewriting grammar (the algae rule A->AB, B->A) into a self-similar symbol string and clocks it out as a stepped CV, generating structured, recursive, never-trivially-periodic melodic patterns.

Lu [MOD] [FX] [SYNTH] in: – out: Audio

Lu (Lu-Chen) attractor (a first as a synth block): integrates x' = a(y-x), y' = -xz + cy, z' = xy - bz - the system that forms the missing BRIDGE between the Lorenz and Chen attractors, morphing continuously from one to the other as its parameter changes. Its double scroll sits between the two, with its own balance of slow orbiting and sudden lobe-switching. It free-runs at Rate, outputting x as a smooth bipolar CV. A transitional double-scroll chaos distinct from both Lorenz and Chen.

LucasSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Lucas numbers: the Fibonacci companion 2,1,3,4,7,11,18,… with L(n)=L(n-1)+L(n-2), stepped live and reduced modulo a base you set - the recurrence is iterated each clock so the modulus knob is fully live, giving a tunable climbing pattern.

LychrelMap [MOD] [FX] [SYNTH] in: Audio out: Audio

Reverse-and-add: take a number, add it to its digit-reversal, and repeat; most numbers quickly reach a palindrome, but suspected Lychrel numbers (like 196) seem never to - this traces that growing iteration, reseeding when a palindrome is hit.

MangoldtSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Von Mangoldt function: Lambda(n) equals log(p) when n is a power of a single prime p and zero otherwise, the weighting that makes the prime-counting explicit formula work; it fires log-scaled spikes only on prime powers - a sparse arithmetic CV.

Marbles [MOD] [FX] [SYNTH] in: – out: Audio

Random voltage source with a loop lock (Mutable Instruments Marbles): on each internal Rate step it generates a new random value around Bias with a width set by Spread, but Deja Vu blends in replay of a stored loop - 1 = a locked, repeating random melody of Length steps, 0 = fresh randomness every step. A stepped CV source between pure chance and a fixed sequence; quantize the output for pitch. Distinct from Random (no loop memory) and Turing (binary shift register, not continuous CV).

MarkovGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Markov sequencer: a small state machine that on each clock step either holds or jumps to a new state by a transition probability, emitting each state as a CV level - structured-random sequences that have memory, unlike pure random.

MarkovMelody [MOD] [FX] [SYNTH] in: Audio out: Audio

Markov chain melody: walks an eight-state chain where the next state is chosen by a near-neighbour transition rule (mostly small moves, occasional leaps), producing stepped pitch sequences that feel coherent yet never quite repeat - a stochastic melody generator.

Martin [MOD] [FX] [SYNTH] in: – out: Audio

Martin-map source (a first as a synth block): iterates the sine variant of Barry Martin's map x <- y - sin(x), y <- A - x, whose orbit hops across a broad, wavy lattice of interleaved bands rather than a compact attractor. The single sine fold gives a wide, drifting, quasi-tiled motion that roams far before folding back. It free-runs at Rate, outputting x as a bipolar CV. A wide-ranging, lattice-like chaos distinct from the compact Hopalong and the trig attractors.

Maths [MOD] [FX] [SYNTH] in: Trigger out: Audio

Function generator (Make Noise Maths / Buchla 281 / Serge DUSG): a rise/fall slope with shapeable curves. Rise and Fall set the up/down times; Curve bends them from logarithmic through linear to exponential. Mode: Cycle self-runs as a variable-shape LFO; Trig fires a one-shot AD envelope on a gate at in0; Gate rises while in0 is held and falls when released (AR). The CV building block behind envelopes, slope LFOs and generative Krell patches. Patch a Clock/Euclid into in0 to trigger it.

Melt [MOD] [FX] [SYNTH] in: Audio out: Audio

Melt: while a note is sustained the pitch slowly sags downward as a growing delay stretches it, so a held tone seems to melt and drip toward the floor, snapping back when you release. A continuous downward droop, not a triggered fall. Rate sets the melt speed, Mix the blend.

MephistoSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Mephisto Waltz sequence: the fixed point of 0->001, 1->110, a self-similar ternary-grouped binary pattern with a swung, waltzing rhythmic feel when clocked out as a gate.

MertensSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Mertens function: the running sum of the Moebius function, a slow random-walk-like staircase whose growth rate is tied to the Riemann hypothesis (the disproved Mertens conjecture said it stays within +/-sqrt(n)) - a meandering number-theoretic CV.

MIDI CC [MOD] [FX] [SYNTH] in: – out: Audio

MIDI continuous-controller mod source: outputs the live value (0..1) of MIDI CC number CC, so any hardware knob, fader, expression or breath controller becomes a modulation source - patch its output into any parameter's mod input. Smooth slews fast controller jumps, Depth scales the range and Bias offsets the centre. Common numbers: 1 mod wheel, 2 breath, 7 volume, 11 expression, 64 sustain, 74 brightness.

MoebiusSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Moebius function: mu(n) is +1, -1 or 0 depending on the prime factorization of n (0 if a square divides it), the sign sequence at the heart of analytic number theory - a sparse three-level CV that flickers between +1, -1 and rest.

MooreSpiegel [MOD] [FX] [SYNTH] in: – out: Audio

Moore-Spiegel oscillator (a first as a synth block): integrates the jerk system x' = y, y' = z, z' = -z - (T - R + Rxx)y - Tx, devised to model the aperiodic flickering of a parcel of gas in a star's atmosphere (thermo-mechanical oscillations under radiation). The cubic stiffness term makes it burst between small and large oscillations chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV. An astrophysical jerk-chaos distinct from the rotational and convective attractors.

Moran [MOD] [FX] [SYNTH] in: – out: Audio

Moran-process source (a first as a synth block): the model of random genetic drift in a fixed population - Balls individuals are each one of two types; at every step one is chosen to reproduce and one (at random) dies, so the share of each type wanders purely by chance. A trickle of Mutation flips the occasional offspring, which keeps the share from ever getting permanently stuck at all-one-type. The output is that share: an unbiased random walk on a population, the discrete cousin of neutral evolution. Rate sets the generation speed. Needs no input.

MoranProcess [MOD] [FX] [SYNTH] in: Audio out: Audio

Moran process: in a fixed population of N, each step one individual reproduces and one dies, so an allele's count does a random walk until it fixes at 0 or N (genetic drift to extinction or takeover); it then reseeds - a drift-to-absorption CV.

MortonScan [MOD] [FX] [SYNTH] in: Audio out: Audio

Morton Z-order scan: de-interleaves the bits of the step index into one coordinate, walking the Z-shaped space-filling curve; unlike Hilbert it has occasional long jumps at quadrant boundaries, giving a self-similar sweep with sudden leaps.

MSEG [MOD] [FX] [SYNTH] in: Audio out: Audio

Multi-segment envelope generator: a gate edge resets a phase that ramps from 0 to 1 over Rate seconds, and the output linearly interpolates between user-defined 'time,level' breakpoints parsed from the node config. Rate sets the total run time of the shape, Loop makes the phase wrap and repeat instead of holding the last level, Level scales the final output, and Sync (1/1..1/16T) locks the cycle length to host tempo instead of the free Rate seconds. Breakpoints come from a 't,l;t,l;…' config string (defaulting to a fast attack and gentle decay when fewer than two are given), giving a flexible Serum/Vital-style modulation source for arbitrary multi-stage curves.

MultiLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Multi-phase LFO: blends two sine waves a settable Phase apart into a single richer, evolving modulation shape - cross between a complex LFO and a quadrature pair. Rate sets the speed, Phase the offset (90 deg = quadrature), Depth the blend of the second phase. A trigger into in0 resets the phase; unpatched = free-run.

NBody [MOD] [FX] [SYNTH] in: – out: Audio

Gravitational N-body source (a first as a synth block): three point masses pull on one another by Newton's inverse-square law inside a box with reflecting walls, tracing the famously unsolvable three-body motion - bodies swing past, sling-shot and tumble in an endless gravitational dance. Gravity sets the attraction strength and Rate the simulation speed; the output follows the first body's horizontal position. A softened-gravity, wall-bounded chaos with its own swooping character, distinct from the pendulum or the strange-attractor maps. Needs no input.

NegBinomialNoise [MOD] [FX] [SYNTH] in: Audio out: Audio

Negative-binomial noise: the count of failures before r successes in repeated trials, a discrete distribution that is more overdispersed (clumpier) than Poisson - the model of contagious or clustered counts, here a stepped integer source.

Newton [MOD] [FX] [SYNTH] in: – out: Audio

Newton-fractal source (a first as a synth block): runs Newton's root-finding method on the equation z^3 = 1 in the complex plane - z <- z - (z^3 - 1)/(3 z^2) - which always homes in on one of the three cube roots of unity, but WHICH one depends with fractal sensitivity on where it starts. The source iterates until it locks onto a root, then reseeds from a fresh random point, so the output jumps between the three roots' values with a swirling transient before each lock. It free-runs at Rate, outputting the real part. A root-finding chaos distinct from the squaring fractals.

NewtonLeipnik [MOD] [FX] [SYNTH] in: – out: Audio

Newton-Leipnik attractor (a first as a synth block): integrates a rigid-body rotation system (from the Euler equations of a tumbling object with feedback) that has the rare property of TWO coexisting strange attractors - depending on where it starts, the trajectory settles onto one of two different chaotic shapes. It free-runs at Rate, outputting x as a smooth bipolar CV that winds through a delicate folded-ribbon attractor. A two-attractor chaos distinct from the single-attractor Lorenz family.

NivenSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Niven (Harshad) gate: high when the step index is divisible by the sum of its own digits (1,2,…,10,12,18,20,…), a pattern that is common among small numbers but thins out - an arithmetic rhythmic gate.

NoiseBreath [MOD] [FX] [SYNTH] in: Audio out: Audio

Breath/texture layer: low-passed deterministic noise scaled by the input envelope adds air, breath or grit that swells with the signal - the 'noise component' of a sound, dynamics-keyed.

NoseHoover [MOD] [FX] [SYNTH] in: – out: Audio

Nose-Hoover oscillator (a first as a synth block): integrates the thermostatted system x' = y, y' = -x + y*z, z' = a - y^2 - the equations a physicist uses to hold a simulated molecule at constant temperature, which famously refuse to settle and instead wander chaotically. Unlike the dissipative attractors it is volume-CONSERVING, so the trajectory never collapses onto a thin sheet but fills a fat tube of phase space, giving a fuller, more even chaotic motion. It free-runs at Rate, outputting x as a smooth bipolar CV. A conservative chaos distinct from the dissipative attractors.

OneEuroSmooth [MOD] [FX] [SYNTH] in: Audio out: Audio

One-euro filter: an adaptive low-pass whose cutoff rises with the input's rate of change, so a slow or still signal is smoothed hard (killing jitter) while a fast move passes with little lag - the Casiez 1-euro algorithm prized for de-noising control signals without the sluggishness of a fixed smoother. MinCut is the at-rest cutoff, Beta how aggressively speed opens it up.

OnsetGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Onset detector: compares a fast and a slow envelope and outputs a short gate whenever the fast one surges past the slow one (an energy-flux onset), turning audio attacks into clean triggers for envelopes or drums.

Orbit [MOD] [FX] [SYNTH] in: – out: Audio

Kepler-orbit LFO (a first as a synth block): a body sweeps an elliptical orbit under gravity, so unlike a sine it lingers slowly at the far point (aphelion) then whips quickly through the near point (perihelion) - the asymmetric speed of a real planet, set by Eccentricity. It solves Kepler's equation each sample for the eccentric anomaly and outputs its cosine. At zero eccentricity it is a pure cosine; cranked up it becomes a lopsided swing that hangs and then lunges. Rate sets the orbital period. A gravitationally-shaped modulation curve, needs no input.

Ornstein [MOD] [FX] [SYNTH] in: – out: Audio

Ornstein-Uhlenbeck source (a first as a synth block): the mean-reverting random process - Brownian motion on a spring, which wanders randomly but is always pulled back toward its centre, so it drifts and meanders without ever running off the way a free random walk does. It is the standard model for a noisy voltage settling, a particle in a fluid, or an interest rate. Pull sets how strongly it springs back to centre (higher = tighter, faster wobble) and Depth the noise amount. A smooth, naturally-bounded random CV distinct from the target-hopping Drift and the free Levy flight. Needs no input.

OrnsteinUhlenbeck [MOD] [FX] [SYNTH] in: – out: Audio

Ornstein-Uhlenbeck source (a first as a synth block): a random walk on an elastic leash - noise kicks it around but a restoring pull always drags it back toward the centre, the textbook mean-reverting process used to model everything from particle velocities to interest rates. The result is smooth, organic, band-limited drift (Brownian motion's well-behaved cousin) with no sharp jumps. Revert sets the strength of the pull, Sigma the noise. A mean-reverting drift distinct from the free random walk. Needs no input.

PadovanSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Padovan sequence: P(n)=P(n-2)+P(n-3) (the recurrence whose growth ratio is the plastic number), the ‘Fibonacci with a gap’ that underlies Padovan spirals; reduced modulo a base it makes a slow, gently-rolling melodic pattern.

PaperFold [MOD] [FX] [SYNTH] in: Audio out: Audio

Paperfolding sequence: the turn directions you get by repeatedly folding a strip of paper in half (which trace the dragon curve), a 2-automatic sequence whose value comes from the bit just above the lowest set bit of the index - a self-similar fold pattern.

PartitionSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Partition sequence: p(n), the number of ways to write n as a sum of positive integers (1,1,2,3,5,7,11,15,…), reduced modulo a base; this deeply-studied sequence (Ramanujan congruences) makes an irregular repeating CV pattern.

PascalMod [MOD] [FX] [SYNTH] in: Audio out: Audio

Pascal mod 2: scans Pascal’s triangle modulo 2 (binomial parity, which draws the Sierpinski triangle); by Kummer’s theorem the bit is 1 only where the row and column indices share no carry - a self-similar triangular gate pattern.

PathLFO [MOD] [FX] [SYNTH] in: — out: Audio

Drawable XY-path LFO: a Rate-driven phase traverses the loop of 2D waypoints set in node config and reads back one coordinate as a bipolar -1 to 1 output, scaled by Level. Rate sets the traversal speed, Axis chooses whether the X or Y coordinate drives the output, and Level scales the result. Pair two nodes set to X and Y on the same drawn path to modulate two targets in a coordinated 2D motion.

PeakPhaser [MOD] [FX] [SYNTH] in: Audio out: Audio

A 6-stage allpass phaser whose sweep is driven by a peak envelope follower on the input, so transients trigger the filter motion instead of an LFO. Soften sets the envelope release time, Feedback sets the allpass regeneration (up to 0.9), Amount scales how far each peak sweeps the phaser, and Mix blends wet against dry. Part of the Traktor Pulse family, it pumps the phaser in time with the signal’s own dynamics.

PeanoScan [MOD] [FX] [SYNTH] in: Audio out: Audio

Peano curve scan: walks Peano’s original base-3 space-filling curve and reads out one coordinate; like the Hilbert scan it keeps consecutive indices spatially adjacent, but its ternary boustrophedon structure gives a different self-similar sweep.

PellSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Pell sequence: P(n)=2P(n-1)+P(n-2) (0,1,2,5,12,29,…), whose ratios converge to the silver ratio and whose terms give the best rational approximations to sqrt(2); modded, a brisk climbing melodic pattern.

Performer [MOD] [FX] [SYNTH] in: Audio out: Audio

Performer (Massive X-style step modulator): a tempo-synced step sequencer that runs a drawn pattern of up to 64 step levels (up to ~8 bars) locked to the host tempo. Draw the pattern in the step-bar editor; Sync sets the per-step note division (1/1..1/16T), Glide slews between steps (hard steps to a smooth contour), Level scales the output, and a gate on in0 restarts the pattern from step 1. Patch its output into cutoff, pitch, level or any mod input for long, evolving rhythmic movement that locks to the transport.

PeriodDoublingSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Period-doubling sequence: the fixed point of 0->01, 1->00, the symbolic itinerary of the logistic map's period-doubling cascade at the Feigenbaum point - a self-similar binary pattern that mirrors the route to chaos.

Perlin [MOD] [FX] [SYNTH] in: – out: Audio

Perlin-noise source (a first as a synth block): smooth coherent gradient noise - the organic, flowing randomness used to generate natural terrain, clouds and fire in computer graphics. Unlike white noise or a sample-and-hold (which jump instantly), Perlin noise glides: nearby moments are always close in value, so the output meanders gently and naturally with no abrupt steps, set by a quintic-smoothed lattice of random gradients. Rate sets how fast it drifts through the noise field. The natural-movement modulator, distinct from the white/pink NoiseLFO and the stepped Random.

PerlinLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Perlin-noise LFO: a smoothly-interpolated coherent-noise modulation source - organic, wandering movement with no audible steps, smoother than sample-and-hold and less regular than an LFO. Rate sets the speed, Smooth the curve, Depth the output. A trigger into in0 resets the phase; unpatched = free-run.

PermutationEntropy [MOD] [FX] [SYNTH] in: Audio out: Audio

Permutation entropy: classifies every consecutive triple in a 16-sample window into one of the 6 possible orderings, then takes the normalized Shannon entropy of that histogram - 0 when the waveform is perfectly ordered (a clean ramp or tone), 1 when the ordering is random (noise). An ordinal-pattern complexity measure that ignores amplitude entirely. Output in 0..1.

PerrinSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Perrin sequence: 3,0,2,3,2,5,5,7,… satisfying P(n)=P(n-2)+P(n-3) with its own seeds (famous because n divides P(n) almost exactly when n is prime - the Perrin pseudoprime test); modded, a quirky climbing pattern.

PersistenceSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Multiplicative persistence: how many times you must replace a number by the product of its digits before reaching a single digit (e.g. 277777788888899 takes 11, the known record); this counts that depth for each index - a sparse spiky CV.

PetrosianFD [MOD] [FX] [SYNTH] in: Audio out: Audio

Petrosian fractal dimension: estimates how 'crinkled' the waveform is from the number of direction changes in its slope across a 16-sample window, via log(n)/(log(n)+log(n/(n+0.4*Nd))). Near 1 for smooth tones, higher for noisy or busy signals - a cheap roughness/complexity control derived from sign changes alone. Output is scaled excess over the smooth baseline.

PhaseDiffuse [MOD] [FX] [SYNTH] in: Audio out: Audio

Phase diffuser: a cascade of first-order all-pass stages whose coefficient drifts under a slow LFO smears phase (decorrelation / shimmer) without any delay line - flat magnitude, moving phase.

PhaseGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Phase-windowed rhythmic gate: an internal LFO opens the gate only during the first Width fraction of each cycle, with a smoothed edge - a tempo-set chopper / trance-gate without a sequencer.

PhaseLFO [MOD] [FX] [SYNTH] in: Reset out: Audio

Phase-offset LFO: a free-running sine LFO with an adjustable phase shift, resettable to zero on a rising edge of in0 - patch one clock into several PhaseLFOs at different Phase settings for quadrature or poly-rhythmic modulation from a single source. Rate sets the speed, Phase the offset (0..1 of a cycle), Depth the output level.

PhaseMod [MOD] [FX] [SYNTH] in: Carrier, Modulator out: Audio

FM / phase-modulation processor: phase-modulates the in0 carrier with the in1 modulator (stack any number of sources into either input) to make classic FM sidebands - the modular way to FM oscillators together without per-oscillator routing. Patch one oscillator into in0 and another into in1 for two-operator FM; stack more cables on in1 to FM with several sources at once. Index sets the modulation depth, Feedback routes the output back into the modulator for harmonic edge toward noise, and Mix blends against the dry carrier. Distinct from the self-contained FM oscillator: this FMs whatever audio you feed it.

Phaser [MOD] [FX] [SYNTH] in: Audio out: Audio

All-pass cascade phaser: an LFO sweeps 2-8 all-pass stages (Stages) to move a series of notches through the spectrum, with Feedback sharpening their resonance. Depth sets the sweep range and Mix the dry/wet blend. More stages add more notches for a deeper, more vocal swirl.

Phaser4 [MOD] [FX] [SYNTH] in: Audio out: Audio

Four-stage phaser: a cascade of four LFO-swept all-pass stages with feedback creates moving notches across the spectrum - the classic sweeping phaser whoosh, a true all-pass (not a flange delay).

PhaserFold [MOD] [FX] [SYNTH] in: Audio out: Audio

Folding phaser: a six-stage allpass chain whose feedback is wavefolded before it re-enters, so the phaser's resonant peaks generate fold harmonics that swirl with the notches - a screaming, harmonically-alive phaser unlike any clean one.

PhaseRotate [MOD] [FX] [SYNTH] in: Audio out: Audio

Phase rotator: a cascade of all-pass stages spins the broadband phase of the signal without touching its magnitude spectrum, so asymmetric waveforms (voice, brass) get their positive/negative peaks evened out - a free headroom/symmetry trick that sounds identical alone but changes how it clips and sums. Amount sets the rotation, Mix the blend.

PhaseScramble [MOD] [FX] [SYNTH] in: Audio out: Audio

Phase scrambler: six allpass sections with incommensurate, slowly LFO-modulated coefficients rotate phase without touching magnitude, smearing and de-correlating transients into a soft wash - a moving phase-only diffuser for thickening or blurring a source.

∿ **Phoenix** [MOD] [FX] [SYNTH] in: – out: Audio

Phoenix-set source (a first as a synth block): iterates z <- zz + c + pz_prev, a complex map with MEMORY - the next value depends not only on the current z but on the PREVIOUS one too, the extra feedback that grows the curling, bird-like Phoenix fractal. That one-step memory makes its bounded orbits curl and spiral more than a plain Julia set; escapes reseed near the centre. It free-runs at Rate, outputting the real part. A memory-fed complex map distinct from the memoryless Julia, Tricorn and Burning Ship.

∿ **PhotoVibe** [MOD] [FX] [SYNTH] in: Audio out: Audio

Photocell vibe: four all-pass stages at staggered offsets are swept by an asymmetric (fast-rise, slow-fall, photocell-shaped) LFO, recreating the warbling, throbbing modulation of a Uni-Vibe rather than a symmetric phaser.

∿ **Pickover** [MOD] [FX] [SYNTH] in: – out: Audio

Pickover attractor source (a first as a synth block): iterates Clifford Pickover's 3-D map x <- sin(ay) - zcos(bx), y <- zsin(cx) - cos(dy), z <- sin(x), whose orbit is drawn into a delicate, three-dimensional thread-like sculpture of looping filaments. The z-coupling pulls the motion out of the plane, giving a richer wander than the 2-D maps. It free-runs at Rate, outputting x as a smooth bipolar CV. A 3-D threaded attractor distinct from the planar De Jong, Clifford and Svensson maps.

∿ **PisanoSeq** [MOD] [FX] [SYNTH] in: Audio out: Audio

Pisano sequence: the Fibonacci numbers reduced modulo M, which are periodic with the Pisano period pi(M); stepped live, changing the modulus knob switches you between different cyclic patterns of varying length - a tunable periodic CV.

∿ **PitchStair** [MOD] [FX] [SYNTH] in: Audio out: Audio

Pitch staircase: a delay line whose length jumps in discrete steps after each onset, so a held note climbs or sits on a STEPPED pitch ladder rather than a smooth glide - a quantized, robotic pitch staircase. Step sets the size of each tread, Rate how often it steps, Mix the blend.

∿ **PitchVibrato** [MOD] [FX] [SYNTH] in: Audio out: Audio

Sine vibrato: a clean sinusoidal LFO sweeps a fractional delay for smooth, periodic pitch modulation (the controlled, musical counterpart to the random wow/drift blocks).

∿ **PLL** [MOD] [FX] [SYNTH] in: Signal out: Audio

Phase-locked loop (Doepfer A-196): an internal square-wave VCO chases the frequency of the in0 input through an XOR phase comparator and a loop filter, locking onto it (or a multiple) - the lock is never perfect, so it warbles, glitches and screams in the classic PLL way. Base sets the VCO centre, Lock the loop tightness (low = sloppy, unstable, vocal; high = quick, tight tracking), and Range how far it can pull (octaves). Feed it audio or a clock for frequency multiplication, robot voices and chaotic tracking tones.

∿ **Poisson** [MOD] [FX] [SYNTH] in: – out: Audio

Poisson-process trigger (a first as a synth block): fires a one-sample impulse at random, memoryless moments at an average Rate of events per second - the canonical model for radioactive decay clicks, photon arrivals or raindrops. The gaps between triggers follow an exponential distribution, so the pulses are genuinely random in time yet hold a steady long-run density. Patch it to gate or trigger anything that wants un-quantised, never-repeating randomness. Distinct from a regular Clock and from the Bernoulli gate (which only thins an existing clock). Needs no input.

PoissonPulse [MOD] [FX] [SYNTH] in: Audio out: Audio

Poisson pulses: random clicks whose gaps follow an exponential distribution (a Poisson point process at a set mean rate), the model of radioactive decay and shot noise - memoryless random impulses with no underlying periodicity.

PolyaUrn [MOD] [FX] [SYNTH] in: – out: Audio

Polya-urn source (a first as a synth block): the canonical 'rich get richer' process - an urn holds red and black balls; draw one at random, then put it back ALONG WITH another of the same colour, so every draw makes that colour likelier still. The proportion drifts and then locks onto a random value that is different every run (the path-dependent limit). Here the urn is periodically emptied and reseeded, so the output settles to one plateau, holds, then jumps and settles on a fresh one - a self-reinforcing sample-and-hold quite unlike uniform randomness. Rate sets the draw speed. Needs no input.

PrimeCountSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Prime-counting function: pi(n), the running count of primes up to the step index, a staircase that climbs by one at each prime and flattens between them; its slope (the prime density) thins logarithmically per the prime number theorem.

PrimeGapSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Prime gaps: the distances between successive prime numbers (2,1,2,4,2,4,2,4,6,…), an irregular yet not-quite-random integer sequence; clocked out, it makes a jagged melodic/CV pattern that drifts upward as primes thin out.

PrimeIndicator [MOD] [FX] [SYNTH] in: Audio out: Audio

Prime gate: outputs high only when the step index is a prime number, so the gate fires on 2,3,5,7,11,… - an irregular, thinning-with-time rhythmic pattern straight from the primes.

PrimeMod6 [MOD] [FX] [SYNTH] in: Audio out: Audio

Primes mod 6: every prime past 3 is congruent to 1 or 5 (i.e. +/-1) modulo 6; this clocks out that residue of successive primes, a near-balanced two-level stream whose subtle biases (Chebyshev's prime-race) are a topic of active research.

PrimeOmegaSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Big-omega Omega(n): counts the prime factors of the step index with multiplicity (so 12 = 223 gives 3), the additive function at the heart of the Erdos-Kac theorem; clocked out it makes a low, jittery integer staircase.

ProbGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Probability gate: on every step it flips a weighted coin and either passes the audio or mutes it for that step, so a steady part is randomly thinned into an ever-changing rhythm. Bernoulli muting, musical-rate, not the audio-rate chop of a ring chopper. Prob sets the pass likelihood, Rate the step clock, Mix the blend.

∿ **ProbSkip** [MOD] [FX] [SYNTH] in: Audio out: Audio

Probability gate: each incoming trigger is passed through with a set probability and dropped otherwise, thinning a steady clock into a sparser, ever-varying rhythm (deterministic RNG).

∿ **Qi** [MOD] [FX] [SYNTH] in: – out: Audio

Qi-Chen four-wing attractor (a first as a synth block): integrates x' = a(y - x) + yz, y' = cx - y - xz, z' = xy - bz, a system famous for growing FOUR wings instead of the usual two, the trajectory hopping chaotically between all four lobes. The extra quadratic cross-terms give it very wide, energetic swings. It free-runs at Rate, outputting x as a smooth bipolar CV with large excursions. A four-wing chaos distinct from the two-wing Lorenz and Chen attractors.

∿ **QuadLFO** [MOD] [FX] [SYNTH] in: Audio out: Audio

Quadrature LFO (Doepfer A-143-9): one rate, four phase-locked outputs 90 degrees apart. Each instance reads the shared transport clock, so several QuadLFOs at the same Rate stay phase-coherent and Phase (0/90/180/270) picks which quadrature tap this one outputs - patch four for circular/3D panning, stereo movement or evolving cross-modulation. Shape selects sine, triangle, saw or square. Distinct from the free-running LFO (no fixed phase relationship between instances). A trigger into in0 resets the phase (trigger several together to re-lock them); unpatched = free-run.

∿ **QuasiCrystal1D** [MOD] [FX] [SYNTH] in: Audio out: Audio

1-D quasicrystal: the cut-and-project construction that tiles the line with two lengths in a golden, non-repeating order (the 1-D Penrose / Fibonacci quasicrystal), giving aperiodic-yet-ordered points - a Sturmian gate with quasicrystalline structure.

∿ **RadicalSeq** [MOD] [FX] [SYNTH] in: Audio out: Audio

Radical rad(n): the product of the distinct prime factors of the step index (the squarefree kernel, central to the abc conjecture); equal to n for squarefree numbers and much smaller for prime-power-rich ones - a log-scaled arithmetic CV.

∿ **Random** [MOD] [FX] [SYNTH] in: – out: Audio

Self-contained random modulation source: latches a new random value at Rate (internal sample-and-hold of a noise source) with Smooth slewing between steps - from hard random steps to smooth wandering drift. Depth scales the swing and Bias offsets the centre. Needs no input; patch its output into any mod input for evolving, never-repeating movement.

∿ **RandomTelegraph** [MOD] [FX] [SYNTH] in: Audio out: Audio

Random telegraph signal: a two-level (+/-1) signal that flips state at random with a fixed per-sample probability (a symmetric two-state Markov chain), the dichotomous noise behind 1/f flicker and burst noise in electronics - a stochastic square wave.

∿ **RandomWalk** [MOD] [FX] [SYNTH] in: – out: Audio

Random-walk source (a first as a synth block): the classic drunkard's walk - each step nudges the value by a random amount, so it wanders with no memory of where it has been, the very model of Brownian diffusion. It free-runs at Rate; reflecting walls at the rails keep it bouncing inside the range forever. Step sets how far each move can carry it. A memoryless diffusion distinct from the mean-pulled Ornstein-Uhlenbeck and the heavy-tailed Levy flight. Needs no input.

∿ Ratchet [MOD] [FX] [SYNTH] in: Gate out: Audio

Ratchet / retrigger: while the in0 gate is high, replaces the held gate with a fast burst of Count sub-gates at the tempo-synced Rate - the classic ‘stutter/roll’ on a step. Count sets how many retriggers fire per gate, Rate the sub-division (1=1/1 .. 8=1/16T), and Width each sub-gate’s length. Patch a clock or sequencer gate into in0 and the ratcheted gate out into an envelope for rolls and rhythmic fills.

∿ RatchetGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Ratchet gate: each bar the gate retriggers at a clock that ACCELERATES through the bar - even subdivisions at the start rushing into a fast machine-gun stutter by the end, then resetting. The drum-fill ratchet as a rhythmic gate. Rate sets the bar speed, Accel how hard it speeds up, Mix the blend.

∿ RecamanSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Recaman’s sequence: start at 0 and at step n jump back by n if that lands on a new non-negative value, else jump forward by n; the result darts unpredictably over the number line (a famous OEIS visualization) - a wide-ranging stepped melody.

∿ RectTremolo [MOD] [FX] [SYNTH] in: Audio out: Audio

Rectified tremolo: the amplitude is modulated by |sin| rather than a sine, so the volume drops to sharp V-shaped dips twice per LFO cycle (double-time) with a rounded top - a more rhythmic, percussive throb than a smooth sine tremolo.

∿ Renewal [MOD] [FX] [SYNTH] in: – out: Audio

Regularised (Erlang renewal) trigger (a first as a synth block): a point process that is MORE evenly spaced than random - it fires only every Regularity-th pulse of a faster internal Poisson source, so the wait between outputs is an Erlang sum of exponentials. Rate sets the output event rate. At Regularity 1 it is plain Poisson (fully random); raise it and the triggers tighten toward a steady metronome while keeping a little jitter - the sub-Poisson, anti-clustering opposite of Hawkes. The natural model for a neuron’s refractory firing or a heartbeat’s near-regular rhythm. Needs no input.

∿ Repressilator [MOD] [FX] [SYNTH] in: – out: Audio

Repressilator gene oscillator (a first as a synth block): the first synthetic gene circuit built to oscillate - three genes wired in a ring where each represses the next (A silences B, B silences C, C silences A), so no state is stable and the three protein levels cycle round in turn, a genetic clock engineered into living bacteria. Rate sets the speed and Strength the repression (raise it to deepen the swing). The output is the first protein’s level. A synthetic-biology oscillation distinct from the metabolic and neural models.

∿ Ridged [MOD] [FX] [SYNTH] in: – out: Audio

Ridged-multifractal source (a first as a synth block): a fractal noise that takes each octave’s absolute value and inverts it (1 - |noise|), so the smooth valleys of fBm become sharp upward RIDGES - the technique that carves mountain crests, canyon walls and lightning veins in procedural terrain. The output rides high and crinkles into sudden creases instead of rolling smoothly. Rate sets the base speed and Octaves the detail. A ridged, peaky coherent-noise modulator distinct from the rounded fBm and Billow. Needs no input.

∿ Rikitake [MOD] [FX] [SYNTH] in: – out: Audio

Rikitake-dynamo attractor (a first as a synth block): integrates the two-disc Rikitake dynamo x’ = -mux + zy, y’ = -muy + (z - a)x, z’ = 1 - x*y - a model of the Earth’s geomagnetic field built from two coupled current discs. Its chaos is the textbook explanation for why the planet’s magnetic poles REVERSE at irregular intervals: the trajectory orbits one polarity for a while then flips to the other, unpredictably. It free-runs at Rate, outputting x as a smooth bipolar CV with long stretches in one sign broken by sudden reversals - a switching chaos distinct from the steadily-wandering attractors.

RingChorus [MOD] [FX] [SYNTH] in: Audio out: Audio

Ring-modulated chorus: a modulated short delay (the chorus voice) is multiplied by a low carrier before it is mixed back in, so the doubled voice takes on a clangorous inharmonic ring on top of the usual chorus widening.

RingDip [MOD] [FX] [SYNTH] in: Audio out: Audio

Ducking ring-mod: the ring modulation depth is keyed INVERSELY to level, so quiet passages turn clangorous and metallic while loud hits ring through nearly clean - the robot tone emerges only in the gaps. Freq sets the carrier, Depth the maximum metallic-ness, Mix the blend.

RingFB [MOD] [FX] [SYNTH] in: Audio out: Audio

Feedback ring modulator: the carrier sine is phase-modulated by the previous output, so the sidebands recursively breed more sidebands - a self-feeding, FM-flavoured ring mod that grows denser with feedback.

RingFollow [MOD] [FX] [SYNTH] in: Audio out: Audio

Pitch-tracking ring modulator: the carrier frequency follows a ratio of the input's own pitch (from its zero crossings), so the ring-mod sidebands stay harmonically related and in tune instead of clashing.

RingMod [MOD] [FX] [SYNTH] in: Audio, Carrier out: Audio

Ring and amplitude modulator that multiplies the input by a carrier on in 2 (plus Bias). Mode morphs from true ring modulation (bipolar carrier, inharmonic metallic sum/difference tones) to amplitude modulation (unipolar, keeps the original plus sidebands), and Amount sets dry/wet. An audio-rate carrier yields clangorous bell tones; a low-rate carrier becomes tremolo.

RingSelf [MOD] [FX] [SYNTH] in: Audio out: Audio

Self ring modulator: multiplies the signal by a short-delayed copy of itself, producing metallic, pitch-tracking sidebands without an external carrier; delay sets the inharmonic flavour.

RissetRhythm [MOD] [FX] [SYNTH] in: Audio out: Audio

Risset rhythm: several click layers at octave-related tempos crossfade so that as fast layers fade in and slow ones fade out the pulse seems to accelerate (or decelerate) forever - the rhythmic analogue of the Shepard tone, as a gate/CV.

Rose [MOD] [FX] [SYNTH] in: – out: Audio

Rose-curve (rhodonea) LFO (a first as a synth block): traces the petal curve r = cos(Petals * angle) and outputs its horizontal sweep, a smooth modulation whose single cycle folds into a rosette of Petals lobes. The result is a repeating shape far richer than a sine - a flower of evenly-spaced swells and dips - whose symmetry is set directly by the petal count. Rate sets the speed. A geometric multi-lobe LFO distinct from the sine, ramp and random sources.

Rossler [MOD] [FX] [SYNTH] in: – out: Audio

Rossler-attractor chaos source (a first as a synth block): integrates the three Rossler equations dx=-y-z, dy=x+ay, dz=b+z(x-c) - a smooth deterministic-chaos system that traces a spiralling band with an occasional fold. The output is x as a slowly wandering CV: gentler and more periodic-looking than Lorenz, with long quasi-cycles broken by sudden excursions. Rate sets the integration speed. A continuous chaos with a different character from the angular Henon or the tumbling DoublePendulum; needs no input.

RotorDoppler [MOD] [FX] [SYNTH] in: Audio out: Audio

Rotary speaker: one LFO drives both a Doppler pitch sweep (modulated delay) and the amplitude tremolo of a spinning horn at once, the intertwined motion a plain chorus or tremolo can't reproduce.

Rucklidge [MOD] [FX] [SYNTH] in: – out: Audio

Rucklidge attractor (a first as a synth block): integrates x' = -kx + ay - y*z, y' = x, z' = -z + y^2, a model of convection in a fluid layer subject to a magnetic field (double-diffusive convection). It produces a symmetric two-lobed attractor as the rolls of fluid reverse chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV with broad excursions. A convective-magnetic chaos distinct from the rigid-body and reduced-Lorenz flows.

RudinShapiroSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Rudin-Shapiro sequence: +/-1 set by the parity of '11' bit-pairs in the index, the automatic sequence famous for its nearly-flat (white-like) Fourier spectrum despite being fully deterministic - a structured yet broadband stepped CV.

RudinShapiroWalk [MOD] [FX] [SYNTH] in: Audio out: Audio

Rudin-Shapiro walk: the running partial sum of the +/-1 Rudin-Shapiro sequence, a deterministic walk that - unlike a random walk - stays provably within order-sqrt(n) of zero (the source of the Rudin-Shapiro polynomials' flat sup-norm) - a tightly-bounded meandering CV.

RulerSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Ruler sequence: the exponent of the highest power of 2 dividing the step index (the tick heights on an imperial ruler: 0,1,0,2,0,1,0,3,…), a self-similar pattern that doubles its tallest mark at every power of two.

Rulkov [MOD] [FX] [SYNTH] in: – out: Audio

Rulkov-map neuron (a first as a synth block): a two-variable MAP (not a differential equation) that reproduces neuron spiking and bursting with just x' = Alpha/(1 + x^2) + y and a slow y - prized because it captures the rich firing of a real cell at a fraction of the cost. A fast x that spikes sits on a slow y that drifts it in and out of bursting, so raising Alpha moves it from silence through tonic spikes into chaotic bursts. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A discrete-map neuron distinct from the integrated Hindmarsh-Rose and FitzHugh-Nagumo models.

Rungler [MOD] [FX] [SYNTH] in: – out: Audio

Rungler (Rob Hordijk): a shift register clocked at Rate is fed a pseudo-random bit and its low bits are read out as a stepped, looping-but-evolving voltage staircase - the gurgling, semi-random heart of the Benjolin. Length sets the loop length / repeat character. Needs no input.

∿ SampleEntropy [MOD] [FX] [SYNTH] in: Audio out: Audio

Sample entropy: counts how many length-2 sub-sequences in a 16-sample window stay within Tolerance of each other and how many of those still match when extended to length 3, returning -log(A/B) - low for regular, repeating signals and high for irregular, unpredictable ones. A self-match-free predictability measure (more robust than approximate entropy). Tolerance sets the matching radius.

∿ SampleHold [MOD] [FX] [SYNTH] in: Audio out: Audio

Clocked sample-and-hold that latches the input value at Rate and holds it until the next tick, producing stepped random or quantized control. Glide slews between steps for a smoother, portamento-like contour. Feed it noise for classic random sample-and-hold modulation, or an audio source to quantize it in time.

∿ SaturateChorus [MOD] [FX] [SYNTH] in: Audio out: Audio

Saturated chorus: the input is tanh-saturated first, then doubled by a modulated short delay, so the chorus voice carries the same harmonic grit as the dry - a thicker, dirtier widening than a clean chorus, the order (drive-then-chorus) giving consistent colour across both layers.

∿ Scan [MOD] [FX] [SYNTH] in: A, B, C, D out: Audio

Scanning four-source mixer: a read position sweeps across in 1..in 4 and crossfades between adjacent sources, wrapping 4 back to 1 for a continuous loop. Pos sets the position by hand; Rate above zero auto-scans the position and loops it; Smooth rounds the crossfade from linear to an S-curve. Feed it four oscillators or signals and scan or automate to morph between them - the sequential companion to the bilinear Vector mixer.

∿ Schelling [MOD] [FX] [SYNTH] in: – out: Audio

Schelling-segregation source (a first as a synth block): models twelve agents of two kinds on a ring; an agent is content only if enough of its neighbours are of its OWN kind, and a discontented one swaps places with a random other - Schelling's Nobel-winning demonstration that even a mild preference for similar neighbours drives a population into sharply segregated blocks. A little random relocation keeps it stirring. The output is the segregation index (how clumped the two kinds are), climbing as the blocks form. A self-sorting source distinct from the opinion models. Needs no input.

∿ SEIR [MOD] [FX] [SYNTH] in: – out: Audio

SEIR epidemic source (a first as a synth block): the Susceptible-Exposed-Infected-Recovered model, which adds a LATENT period - the newly-infected are Exposed but not yet contagious for a while before they become Infectious, the more realistic model used for measles and COVID. That incubation delay slows and stretches the epidemic wave and makes its peaks lag behind the cause. The output is the infected fraction, a delayed rising-and-falling curve. R0 sets the contagiousness, Season the seasonal swing. A latent-period epidemic distinct from the instantaneous SIR. Needs no input.

∿ SEIRS [MOD] [FX] [SYNTH] in: – out: Audio

SEIRS epidemic source (a first as a synth block): the full Susceptible-Exposed-Infected-Recovered-Susceptible cycle, combining BOTH a latent incubation period AND waning immunity - the most complete of the classic compartment models, used for endemic seasonal diseases. The exposed delay and the loss of immunity together produce rich, sustained, sometimes multi-year oscillation cycles in the infection level. The output is the infected fraction, oscillating with a layered rhythm. R0 sets the contagiousness, Season the seasonal forcing. The full-cycle epidemic distinct from the simpler SIR, SIRS and SEIR. Needs no input.

SelfNumberSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Self-number gate: high when the step index is a self (Colombian) number - one that cannot be written as some smaller number plus that number's digit sum (1,3,5,7,9,20,31,…) - a sparse, self-referential arithmetic gate.

Selkov [MOD] [FX] [SYNTH] in: – out: Audio

Sel'kov glycolytic oscillator (a first as a synth block): the model of glycolysis - the pathway cells use to burn sugar - which does not proceed steadily but PULSES, the concentration of an intermediate rising and falling in a self-sustained rhythm (the glycolytic oscillations seen in yeast extract). Rate sets the speed and Feed the substrate supply. The output is the oscillating concentration. A metabolic oscillation distinct from the Brusselator chemistry and the neuron models.

ShiftMelody [MOD] [FX] [SYNTH] in: Audio out: Audio

Shift-register melody: a looping bit register clocks out a stepped CV; a probability knob occasionally flips the recirculating bit, morphing a locked loop into a slowly-mutating one - the Turing-machine sequencer in a block.

ShimizuMorioka [MOD] [FX] [SYNTH] in: – out: Audio

Shimizu-Morioka attractor (a first as a synth block): integrates x' = y, y' = x - ay - xz, z' = -b*z + x^2, a simplified model of the Lorenz system near the onset of chaos. It shows a symmetric butterfly that is born through a sequence of bifurcations as the parameters change, a clean laboratory for studying how order gives way to chaos. It free-runs at Rate, outputting x as a smooth bipolar CV that flips between the two wings. A reduced-Lorenz chaos distinct from the full Lorenz attractor.

SideLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Sidechain-curve mod source: outputs the classic pump/duck envelope as a control signal (no audio through-path) so one ducking shape drives many destinations - patch it into a Mixer level, a reverb/delay send or a filter cutoff to sidechain them all to the same beat without a key signal. Rate sets the duck cycle in Hz (modulate or tempo-sync it to the kick), Depth how far the curve dips, Curve shapes the recovery ramp from snappy to gradual, and Bias offsets the output centre. Shares the Pump curve, exposed as CV instead of an inline gain. A trigger into in0 resets the duck cycle (sync it to the kick); unpatched = free-run.

SierpinskiArrowhead [MOD] [FX] [SYNTH] in: Audio out: Audio

Sierpinski arrowhead: the turn sequence of the space-filling curve that traces out the Sierpinski triangle, a self-similar left/right turn pattern (derived from the ruler-like 2-adic structure of the index) - a fractal gate rhythm.

SineMap [MOD] [FX] [SYNTH] in: – out: Audio

Sine-map source (a first as a synth block): iterates x <- Rsin(pix), the trigonometric twin of the logistic map - a single smooth hump that, as R approaches 1, period-doubles its way into chaos by the very same Feigenbaum route, which is why it is a favourite engine for chaotic encryption. It free-runs at Rate, stepping the map and outputting x as a 0..1 CV. R sets the position on the road to chaos: low and it settles, near 1 and it never repeats. A sinusoidal unimodal map distinct from the polynomial logistic. Needs no input.

SIR [MOD] [FX] [SYNTH] in: – out: Audio

SIR epidemic source (a first as a synth block): integrates the textbook Susceptible-Infected-Recovered model - the equations Kermack and McKendrick wrote in 1927 that underlie every epidemic forecast. A pathogen burns through the susceptible population in a wave that peaks and fades as the recovered build up herd immunity; births slowly refill the susceptible pool and a seasonal swing in contagion keeps the waves coming. The output is the infected fraction, a rising-and-falling epidemic curve. R0 sets the basic reproduction number (how contagious), Season the seasonal swing. A disease-dynamics source distinct from the chemical and predator-prey oscillators. Needs no input.

∿ SIRS [MOD] [FX] [SYNTH] in: – out: Audio

SIRS epidemic source (a first as a synth block): the Susceptible-Infected-Recovered-Susceptible model, where immunity WANES - the recovered slowly become susceptible again and can be reinfected, so instead of a single epidemic the disease settles into recurring waves, the dynamics behind seasonal flu and the common cold. The output is the infected fraction, oscillating endlessly as immunity builds and fades. R0 sets the contagiousness, Season the seasonal forcing. A waning-immunity epidemic distinct from the one-shot SIR. Needs no input.

∿ SIS [MOD] [FX] [SYNTH] in: – out: Audio

SIS epidemic source (a first as a synth block): the Susceptible-Infected-Susceptible model, where there is NO immunity at all - the infected recover straight back to susceptible and can immediately catch it again, the model for diseases like the common cold or many bacterial infections that confer no lasting protection. Above its threshold the disease never dies out but stays endemic at a high steady level, breathing up and down with the seasons. The output is the infected fraction, hovering high. R0 sets the contagiousness, Season the swing. A no-immunity epidemic distinct from the SIR and SIRS. Needs no input.

∿ SkellamNoise [MOD] [FX] [SYNTH] in: Audio out: Audio

Skellam noise: the signed difference of two independent Poisson counts, the integer distribution of net change when arrivals and departures both happen randomly (goal differences, net votes) - a discrete, signed, mean-zero stepped source.

∿ SkewLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Variable-skew LFO: a triangle whose rising/falling balance is continuously adjustable, morphing from a ramp-down through a symmetric triangle to a ramp-up - one knob covers saw, triangle and reverse-saw modulation shapes. Rate sets the speed, Skew the rise/fall ratio, Depth the output level. A trigger into in0 resets the phase; unpatched = free-run.

∿ SkewnessDetector [MOD] [FX] [SYNTH] in: Audio out: Audio

Skewness detector: computes the 3rd standardized moment of a 7-sample window, which is zero for a symmetric waveform and swings positive or negative when the local shape leans one way - so it tracks waveform asymmetry and even-harmonic / DC-offset character. A different statistical feature from kurtosis: lopsidedness rather than peakedness. Useful as a modulation source.

∿ Slew [MOD] [FX] [SYNTH] in: Audio out: Audio

Slew-rate limiter that caps how fast the signal can rise and fall, with independent Rise and Fall rates. Use it as portamento/glide on pitch CV, a smoother on stepped modulation, or a lag to de-zipper control changes. Asymmetric Rise/Fall gives simple envelope-like attack and decay shaping.

∿ SlewNoise [MOD] [FX] [SYNTH] in: – out: Audio

Slew-limited random LFO: picks a new random target at a set rate and glides toward it with a slew time, producing smooth, organic, non-repeating modulation (a continuous cousin of sample-and-hold).

SlewSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Slewed step sequencer: a fixed-length loop of deterministic pseudo-random step values glided together by a slew limiter, a repeating melodic/CV contour that you can smooth from stepped to legato.

SlideUp [MOD] [FX] [SYNTH] in: Audio out: Audio

Slide up: each onset starts the pitch below target and slides up into place as a shrinking delay closes the gap - the scoop-up / fall-up portamento into every note. Time sets how long the slide takes, Depth how far below it starts, Mix the blend.

SmoothNumberSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Smooth-number gate: high when the step index's largest prime factor does not exceed the bound B (a B-smooth number), the property that makes integers easy to factor and underlies sieve algorithms; raising B opens the gate on more indices.

SmoothRandom [MOD] [FX] [SYNTH] in: – out: Audio

Smooth random modulation: latches a new random target at Rate and continuously glides toward it, for organic wandering motion with no steps. Depth scales the swing. The smooth cousin of a sample-and-hold random - free-running, needs no clock.

Spirograph [MOD] [FX] [SYNTH] in: – out: Audio

Spirograph (hypotrochoid) LFO (a first as a synth block): traces the looping curve a small circle draws as it rolls inside a larger one - the toy-spirograph pattern - and outputs its sweep, the sum of two circular motions whose speeds are set by Ratio. The result is an epicyclic LFO with a fast inner wobble riding on a slow outer swing, Depth balancing the two. A geometric two-rate modulation distinct from a plain sine or a synced pair. Rate sets the speed.

SprottB [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-B attractor (a first as a synth block): integrates x' = yz, y' = x - y, z' = 1 - xy, one of the algebraically SIMPLEST possible chaotic systems - J.C. Sprott found, by computer search, the minimal sets of terms that still produce chaos, and this is among them. Despite having only five terms it folds a genuine strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-form chaos distinct from the larger Lorenz and Chen systems. Needs no input.

SprottC [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-C attractor (a first as a synth block): integrates x' = yz, y' = x - y, z' = 1 - x^2, a sibling of the Sprott-B system with the cross-term xy swapped for the square x^2 - another of Sprott's minimal five-term chaotic flows. The small change reshapes the attractor into its own distinct double-lobed fold. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal quadratic chaos distinct from Sprott-B and the classical attractors. Needs no input.

SprottE [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-E attractor (a first as a synth block): integrates x' = y*z, y' = x^2 - y, z' = 1 - 4x, another of Sprott's algebraically-minimal chaotic systems, with a square nonlinearity in the y equation and a linear ramp in z. Its orbit winds through a compact, tightly-folded strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-form chaos distinct from the Sprott-B and Sprott-C variants. Needs no input.

SprottG [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-G attractor (a first as a synth block): integrates x' = 0.4x + z, y' = xz - y, z' = -x + y, a minimal chaotic flow with a small linear self-feedback that lets it spiral outward before the nonlinear term folds it back. Its attractor is a delicate scrolled spiral. It free-runs at Rate, outputting x as a smooth bipolar CV that winds and folds. A spiral minimal chaos distinct from the other Sprott systems. Needs no input.

SprottJ [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-J attractor (a first as a synth block): integrates x' = 2z, y' = -2y + z, z' = -x + y + y^2 - one of the 19 algebraically SIMPLEST chaotic flows that Julien Sprott found by computer search, each with the bare minimum of terms needed for chaos. Case J folds a single quadratic term into an otherwise linear system. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-quadratic chaos distinct from the elaborate named attractors.

SprottK [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-K attractor (a first as a synth block): integrates x' = xy - z, y' = x - y, z' = x + 0.3z - another of Sprott's minimal chaotic systems, this one driven by the single product term x*y. Despite having only one nonlinearity it produces a fully chaotic orbit. It free-runs at Rate, outputting x as a smooth bipolar CV. A single-product chaos distinct from the other minimal Sprott cases.

SprottN [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-N attractor (a first as a synth block): integrates x' = -2y, y' = x + z^2, z' = 1 + y - 2z - a minimal chaotic flow whose only nonlinearity is the squared z term. It traces a folded sheet that the orbit wanders chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV. A squared-term minimal chaos distinct from the product-driven Sprott cases.

SprottR [MOD] [FX] [SYNTH] in: – out: Audio

Sprott-R attractor (a first as a synth block): integrates x' = 0.9 - y, y' = 0.4 + z, z' = xy - z - a minimal chaotic system with constant forcing terms and a single product nonlinearity. The constants push the orbit steadily while the xy coupling folds it into chaos. It free-runs at Rate, outputting x as a smooth bipolar CV. A forced minimal chaos distinct from the unforced Sprott cases.

SquareFreeSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Squarefree gate: high when the step index has no repeated prime factor (1,2,3,5,6,7,10,…), the indicator of the squarefree numbers (density 6/pi^2) - a sieve-flavoured aperiodic gate.

Stages [MOD] [FX] [SYNTH] in: Trigger out: Audio

Segment generator (Mutable Instruments Stages): builds a function from a chain of Segs segments, each ramping to a new level over Time. Shape morphs every segment from a hard step, through a linear ramp, to a smooth S-curve; Loop makes the chain cycle as an evolving LFO, or one-shot it from a gate at in0 as a complex envelope. The flexible CV building block between an envelope, an LFO and a slow sequencer. Distinct from the breakpoint MSEG.

Standard [MOD] [FX] [SYNTH] in: – out: Audio

Chirikov standard-map source (a first as a synth block): the kicked-rotor map p <- p + K*sin(theta), theta <- theta + p on a torus - the textbook model of Hamiltonian chaos and the onset of turbulence. Below the critical kick the motion is smooth and quasi-periodic; above it (raise K) the phase space dissolves into a chaotic sea. It free-runs at Rate, stepping the map and outputting sin(theta) as a bipolar CV - regular and rolling at low K, wildly scattered at high K. A momentum-kicked chaos distinct from the dissipative attractors.

StandardMap [MOD] [FX] [SYNTH] in: – out: Audio

Chirikov standard-map chaos: iterates the area-preserving kicked-rotor map (p += K*sin(theta); theta += p) at Rate. At low K the output is smooth and quasi-periodic; past K~1 it breaks into hard chaos - a tunable bridge from orderly to wild stepped modulation. Needs no input.

StanleySeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Stanley sequence: the greedy set containing no three-term arithmetic progression (0,1,3,4,9,10,12,13,…), which turns out to be exactly the integers whose base-3 representation uses only 0s and 1s - a Cantor-set-flavoured stepped pattern.

SteppedLFO [MOD] [FX] [SYNTH] in: Audio out: Audio

Stepped LFO: a ramp quantized to a chosen number of discrete levels, producing a rising (or shaped) staircase control voltage for arpeggiated, sequenced-feeling modulation rather than a smooth sweep.

SteppedRandom [MOD] [FX] [SYNTH] in: Clock out: Audio

Stepped random (sample & hold): on each rising edge of the in0 clock it latches a new random voltage and holds it until the next clock - the hard-stepped random for melodic and rhythmic modulation. Range scales the swing.

Stepper [MOD] [FX] [SYNTH] in: – out: Audio

Tempo-synced step-sequencer modulator: steps through S1..S5 levels locked to the host tempo, with Glide smoothing the jumps from hard sample-and-hold steps to a slewed contour. Sync sets the per-step note division (1/1..1/16T), Steps how many levels are used, and Glide the slew time. Per-voice (each instance runs independently); patch it into cutoff, pitch or level for rhythmic, tempo-locked movement. Distinct from StepSeq, which free-runs in Hz with no glide or sync.

StepSeq [MOD] [FX] [SYNTH] in: – out: Audio

Step sequencer modulation source: cycles through up to 6 step levels (S1-S6) at Rate steps per second, outputting the held value of the current step. Use it as a rhythmic mod source for cutoff, pitch or level, or to chop gain into patterns. Modulate Rate to lock to tempo; see TranceGate for a simple on/off duty gate.

SternBrocotSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Stern diatomic sequence: the ‘fusc’ function that builds the Stern-Brocot tree of all rationals (a(2n)=a(n), a(2n+1)=a(n)+a(n+1)), giving a fractal sawtooth-of-sawtooths whose every value is a rational’s numerator - structured self-similar CV.

SumOfDivisorsSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Divisor sum sigma(n): the total of all divisors of the step index (1+2+3+6=12 for n=6), the multiplicative function whose comparison to 2n defines perfect/abundant numbers; log-scaled it makes a lumpy arithmetic CV that peaks on highly-composite indices.

~ SumOfSquaresSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Sum-of-two-squares count: r2(n), how many ways the step index is a sum of two (signed, ordered) squares, governed by its primes congruent to 1 vs 3 mod 4 (Gauss circle problem); a sparse pattern that is zero for many indices.

~ Superformula [MOD] [FX] [SYNTH] in: – out: Audio

Superformula LFO (a first as a synth block): Johan Gielis's single equation that draws a vast family of natural shapes - circles, polygons, stars, flowers, leaves - from just two knobs, by setting the radius from |cos(Symt/4)| and |sin(Symt/4)| raised to a power. Symmetry sets how many lobes (the polygon/star order) and Shape how sharp or rounded they are, so sweeping them MORPHS the LFO contour continuously from a smooth blob to a spiky star. The most shape-versatile single LFO, distinct from every fixed-curve source. Rate sets the speed.

~ Svensson [MOD] [FX] [SYNTH] in: – out: Audio

Svensson attractor source (a first as a synth block): iterates Johnny Svensson's map x <- Dsin(Ax) - sin(By), y <- Ccos(Ax) + cos(By), a sine-and-cosine attractor that draws tight rosette and starburst figures with sharp radial symmetry. The mix of large and small coefficients pulls the orbit into spiked petals quite unlike the woven webs of De Jong. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A radial-symmetry attractor distinct from the other trigonometric maps. Needs no input.

~ Swarm [MOD] [FX] [SYNTH] in: – out: Audio

Flocking modulation source (a first as a synth block): a swarm of six agents drift under boids rules - steering toward the flock centre (Cohesion), away from crowding neighbours (Separation), toward the average heading, plus a little wander - and the output is the swarm's collective centre as a slowly evolving, organic CV. Rate sets the motion speed. Unlike an LFO (a fixed shape) or the memoryless random of Marbles/Wogglebug, the swarm self-organises: it flocks, splits and regroups, moving smoothly but never repeating. Needs no input.

~ SwitcherLFO [MOD] [FX] [SYNTH] in: – out: Audio

Massive X morphing LFO: runs one phase at Rate through a bank of 8 bipolar shapes (sine, triangle, saw up/down, square, narrow pulse, s-curve, double sine), crossfading between adjacent shapes and mapping the result to a 0 to 1 output. Rate sets the cycle speed, Shape selects and morphs continuously across the bank, and Phase offsets the read position. Sweeping Shape gives a continuous waveform-morph modulator rather than a fixed LFO shape.

~ TapeWow [MOD] [FX] [SYNTH] in: Audio out: Audio

Tape wow / flutter: a slow LFO sweeps a fractional delay so pitch wobbles like a slipping tape transport - the analog instability a clean digital path lacks.

~ TeagerEnergy [MOD] [FX] [SYNTH] in: Audio out: Audio

Teager-Kaiser energy operator: computes psi(x)=x[n-1]^2 - x[n-2]*x[n], which tracks the instantaneous energy of an oscillation (proportional to amplitude^2 times frequency^2) using only three samples - it spikes on transients and follows combined amplitude/frequency modulation far faster than an RMS detector. Gain scales the output. Adds 1 sample latency.

~ Telegraph [MOD] [FX] [SYNTH] in: – out: Audio

Random-telegraph-signal source (a first as a synth block): the dichotomous noise of physics - a two-level signal that flips between 0 and 1 at random (memoryless) moments, the model for a trapped electron blinking or a bistable circuit jittering. Rate sets the mean number of flips per second; because the wait between flips is exponentially distributed the result is a random square wave with no fixed period, holding each level for an unpredictable time. A genuine stochastic gate, distinct from the random-but-clocked Sample-and-Hold sources. Needs no input.

TelegraphProcess [MOD] [FX] [SYNTH] in: – out: Audio

Random-telegraph source (a first as a synth block): a two-state signal that sits at one rail and then randomly flips to the other, the dichotomous noise that models a flickering switch, burst noise in electronics, and ion channels snapping open and shut. The hold times are exponentially distributed, so it is a stepped square wave with utterly unpredictable timing. Switch sets the flip probability per step (the average rate of jumps). A two-level jump process distinct from the continuous random walks. Needs no input.

Thomas [MOD] [FX] [SYNTH] in: – out: Audio

Thomas cyclically-symmetric attractor (a first as a synth block): integrates x' = sin(y) - b*x and its two cyclic rotations - a flow built entirely from sines, completely symmetric under swapping x->y->z. As the damping b is lowered the trajectory unwinds from a simple loop into a labyrinthine chaotic walk that drifts through space like a particle in a turbulent crystal lattice. It free-runs at Rate, outputting x as a smooth, gently-wandering bipolar CV. A trig-flow chaos distinct from the polynomial Lorenz or Chua systems.

ThreeScroll [MOD] [FX] [SYNTH] in: – out: Audio

Three-scroll attractor (a first as a synth block): integrates the Pan-Xu-Zhou system x' = a(y - x) + dxz, y' = cy - xz, z' = bz + xy - exx, a unified chaotic flow whose trajectory weaves through THREE separate scroll centres instead of the usual two, hopping between all three lobes. It free-runs at Rate, outputting x as a smooth bipolar CV with wide swings. A three-lobe chaos distinct from the two-wing Lorenz and four-wing Qi attractors.

ThruZero [MOD] [FX] [SYNTH] in: Audio out: Audio

Through-zero flanger: two delay taps sweep symmetrically about a short centre delay - one lengthening while the other shortens - so as the LFO crosses zero the taps pass through each other, producing the deep notch that sweeps all the way down to DC and momentarily cancels, the unmistakable tape-flanger 'through-zero' sound a normal flanger (with its fixed minimum delay) can't make. Rate sets the sweep speed, Depth the excursion, Feedback the resonance, and Mix the wet blend.

ThruZeroFlange [MOD] [FX] [SYNTH] in: Audio out: Audio

Through-zero flanger: a wet delay is modulated against a fixed dry delay so their difference passes through (and beyond) zero, producing the deep tape-flange null sweep and the brief reversal that only true through-zero flanging gives.

ThueMorseSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Thue-Morse sequence: each step's value is the parity of the number of 1-bits in the step index, the famously cube-free, self-similar, aperiodic binary sequence - a non-repeating yet highly-structured gate/CV pattern.

ThueMorseTernary [MOD] [FX] [SYNTH] in: Audio out: Audio

Ternary Thue-Morse: the sum of the base-3 digits of the index taken mod 3, a three-symbol automatic sequence generalizing the binary Thue-Morse, giving a self-similar three-level stepped CV that avoids short repetitions.

Tides [MOD] [FX] [SYNTH] in: – out: Audio

Tidal modulator (Mutable Instruments Tides): a slope source whose shape morphs continuously. Rate sets the speed (LFO into audio range); Slope skews the ramp from a fast-rise/slow-fall sawtooth through a symmetric triangle to slow-rise/fast-fall; Smooth rounds the contour from a sharp triangle toward a smooth sine swell. Use it as an evolving LFO, an envelope, or a raw oscillator. Distinct from the stepped MSEG and the fixed LFO shapes.

Tinkerbell [MOD] [FX] [SYNTH] in: – out: Audio

Tinkerbell-map source (a first as a synth block): iterates x <- x^2 - y^2 + Ax + By, y <- 2xy + Cx + Dy - a quadratic map of the plane whose orbit traces a comma-shaped fractal said to resemble Tinkerbell's flight path. Its mix of squaring and cross-coupling folds the trajectory into a tight curl with delicate internal banding. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A quadratic-plane attractor distinct from the trig-based De Jong and Clifford. Needs no input.

Toda [MOD] [FX] [SYNTH] in: – out: Audio

Toda-lattice source (a first as a synth block): a ring of sixteen masses joined by springs whose force grows EXPONENTIALLY with compression - the Toda lattice, one of the rare nonlinear systems that is exactly solvable and supports SOLITONS, lumps of energy that travel through the chain and pass through each other unchanged. Kick it and solitons circulate the ring forever, colliding and re-emerging. The output is one mass's momentum, kicking each time a soliton passes through. An integrable-soliton source distinct from the chaotic lattices. Needs no input.

TotientSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Euler totient: phi(n), the count of integers up to n coprime to it, an arithmetic function full of multiplicative structure; normalized by n it hovers and dips in a fractal-looking pattern, here a stepped control source.

TranceGate [MOD] [FX] [SYNTH] in: Audio out: Audio

Rhythmic hard gate: an LFO at Rate chops the signal on and off with a Duty cycle, smoothed by Smooth and scaled by Depth. Modulate Rate to lock it to tempo for the classic trance/EDM stutter. See StepSeq for arbitrary per-step patterns rather than a fixed duty.

Tremble [MOD] [FX] [SYNTH] in: Audio out: Audio

Tremble: a nervous, fluttering tremolo - the amplitude jitters on a fast smoothed-random envelope rather than a steady LFO, so the sound shivers and trembles unevenly, like a held note played by a shaking hand. Rate sets the flutter speed, Depth the shake, Mix the blend.

TremHarmonic [MOD] [FX] [SYNTH] in: Audio out: Audio

Harmonic tremolo: the signal is split into low and high bands whose amplitudes are modulated in anti-phase by one LFO, producing the swirling, phase-shifting throb of a brownface amp tremolo rather than flat volume wobble.

Tremolo [MOD] [FX] [SYNTH] in: – out: Audio

Amplitude tremolo: an LFO at Rate modulates the signal level by Depth, switchable between a smooth sine and a choppy square (Shape). Classic for rhythmic volume pulsing, surf-guitar wobble and rotary-style throb. At full Depth with the square shape it becomes a hard on/off gate.

TribonacciSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Tribonacci sequence: T(n)=T(n-1)+T(n-2)+T(n-3), the three-term generalization of Fibonacci whose growth ratio is the tribonacci constant; reduced modulo a base it gives a faster-climbing, busier melodic pattern than Fibonacci.

TribonacciWordSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Tribonacci word: the fixed point of the substitution a->ab, b->ac, c->a, a three-letter aperiodic sequence whose letter frequencies follow the tribonacci constant (the 3-symbol analogue of the Fibonacci word) - a structured three-level CV.

TriceraChorus [MOD] [FX] [SYNTH] in: Audio out: Audio

Tri-voice chorus with a micro-detune swirl: three LFO-modulated delay taps at spread phases plus a slow pitch-detuned voice for a rich animated widening.

Tricorn [MOD] [FX] [SYNTH] in: — out: Audio

Tricorn (Mandelbar) source (a first as a synth block): iterates z <- conj(z)^2 + c, the same squaring as a Julia set but on the CONJUGATE of z (flipping the sign of its imaginary part each step). That conjugation gives the fractal a three-cornered symmetry and makes its boundary fractal in a different way, so the bounded orbits twist with a distinctive three-fold pull; escapes reseed near the centre. CReal and CImag pick the region. It free-runs at Rate, outputting the real part. A conjugate complex map distinct from the Julia and Burning Ship sources.

TrigDelay [MOD] [FX] [SYNTH] in: Audio out: Audio

Trigger delay: outputs the gate plus a time-shifted copy of itself, so one hit becomes a flam (two close hits) or a rhythmic echo of triggers - timing displacement for drums and envelopes.

TripleCorrelation [MOD] [FX] [SYNTH] in: Audio out: Audio

Triple correlation: a leaky average of the triple product x[n]x[n-1]x[n-2], the third-order autocorrelation at lags 1 and 2 - unlike variance (lag-zero, 2nd order) or skewness it captures lagged, time-asymmetric structure and quadratic phase coupling, staying near zero for symmetric/Gaussian signals and growing for waveforms with consistent directional shape. A higher-order-statistics modulation source.

Turing [MOD] [FX] [SYNTH] in: Clock out: Audio

Turing-machine sequencer (looping shift register): on each rising edge of the in0 clock it shifts an internal register of step values and feeds the end back to the front - but with probability set by Lock it injects a fresh random value instead, so Lock fully up gives a locked repeating loop, fully down a new random sequence each pass, and in between a slowly mutating pattern. Length sets the loop length (2-16 steps), Lock the lock/mutation amount, Range the CV span, and Quantize snaps each step to a chromatic grid. The classic generative Eurorack CV source - patch its output into an oscillator pitch or any mod input.

TuringMachineSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Turing-machine sequencer: a 16-bit loop is rotated each step and the recirculating bit is flipped with an adjustable probability - at zero the loop locks into a repeating melody, at one-half it is fully random, and in between it slowly mutates; a window of bits forms the CV.

TZFlanger [MOD] [FX] [SYNTH] in: Audio out: Audio

Through-zero flanger that reads two short delay taps swept in opposite directions by a cosine LFO, so the offsets cross and produce a deep tape-style null. Rate sets the LFO speed, Depth scales the sweep range, Feedback (up to 0.95) regenerates one tap for sharper resonance, and Mix sets dry/wet. The counter-swept design gives the dramatic zero-crossing whoosh that fixed flangers cannot reach.

∿ UlamSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Ulam sequence: 1,2,3,4,6,8,11,13,… where each new term is the smallest integer that is the sum of two distinct earlier terms in exactly one way; mysteriously near-periodic in its density, it makes an intriguing climbing stepped CV.

∿ Undulate [MOD] [FX] [SYNTH] in: Audio out: Audio

Undulate: amplitude and timbre roll together on one slow LFO - as the volume swells the tone opens and brightens, as it dips the tone closes and darkens - so the sound undulates like a wave rather than just pulsing. Rate sets the swell speed, Depth the range, Mix the blend.

∿ Undulator [MOD] [FX] [SYNTH] in: Audio out: Audio

Modulated short delay combined with an envelope swell and a tremolo running at twice the LFO rate for H3000-style undulation. Rate sets the LFO speed, Depth sets how far the delay time is swept for pitch-wobble and chorus, and Swell controls the tremolo depth driven from a slow input follower. Mix blends the moving, detuned voice against the dry signal for shimmering, breathing modulation.

∿ UnitaryDivisorSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Unitary divisor sum: sigma*(n) adds only the divisors d of the step index that are coprime to n/d (the unitary divisors), the product of (p^a + 1) over prime powers; a multiplicative function distinct from the ordinary divisor sum.

∿ VanDerCorputSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Van der Corput sequence: each step's value is the index's binary digits mirrored after the point (the base-2 radical inverse), which fills the unit interval by repeated bisection - the 1-D foundation of quasi-Monte-Carlo low-discrepancy sampling.

∿ VanDerPol [MOD] [FX] [SYNTH] in: — out: Audio

Van-der-Pol relaxation oscillator (a first as a synth block): integrates x'' - Mu(1 - x^2)x' + x = 0, the classic self-sustaining oscillator from early radio valves and the heartbeat. Unlike a sine it has nonlinear damping that pumps energy in when the swing is small and bleeds it off when large, so it settles into a fixed-amplitude limit cycle. At low Mu it is nearly sinusoidal; raise Mu and it snaps into a sharp relaxation wave - slow charge, sudden flip - the sawtooth-like pulse of a blinking neon or a firing neuron. Rate sets the speed. A self-stabilising oscillator distinct from the chaotic attractors.

∿ Vector [MOD] [FX] [SYNTH] in: A, B, C, D out: Audio

Bilinear XY vector mixer: blends four inputs across an X/Y plane (in 1/in 2 the top corners, in 3/in 4 the bottom), with X and Y setting the morph position. Classic vector synthesis (Prophet VS / Wavestation) - feed it four oscillators or sources and automate X/Y to morph between them.

∿ Vibe [MOD] [FX] [SYNTH] in: Audio out: Audio

Photocell-style vibe that runs the signal through four staggered first-order all-pass stages whose coefficients are swept by a cosine LFO, each stage offset slightly to mimic mismatched optical cells. Rate sets the swirl speed, Depth scales how far the all-pass coefficients move, and Mix blends the phase-shifted output against the dry signal. The uneven staging gives the watery, throbbing swirl associated with classic vibe pedals.

VibratoFold [MOD] [FX] [SYNTH] in: Audio out: Audio

Folded vibrato: the signal is pitch-modulated by a short swept delay (vibrato) and then wavefolded, so the fold harmonics shift in pitch with the vibrato sweep - a warbling, animated brightness on top of the pitch waver.

Vicsek [MOD] [FX] [SYNTH] in: – out: Audio

Vicsek-flocking source (a first as a synth block): models twelve self-propelled particles on a ring, each steering its heading toward the AVERAGE direction of its neighbours plus a dash of random Noise - the minimal model of how birds flock, fish school and bacteria swarm. At low noise they all line up and move as one (high order); raise the noise past a critical point and the flock shatters into disorder. The output is the order parameter (how aligned the flock is), which hovers and flickers near the flocking transition. A collective-motion source distinct from the single-body chaos. Needs no input.

VocoderBank [MOD] [FX] [SYNTH] in: In A, In B out: Audio

Channel vocoder: a modulator (e.g. voice) is split into four bands whose envelopes gate the matching bands of a carrier, stamping the modulator’s moving spectrum onto the carrier - the classic robot-voice/talking-synth effect.

Voter [MOD] [FX] [SYNTH] in: – out: Audio

Voter-model consensus source (a first as a synth block): a 4x4 grid of cells each holding one of two opinions; each step a random cell simply copies a random neighbour, the simplest model of imitation, gossip and herd behaviour. Domains coarsen and fight along their borders until - on a finite grid - one opinion wins and the whole field falls into consensus, at which point it freezes. The output is the net opinion, a CV that wanders then locks. Unlike the Ising model there is no temperature, only blind copying. Needs no input.

Walk [MOD] [FX] [SYNTH] in: Clock out: Audio

Random walk (drunk): on each rising edge of the in0 clock the output takes a random step of up to +/- Step, bounded to 0..1 - a wandering melodic/modulation CV that meanders rather than jumping, the classic ‘drunk’ generator. Hold the clock high-rate for a smooth wander, slow for stepped motion.

WangSun [MOD] [FX] [SYNTH] in: – out: Audio

Wang-Sun attractor (a first as a synth block): integrates a four-term quadratic system x’ = ax + cyz, y’ = bx + dy - xz, z’ = ez + fx*y with an unusual mix of small and large coefficients, one of the gallery of distinct chaotic flows catalogued in the search for new attractor shapes. Its orbit folds into a lopsided, asymmetric scroll. It free-runs at Rate, outputting x as a smooth bipolar CV. A custom-coefficient chaos distinct from the named classical attractors.

WeylSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Weyl sequence: adds a chosen irrational increment each step and wraps; by Weyl’s equidistribution theorem the values fill the range uniformly without ever repeating - a tunable quasi-periodic CV whose step size sets how it scans the range.

WeylTwoTone [MOD] [FX] [SYNTH] in: Audio out: Audio

Two-tone Weyl: two additive-irrational (Weyl) sequences at incommensurate increments summed and wrapped, filling the range with a quasi-periodic, never-exactly-repeating two-frequency beat - an equidistributed dual-rate control source.

WilsonCowan [MOD] [FX] [SYNTH] in: – out: Audio

Wilson-Cowan neural-mass oscillator (a first as a synth block): models a whole population of brain cells, not one neuron - a pool of excitatory cells E that drives itself and an inhibitory pool I that reins it back, each passed through a sigmoid firing curve. The push-pull between excitation and delayed inhibition makes the population activity oscillate, the mechanism behind brain rhythms like the alpha and gamma waves on an EEG. The output is the excitatory activity, a smooth 0..1 rhythmic swell. Rate sets the speed and Drive the background input. A network-level oscillation distinct from the single-cell neuron models.

Wobbulate [MOD] [FX] [SYNTH] in: Audio out: Audio

Wobbulator: a resonant band sweeps wildly across the spectrum driven by a sine LFO blended with a smoothed random walk, so the colour lurches and wobbles unpredictably between hollow vowels and shrill peaks - the lab-bench wobbulator turned into an effect. Rate sets the wobble speed, Depth the sweep range, Mix the blend.

Wogglebug [MOD] [FX] [SYNTH] in: – out: Audio

Source of uncertainty (Make Noise Wogglebug / Buchla 266): an irregular internal clock latches a new random value each tick; Smooth blends from hard stepped voltages toward a smoothly wandering output, and Woggle rings each step with a decaying sinusoid for the burbling Buchla character. A self-running random modulation source - no input needed. Distinct from Random (steady clock, no woggle) and Chaos/LorenzLFO (deterministic attractors).

Worley [MOD] [FX] [SYNTH] in: – out: Audio

Worley (cellular) noise source (a first as a synth block): scatters random feature points along the timeline and outputs the distance from the moving read-head to the NEAREST one - the cellular/Voronoi noise behind stone, scales, cracked mud and water-caustic textures. The value falls to zero each time the read-head passes a feature point and swells up in the gaps between them, giving a bumpy, organic ramp-and-dip motion quite unlike the smooth hills of Perlin. Rate sets how fast it travels between cells. A cellular coherent-noise modulator distinct from the gradient-noise family. Needs no input.

WowFlutter [MOD] [FX] [SYNTH] in: Audio out: Audio

Tape pitch instability emulated by a short delay line (centred near 5 ms) whose read time is modulated by two sine LFOs, a slow ~0.6 Hz wow and a fast ~7 Hz flutter, producing continuous pitch drift. Wow scales the slow drift depth and Flutter scales the fast wobble, while Mix blends the warped signal against the dry input. Use low amounts for subtle tape warmth and higher settings for seasick, detuned movement.

WrightFisher [MOD] [FX] [SYNTH] in: Audio out: Audio

Wright-Fisher model: each generation the whole population is resampled, so the next allele count is binomial in the current frequency, driving faster drift to fixation than the Moran process; it reseeds on fixation - a jumpier generational drift CV.

WythoffSeq [MOD] [FX] [SYNTH] in: Audio out: Audio

Wythoff sequence: floor(nphi), the lower Wythoff sequence whose pairs with floor(nphi^2) are the losing positions of Wythoff's nim game; its values climb in a golden, self-similar staircase, here a stepped control source.

Yu [MOD] [FX] [SYNTH] in: – out: Audio

Yu-Wang attractor (a first as a synth block): integrates x' = a(y - x), y' = bx - cxz, z' = exp(xy) - d*z - unusual for its EXPONENTIAL term, which makes the z dynamics snap rather than curve. The orbit forms a tight two-scroll butterfly with sharp, fast reinjections. It free-runs at Rate, outputting x as a smooth bipolar CV. An exponential-nonlinearity chaos distinct from the purely polynomial attractors.

Oscillator (1813)

AbelianSandpile [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Abelian sandpile (Bak-Tang-Wiesenfeld): grains drop on a ring of cells and any cell reaching the critical height topples, sending grains to its neighbours and possibly triggering a chain-reaction avalanche; the avalanche sizes follow a power law - the founding model of self-organized criticality.

Accordion [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Accordion voice: a gate on in0 sounds a stack of two or three slightly-detuned reed voices into the warm wheezy musette of an accordion or bandoneon. Freq sets the note, Musette the detuning between the reed banks (dry to wide wet musette), Tone the brightness, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

AccordionOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Accordion oscillator: two or three free reeds sounding the same note slightly detuned (the 'musette' tuning) under steady bellows pressure, producing the warm, wavering, chorused buzz of an accordion or bandoneon; the detune knob sets the musette wetness.

AcidBass [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

TB-303 acid bassline voice: a single saw through a resonant 4-pole-style low-pass whose cutoff and amp share one decay envelope, so each note opens with the squelchy 'wow' and snaps shut. Slide glides the pitch between overlapping notes (303 portamento) and Accent boosts the envelope depth and level for the accented-step bite. Tune sets the pitch (in1 = Pitch CV), Cutoff the filter floor, Reso the squelch (high = self-oscillating scream), EnvMod how far the envelope opens the filter, Decay the note length, Accent the accent intensity, Slide the glide time, and Level the output. Distinct from Acido, which is the filter alone - this is the whole mono voice. Patch a per-step CV into the Accent mod input for classic accenting.

Additator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Additive oscillator: sums up to Partials sine harmonics of the in1 pitch, with Tilt rolling off the upper partials (0 = bright/full spectrum, 1 = steep, near-sine) and Balance fading between the odd and even harmonics - organ-like and vocal timbres built straight from the harmonic series (the Bogaudio Additator concept).

Additive [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Additive oscillator that sums up to 16 sine partials (Partials) with a spectral Tilt: negative Tilt rolls off the upper harmonics for a soft, mellow tone, positive brightens toward a buzzy saw. Partials above Nyquist are skipped, so it stays alias-free. A clean, controllable harmonic source from pure sine to bright buzz.

Agogo [OSC] [FX] [SYNTH] in: Trigger out: Audio

Agogo bells: a pair of tuned metal bells (high and low) struck on the trigger, the bright Latin/samba percussion ping. Tune sets the pitch, Bell selects high vs low, Level the output. Trigger on in0.

AiharaNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Aihara chaotic neuron: a single-unit map with a graded sigmoid output and an exponentially-decaying refractory feedback, which fires in chaotic, non-periodic spike trains - the model behind chaotic neural networks.

AirHorn [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Reggae sound-system air horn: the 'BWAAAH' drop blast - two stacks of detuned saws a fixed Interval apart (the dancehall horn dyad) honking through a band-pass formant, with a pitch-up bend on the attack. The rewind / drop signal of dancehall, jungle, dubstep, trap and festival EDM. Tune sets the pitch (in1 = Pitch CV), Interval the second horn's offset in semitones (the dyad/cluster dissonance), Detune the beating spread per stack, Bend the attack pitch-rise, Tone the formant brightness, and Level the output. A gate on in0 sounds the blast; hold for the long rewind horn. Distinct from Siren, which sweeps its pitch - this is a fixed-pitch detuned dyad honk.

AirHornOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Air horn: the obnoxious blast of a compressed-air stadium horn (or dancehall siren), a mid tone thick with bright harmonics and a hard edge, blaring in held blasts; the rate sets the honk pattern.

AiryOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Airy oscillator: each cycle traces the Airy function Ai(-t) over its oscillatory region via the asymptotic form, so the waveform sweeps upward in frequency and tapers in amplitude within every period - a built-in intra-cycle chirp no static waveform has. Range sets how far up the tail it reads.

AizawaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Aizawa attractor: a chaotic flow that wraps around a torus and punches through its axis, producing a complex orbit with both rotational and spiking character - a richly-structured strange attractor.

AkashOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Akash oscillator: each cycle is the Akash PDF proportional to (1+x$^{2)e}$(-Theta*x), a lifetime distribution whose quadratic factor gives a more pronounced hump before the exponential tail than Lindley - a rounder, later-peaking skewed bump. Peak-normalized, bounded.

AkgulFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Akgul flow: a representative three-dimensional chaotic system from the chaos-based-encryption literature, with an exponential nonlinearity in one equation folding an asymmetric scroll - a hardware-RNG-oriented chaotic source.

Alarm [OSC] [FX] [SYNTH] in: – out: Audio

Alarm - an insistent two-tone or pulsing siren for warnings and tension. Rate sets the pulse speed, Pitch the tone, Level the output.

AlarmBeepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Alarm beep: the urgent insistent beeping of an alarm or alert (clock, smoke detector, warning), a piercing tone hard-gated into fast even beeps, the wake-up / attention-required sound; the beep rate sets the urgency.

AllpassPdOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Allpass phase-distortion oscillator: a sine is passed through a first-order allpass whose coefficient is itself modulated by a second oscillator at a set ratio, warping the phase at audio rate to generate FM/PD-like sidebands - a distinct distortion-synthesis path between phase distortion and through-zero FM. Ratio sets the modulator pitch, Depth the warp amount.

Almglocken [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Almglocken (tuned cowbells): pitched alpine bells with a metallic, clanking attack between a cowbell and a bell, and a medium decay. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

AmariField [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Amari neural field: a continuum model of cortical activity with local excitation and broader inhibition, reduced to one bump-amplitude mode that pulses and (with adaptation) oscillates - working-memory/attention dynamics as a slow source.

Angklung [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Angklung: shaken Indonesian bamboo tubes - a soft, woody pitched tone with the rattling buzz of the tubes knocking in their frame. Tune sets the pitch, Rattle the shake noise, Level the output. Trigger on in0, in1 = Pitch CV.

AngleGrinderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Angle grinder: a power grinder cutting metal, a very high-pitched motor whine over a gritty abrasive grind, throwing a crackle of sparks, the metal-shop scream; the spark knob sets the sparking.

AnishchenkoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Anishchenko-Astakhov oscillator: a self-sustained chaotic generator from radiophysics with a switching nonlinearity, producing noisy-periodic signals on the boundary between an oscillator and a noise source.

AntPheromone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ant pheromone trail: an ant walks a ring biased toward stronger pheromone, deposits scent on each cell it visits, and the whole field slowly evaporates and diffuses - the stigmergic feedback that lets ant colonies self-organize trails; a probe cell’s concentration is the output.

Anvil [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal anvil: a trigger on in0 strikes iron - a hard, bright, dense burst of high inharmonic metal partials with a fast attack and a ringing metallic clang, the blacksmith’s hammer or industrial-percussion hit. Tune sets the pitch, Decay the ring, Ring the metallic richness, Level the output. Trigger from a Clock, Euclid or sequencer.

Applause [OSC] [FX] [SYNTH] in: – out: Audio

Applause - a dense shower of random claps with a soft noise wash. Intensity sets the clap density (from polite to thunderous ovation), Tone the brightness, Level the output.

ApplauseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Applause: a roomful of clapping hands, a dense random scatter of short broadband clap transients whose rate sets the size and enthusiasm of the crowd, the wash of an audience ovation.

AradhanaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Aradhana oscillator: each cycle is the Aradhana PDF proportional to (1+x)^2 e^(-Theta*x), a lifetime distribution whose squared-linear factor gives a stronger, later hump than Lindley before the exponential tail. Peak-normalized, bounded.

ArcFurnace [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arc furnace: the chaotic V-I characteristic of an electric arc (a negative-resistance plasma whose length wanders) feeding a reactive supply, producing the broadband voltage flicker that arc furnaces inflict on the grid.

ArchimedeanSpiralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Archimedean-spiral oscillator: the projection theta*cos(theta) of the spiral r=theta over several turns per cycle - a cosine whose amplitude rises linearly from zero across the period, a ramped swell with a clean zero start distinct from the Involute and Fermat spirals.

ArcsineDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arcsine oscillator: each cycle is the arcsine PDF 1/(pi*sqrt(x(1-x))), a U-shaped curve that dips in the middle and rises toward both edges (the long-run fraction-of-time distribution of a random walk) - a hollow, edge-weighted waveform unlike the centre-peaked bumps.

ArcWeldOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arc welder: the harsh buzzing of an electric welding arc, a gritty mains-frequency-pulsed broadband noise spitting random bright spatter clicks as molten metal flies, the fierce sizzle of a struck arc; the density sets the spatter rate.

ArneodoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arneodo attractor: a jerk system with a cubic term whose orbit spirals through a screw-like attractor, a smooth chaotic source related to the Coullet/Genesio family but with its own fold.

ArnoldCat [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arnold's cat map: a classic area-preserving toral automorphism that stretches and re-wraps the state every step, a maximally-mixing chaotic source whose two coordinates scramble into hard, modular sawtooth jumps.

ArnoldTongue [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Arnold tongue: the sine circle map theta -> theta + Omega - (K/2pi) sin(2pi theta), whose average rotation locks onto rational winding numbers over ranges of the drive (the Arnold tongues), so sweeping the bias traces the devil's-staircase of mode-locking.

AssocLegendreOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Associated-Legendre oscillator: each cycle is an associated Legendre function P_n^m(x) (the angular part of spherical harmonics) - polynomials multiplied by powers of sqrt(1-x^2), giving lobed waveforms that taper to zero at the edges, distinct from the plain Legendre/Chebyshev families. Peak-normalized, bounded.

AsymLaplaceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Asymmetric-Laplace oscillator: each cycle is a two-sided exponential cusp whose left and right decay rates differ by Skew - a sharp peaked bump leaning to one side, the skewed cousin of the Laplacian, with a hard cusp at the centre and exponential tails.

AsymSechOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Asymmetric-sech oscillator: each cycle is a hyperbolic-secant bump whose left and right skirts decay at different rates set by Skew - a smooth, click-free pulse leaning to one side, the soft-tailed counterpart of the asymmetric-Laplace cusp. Bounded.

AuroralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Auroral pulsation: the slow magnetic-field oscillations (Pc continuous pulsations) that ripple through the magnetosphere during auroral activity, modeled as two near-equal frequencies beating into a waxing-and-waning quasi-sine - the magnetosphere's gentle ring.

AutocatalatorChem [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Autocatalator: a two-variable cubic-autocatalysis chemical model (a precursor decays into an activator that catalyses its own production), giving sharp relaxation oscillations and, when driven, chaos - a minimal self-accelerating reaction clock.

AvalancheOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Avalanche: a mountainside of snow letting go, a low broadband rumble that builds and roars to a thunderous peak with grinding debris before slowly subsiding, looping at the slow event rate; a force-of-nature wall of noise.

AYChip [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

AY-3-8910 / YM2149 oscillator: a square plus a noise channel shaped by the chip's hardware sawtooth envelope - the classic ZX-Spectrum / Amstrad chiptune voice. Noise sets the noise mix, EnvRate the hardware-envelope speed, Level the output. in0 = Gate, in1 = Pitch CV.

AYToneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

AY tone: the General Instrument AY-3-8910 / Yamaha YM2149 PSG (ZX Spectrum, Atari ST, MSX) square tone shaped by its hardware volume envelope, which can repeat fast enough to act as a buzzy sawtooth-like timbre - the chip's signature 'buzzer bass' voice.

BackEaseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Back-ease oscillator: each cycle is the easeOutBack animation curve, which shoots past the top then settles back - an S-ramp with a single overshoot whose size Overshoot controls, giving a waveform between a sigmoid and a damped step, distinct from any plain ramp.

Bagpipes [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Bagpipes: a constant droning reed under a buzzy chanter melody - the fixed low drone plus a pitched reed voice that never stops between notes. Drone sets the drone level, Reed the chanter buzz, Level the output. in0 = Gate, in1 = Pitch CV.

BakerMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Baker's map: the dough-kneading stretch-cut-and-stack transform that is the archetype of chaotic mixing, iterated at a chosen rate for a hard, fractal, modular-jump texture.

BakSneppenEvo [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bak-Sneppen model: a ring of species each with a fitness; each step the least-fit species and its two neighbours are replaced with fresh random fitnesses, driving the system to self-organized criticality with avalanches of extinction - the minimum fitness rises in punctuated jumps.

Balafon [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Balafon: a struck wooden-bar xylophone over gourd resonators that add a characteristic membrane buzz - the bright, woody West-African mallet voice. Tune sets the pitch, Buzz the gourd rattle, Level the output. Trigger on in0, in1 = Pitch CV.

BandedWG [OSC] [FX] [SYNTH] in: Strike, Pitch CV out: Audio

Banded waveguide resonator (Essl/Cook/Serafin): a strike on in0 excites a bank of four high-Q modal resonators tuned to inharmonic ratios, so it rings like a struck bar, bowl or glass rather than a plucked string - the model behind physically-voiced marimbas, glass harmonicas and Tibetan singing bowls. Tune sets the fundamental (in1 = Pitch CV), Material morphs the ratios from wooden bar toward glass bowl, Decay the ring time, Level the output.

BandNoiseTuned [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tuned noise band: white noise driven through a steep resonant band-pass at the set pitch, so the noise rings into a breathy, whistling pitched tone - wind, flute-edge and surf-like timbres that a clean oscillator cannot make; raise Reso toward self-oscillation for a pure whistle.

BanjoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Banjo: a bright, lightly-damped Karplus-Strong string whose output is coupled into a quick drum-head resonance, giving the twangy, plinky, fast-decaying attack of a banjo; re-plucked at the set rate for rolls.

Bansuri [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bansuri: the Indian bamboo transverse flute - an open-bore air-jet waveguide voiced warmer and breathier than a metal Western flute, with a soft woody tone and heavy breath noise, the meditative voice of Hindustani classical music. Breath pushes it overblown, Air sets the breathiness.

BaoMemristor [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bao memristive system: a Lorenz-like flow with a flux-controlled memristor whose state-dependent conductance adds a fourth dimension of memory, producing memristor-flavoured chaos with extreme multistability.

BarModeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bar-mode oscillator: sines at the bending-mode ratios of a free-free bar (1, 2.76, 5.40, 8.93, 13.34) - the sharply-inharmonic, fast-thinning overtones of a xylophone or marimba bar (which tuners flatten by undercutting); a struck-metal pitched tone.

BarnsleyFern [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Barnsley fern: the chaos game over four affine contraction maps (stem, successively smaller leaflets) whose probabilistically-chosen iteration converges to the famous self-similar fern; the wandering point's coordinate is read out as a fractal-distributed CV.

BartlettHannOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bartlett-Hann oscillator: each cycle is the Bartlett-Hann window 0.62 - 0.48|p-0.5| - 0.38cos(2pip), a hybrid of the triangular and Hann windows - a rounded bump with slightly straighter flanks than a pure cosine window, a mellow tone with its own harmonic balance.

Basimilus [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Harmonic percussion synth (Noise Engineering Basimilus Iteritas): a trigger on in0 fires a stack of six sine partials whose spacing morphs from harmonic toward inharmonic, run through a wavefolder and shaped by a fast percussive decay - the digital, metallic counterpart to the analog drum voices, good for kicks, snares, zaps and bells. Tune sets the fundamental, Harmonics the partial spread, Fold the wavefolder crunch, Decay the length, and Level the output. Gate it from a Clock, Euclid or sequencer.

BassGuitarPluckOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bass guitar: a low, thick Karplus-Strong string with heavier damping for a round finger-style tone, plus an optional sharp attack transient for slap-and-pop; re-plucked at the set rate for basslines.

Bassoon [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Bassoon: the woody, reedy low double-reed - a sawtooth through a low formant and a warm lowpass with reed buzz, played by holding in0 as breath. Reed sets the buzz, Breath the air, Level the output. in0 = Breath gate, in1 = Pitch CV.

BeadHoop [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bead on a rotating hoop: a bead sliding on a vertical hoop spun about its axis; centrifugal force fights gravity in a pitchfork bifurcation, and a wobble in the spin rate drives the bead's angle chaotically.

BeddingtonMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beddington map: a host-parasitoid model that adds host density-dependence to Nicholson-Bailey, taming the divergence into bounded cycles and chaos - a more realistic ecological two-species oscillator.

BedheadMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bedhead attractor: an iterated map mixing a product term with a sine, tracing wispy, hair-like fractal strands (the 'bedhead' pattern), a fine, scratchy chaotic texture.

BeeBuzzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bee: the wingbeat buzz of a honeybee, a low ~200 Hz fundamental rich in harmonics, gently amplitude- and pitch-fluttered as the insect hovers and turns; the wingbeat knob sets the species pitch.

BeetleScuttleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beetle scuttle: the dry skittering of a beetle's legs scrambling across a hard surface, a sparse irregular spatter of tiny high noise micro-bursts, fast and stop-start; the speed knob sets how frantically it scurries.

BellFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-operator FM bell: a sine carrier is phase-modulated by a sine modulator running at a non-integer multiple of the carrier frequency, producing inharmonic metallic partials. Freq sets the carrier pitch, Ratio sets the modulator-to-carrier frequency ratio (typically fractional for bell tones), Index sets the modulation depth (brightness and partial spread), and Level scales the output. Higher Index and non-integer Ratio yield clangorous, bell-like timbres.

BellLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

FM bell lead: a bright, ringing two-operator FM bell tone playable as a lead - metallic, glassy and clear. Ratio sets the modulator interval, Bright the FM index, Level the output. in0 = Gate, in1 = Pitch CV.

BellPartialOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bell-partial oscillator: the named partials of a tuned church bell - hum (0.5), prime (1), the minor-third tierce (1.2), quint (1.5), nominal (2) and beyond - whose deliberate minor-third tierce gives a bell its characteristic plaintive, slightly-dissonant ring.

BeltSanderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Belt sander: a belt sander running flat on wood, a steady mid motor drone overlaid with the continuous grit-hiss of the abrasive belt looping and a periodic seam thump where the belt joins, the workshop-sanding sound; the belt rate sets the loop.

Bendir [OSC] [FX] [SYNTH] in: Trigger out: Audio

Bendir: a North-African frame drum with gut snares stretched across the skin that add a buzzing rattle on every hit - a membrane tone under a snare-like buzz. Tune sets the pitch, Buzz the snare rattle, Level the output. Trigger on in0.

BeniniOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Benini oscillator: each cycle is the Benini PDF $x^{(-Alpha)*e}$(-Betaln(x)^2)(Alpha+2Betaln x)/x over x>=1, a heavy-tailed income distribution blending Pareto power-law and log-normal curvature - Alpha sets the power tail and Beta the log-normal bending. Peak-normalized, bounded.

Benjolin [OSC] [FX] [SYNTH] in: – out: Audio

Benjolin (Rob Hordijk): a self-patched chaos voice - two cross-modulating triangle oscillators feeding a rungler shift register that feeds their pitch back, producing wild, generative, never-repeating drones and sequences from no input. Rate sets the base speed, Chaos the rungler feedback depth.

Berimbau [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal berimbau: a trigger on in0 strikes the steel string of the Brazilian musical bow - a bright metallic plucked tone (near-harmonic partials) with a gourd-resonated body and the buzz of the caxixi rattle, the hypnotic pulse of capoeira. Tune sets the string pitch (open/pressed via Pitch CV), Decay the ring, Buzz the rattle amount, Level the output. Trigger from a Clock, Euclid or sequencer.

BernoulliPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bernoulli-polynomial oscillator: each cycle is the nth Bernoulli polynomial B_n(x) on [0,1] (the polynomials behind the Euler-Maclaurin formula and sums of powers) - an oscillating polynomial with characteristic end symmetry, distinct from the orthogonal-polynomial families. Peak-normalized, bounded.

BernoulliShift [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bernoulli shift: the doubling map x -> 2x mod 1 is the dynamical-systems picture of shifting a number's bits left, producing a sharp, deterministic noise stream - chaos as a bit-shift register.

BernsteinOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bernstein oscillator: each cycle is a Bernstein basis polynomial $x^{K(1-x)}$(N-K) on [0,1] (the building block of Bezier curves), a single compact bump whose position is set by K/N and whose width narrows as N grows - a placeable, scalable pulse. Peak-normalized, bounded.

BertramBurst [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bertram burster: a phenomenological polynomial model of bursting electrical activity with two slow variables on different timescales, capable of square-wave, parabolic and chaotic bursting - a flexible burster for rhythmic clusters.

BesselPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bessel-polynomial oscillator: each cycle is a Bessel polynomial y_n(x) (1+x, 1+3x+3x^2, 1+6x+15x^{2+15x}3) - a monotonically accelerating rise whose curvature steepens with Order, a polynomial ramp distinct from the orthogonal Chebyshev/Legendre families. Peak-normalized, bounded.

BetaNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beta noise: samples on [0,1] formed as X/(X+Y) from two gamma variables, whose two shape knobs sculpt the density from U-shaped (peaks at the edges) through uniform to a central bell - a bounded, fully-shapeable random source.

BetaPdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beta-PDF oscillator: each cycle is the Beta distribution shape $x^{(Alpha-1)(1-x)}$(Beta-1) on [0,1], normalized to its mode - two knobs slide the bump from left-skewed to symmetric to right-skewed and from narrow to broad, a continuously morphable compact waveform.

BetaPrimeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beta-prime oscillator: each cycle is the beta-prime PDF $x^{(a-1)(1+x)}$(-a-b) on x>=0, a right-skewed bump with a polynomial tail (the distribution of a ratio of gammas) - Alpha sets the rise, Beta the tail decay; peak-normalized and bounded.

BezierOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bezier-curve oscillator: one cycle is a cubic Bezier whose two middle control points (Shape1/Shape2) bend the waveform between the fixed endpoints, sweeping from a soft pulse to a skewed bell to a near-saw - an arbitrary smooth waveshape from two knobs.

BhalekarGejji [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bhalekar-Gejji attractor: a three-dimensional chaotic system (introduced in the fractional-calculus literature) whose single product nonlinearity and constant drive fold a compact scroll, here integrated as an integer-order flow.

BhaskaraOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bhaskara oscillator: generates its cycle from Bhaskara I's 7th-century rational sine approximation 16x(pi-x)/(5*pi^2-4x(pi-x)), mirrored for the negative half - a near-perfect sine whose tiny approximation error contributes a faint, fixed upper-harmonic warmth. A 'soft sine' variant distinct from a pure lookup sine.

BicornOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bicorn oscillator: the y-coordinate of the bicorn (cocked-hat) curve, cos^2 t (Shape + cos t)/(3 + sin^2 t), a smooth two-humped waveform whose symmetry and harmonic content shift with Shape - a rounded twin-peak tone distinct from any sine or pulse.

BifoliumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bifolium oscillator: the radial curve sin(theta)*cos^2(theta), which expands to an equal-weight in-phase blend of the fundamental and third harmonic - a hollow, double-petalled waveform distinct from the nephroid's 3:1 weighting, mellow and woody.

BinaryEntropyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Binary-entropy oscillator: each cycle is the Shannon binary entropy H(p) = -plog2(p)-(1-p)log2(1-p), a symmetric hump that is zero at both ends and peaks at the centre with steep, near-vertical shoulders - a flat-topped-yet-pointed bump distinct from the polynomial and cosine windows.

BinaryNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Binary noise: a one-bit random telegraph signal that flips between +1 and -1 at a decimated rate, the harsh, hollow hiss of single-bit quantized noise - Rate sets the sample-and-hold step (lower = chunkier, more tonal) and Bias skews the duty between the two levels.

∿ BinaryStarOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Binary-star oscillator: the radial-velocity curve of a star orbited in an eccentric orbit (true anomaly through Kepler's equation), the skewed, asymmetric wobble astronomers measure to find companions and exoplanets - sine-like when circular, sharply lopsided when eccentric.

∿ BinauralBeat [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Binaural beat source: two sine tones a few Hz apart summed to one channel, so they alternately reinforce and cancel at the difference frequency - the slow amplitude throb used for entrainment tones (true binaural needs one tone per ear).

∿ Birds [OSC] [FX] [SYNTH] in: — out: Audio

Bird ambience: a bed of randomly-timed bird tweets - each a fast frequency-swept sine whistle (the chirp glide) at a scattered pitch - the dawn-chorus / forest / garden ambience of film sound design. Distinct from the Insects stridulation bed: birds are gliding whistles. Self-running. Density sets how often tweets fire, Pitch the whistle base frequency, Sweep the chirp glide depth, Variety the random pitch range between calls, and Level the output.

∿ Birdsong [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Birdsong: random whistled chirps - each a short FM sweep that rises or falls at a random pitch and rate, the way a songbird trills. Sparse and pretty as an ambience layer, dense and frantic as a dawn chorus. Rate sets the chirp frequency, Mix the blend.

∿ BirdsongOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Birdsong oscillator: a high whistle that rapidly glides up and down in pitch and is grouped into trilled chirp phrases with gaps, the bright frequency-swept warble of a songbird; the chirp rate and glide depth shape the species' call.

∿ BirnbaumSaundersOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Birnbaum-Saunders oscillator: each cycle is the fatigue-life PDF (a reliability distribution), a sharply-rising right-skewed bump - Beta sets the scale/peak position and Gamma the spread, peak-normalized to a bounded waveform with a distinctive crack-propagation skew.

∿ BirthDeathProcess [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Birth-death process: a population whose per-capita birth and death rates make it grow or shrink by one at random, fluctuating and occasionally going extinct then reseeding (the basic stochastic population/queue model) - a jittery boom-and-bust integer CV.

∿ BitGlitchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bit glitch: a corrupted digital signal, a tone that randomly drops into harsh bit-mangled aliased bursts (bit-flips and sample-rate errors) between clean moments, the broken-machine sound of datamosh; the error rate sets the corruption.

∿ BiweightOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Biweight oscillator: each cycle is the biweight (quartic) kernel $(1-x^{2)}2$, a compact bump flatter on top and steeper at the edges than the parabolic Welch - a smooth pulse with a rounder shoulder and faster-decaying harmonics.

BlackbodyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blackbody oscillator: each cycle is Planck's law spectral curve $x^{3/(e}x-1)$, the universal blackbody radiation shape - a smooth bump that rises as a cubic and falls exponentially with the characteristic Planck skew; Spread sets how much of the spectrum the cycle traverses. Peak-normalized, bounded.

BlackHole [OSC] [FX] [SYNTH] in: – out: Audio

Black hole - a crushing sub-bass gravitational drone slowly bending downward into the abyss, with a faint event-horizon shimmer. Pull sets the intensity, Tone the shimmer, Level the output.

BlackmanHarrisOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blackman-Harris oscillator: each cycle is the 4-term Blackman-Harris window, a cosine-sum bump with extremely low sidelobes - a very smooth, rounded, near-Gaussian pulse whose steep symmetric taper gives a mellow tone with fast-decaying even harmonics.

Blancmange [OSC] [FX] [SYNTH] in: – out: Audio

Blancmange (Takagi) oscillator (a first as a synth block): synthesises the blancmange curve - a fractal made by stacking ever-smaller TRIANGLE waves, each half the height and twice the rate of the last. Like the Weierstrass function it is continuous everywhere but jagged at every scale, yet its triangle (rather than cosine) building blocks give a softer, more granular roughness - a self-similar timbre that thickens as Detail adds layers. Detail sets how many triangle octaves to sum. A triangle-based fractal waveform distinct from the cosine-based Weierstrass. Needs no input.

BlancmangeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blancmange (Takagi) curve: a sum of triangle waves at doubling frequencies and halving amplitudes, the pudding-shaped fractal that is continuous everywhere but has a corner at every scale - a soft, self-similar, octave-stacked waveform.

Blaster [OSC] [FX] [SYNTH] in: Trigger out: Audio

Sci-fi blaster: a noisy, metallic energy-weapon shot - a fast pitch-swept tone mixed with ring-modulated noise and a short ringing tail. Tune sets the pitch, Grit the noise, Level the output. Trigger on in0.

Bleep [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Bleep-techno tone (Warp / Sheffield 'Sweet Exorcist' sound): a simple root-plus-fifth motif from sine-to-square waves with a hard on/off organ-style amp gate and an octave-down sub - the ultra-deep, dub-system bleep of early Warp techno. Tune sets the pitch (in1 = Pitch CV), Wave morphs sine toward square, Fifth sets the level of the +7-semitone partial (the signature root/5th bleep), Sub the octave-down sub-bass weight, and Level the output. A gate on in0 switches it on and off with organ snap - no slow attack or release.

BlenderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blender: a kitchen blender grinding, a buzzy motor tone that surges and bogs as the blades catch and free, with a rattly chopping modulation, the chunky roar of crushing ice; the load knob sets how much it labours.

BlinkingVCO [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blinking oscillator: two oscillators whose coupling link switches on and off at a fast clock (a ‘blinking network’); the on-average coupling still synchronizes them, but the switching injects chaotic jitter - stochastic-coupling dynamics.

BlitOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

BLIT sawtooth: a band-limited impulse train (the closed-form sinc-sum) leaky-integrated into a sawtooth, giving the bright edge of a saw with the high harmonics correctly band-limited - alias-free even up high.

BLITSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Band-limited sawtooth built additively (summed harmonics up to Nyquist) so it stays alias-free even at high pitch - the clean, classic analog saw. Bright sets the harmonic count from mellow to razor-sharp.

BlitSquare [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

BLIT square: a bipolar band-limited impulse train (two opposite-sign sinc sums a half-cycle apart) leaky-integrated into a square wave, keeping the high harmonics correctly band-limited - an alias-free square even up high.

Blizzard [OSC] [FX] [SYNTH] in: – out: Audio

Howling blizzard - storm-wind noise driven through a sweeping resonant peak that wails and gusts. Howl sets the wail depth, Tone the iciness, Level the output.

BlizzardWindOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blizzard: a howling winter gale, a resonant band of wind noise whose pitch and loudness surge in long gusts, over a fine high hiss of driven snow; the gust rate sets how violently the storm heaves.

BlochDriven [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Driven Bloch equations: the magnetization (Mx, My, Mz) of a spin ensemble under an oscillating RF field with T1/T2 relaxation; strong driving makes the nutation chaotic - the model behind NMR and maser instabilities.

BlochOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bloch oscillation: an electron in a periodic lattice under a constant force does not accelerate away but oscillates in real space (the Wannier-Stark effect), tracing a sawtooth-like real-space position from the integral of a cosine band - solid-state physics as a waveform.

BlockingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blocking oscillator: a transformer-coupled circuit that fires a brief, sharp pulse then blocks itself off for a long recovery while a capacitor recharges - the spiky, low-duty waveform of flyback supplies, fence chargers and early sweep generators.

BlowflyNicholson [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nicholson’s blowflies: a delay-differential population model (recruitment depends on the egg-laying population one maturation-time ago: x’ = P x(t-tau) e^{-x(t-tau)} - d x) that famously matches lab fly cultures crashing into chaotic cycles.

Blown [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Waveguide wind instrument (STK-clarinet): a bore delay line with a nonlinear reed reflection, driven by breath from the gate. The blown counterpart to Karplus-Strong.

BlownBottle [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Blown bottle (Helmholtz resonator): a breathy hoot from white-noise breath excited through a single sharp resonant peak at the pitch - the hollow, woody tone of blowing across a bottle. Tune sets the pitch, Breath the air drive, Level the output. in0 = Breath gate, in1 = Pitch CV.

BlueNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blue noise: white noise passed through a first difference, tilting the spectrum up at +3 dB/octave so high frequencies dominate; the airy hiss used for dithering and high-frequency texture (deterministic RNG).

BlumBlumShub [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Blum-Blum-Shub: the cryptographically-secure generator x -> x^2 mod N (N a product of two primes), whose least-significant bit is provably hard to predict; its bit-stream is a high-quality deterministic white-noise source.

Bodhran [OSC] [FX] [SYNTH] in: Trigger out: Audio

Bodhran: the Irish frame drum played with a tipper - a tunable low-mid membrane with a dry, woody skin tone and a short decay. Tune sets the pitch, Decay the length, Level the output. Trigger on in0.

BogdanovMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bogdanov map: the discrete normal form of the Bogdanov-Takens bifurcation, whose orbit spirals and folds through an annular chaotic ring - a 2-D map capturing the birth of chaos from a double-zero eigenvalue.

BohmanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bohman oscillator: each cycle is the Bohman window (1-|x|)cos(pi|x|) + sin(pi|x|)/pi, a cosine-product bump with a distinctive gently-shouldered shape - smoother-edged than Welch but brighter than Blackman-Harris, a mid-character rounded pulse.

BoidsFlock1D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Boids (1-D): a dozen agents on a line steer by Reynolds' three rules - move toward the flock centre, match neighbours' velocity, and avoid crowding - producing emergent swarming, splitting and merging; the flock centroid drifts as a smooth organic CV.

BoilingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Boiling oscillator: random bubble events, each a short rising-pitch chirp (a collapsing cavity's note glides up as it shrinks), at a rate that climbs from a slow simmer to a rolling boil - granular bubble synthesis of boiling water.

BoingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cartoon boing: the springy 'boing' of a bouncing object, a tone whose pitch wobbles rapidly up and down with a decaying amplitude (a vibrating spring), the classic comedy bounce; fired at the set rate.

Bonang [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Gamelan bonang: a bossed bronze kettle-gong with a clear, bright pitch over inharmonic partials and a medium shimmering decay - the melodic voice of the Indonesian gamelan. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

Bongo [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal bongo: a trigger on in0 strikes a small tuned hand drum - a tight high-pitched membrane whose inharmonic modes (about 1 : 1.6 : 2.3) ring briefly with a quick downward pitch drop on the hit and a touch of finger-slap noise. Tune sets the head pitch (high or low bongo), Decay the short ring, Slap the attack-noise amount, Level the output. Trigger from a Clock, Euclid or sequencer.

BonkOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bonk: a comedic hollow head-bonk, a woodblock-like resonant knock with a fast downward pitch drop and a quick decay, the cartoon clonk-on-the-noggin; fired at the set rate.

BooOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Boo: a crowd voicing its displeasure, a low 'oooo' vowel drone of many detuned voices with a rough beating edge, swelling and wavering, the disapproving rumble of a booing audience.

BoostConverter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Boost converter chaos: a step-up switching regulator whose discontinuous-conduction switching gives border-collision bifurcations straight into chaos, a hallmark of piecewise-smooth power-electronic systems - read as the inductor ripple.

BoualiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bouali attractor: a chaotic flow modelling economic/predator-prey-style dynamics, tracing a folded ribbon orbit - a distinctive strange attractor for slow, evolving modulation.

BoubakerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Boubaker oscillator: each cycle is the nth Boubaker polynomial (B_1=x, B_2=x^2+2, B_n=x*B_{n-1}-B_{n-2}; the +2 in B_2 breaks the pure-Chebyshev pattern) - a physics polynomial family giving lobed waveforms with a distinct low-order offset. Peak-normalized, bounded.

BounceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bounce oscillator: each cycle is the easeOutBounce animation curve - a series of decreasing parabolic bounces rising to the top - so the waveform climbs in three settling hops, a rhythmic, percussive contour unlike any smooth or ramp wave.

∿ **BouncingBall** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bouncing ball: a ball in free fall repeatedly striking a vibrating table, where the table's phase at each impact randomizes the next flight; strong shaking gives chaotic, period-doubled bounce heights - the classic impact-map chaos.

∿ **BouncyBallOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bouncing ball: a rubber ball dropped and left to bounce, a series of short pitched impacts whose gaps shrink and pitch drops as the ball settles, then the cycle restarts, the playful Galileo-bounce; the rate sets how often it is dropped.

∿ **Bowed** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bowed-string waveguide (STK-bowed): two delay lines for the string halves with a stick-slip bow-table at the bow point, driven by the gate. Violin/cello family.

∿ **BowedBar** [OSC] [FX] [SYNTH] in: Bow, Pitch CV out: Audio

Bowed metal bar / singing bowl: a sustained, shimmering inharmonic metal tone with a slow swell, played by holding in0 as bow pressure - the eerie, ringing bowed-vibraphone or Tibetan-bowl voice. Tune sets the pitch, Shimmer the beating, Level the output. in0 = Bow gate, in1 = Pitch CV.

∿ **BowedBarVoice** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bowed bar: three inharmonic bar modes are continuously re-energized by a stick-slip friction drive, so they sustain instead of decaying - the eerie sustained tone of a bowed vibraphone bar or musical saw.

∿ **BowedString** [OSC] [FX] [SYNTH] in: Bow, Pitch CV out: Audio

Bowed string (violin / cello): a sawtooth Helmholtz motion with a slow bow attack, bow-noise scrape and a body resonance, expressively controlled by holding in0 as bow pressure. Tune sets the pitch, Bow the scrape amount, Level the output. in0 = Bow gate, in1 = Pitch CV.

∿ **BowedStringChaos** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Over-bowed string: a bowed-string waveguide pressed past the Helmholtz regime, where the stick-slip friction and string feedback lock into the raucous subharmonic and chaotic multiphonics players call 'surface sound' or wolf-tones.

∿ **BowFriction** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bowed-string model: a delay-line string driven by a velocity-dependent stick-slip friction curve self-oscillates into a sustained bowed tone whose bite depends on bow force - physical bowed synthesis.

∿ **BoxingBellOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ring bell: the bright clang of a boxing-ring bell signalling the round, a struck metal bell with a strong fundamental and inharmonic shimmer ringing for a long time, fired at the set rate; the call to fight.

∿ **Braaam** [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Inception 'BRAAAM': the massive low brass blast that defined trailer scoring - a stack of detuned saws in octaves and a fifth (the brass cluster) over a deep sub sine, swelling in over a slow attack and driven through saturation for the menacing low-end roar. Tune sets the fundamental (in1 = Pitch CV), Swell the attack fade-in, Detune the cluster beating (the dread), Sub the deep sub-sine weight, Drive the saturation, and Level the output. A gate on in0 fires the blast; hold for the long swell.

BrakeJudder [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brake judder: a disc brake where normal and tangential pad motions couple through friction, so above a friction threshold the two modes lock unstably and the brake grabs and judders - the mode-coupling mechanism behind brake squeal.

Brass [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Synth brass (OB-X / Jupiter brass): two detuned saws through a resonant low-pass driven by a filter envelope - the cutoff snaps bright on the attack then drops to a mellower sustain, the timbral movement that makes a saw sound brassy. The eurodance / trance / funky-house brass stab and swell. Tune sets the pitch (in1 = Pitch CV), Detune the saw spread (thickness), Cutoff the base brightness, FiltEnv how far the attack opens the filter, Decay how fast the bright attack drops to the sustain, and Level the output. A gate on in0 plays the note (hold for the sustain).

BrassEnsemble [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Analog brass ensemble: five detuned sawtooths with a slow filter-swept attack and ensemble chorus, the bold, swelling ARP/Solina brass-section pad. Cutoff sets the brightness, Attack the swell time, Level the output. in0 = Gate, in1 = Pitch CV.

BrassLip [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brass lip-reed model: a delay-line bore closed by a pressure-controlled lip valve nonlinearity self-oscillates into a buzzing brass tone whose brightness blooms with input drive - physical brass synthesis.

BrassWGOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brass waveguide: a bore driven by a lip-reed (a pressure-controlled valve with its own resonance) and shaped by a flared-bell brightness, producing the buzzy, harmonically-rich tone of a trumpet or trombone that blares harder as you blow.

BrayLiebhafsky [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bray-Liebhafsky reaction: the first discovered homogeneous chemical oscillator (hydrogen peroxide decomposing via iodate/iodine), reduced to its autocatalytic core, ticking out relaxation oscillations and bursting.

Breathing [OSC] [FX] [SYNTH] in: – out: Audio

Slow human breathing - filtered noise swelling brighter on the inhale and darker on the exhale. Rate sets the breath pace, Tone the airiness, Level the output.

BreathingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Breathing: the slow rhythm of human breath, a band of airy noise swelling on the inhale and falling on the exhale once per breath, the soft hiss of lungs filling and emptying; the rate sets the breaths per minute.

BreathyVoiceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Breathy voice: a soft glottal pulse train mixed with continuous aspiration noise (the glottis never fully closes), the intimate, airy vocal quality of a whispered-sung tone; the breath knob crossfades from clear voice to pure air.

BreitWignerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Breit-Wigner oscillator: each cycle is the relativistic Breit-Wigner resonance $1/((x^{2-M}2)^{2+M}2*W\hat{}2)$, a sharp resonance peak at Mass M with a quartic (steeper-than-Lorentzian) roll-off - Width sets the linewidth; peak-normalized to a bounded, spiky resonance waveform.

BriansBrain [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brian's Brain: a three-state automaton (off, on, dying) where an off cell turns on only with exactly two on neighbours, every on cell next becomes dying, and dying cells go off; almost every pattern explodes into a sky of drifting gliders. On-cell density is the output.

BriggsRauscher [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Briggs-Rauscher reaction: the iodine-clock oscillator that flips amber-clear-blue, reduced to its activator-inhibitor-iodide core, firing sharp relaxation pulses with a long quiescent interval - a chemical-clock spike source.

BrockHommes [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brock-Hommes market: fundamentalist and trend-chasing traders switch strategies by past profit, so the price deviation from fundamentals oscillates and chaotically bubbles-and-crashes - behavioural-finance dynamics as a control source.

BrookOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brook: a babbling stream, a continuous bed of band-passed flow noise sprinkled with random little rising bubble tones, the gentle chatter of water running over rocks; density sets how lively the brook.

BrownianBridgeWalk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brownian bridge: a random walk constrained to start and end at zero over each period (the walk minus its linear drift to the endpoint), giving wandering shapes that always return home - the stochastic process used to interpolate between fixed points.

BrownianNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brown noise generator: white noise integrated into a bounded random walk rolls off at 6 dB/octave, the deep, rumbly low-weighted noise of distant surf or thunder.

BrownNoise [OSC] [FX] [SYNTH] in: – out: Audio

Brown/red noise - a deep -6 dB/oct rumble (integrated white noise), the heaviest of the noise colours for thunder, wind and sub texture. Tone opens the top, Level sets the output.

BrusselatorDiffusion [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled Brusselators: two Brusselator reaction cells joined by diffusion, the minimal Turing system where unequal diffusion of activator and inhibitor breaks symmetry into out-of-phase chemical oscillation and spatiotemporal chaos.

BrusselatorOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Brusselator: the textbook autocatalytic chemical-reaction oscillator (x'=a-(b+1)x+x^2 y, y'=bx-x^2 y); a periodic forcing of its feed rate pushes its limit cycle into chaos - reaction-kinetics rhythms as sound.

Bubble [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bubbles: a sparse scatter of underwater bubbles, each a short sine that RISES in pitch as it shrinks and pops - the inverse of a falling rain drop. Sprinkle it for an aquatic ambience or a playful blip layer. Density sets the bubble rate, Mix the blend.

BubbleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bubble: the iconic underwater 'bloop' of a rising air bubble, a short sine burst whose pitch sweeps upward as the bubble shrinks toward the surface (the physically-modelled bubble acoustic), fired at the set rate with random size variation.

BubbleOscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Acoustic bubble: a gas bubble in a sound field expanding and violently collapsing per the Rayleigh-Plesset equation; strong driving gives the period-doubling, chaotic radial pulsation behind single-bubble sonoluminescence.

Bubbles [OSC] [FX] [SYNTH] in: — out: Audio

Underwater bubbles - random rising-pitch blips for a fish-tank or undersea texture. Rate sets the bubble density, Pitch the blip range, Level the output.

Buchla259 [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Complex oscillator (Buchla 259 / West Coast): a principal sine is frequency-modulated by a second oscillator and then run through a wavefolder, so one voice spans pure tones, metallic FM and bright folded harmonics with no subtractive filter. Ratio sets the modulator interval, FM the modulation depth, Fold the wavefold amount. in0 = Pitch CV.

BuckConverter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Buck converter chaos: a switching DC-DC buck regulator under current-mode control, where the switching ramp interacts with the inductor current to period-double into chaos - a real power-supply instability, read as the ripple waveform.

BudykoSellers [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Budyko-Sellers model: a global energy-balance climate model whose ice-albedo feedback creates a bistable warm/snowball Earth; a periodic solar forcing makes the temperature flicker between the two states - tipping-point climate dynamics.

Bullroarer [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bullroarer: the ancient whirled aerophone - a slat on a cord spun overhead roars as it rotates, the tone swelling in pitch and amplitude on each pass and fluttering as the slat spins on its own axis; the ritual sound of many archaic cultures. Whirl sets the rotation speed, Roar the tonal-vs-noise balance.

BumpWaveletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bump function: the classic compactly-supported smooth function exp(-1/(1-x^2)), which is infinitely differentiable yet exactly zero outside one period (a true mollifier), modulating a tone into a grain that starts and ends in perfect silence with no discontinuity at any derivative.

BurgersMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Burgers map: a 2-D discretization of the Burgers turbulence equation whose coupled quadratic terms give a shearing, eddy-like chaotic orbit - turbulence-flavoured chaos as a sound source.

BurkeShawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Burke-Shaw attractor: a tightly-wound spiral chaotic flow that fills a thin shell, producing a fast, hissy, almost-tonal chaos useful for noisy oscillator textures.

BurningShipOrbit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Burning Ship: the Mandelbrot iteration but with the real and imaginary parts made absolute before squaring, whose folding produces the famous ship-and-antenna fractal; iterating the orbit and slowly scanning the parameter, the real part (reseeded on escape) is the output.

BurrQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Burr-quantile oscillator: each cycle is the Burr-XII inverse-CDF ((1-p)$^{(-1/K)-1)}$(1/C), a doubly-parametrized heavy-tailed ramp - C shapes the early rise and K the tail weight, a more flexible blowup than the single-parameter Lomax. Bounded.

BurrXOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Burr-X oscillator: each cycle is the Burr type-X PDF 2alphaxe$^{(-x}$2)(1-e$^{(-x}$2))^(alpha-1), a Rayleigh-based exponentiated bump whose skew and rise sharpen with Alpha - a generalized-Rayleigh shape distinct from the Weibull-based families. Peak-normalized, bounded.

BusySignalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Busy-signal oscillator: the engaged tone, a 480 + 620 Hz sine pair gated in the steady half-second-on, half-second-off cadence of a busy line (or the faster reorder/fast-busy when Rate is raised).

ButterflyCurveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Butterfly-curve oscillator: the x-coordinate of Fay's transcendental butterfly curve, sin(t)(e^cos t - Wingscos 4t), giving a harmonically rich, asymmetric waveform with sharp wing-beats whose spikiness the Wings knob controls - a complex timbre from one closed-form curve.

ButterworthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Butterworth oscillator: each cycle is the Butterworth lowpass magnitude response 1/sqrt(1+x^(2*Order)) - a maximally-flat plateau that drops to a steep cliff whose sharpness Order sets - a distinct flat-topped-then-falling waveform borrowed from filter design.

BuzzerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Buzzer: the harsh low buzz of a game-show wrong-answer or quiz-show timeout, a gritty saw-and-square tone with a mains-frequency rasp, blaring in long held blasts; the rate sets the buzz cadence.

ByteBeat [OSC] [FX] [SYNTH] in: – out: Audio

Bytebeat oscillator (a first as a synth block): the famous 'one-line music' - feed a running sample counter through a short expression of bit-shifts, ORs, ANDs and multiplies, keep the low 8 bits, and a whole chiptune-like texture falls out of pure integer arithmetic. The bitwise interference of the counter against its own shifted copies produces self-similar, glitchy, rhythmic patterns for free. Formula selects among classic bytebeat expressions and Speed sets the counter rate (its pitch and tempo). A bit-arithmetic waveform unlike any oscillator or sample. Needs no input.

BytebeatOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bytebeat: the famous one-line music genre where an 8-bit sample is a bitwise/arithmetic formula of a sample counter t, producing complex self-similar fractal rhythms and timbres from a single expression; several classic formulas are selectable.

Cabasa [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cabasa: a trigger on in0 fires the steel-bead cabasa - a brighter, raspier, grittier rattle than the Shaker, with a sharp metallic-bead attack and a fast scrape-like decay. Tone shifts the bead brightness, Decay the rasp length, Level the output. Needs only a trigger.

CaiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cai attractor: a chaotic system with a squared term in its third equation that broadens the Lorenz butterfly into a fuller, rounder twin-wing orbit, another distinct Lorenz-family flow.

Cajon [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal cajon: a trigger on in0 plays the box drum - a low wooden front-plate resonance (the bass 'thump' when struck in the centre) layered with a bright buzzing rattle of internal snare wires that rings on after the tone. Tune sets the box pitch, Decay the body, Snare the wire-buzz amount and brightness, Level the output. Trigger from a Clock, Euclid or sequencer.

CalciumSpike [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Calcium oscillator: Goldbeter's two-pool model of calcium-induced calcium release, where cytosolic and store calcium pump and leak between each other to fire the spiky Ca2+ oscillations that carry signals inside living cells.

CalogeroOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Calogero-Moser oscillator: two particles in a harmonic trap repelling through an inverse-cube force (the integrable Calogero model) plus a symmetry-breaking perturbation that turns the regular motion chaotic - many-body-physics chaos.

CameraShutterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Camera shutter: the snap of an SLR taking a photo, a two-part mechanical click-clack (mirror up, then the shutter curtain) with a tiny motor whirr of the film advance, fired at the set rate; the iconic ka-chunk.

Campfire [OSC] [FX] [SYNTH] in: – out: Audio

Crackling campfire - a warm noise bed punctuated by random spitting crackles and pops. Crackle sets the pop activity, Tone the brightness, Level the output.

CantorOsc [OSC] [FX] [SYNTH] in: – out: Audio

Cantor-spectrum oscillator (a first as a synth block): an additive tone that keeps harmonic k only when k written in base 3 contains NO digit 1 - the membership rule of the Cantor set, which leaves overtones at 2, 6, 8, 18, 20, 24… and removes the fundamental entirely. The result is a lacunary (gap-riddled) fractal spectrum, the audible cousin of the Cantor middle-thirds set, giving a pitched-but-rootless, metallic-hollow timbre with self-similar holes. Partials sets the highest harmonic considered. A fractal-set spectrum distinct from the other number-theoretic oscillators.

CantorStaircase [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Devil's staircase: the Cantor function, which climbs only on the Cantor set and is flat almost everywhere; sampled around a pitch period it makes a self-similar stepped ramp with plateaus at every scale - fractal staircase modulation/timbre.

CapacitorWhineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Capacitor whine: the piercing high-pitched squeal of a stressed switching power supply or ceramic capacitor (coil whine), a near-sine tone with a thin edge that wavers slightly with load, the maddening whine of cheap electronics.

CardiacAlternans [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cardiac alternans: the action-potential-duration restitution map, where each beat's duration depends on the preceding rest interval; pacing faster period-doubles into long-short-long alternans, the arrhythmia precursor - heartbeat-timing chaos.

CardiacPurkinje [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Purkinje fiber: a reduced model of the heart's Purkinje conduction-fiber action potential with a slow pacemaker current, firing the long-plateau cardiac spikes and, when over-driven, early-afterdepolarization chaos.

CardioidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cardioid oscillator: the heart-shaped curve r = 1 - cos(theta) (an epicycloid with one cusp, and the shape of a directional-microphone pattern) projected to one axis, giving an asymmetric single-lobe waveform with a soft cusp.

CarHornOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Car horn: two tones a small interval apart (typically a major third around A and C-sharp) blown together with a buzzy edge, the brash, beating honk of an automobile horn; the interval knob tunes the dyad.

Carillon [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Carillon bell: a large struck tower bell with rich inharmonic partials and a long, humming decay - the deep, resonant cast-bronze voice. Tune sets the pitch, Decay the ring (several seconds), Level the output. Trigger on in0, in1 = Pitch CV.

CashRegisterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cash register: the cheerful ka-ching of a till ringing up a sale, a bright two-note bell ding followed by the wooden thunk of the drawer popping open, fired at the set rate; the sound of a prize won.

CasioCZ [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Casio CZ-style phase-distortion oscillator: a sine whose phase is warped toward a resonant window, producing a sweepable formant peak. Mod sets the distortion depth and Res the resonant-peak harmonic - classic 80s digital reso without a filter.

CassiniOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cassini-oval oscillator: traces the x-coordinate of a Cassini oval as the angle sweeps a cycle; the Pinch knob squeezes the oval toward its lemniscate limit, narrowing the waveform's lobes and adding harmonics - a smooth two-lobed shape that sharpens into a pinched waist.

Castanets [OSC] [FX] [SYNTH] in: Trigger out: Audio

Castanets: a sharp pair of wooden clicks - two fast highpassed noise bursts a few milliseconds apart, the flamenco hand-percussion snap. Tone sets the brightness, Decay the click length, Level the output. Trigger on in0.

CatalanOsc [OSC] [FX] [SYNTH] in: — out: Audio

Catalan-number oscillator (a first as a synth block): an additive tone whose overtones land on the Catalan numbers - the 1st, 2nd, 5th, 14th, 42nd… harmonics. Because the Catalan numbers explode almost as fast as 4^n, only a few partials fit below Nyquist and they are spaced ever further apart, so the tone is nearly a pure sine with two or three widely-separated, ringing upper partials - a strikingly sparse, bell-clear spectrum. Partials sets how many Catalan overtones to attempt, alias-free up to Nyquist. A combinatorial spectrum distinct from the other number-theoretic oscillators. Needs no input.

CatenaryOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Catenary oscillator: each cycle is one arc of the catenary curve (the shape of a hanging chain, cosh x), normalized to a bipolar waveform - a smooth U with curvature set by Width, giving a soft, rounded timbre between a triangle and a parabola but with the catenary's distinct droop.

CatmullRomOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Catmull-Rom oscillator: each cycle is the Catmull-Rom cubic interpolation kernel - a central lobe flanked by small negative side-lobes (the kernel behind smooth spline interpolation) - giving a rounded pulse with a touch of ringing, distinct from the strictly-positive smoothing windows.

CatPurrOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cat-purr oscillator: a low rumbling tone amplitude-modulated at the ~25 Hz purr frequency with a gentle inhale/exhale swell, the contented buzzing of a purring cat produced by rhythmic laryngeal muscle twitches.

CauchyNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cauchy noise: each sample is drawn from the Cauchy (Lorentzian) distribution via tan(pi(u-1/2)), whose undefined mean and variance produce frequent large outliers among small values - a spiky, heavy-tailed alternative to Gaussian white noise (clamped).

CayleyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cayley-sextic oscillator: the projection cos^3(theta/3)*cos(theta) of Cayley's sextic, where the slow cos^3(theta/3) term shapes a lobed amplitude contour over the fast cosine - a sixth-order algebraic curve giving a distinctive multi-lobed, partially-rectified tone.

CeilingFanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ceiling fan: the soft rhythmic whoosh of a ceiling fan, a band of airflow noise pulsed once per blade pass with a faint motor hum and a slow wobble from a bent blade; the speed knob sets the blade-pass rate.

CelestaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Celesta: a small steel plate struck by a felt hammer over a wooden resonator box, a soft, sweet, bell-like tone with a gentle attack and a pure near-harmonic spectrum (the celesta of the Sugar-Plum Fairy); re-struck at the set rate.

CelloBowOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cello: the bowed-string waveguide tuned to the cello's low range with a darker bow filter and heavier string, giving the deep, woody, resonant sustain of a bowed cello; bow force sets the grip from gentle to gritty.

CellularNeuralNet [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cellular neural network (Chua-Yang): a ring of analog cells coupled to their neighbours through a fixed feedback template and a piecewise-linear output, the architecture that performs image processing in hardware; with an active template it self-organizes into striped or oscillating patterns.

CesOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

CES oscillator: each cycle is the constant-elasticity-of-substitution form (0.5x^Rho + 0.5(1-x)$^{\text{Rho)}}$(1/Rho) over [0,1], whose Rho morphs the curve from a U (negative Rho, min-like) through geometric to an arithmetic blend - a continuously-bending symmetric shape from economics. Bounded.

ChainRattleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rattling chains: the metallic clanking of dragged or shaken chains, a dense scatter of random short bright metal clinks at varying pitches, the ghost-dragging-his-chains sound of a haunting; density sets how violently they rattle.

ChainsawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chainsaw: a two-stroke chainsaw revving, a harsh high buzzing engine with a bright sawtooth snarl and a chain rattle, the lumberjack roar; the rev knob raises the pitch as the throttle opens.

ChantOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chant: a crowd chanting in unison, a rough voiced-vowel tone gated into a steady rhythmic shout pattern (the two-syllable terrace chant of a football crowd); the chant rate sets the tempo of the slogan.

ChaosOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Audio-rate chaotic oscillator built on the logistic map blended with a sine carrier: Chaos pushes it from a clean tone into noisy, glitchy, semi-pitched textures, and Tone tames the top end. A digital-noise sound source.

ChaoticFiberLaser [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chaotic fiber laser: an erbium-doped fiber laser whose long upper-state lifetime gives slow relaxation oscillations that, under pump modulation or feedback, period-double into the spiking intensity chaos used in chaos-communication links.

ChaoticReed [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chaotic reed: a clarinet-style reed-and-bore waveguide blown past its stable regime, where the reed's nonlinear flow valve and the bore feedback lock into the squealing multiphonic and chaotic oscillations real players fight to avoid.

ChapmanRichardsOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chapman-Richards oscillator: each cycle is the forestry growth curve $(1-e^{(-K*x))}P$, a flexible S/ramp where K sets the rise rate and P the shape - from a saturating exponential (P=1) to a sigmoid with a slow start (P>1), a tunable growth-shaped ramp.

Charango [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Charango: the tiny high-pitched Andean lute (traditionally an armadillo-shell body) - five courses of paired strings give a bright, shimmering, chorused pluck with the courses' slight detune beating against each other; the sparkling voice of Andean folk. Shimmer sets the course detune.

ChayNeuron [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chay neuron: a three-variable model of a pancreatic-beta-cell membrane whose calcium dynamics gate bursts of spikes, producing complex chaotic bursting patterns - rhythmic clusters of activity.

Cheby [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chebyshev waveshaping oscillator: a sine is shaped by a Chebyshev polynomial so Order directly selects which harmonic dominates, and Drive blends from pure sine to the full harmonic. Precise additive-style timbres with one knob.

ChebyMap [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Chebyshev map: iterates x -> cos(n * arccos(x)), the degree-n Chebyshev polynomial, which is exactly, provably chaotic on [-1,1] for n>=2 and uniformly fills the interval - mathematically pure deterministic noise.

ChebyshevRippleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chebyshev-ripple oscillator: each cycle is the Chebyshev type-I lowpass magnitude response 1/sqrt(1+Ripple$^{2*T_n(x)}$2), a passband that ripples up and down Order times then drops in a steep cliff - a wavy flat-top with a sharp cutoff, distinct from the smooth Butterworth response.

ChebyshevUOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chebyshev U wave: the second-kind Chebyshev polynomial U_n, which equals sin((n+1)theta)/sin(theta) - a comb of n+1 evenly-weighted partials whose order sets the brightness; the second-kind sibling of the harmonic-stacking first kind.

ChebyshevVOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chebyshev-V oscillator: each cycle is a third-kind Chebyshev polynomial V_n(x) (2x-1, 4x^2-2x-1, 8x^{3-4x}2-4x+1) over [-1,1] - an asymmetric orthogonal-polynomial waveform distinct from the first-kind (T) and second-kind (U) Chebyshev shapes. Order picks the degree.

ChebyshevWOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chebyshev-W oscillator: each cycle is a fourth-kind Chebyshev polynomial W_n(x) (2x+1, 4x^2+2x-1, 8x^{3+4x}2-4x-1) over [-1,1] - the mirror partner of the third-kind V, an asymmetric orthogonal-polynomial waveform with the opposite endpoint behaviour. Order picks the degree.

ChemostatChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forced chemostat: nutrient feeding a microbe feeding a predator in a flow reactor; a periodic dilution rate pushes the three-trophic chain into chaos - a lab-realizable ecological chaos with Monod-type uptake kinetics.

ChenCelikovsky [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chen-Celikovsky attractor: a Lorenz-family system at the opposite end of the Vanecek-Celikovsky classification, with the cross-term moved to a different equation, producing a distinct twin-scroll orbit.

ChenDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chen oscillator: each cycle is the Chen PDF Beta*x$^{(Beta-1)*e}$(x$^{Beta)*e}$(-Lambda(e$^{(x}$Beta)-1)), a lifetime distribution with a bathtub-shaped hazard - it produces a sharply-rising, hard-cut-off bump whose abruptness Beta and Lambda set. Peak-normalized, bounded.

ChenLeeHyper [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Chen-Lee: the rigid-body Chen-Lee attractor extended with a fourth state and feedback into hyperchaos, combining the symmetric multi-lobe orbit with two-directional chaotic expansion.

ChenLeeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chen-Lee attractor: a model of rigid-body rotation with feedback that is also studied in anti-control of chaos, tracing a smooth, symmetric multi-lobed orbit.

ChenMomeni [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chen-Momeni attractor: a chaotic system engineered to produce a four-wing orbit through balanced quadratic couplings, switching between four lobes for jumpy, multi-state modulation.

ChenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chen attractor: a double-scroll chaotic flow (a cousin of Lorenz with a different fold), producing a denser, more energetic chaotic signal for harsh tones and wild modulation.

ChialvoMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chialvo neuron map: a two-variable iterated model of an excitable neuron that spikes and bursts via an exponential recovery term, clocked at a chosen rate for sharp neural firing patterns.

Chip [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chiptune / 8-bit lead (NES-style): a pulse wave at a selectable duty cycle, quantized down to the lo-fi console DAC crunch, with the classic fast ‘arpeggio’ that machine-guns the pitch between chord tones to fake a chord on a single channel - the happy-hardcore / 8-bit-rave lead. Tune sets the pitch (in0 = Pitch CV), Duty the pulse width (0 = thin 12.5 percent .. 1 = 50 percent square), Arp the arpeggio speed in Hz (0 = off; when on it fakes a major chord), Vibrato the pitch wobble, and Level the output.

ChipOsc [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Chiptune pulse oscillator (NES / Game Boy voice): a square with four selectable duty cycles - 12.5%, 25%, 50%, 75% - for the classic 8-bit pulse timbres, from thin and reedy to full and hollow. Duty steps through the four widths, Level scales the output. in0 = Pitch CV.

ChirikovStd [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Standard map: the Chirikov kicked-rotor, the textbook gateway from regular to chaotic motion; raising the kick strength K dissolves smooth orbits into global chaos - tunable order-to-chaos as a sound source.

Chirp [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

Chirp / sweep oscillator: a trigger on in0 sweeps the pitch from the base Freq up (or down) by Range octaves over Time, then holds - the rising ‘pew’ zap and falling laser drop. in1 transposes the base pitch. A one-shot frequency sweep, not the continuous Generative walk.

ChiSquaredNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chi-squared noise: the sum of the squares of k Gaussian samples (degrees of freedom k), the distribution behind variance estimates and goodness-of-fit tests; right-skewed for small k and approaching Gaussian as k grows - a positive, skew-tunable source.

ChlorineArsenite [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Iodate-arsenite reaction: an autocatalytic bistable chemistry in a stirred-flow reactor whose feed-vs-reaction competition flips it between low and high states; a slow inflow modulation gives relaxation switching between the two branches.

ChoirPad [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Choir 'aah' pad: a lush vocal ensemble - several detuned voices through an open vowel formant with a gentle vibrato, the synth-choir pad. Vowel morphs the vowel, Width the detune, Level the output. in0 = Gate, in1 = Pitch CV.

ChoppingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chopping: a knife chopping vegetables on a wooden board, a series of short woody thwack impacts at a steady rhythm each a damped low-mid resonance, the rhythmic prep-cook sound; the chop rate sets the tempo.

ChorusUnisonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chorus-unison oscillator: seven sine voices detuned around the pitch and each slowly wandering in tuning, the rich, swirling 'super-unison' of a thick analog stack or a choir singing in not-quite-unison; the spread sets how lush the swarm.

ChossatGolubitsky [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chossat-Golubitsky icon: an iterated complex map built to commute with a dihedral symmetry group, so its chaotic orbit fills a perfectly symmetric mandala-like attractor - 'symmetry in chaos' as a structured source.

ChuaArray [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled Chua circuits: two Chua double-scroll oscillators linked through their voltages, which synchronize, phase-lock or desynchronize into spatiotemporal chaos as the coupling changes - a minimal cell of a chaotic-circuit array.

ChuaCircuit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chua's circuit: the canonical analog chaos generator, three states coupled through a piecewise-linear (Chua diode) nonlinearity that produces the famous double-scroll attractor - the most-studied chaotic oscillator in electronics.

ChuaCubic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cubic Chua: Chua's circuit with the piecewise-linear diode replaced by a smooth cubic characteristic, which removes the corners and gives a rounder, smoother double-scroll than the classic piecewise Chua.

ChuaInductorless [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inductor-free Chua: an op-amp/gyrator realization of Chua's circuit that replaces the bulky inductor with a synthetic-impedance block, giving the same double-scroll chaos in a fully-active circuit - read as the capacitor voltage.

ChuaMemristor [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Memristive Chua: Chua's circuit with the Chua diode replaced by a flux-controlled memristor, adding a fourth (flux) state and extreme multistability to the classic double-scroll - a canonical memristive chaos circuit.

ChuaMultiScroll [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Multi-scroll Chua: Chua's circuit with a staircase (multi-breakpoint) nonlinearity instead of the single Chua diode, so the attractor grows several stacked scrolls the orbit wanders between - richer chaos than the classic double-scroll.

ChuaSine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sine Chua: Chua's circuit with the diode replaced by a sinusoidal characteristic, so the single double-scroll becomes a long row of scrolls the orbit can hop along - a grid-multi-scroll generalization of Chua.

ChuaTorus [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Torus-breakdown Chua: Chua's circuit parameterized in its two-frequency quasiperiodic regime, where an invariant torus wrinkles and breaks into chaos as the coupling shifts - the Ruelle-Takens route inside the canonical chaos circuit.

CicadaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cicada oscillator: a piercing high drone built from a very fast train of tymbal clicks (the buckling ribs of the cicada's sound organ) under a slow intensity swell, the deafening summer wall of sound a chorus of cicadas makes.

Cicadas [OSC] [FX] [SYNTH] in: – out: Audio

Cicada drone - the dense, buzzing, amplitude-pulsing wall of a summer afternoon. Buzz sets the rasp rate, Pitch the brightness, Level the output.

Cimbalom [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cimbalom: the Hungarian hammered dulcimer - metal strings STRUCK by soft mallets (a sharp struck excitation, not a pluck) ringing into a long, bright, shimmering metallic sustain; the ringing, reverberant voice of Romani and Eastern-European music; re-struck at the set rate.

CircadianGoodwin [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Goodwin circadian oscillator: the original three-variable enzyme model (mRNA -> protein -> repressor, the repressor inhibiting the gene through a steep Hill term) that gives the ~24-hour molecular oscillation at the core of circadian clocks.

CircadianOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Circadian oscillator: a smooth, asymmetric daily-rhythm curve (a fundamental plus a second harmonic to model the steep morning rise and gentle evening decline) of the body's internal clock, rendered as a slow, organic, sun-like swell.

CircEaseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Circular-ease oscillator: each cycle is the easeOutCirc curve sqrt(1-(t-1)^2), a quarter-circle arc that rises steeply at first then flattens to a horizontal tangent at the top - a geometric ramp with a distinctly rounded shoulder unlike a parabola or cosine.

CIRProcess [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cox-Ingersoll-Ross process: dr = a(b-r)dt + sigmasqrt(r)dW, a mean-reverting random walk whose square-root diffusion keeps it non-negative (used for interest rates and as Heston's variance) - colored noise that hugs a floor and reverts to a level.

Clap [OSC] [FX] [SYNTH] in: Trigger out: Audio

Analog hand-clap: a trigger on in0 fires band-passed noise shaped by a multi-burst envelope - three quick retriggers then a diffuse tail - the characteristic 909 clap. Tone sets the band-pass colour, Spread the gap between the bursts, Decay the tail length, and Level the output.

Clap707 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-707 hand clap: three rapid noise bursts plus a short bright tail through a highpass - tighter and crisper than the 808 clap, the 707's house clap. Tone sets the brightness, Decay the tail, Level the output. Trigger on in0.

Clap808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 hand clap: a trigger fires three rapid bursts of bandpassed noise followed by a longer diffuse tail - the stacked-hands attack and room ring of the iconic 808 clap. Tone sets the noise colour, Decay the tail length, Level the output. Trigger on in0.

Clap909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

Handclap: a trigger on in0 fires three quick bursts of band-passed noise (the multiple hands) followed by a longer diffuse tail - the classic drum-machine clap. Spread sets the gap between bursts, Decay the tail length, Tone the noise colour, Level the output.

ClappOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Clapp oscillator: the Clapp RF oscillator (a Colpitts with an extra series capacitor) driven by its transistor nonlinearity into chaos, a three-state circuit with a tighter frequency-defining loop than the Colpitts.

ClapSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth hand clap: a gate fires a cluster of fast high-passed noise bursts with a reverb-like tail, recreating the multi-slap transient of several hands clapping - a drum-machine clap voice.

ClapSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hand-clap voice: three quick stuttered noise bursts followed by a diffuse decaying tail, the 808 clap trick that mimics many hands not quite clapping together; re-triggered at the set rate, the spread knob sets the burst spacing.

Clarinet [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Clarinet: a closed-pipe reed voice with only odd harmonics (a softened square through a woody lowpass) plus breath noise, played by holding in0 as breath. The hollow, reedy clarinet timbre. Tune sets the pitch, Breath the air noise, Level the output. in0 = Breath gate, in1 = Pitch CV.

ClarinetWGOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clarinet waveguide: a delay-line bore with a phase-inverting reflection (giving the odd-harmonic, hollow timbre of a closed-open cylinder) driven by a clipped single-reed nonlinearity; the breath knob blows it from a pure tone into a squawk.

ClausenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clausen oscillator: sums sin(k*theta)/k^Order over the harmonics (the Clausen function for Order=2) - the sine-series counterpart to the polylog's cosine series, giving a smooth odd-symmetric waveform whose harmonic roll-off Order sets, band-limited to Nyquist.

Clav [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Clavinet (Hohner D6): a Karplus-Strong plucked string struck by a gate, run through a pickup comb filter that notches the spectrum - the bright, percussive, slap-funk voice of funk, disco, French house and funky house. A rubber-tip excitation burst fills the string, the feedback loop rings at the pitch, and the comb pickup gives the signature hollow bite. Tune sets the pitch (in1 = Pitch CV), Decay the string sustain, Bright the string damping and attack brightness, Pickup the comb depth (the clav twang), and Level the output. A gate on in0 plucks it; play it staccato.

Clave [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal clave / woodblock: a trigger on in0 strikes a solid piece of wood - two tight high resonant modes (about 1 : 1.5) that ring for only a few milliseconds, the bright dry 'tok' that cuts through any mix. Tune sets the pitch (clave up high, woodblock lower), Decay the very short ring, Level the output. No pitch bend, no body - just wood. Trigger from a Clock, Euclid or sequencer.

Clave808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 clave / rimshot click: a single high resonant sine ping with a very fast decay - the dry, sharp woodblock 'tick' that cuts through a mix. Tune sets the pitch, Decay the click length, Level the output. Trigger on in0.

ClaveHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clave: a single very-high-Q resonant mode rings briefly when struck, the sharp, bright, dry crack of a clave or rimshot - a one-mode percussion voice excited by the input.

ClaveSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clave voice: a single high, dry, quickly-decaying resonant tone, the crisp 'tok' of two claves or a woodblock struck together, re-triggered at the set rate; the classic Latin-percussion click.

Clavinet [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Clavinet: a hammered string struck against a metal anvil - a bright, percussive Karplus-Strong pluck with a sharp funky attack, the Hohner D6 wah-bait. Bright sets the string brilliance, Decay the damping, Level the output. Trigger on in0, in1 = Pitch CV.

ClavinetVoice [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Clavinet voice: a bright, tightly-damped Karplus string struck by a gate and read through a pickup comb filter, the funky, percussive, wah-friendly tone of a Hohner Clavinet.

Clavioline [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Clavioline: the 1947 valve monosynth (the riff on the Tornados' 'Telstar') - a buzzy reed-organ square with octave-thick harmonics and a slight vibrato. Buzz sets the harmonic edge, Vibrato the wobble, Level the output. in0 = Gate, in1 = Pitch CV.

ClickConsonantOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Click oscillator: very brief, sharply-band-passed transients produced by a sudden release of an oral vacuum, the click consonants of Khoisan and Bantu languages; the tone knob shifts the click from a deep 'pop' to a high 'tsk'.

CliffordAttr [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clifford attractor: a 2-D trigonometric iterated map (sin/cos of cross terms) that traces delicate, swirling fractal curves - a distinctive strange-attractor texture, sampled at a chosen rate.

ClockEscapementOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clock escapement: the tick-tock of a mechanical clock, two slightly-different short resonant clicks alternating at the pendulum rate (the escapement releasing one tooth per swing); the rate sets the pendulum length.

ClockTick [OSC] [FX] [SYNTH] in: – out: Audio

Clock - a steady tick-tock with two alternating click pitches, the metronome/grandfather-clock pulse. Rate sets the tempo, Tone the click character, Level the output.

CO2Laser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modulated CO2 laser: a single-mode laser whose cavity loss is periodically modulated drives the intensity, population inversion and polarization into chaotic spiking - the experimentally-classic route to laser chaos.

CobwebModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cobweb market map: prices set from last period's expectation against a nonlinear supply curve; when supply is steep enough the price iterates period-double into chaotic boom-bust swings - the cobweb theorem gone wild.

Coin [OSC] [FX] [SYNTH] in: – out: Audio

Arcade coin pickup - the classic two-note 'bling' reward chime, auto-repeating. Rate sets the repeat interval, Pitch the note, Level the output.

CoinPickupOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coin pickup: the iconic platformer coin-grab, a square-wave tone that snaps up a major sixth partway through (the cheerful two-note 'bling') and decays, fired at the set rate; the reward sound of arcade games.

CollatzAudio [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Collatz sequence: the famous 3n+1 iteration (halve if even, else 3n+1) restarted from a growing seed, whose wildly-varying hailstone trajectories are clocked out as an erratic stepped control voltage - number theory you can hear.

CollatzFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Collatz fractal: the 3n+1 rule extended to the complex plane by its smooth interpolation z -> (2+7z-(2+5z)cos(pi z))/4, whose iteration draws the fractal boundary of the Collatz basins; the reseeded orbit's real part makes a spiky CV.

ColorNoise [OSC] [FX] [SYNTH] in: – out: Audio

Noise generator producing white, pink or brown noise from an xorshift source filtered to taste. Color selects the spectrum: flat white, a one-pole pink-tilted blend, or a heavily integrated brown noise with a steeper low-frequency emphasis. Level sets the output amplitude, making it a self-running source for textures, test signals or modulation.

ColpittsHyper [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Colpitts: the chaotic Colpitts LC oscillator augmented with an extra LC tank and coupling, lifting it to four dimensions with two positive Lyapunov exponents - a broadband RF hyperchaos generator.

ColpittsOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Colpitts: the classic Colpitts LC oscillator pushed into chaos by its transistor's exponential nonlinearity, a real radio-frequency circuit whose three state variables (two capacitor voltages, one inductor current) tumble chaotically.

CombFmOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Comb-resonated FM: a 2-operator FM tone is run through a feedback comb tuned to the same fundamental, so the comb reinforces the FM's harmonic series into a sharp, resonant, almost-physical timbre that neither FM nor a comb makes alone.

CombHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Comb-harmonic oscillator: a full harmonic series whose partial amplitudes are scalloped by a raised-cosine comb, carving evenly-spaced spectral notches like a comb/flanger frozen onto an oscillator; the spacing knob slides the teeth through the spectrum.

CombResonanceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Comb-resonance oscillator: a periodic impulse fed into a tuned feedback comb filter, whose evenly-spaced resonant peaks ring out a hollow, metallic, pitched timbre - the sound of plucked-string and flanger resonance frozen into a tone.

CombustionInstability [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Combustion instability: an acoustic chamber mode coupled to a time-lagged, saturating flame heat-release (the Crocco n-tau model), which self-amplifies into the screeching limit-cycle and chaotic howl that plagues rockets and gas turbines.

ComplexOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Complex oscillator (Make Noise DPO / Buchla 259): a modulation oscillator thru-zero-FMs a primary oscillator, whose output is then run through a sine wavefolder for rich, evolving harmonics. Freq sets the primary pitch, Ratio the modulator's frequency relative to it, FM the thru-zero modulation depth (inharmonic clang at high settings), and Fold the wavefolder drive (adds bright folded harmonics). One voice spanning clean tones to metallic, animated timbres. Distinct from the separate ThruZero and Wavefolder blocks.

CompoundPoisson [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Compound Poisson process: a running total that jumps by a random amount each time a Poisson event fires and holds flat in between, the staircase model of accumulated insurance claims or queue arrivals - a piecewise-constant random ramp, reset on overflow.

ComputerBootOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Computer boot: the cheerful ascending run of data tones a retro computer plays on power-up, a square-wave stepping upward through a scale then looping, the bleeping startup chime of a vintage machine; the rate sets the boot speed.

ConfirmBlipOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Confirm blip: the short positive blip of a confirmed action or accepted input, a single clean tone with a quick upward pitch chirp, the OK / accepted feedback; fired at the set rate.

Conga [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal conga: a trigger on in0 strikes a tall hand drum - a deeper, woodier membrane than the Bongo, with modes near 1 : 1.6 : 2.2, a longer open ring and a pronounced pitch drop. Tune sets the head pitch, Decay the ring length, Tone moves from an open tone toward a tighter heel-slap, Level the output. Trigger from a Clock, Euclid or sequencer.

Conga808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 conga: a tuned, resonant sine drum with a short pitch drop and medium decay - the round, woody 808 conga/tom voice used for melodic fills. Tune sets the pitch, Decay the length, Level the output. Trigger on in0.

CongaHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conga/bongo: two pitched membrane modes with a short decay ring when struck by the input, the round, hand-drum tone of a tuned skin - feed an impulse or gate to play.

CongaSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conga voice: a clear pitched membrane tone with a slight upward pitch snap and a short skin-like decay, the warm open-hand-drum hit of a conga or bongo, re-triggered at the set rate; tune from a low tumba to a high quinto.

ConservativeJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conservative jerk: a third-order jerk system with zero net damping (the divergence vanishes), so it preserves phase-space volume and its chaos is conservative - dense, non-attracting, sticky orbits rather than a thin attractor.

ContopoulosOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Contopoulos system: a Hamiltonian model of a star orbiting in a galactic potential with two coupled anharmonic oscillators, whose ordered orbits dissolve into chaos as the coupling grows - conservative galactic-dynamics chaos.

ConveyorRumbleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conveyor rumble: the low continuous rumble of a running conveyor belt over its rollers, with a periodic bump each time the belt seam passes a roller, the industrial drone of a moving belt; the bump rate sets the belt speed.

CorkPopOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cork pop: the hollow 'pop' of a cork leaving a bottle (or a finger-in-cheek pop), a short resonant blip with a fast upward pitch chirp and a soft thump, fired at the set rate; the celebratory uncorking.

CoronaDischargeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Corona discharge: the faint hiss of ionised air leaking off a high-voltage line, a band of high-frequency noise with a continuous fizz of tiny micro-discharges, the eerie crackle around power transmission gear in damp weather.

CoronaTrichel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trichel-pulse corona: the self-quenching current pulses of a negative corona discharge, where each avalanche builds space charge that chokes itself off until it clears; raising the voltage drives the pulse train from regular into chaotic.

CosineMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cosine map: iterates x -> a cos(x), whose single smooth fold yields period-doubling and chaos as the amplitude rises - a transcendental cousin of the logistic map with a rounder, symmetric hump.

CosmicNoiseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cosmic noise: the steady background hiss of the radio sky - galactic synchrotron emission and the cosmic microwave background - modelled as a tunable power-law colored noise from white toward deep red; the color knob sweeps the spectral tilt of the cosmos.

CoughOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cough: a sudden glottal burst - a short voiced thump opening into a rasp of turbulent throat noise that quickly fades, the explosive clearing of a cough, fired at the set rate.

CoulletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coullet attractor: a jerk-equation chaotic system with a cubic restoring term, producing a smooth single-scroll orbit - a clean, low-cost chaos source related to the Genesio-Tesi family.

CountdownBeepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Countdown beep: a timer ticking down, a short clean beep repeated at a steady rate then a long final tone, the tense countdown of a game show or bomb timer; the rate sets the beep interval.

CoupledFitzHugh [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled FitzHugh-Nagumo: two excitable cells linked by diffusion that synchronize in-phase, lock anti-phase or fire chaotically out of step as the coupling and excitability change - a minimal model of neural-pair rhythm coordination.

CoupledFormOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled-form oscillator: two state variables rotate around a circle via the magic-circle recursion s1+=epss2; s2-=epss1 with eps=2sin(pif/fs), producing a sine by recursive feedback rather than a phase accumulator. Magnitude-stable, self-starting, and naturally in quadrature - the resonant-oscillator structure used in Chamberlin SVFs.

CoupledLogistic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled logistic maps: two logistic-map oscillators that exchange a fraction of their state each step (a coupled-map-lattice cell), which can synchronize, beat or desynchronize into spatiotemporal chaos as coupling changes.

CoupledLorenz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled Lorenz: two Lorenz oscillators linked through their x-variables, sweeping from independent chaos through on-off intermittent synchronization to full identical sync as coupling rises - the textbook chaos-synchronization demonstration.

CoupledMapLattice [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled map lattice (Kaneko): a ring of logistic maps each diffusively coupled to its two neighbours, producing spatiotemporal chaos, traveling waves and frozen-random domains as the coupling and nonlinearity vary - local coupling, unlike the mean-field global version.

CoupledPendulumPair [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled pendulum pair: two pendula linked by a torsion spring trade energy back and forth (the classic beating-modes demo); driven hard, their angles fall out of step into chaotic, never-repeating swings.

CoupledRossler [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled Rossler: two Rossler oscillators linked through their x-variables, which transition between phase synchronization, lag synchronization and high-dimensional hyperchaos as the coupling and detuning change - a classic chaos-synchronization testbed.

CournotDuopoly [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cournot duopoly: two firms adjust output toward their best response against the other's last move; with nonlinear costs and fast adjustment their outputs period-double into chaotic quantity wars - economic game-theory chaos.

Cowbell [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog cowbell (808-style): a trigger on in0 fires two detuned pulse oscillators a fixed ratio apart, band-passed and shaped by a short percussive decay for the classic metallic clank. Tune sets the lower oscillator, Detune the ratio of the upper one (the cowbell interval), Decay the ring length, and Level the output. Patch a Clock, Euclid or sequencer gate into in0.

Cowbell808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 cowbell: a trigger on in0 fires the two-square-oscillator ‘clonk’ (640 Hz + 800 Hz) through a bandpass with a medium decay - the unmistakable 808 cowbell. Tune scales the pitch, Decay the length, Level the output.

CowbellSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth cowbell: two square oscillators a dissonant interval apart, band-passed and shaped by a sharp percussive envelope on each gate - the unmistakable 808 cowbell voice.

CowbellSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cowbell voice: two square oscillators a non-harmonic interval apart (the famous 808 cowbell ratio near 845/540 Hz) gated by a sharp decay, the clanky percussion that demands more cowbell; re-triggered at the set rate.

CowMooOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cow moo: the long low lowing of a cow, a deep voiced ‘moo’ vowel that swells and falls with a slight pitch droop and a breathy edge, fired at the set rate; the placid sound of the pasture.

CPGNetwork [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Central pattern generator: three mutually-inhibiting neural oscillators that settle into rotating phase patterns (the circuits that produce walking, swimming and breathing gaits); detuned, they wander chaotically between gait patterns.

CR78Hat [OSC] [FX] [SYNTH] in: Trigger out: Audio

Roland CR-78 hi-hat: the lo-fi metallic hat of the CompuRhythm - a thin cluster of detuned squares mixed with noise through a sharp highpass, with a fast decay. Tone sets the brightness, Decay the length, Level the output. Trigger on in0.

Crackle [OSC] [FX] [SYNTH] in: – out: Audio

Vinyl/dust crackle generator - random filtered pops and ticks for lo-fi, vinyl and analog-warmth textures. Density sets how busy the crackle is, Tone the pop brightness.

CraneSway [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crane sway: a payload hanging from a trolley that is driven back and forth; the pendulum’s nonlinear response to the trolley acceleration swings, beats and chaotically slews - the load-sway problem crane controllers fight.

Crash808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 crash cymbal: a long, splashy wash of six inharmonic squares plus noise through a highpass, decaying over a second - the 808’s big accent cymbal. Tone sets the brightness, Decay the wash length, Level the output. Trigger on in0.

Crash909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

Crash cymbal: a long, bright wash of inharmonic squares plus white noise through a highpass, decaying over a second or more - the splashy crash for accents and builds. Tone sets the brightness, Decay the wash length, Level the output. Trigger on in0.

CreakyFloorOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Creaky floor: an old wooden floorboard groaning under a footstep, a low woody resonance that bends in pitch as the weight shifts with a stick-slip creak, fired at the set rate; the giveaway of sneaking through an old house.

CreakyVoiceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Creaky voice (vocal fry): very low, irregularly-spaced glottal pulses with jittered timing, the popping, rattling register at the bottom of the voice; the jitter knob sets how erratic and gravelly the creak becomes.

Creature [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Monster / creature roar: a vocal-cord source (saw plus a sub-octave for size) roughened by a vocal-fry flutter, shaped by a sweeping formant band-pass (the vowel of the roar) and driven through distortion - the layered beast vocalization of film and game sound design. Tune sets the pitch (in1 = Pitch CV), Growl the vocal-fry roughness (the guttural rasp), Formant the vowel centre, Size the sub-octave weight and how big the creature reads, Drive the aggression, and Level the output. A gate on in0 sounds the roar with a pitch-up attack. Distinct from the musical Growl bass - this is an organic vocalization.

CreepyLaughOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Creepy laugh: a distant unsettling cackle, a low voiced tone chopped into a ‘heh-heh-heh’ laugh pattern with a hollow distorted edge, the disembodied villain laugh of horror; the laugh rate sets the cackle speed.

CricketChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cricket chirp: a high carrier tone gated into short pulses grouped into chirps with silent gaps, the stridulation of a cricket whose chirp rate famously tracks temperature (Dolbear’s law); a pulsed bioacoustic insect tone.

Crickets [OSC] [FX] [SYNTH] in: – out: Audio

Night crickets - rhythmic high chirps over a faint air bed for an evening/forest ambience. Rate sets the chirp tempo, Pitch the chirp frequency, Level the output.

CristalBaschet [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cristal Baschet: the Baschet brothers’ glass-rod organ - wetted fingers stroke glass rods coupled to metal resonators, a friction-driven, extremely high-Q glassy tone with a slow shimmer and long ring, the ethereal sound of Forbidden Planet / experimental film scores.

CrookedEggOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crooked-egg oscillator: each cycle is cos$^{3(theta)+sin}$3(theta), the radial form of the crooked-egg curve - a lopsided wave that factors into (cos+sin)(1-cos*sin), giving an asymmetric fundamental-plus-third-harmonic shape with a distinct skew. Bounded.

CrotaleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crotale: a small thick tuned bronze disc struck with a mallet, ringing a high, bright, slowly-decaying tone with shimmering inharmonic upper partials (the antique-cymbal sound of orchestral crotales); re-struck at the set rate.

Crotales [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Crotales (antique cymbals): small tuned bronze discs with a bright, high, shimmering inharmonic ring and a very long decay - the sparkling orchestral bell-cymbal. Tune sets the pitch, Decay the shimmer, Level the output. Trigger on in0, in1 = Pitch CV.

Crowd [OSC] [FX] [SYNTH] in: — out: Audio

Crowd murmur - band-passed noise with a slow random swell for the ambient hubbub of a room or stadium. Energy sets the activity, Tone the brightness, Level the output.

CrowdGaspOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crowd gasp: a sudden collective intake of breath as an audience reacts in shock, a swelling band of airy breath noise that rushes in and tensely holds before fading, fired at the set rate; the held-breath of a stunned crowd.

CrowdLaughOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crowd laugh: an audience erupting in laughter, many overlapping voiced ‘ha-ha’ bursts at scattered pitches swelling to a peak then dying down, the sitcom laugh-track wash; the rate sets the laughter’s pace.

CrowdMurmurOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crowd murmur: the indistinct babble of a room full of talking people, a band of voiced-formant noise that swells and ebbs with the ebb and flow of conversation, the hum of a busy hall before a show.

CrowdWhistleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Crowd whistle: the piercing scattered whistles of an excited or jeering crowd, randomly-timed high near-sine chirps at varying pitches over a faint murmur, the catcalls of a stadium; the density sets how many are whistling.

CrtMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chua cubic map: a discrete two-dimensional map with the cubic characteristic of a Chua circuit, folding the plane through an S-shaped nonlinearity into a double-scroll-flavoured fractal attractor.

Cubic1DMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cubic map: the odd-symmetric iterated map x -> a x - x^3, whose two humps give a period-doubling cascade with symmetry-breaking and coexisting attractors - a bipolar cousin of the logistic map.

Cuica [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal cuica: a trigger on in0 plays the Brazilian friction drum - a resonant tone whose pitch swoops up or down as a wet thumb drags the internal stick, the squeaking, talking, almost human voice of a samba bateria. Tune sets the base pitch, Decay the ring, Glide the direction and depth of the pitch swoop (bipolar), Level the output. Trigger from a Clock, Euclid or sequencer; modulate Glide for talking lines.

CurseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Curse: a dark malediction, a low cluster of dissonant detuned partials sliding slowly downward with a menacing beating growl, the ominous swell of a hex or debuff; the slide rate sets how fast the dread descends.

CuspMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cusp map: an iterated map with a square-root nonlinearity (x -> 1 - a*sqrt(|x|)) that folds the line through a cusp, giving a spiky chaotic stream distinct from the polynomial maps.

CutleryClinkOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cutlery clink: metal forks, knives and spoons clinking as they are sorted or dropped in a drawer, a random scatter of short bright high-pitched metallic clinks, the table-setting jingle; density sets how busy.

CyclicCARing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cyclic CA: a ring of cells with N colours where a cell advances to its next colour if a neighbour already holds that colour; this ‘eat-the-next-colour’ rule spontaneously forms rotating waves and spirals that lock into a periodic pattern - read out as a mean-colour CV.

CycloidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cycloid oscillator: the arch traced by a point on the rim of a rolling wheel (the brachistochrone and tautochrone curve), here the height profile per revolution - a series of smooth arches with cusps where the point momentarily touches the ground.

Cymbal [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog cymbal (808-style metal): a trigger on in0 fires six square oscillators tuned to a fixed inharmonic ratio set, summed and high-passed, with a fast onset plus a long metallic tail. Distinct from the Hat (which filters noise) - this is the additive square-oscillator method for ride/crash timbres. Tune sets the base frequency of the cluster, Decay the tail length, Tone the high-pass brightness, and Level the output. Patch a Clock, Euclid or sequencer gate into in0; patch a Note source into in1 to keytrack the cluster.

Cymbal808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808-style cymbal/crash: like the 808 hat but with six inharmonic square oscillators ringing through a long, shimmering, slowly-decaying highpassed tail. Tone sets the brightness, Decay the (long) ring length, Level the output.

CymbalSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth cymbal: six square oscillators at inharmonic ratios are summed, band-passed bright and mixed with high-passed noise, then struck by a gate with a long decay - the metallic 808-style cymbal/ride voice.

CymbalSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cymbal voice: a dense bank of inharmonic square oscillators high-passed into a shimmering metallic wash with a long decay (the 909 crash recipe), re-triggered at the set rate; the decay knob sets ride versus crash.

CzPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

CZ phase-distortion pulse: the cosine phase ramps to the half point, holds flat, then ramps on - the held section pins the carrier at its trough, producing the hollow, square/pulse-like Casio CZ tone. Width sets the ramp portion (narrow = squarer/brighter).

CzSawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

CZ phase-distortion saw: a cosine carrier is read through a two-segment phase ramp with an inflection at Bend, speeding the phase early then slowing it - the classic Casio CZ sawtooth, morphing from a pure sine (Bend 0.5) to a bright buzzy saw (Bend near 0) with no aliasing-prone hard edges.

DadrasOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dadras attractor: a four-parameter chaotic flow that sweeps out a wide, multi-scroll orbit, a high-energy strange attractor useful for broad, unpredictable modulation and rough tones.

DagumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dagum oscillator: each cycle is the Dagum (Burr-III) PDF $apx^{(a*p-1)/(1+x}a)\^(p+1)$, a flexible skewed income-distribution bump - A controls the rise sharpness and P the tail, peak-normalized to a bounded asymmetric waveform distinct from the Beta/Wald skews.

DagumQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dagum-quantile oscillator: each cycle is the Dagum inverse-CDF ($p^{(-1/P)-1)}(-1/A)$, an income-distribution quantile that stays low then accelerates into a heavy upper tail - A and P jointly set the rise and the tail steepness. Peak-normalized, bounded.

DaisyWorld [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Daisyworld: Lovelock and Watson's model of black and white daisies regulating a planet's temperature through albedo feedback; under a rising forcing it gives slow self-stabilizing population oscillations - Gaia-hypothesis dynamics as a modulator.

DansgaardOeschger [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dansgaard-Oeschger events: the abrupt millennial warm/cold flips recorded in ice cores, modelled as a bistable climate potential with a slow ramp that triggers stochastic-resonance-like jumps between glacial states - very-slow climate flicker.

Darbuka [OSC] [FX] [SYNTH] in: Trigger out: Audio

Darbuka / goblet drum: the sharp Middle-Eastern hand drum with a deep 'dum' at the centre and a crisp 'tek' at the rim - Rim morphs from the low boom to the bright edge slap. Tune sets the pitch, Rim the dum-to-tek balance, Level the output. Trigger on in0.

Daxophone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Daxophone: Hans Reichel's bowed wooden tongue - a strip of wood bowed at one end, its pitch set by where it is clamped, producing uncannily vocal, animal-like cries and barks; a friction-driven waveguide through a movable formant resonance that gives it the talking, throaty character. Voice sweeps the formant (the clamp position).

DayGrowth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Day's growth map: Richard Day's classical model where capital grows by saving but is dragged down nonlinearly by pollution/diminishing returns, giving a hump-shaped map that period-doubles into chaotic boom-bust growth.

DblPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double pendulum: simulates two coupled pendulums whose motion is famously chaotic and exquisitely sensitive to initial conditions, reading the bob angles as the output - swinging, organic chaos with sudden flips.

DCMotorChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

DC motor chaos: a brushed DC motor under PWM supply whose Stribeck (velocity-weakening) friction and back-EMF coupling make the shaft speed stick-slip and chaotically ripple - a real drive-system instability.

DeathwatchBeetleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Deathwatch beetle: the eerie tapping of a deathwatch beetle knocking its head against old timber to attract a mate, a short series of dry resonant ticks repeated after a pause, the ominous tick of quiet old houses.

DeffuantOpinion [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Deffuant model: agents holding continuous opinions meet in random pairs and move toward each other only if they already agree within a confidence threshold; wide thresholds collapse everyone to consensus, narrow ones freeze into separate opinion clusters - the spread tracks that.

DegaussOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Degauss: the deep wobbling thunk of a CRT monitor degaussing, a low mains-frequency tone that surges loud as the coil energises then rings down and decays to nothing over a couple of seconds, looping at the trigger rate; the classic booiiing of an old screen.

DeJongAttr [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

De Jong attractor: a four-parameter trigonometric iterated map famous for intricate web-like fractal forms, producing a busy, fine-grained chaotic texture as an audio/CV source.

DelayDuffing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Delayed Duffing: the twin-well Duffing oscillator whose restoring force is augmented by a time-delayed copy of its own position, so the feedback delay alone (no external forcing) destabilizes it into chaos.

DelayedHenon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Delayed Henon map: the Henon recurrence rewired to feed back a delayed value ($x_n = 1 - a x_{n-2}^2 + b x_{n-1}$), which raises the attractor's dimension and can make it hyperchaotic - delay-boosted map chaos.

DengToggle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Deng toggling system: a chaotic flow whose nonlinearity is a toggling (sign-switching) feedback that flips the attractor between scroll regions, generating a grid of scrolls the orbit hops across - a multi-scroll chaos generator.

DequanLiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dequan Li attractor: a chaotic flow with several quadratic couplings producing a dense, fast, multi-scroll orbit - a busy, energetic strange-attractor source.

DerivativeGaussianOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Derivative-of-Gaussian wavelet: the nth derivative of a Gaussian (order 1 is an edge-detecting odd bump, order 2 is the Mexican hat, higher orders add lobes), a smooth family of DC-free packets whose order sets the number of oscillations.

DeskBellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Service bell: the little dome bell on a hotel or shop counter, a bright struck metallic ding with shimmering inharmonic partials and a long ring-out, fired at the set rate, the 'ring for service' sound; the rate sets the dinging.

Devil [OSC] [FX] [SYNTH] in: — out: Audio

Devil's-staircase oscillator (a first as a synth block): synthesises the Cantor function - the 'devil's staircase' that climbs from 0 to 1 entirely on the Cantor set, staying perfectly FLAT across every gap yet still rising overall, with no jumps. As a repeating waveform it is a fractal ramp: a sawtooth assembled from infinitely many flat plateaus and self-similar steps, a singular curve whose spectrum is unlike any band-limited ramp. A monotone fractal staircase distinct from the jagged Weierstrass and Blancmange. Needs no input.

Dholak [OSC] [FX] [SYNTH] in: Trigger out: Audio

Dholak: the two-headed Indian/folk hand drum - a low bass head and a higher treble head struck together, with a warm, woody tone. Tune sets the pitch, Treble the high-head mix, Level the output. Trigger on in0.

DialToneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dial-tone oscillator: the continuous North-American dial tone, the precise sum of a 350 Hz and a 440 Hz sine that says the line is ready; the Detune knob nudges both for that slightly-beating analog-exchange character.

Didgeridoo [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Didgeridoo voice: a gate on in0 drones a deep buzzing tone whose overtones swell and fade under a slow wah-like formant sweep, the circular-breathing rhythm of the instrument. Freq sets the drone pitch (very low), Wobble the speed and depth of the formant sweep, Tone the overall brightness, Level the output. Hold the gate to drone; in1 is the Pitch CV.

DidgeridooOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Didgeridoo oscillator: a low lip-buzzed drone over which a slowly-moving vocal-tract formant carves the wah-wah overtone shapes and rhythmic vocalizations of circular-breathing didgeridoo playing; the wobble knob animates the formant.

DieselEngineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Diesel engine: the gruff idle of a diesel motor, a low rumble pulsed by the knocking firing of its cylinders, the chugging clatter of a truck or generator at rest; the RPM knob raises the idle speed and pitch.

DifferenceToneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Difference-tone oscillator: two upper sine carriers passed through a squaring nonlinearity that generates the Tartini combination tone at their frequency difference, so a phantom low pitch (heard but not present in the inputs) emerges from the beating pair.

DiffusionAggregation [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Diffusion-limited aggregation: a random walker wanders the ring until it touches the growing cluster and sticks, then a new walker is released, building the branching fractal of mineral dendrites and electrodeposits; the cluster's growing size is the output, resetting when it fills.

DigammaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Digamma oscillator: each cycle traces the digamma psi(x) ~ ln(x) - 1/2x - 1/12x^2 over a span, a logarithmically-curved ramp that rises fast then flattens - a soft asymmetric saw with a distinct log knee, brighter at the start of each period than a linear ramp.

DingCorrectOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Correct ding: the bright reward bell of a right answer, a clear high bell tone (fundamental plus a sweet octave-ish partial) ringing out and fading, fired at the set rate; the cheerful affirmation chime.

DiodeChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Diode chaos circuit: a sinusoidally-driven RLC loop with a diode (half-wave) nonlinearity, one of the simplest physically-built chaotic circuits, whose rectifying kink folds the driven resonance into chaos.

DiodeOsc [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Diode-shaped oscillator: a sawtooth core pushed through an exponential diode curve, going from a soft sine-ish tone to a bright, brassy, slightly-clipped buzz as Shape rises - the warm, organic waveshape of West Coast and vintage analog timbres. in1 = Pitch CV.

DiphthongOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Diphthong oscillator: a buzzy harmonic source whose formant emphasis glides from one vowel to another and back each cycle (as in the 'oy' of boy or 'ay' of day), the moving-resonance gesture that turns a steady vowel into a diphthong.

DirichletKernelOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dirichlet kernel: the periodic sinc sin((N+1/2)theta)/sin(theta/2), the sum of the first N cosine harmonics at equal amplitude (the partial-sum kernel of a Fourier series); it makes a bright band-limited pulse that rings with Gibbs-like sidelobes.

DishClatterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dish clatter: plates and bowls clattering as they are stacked or washed, a random scatter of short resonant ceramic clinks and dull clacks at varying pitches, the kitchen-sink bustle; density sets how busy the clatter.

DismissSwooshOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dismiss swoosh: the soft downward swoosh of dismissing a notification or closing a panel, a filtered-noise band sweeping quickly downward in pitch, the swipe-away / dismiss feedback; fired at the set rate.

DistantScreamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Distant scream: a far-off human scream muffled by distance, a high voiced tone with a rising-then-falling pitch arc and a rough strained edge, band-limited and reverberant as if heard from far away; fired at the set rate.

DistantStormOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Distant storm: the continuous low rolling rumble of thunder far across the horizon, a deeply low-passed noise bed that swells and recedes in slow waves, never the sharp crack of nearby lightning; the swell rate sets how restless the storm.

DixonEllipticOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dixon elliptic oscillator: integrates the defining cubic-symmetry ODE sm'=cm^2, cm'=-sm^2 (with sm^{3+cm}3=1) to produce the sm waveform - a sine relative of the lemniscate functions but built on the Fermat cubic, giving a distinctly lopsided, three-fold-flavoured periodic shape unlike sine or the Jacobi elliptic waves. RK2-integrated with a per-period reset for stability.

Dizi [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dizi: the Chinese bamboo transverse flute - an open-bore air-jet flute with the signature dimo, a thin reed membrane glued over a hole that BUZZES sympathetically, adding a bright, reedy, almost vocal rasp no Western flute has. Breath drives the jet, Membrane sets the dimo buzz.

Djembe [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal djembe: a trigger on in0 strikes a West-African goblet drum - a deep resonant bass membrane (modes near 1 : 1.5 : 2.1) with a sharp bright finger-slap on top, so it spans from a booming bass tone to a cracking slap. Tune sets the head pitch, Decay the ring, Slap the bright attack-noise amount, Level the output. Trigger from a Clock, Euclid or sequencer.

DMXKick [OSC] [FX] [SYNTH] in: Trigger out: Audio

Oberheim DMX kick: a punchy, tight sampled-style bass drum with a fast pitch sweep and a short, controlled decay - the classic 80s electro/hip-hop kick. Tune sets the pitch, Decay the length, Level the output. Trigger on in0.

DolphinClickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dolphin oscillator: a fast train of broadband echolocation clicks (a buzzy click rate that accelerates into a 'creak') layered with a gliding signature whistle, the dual voice of a dolphin's biosonar and social calls.

Donk [OSC] [FX] [SYNTH] in: Trigger out: Audio

Scouse-house donk: a trigger on in0 fires a short resonant blip with a fast downward pitch snap through a sharp band-pass - the hollow 'donk' bounce of UK bounce / donk house. Tune sets the blip pitch, Snap the pitch-drop speed (the bounce), Reso the band-pass sharpness (how hollow and ringing), Decay the blip length, and Level the output.

DonkBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Donk bass: the bouncy, hard-plucked UK bassline 'donk' - a short pitched pluck pinged through a high-resonance filter for the rubbery, popping attack. Decay sets the bounce, Reso the donk, Level the output. in0 = Gate, in1 = Pitch CV.

DonkeyBrayOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Donkey bray: the hee-haw of a donkey, a rough rasping voiced tone alternating between a high inhale 'hee' and a low exhale 'haw' with a buzzy braying edge, fired at the set rate; the comical ass.

Doorbell [OSC] [FX] [SYNTH] in: — out: Audio

Doorbell - the classic two-note 'ding-dong' chime with a metallic ring, auto-repeating. Rate sets the interval, Pitch the chime, Level the output.

DoorbellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Doorbell: two struck resonant chime tones sounded in sequence (the descending 'ding-dong' of a two-note door chime), each a decaying sine, re-triggered at the set rate.

DoorCreakOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Door creak: the slow eerie squeal of a swinging door, a resonant tone whose pitch rises and wavers irregularly as the hinge sticks and slips, fired at the set rate; the haunted-house door.

DoorKnockOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Door knock: knuckles rapping on a wooden door, a series of short dull woody knocks (a quick triple-knock pattern with a gap) each a damped low-mid resonance, fired at the set rate.

DoorSlamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Door slam: a door banged shut, a hard low broadband impact followed by the brief boxy resonance of the wood panel and frame ringing, fired at the set rate; the angry exit.

DotMatrixOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dot-matrix printer: the screeching buzz of a dot-matrix print head, a pin-fire tone that sweeps in pitch as the head accelerates across each line then snaps back for the next pass; the line rate sets the carriage speed.

DoubleFoliumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double-folium oscillator: the radial double-folium curve 4cos(t)sin^2(t), which reduces to cos(t)-cos(3t) - a fixed cosine blend of the fundamental and third harmonic with a hollow, twin-lobed shape distinct from the sine-phase nephroid and bifolium.

DoubleParetoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double-Pareto oscillator: each cycle rises as a power law x^(Beta-1) up to a peak then falls as x^(-Alpha-1) (the double-Pareto / log-Laplace shape) - so both flanks are power laws with independently-set exponents, a heavy-tailed bump with a sharp peak distinct from the exponential-tailed distributions. Peak-normalized, bounded.

DoubleScrollJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double-scroll jerk: a single-equation jerk circuit with a piecewise-linear feedback that reproduces a Chua-style double-scroll attractor from just three op-amp integrators - the minimal double-scroll generator.

DoubleScrollSat [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saturated double-scroll: a third-order linear system closed by a saturation function (the Chua-Lin saturated-nonlinearity circuit), which generates the Chua double-scroll - and, by stacking saturation steps, n-scroll - attractors without a Chua diode.

DoubleWellHam [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double-well Hamiltonian: an undamped particle in a twin-well potential (x'=p, p'=x-x^3) that conserves energy, orbiting within one well or figure-eighting across both depending on its energy - a clean conservative nonlinear oscillator.

DoublingComplex [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Angle-doubling map: z -> z^2 on the unit circle, which doubles the phase angle each step (the complex face of the Bernoulli shift); the real part is a chaotically-scrambled cosine, deterministic yet broadband.

Downlifter [OSC] [FX] [SYNTH] in: Trigger out: Audio

Down-sweep transition: a trigger on in0 fires a descending sweep that falls over Time seconds - the post-drop 'whoosh down'. A pitch-falling sine and band-passed noise slide from bright/high to dark/low while the level decays, marking the move into a breakdown. Time sets the fall length, Tone balances the noise against the falling sine, Reso the band-pass sharpness, and Level the output. The inverse of Riser - patch a transition gate into in0.

DpwSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

DPW sawtooth: squares a trivial ramp into a parabola then differentiates it, which suppresses the high-harmonic aliasing of a naive saw cheaply (the differentiated-parabolic-waveform method) - smoother than a raw saw, lighter than oversampling.

DpwSquare [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

DPW square: subtracts two differentiated-parabolic sawtooths a half-cycle apart to build an alias-reduced square wave, the band-limiting trick of the DPW family applied to a pulse rather than a ramp.

DR110 [OSC] [FX] [SYNTH] in: Trigger out: Audio

Boss DR-110 kick: a tight, dry analog bass drum with a short pitch blip and a clicky attack - the small, punchy Dr Rhythm kick. Tune sets the pitch, Click the attack snap, Level the output. Trigger on in0.

DragonflyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dragonfly: the dry papery clatter of a dragonfly's four stiff wings, a buzzy tone chopped by a fast secondary wing-rattle so it shimmers and crackles, unlike the smooth bee buzz; the rattle rate sets the wing speed.

Drawbar [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nine-drawbar tonewheel organ oscillator: stacks the classic Hammond harmonics (sub, fifth, octaves, third, etc.). Tilt rolls the registration from foundation-heavy to bright upper partials and Click adds the key-on percussion transient.

DrawbarOrgan [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Drawbar organ: nine sine partials at the Hammond drawbar footages (sub-octave, fifth, unison, octave, twelfth, fifteenth and so on) mixed by a single registration morph from mellow flutes to bright full-organ - tonewheel additive synthesis.

DreadDroneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dread drone: the unsettling low drone of a horror score, a cluster of slightly-detuned low partials with a slow beating dissonance and a creeping upper overtone, the bed of tension under a scary scene; the beat rate sets the unease.

DrillString [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Drill-string stick-slip: a kilometers-long elastic drill pipe driven from the top while the bit grabs the rock with velocity-weakening friction, so the bit alternately sticks and slips - the torsional vibration that destroys drilling tools.

DripOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Drip: the resonant 'plink' of a water drop hitting a surface, a sine ping that drops slightly in pitch as the cavity resonance settles, with a soft tail, fired at the set rate; the patient drip of a leaky tap.

DrippingFaucet [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dripping faucet: a hanging drop grows from inflow until its mass breaks the surface-tension threshold and detaches, leaving a recoiling residual; the interval between drops period-doubles into chaos - Robert Shaw's famous kitchen-sink experiment.

DrivenPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Driven pendulum: a damped pendulum kicked by a periodic torque, the canonical single-degree-of-freedom system that tumbles between swinging and full rotation chaotically as the drive strength rises.

DrizzleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Drizzle: a fine soft rain, a gentle stream of tiny high-frequency droplet ticks over a faint wet hiss, the quiet patter of light rain on leaves; the density knob raises it from a sprinkle to a steady shower.

DroidChatterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Droid chatter: the emotional warble of an astromech droid, short sine blips that randomly glide up or down in pitch with expressive little swoops, stringing together into excited mechanical conversation; the blip rate sets the talkativeness.

DroneOsc [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

The most powerful drone generator in existence. Up to 16 morphable voices (sine through saw to PWM pulse) detuned and stacked into octaves, fifths or chords; a two-octave sub stack for deep weight; an octave-plus-fifth shimmer chord on top; a filtered air-noise layer; chaotic Lorenz-attractor drift for organic never-repeating motion; a wavefolder-plus-saturation Drive stage for enormous harmonic content; and a Motion-driven resonant low-pass that breathes. Free-runs at Freq as a standalone drone, or tracks the keyboard via a Note (Pitch CV) on in0; in1 adds to the tone. Twelve controls: Voices, Detune, Sub, Shimmer, Stack, Drift, Tone, Shape (waveform morph), Drive (harmonic fold and saturate), Motion (chaotic movement) and Air. The cinematic, doom, ambient and sound-design powerhouse.

DropoutOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dropout: an intermittent signal cutting in and out, a steady tone randomly muted by silence gaps as if a loose cable or weak radio link keeps failing, the glitchy stutter of a broken connection; the dropout rate sets how often it cuts.

DrumMembraneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Drum-membrane oscillator: a fixed sum of sines at the inharmonic ratios of an ideal circular membrane (the zeros of the Bessel functions: 1, 1.59, 2.14, 2.30, 2.65, 2.92, 3.16), the partial structure that gives a timpani or tom its pitched-but-not-quite-harmonic tone.

DSF [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Discrete summation formula oscillator (Moorer 1976): a closed-form band-limited buzz - the whole geometric sum of harmonics evaluated in one expression, so Brightness sets the harmonic roll-off (0 = pure sine, near 1 = bright sawtooth-like) with no aliasing, and Ratio spaces the partials (1 = integer harmonics, off-integer = clangorous inharmonic spectra). Freq sets the pitch (in0 = Pitch CV), Level the output.

DsfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

DSF oscillator: Moorer's discrete summation formula sums a geometric series of harmonics in closed form, so one 'brightness' knob smoothly fades from a sine to a buzzy band-limited pulse - alias-controlled spectral morphing.

DTMFOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

DTMF oscillator: the touch-tone of a telephone keypad, the sum of one low-group and one high-group sine chosen by the key (the standard dual-tone multi-frequency signalling), cycling 0-9 and the symbol keys.

DuckQuackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Duck quack: the nasal quack of a duck, a short buzzy voiced burst with a fast amplitude flutter and a comic nasal honk, repeated, fired at the set rate; the pond's resident comedian.

∿ **Duduk** [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Armenian duduk (Gladiator / Dune cinematic wind): a cylindrical double-reed voice - odd-harmonic dominant like a cor anglais, warm and woody and mournful - with breath noise and an expressive delayed vibrato, the soul-instrument used for grief and longing in film scores. Tune sets the pitch (in1 = Pitch CV), Breath the airy reed noise, Vibrato the expressive pitch waver (it swells in as the note is held), Tone the brightness, and Level the output. A gate on in0 sounds the note with a soft reed attack; hold for the long mournful sustain.

∿ **DuffingForced** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forced Duffing oscillator: a damped twin-well spring driven by a sine force jumps chaotically between its two wells, a textbook nonlinear resonator whose response is rich, hysteretic and period-doubling.

∿ **DuffingMap** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Duffing map: the discrete Holmes form of the Duffing oscillator (a cubic 2-D map), tracing a stretched chaotic loop - the iterated-map counterpart to the forced Duffing flow, with a crisper digital character.

∿ **DuffingOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Duffing chaotic oscillator: a driven nonlinear spring that ranges from a pure pitched tone to wild bifurcating chaos. Chaos sets the drive into instability and Damp reins it back - an organic, unpredictable analog-style source.

∿ **DuffingVanDerPol** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Duffing-van-der-Pol oscillator: a single resonator carrying both a cubic stiffness (Duffing twin-well) and a van-der-Pol negative-damping term, so it self-starts and, when forced, roams chaotically across its wells.

∿ **DumbbellOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dumbbell oscillator: sweeps the dumbbell curve $x^{2*\mathrm{sqrt}(1-x}2)$ with the sign of x, giving an odd waveform with two rounded lobes that pinch to zero at the centre and the edges - a hollow, weighted-at-the-ends shape distinct from triangle or sine.

∿ **DustDevilOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dust devil: a small whirling vortex of wind, a resonant whistling noise band whose pitch spirals up and down as the column spins, swelling as it passes and fading as it wanders off; the spin rate sets the vortex speed.

∿ **DustOsc** [OSC] [FX] [SYNTH] in: — out: Audio

Dust - randomly-timed unipolar impulses (a stochastic click source). Density sets the average rate from sparse ticks to a dense crackle. Classic for granular triggers and percussive texture.

∿ **DustRandom** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Random impulse generator (dust): fires impulses at random times whose average rate is set by a density control, the classic granular/stochastic trigger source for clouds, crackle and sparse percussion.

DustyPlasma [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dusty-plasma oscillation: charged dust grains levitated in a plasma sheath bob on the dust-acoustic mode, whose charge-fluctuation feedback and nonlinear sheath force drive self-excited and chaotic grain oscillations.

Earthquake [OSC] [FX] [SYNTH] in: – out: Audio

Earthquake - a sub-bass brown-noise rumble that swells and recedes, felt more than heard. Intensity sets the magnitude, Tone the upper grind, Level the output.

ECGSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synthetic ECG: a point moving around a circular limit cycle gets pushed up and down by Gaussian bumps placed at the P-Q-R-S-T angles, tracing a realistic electrocardiogram waveform (the McSharry-Clifford model) you can detune into arrhythmia.

EdgeTone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Edge tone: an air jet striking a sharp wedge flips from side to side with a feedback delay set by the jet's travel time (the sound source in flutes, whistles and organ flue pipes), here driven into its chaotic over-blown mode.

EEGRhythmOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

EEG rhythm oscillator: blends the five brainwave bands - slow delta, theta, alpha, beta and fast gamma at their characteristic frequency ratios - into one composite, the Band knob crossfading from a drowsy delta throb to alert gamma chatter.

EggCrackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Egg crack: an eggshell cracked against a bowl, a sharp brittle crack transient followed by a brief wet splat as the contents drop, fired at the set rate; the cooking-prep detail.

EhrenfestUrn [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ehrenfest urn: N balls split between two urns, each step a random ball changes urn, so the count mean-reverts toward N/2 (a discrete model of heat diffusion and entropy); the fluctuations make a self-centering integer wander.

ElasticPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Elastic pendulum: a mass on a spring that also swings, whose bounce and swing modes exchange energy through their 2:1 resonance and tumble chaotically - the classic 'spring pendulum' demonstration of coupled-mode chaos.

Electric [OSC] [FX] [SYNTH] in: Note, Gate out: Audio

Electric piano (Rhodes/Wurlitzer-style physical model): a gate on in1 strikes a tuned tine - an exponentially-decaying sine fundamental plus a bright inharmonic 'bark' partial that decays faster - run through a soft pickup nonlinearity, the classic electric-piano bell-and-body tone. in0 is the note (keyboard-tracked), Transpose tunes it, Decay the note length, Bark the metallic attack, Tone the pickup brightness, and Level the output.

Elements [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal synthesis voice (Mutable Instruments Elements): an exciter feeds a tuned modal resonator bank. A gate on in0 fires the exciter, which Exciter morphs from blown (filtered noise) to struck (impulse); the six-mode resonator imposes the ringing of a physical object, with Geometry morphing its partials from a harmonic series toward inharmonic bell/plate ratios and Damping setting the ring time. Tune sets pitch, Level the output. Bowed strings, struck bars, blown tubes and alien resonances from one voice. Distinct from the self-struck Modal and the resonator-only Rings.

ElevatorDingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Elevator ding: the polite arrival chime of an elevator reaching a floor, a soft clean single (or double) bell tone ringing briefly and fading, fired at the set rate; the floor-has-arrived sound.

EnergyChargeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Energy charge: a weapon or spell charging up, a tone that rises steadily in pitch and intensity to a peak then snaps back to recharge, with a faint crackling energy edge, the building hum before a big release; the rate sets the charge time.

Engine [OSC] [FX] [SYNTH] in: – out: Audio

Combustion engine: a pulse train of sharp combustion transients fired at the engine's cylinder rate through a resonant exhaust filter, with a noise rasp - the procedural car / motorcycle / machine engine of film and game sound design. RPM sets the engine speed (the firing rate), Cylinders the number of cylinders (rhythm and harmonic spacing), Tone the exhaust resonance frequency, Growl the resonance and grit, and Level the output. Automate RPM for revs and gear changes; self-running.

EngineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Engine oscillator: a train of combustion thumps fired at the cylinder rate of a running engine (RPM times cylinders over two revolutions), each a short decaying pulse, so raising the RPM sweeps from a lumpy idle to a smooth high-rev drone - a motor as a tonal source.

EnvToPitch [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Envelope-to-pitch oscillator: tracks the input loudness and maps it to the frequency of an internal sine, so dynamics play a melody - a self-contained envelope-controlled drone voice.

EpicycleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Epicycle oscillator: the Ptolemaic deferent-plus-epicycle model of planetary motion - a small circle whose centre rides a large one - which traces looping retrograde paths; the x-projection makes a waveform with little backward kinks where the planet appears to reverse.

EpicycloidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Epicycloid oscillator: the curve traced by a point on a small circle rolling around the outside of a larger one, whose cusp count is set by the radius ratio; the x-coordinate makes a cusped, gear-like waveform rich in harmonics.

EpidemicSEIRS [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seasonal SEIRS: an SEIR epidemic where recovered hosts lose immunity and return to susceptible (the extra loop), which under seasonal transmission sustains rich multi-annual and chaotic outbreak cycles - recurrent-epidemic dynamics.

EpidemicSIR [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seasonal SIR epidemic: a susceptible-infected-recovered model whose transmission rate is modulated annually (school terms, seasons); strong seasonality drives the infected fraction into chaotic, irregular outbreak cycles.

Epileptor [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Epileptor: Jirsa's phenomenological model of an epileptic-seizure cycle, where a very slow permittivity variable drags a fast spiking subsystem in and out of the seizure state - the autonomous build-up, discharge and recovery of a seizure.

ErfcOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Erfc oscillator: each cycle is the complementary error function erfc(x), a falling S-curve with Gaussian (super-exponential) tails - it stays near the top, drops steeply through the centre, then flattens fast, a sharper-shouldered descending ramp than the logistic or arctangent sigmoids.

Erhu [OSC] [FX] [SYNTH] in: Bow, Pitch CV out: Audio

Chinese erhu: a bowed two-string fiddle with an intense, vocal vibrato and a nasal, reedy tone - a bowed sawtooth through a nasal formant, played by holding in0 as bow pressure. Vibrato sets the wobble depth, Nasal the formant emphasis, Level the output. in0 = Bow gate, in1 = Pitch CV.

ErrorBuzzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Error buzz: the harsh dissonant buzz of an invalid action or denied input, two close detuned square tones beating against each other in a short blast, the access-denied / wrong-input sound; fired at the set rate.

EspressoSteamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Milk steamer: an espresso machine steam wand frothing milk, a high-pressure steam hiss with a gurgling burble that rises in pitch as the milk foams, the barista's froth; the burble knob sets the foaming.

EulerPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Euler-polynomial oscillator: each cycle is the nth Euler polynomial E_n(x) on [0,1] (the alternating-sum companion of the Bernoulli polynomials) - a related but distinct oscillating polynomial with different coefficients and symmetry. Peak-normalized, bounded.

EulerRigid [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rigid-body tumble: the Euler equations for a torque-free asymmetric rigid body (the intermediate-axis 'tennis-racket' instability) with weak feedback, so the angular-velocity vector flips and tumbles chaotically.

EvenHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Even-harmonic oscillator: a fundamental plus only its even partials (2, 4, 6, 8, 10, 12), an octave-stacked spectrum with the empty-twelfths quality of certain organ stops and the warmth even-order tube distortion adds.

ExcitableMedium [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Excitable medium: a ring of cells that fire when a neighbour fires, then enter a refractory period before they can fire again (the cardiac/Belousov wave rule); periodic stimulation launches pulses that can circulate as self-sustaining reentry - a probe cell's excitation is the output.

ExpLogarithmicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential-logarithmic oscillator: each cycle is proportional to (1-P)betae^(-beta x)/(1-(1-P)e^(-beta x)), a decreasing-hazard distribution (mixture of exponentials over a logarithmic series) - a heavy-front, slow-tailing decay shaped by P. Peak-normalized, bounded.

Explosion [OSC] [FX] [SYNTH] in: – out: Audio

Explosion - a deep boom (pitch-dropping body + noise blast) with a rumbling tail, auto-repeating. Rate sets the interval, Tone the debris brightness, Level the output.

ExplosionOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Explosion: the boom of a destroyed enemy, a burst of white noise filtered downward over a low rumble as it decays, the chunky 8-bit kaboom, fired at the set rate.

ExpMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential map: iterates x -> r x e^(-x), a transcendental map (related to the Ricker model) whose single hump drives a period-doubling cascade into chaos with a smooth, asymmetric bifurcation structure.

Expo [OSC] [FX] [SYNTH] in: – out: Audio

Exponential-ramp oscillator (a first as a synth block): a sawtooth whose ramp is not straight but EXPONENTIALLY curved - it rises (or, with the curve reversed, decays) like a charging capacitor rather than a linear slope, the natural shape of an analogue envelope or an RC circuit. Curve sets how bent the ramp is: near zero it is an ordinary saw, high and it hugs the floor then snaps up at the end. The curvature reshapes the harmonic rolloff, giving a rounder, more organic edge than the buzzy linear saw. A capacitor-curve waveform distinct from the geometric saw. Needs no input.

ExpoEaseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential-ease oscillator: each cycle is the easeOutExpo curve 1-2^(-Rate*t), which snaps up fast then asymptotically approaches the top - a soft sawtooth with exponential curvature (sharp attack, long flat tail), Rate setting how abrupt the rise.

ExponentialNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential noise: samples -ln(u)/lambda from the memoryless exponential distribution (the gaps of a Poisson process), strongly peaked at zero with a single light tail - a one-sided source distinct from symmetric white noise, here centred.

ExponentiatedExpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponentiated-exponential oscillator: each cycle is Alpha*e$^{(-x)*(1-e}$(-x))^(Alpha-1), a flexible distribution that is a pure decay for Alpha<=1 but grows a rising hump for Alpha>1 - one knob morphs from a falling ramp to a skewed bump. Peak-normalized, bounded.

ExpQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential-quantile oscillator: each cycle is the exponential inverse-CDF -ln(1-p), a ramp that climbs almost linearly at first then accelerates ever more steeply toward the top - the pure-logarithmic member of the quantile family, a convex accelerating sweep. Bounded.

ExpWeibullOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponentiated-Weibull oscillator: each cycle is alphabetax$^{\text{(beta-1)*e}}$(-x$^{\text{beta)*(1-e}}$(-x$^{\text{beta))}}$(alpha-1), a flexible lifetime PDF that can be unimodal, decreasing or bathtub-shaped - Alpha and Beta together set the rise and tail, a richly-morphable skewed bump. Peak-normalized, bounded.

ExtraLifeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

1-up: earning an extra life, a short cheerful jingle of square-wave notes climbing a major arpeggio, the happy reward chime; the rate sets how often it plays.

Factory [OSC] [FX] [SYNTH] in: — out: Audio

Factory floor - rhythmic mechanical clanks and presses over a motor hum and steam hiss. Rate sets the machine tempo, Tone the metallic brightness, Level the output.

FaradayWave [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Faraday wave: a fluid surface shaken vertically grows standing waves at half the drive frequency (a parametric instability); a single mode with cubic saturation captures the onset and the chaotic competition of these ripples.

FastRadioBurstOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fast radio burst: the dispersed millisecond flash of an FRB, where interstellar plasma makes the high frequencies arrive before the low ones, so each repeat sweeps rapidly downward in pitch; an enigmatic cosmic radio signal rendered as a falling chirp burst.

FaxToneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fax handshake: a fax machine negotiating a connection, a pair of alternating pure carrier tones over a band-limited modem hiss, the unmistakable screech of a fax dialling in, the retro-telecom sound; the swap rate sets the warble.

FBMNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fractional Brownian motion: a self-affine random walk whose Hurst exponent sets the long-range correlation (persistence or anti-persistence) between ordinary Brownian motion and smooth drift - the fractal landscape of natural terrains and fGn.

FDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

F-distribution oscillator: each cycle is the Fisher-Snedecor F PDF, shaped by two degrees-of-freedom (D1 rise, D2 tail) - a right-skewed bump from variance-ratio statistics whose skew and tail weight the two knobs set independently; peak-normalized and bounded.

FeedbackFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Single-operator FM with self-feedback (the DX7 feedback operator): a sine modulates its own phase, sweeping smoothly from pure tone toward a saw-like buzz as Feedback rises. Tone tames the top end.

FeedbackFMOp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Feedback FM operator: a sine oscillator phase-modulated by its own previous output (the DX7 feedback operator), which morphs continuously from a pure sine through a sawtooth-like spectrum into noise as the feedback amount rises - one operator, a whole timbral range.

FejerKernelOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fejer kernel: the average of Dirichlet kernels (a squared sinc), always non-negative and with triangular harmonic weighting, giving a smooth single bump per cycle with far gentler sidelobes than the Dirichlet kernel - a clean unipolar pulse.

FermatPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fermat-polynomial oscillator: each cycle is the nth Fermat polynomial (F_n = 3xF_{n-1} - 2F_{n-2}, giving 3x, 9x^2-2, 27x^3-12x…) - a steeper recurrence than Fibonacci, so its lobes grow and sharpen faster, a distinct polynomial-sequence waveform. Peak-normalized, bounded.

FermatSpiralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fermat-spiral oscillator: the radial projection sqrt(theta)*cos(theta) of Fermat's parabolic spiral over several turns per cycle - a sine whose amplitude grows as a square root then resets, a gentler swell than the linear Involute or exponential LogSpiral.

FermiAccelMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fermi-Ulam model: a ball bounces between a fixed wall and an oscillating one, gaining or losing speed at each kick depending on the wall phase - Fermi's mechanism for cosmic-ray acceleration; the simplified velocity-phase map mixes accelerating chaos with trapping islands.

FermiMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fermi-Ulam map: models a ball bouncing between a fixed and a vibrating wall (Fermi's cosmic-ray acceleration idea), whose velocity gains and loses energy chaotically - a stochastic-acceleration source with bursts of speed.

FerroResonance [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ferroresonance: a power transformer's saturable iron core resonating with line capacitance, whose cubic magnetization nonlinearity under AC drive jumps between low and high overvoltage modes and chaotically - a dangerous grid instability.

FibonacciHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fibonacci-harmonic oscillator: partials placed at the Fibonacci harmonic numbers (1, 2, 3, 5, 8, 13, 21), so the overtones spread ever wider in the golden cadence - a sparse, increasingly-inharmonic-feeling spectrum with a shimmering top.

FibonacciOsc [OSC] [FX] [SYNTH] in: — out: Audio

Fibonacci-harmonic oscillator (a first as a synth block): an additive tone whose overtones land only on the Fibonacci numbers - the 1st, 2nd, 3rd, 5th, 8th, 13th, 21st… harmonics. The gaps between partials widen by the golden ratio as you climb, so the low end is dense and the top sparse, giving a warm body with a thin, glassy shimmer on top - a spectrum tuned to phi rather than the even comb of a saw. Partials sets how many Fibonacci overtones to stack, alias-free up to Nyquist. Distinct from the prime-harmonic and tilt-based oscillators.

FibonacciPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fibonacci-polynomial oscillator: each cycle is the nth Fibonacci polynomial F_n(x) (F_n = x*F_{n-1}+F_{n-2}) - a polynomial whose oscillating lobes increase with Order, a recurrence-defined waveform distinct from the orthogonal Chebyshev/Legendre families. Peak-normalized, bounded.

FibreString [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Two coupled Karplus-Strong plucked strings tuned a hair apart for a rich, chorused acoustic pluck/pad. Damp sets the decay (short pluck to long sustain) and Detune the beating between the two strings.

FifthLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Power-fifth lead: a saw doubled a perfect fifth above for the bold, open parallel-fifths organ/rock lead. Mix sets the fifth level, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

FinanceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Finance chaos attractor: a three-state model of interest rate, investment demand and price index whose coupled nonlinear dynamics are chaotic, repurposed as a slow, organic strange-attractor modulation source.

Fire [OSC] [FX] [SYNTH] in: — out: Audio

Fire / combustion texture: three procedural layers - a brown-noise roar (the steady burn), an airy high-frequency hiss (combustion), and randomly-timed crackle pops (wood splitting) - the campfire / torch / burning-building ambience of film sound design. Self-running. Roar sets the low rumble level, Crackle the density of the popping, Hiss the airy combustion noise, Bright the crackle brightness, and Level the output. Layer two instances for a richer fire; automate Crackle for flare-ups.

FireballOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fireball: a hurled ball of flame, a broadband noise roar swept by a fast-opening then closing lowpass (the whoosh of the fireball passing) with a crackling fiery edge, fired at the set rate; the magic-projectile staple.

FireCrackle [OSC] [FX] [SYNTH] in: Amount out: Audio

Fire: a crackling campfire - a low filtered-noise roar speckled with random snaps and pops. Crackle sets the snap rate, Tone the brightness, Level the output. in0 = Amount gate.

FireCrackleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fire oscillator: a soft filtered-noise bed of flame hiss sprinkled with random sharp crackle pops, the comforting roar-and-snap of a campfire or hearth; the crackle knob sets how busy the pops are.

FireflySync [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Firefly synchronization: a dozen integrate-and-fire oscillators each charge toward a flash threshold and, when one fires, nudge all the others forward (Mirollo-Strogatz pulse coupling), so an initially-random swarm spontaneously locks into unison flashing. The flash count is the output.

FishCurveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fish-curve oscillator: the x-coordinate cos(t) - sin^2(t)/sqrt(2) of the fish curve, DC-removed, which is a fixed in-phase blend of the fundamental and second harmonic - an asymmetric waveform with a single bulged lobe and strong even-harmonic colour.

FitzHughForced [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forced FitzHugh-Nagumo: the excitable two-variable neuron driven by a periodic current, which mode-locks, skips beats and fires chaotic spike patterns (the devil's-staircase response of a forced excitable cell).

FitzNagumo [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

FitzHugh-Nagumo neuron: a two-variable excitable model that fires voltage spikes when its current crosses threshold, giving clicky, pulsing, neuron-like rhythms a wavetable can't produce.

FizzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fizz: the effervescent hiss of a freshly-poured carbonated drink, a dense random crackle of countless tiny high-frequency bubble bursts, slowly thinning as the soda goes flat; density sets the carbonation.

FKNReaction [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

FKN reaction: the Field-Koros-Noyes mechanism behind the Belousov-Zhabotinsky reaction, reduced to its activator-inhibitor-catalyst core, firing sharp relaxation pulses with a long refractory interval - the chemical-clock spike train.

FlagFlutter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Flag flutter: a flexible panel in a flow whose bending and torsion modes lock together past a critical wind speed and flutter, then chaotically flap; aeroelastic mode-coupling instability, the flapping-flag and bridge-deck mechanism.

FlameOscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Flame instability: a premixed flame front whose heat-release responds to its own curvature and a downstream acoustic feedback, pulsating and chaotically flickering - the thermo-diffusive instability behind combustion roar and flame chatter.

FlexibleWeibullOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Flexible-Weibull oscillator: each cycle is (Alpha+Beta/$x^{2)*e}$(Alphax-Beta/x)$e^{(-e}$(Alpha*x-Beta/x)), the flexible-Weibull extension whose Beta/x term gives a steeper left rise than a plain Weibull - a sharply front-loaded bump with an exponential cut-off. Peak-normalized, bounded.

FluorescentBuzzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fluorescent buzz: the gritty 100/120 Hz buzz of a magnetic-ballast fluorescent tube (twice the line frequency, since the lamp fires on both half-cycles) overlaid with random flicker crackle from an aging starter.

Flute [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Breathy flute voice: a gate on in0 sounds a soft nearly-pure tone (fundamental with a whisper of upper harmonics) wrapped in breath noise and a gentle vibrato, with a slow soft attack. Freq sets the note, Breath the airy-noise amount, Vibrato the pitch waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

FluteWGOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Flute waveguide: an open bore excited by an air jet modelled as a cubic nonlinearity plus breath noise, giving the breathy, pure, even-and-odd-harmonic tone of a flute or whistle; the breath knob adds air hiss and pushes it overblown.

FM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-operator FM voice (the DX7 building block): a sine carrier is phase-modulated by a sine modulator at Ratio times the frequency, with Index setting depth and modulator self-Feedback adding harmonic edge toward noise. Integer ratios give harmonic, organ/bass timbres; non-integer ratios give bells and metallic tones.

FMBass [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Two-operator FM bass: a sine carrier frequency-modulated by a second sine at an integer ratio, the punchy, metallic DX-style bass with a bright attack-to-body movement. Ratio sets the modulator interval, Index the FM brightness, Decay the index falloff. in0 = Pitch CV.

FMDrum [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

FM percussion voice: a trigger on in0 fires a 2-operator FM tone with a fast decay - metallic toms, bells, blips and zaps depending on Ratio. Tune sets the carrier pitch, Ratio the modulator ratio (inharmonic at non-integers), Decay the length, Level the output. in1 = Pitch CV.

FOF [OSC] [FX] [SYNTH] in: – out: Audio

Formant-wave-function synthesis (CHANT): each fundamental period retriggers a bank of decaying formant grains, generating vocal/formant tones. Vowel morphs A-E-I-O-U.

FofGrain [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

FOF formant synthesis: each pitch period triggers a grain that is an exponentially-decaying sine at a formant frequency (the IRCAM Chant technique), building voiced, formant-rich tones one excitation grain at a time.

FofSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

FOF synthesis (IRCAM CHANT): each fundamental period triggers a formant wave function - a sine at the formant frequency wrapped in a sharp-attack, exponential-decay grain; summing several gives the ringing vowel formants of the singing voice.

FogHorn [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fog horn: a low two-tone horn (a fundamental plus a fifth above) that swells in and out on a slow breath envelope - the mournful blast of a lighthouse or a ship in fog, ducking the dry signal under it. Pitch sets the horn fundamental, Mix the blend.

FoldedNormalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Folded-normal oscillator: each cycle is the folded-normal PDF phi(x-Mu)+phi(x+Mu) (the distribution of |N(mu,1)|), which is a half-Gaussian at Mu=0 and develops a shifted, shouldered bump as Mu grows - peak-normalized, a distinct one-sided shape from the standard Gaussian.

FoldedTowel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Folded-towel map: Rossler's three-dimensional 'folded towel' map, an iconic hyperchaotic system whose attractor looks like a folded towel and expands chaotically in two directions at once.

FoldFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Folded FM: a 2-operator FM carrier is driven into a sine wavefolder, stacking FM sidebands and fold harmonics into a screaming, complex-oscillator timbre that neither FM nor folding makes alone.

FoliumDescartesOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Folium-of-Descartes oscillator: each cycle is the y-coordinate $3t^{2/(1+t}3)$ of the folium of Descartes, a fast-rising loop that peaks early then trails off down its asymptotic tail - an asymmetric bump from the classic cubic curve. Peak-normalized, bounded.

Footstep [OSC] [FX] [SYNTH] in: Trigger out: Audio

Footstep foley: a trigger on in0 fires a layered foot-floor impact - a damped low resonance (the body of the step) plus a band-passed noise contact - with a Surface control that morphs from a tight solid hit (wood / asphalt) to a scattered cluster of stochastic micro-pops (gravel / aggregate). The walk, run and stomp foley of film and game sound design. Weight sets the step body pitch, Surface the solid-to-gravel morph, Tone the contact brightness, Hardness the impact sharpness, and Level the output. Trigger per step; vary Weight / Surface per footfall.

Footsteps [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Footsteps: a steady tread of soft low thumps - filtered noise bursts with a quick body-and-decay, paced like walking. Useful as a rhythmic foley/percussion bed or a sub-pulse generator. Rate sets the steps per minute, Mix the blend.

Forcefield [OSC] [FX] [SYNTH] in: – out: Audio

Energy force-field - a detuned electric hum pulsing with a slow tremolo and a faint crackle, the protective barrier. Power sets the intensity, Tone the brightness, Level the output.

ForceFieldOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Force field: the angry buzzing crackle of an energy barrier, a harsh saw-and-square drone amplitude-modulated by a fast ring buzz and sprinkled with random electric crackle, the sound of a shield you should not touch.

ForestFireSOC [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forest-fire model: cells are empty, growing trees, or burning; fire spreads to neighbouring trees, burnt cells clear, empty cells regrow, and rare lightning ignites new fires - the Drossel-Schwabl self-organized-criticality model whose fire sizes are scale-free; fire density is the output.

FormantBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Formant (talking) bass: a sawtooth shaped by a two-formant vowel filter that morphs through vowels, for a bass that says ‘wow’ and ‘yoy’. Vowel morphs the vowel, Sub the sub-octave weight, Level the output. in0 = Gate, in1 = Pitch CV.

FormantBuzz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Formant buzz voice: a harmonic-rich sawtooth excites two resonant band-pass formants whose centre frequencies morph with the Vowel knob (ah -> ee -> oo), giving a buzzy talking/vocal timbre from a single oscillator - a poor-man’s vocal synth.

FormantGrainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Formant grain: FOF synthesis (fonction d’onde formantique), where each pitch-period grain is a windowed sine burst at a formant frequency, the technique behind CHANT vocal synthesis; the fundamental sets pitch while the formant and decay shape the vowel.

FormantStackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Formant-stack oscillator: a full harmonic series whose partials are weighted by a resonant Gaussian peak at a fixed formant frequency, so the timbre keeps a vowel-like resonance at one absolute pitch regardless of the note played - source-and-formant synthesis in one block.

FourModeNS [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Truncated Navier-Stokes: a four-mode spectral truncation of the two-dimensional incompressible flow equations (Franceschini-Tebaldi), whose energy cascades chaotically among Fourier modes - a minimal model of fluid turbulence.

FourOpFM [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Four-operator FM: a stack of four sine operators (4 modulates 3 modulates 2 modulates the carrier), giving deep, evolving FM timbres from bells to brass to clangour. Ratio sets the operator interval, Index the FM depth, Decay the index falloff. in0 = Gate, in1 = Pitch CV.

FourOpFMVoice [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Four-operator FM: a serial stack of four FM operators (op4 modulates op3 modulates op2 modulates op1 carrier) at independent ratios, the deep-FM algorithm behind DX-style metallic and evolving timbres.

FourWingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Four-wing attractor: a chaotic flow whose orbit jumps between four lobes (‘wings’), giving abrupt, multi-state switching dynamics - chaos with a distinct sense of discrete regions.

FractalDreamMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fractal-dream map: a sine-based 2-D iterated map (Pickover) that traces dreamy, swirling, doubly-symmetric fractal curves, a distinctive flowing chaotic texture as an audio/CV source.

FractalFlame [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fractal flame (Scott Draves): an iterated function system whose affine maps are post-composed with nonlinear ‘variations’ (sinusoidal, spherical, swirl), the algorithm behind Electric Sheep; the warped chaos-game orbit traces a flame-like attractor read out as a CV.

FractalNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fractal noise: a bank of octave-spaced filtered noises summed with a slope control, sweeping the spectrum continuously from white (slope 0) through pink (1) to brown/red (2) - any 1/f^beta colour from one knob.

FrameDrum [OSC] [FX] [SYNTH] in: Trigger out: Audio

Frame drum: a shallow hand drum with a warm mid-pitched membrane tone and a soft skin slap - the open, resonant world-percussion drum. Tune sets the pitch, Decay the length, Level the output. Trigger on in0.

FrechetOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Frechet oscillator: each cycle is the Frechet extreme-value PDF x^(-1-a) $e^{(-x}$-a), which slams up from zero to a sharp peak then trails off in a heavy power-law tail - a spiky, front-loaded waveform with a long decay, Alpha setting the sharpness.

FrechetQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Frechet-quantile oscillator: each cycle is the Frechet inverse-CDF (-ln p)^(-1/Alpha), a ramp that lingers near zero then accelerates sharply toward the top as the extreme-value tail dominates - Alpha sets the violence of the late rise. Bounded.

FreeReed [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Free-reed model: a mass-on-spring reed driven past an asymmetric airflow valve self-oscillates with the buzzy, harmonically-rich, slightly-detuned tone of a harmonica or accordion reed under air pressure.

FreitasMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Freitas map: a two-dimensional piecewise map studied for its robust chaos (chaotic over a whole parameter window with no periodic windows), giving a reliably noise-like deterministic stream.

FrenchHorn [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

French horn: the warm, round, mellow brass - a sawtooth through a softer formant and a gentle lowpass with a slow swell, played by holding in0. Brass sets the formant bite, Breath the air, Level the output. in0 = Breath gate, in1 = Pitch CV.

FricativeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Fricative oscillator: turbulent noise high-passed and shaped by a single broad resonant peak, the hiss of an unvoiced fricative like sh, s or f; the peak frequency sweeps the sound from a dark 'sh' to a bright 's'.

Fridge [OSC] [FX] [SYNTH] in: — out: Audio

Refrigerator hum - a low compressor drone with a faint mechanical buzz and the occasional thermostat click, the quintessential room-tone. Tone sets the brightness, Buzz the rattle, Level the output.

FrogCroakOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Frog-croak oscillator: a low buzzy tone chopped into a fast pulse train and gated into croak bursts, the ribbiting call of a frog or toad whose pulse rate and croak rate set the species; a comb-buzzy amphibian voice.

Frogs [OSC] [FX] [SYNTH] in: — out: Audio

Frog pond - random low croaks with a watery resonance for a swamp/marsh night ambience. Activity sets how busy the pond is, Pitch the croak depth, Level the output.

FrostCrackleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Frost crackle: the delicate snapping of ice crystals forming and contracting in the cold, a sparse field of very short, very high ticks with a faint glassy ring, the near-silent sound of a deep freeze; the density knob sets how fast the frost spreads.

FroudePendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Froude pendulum: a pendulum mounted on a steadily-rotating shaft whose velocity-dependent friction pumps energy in, self-oscillating and going chaotic - the mechanism behind machinery shudder and stick-slip vibration.

FutureBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Future-bass lead-bass: bright detuned supersaws with a pitch-glide attack and a lowpass, the gliding, vocal-ish drop bass of future bass. Glide sets the attack slide, Detune the width, Cutoff the brightness. in0 = Gate, in1 = Pitch CV.

Gabber [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Gabber / hardcore kick: a 909-style kick - a sine body with a downward pitch sweep - overdriven hard into clipping until it becomes a distorted, square-ish melodic tone, the way the classic gabber kick was made by feeding a 909 through an overdriven mixer channel. The pitch tail falls off slower than a clean kick so the body stays melodic. Tune sets the body pitch (in1 = Pitch CV), PSweep the pitch-tail length, Decay the body length, Distort the overdrive amount (the gabber character), Tone the final clip hardness (soft -> square), and Level the output. A trigger on in0 strikes it.

GaborAtomOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gabor atom: a sinusoid at a chosen carrier ratio windowed by a Gaussian, the minimum-uncertainty grain at the heart of granular synthesis and the Gabor transform; the carrier and window-width knobs set the grain's pitch and its time-frequency spread.

GallopingLine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conductor galloping: an ice-coated transmission line whose asymmetric cross-section gives negative aerodynamic damping (the Den-Hartog criterion), self-exciting large slow vertical oscillations that can clash the phases - power-grid aeroelastic chaos.

GaltonBoard [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Galton board: a ball bouncing left/right down N rows of pegs lands in a binomial bin, and by the central limit theorem the heap of many balls approaches a bell curve; summing N Bernoulli bounces makes a quantized, near-Gaussian source.

GameBoyWaveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Game Boy wave channel: a 32-sample, 4-bit wavetable (the DMG's user-definable voice), here morphing between a few classic LSDj-style wave shapes; its coarse quantization gives the gritty, characterful timbre of handheld chiptune.

GameOfLife1D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

1-D life: a 32-cell ring where a cell lives next step only if exactly one of its two neighbours is alive (a one-dimensional birth/survival rule), breeding little gliders and blinkers; the live-cell density is read out as a fluttering CV.

GameOver [OSC] [FX] [SYNTH] in: – out: Audio

Game-over jingle - a descending square-wave arpeggio that tumbles down, auto-repeating. Rate sets the replay interval, Pitch the key, Level the output.

GameOverOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Game over: the deflating defeat sound, a square-wave tone sliding mournfully downward in pitch (often with a final low note), the dreaded end-of-game cue; the rate sets how often it repeats.

GammaFuncOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gamma-function oscillator: each cycle traces Gamma(x) (the continuous factorial, via Stirling) over a range spanning its minimum near x=1.46 - so the waveform dips to a soft trough then rises steeply, a convex U-with-a-steep-side shape distinct from the polynomial and bell curves. Peak-normalized, bounded.

GammaNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gamma noise: the sum of k independent exponential variables (the Erlang distribution), the waiting time for k Poisson events; increasing the integer shape k pulls the distribution from a sharp exponential toward a symmetric bell - a tunable-skew positive source.

GammaPdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gamma-PDF oscillator: each cycle is the Gamma distribution shape x^(K-1) e^(-x/Theta), a skewed bump whose rise sharpness (K) and decay length (Theta) are independent - peak-normalized to a bounded asymmetric waveform spanning near-exponential (low K) to near-Gaussian (high K).

GammatoneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gammatone: the impulse response of the auditory filters in the cochlea - a gamma-distribution envelope times a tone - widely used to model the ear's frequency analysis; looped per period it gives a soft attack-and-ring grain shaped like a hair-cell response.

GargleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gargle: a low throat tone whose saturation drive and amplitude burble on a smoothed random signal, so the timbre gurgles and bubbles wetly the way a gargle or a mud-pot does - Tone sets the pitch, Burble how violently it churns.

GarimaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Garima oscillator: each cycle is the Garima PDF proportional to (1+Theta+Thetax)e^(-Thetax), a lifetime distribution whose linear lift is itself scaled by Theta - so the rise and the decay rate move together, a tightly-coupled skewed bump. Peak-normalized, bounded.

GaussCFMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gauss map: the continued-fraction shift x -> {1/x} (fractional part of the reciprocal), whose iterates are the continued-fraction digits of the seed; ergodic with the Gauss-Kuzmin measure, it is a chaotic source biased toward small values.

GaussianPulseTrain [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gaussian pulse train: one Gaussian bump per period whose Width sets the spread (narrow gives a bright impulse-like click, wide gives a near-sine), the maximally time-frequency-compact pulse - a clean band-shaping source for grains and clicks.

GaussMixtureOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gaussian-mixture oscillator: each cycle sums two Gaussians placed symmetrically at +/-Sep, so Sep morphs the waveform from a single centred bump (Sep~0) into a clearly bimodal twin-peak shape - a two-hump waveform unlike the single-peaked bells. Peak-normalized, bounded.

GaussMouse [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gauss map: an exponential 'mouse' map whose bifurcation diagram folds back on itself, producing period-doubling and chaotic regimes from a bell-shaped iterator - smooth 1-D chaos with a distinctive flavor.

GBMProcess [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Geometric Brownian motion: the multiplicative random walk dS = muS dt + sigmaS dW underlying the Black-Scholes model, where proportional drift and volatility make a positive, exponentially-trending, never-negative path - a stock-price-style slow CV.

GearRattle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gear rattle: a driven gear pair with backlash, where the teeth lose and regain contact across a dead-zone, producing the impacting, chattering chaos of an unloaded gearbox - piecewise-stiffness vibro-impact dynamics.

GearWhineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gear whine: the pitched whine of meshing gear teeth (the tooth-pass frequency and its harmonics) with a touch of irregular clatter from backlash, the sound of a gearbox under load; the mesh frequency sets the gear speed.

GegenbauerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gegenbauer wave: the ultraspherical polynomial C_n^alpha on cos(theta), the family that interpolates between Chebyshev (alpha small) and Legendre (alpha = 1/2) and beyond, so alpha continuously reshapes the partial weighting of an n-lobe orthogonal-polynomial tone.

Geiger [OSC] [FX] [SYNTH] in: – out: Audio

Sparse-impulse noise: each sample has a probability set by Rate of firing a random-amplitude impulse, which then decays exponentially, all scaled by Level. Rate sets the average clicks per second and Decay sets how long each impulse rings before fading. Low Rate with short Decay gives sparse ticks like a Geiger counter, while higher Rate and longer Decay thicken it toward a crackling texture.

GeigerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Geiger oscillator: sharp clicks arriving as a memoryless Poisson process, the random spacing of radioactive-decay events on a Geiger counter; the count-rate knob sweeps from a slow tick to the menacing buzz of a hot source.

Gendy [OSC] [FX] [SYNTH] in: – out: Audio

Dynamic-stochastic (GENDYN) oscillator (a first as a synth block): Xenakis's radical idea of synthesising sound directly from probability - the waveform is a handful of breakpoints joined by straight lines, and after every cycle each breakpoint takes a small RANDOM WALK in height. So the wave keeps its rough shape but never repeats exactly, slowly mutating from one cycle to the next - the engine behind his GENDY pieces, sound built bottom-up from chance rather than from notes. Points sets the number of breakpoints and Step how far they wander each cycle (0 = frozen, high = noisy). A stochastic waveform distinct from every fixed-shape oscillator. Needs no input.

GeneralizedExtremeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Generalized-extreme-value oscillator: each cycle is the GEV PDF, the single family unifying the Gumbel, Frechet and reversed-Weibull extreme-value laws - the shape parameter Xi morphs the tail from light (Xi<0, bounded) through Gumbel (Xi~0) to heavy (Xi>0), one knob across all three. Peak-normalized, bounded.

GeneralizedGammaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Generalized-gamma oscillator: each cycle is $x^{(D-1)*e}$(-(x/1.5)^P), a three-way family that becomes exponential, Weibull or gamma depending on D and P - so two knobs span a wide range of skewed bump shapes from sharp spike to broad hump. Peak-normalized, bounded.

GenesioHyper [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Genesio: the Genesio-Tesi jerk system extended to a fourth state with feedback into hyperchaos, layering a second positive Lyapunov exponent onto its smooth single-scroll jerk dynamics.

GenesioTesi [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Genesio-Tesi attractor: a third-order jerk system (one variable, its first three derivatives) with a single quadratic term, the minimal chaotic 'jerk' circuit - smooth, single-scroll chaos.

GeneToggle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Genetic toggle switch: two genes each repressing the other form a bistable memory (Gardner-Collins synthetic biology); a slow forcing of one promoter flips it back and forth, giving a relaxation square-wave between cell states.

GenGaussianOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Generalized-normal oscillator: each cycle is exp(-|x|^Beta), so Beta morphs the symmetric bump from a sharp Laplacian cusp (Beta<1) through a pointed exponential (1) to a rounded Gaussian (2) - the cusp/peaked regime that complements the flat-top SuperGauss, rich in odd-order spectral character.

GenLogisticOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Generalized-logistic oscillator: each cycle is the type-I skew-logistic PDF a*e$^{-x/(1+e}$-x)^(a+1), a logistic bell that Alpha skews to one side (a=1 is symmetric logistic) - a continuously-lopsided bump with logistic tails, peak-normalized and bounded.

Geyser [OSC] [FX] [SYNTH] in: – out: Audio

Geyser - a cyclic eruption that builds from a low gurgle into a hissing high-pressure blast, then subsides. Cycle sets the eruption period, Tone the spray brightness, Level the output.

GeyserOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Geyser: a periodic eruption, a low rumble that swells and then bursts into a roaring jet of pressurised steam and water before subsiding, looping at the eruption rate; the violent breath of a hot spring.

GFSRNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

GFSR noise: a generalized feedback shift register XORs the word with shifted copies of itself at two tap positions (the family behind the Mersenne Twister's recurrence), producing fast maximal-length noise with a distinct broadband spectrum.

GhostWailOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ghost wail: the eerie wailing moan of a ghost, a pure tone slowly gliding up and down in pitch with a wide tremulous vibrato and a hollow breathy edge, the woo-oo-oo of a haunting; the glide rate sets the moan.

GingerMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gingerbreadman map: a piecewise-linear 2-D map (x' = 1 - y + |x|) whose orbit fills a six-pointed chaotic region - angular, glitchy chaos with hard corners, unlike the smooth attractors.

GinzburgLandau [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Complex Ginzburg-Landau: the universal amplitude equation near a Hopf bifurcation, here a single mode whose amplitude saturates while its phase winds - the building block of pattern-forming and spatiotemporal-chaos models.

GlacialPaillard [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Paillard ice-age model: ice volume grows slowly then collapses when insolation crosses a threshold, switching between glacial regimes - a multiple-state relaxation oscillator that reproduces the sawtooth of the Pleistocene ice ages.

GlassFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-operator FM with bright, inharmonic ratios for bell, glass and electric-piano tones. Ratio sets the modulator-to-carrier frequency ratio and Index the modulation depth (brightness and clang).

GlassHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glass-harmonic oscillator: the widely-spaced ringing modes of a rubbed glass shell (ratios near 1, 2.3, 4.2, 6.6, 9.4), the pure, slowly-decaying, almost-sine-with-a-halo tone of a wine glass or Franklin glass armonica.

GlassHarp [OSC] [FX] [SYNTH] in: Pressure, Pitch CV out: Audio

Rubbed glass (glass harmonica / wine glass): a sustained, almost-pure tone with a slow swell and a faint shimmering second partial, played by holding in0 as the 'finger pressure'. The ethereal, breathy crystal-glass voice. Tune sets the pitch, Shimmer the vibrato depth, Level the output. in0 = Pressure gate, in1 = Pitch CV.

GlassMode [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glass resonator: three high, sparsely-spaced inharmonic modes with a very long decay ring like a wet finger on a wineglass or a singing bowl when excited by the input.

GlissonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glisson: a granular stream of glissons - grains that each sweep their own pitch from start to end (Roads's gliding grain) - producing a streaky, smeared texture full of tiny internal portamenti; the glide knob sets how far each grain slides.

GlitchStutterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glitch stutter: a digital stutter effect, a tone hard-gated into a fast rhythmic on/off pattern that occasionally jumps pitch, the buffer-repeat glitch of IDM and breakcore; the stutter rate sets the chop speed.

GlobalCoupledMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Globally-coupled maps: three logistic maps each pulled toward the population mean (Kaneko's GCM), which spontaneously form and dissolve synchronized clusters - the simplest model of collective chaos and chimera-like clustering.

Glockenspiel [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal glockenspiel: a trigger on in0 strikes a metal bar modelled as bright, long-ringing inharmonic partials (1 : 2.7 : 5.4 : 8.9), the clear bell-like tone of a glock/celesta. Tune sets the pitch, Decay the (long) ring, Mallet the brightness, and Level the output.

GlottalPulse [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glottal pulse source: a Rosenberg-shaped glottal flow pulse (asymmetric open phase + sharp closure) locked to the input pitch - the raw vocal-fold excitation you feed into a formant filter to synthesize voice.

GlottalPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glottal pulse: the Rosenberg model of airflow through the vibrating vocal folds - a smooth rising open phase and a sharper closing return - which is the raw source signal that vocal-tract formants then shape into speech; the open-quotient knob sets vocal effort from breathy to pressed.

GlowDischarge [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glow-discharge oscillator: the ionization-and-recombination dynamics of a neon glow tube biased near breakdown, whose negative-resistance plasma self-oscillates and, under a series capacitance, chaotically flickers.

GlucoseInsulin [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Glucose-insulin loop: a model of the body's blood-sugar control where insulin suppresses glucose and glucose stimulates insulin with a delay, producing the ~2-hour ultradian oscillations (and, when forced, irregular swings) of metabolism.

GlycolyticHopf [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Higgins glycolytic oscillator: the substrate-product feedback of the phosphofructokinase reaction whose autocatalysis crosses a Hopf bifurcation into the sustained glycolytic oscillations measured in yeast extracts - metabolic limit-cycle rhythm.

GoatBleatOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Goat bleat: the maaa of a goat, a voiced vowel with the famously fast, deep, almost-comic vibrato (faster and wider than a sheep's) that gives the screaming-goat character, fired at the set rate.

GoldenDragon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Golden dragon: a dragon-curve variant whose two contraction ratios are set by the golden ratio (r and r^2 with r the plastic-like root), giving an asymmetric self-similar fractal subtly different from the classic Heighway dragon - a uniquely-textured coordinate CV.

GoldenOsc [OSC] [FX] [SYNTH] in: – out: Audio

Golden-ratio oscillator (a first as a synth block): an INHARMONIC tone whose partials are spaced not by whole numbers but by powers of the golden ratio (1, 1.62, 2.62, 4.24, 6.85…). Because no partial is a simple multiple of another, nothing fuses into a clear pitch the way a harmonic tone does - the result is a shimmering, metallic, bell-like timbre that sits between a pitch and a noise, the most irrationally-spaced spectrum there is. Partials sets how many golden overtones to stack, alias-free up to Nyquist. A phi-spaced inharmonic spectrum distinct from the integer-harmonic oscillators. Needs no input.

GoldenStretchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Golden-stretch oscillator: partials placed at powers of the golden ratio (1, 1.618, 2.618, 4.236 …) instead of integers, the maximally-irrational spacing that minimizes any sense of a shared fundamental - a smooth, perfectly-inharmonic, bell-without-beating tone.

GompertzMakehamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gompertz-Makeham oscillator: each cycle is the mortality hazard Alphae^(Betax)+lambda - a flat constant-risk floor that bends upward into an exponential climb - giving a waveform that hugs a baseline then accelerates skyward, the shape behind actuarial life tables. Peak-normalized, bounded.

GompertzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gompertz oscillator: each cycle is the asymmetric Gompertz double-exponential sigmoid e^(-3 e^(-c x)), which rises slowly then snaps - a lopsided soft sawtooth rich in even harmonics, distinct from the symmetric Gudermannian/tanh ramps; Steep sets the asymmetry.

GompertzPdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gompertz-PDF oscillator: each cycle is the Gompertz density betae$^{(bx)e}$(-eta(e^(b*x)-1)), an increasing-hazard distribution that ramps up then is cut off sharply by the double-exponential - a right-leaning bump with a soft rise and an abrupt fall, distinct from the Gompertz sigmoid. Peak-normalized, bounded.

GompertzQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gompertz-quantile oscillator: each cycle is the Gompertz inverse-CDF (1/B)*ln(1 - ln(1-p)/Eta), a log-of-log ramp with a gentle start and a hard-cut-off finish (the inverse of the increasing-hazard Gompertz law) - B and Eta jointly shape the curvature. Bounded.

Gong [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal gong: a trigger on in0 strikes a large gong - a dense cluster of low inharmonic metal partials that wash together into a shimmering, slowly-decaying roar with a long sustain. Tune sets the pitch, Decay the long tail, Shimmer the upper-partial brightness and clangour, Level the output. Distinct from the noise-based Cymbal - this is tuned, blooming metal. Trigger from a Clock, Euclid or sequencer.

GongHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gong: three densely-spaced low inharmonic modes plus a noisy strike transient ring with a long shimmering decay, the crashing, slowly-blooming wash of a struck gong or tam-tam.

GongModeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gong oscillator: a dense cluster of low inharmonic modes (ratios near 1, 1.46, 1.97, 2.44, 3.16, 4.0) with a slow detuning shimmer that mimics the nonlinear mode-coupling ‘build-up’ of a struck gong or tam-tam - a roaring, evolving metallic wash.

GongShowOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Big gong: the dramatic crash of a large struck gong (the game-show 'you're done' hit), a dense cluster of low inharmonic partials that shimmer up after the strike then slowly fade, fired at the set rate.

GoodwinCycle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Goodwin growth cycle: an economic predator-prey model where employment and the wage share chase each other (workers as predator, profit as prey), giving the self-sustaining boom-and-slump business cycle - econophysics as a slow modulator.

GooseHonkOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Goose honk: the loud aggressive honk of a goose, a strident harmonic-rich voiced blast with a hard nasal edge, blaring in repeated honks, fired at the set rate; the warning of the gander.

GopyAttractor [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Strange nonchaotic attractor (Grebogi-Ott-Pelikan-Yorke): a map quasiperiodically forced at the golden-ratio frequency, x' = 2sigmatanh(x)*cos(2 pi theta), whose attractor is geometrically fractal yet has no positive Lyapunov exponent - structured but not chaotic, a rare dynamical regime.

GrainCloudOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Grain cloud: an asynchronous granular texture where short windowed sine grains are sprayed at random times around a center pitch (Curtis Roads's grain cloud), building a shimmering, breathy cloud whose thickness rises with grain density.

GrainletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Grainlet: a granular stream where each grain's active window shortens as its frequency rises (Roads's grainlet, after wavelet logic) so high grains are brief and low grains stretch, yielding a self-scaling texture; the link knob sets how tightly duration tracks pitch.

GrainOsc [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Granular formant oscillator: emits one Hann-windowed sine grain per cycle of the fundamental, with the grain's internal frequency set by Formant - so pitch follows Freq while a fixed formant peak stays put, giving vocal, buzzy, FOF-style timbres. Freq sets the pitch, Formant the grain harmonic, Level the output. in0 = Pitch CV.

Granular [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Granular oscillator/texture: continuously captures the input into a ~2 s ring buffer and sprays overlapping Hann-windowed grains read back from it. Size sets grain length in ms, Density the spawn rate in grains/sec, Pitch transposes each grain in semitones, and Jitter randomizes the read origin for a diffuse, smeared texture. Mix blends the grain cloud against the dry input - low density gives a sparse stutter, high density a thick wash.

GranularGas [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Granular gas: a shaken collection of inelastically-colliding grains whose energy input fights collisional cooling; the balance oscillates and, between clustered and gaseous states, jumps chaotically - non-equilibrium granular dynamics.

GrasshopperOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Grasshopper: stridulation, where the insect rasps a leg across a wing to make a fast train of dry noisy pulses grouped into short buzzing phrases with gaps; the pulse rate sets the rasp tempo.

GravWaveChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gravitational-wave chirp: the audio analogue of a black-hole/neutron-star inspiral, where frequency and amplitude both sweep upward ever faster to a peak (merger) then cut to a brief ringdown, the LIGO ‘chirp’ that swept up and was heard in 2015; the rate sets the inspiral speed.

GrayScott [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gray-Scott reaction: the two-species reaction kinetics behind Turing-pattern formation, run as a point oscillator that pulses and oscillates between chemical concentrations - organic, cell-like rhythmic modulation.

GreenbergHastings [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Greenberg-Hastings model: the canonical excitable cellular automaton with rest, excited and graded refractory states, where a resting cell fires only if a neighbour is excited and then cycles through refractoriness before it can fire again - producing clean traveling fronts and reentrant waves.

GreyNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Grey noise: white noise shaped by an inverted equal-loudness curve (boosted lows and highs, scooped mids) so it sounds equally loud at every frequency - the perceptually-flat noise used in hearing tests.

GrindCoffeeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coffee grinder: beans crunching through a burr grinder, a harsh low motor whir overlaid with a gritty crackling crunch as beans shatter, the morning-ritual grind; the grit knob sets how many beans crunch.

GrossPitaevskii [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gross-Pitaevskii mode: a single mode of a Bose-Einstein condensate whose self-interaction (cubic nonlinearity) and a modulated trap make its amplitude and phase oscillate and, under strong driving, go chaotic - matter-wave nonlinear dynamics.

Growl [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dubstep / brostep growl bass: an FM voice whose modulation index and a midrange formant band-pass are swept together by one LFO, then pushed through heavy distortion - moving the FM amount and the formant at once is what gives the Skrillex-style ‘talking’ growl, distinct from a plain filter wobble. Tune sets the pitch (in0 = Pitch CV), Rate the growl LFO speed (tempo-sync it), Ratio the FM modulator ratio, Index the FM depth, Formant the band-pass centre the LFO sweeps, Drive the distortion, and Level the output. Layer a clean Sub808 under it for the low end.

GrowlBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Growl bass: a sawtooth through a sharp formant bandpass swept by a built-in LFO, giving a vocal, growling, talking-bass character. Growl sets the LFO rate, Formant the centre, Drive the grit. in0 = Gate, in1 = Pitch CV.

GudermannianOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gudermannian oscillator: each cycle is the Gudermannian gd(x)=2*atan(e^x)-pi/2 over a symmetric span, a transcendental S-curve that links the circular and hyperbolic functions - a soft sawtooth with a gentler-than-tanh knee, smoothly sweeping from near-sine to near-saw with Width.

Guiro [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Guiro scraper: a trigger on in0 plays one stroke across the ridges of a guiro - a ‘rrrip’ of band-passed noise rhythmically chopped by the teeth at Rate, the rasping scrape of Latin and cumbia percussion. Rate sets how fast the ridges pass, Length the stroke duration, Tone the brightness, Level the output. Re-trigger for the up-down stroke pattern.

GuitarPluckOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Guitar pluck: a Karplus-Strong delay line excited by a noise burst and damped by a lowpass in the feedback, the classic plucked-string algorithm; re-plucked at the set rate, the damping knob takes it from a bright nylon ping to a dull muted thunk.

GulpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gulp: the sound of swallowing, a short downward liquid glug ending in a soft throat click, the cartoonish ‘gulp’ of a hard swallow, fired at the set rate.

GumbelCdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gumbel-CDF oscillator: each cycle is the Gumbel (extreme-value) cumulative $e^{(-e}(-x))$, a strongly asymmetric S-curve that rises sharply on the left then crawls toward the top - a lopsided ramp distinct from the symmetric logistic/erf sigmoids. Bounded.

GumbelNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gumbel noise: samples -ln(-ln u) from the type-I extreme-value distribution, the limiting law of the maximum of many samples (flood levels, record temperatures); right-skewed with a long upper tail - a maxima-flavoured asymmetric noise.

GumbelQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gumbel-quantile oscillator: each cycle is the Gumbel (extreme-value) inverse-CDF -ln(-ln(p)), a doubly-logarithmic ramp that crawls up slowly then shoots toward the top - a strongly right-skewed monotone curve distinct from the symmetric quantiles. Bounded.

GumowskiMiraMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gumowski-Mira map: a 2-D map from particle-accelerator physics whose rational nonlinearity produces ornate, organic, almost-biological fractal orbits - a richly-textured chaotic source unlike the polynomial maps.

GunnDiode [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gunn diode: a bulk-semiconductor device whose electron transfer between energy valleys gives a negative-differential-resistance (N-shaped) velocity curve, self-oscillating into the relaxation/chaotic regime of a Gunn microwave source.

Gunshot [OSC] [FX] [SYNTH] in: Trigger out: Audio

Gunshot SFX: a trigger on in0 fires the layered crack of a weapon - a bright transient snap, a band-passed noise body for the weight, a sub LFE thump with a fast pitch drop, and a noise tail for the room reflection. The film / game firearm report. Tone sets the body and transient brightness, Body the low-mid weight, Sub the LFE thump level, Tail the room-decay length, and Level the output. Trigger from a sequencer or one-shot.

Guqin [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Guqin: the ancient Chinese 7-string fretless zither of the scholar tradition - a soft silk-string pluck with a long, quiet, intimate sustain and the famous sliding portamento (no frets) plus flageolet harmonics; the meditative voice of Chinese literati music. Slide sets the attack glide, Harmonic the flageolet shimmer.

GurgleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gurgle: water draining down a plughole, a low resonant bubbling tone whose pitch and loudness lurch irregularly as air glugs back through the vortex, the throaty gurgle of an emptying sink.

Guzheng [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Guzheng: the Chinese plucked zither - a bright, ringing metallic Karplus string with the signature attack pitch-bend (the player's press behind the bridge), giving the shimmering, pentatonic, glissando-rich voice of Chinese classical music; re-plucked at the set rate.

GyroNutation [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gyroscope nutation: a fast-spinning heavy top whose axis precesses while nodding (nutating); weak spin and a forcing torque make the nodding angle wander chaotically between precession and tumbling - rotational rigid-body dynamics.

GyrostatOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Gyrostat: a rigid body carrying a spinning internal rotor whose Euler equations, with a gyrostatic momentum bias, tumble chaotically - the attitude dynamics of a spinning spacecraft losing control.

HaarWaveletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Haar wavelet: the first and simplest wavelet, +1 on the first half of the period and -1 on the second (a single-cycle square step), the orthonormal basis behind the fastest wavelet transform - a hard, hollow, square-edged tone.

HadleyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hadley attractor: a low-order model of a Hadley convection cell in the atmosphere, whose chaotic orbit drifts on a slow, weather-like timescale - organic long-form modulation.

Hail [OSC] [FX] [SYNTH] in: – out: Audio

Hailstorm - a dense barrage of sharp icy ticks over a rain bed. Intensity sets the hail density, Tone the hardness of each strike, Level the output.

HailOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hail: a dense random scatter of hard icy impacts hammering a surface, each a short bright high-passed click, building from occasional taps to a deafening clatter; the density knob sets the storm's intensity.

HalfCauchyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Half-Cauchy oscillator: each cycle is the half-Cauchy PDF 2/(pi(1+x^2)) over x>=0, a one-sided Lorentzian that starts at its peak and decays with a very heavy (power-law) tail - a front-loaded falling waveform with far more tail energy than a half-normal.

HalfLogisticOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Half-logistic oscillator: each cycle is the half-logistic PDF $2e^{-x/(1+e}$-x)^2 over x>=0, a one-sided bump that starts at its peak and decays with logistic (heavier-than-Gaussian) tails - a front-loaded falling waveform distinct from the half-normal.

HalleyFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Halley fractal: the cubically-convergent Halley root-finding iteration applied to z^3-1 in the complex plane, whose basins of attraction interlock in a fractal pattern with thinner boundaries than Newton's; the reseeded orbit's real part is the output.

HalvorsenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Halvorsen attractor: a symmetric three-equation chaotic flow with quadratic cross-terms that traces a tight three-lobed spiral - a denser, faster-folding chaos source than the Lorenz family.

HammerDulcimerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hammered dulcimer: a bright metal string struck by a light mallet, modelled as a quick-attack Karplus pluck with a faint detuned sympathetic shimmer (the neighbouring courses ringing), the silvery sustain of a hammered dulcimer or cimbalom.

Hammond [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Tonewheel organ (Hammond): nine sine drawbars summed into one rich sustained voice, with a percussive key-click on attack and a Leslie-style vibrato - the gospel/rock B3 sound. Bright mixes in the upper drawbars, Leslie the rotary depth, Level the output. in0 = Gate, in1 = Pitch CV.

Handbell [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Handbell: a clear, ringing tuned handbell - a strong fundamental with an octave and a twelfth, a bright strike and a medium shimmering decay. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

Handpan [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Struck handpan / hang drum: mellow near-harmonic partials (1 : 2 : 3 : 4) with a soft attack and medium-long sustain - the warm, meditative hang voice. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

HandpanHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Handpan voice: three modes tuned to a fundamental, its octave and its twelfth (the near-harmonic ratios of a hang/handpan dimple) ring with a warm, bell-meets-marimba tone when struck by the input.

HandpanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Handpan: a struck note-dimple of a hang/handpan, voiced as a soft fundamental locked with its octave and compound-fifth overtones (the deliberate handpan tuning), giving a warm, meditative, bell-like ding; re-struck at the set rate.

HannPoissonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hann-Poisson oscillator: each cycle is a Hann window multiplied by a two-sided exponential (Poisson) taper, so Alpha sharpens the bump's decay - a cosine bell with exponentially-steepened skirts, between a Hann and a sech pulse in character.

HannSquaredOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hann-squared oscillator: each cycle is sin^4(pi*p), the Hann window squared - a narrower, more sharply-peaked bump than the Hann with even softer edges, giving a focused pulse with a fast-decaying harmonic series.

HardLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Hardstyle lead: detuned sawtooths slammed through drive into a bright filter for the screaming, euphoric rawstyle lead. Detune sets the width, Drive the distortion, Tone the brightness. in0 = Gate, in1 = Pitch CV.

HardSync [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hard-sync oscillator: a slave saw is reset every cycle of the master pitch, and Sync sweeps the slave frequency up for the classic tearing, vowel-like sync lead. Shape blends saw toward square.

HardSyncBlep [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Band-limited hard sync: a slave saw is reset by a master oscillator and the discontinuity at each sync reset is smoothed with a PolyBLEP, giving the screaming sync-sweep timbre without the heavy aliasing of a naive reset.

HardSyncOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hard sync: a bright slave sawtooth is force-reset to zero every time the master oscillator completes a cycle, so sweeping the slave-to-master ratio sweeps a formant peak through the spectrum - the screaming sync-lead timbre.

HardSyncSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hard-sync sawtooth: a free-running internal saw is reset to phase zero on each input zero crossing, so its (independent) Freq sweeps the classic hard-sync formant tearing while staying locked to the input pitch.

Harmonica [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Harmonica voice: a gate on in0 sounds two slightly-detuned free reeds beating together into the bright wheezing tone of a blues harp, with vibrato. Freq sets the note, Detune the reed beating (thin to thick), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

HarmonicaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Harmonica oscillator: a free reed whose buzzy, slightly-detuned tone (a sawtooth-rich source through the reed's own resonance) plus a wavering airflow models the reedy, bluesy timbre of a blues harp; the bend knob droops the pitch like a draw-bend.

HarmSweepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sweeping-formant additive: eight harmonics of the pitch are summed under a Gaussian emphasis window that an LFO slides up and down the harmonic series, so a bright resonant peak travels through the spectrum - a vowel-like wah that is built into the oscillator, not a filter.

HarperMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kicked Harper map: an area-preserving map $x' = x + K \sin(p)$, $p' = p - L \sin(x')$ derived from the Harper (almost-Mathieu) Hamiltonian behind the Hofstadter butterfly; it mixes regular tori and chaotic seas as the kick strengths vary - a quantum-chaos-flavoured source.

HarpPluckOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Harp pluck: a Karplus-Strong string with a softened (lowpassed) excitation and very light damping, giving the gentle attack and long, shimmering decay of a concert-harp string; re-plucked at the set rate for arpeggiated runs.

Harpsichord [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Harpsichord: a plectrum-plucked string pair (8' + 4' octave) with a bright, thin, buzzy attack and a quick decay - the baroque keyboard voice. Bright sets the upper-octave mix, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

HarpsichordOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Harpsichord: a metal string plucked by a quill plectrum - a very bright, lightly-damped Karplus string preceded by a sharp pluck click - giving the crisp, jangly, uniform-dynamic tone of a harpsichord; re-plucked at the set rate.

HarpsichordVoice [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Harpsichord voice: a bright, lightly-damped Karplus string plucked sharply by a gate (the quill jack) with a touch of body resonance, the crisp, metallic, even-decaying tone of a harpsichord.

HartleyChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Hartley oscillator: the tapped-inductor Hartley RF oscillator pushed past its limit cycle by the transistor's exponential transfer, a three-state circuit that tumbles into chaos - companion to the chaotic Colpitts.

HassellMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hassell map: the population model x -> r x/(1+x)^b, whose tunable competition exponent moves it between stable, cyclic and chaotic regimes - a rational-feedback population chaos with its own bifurcation route.

HastingsPowell [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hastings-Powell food chain: resource, consumer and top predator linked by saturating (Holling type-II) feeding, whose three-level interaction produces the famous ‘tea-cup’ chaotic attractor of population ecology.

Hat [OSC] [FX] [SYNTH] in: Trigger out: Audio

Analog hi-hat: a trigger on in0 fires bright high-passed metallic noise with a short percussive decay. Tone sets the brightness, Decay the length (short = closed, long = open hat), Metal blends in inharmonic ringing for a more cymbal-like tone, and Level the output.

Hat606 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-606 hi-hat: a bright metallic hat built from four detuned squares through a sharp highpass with a fast decay - thinner and brasher than the 808 hat, the sizzling Drumatix top end. Tone sets the brightness, Decay the length (short = closed, long = open), Level the output. Trigger on in0.

Hat808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 hi-hat: a trigger on in0 fires six detuned square-wave oscillators (the 808’s metal-on-metal ring) through a bright highpass with a fast decay. Tone sets the brightness, Decay the length (short = closed, long = open), Level the output.

Hat909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-909 hi-hat: six detuned squares blended with a burst of white noise through a sharp highpass - noisier and more sizzling than the 808 hat, the bright house/techno cymbal top. Tone sets the brightness, Decay the length (short = closed, long = open), Level the output. Trigger on in0.

HatSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth hi-hat: a gate fires a bright high-passed noise burst with a fast decay (open vs closed set by Decay), the sizzle on top of a drum-machine kit - a self-contained hat voice.

HealOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Healing glow: a restorative spell, a warm major chord of soft sine partials that swells in gently with a slow shimmering tremolo and a faint rising sparkle, the comforting glow of a heal/buff; the rate sets the pulse.

Heartbeat [OSC] [FX] [SYNTH] in: – out: Audio

Heartbeat: a self-clocking ‘lub-dub’ double sub-thump - two soft low pulses per beat (the first louder), each a sine with a fast pitch drop and a muffled decay - the tension pulse of film and game sound design. Rate sets the beats per second (resting ~1, racing ~2.5), Tune the thump pitch, Decay the thump length, Tone the muffled softness (low-pass), and Level the output. Self-running; automate Rate to race the pulse for suspense.

HeartbeatOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Heartbeat: the lub-dub of a beating heart, two soft low thumps per cycle (the first louder, the second quieter and close behind) repeating at the pulse rate, the muffled cardiac pulse heard through a stethoscope; the BPM knob sets the heart rate.

HeartCurveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Heart-curve oscillator: the y-coordinate 13cos t - 5cos2t - 2cos3t - cos4t of the classic heart curve, a fixed-phase blend of the first four harmonics with decreasing weights - a warm, slightly hollow tone with a built-in spectral tilt, no filtering needed.

HedgeTrimmerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hedge trimmer: an electric hedge trimmer cutting, a buzzing motor under a fast reciprocating blade clatter that rattles as the teeth chatter back and forth, the garden-hedge sound; the stroke rate sets the blade speed.

HeighwayDragon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Heighway dragon: the chaos game over the two rotate-and-scale maps that generate the dragon curve (the paper-folding fractal), whose 45-degree contractions tile the plane; the iterate’s coordinate is a self-similar, square-cornered fractal CV.

Helicopter [OSC] [FX] [SYNTH] in: — out: Audio

Helicopter - a deep rotor thump (blade-rate amplitude chop) under an engine whine. Rotor sets the blade rate, Tone the engine brightness, Level the output.

HelicopterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Helicopter oscillator: periodic blade-slap thumps at the blade-passage frequency layered with a steady rotor-whine tone, the unmistakable chop-chop-whine of a helicopter; blade count and RPM set the rhythm of the slaps.

Helmholtz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Helmholtz resonator: a single high-Q resonance (the air-spring of a bottle or cavity) excited by breath noise whistles into a pure, breathy tone, the physics of blowing across a bottle top.

HelmholtzDuffing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Helmholtz-Duffing oscillator: a forced oscillator with both quadratic and cubic stiffness (an asymmetric potential with a finite escape barrier), used to model capsize and snap-through; near the escape it is chaotic and intermittent.

Henon3D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

3-D Henon map: a three-dimensional generalization of the Henon map (x’=a-y^2-b z, y’=x, z’=y) whose extra delayed coordinate gives a thicker, more folded fractal attractor than the planar Henon.

HenonArea [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Area-preserving Henon map: the conservative quadratic twist map (a rotation followed by a parabolic shear) whose phase plane is a fractal sea of chaos riddled with stable islands - Hamiltonian map chaos.

HenonHeiles [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Henon-Heiles system: the energy-conserving Hamiltonian model of a star orbiting in a galactic potential, whose coupled quadratic potential makes bounded orbits go chaotic - conservative (non-decaying) chaos with rich quasi-periodic structure.

HenonHeilesSection [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Henon-Heiles system: the two-degree-of-freedom Hamiltonian devised for stellar orbits in a galactic potential, whose cubic coupling turns regular orbits chaotic above a critical energy; integrated here as a conservative oscillator, one position coordinate is the output.

HenonMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Henon map: the canonical 2-D quadratic chaotic map (x'=1-a x^2+y, y'=b x) whose attractor is the textbook stretched-and-folded fractal, iterated at a chosen rate as a crisp digital chaos source.

HenonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Henon attractor source: the two-variable Henon map (x' = 1 - a x^2 + y, y' = b x) produces a folded, banded chaotic signal - a 2-D strange attractor turned into sound, distinct from 1-D chaos.

HermiteGaussOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hermite-Gaussian wave: the energy eigenstates of the quantum harmonic oscillator - a Hermite polynomial times a Gaussian - swept across the phase; the Gaussian taper makes the cycle ends meet smoothly while the order sets the lobe count, a physics-derived band-limited pulse.

HestonVol [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Heston model: a price driven by a volatility that is itself a randomly-varying (CIR) process, so calm and turbulent regimes alternate - the two coupled SDEs that give the volatility clustering and fat tails real markets show.

HiccupOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hiccup: the abrupt ‘hic’ of a diaphragm spasm snapping the glottis shut, a very short voiced chirp cut off by silence, repeating at the slow irregular hiccup rate.

HicksCycle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hicks trade cycle: a multiplier-accelerator economy whose investment is clipped between a full-employment ceiling and a scrappage floor; those piecewise limits turn an explosive oscillation into a bounded, irregular business cycle.

HiddenLineEq [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Line-equilibrium flow: a three-dimensional chaotic system whose equilibria form an entire line rather than isolated points; its attractor is hidden because that line is unstable everywhere, a modern class of exotic chaos.

HiHatSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hi-hat voice: six inharmonically-tuned square oscillators summed and high-passed into a metallic hiss, gated by a fast decay (the 808 hi-hat recipe), re-triggered at the set rate; the decay knob switches between closed and open hats.

HillEquationOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hill-equation oscillator: each cycle is the Hill dose-response sigmoid $x^{N/(K}$N+x^N) (cooperative-binding pharmacology) - the Hill coefficient N sets the steepness of the switch and K its midpoint, a sigmoid ramp whose sharpness and threshold are independently controllable.

HindmarshNetwork [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled HR neurons: two Hindmarsh-Rose bursting cells joined by a fast chemical synapse, which synchronize, fire in anti-phase or burst chaotically depending on coupling - a minimal model of neural-network rhythm generation.

HindmarshOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hindmarsh-Rose neuron: a three-variable model that fires bursts of spikes separated by quiet intervals (bursting), giving rhythmic clusters of clicks - neural burst patterns no LFO produces.

HindmarshRose2D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hindmarsh-Rose fast subsystem: the two-variable core of the HR neuron (voltage + recovery, without the slow adaptation), a compact relaxation oscillator that fires fast spikes or sits bistable depending on the current.

HingeSqueakOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hinge squeak: a sharp rusty metal-on-metal squeal, a high resonant tone modulated by a fast irregular stick-slip chatter, the piercing squeak of an unoiled hinge; the squeal pitch sets the metal size.

HippopedeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hippopede oscillator: the x-coordinate of Booth's hippopede r^2 = 4b(1 - b*sin^2(theta)); as B approaches 1 the curve tends to a figure-eight and the waveform sharpens toward a rectified-cosine shape with strong even harmonics - a smoothly morphable lobed tone.

HitzlZele [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hitzl-Zele map: a three-dimensional quadratic iterated map (one of the early computer-search-discovered 3-D chaotic maps) whose coupled squared terms fold a compact strange attractor - crisp, high-rate map chaos.

HIVModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

HIV dynamics: healthy and infected T-cells and free virus in the classic Perelson-style model, whose infection-and-clearance feedback can oscillate and (with immune-response delay) go chaotic - viral-load dynamics as a modulator.

HodgepodgeMachine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hodgepodge machine: a chemical cellular automaton where cells carry an infection level - healthy cells catch it from infected neighbours, infected cells intensify then suddenly recover - reproducing the rotating spiral waves of the Belousov-Zhabotinsky reaction. Mean infection is the output.

HodgkinHuxley [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hodgkin-Huxley neuron: the Nobel-winning four-variable conductance model (membrane voltage plus sodium m/h and potassium n gating) fires biophysically-accurate action-potential spike trains under injected current - the gold-standard spiking-neuron source.

HodgkinHuxleyBurst [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bursting Hodgkin-Huxley: the full HH spiking neuron augmented with a slow calcium-activated potassium current that periodically silences the fast spikes, producing biophysically-detailed bursting and, when forced, chaos.

HopalongB [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hopalong (variant B): a three-parameter form of Barry Martin's hopalong iterated map, whose sign-and-sqrt fold lays down a denser, more lattice-like fractal than the basic hopalong - jumpy, granular chaos.

HopalongMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hopalong attractor: Barry Martin's iterated map (using sign and square-root of the running coordinate) that hops around building a layered fractal, a distinctive jumpy chaotic texture popularised in computer-recreations columns.

HopfieldChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Hopfield network: four continuous-time neurons connected by an asymmetric weight matrix (which breaks the usual convergence guarantee) wander chaotically through state space instead of settling - an artificial-neural-network chaos source.

HopfNetwork3 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Three-neuron Hopfield net: three sigmoidal neurons wired with an asymmetric weight matrix that defeats the usual convergence, so the network wanders chaotically through state space instead of settling on a memory - a compact neural-network chaos source.

HopfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hopf oscillator: a normal-form Hopf limit cycle whose amplitude self-stabilizes to sqrt(mu) - a self-sustaining sine that grows from silence and is amplitude-locked, the model behind cochlear and active-resonator synthesis.

HorseNeighOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Horse neigh: the whinny of a horse, a bright voiced tone that starts high, breaks into a fast fluttering warble and descends, fired at the set rate; the spirited call of a horse.

HouseflyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Housefly: the irritating erratic buzz of a fly, a mid-frequency wingbeat tone that randomly jumps in pitch as the insect darts and changes direction, never settling; the jumpiness knob sets how frantic the flight.

HulthenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hulthen oscillator: each cycle is the Hulthen screened-potential e$^{(-Theta*r)/(1-e}$(-Theta*r)), which diverges to a sharp spike near the origin (like a Coulomb tail) then decays exponentially - a very front-loaded, spiky-then-falling waveform. Peak-normalized, bounded.

Hum [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Closed-mouth hum: the soft, nasal ‘mmm’ of humming - a mellow sine with a low nasal formant and a gentle second harmonic. Nasal sets the formant emphasis, Vibrato the wobble, Level the output. in0 = Gate, in1 = Pitch CV.

HurtOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hurt: taking a hit, a short harsh buzz of square wave mixed with noise and a quick downward pitch drop, the unpleasant damage zap, fired at the set rate.

Hyper4Wing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic four-wing: a four-dimensional system whose orbit jumps between four wings while expanding chaotically in two directions, combining multi-lobe switching with hyperchaos for especially wild modulation.

HyperbolicChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperbolic-chirp oscillator: within each cycle the phase warps hyperbolically (via ln(1-Bend*t)), so the frequency sweep accelerates even more sharply toward the cycle end than a log chirp - a dramatic rising glide that snaps back. Bend sets how extreme the late acceleration is.

HyperbolicDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperbolic-distribution oscillator: each cycle is exp(-Alphasqrt(1+x^2)+Betax), the hyperbolic PDF whose log-density is a hyperbola - so it has a sharper peak and heavier, exponentially-decaying tails than a Gaussian, with Beta skewing it; peak-normalized and bounded.

HyperbolicSpiralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperbolic-spiral oscillator: the projection (1/theta)*cos(theta) of the spiral r=1/theta, a cosine whose amplitude decays as the cycle progresses then snaps back - a reverse-ramp swell (loud attack, fading body) opposite to the growing Fermat/Involute spirals.

HyperChen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Chen: the Chen attractor augmented with a fourth controller state and feedback into hyperchaos (two positive Lyapunov exponents), an especially turbulent broadband chaotic source.

HyperChen5D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

5-D hyperchaotic Chen: the Chen attractor lifted to five dimensions with two controller states, yielding multiple positive Lyapunov exponents and a dense, maximally-unpredictable chaotic signal.

HyperexpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperexponential oscillator: each cycle is a mixture of two exponential decays at different rates, Mix blending a fast and a slow falloff - a front-loaded waveform with a compound (knee-then-tail) decay distinct from a single exponential; peak-normalized and bounded.

HyperJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperjerk (snap) system: a single fourth-order differential equation in one variable and its first three derivatives (a ‘snap’ circuit), the minimal scalar route to hyperchaos with two positive Lyapunov exponents.

HyperJerk5D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

5-D hyperjerk: a single fifth-order scalar equation (x and its first four derivatives - ‘snap’ and ‘crackle’) with a tanh nonlinearity, the minimal one-variable route to multi-exponent hyperchaos.

HyperJha [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Jha: a four-state Lorenz-derived hyperchaotic system with an added linear feedback loop, producing dense high-dimensional chaos for the most unpredictable modulation.

HyperLorenz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Lorenz: the Lorenz system augmented with a fourth feedback state so it has two positive Lyapunov exponents, expanding chaotically in two directions for an especially unpredictable, broadband signal.

HyperLorenz5D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

5-D hyperchaotic Lorenz: the Lorenz system extended with two extra feedback states, supporting up to three positive Lyapunov exponents - an extremely high-dimensional, broadband chaotic source.

HyperLu [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Lu: the Lu attractor extended with a fourth state and feedback gain into hyperchaos, a dense, high-dimensional chaotic source for the most turbulent modulation and tones.

HyperPang [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Pang: a Lorenz-derived four-dimensional system with an extra state coupled into the first two equations, yielding hyperchaos with a broad, twin-scroll-plus-drift orbit.

HyperRabinovich [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Rabinovich: the plasma-wave Rabinovich system extended with a fourth state and feedback into hyperchaos, combining the spiky cubic dynamics of Rabinovich-Fabrikant with two-directional chaotic expansion.

HyperRikitake [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Rikitake: the two-disk geodynamo model extended with a fourth feedback state into hyperchaos, intensifying the chaotic field reversals of the Rikitake dynamo - turbulent geomagnetic-style switching.

HyperRossler [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rossler hyperchaos: the four-variable Rossler system with two positive Lyapunov exponents, so it expands chaotically in two directions at once - a more violently unpredictable signal than 3-D chaos.

HyperSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seven detuned saw voices in a tight unison (a classic supersaw): Detune spreads the voices and Blend balances the centre voice against the spread sides for anything from a subtle thickening to a screaming trance lead.

HyperSprott5D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

5-D hyperchaotic Sprott flow: a minimal five-variable chain with a single quadratic nonlinearity and cyclic linear coupling tuned for two positive Lyapunov exponents - an elegant, high-dimensional hyperchaos generator.

HyperWang [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Wang: a four-state system with two quadratic couplings and a linear controller that produces two positive Lyapunov exponents, a dense broadband hyperchaotic source.

HyperYu [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Yu: a four-state Lorenz-derived system with a fourth feedback variable producing two positive Lyapunov exponents, a dense, high-dimensional chaotic source for broadband turbulence.

HypocycloidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hypocycloid oscillator: the curve from a point on a circle rolling inside a larger one (the family of the astroid and deltoid), whose ratio sets the number of inward cusps; the x-coordinate gives a pinched, star-like waveform.

IceCrackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cracking ice: a frozen lake or glacier under stress, sharp snapping cracks each followed by a low eerie groan as the fracture propagates and the sheet resonates, fired at the set rate; the haunting voice of ice.

IceSpellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ice spell: a freezing frost magic, a glassy cluster of high inharmonic partials with a crystalline shimmer and scattered shattering tinkles, the cold crackle of an ice attack; the rate sets the cast frequency.

IkedaDelay [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ikeda delay system: a delay-differential model of light in a nonlinear ring cavity (x' = -x + mu sin(x(t-tau))), whose feedback through a sine of its delayed self generates broadband delay-induced chaos.

IkedaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ikeda map: the discrete laser-ring-cavity map whose rotation angle depends on the current radius, iterated at a chosen rate - a tightly-folded 2-D chaos with a very different texture from the flows.

Impact [OSC] [FX] [SYNTH] in: Trigger out: Audio

Cinematic drop impact: a trigger on in0 fires a deep sub-sine whose pitch plunges from a high boom down to Tune, layered with a short noise crack on the transient and a long decaying sub tail - the 'boom' that lands on a drop or downbeat. Tune sets the final sub pitch, Boom the pitch-plunge depth and time, Decay the sub-tail length, Noise the transient crack level, and Level the output. Distinct from Kick: a deep cinematic hit, not a tuned drum.

ImpactBoomOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Impact boom: a cinematic trailer hit, a deep sub-frequency thump with a downward pitch drop layered with a bright noise crack at the transient, the chest-thumping BRAAAM-impact of film sound design; fired at the set rate.

ImpactOscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Impact oscillator: a sinusoidally-driven spring-mass that bounces off a rigid wall with partial restitution; near-grazing impacts trigger the discontinuity-induced chaos studied in rattling machinery and vibro-impact systems.

Impulse [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Band-limited impulse/pulse train (a BLIT): a stream of clean clicks at the pitch rate, the raw material for subtractive synthesis and excitation. Width morphs the impulse toward a narrow pulse.

InharmBellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inharmonic bell: five sine partials tuned to non-integer ratios (the clangorous spectrum of a struck bell or metal bar), with a Stretch knob that pulls the upper partials further from harmonicity - from chime to dissonant gong, sustained rather than struck.

Insects [OSC] [FX] [SYNTH] in: – out: Audio

Insect night ambience: a bed of randomly-timed cricket chirps - a high tone (the stridulation pitch) pulsed by a fast wing-stroke amplitude modulation and gated into short chirps - the field, jungle and night-time critter ambience of film sound design. Self-running. Density sets how often the chirps fire, Pitch the stridulation tone, Rate the wing-stroke pulse speed inside each chirp, Spread the random pitch variation between insects, and Level the output.

InsectSwarmOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Insect-swarm oscillator: a cluster of detuned buzzing tones (a cloud of flying insects), each randomly fluttering in level as individuals weave nearer and farther, summing into the restless drone of a swarm; spread sets how wide the detune.

IntegrateFireChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Driven integrate-and-fire: a leaky integrate-and-fire neuron with a sinusoidal drive whose firing times become chaotic (devil's-staircase / mode-locking structure), emitting an irregular pulse train locked loosely to the drive.

IntercomBuzzOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Intercom buzzer: the harsh rasping buzz of an apartment door-entry intercom releasing the lock, a gritty square-wave tone gated in short blasts with a mains-frequency rasp; the let-me-in buzz.

IntermodOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Intermodulation oscillator: two tones at a non-integer ratio passed through a cubic nonlinearity, which breeds sum-and-difference intermodulation product tones (2f1-f2, 2f2-f1, …) - the clangorous sideband cluster of overdriven mixers and amps.

InverseExponentialOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inverse-exponential oscillator: each cycle is the PDF of 1/X for an exponential X, $(1/x^{2)e}(-1/x)$, which is zero at the origin, rises to a sharp off-centre peak near x=0.5, then decays with a heavy tail - a hollow-then-spiked shape distinct from the centre-peaked bells. Peak-normalized, bounded.

InverseGammaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inverse-gamma oscillator: each cycle is the inverse-gamma PDF x^(-a-1) e^(-b/x), a right-skewed bump with a hard zero approach at the origin and a heavy polynomial tail - Alpha sets the tail weight, Beta the peak position; peak-normalized and bounded.

InverseRayleighOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inverse-Rayleigh oscillator: each cycle is the inverse-Rayleigh PDF $(1/x^{3)e}(-1/(2*sigma^{2*x}2))$, which is zero at the origin, rises to a sharp off-centre peak, then decays with a heavy power-law tail - Sigma sets the peak position, a distinctly front-skewed spike. Peak-normalized, bounded.

InversiveCongruential [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Inversive congruential noise: a nonlinear PRNG that uses the modular multiplicative inverse (x = a*x^-1 + c mod p) instead of a plain multiply, breaking the lattice structure of linear generators - a noise free of the planes that plague LCGs.

InvoluteOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Involute oscillator: the x-coordinate of the involute of a circle, cos(t)+t*sin(t), swept over several turns per cycle - a sine whose amplitude grows linearly across the period then resets, giving a built-in ramp-shaped swell distinct from the exponential LogSpiral.

IsingMeanField [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mean-field Ising magnet: the average magnetization of a spin system relaxing toward its self-consistent (tanh) value under an oscillating applied field; near the Curie point the dynamic hysteresis loop pulses and chaotically flips - phase-transition dynamics.

ISLMModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

IS-LM cycle: the textbook macroeconomic model of goods and money markets, where a nonlinear investment response coupled to interest-rate dynamics self-oscillates and (with capital accumulation) goes chaotic - aggregate-economy cycles.

Izhikevich [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Izhikevich spiking-neuron oscillator (2003): integrates the two-variable model whose membrane voltage fires spikes and bursts, taken straight as a sound/control source - the Regular, Bursting, Chattering and Fast firing modes give clicky, gurgling and rhythmic textures no wavetable makes. A biological neuron model repurposed for synthesis. Rate scales the firing speed (in0 = Pitch CV), Mode picks the pattern, Current the drive (more = faster, denser firing), Level the output.

IzhikevichNet [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Izhikevich network: a dozen of the efficient two-variable spiking neurons wired in a ring with excitatory coupling, so spikes propagate and the population settles into synchronous bursts or asynchronous chatter depending on drive - the summed firing makes a lively neural CV.

IzhikevichNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Izhikevich neuron: a two-variable model with a reset rule that reproduces a huge range of cortical firing patterns (regular, fast-spiking, chattering) from two knobs - efficient biological spiking dynamics.

JacobiCnOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jacobi elliptic cn: the elliptic cousin of the cosine from its q-series, which sharpens from a smooth cosine toward a narrowing pulse as the nome increases - the 90-degree partner of sn with a distinct, more peaked harmonic spectrum.

JacobiDnOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jacobi elliptic dn: the strictly-positive elliptic function (the delta amplitude) from its q-series, which rises from a flat DC level toward a train of sharp positive humps as the nome grows - a unipolar, formant-like elliptic source.

JacobiSnOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jacobi elliptic sn: the doubly-periodic generalization of the sine, built from its q-series (nome) so that as the nome rises the smooth sine continuously hardens toward a square wave - one knob morphs sine to square through every elliptic shape between.

JacobsLadderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jacob's ladder: the classic climbing spark between two diverging rods, a buzzing arc tone that rises in pitch and thins as it travels up the widening gap, then snaps back to the bottom to climb again; the climb rate sets the ascent speed.

JafariNE1 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

No-equilibrium flow I: a minimal three-state chaotic system from the Jafari-Sprott search that has no fixed points at all, so its attractor is 'hidden' - the orbit stays chaotic with nowhere to converge.

JafariNE17 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

No-equilibrium flow XVII: a later entry in the Jafari-Sprott catalogue of fixed-point-free chaotic systems, with a quadratic-plus-constant third equation yielding a distinct hidden attractor.

JafariNE2 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

No-equilibrium flow II: a second Jafari-Sprott system with no fixed points, whose product-and-constant structure hides a chaotic attractor that exists without any equilibrium to organize it.

JafariNE5 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

No-equilibrium flow V: another fixed-point-free Jafari-Sprott system whose absolute-value nonlinearity sustains a hidden chaotic attractor, found only by scanning initial conditions rather than locating equilibria.

JafariNE9 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

No-equilibrium flow IX: another of the Jafari-Sprott no-equilibrium systems, with a different nonlinear coupling that yields a distinct hidden chaotic attractor (no fixed points, found only by initial-condition search).

JanoschekOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Janoschek oscillator: each cycle is the Janoschek growth curve 1-(1-Beta)e$^{(-x}$Gamma), a sigmoid rise that starts from a raised floor Beta (not zero) and whose inflection sharpness Gamma is independent - a growth ramp with a built-in lower asymptote. Bounded.

JansenRit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jansen-Rit neural mass: a cortical-column model of coupled excitatory and inhibitory populations whose sigmoid feedback generates alpha-band EEG-like oscillations and, with strong input, chaotic and epileptiform rhythms.

JawHarp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jaw harp: a fixed-pitch metal-tongue twang whose bright resonant formant is swept by an envelope on each pluck, the boing-wah ‘mouth harp’ sound where the drone stays put and the formant moves.

JawHarpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jaw-harp oscillator: a fixed-pitch buzzing lamella (a decaying-then-replucked twang) filtered by a sweepable mouth-cavity formant that selects which overtones ring, the ‘boing-wow’ twang of a jew’s-harp shaped by the player’s mouth.

JeffcottRotor [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jeffcott rotor: a spinning unbalanced shaft whose whirl orbit intermittently rubs the stator, the nonlinear contact stiffness folding the synchronous whirl into chaotic, rattling motion - a rotating-machinery fault model, read as the shaft displacement.

JerkAbs [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Absolute-value jerk: the elegant chaotic jerk system x’’‘=-a x’’ - x’ - |x| + 1, one of the simplest electronic circuits that produces chaos, with a sharp piecewise-linear fold.

∿ JerkCosine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cosine jerk: a jerk system whose nonlinearity is a cosine of the velocity (x''), an unusual feedback that bends the flow into a distinctive folded chaotic attractor unlike the polynomial jerks.

∿ JerkExp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Exponential jerk: a jerk system x'''=-a x'' - x' - (e^x - 1) whose one-sided exponential restoring force breaks the symmetry, giving an asymmetric, spiky chaotic flow.

∿ JerkMemristive [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Memristive jerk: a third-order jerk circuit whose nonlinearity is a flux-controlled memristor (cubic memductance), so the chaotic flow carries the pinched-hysteresis character of a memristor element.

∿ JerkMult [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Multiplicative jerk: a jerk system x'''=-a x'' + x x' - x whose product-of-state nonlinearity (a multiplier element) drives a smooth single-scroll chaotic flow distinct from the polynomial-in-x jerks.

∿ JerkPiecewise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Piecewise jerk: a jerk system whose nonlinearity is a hard min/max clamp of x, the easiest chaotic circuit to build from op-amp comparators - producing a crisp, angular chaotic flow.

∿ JerkQuad [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quadratic jerk: the Sprott jerk system x'''=-a x'' + (x')^2 - x, whose squared velocity term folds a smooth single-scroll chaotic attractor from a minimal third-order equation.

∿ JerkSign [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Signum jerk: a jerk system x'''=-a x'' - x' + x - 2 sgn(x) whose hard relay nonlinearity switches the restoring force, giving an abruptly-switching, square-edged chaotic flow.

∿ JerkSine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sine jerk: a jerk system x'''=-a x'' - x' - sin(x), the third-order pendulum-like flow whose bounded sinusoidal restoring force folds a smooth chaotic attractor - the autonomous cousin of the driven pendulum.

∿ JerkTanh [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tanh jerk: a jerk system x'''=-a x'' - x' - 2 tanh(x) whose smooth saturating nonlinearity yields a gently-folding chaotic flow - the soft-clip cousin of the signum jerk.

∿ Jet [OSC] [FX] [SYNTH] in: – out: Audio

Jet engine - a massive filtered-noise roar topped by a screaming turbine whine. Thrust sets the power, Tone the whine pitch, Level the output.

JetEngineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jet engine: the scream of a turbofan, a high turbine whine and its harmonics riding on a massive broadband combustion roar, building with throttle; the throttle knob spools the engine up in pitch and intensity.

JetFlute [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Air-jet flute model: a delay-line waveguide closed by a cubic jet nonlinearity and excited by breath noise self-oscillates into a breathy flute tone - physical wind synthesis from a delay and a polynomial.

JetMidlatitude [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Charney-DeVore jet: a low-order model of the mid-latitude atmospheric jet that flips between a fast 'zonal' flow and a stuck 'blocking' pattern; topographic and thermal forcing make the regime transitions irregular - weather-regime chaos.

JewsHarp [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Jew's harp / mouth harp: a fixed low twang reed whose harmonics are shaped by a sweeping mouth-cavity formant, retriggered on in0 - the wobbling 'boing-wow' of a plucked lamellophone. Tune sets the twang pitch, Sweep the formant motion, Level the output. Trigger on in0, in1 = Pitch CV.

JohnsonSuOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Johnson-SU oscillator: each cycle is the Johnson SU PDF, a Gaussian warped through an inverse-hyperbolic-sine transform so Gamma skews it and Delta sets its tail weight independently - an extremely flexible bell spanning sharp-and-heavy-tailed to broad, peak-normalized and bounded.

JonesPewseyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jones-Pewsey oscillator: each cycle is the Jones-Pewsey circular family (cosh(kpsi)+sinh(kpsi)cos(theta))^(1/psi), a single shape that contains the von Mises, cardioid and wrapped-Cauchy as special cases - Kappa sets concentration, Psi morphs the peak character continuously. Peak-normalized, bounded.

JosephsonArray [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled Josephson junctions: two driven superconducting junctions sharing a load resistor, whose phases compete between synchronization and chaos - the building block of Josephson-junction arrays used as voltage standards and chaotic emitters.

JosephsonJunction [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Josephson junction: a superconducting junction whose phase obeys a damped, driven pendulum equation (phi'' + beta phi' + sin phi = I + A sin wt); the tilted washboard potential makes the voltage chaotic - quantum-electronics chaos.

JosephsonStack [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stacked Josephson junctions: three series junctions sharing a bias current and coupled through their phases (as in intrinsic-junction THz emitters), whose competition between synchronization and chaos shapes the emitted spectrum.

JovianBurstOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jovian S-burst: the fast staccato ‘popcorn’ of Jupiter’s decametric radio storms (driven by the Io flux tube), short noisy crackle bursts at a rapid random rate, the sound radio astronomers hear from Jupiter on a shortwave receiver.

JuliaOrbit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Julia orbit: iterates the complex map z -> z^2 + c (for a fixed c on a Julia-set boundary) and reseeds whenever the orbit escapes, so the real part wanders chaotically through the fractal’s filigree - complex-plane dynamics as sound.

JuliaQuartic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quartic Julia orbit: iterates the complex map z -> z^4 + c, whose four-fold symmetry folds the orbit through a busier, higher-symmetry fractal boundary than the quadratic Julia set - denser complex-plane chaos.

JumpDiffusion [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Merton jump-diffusion: a Brownian-motion price path with occasional sudden Poisson-timed jumps of random size, capturing the crashes and gaps that pure diffusion misses - a smooth wander punctuated by abrupt discontinuous leaps.

JumpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jump: the springy hop of a platform hero, a square-wave tone that sweeps quickly upward in pitch as the character leaps, fired at the set rate; the boing of arcade jumping.

JunoVoice [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Juno-style voice: a single sawtooth with a square sub-oscillator an octave down and PWM movement, run through a lowpass and lightly detuned for the lush Roland Juno-106 character. Cutoff sets the filter, Sub the sub-oscillator level, Level the output. in0 = Gate, in1 = Pitch CV.

KaiserPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kaiser pulse: one cycle shaped by the Kaiser window I0(beta*sqrt(1-x^2))/I0(beta), the near-optimal adjustable window built from the modified Bessel function; beta trades main-lobe width against sidelobe level, morphing the pulse from rectangular toward Gaussian.

KaldorModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kaldor business cycle: a macroeconomic model where the sigmoidal investment response to income, coupled to capital accumulation, self-oscillates and (when forced) goes chaotic - cyclical economic dynamics as a control source.

KaleckiCycle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kalecki cycle: the first delay-based business-cycle model, where investment decisions take a gestation time to become capital; that delay alone destabilizes the economy into sustained (and, when nonlinear, chaotic) cycles.

Kalimba [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal kalimba (thumb-piano tine): a trigger on in0 plucks a metal tine modelled as a strong fundamental plus a couple of mellow inharmonic partials (1 : 2.0 : 3.9) with a soft buzzy onset. Tune sets the pitch, Decay the ring length, Buzz the attack-noise transient, and Level the output.

KalimbaModeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kalimba oscillator: a clamped cantilever tine whose few stretched bending modes (ratios near 1, 6.27, 17.5) give the mbira/kalimba its plucky, slightly-buzzy, fast-decaying ping dominated by the fundamental with sparse high overtones.

KapitzaPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kapitza pendulum: a pendulum whose pivot vibrates so fast that the normally-unstable inverted (upright) position becomes stable; near the stability boundary it wobbles chaotically about the top - the counterintuitive dynamic-stabilization effect.

KaplanYorke [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kaplan-Yorke map: a two-dimensional map coupling a modular doubling of one coordinate with an exponentially-damped cosine drive of the other, producing a banded fractal chaos used to study chaotic dimension.

KappaDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kappa oscillator: each cycle is the kappa distribution $(1+x^{2/Kappa})(-Kappa-1)$, a symmetric bell (from space-plasma physics) whose tails are heavy for small Kappa and approach a Gaussian as Kappa grows - one knob continuously sweeping tail weight between Lorentzian-like and Gaussian-like. Peak-normalized, bounded.

KarplusStretch [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Stretched Karplus-Strong: a gate plucks a tuned delay loop whose damping filter is blended toward all-pass by a Stretch knob, extending sustain from a dull thud to an endless drone - extended string synthesis.

KatydidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Katydid: the raspy ‘katy-did’ night call, a pair of short high broadband rasps repeated steadily with a gap, the relentless summer-night chorus of the bush cricket; the call rate sets the species and temperature.

KauffmanNK [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kauffman NK network: sixteen Boolean nodes each updated by a fixed random logic function of two fixed inputs (wiring and rules derived deterministically from node index); the network settles onto one of its attractor cycles - a model of gene-regulatory order at the edge of chaos.

KazooOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kazoo: humming through a kazoo’s membrane, a buzzy nasal tone where a tuned drone is broken up by a fast membrane rattle into the classic comb-and-paper raspberry, the party-favour buzz; the pitch sets the hum.

KeenModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Keen model: Steve Keen's extension of Goodwin's growth cycle with private debt, whose nonlinear investment-and-interest feedback produces Minsky-style booms followed by debt-driven crashes - financial-instability dynamics.

Kenong [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Gamelan kenong: a large bronze kettle-gong with a deep, round, resonant tone and a long humming decay - the time-keeping voice of the gamelan. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

KeplerOrbitOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kepler-orbit oscillator: solves Kepler's equation each cycle for the eccentric anomaly of an orbit and reads out the radial position, so a circular orbit (eccentricity 0) gives a pure sine and a comet-like ellipse gives a slow drift with a sharp swing at perihelion.

Kettle [OSC] [FX] [SYNTH] in: – out: Audio

Boiling kettle - a bubbling hiss that rises into a building steam whistle, then resets. Cycle sets the boil time, Pitch the whistle height, Level the output.

KettleWhistleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kettle whistle: a stovetop kettle coming to the boil, a high near-sine whistle that rises in pitch as the steam builds then holds steady, with a breathy noise edge and a faint warble; the boil knob sets how close to full boil.

KeyboardTypeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mechanical keyboard: fingers clattering across a keyboard, a stream of short clicky key-press impacts fired at the typing rate, each a brief filtered tick with a hard onset, the office-desk typing sound; the rate sets the words-per-minute.

KeyClickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Key click: the short mechanical tick of a keypress (a keyboard tap or UI button), a very brief filtered-noise click with a faint resonant body, fired at the set rate; the tactile feedback blip.

KeyJingleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jingling keys: a keyring being shaken or picked up, a random scatter of short bright metallic clinks at varying high pitches as the keys knock together, the pocket-jangle of keys; density sets how vigorously they shake.

Kick [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog kick drum: a trigger on in0 fires a sine whose pitch sweeps from a high click down to the body Tune, with a percussive amp Decay. Tune sets the fundamental, Punch the pitch-drop time (snap), Decay the body length, Drive adds saturation for harder kicks, and Level the output - the synthesized 808/909-style kick. Patch a Clock, Euclid or sequencer gate into in0.

Kick606 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-606 kick drum: a short, punchy bass drum with a fast pitch sweep and a tight decay - brighter and snappier than the 808's long boom, the driving Drumatix kick. Tune sets the base pitch, Decay the length, Level the output. Trigger on in0.

Kick707 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-707 kick drum: a tight, clicky digital-flavoured bass drum - a short pitch-swept sine with an added transient click at the attack, the dry 707 thump used across early house. Tune sets the pitch, Click the attack snap, Level the output. Trigger on in0.

Kick808 [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

TR-808-style kick drum: a trigger on in0 fires a long, deep sine boom with a downward pitch sweep and a soft saturated tail - the classic 808 sub-kick. Tune sets the body pitch, Decay the (long) length, Punch the attack pitch-sweep amount, Level the output. in1 = Pitch CV.

Kick909 [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

TR-909-style kick drum: punchier and clickier than the 808 - a trigger on in0 fires a sine body with a fast, snappy pitch envelope plus a sharp transient click for cut. Tune sets the body pitch, Decay the length, Click the attack-click amount, Level the output. in1 = Pitch CV.

KickedTop [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kicked top: an angular-momentum vector that precesses freely then receives a periodic nonlinear twist (the classical limit of a quantum-chaos paradigm), tumbling chaotically over the sphere - read as one spin component.

KickSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth kick drum: a gate triggers a sine whose pitch drops from a punchy transient down to the tuned fundamental while the amplitude decays - a self-contained 808/909-style kick voice.

KickSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kick-drum voice: a sine whose pitch drops sharply from a click down to a low boom, re-triggered at the set rate, the classic 808/909 analog bass-drum synthesis; the drop and decay knobs shape it from a tight thud to a long sub.

KingsDream [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

King's Dream attractor: a sine-and-power iterated map from Pickover's collection that traces sweeping, dreamlike fractal arcs, a flowing chaotic texture with broad, smooth orbits.

Knock [OSC] [FX] [SYNTH] in: — out: Audio

Door knock - a periodic double or triple rap on wood with a hollow resonance. Rate sets how often, Tone the wood resonance, Level the output.

KochWave [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Koch waveform: the height profile of a Koch snowflake edge, built by recursively replacing each segment with a bumped four-segment zig-zag, gives a crinkled fractal waveform whose jaggedness deepens with the iteration depth.

KohonenSOM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Self-organizing map (Kohonen): a chain of nodes with adjustable weights is shown random inputs; the closest-matching node and its neighbours are nudged toward each input, so the initially-jumbled weights untangle into a smooth ordered map - a probe node's weight settles as it self-organizes.

KolmogorovFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kolmogorov flow: a three-mode Galerkin truncation of two-dimensional shear-driven fluid turbulence, whose energy sloshes chaotically between large- and small-scale modes - a toy model of the route to turbulence.

Kora [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kora: the West African 21-string harp-lute - a warm nylon-string Karplus pluck rung through a calabash-gourd body resonance, the soft, rolling, harp-like tone of griot music; re-plucked at the set rate.

Koto [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Japanese koto: a plucked zither string with a bright woody attack and the characteristic upward pitch bend as the note settles (the ato-oshi press), over a Karplus-Strong string. Decay sets the ring, Bend the attack pitch rise, Level the output. Trigger on in0, in1 = Pitch CV.

Krell [OSC] [FX] [SYNTH] in: – out: Audio

Self-generating Krell patch: a self-running voice that, at random intervals, jumps to a new random pitch and glides its amplitude toward a new random level - the endlessly-evolving generative tone (after the Forbidden Planet patch). Rate sets the event speed, Range the pitch spread in octaves, Level the output. No input needed; patch it and walk away.

KumaraswamyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kumaraswamy oscillator: the closed-form Beta-like PDF $abx^{(a-1)(1-x}$a)^(b-1) on [0,1], normalized to its mode - similar to the Beta bump but with distinctly different edge tails, giving a related-yet-separate family of skewable compact waveforms.

KumaraswamyQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kumaraswamy-quantile oscillator: each cycle is the closed-form Kumaraswamy inverse-CDF $(1-(1-p)^{(1/B))}$(1/A), a bounded [0,1] S whose two exponents bend the early and late curvature independently - a fully-bounded, doubly-skewable ramp distinct from the unbounded heavy-tail quantiles.

KuramotoChain [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kuramoto chain: four phase oscillators at spread frequencies coupled all-to-all; as coupling rises they pass through chimera-like partial synchronization into full lock, the mean field tracing a rich evolving waveform.

KuramotoChainNN [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kuramoto chain: a line of phase oscillators with spread-out natural frequencies coupled only to their immediate neighbours (not all-to-all), so synchronization spreads locally and can form travelling phase waves and partially-locked clusters - the order parameter tracks the local sync.

KuramotoSivashinsky [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Kuramoto-Sivashinsky modes: a low-mode truncation of the equation governing unstable flame fronts and falling liquid films, whose competing growth and damping of two Fourier modes produce spatiotemporal-chaos-flavoured signals.

LacOperon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lac operon: the classic bacterial gene switch where lactose induces the enzymes that import it, a positive feedback giving bistability and, with degradation delays, oscillatory induction - molecular-biology dynamics as a control source.

LaggedFib [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lagged-Fibonacci generator: a PRNG that combines two earlier outputs from a ring of past values (x_n = x_{n-j} + x_{n-k} mod 1), a long-period additive generator whose noise has a flatter spectrum than a simple linear-congruential source.

LaguerreFuncOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laguerre-function oscillator: each cycle is the nth Laguerre function L_n(x)*e^(-x/2) (the radial hydrogen-atom / quantum form), which crosses zero n times then decays - so Order sets how many oscillating lobes appear before the exponential tail, a damped multi-lobe waveform. Magnitude-normalized, bounded.

LaguerreGaussOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laguerre-Gaussian wave: a Laguerre polynomial multiplied by a decaying exponential (the radial profile of optical Laguerre-Gaussian beams and hydrogen orbitals), swept over the phase as an asymmetric, one-sided-decaying pulse whose order sets the ripple count.

LaiSystem [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lai system: a three-dimensional autonomous chaotic flow with two quadratic terms producing a compact multi-scroll attractor, one of the systematically-constructed systems used in chaos-based secure communication.

LanczosSincOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lanczos-sinc oscillator: each cycle is a sinc multiplied by a wider sinc window, sinc(x)*sinc(x/Lobes), the Lanczos resampling kernel - a central lobe with a controlled number of decaying side-lobes that taper smoothly to zero at the edges, distinct from the infinite sinc and the sinc-squared. Bounded.

LandauLifshitz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Landau-Lifshitz spin: a magnetization vector precessing about an effective field with Gilbert damping; an oscillating drive field tips it into chaotic precession on the unit sphere - the physics of spin-torque oscillators.

LangKobayashi [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lang-Kobayashi laser: a semiconductor laser whose field re-enters after a delay (optical feedback) destabilizes into coherence-collapse chaos; here a delayed self-feedback of the intensity drives broadband chaotic fluctuation - the physics of a laser pointed at a mirror.

LaplaceNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laplace noise: samples from the double-exponential distribution (a sharp peak at zero with exponential tails on both sides), drawn via the inverse-CDF -sign(u)*ln(1-2|u|); spikier than Gaussian but lighter-tailed than Cauchy.

LaplaceQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laplace-quantile oscillator: each cycle is the Laplace inverse-CDF (ln(2p) below the midpoint, -ln(2(1-p)) above), a two-sided logarithmic ramp with a sharp cusp at the centre and steep ends - a distinctly kinked S unlike the smooth quantiles. Bounded.

LaplaceResonanceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laplace-resonance oscillator: three voices locked in the 1:2:4 orbital resonance of Jupiter’s moons Io, Europa and Ganymede (whose periods never let all three align), summed into a perfectly-periodic but intricately-phased composite - celestial mechanics as an additive tone.

LarsenHowl [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Larsen feedback howl: a delayed feedback loop with a resonant peak and amplitude limiting models a microphone pointed at a speaker, building up the squealing PA-feedback tone that locks to the loop’s strongest resonance.

Laser [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Sci-fi laser blaster: a trigger on in0 fires a square tone with a steep attack and a fast descending pitch drop, ring-modulated by a carrier for the metallic ‘pew’, through a resonant low-pass that tracks the pitch, with a noise-grit layer. The blaster bolt and beam of sci-fi sound design. Tune sets the start pitch (in1 = Pitch CV), Drop the pitch-envelope descent depth, Ring the ring-mod carrier (metallic colour), Length the decay (short = bolt, long = beam), Grit the noise layer, and Level the output. Distinct from Zap: ring-modulated and metallic, with a beam length.

LaserShootOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laser shot: the pew of a space-shooter blaster, a square-wave tone diving rapidly in pitch with a touch of buzz, fired at the set rate; the 8-bit weapon staple.

Latoocarfian [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Latoocarfian attractor: Clifford Pickover’s four-sine iterated map (x’=sin(by)+c sin(bx), y’=sin(ax)+d sin(ay)), which folds into delicate, organic, often bird- or leaf-like fractal forms - a flowing chaotic texture.

LatticeBoltzmann1D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lattice-Boltzmann (D1Q3): a mesoscopic fluid model where left, rest and right particle populations stream to neighbours and relax toward local equilibrium each step (the BGK collision), recovering wave-and-diffusion behaviour; the cell density sloshes as a fluid pulse circulates.

LaueOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laue oscillator: each cycle is the N-slit diffraction intensity (sin(N*theta)/sin(theta))^2 normalized - a train of sharp principal maxima separated by Slits-1 tiny subsidiary peaks - so raising Slits concentrates the energy into ever-narrower spikes, a grating-interference waveform distinct from any single bump.

LaughOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Laughter: a string of voiced ‘ha-ha-ha’ bursts, a vowel tone chopped into rhythmic syllables that drift slightly in pitch, the contagious rhythm of a laugh; the syllable rate sets how fast the laughter tumbles.

LawnmowerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lawnmower: a petrol push-mower running, a low four-stroke engine putt-putt with a steady blade whir and exhaust chuff, the summer-yard drone; the RPM sets the engine speed.

LeafBlowerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Leaf blower: a handheld leaf blower at full throttle, a loud high-pressure air whoosh of broadband noise over a whining impeller motor, the autumn-yard roar; the throttle shapes the air rush.

LegendreOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Legendre wave: the Legendre polynomial P_n evaluated on cos(theta) as the phase sweeps, an orthogonal-polynomial waveform whose order sets the number of lobes per cycle - the angular harmonics of a sphere, rendered as a band-limited tone.

LemniscateOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lemniscate sine: the elliptic sine sl at the lemniscatic modulus (k^2 = 1/2, the case Gauss studied that connects pi to the lemniscate constant), giving a fixed, slightly-squared sine with the characteristic lemniscatic overtone balance.

LengyelEpstein [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lengyel-Epstein model: the two-variable kinetics of the chlorite-iodide-malonic-acid reaction (the first lab demonstration of Turing patterns), an activator-inhibitor oscillator with sharp relaxation pulses.

LennardJonesOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lennard-Jones oscillator: each cycle traces the 12-6 intermolecular potential $4((1/r)^{12-(1/r)}6)$ - a steep repulsive wall, a sharp attractive well, then a flat tail - giving a strongly asymmetric waveform with a hard spike and a rounded dip, the shape of a diatomic bond. Magnitude-normalized, bounded.

LevelClearOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Level clear: completing a stage, a triumphant rising fanfare of square-wave notes ending on a held high chord-tone, the victorious end-of-level flourish; the rate sets how often it plays.

LevyDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Levy oscillator: each cycle is the Levy stable PDF sqrt(c/2pi)*$e^{(-c/2x)/x}$1.5, which stays near zero, spikes up sharply, then decays in an extremely heavy power-law tail - the most front-loaded, long-tailed of the distribution waveforms; C sets where the spike sits.

∿ LevyFlightWalk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Levy flight: a random walk whose step sizes are drawn from a heavy-tailed (power-law) distribution, so long quiet drifts are punctuated by sudden large jumps - the super-diffusive motion of foraging animals and turbulent tracers, here a bounded slow CV.

∿ LevyNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Levy-flight noise: a leaky random walk driven by Cauchy-distributed steps, so the signal mostly drifts slowly but occasionally takes a sudden large jump - a heavy-tailed, intermittent control/noise source quite unlike smooth Gaussian or flat white noise. Jump sets the step scale.

∿ LFGlottal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Liljencrants-Fant glottal source: synthesizes the glottal-flow derivative directly - an exponentially-growing sinusoid through the open phase that snaps to a sharp negative spike at glottal closure, then an exponential return to zero. The most widely-used physiological voice-source model, richer and more controllable than the Rosenberg pulse: Open sets the open quotient (breathy to pressed), Return the closure abruptness (which sets the spectral tilt / brightness). Feed it through a formant filter for vowels.

∿ LibrationOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Libration oscillator: the slow pendulum-like rocking of a body trapped in an orbital resonance (like the Trojan asteroids swinging about Jupiter's Lagrange points, or the Moon's face nodding), modeled as a resonant pendulum whose amplitude sets a gentle sway or a wide tumble.

∿ LightSwitchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Light switch: the crisp snap of a wall light switch toggling, a single short mechanical click with a tiny resonant tick, fired at the set rate; the small everyday clunk of on and off.

∿ LimaconOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Limacon oscillator: the limacon of Pascal r = b + a*cos(theta), which morphs from a convex oval through a dimpled curve to a self-intersecting inner loop as the a/b ratio grows (the cardioid is the boundary case); the x-projection gives a tunably-asymmetric waveform.

∿ LindleyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lindley oscillator: each cycle is the Lindley PDF proportional to (1+x)e^(-Theta*x), a right-skewed lifetime-distribution bump - a near-exponential decay lifted by a linear factor so it rises briefly before falling; Theta sets the decay rate. Peak-normalized, bounded.

∿ LinearFailureOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Linear-failure-rate oscillator: each cycle is (A+Bx)e$^{(-(Ax+Bx}$2/2)), the distribution whose hazard rises linearly (constant A plus slope B) - blending an exponential (B=0) and a Rayleigh (A=0) into one shape, a smoothly-tunable front-loaded decay. Peak-normalized, bounded.

LinnSnare [OSC] [FX] [SYNTH] in: Trigger out: Audio

LinnDrum snare: a tight, punchy sampled-style snare - a short tonal body under a crisp, highpassed noise crack, the early-80s drum-machine backbeat. Tone sets the body-vs-noise balance, Decay the length, Level the output. Trigger on in0.

LissajousOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lissajous oscillator: traces the classic two-frequency figure x=sin(theta), y=sin(ratio*theta + phase) and reads out the y-axis, so integer ratios give harmonically-related multi-lobe waves and the phase knob rotates the figure - the oscilloscope-art curve as a tone.

LiSystem [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Li system: a three-dimensional autonomous chaotic flow (Li, 2009) with three quadratic cross-terms that produces a distinctive multi-wing attractor, studied for its rich coexisting-attractor structure.

LituusOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lituus oscillator: the projection (1/sqrt(theta))*cos(theta) of the lituus spiral r=1/sqrt(theta), a cosine whose amplitude falls off as an inverse square root - a gentler decay than the hyperbolic spiral, giving a long-tailed fading swell per cycle.

LiuChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Liu attractor: a chaotic system with a single quadratic and a squared term that traces a compact double-wing orbit, a tidy strange attractor for smooth chaotic modulation.

LiuChenLu [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Liu-Chen-Lu system: a Lorenz-family chaotic flow with two quadratic cross-terms positioned to give a wide, flat twin-scroll, one of the systematically-generated members of the generalized Lorenz canonical family.

LiuSuOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Liu-Su attractor: a chaotic flow whose third equation uses an exponential of a product term, giving a sharply asymmetric, rapidly-folding orbit distinct from the purely-polynomial Lorenz family.

LoadingPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Loading pulse: the gentle breathing pulse of a loading or processing indicator, a soft tone whose amplitude swells in and out at a steady rate, the please-wait heartbeat of a UI spinner; the pulse rate sets the tempo.

LockClickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lock click: a deadbolt or key lock turning, a short two-stage mechanical clunk (the bolt sliding then seating) with a metallic tick, fired at the set rate; the reassuring sound of locking up.

LocustSwarmOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Locust swarm: the ominous wall of sound of a locust plague, many overlapping wingbeat buzzes at slightly different pitches blurring into a dense seething drone with a slow swell; density sets how thick the swarm.

LogCauchyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-Cauchy oscillator: each cycle is the log-Cauchy PDF (1/(xpi))sigma/((ln x)$^{2+sigma}$2), a right-skewed bump with an even heavier (log-Cauchy) tail than the log-normal - Sigma sets the width; peak-normalized and bounded.

LogChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-chirp oscillator: within each cycle the instantaneous frequency rises exponentially (the phase follows (e^(Rate*t)-1)), so the pitch glides up faster and faster then resets - the constant-Q sweep used in acoustic measurement, repeated at the fundamental. Rate sets the sweep curvature.

LogisticDelay [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Delayed logistic map: a logistic recurrence that also feeds back its previous value (a delay-coupled population model), whose extra memory produces invariant-circle and torus-breakdown routes to chaos distinct from the plain logistic map.

LogisticHenon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Logistic-Henon hybrid: a two-dimensional map that drives a Henon-style delayed feedback with a logistic nonlinearity, combining the period-doubling of the logistic map with the folding of Henon - a denser, more tunable map chaos.

LogisticKernelOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Logistic-kernel oscillator: each cycle is the logistic density kernel 1/(2+2cosh(x)) = sech^2(x/2)/4, a smooth symmetric bump with a flatter peak and heavier-than-Gaussian exponential tails - the derivative-of-a-sigmoid shape, distinct from the plain sech pulse.

LogisticNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Logistic noise: samples ln(u/(1-u)) from the logistic distribution, bell-shaped like the Gaussian but with slightly heavier tails (the law behind logistic regression and the sigmoid) - a symmetric noise with gentle outliers.

LogisticOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Logistic-map chaos source: iterates x = rx(1-x) at a chosen step rate, sweeping from periodic to fully chaotic as r climbs past 3.57 - structured noise / a strange-attractor oscillator, not a wavetable.

LogitNormalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Logit-normal oscillator: each cycle is the logit-normal PDF on (0,1) - a Gaussian in logit space mapped back - so Sigma morphs it from a single centred bump (small Sigma) to a U-shaped bimodal curve with peaks near both edges (large Sigma), a bounded shape no plain distribution offers. Peak-normalized, bounded.

LogLogisticOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-logistic oscillator: each cycle is the Fisk (log-logistic) PDF (b/a)(x/a)$^{(b-1)/(1+(x/a)}$b)^2, a skewed bump with a heavier tail than the gamma - Beta sets the peakedness and Alpha the scale, peak-normalized to a bounded asymmetric waveform.

LogLogisticQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-logistic-quantile oscillator: each cycle is the Fisk inverse-CDF (p/(1-p))^(1/Beta), an odds-ratio ramp that is gentle in the middle and blows up toward both ends (heaviest at p->1) - Beta sets the steepness. Bounded.

LogNormalCdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-normal-CDF oscillator: each cycle is the log-normal cumulative Phi(ln(x)/Sigma), an S-curve in the log domain - so it rises gently, steepens, then tails off asymmetrically, Sigma setting the spread. A skewed ramp distinct from the linear-domain sigmoids. Bounded.

LogNormalNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-normal noise: the exponential of a Gaussian, a right-skewed positive distribution arising wherever many small multiplicative effects combine (particle sizes, incomes, biological growth) - clustered low values with occasional large excursions.

LogSpiralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Log-spiral oscillator: the radial projection of a logarithmic spiral - a sine of several turns whose amplitude grows exponentially across each cycle then resets - giving a built-in sawtooth-shaped tremolo on the tone, a rhythmic swell that repeats at the pitch period.

LomaxOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lomax oscillator: each cycle is the Lomax (Pareto-II) PDF (a/lambda)(1+x/lambda)^(-a-1), a monotonically decaying heavy-tailed shape that starts high and trails off slowly - a front-loaded sawtooth-like waveform whose tail weight Alpha controls, peak-normalized and bounded.

LomaxQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lomax-quantile oscillator: each cycle is the Lomax (Pareto-II) inverse-CDF (1-p)^(-1/Alpha)-1, a ramp that starts at zero and accelerates into a heavy power-law tail toward the top - Alpha sets the tail violence. Bounded.

Lorentzian [OSC] [FX] [SYNTH] in: – out: Audio

Lorentzian-pulse oscillator (a first as a synth block): emits the Lorentzian (Cauchy) bell - 1 / (1 + (t/Width)^2) - once per cycle, a peak with a tall sharp tip and long, slowly-fading shoulders, the lineshape of a resonance in physics. Its fatter tails (compared with a Gaussian or cosine pulse) give a brighter, buzzier spectrum that rolls off exponentially, while Width sets how narrow and piercing the spike is. A resonance-shaped pulse train distinct from the Theta and window pulses. Needs no input.

Lorenz84Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lorenz-84 attractor: a low-order model of the atmosphere's global circulation whose chaotic orbit drifts on a slow, weather-like timescale - organic, long-form modulation that never quite repeats.

Lorenz96 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lorenz-96 model: a ring of five atmospheric sites each advected by its neighbours and pushed by a uniform forcing, the standard testbed for weather-prediction and data-assimilation research - spatiotemporal weather-like chaos.

LorenzChemical [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chemical Lorenz: a mass-action reaction network engineered to realize Lorenz-type dynamics in concentration space (all variables non-negative), showing that the Lorenz butterfly can in principle be built from chemistry.

LorenzCurveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lorenz-curve oscillator: each cycle is the exponential-distribution Lorenz curve p + (1-p)*ln(1-p), a convex ramp that hugs zero at the start and accelerates to one at the end (the economics inequality curve) - a distinctly bowed ramp sitting below the diagonal. Bounded.

LorenzMod1 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modified Lorenz I: a reformulation of the Lorenz equations with a cubic cross-term that opens the butterfly into a broader, smoother two-lobed attractor - a distinct flavour of the classic chaos.

LorenzMod2 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modified Lorenz II: another quadratic reformulation of Lorenz with symmetric squared terms, tracing a four-lobed attractor that the orbit threads in a figure-eight - a higher-symmetry chaotic flow.

LorenzStenflo [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lorenz-Stenflo attractor: a four-variable extension of Lorenz from atmospheric acoustic-gravity waves, with an extra rotational mode that gives a richer, hyperchaotic orbit.

LorenzWaterwheel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Malkus waterwheel: a tilted wheel with leaky water cups that speeds up, stalls and chaotically reverses; its equations of motion reduce exactly to the Lorenz system - chaos you can build out of plumbing, read as the wheel's spin rate.

LotkaChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lotka-Volterra chaos: three competing populations whose nonlinear competition becomes chaotic at Vano's parameters, producing slow, organic, boom-and-bust oscillations - ecosystem dynamics as a modulation source.

LotkaCompetition [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Competitive Lotka-Volterra: three species competing asymmetrically for a shared resource (May-Leonard dynamics), which cycle heteroclinically around the single-species states and, at the right asymmetry, become chaotic - competition-driven population chaos.

Lozi3D [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

3-D Lozi map: a three-dimensional piecewise-linear map (Lozi's absolute-value fold extended with a third delayed coordinate), giving sharp-edged, angular fractal chaos with more depth than the 2-D Lozi.

LoziMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lozi map: a piecewise-linear 2-D map (Henon with |x| in place of x^2) whose chaotic attractor has sharp, straight-edged structure - angular, glitchy chaos with hard corners, iterated at a chosen rate.

LucasOsc [OSC] [FX] [SYNTH] in: – out: Audio

Lucas-number oscillator (a first as a synth block): an additive tone whose overtones land on the Lucas numbers - 1, 3, 4, 7, 11, 18, 29… - the Fibonacci sequence's companion, built on the same golden-ratio recurrence but from a different start. Its partials share Fibonacci's phi-spaced widening yet sit on different harmonics, so it has a related warm-but-glassy character with its own distinct colour - a sibling timbre to the FibonacciOsc. Partials sets how many Lucas overtones to sound, alias-free up to Nyquist. Needs no input.

LucasPolyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lucas-polynomial oscillator: each cycle is the nth Lucas polynomial L_n(x) (L_0=2, L_1=x, L_n = x*L_{n-1}+L_{n-2}) - the Fibonacci-polynomial companion with different seed values, giving a related but distinct family of recurrence-defined polynomial waveforms. Peak-normalized, bounded.

LuChenCheng [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lu-Chen-Cheng system: a chaotic flow whose nonlinear cross-coupling and constant term build a generalized multi-scroll attractor, the orbit threading several stacked lobes for rich switching dynamics.

LuChenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lu attractor: the bridge system between the Lorenz and Chen attractors, a double-scroll chaotic flow whose fold sits between the two - a distinct chaotic source for tones and modulation.

LugiatoLefever [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lugiato-Lefever cavity: a driven, detuned Kerr-nonlinear optical resonator (the equation behind microresonator frequency combs), whose intracavity field is bistable and breaks into chaotic modulation instability under strong pumping.

LyapunovFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lyapunov (Markus) fractal: drives the logistic map with a periodic A/B sequence of two growth rates and measures the Lyapunov exponent; scanning the two rates paints the Zircon-Zity fractal of stable (negative) and chaotic (positive) regions - the running exponent is the output.

MachiningChatter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Machining chatter: a turning tool cuts a surface left wavy by its previous pass, so the cutting force depends on the tool position one revolution ago; that delayed feedback self-excites the screeching chatter vibration that ruins finishes.

MackeyGlass [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mackey-Glass: a delay-differential equation from blood-cell production whose feedback acts on a delayed copy of itself (x' = beta x(t-tau)/(1+x(t-tau)^10) - gamma x); long delays push it from periodic to high-dimensional chaos - the classic time-delay chaos source.

MagicSparkleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Magic sparkle: the glittering twinkle of pixie dust, random short high bell-like blips scattered across a bright pitch band each with a quick decay, the fairy-dust shimmer of fantasy games; density sets how busy the sparkle.

MagnetarFlareOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Magnetar flare: a magnetar giant flare, a sudden hard spike of energy followed by a decaying tail that rings at the star's quasi-periodic seismic oscillation frequencies (QPOs detected after the 2004 SGR event); the burst rate sets how often the neutron star quakes.

MagnetFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Magnet fractal: an escape-time iteration of the rational map $((z^{2+c-1)/(2z+c-2))}2$ that arose from renormalizing the Ising model of magnetism, whose fixed point at z=1 gives it a distinctive lobed shape; the reseeded orbit's real part is the output.

MagneticDipole [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Coupled magnetic dipoles: two pivoted compass needles that both attract each other and feel an oscillating external field; their nonlinear magnetic torques tangle into chaotic flipping - a desktop magnetic-pendulum chaos.

MagnetronHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Magnetron hum: the gritty droning buzz of a microwave oven's magnetron, a low mains-derived tone thick with harmonics and a faint high transformer whine, the angry hum of cooking; the harmonics knob sets the grit.

MagPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Magnetic pendulum: a damped pendulum bob swinging over three attracting magnets, whose decision of which magnet to settle on is famously fractal and chaotically sensitive, reading the bob's x/y path as the output.

MainsHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mains hum: the ground-loop drone of AC power, a line-frequency fundamental (50 or 60 Hz) stacked with the odd harmonics that transformer saturation adds, the ever-present buzz behind unbalanced audio gear.

MajorityRule [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Majority rule: each cell on the ring synchronously becomes the majority opinion of itself and its two neighbours; this deterministic coarsening rapidly grows aligned domains until it freezes into stripes - a self-organizing magnetization that snaps to a fixed pattern.

MandelOrbit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mandelbrot orbit tour: iterates z -> z^2 + c from zero while slowly sweeping c along the real axis, so the orbit cycles through the period-doubling and chaotic regimes mapped by the Mandelbrot set - a self-scanning bifurcation source.

Mandolin [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mandolin: the 8-string (4 double-course) mandolin - bright steel strings in slightly-detuned pairs picked with a sharp plectrum and sustained by the fast tremolo roll; the chiming, shimmering courses give it a brilliance distinct from a single-string lute. Tremolo sets the roll rate, Course the pair detune.

Maracas [OSC] [FX] [SYNTH] in: Trigger out: Audio

Maracas / shaker: a trigger on in0 fires a short burst of bright high-passed noise with a fast decay - the crisp shaker tick. Tone sets the brightness, Decay the length, Level the output.

MarbleRollOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Marbles: glass marbles rolling and knocking on a hard floor, a random scatter of short bright glassy clicks at varying pitches, the marble-game clatter; density sets how busy the rolling.

MarchingFeetOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Marching feet: a column of soldiers marching in step, a low broadband thud on each beat with a slight scatter (not perfectly synchronised) and a dusty scuff, the ominous tramp of boots; the step rate sets the march tempo.

MargolusGas [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lattice gas: a one-dimensional HPP-style automaton where cells hold left- and right-moving particles that stream and bounce on head-on collision, conserving particle number - the discrete microworld behind fluid simulation. The net momentum flux is the output.

Marimba [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal marimba: a trigger on in0 strikes a wooden bar modelled as inharmonic decaying sine partials (the characteristic 1 : 3.9 : 9.2 bar modes), warm with a soft mallet onset. Tune sets the pitch, Decay the ring length, Mallet the brightness (how long the upper modes sustain), and Level the output. A melodic mallet voice to pair with the drum kit; gate it from a Clock, Euclid or sequencer.

MarimbaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Marimba: a struck rosewood bar voiced from its characteristic stretched partials near the 1:4:10 ratio (the deep-cut tuning of a marimba) plus a tuned resonator-tube hum under the fundamental, re-struck at the set rate; warm, woody and round.

MarshallOlkinOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Marshall-Olkin oscillator: each cycle is the Marshall-Olkin exponential PDF Alphalambdae^(-lambda x)/(1-(1-Alpha)e^(-lambda x))^2, a tilt-stretched exponential whose Alpha reshapes the early decay (a stability/proportional-odds add-on to the exponential). Peak-normalized, bounded.

MathieuChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parametric Mathieu oscillator: a resonator whose stiffness is modulated periodically (the Mathieu equation), so parametric pumping at the right depth pushes it through instability tongues into chaotic motion - the physics of a child pumping a swing, taken to extremes.

MathieuOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mathieu oscillator: integrates the Mathieu equation y'' + (a - 2q cos 2t) y = 0, the parametric oscillator behind vibrating elliptical drumheads and ion-trap stability; near the stability-band edges the amplitude swells into rich beating modes before the limiter holds it bounded.

MaxwellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Maxwell oscillator: each cycle is the Maxwell-Boltzmann speed PDF x^2 $e^{(-x}$2/2a^2), a skewed bump that rises as a parabola and falls as a Gaussian - an asymmetric pulse with a soft front and a long tail, Spread setting the skew and harmonic content.

MaynardSmith [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Maynard-Smith map: the population recurrence x -> r x/(1 + (x/k)^b), whose tunable density-dependence exponent moves it from a stable point through cycles into chaos - a flexible ecology-derived bifurcation source.

Mbira [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Mbira (thumb piano): a plucked metal tine with the characteristic buzzing rattle of its bottle-cap resonators - the warm, percussive African lamellophone. Tune sets the pitch, Buzz the rattle, Level the output. Trigger on in0, in1 = Pitch CV.

MegastableSine [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Megastable sine oscillator: a linear oscillator whose damping is a sine of position, creating a countable infinity of nested coexisting limit cycles, so which orbit (and thus which timbre) it settles into depends on the starting energy.

Mellotron [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Mellotron: the tape-replay keyboard's wobbly string/choir ensemble - lightly-detuned sawtooth voices with tape wow and a breath of hiss, the lo-fi orchestral pad of prog and pop. Wobble sets the tape flutter, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

Melodica [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Melodica: the buzzy, sustained free-reed of the breath-blown keyboard harmonica - lightly-detuned reed sawtooths through a soft lowpass, played by holding in0. Reed sets the detune buzz, Tone the brightness, Level the output. in0 = Breath gate, in1 = Pitch CV.

MemristorChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Memristor oscillator: a minimal circuit built around a memristor whose state-dependent conductance (z-squared term) provides negative damping, self-oscillating into chaos - the simplest chaotic circuit with a memristor.

MemristorHindmarsh [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Memristive Hindmarsh-Rose: the bursting HR neuron coupled to a memristor that models the magnetic-flux feedback across the membrane, enriching the burst patterns and adding hidden, flux-dependent chaotic firing.

MemristorTwin [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Twin-memristor oscillator: an LC loop loaded by two flux-coupled memristors whose state-dependent conductances interact, producing the coexisting-attractor 'extreme multistability' that makes memristive circuits prized chaos sources.

MenuBlipOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Menu blip: the tiny beep of moving the cursor in a game menu, a very short clean square-wave tick, fired at the set rate; the crisp UI navigation sound.

MetalFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Three-operator FM tuned to inharmonic ratios for metallic, gong and bell percussion. Ratio sets the (deliberately non-integer) modulator spacing and Index the clang/brightness.

MeteorEchoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Meteor echo: the brief Doppler-shifted ping of a radar/forward-scatter reflection off the ionized trail of a meteor (the 'pings' of meteor-scatter radio), a short tone that drops in pitch as the meteor recedes, scattered randomly across the sky at the set rate.

MeyerWaveletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Meyer wavelet: a smooth, band-limited wavelet defined by a raised-cosine roll-off in the frequency domain (infinitely differentiable, unlike the abrupt Shannon), giving a gently-ringing packet with no harsh sidelobes - approximated here from its smooth spectral window.

MichaelisMentenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Michaelis-Menten oscillator: each cycle is the enzyme-kinetics saturation curve x/(K+x), a hyperbolic rise that climbs fast then asymptotes - K sets the half-saturation point and thus the curvature, a smooth saturating ramp distinct from exponential or power approaches.

Microwave [OSC] [FX] [SYNTH] in: – out: Audio

Microwave oven - a steady transformer hum and turntable whir, ending each cycle with the finished ding. Cycle sets the run length, Tone the hum brightness, Level the output.

MicrowaveHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Microwave hum: a running microwave oven, a steady mains-frequency magnetron drone with harmonics over the faint rumble of the rotating turntable motor, the kitchen-appliance hum; the pitch sets the mains base.

MiddleSquarePRNG [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Middle-square PRNG: von Neumann's 1949 method - square the seed and keep the middle digits as the next seed; famously prone to collapsing into short cycles or zero, its quirky, sometimes-stalling output is a vintage deterministic noise.

MidpointFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Midpoint displacement: the diamond-square algorithm that builds fractal mountain terrain by recursively offsetting segment midpoints by shrinking random amounts; precomputed into one cycle, it makes a jagged self-affine waveform.

Minkowski [OSC] [FX] [SYNTH] in: – out: Audio

Minkowski question-mark oscillator (a first as a synth block): synthesises Minkowski's ?(x) function - the 'question-mark' curve that warps the number line by walking the Stern-Brocot tree, sending the quadratic irrationals to the rationals and flattening the golden ratio's run of 1s into a plain binary fraction. As a waveform it is a singular staircase like the devil's staircase but woven from continued fractions instead of base 3, giving a different self-similar ramp with its own fractal spectrum. A continued-fraction singular function distinct from the Cantor (Devil) and the smooth ramps. Needs no input.

MinkowskiQ [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Minkowski question-mark function: the singular, strictly-increasing map that sends quadratic irrationals to rationals and flattens on every rational, built from the continued-fraction expansion - a fractal staircase-like waveform with structure at all scales.

MinorityGame [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Minority game (El Farol bar problem): many agents each decide whether to 'attend' based on recent winning history, and the minority side wins; their competing adaptive strategies make the attendance fluctuate around the half mark with self-organized, never-settling deviations.

MinskyCircle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Minsky circle: the famous PDP-1 rotation hack x -= ey; y += ex (updating x before y), which draws an almost-circle - actually a slightly-eccentric ellipse that never quite closes; it makes a clean, slowly-precessing quadrature sine pair.

MinstdNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

MINSTD noise: the Park-Miller 'minimal standard' multiplicative generator (x = 16807*x mod 2^31-1), the reference LCG that fixed RANDU's flaws and became the textbook default; a clean, well-tested deterministic white noise.

MiraMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mira map: a 2-D iterated map with a rational nonlinearity (the basis of the Gumowski-Mira family) tuned to its embroidery regime, where the orbit stitches ornate, lace-like fractal loops - a richly-patterned chaotic texture.

MirandaStone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Miranda-Stone attractor: a refined parameterization of the Lorenz-84 atmospheric model that exposes a delicate, symmetric chaotic flow with slow recurrences - organic, weather-like long-form modulation.

MissingFundamentalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Missing-fundamental oscillator: stacks only the 2nd through 6th harmonics of a pitch while omitting the fundamental itself, yet the ear still hears that absent fundamental (the residue-pitch phenomenon that lets tiny speakers seem to produce bass).

MitchellNetravaliOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mitchell-Netravali oscillator: each cycle is the Mitchell-Netravali cubic reconstruction kernel (B=C=1/3, the imaging-standard balance of sharpness and ringing) - a rounded central lobe with gentler side-lobes than Catmull-Rom, a softer ringing pulse.

MLCircuit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Murali-Lakshmanan-Chua circuit: a sinusoidally-driven series circuit with a single Chua-diode (piecewise-linear) element, the simplest non-autonomous chaotic circuit - just one nonlinearity plus a forcing term.

MobiusOsc [OSC] [FX] [SYNTH] in: — out: Audio

Mobius oscillator (a first as a synth block): an additive tone where each harmonic k is weighted by the Mobius function mu(k) - so harmonics on a repeated-prime multiple (4, 8, 9, 12…) are silenced entirely, while the squarefree ones are added with a PLUS or MINUS sign depending on whether they have an even or odd number of prime factors. The mix of dropped partials and phase-flipped (inverted) partials carves a spectrum found nowhere in ordinary synthesis - a signed, squarefree-only comb. Partials sets the highest harmonic considered. Distinct from the other number-theoretic oscillators.

Modal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modal synthesis: a bank of 6 tuned decaying resonators struck by a gate (bells, mallets, struck metal). Structure morphs harmonic to inharmonic-bar partials.

ModalBar [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modal bar resonator: three tuned two-pole resonators at the inharmonic ratios of a struck bar ring when excited by the input (feed it an impulse, gate or noise) - mallet-percussion bodies by modal synthesis.

ModeBeatLaser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mode-beating laser: two longitudinal cavity modes drawing on one gain medium, whose cross-gain saturation makes their intensities flicker antiphase and chaotically as the pump rises - the multimode intensity dynamics of a free-running laser.

ModemOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modem oscillator: frequency-shift keying that hops between a mark and a space tone driven by a pseudo-random bit stream, the screeching data warble of a dial-up modem handshake; the baud rate sets how frantically it chatters.

ModFM [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Modified-FM oscillator (Lazzarini & Timoney 2009): a closed-form formant generator - a carrier sine is shaped by exp(Index*(cos(modulator)-1)), placing one resonant formant peak at the carrier whose bandwidth widens with Index, while Ratio (modulator/carrier) sets the harmonics under it. The FOF/CHANT formant sound from a single modified-FM expression - vocal 'aah/ooh', brass and resonant-sweep timbres. Freq sets the pitch (in0 = Pitch CV), Ratio the formant spacing, Index the bandwidth, Level the output.

MoireOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Moire oscillator: two harmonic series whose spacings differ slightly, so their partials drift through each other and beat in a shifting interference pattern - the auditory equivalent of the moire pattern seen through two overlaid grids.

Monsoon [OSC] [FX] [SYNTH] in: – out: Audio

Torrential downpour - dense high rain hiss over a deep rumble, with random thunder swells. Downpour sets the rain density, Tone the brightness, Level the output.

MonsoonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Monsoon: a torrential tropical downpour, a thick roaring wall of dense rain noise filling the mid band with countless overlapping droplet impacts and a low water-sheet rumble underneath, the relentless drum of heavy rain on a roof.

MoogVoice [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Minimoog-style voice: three slightly-detuned sawtooths stacked and pushed through a self-resonant four-pole Moog ladder filter for the fat, creamy mono-synth lead/bass. Cutoff sets the filter, Reso the ladder resonance, Level the output. in0 = Gate, in1 = Pitch CV.

MooreSpiegelOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Moore-Spiegel oscillator: a model of an ionized gas parcel oscillating in a star's atmosphere, a third-order system whose amplitude swells and collapses chaotically - astrophysical chaos as sound.

MorganMercerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morgan-Mercer-Flodin oscillator: each cycle is the MMF growth sigmoid (floor*Beta + x^Delta)/(Beta + x^Delta), a rational S-curve rising from a nonzero floor to a ceiling - Delta sets the steepness and Beta the half-point, a rational alternative to the exponential sigmoids. Bounded.

MorganVoyceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morgan-Voyce oscillator: each cycle is the nth Morgan-Voyce polynomial B_n (B_n = (x+2)*B_{n-1} - B_{n-2}, the polynomials describing resistor-ladder networks) - a recurrence with a +2 offset that produces lobed waveforms distinct from the Chebyshev family. Peak-normalized, bounded.

Morinkhuur [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morin khuur: the Mongolian horsehead fiddle - two horsehair string bundles bowed together (not pressed to a fingerboard) give a wide, breathy, harmonic-rich bowed tone with a characteristic flat, overtone-laden voice that pairs with khoomei throat-singing. BowForce sets the grip, Overtones the harmonic richness.

MorletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morlet wavelet: a sinusoid of a chosen number of cycles wrapped in a Gaussian envelope (the workhorse of continuous wavelet analysis), here looped per period as a wave packet whose Cycles knob trades a near-sine for a tight burst - a time-frequency-balanced grain.

MorphOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

A single oscillator that morphs continuously through the four classic waveforms - sine, triangle, saw, square - as Morph sweeps 0 to 1, with a Warp control that bends the phase for extra harmonic motion.

MorrisLecar [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morris-Lecar neuron: a two-variable conductance-based membrane model (calcium + potassium gates) that fires smooth, rounded action potentials - a biophysically-grounded spiking oscillator distinct from the cubic neuron models.

MorrisLecarForced [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forced Morris-Lecar: the calcium-potassium membrane oscillator driven by a periodic current, whose interplay of intrinsic and forcing frequencies produces Arnold-tongue mode-locking and chaotic firing - excitable-cell forced dynamics.

MorseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morse oscillator: a steady tone keyed on and off in dots and dashes, here endlessly transmitting 'SOS' (the dot-dot-dot dash-dash-dash dot-dot-dot distress call); the WPM knob sets the keying speed.

MorsePotentialOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morse-potential oscillator: each cycle traces the Morse anharmonic molecular potential $(1-e^{(-A*x)})2$ - a steep repulsive wall on one side flattening to a dissociation plateau on the other - giving a strongly asymmetric waveform with a hard edge and a soft tail, A setting the wall steepness. Peak-normalized, bounded.

MosquitoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mosquito oscillator: a thin, high, slightly-detuned whine that wavers in pitch with the insect's wingbeat, the maddening drone of a mosquito near your ear; the vibrato depth and rate make it dart and waver.

MothFlutterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Moth: the soft papery flutter of a moth batting against a lampshade, a low breathy buzz that swells and fades irregularly with random soft thumps as it bumps the surface; the flutter rate sets the wing speed.

MotorcycleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Motorcycle: the throaty exhaust of a revving motorbike, a buzzy firing tone whose pitch surges up and falls back as the rider blips the throttle, with a hard distorted edge; the rev rate sets how often it revs.

MouseClickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mouse click: a computer mouse button pressed, a short two-stage tick (a bright down-click immediately followed by a softer release tick), fired at the set rate, the point-and-click office sound; the rate sets the clicking.

MoyalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Moyal oscillator: each cycle is the Moyal PDF exp(-(x+e^-x)/2), a strongly right-skewed bump (the classic analytic approximation to the Landau energy-loss distribution) with a sharp leading edge and a long exponential tail - a distinctly lopsided waveform.

MridangamHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tuned hand-drum: a pitched membrane with a fast downward pitch glide and harmonically-loaded overtones gives the singing, bending ‘tun/dhin’ strokes of a mridangam or tabla when struck.

MultifractalNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Multifractal noise: white noise multiplied by a slowly-wandering log-normal envelope (a multiplicative cascade), so the loudness clusters into bursts and lulls with heavy-tailed statistics - the intermittent texture of turbulence and rainfall.

MultiplyWithCarry [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Multiply-with-carry noise: Marsaglia’s MWC generator packs a 16-bit value and its carry into one word and updates x = a*(x low) + (x high), a fast generator with an enormous period and better equidistribution than a plain LCG.

Multisampler [OSC] [FX] [SYNTH] in: Note, Gate, Velocity out: Audio

Multisampler: on each note-on (from the note, gate, and velocity inputs) it strikes every region whose key and velocity zones match, resampling each by note-vs-root plus fine tune so it plays at the right pitch. Slot selects which loaded .sfz region set to use (0 to 7), Loop forces looping of the regions’ loop or play windows, and Level scales output; velocity crossfades and round-robin sequencing come from the region data. Use it for realistic multisampled instruments where timbre tracks key and velocity.

MultiScrollJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Multi-scroll jerk: a jerk system whose nonlinearity is a sine of x, so its restoring force has infinitely many zero crossings and the orbit can wander across a row of stacked scrolls - a grid-of-scrolls chaotic flow.

MultivibratorOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Astable multivibrator: the two-transistor square-wave circuit whose output is not an ideal square but shows the exponential RC capacitor-charge curve on each transition, giving a softer, rounded-edged square with a characteristic analog slump.

MunmuangsaenSri [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Munmuangsaen-Srisuchinwong system: an unusually simple chaotic flow with only one nonlinear term and no equilibrium point, designed as a minimal hidden-attractor chaos generator for hardware random-number use.

MusicalSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Musical saw: a hand saw bowed on its flexible edge - a bowed-string waveguide filtered very dark so the tone collapses toward a pure, eerie, voice-like sine (the bending blade sets the pitch); the ghostly, theremin-adjacent timbre, but produced by stick-slip bow friction rather than a heterodyne oscillator.

MusicalSawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Musical-saw oscillator: the bowed flexible saw blade produces an almost-pure sine (the flat sweet spot of the bend vibrates cleanly) with a wide, slow, expressive vibrato and a faint metallic shimmer - the ghostly, vocal wail of the singing saw.

MusicBox [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal music box: a trigger on in0 plucks a steel comb tine - a bright bell-like tone from a strong fundamental plus high inharmonic partials (1 : 5.4 : 13) with a fast tinkling attack and medium decay, the unmistakable music-box timbre. Tune sets the pitch, Decay the ring, Tone the brightness of the upper partials, and Level the output.

MusicBoxOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Music box: the fragile haunting tinkle of a wind-up music box, a thin bell-like tone with a slightly-detuned shimmer and a quick decay, the eerie nursery melody of horror films; fired at the set rate.

NadarajahHaghighiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nadarajah-Haghighi oscillator: each cycle is alphalambda(1+lambdax)$^{(alpha-1)*e}$(1-(1+lambdax)^alpha), a generalized-exponential lifetime PDF built on (1+lambda*x) rather than x^beta - giving a distinct rise/cut-off balance with a true zero density at the origin for alpha>1. Peak-normalized, bounded.

NagelSchreckenberg [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nagel-Schreckenberg model: cars on a ring accelerate, brake to avoid the car ahead, randomly dawdle, then advance by their speed - the minimal traffic automaton that spontaneously breeds phantom stop-and-go jams out of free flow. The average speed (traffic flow) is the output.

NailGunOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nail gun: a pneumatic framing nailer firing, a sharp percussive punch immediately followed by a short blast of escaping compressed air, fired at the set rate, the construction-site sound; the rate sets the firing.

NakagamiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nakagami oscillator: each cycle is the Nakagami-m PDF x^(2m-1) e^(-m x^2), the fading-channel distribution; the shape M morphs from a sharply-rising one-sided spike (low m) to a more symmetric Gaussian-like bump (high m) - a smoothly tunable asymmetric pulse.

NasalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nasal oscillator: a glottal buzz routed through the heavily-damped, low resonance of the nasal cavity with an antiresonance notch above it, the muffled hum of m, n and ng where sound radiates through the nose rather than the mouth.

Nebula [OSC] [FX] [SYNTH] in: – out: Audio

Nebula - a slowly shifting cloud of detuned high sines over a soft noise wash, an icy interstellar shimmer. Drift sets the movement, Tone the brightness, Level the output.

NeimarkSackerMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Neimark-Sacker map: the discrete analogue of a Hopf bifurcation, where a fixed point sheds an invariant circle (quasiperiodicity) that wrinkles and breaks into chaos as the gain rises - the map route to torus-breakdown chaos.

NeonRelaxation [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Neon relaxation network: capacitors charging through resistors until neon bulbs strike and dump them, two such relaxation oscillators coupled so their firing times interfere into chaotic, irregular blinking - vintage relaxation-oscillator chaos.

NeonRelaxOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Neon-lamp relaxation oscillator: a capacitor charges until it reaches the lamp's strike voltage, then the lamp fires and dumps it down to the lower extinguish voltage before going dark again - a two-threshold hysteresis sawtooth, the simplest possible oscillator.

NephroidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nephroid oscillator: the coordinate 3sin(t) - Shapesin(3t) traced from the kidney-shaped nephroid, a clean blend of the fundamental and its third harmonic in fixed phase - Shape dials the third-harmonic weight from a pure sine to a hollow, clarinet-like tone.

NESPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

NES pulse: the Nintendo 2A03's pulse channel with its four hard-coded duty cycles (12.5%, 25%, 50%, 75%), the thin, nasal square-wave voice behind countless NES melodies and basslines.

NESTriangleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

NES triangle: the 2A03's triangle channel quantized to its 16 discrete steps and locked at full volume (the chip gives it no amplitude control), the slightly-staircased bass-and-lead triangle that anchored NES soundtracks.

NeuralAvalanche [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Neuronal avalanches: a self-organizing branching network where each firing neuron triggers, on average, a critical number of others; tuned near criticality it emits scale-free, power-law-distributed bursts of activity (deterministic RNG) - cortical SOC.

NeuralSpikeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Neural spike train: a leaky integrate-and-fire neuron whose membrane ramps toward threshold, fires a sharp spike and then sits in a refractory dip before recharging - the all-or-nothing pulse train of a real neuron, its rate set by the input current.

NeuroBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Neurofunk bass: a detuned-saw sub pushed through cascaded distortion and a resonant lowpass for the screaming, metallic, multi-layered DnB bass. Detune sets the beating, Drive the distortion, Cutoff the filter. in0 = Gate, in1 = Pitch CV.

NewtonLeipnikHyper [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hyperchaotic Newton-Leipnik: the rigid-body-with-torque Newton-Leipnik attractor extended with a fourth feedback state into hyperchaos, intensifying its twin-lobed orbit into a denser, higher-dimensional tumble.

NewtonLeipOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Newton-Leipnik attractor: a rigid-body rotation model with feedback torque that supports two coexisting strange attractors, tracing a complex twin-lobed chaotic orbit.

NewtonMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Newton fractal map: iterates Newton's root-finding step for z^3-1 in the complex plane and reseeds at a slowly-rotating start each time it converges, so the output hops chaotically between the three roots' basins - the Newton-fractal boundary as a stepped source.

Ney [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ney: the end-blown reed flute of Persian/Turkish/Arab music - blown across the rim so a large fraction of the tone is turbulent breath, giving an intensely airy, hollow, human voice with a husky attack; Breath drives the tone, Air the prominent breath component.

NicholsonBailey [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nicholson-Bailey model: the classic host-parasitoid map whose coupled exponential search terms produce diverging, oscillating boom-crash cycles - ecological instability iterated into a jagged chaotic stream.

NishioOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nishio oscillator: a chaotic circuit built from two LC stages coupled through a diode bridge, whose piecewise conduction folds the coupled resonances into chaos - a classic coupled-oscillator chaos demonstrator.

Noise [OSC] [FX] [SYNTH] in: — out: Audio

Colored-noise generator: Color morphs from white (flat spectrum) toward a low-passed pink-ish tone, scaled by Level. Use it as a percussion/wind source, a sample-and-hold input, or to add air and dither. It takes no signal input - it is a pure generator.

NoiseBlendOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tone-to-breath oscillator: blends a pure sine at the pitch with a low-passed noise whose colour tracks the same frequency, sweeping from a clean tone to a breathy, airy, pitched hiss - flute-attack, vocal-breath and wind textures from one knob.

NoiseColor [OSC] [FX] [SYNTH] in: – out: Audio

Tunable coloured noise: one knob sweeps the spectral tilt from deep brown (-6 dB/oct, rumbling) through pink (-3 dB/oct), white (flat), blue (+3 dB/oct) to violet (+6 dB/oct, hissy). A single source for every noise colour, where the other noise blocks are fixed. Level scales the output.

NoiseFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Noise-FM oscillator: a sine carrier is phase-modulated by band-limited noise, scattering its energy into a gritty, breathy, partly-pitched tone - between an oscillator and colored noise.

NoseHooverMod [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modified Nose-Hoover: the time-reversible Nose-Hoover oscillator with an added cubic restoring term that breaks its near-conservative symmetry into a dissipative attractor - a bridge between conservative and dissipative chaos.

NoseHooverOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nose-Hoover oscillator: a time-reversible chaotic system from molecular-dynamics thermostats, producing a smooth, breathing chaotic tone with long quasi-periodic stretches.

NotificationPingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Notification ping: the soft pleasant two-tone chime of a phone or app notification, a clean bell tone followed by a higher partner tone, both ringing briefly and fading, fired at the set rate; the you-have-a-message alert.

Novachord [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Novachord: the 1939 vacuum-tube polyphonic - a soft, divide-down organ tone with a slow vibrato and a gentle, hollow voice, the first commercial synthesizer. Vibrato sets the wobble, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

NovaFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Nova fractal: Newton's method for z^3-1 with a relaxation factor and an added constant (z -> z - R*($z^{3-1)/(3z}$2) + c), blending Newton basins with Mandelbrot-like structure into glowing nova filaments; the orbit's real part is the output.

NTSCSyncOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

NTSC oscillator: the high-pitched whine of an analog television, the 15.734 kHz horizontal-line-scan frequency shaped as the sawtooth-with-sync-pulse of the flyback, plus the audible field-rate flicker - the sound of an old CRT left on.

Oboe [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Oboe: a bright, nasal, penetrating double-reed - a narrow pulse driven through a strong resonant formant with reed buzz, played by holding in0 as breath. Reed sets the buzz, Breath the air, Level the output. in0 = Breath gate, in1 = Pitch CV.

Ocarina [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Ocarina: a vessel flute with a nearly pure tone plus a soft second harmonic and a touch of breath, played by holding in0. The round, whistle-like clay-flute timbre. Tune sets the pitch, Breath the air noise, Level the output. in0 = Breath gate, in1 = Pitch CV.

Ocean [OSC] [FX] [SYNTH] in: – out: Audio

Ocean surf - filtered noise swelling in and out like waves rolling onto a beach. Rate sets the wave period, Tone the brightness (spray vs deep roll), Level the output.

OceanSwellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ocean swell: filtered noise shaped to the Pierson-Moskowitz spectrum of a fully-developed sea, giving the slow, narrow-banded heave of ocean swell whose peak period the wind-speed knob sets - a rolling, randomized low-frequency surge.

OceanWave [OSC] [FX] [SYNTH] in: Amount out: Audio

Ocean waves: slow swells of surf - filtered noise that rises and breaks in a long, repeating crash-and-recede cycle. Rate sets the swell period, Tone the spray brightness, Level the output. in0 = Amount gate.

OctaveOsc [OSC] [FX] [SYNTH] in: – out: Audio

Octave-stack oscillator (a first as a synth block): an additive tone whose overtones are only the powers of two - the 1st, 2nd, 4th, 8th, 16th… harmonics, i.e. the fundamental plus a stack of pure octaves above it. Because every partial is an octave of the one below, they fuse into a single, enormously reinforced pitch with the bright, hollow body of a full organ stop or a Shepard-tone layer, no in-between harmonics to muddy it. Partials sets how many octaves to stack, alias-free up to Nyquist. A pure-octave spectrum distinct from the dense harmonic saw. Needs no input.

OddHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Odd-harmonic oscillator: sums only the odd partials (1, 3, 5, 7, 9, 11, 13) with an adjustable rolloff, the spectrum of a square wave or a stopped pipe like the clarinet - hollow and woody, with no even overtones to brighten it.

OminousTickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ominous tick: the slow dread-laden tick of a clock counting down to something terrible, a hard dry resonant tick repeated at a steady slow rate over a faint low rumble, the time-is-running-out tension; the tick rate sets the tempo.

OndesMartenot [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Ondes Martenot: the haunting, vocal early-electronic voice - a sine with a few harmonics and a deep, expressive vibrato (the classic 'gliding' theremin-cousin), played by holding in0. Vibrato sets the wobble, Timbre the harmonic richness, Level the output. in0 = Gate, in1 = Pitch CV.

Ondioline [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Ondioline: the buzzy, intensely-vibratoed reed-organ keyboard (Jean-Jacques Perrey's voice) - a harmonically-rich pulse with a fast, deep vibrato and a percussive attack. Vibrato sets the wobble, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

OpenHat909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-909 open hi-hat: a long, sizzling wash of six detuned squares and white noise through a sharp highpass, the bright, ringing open hat of house and techno. Tone sets the brightness, Decay the length, Level the output. Trigger on in0.

OPOChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Optical parametric oscillator: a pumped chi-2 cavity converting pump into signal and idler photons, whose three-wave mixing with detuning produces self-pulsing and chaotic intensity dynamics above threshold - nonlinear-optics chaos.

OpticalBistable [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Optical bistability: a nonlinear optical cavity whose transmitted field depends on its own intensity-shifted phase, giving a hysteretic input-output map that period-doubles into chaos as the drive rises - the foundation of all-optical switching.

OrchHit [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Orchestra hit (Fairlight ORCH5 / Stravinsky stab): a trigger on in0 fires a dense detuned cluster of saws stacked in octaves and fifths - strings and brass hitting one staccato chord - with a sharp attack, a short decay and 8-bit-style bit-crush for the lo-fi digital crunch that made the sound iconic in rave, hip-hop, hardcore and techno. Tune sets the pitch (in1 = Pitch CV), Spread the detune width of the cluster, Decay the staccato length, Bits the lo-fi crunch (low = crunchier), Tone the brightness, and Level the output.

OregonatorOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Oregonator: the three-variable model of the Belousov-Zhabotinsky oscillating chemical reaction, whose relaxation kinetics produce sharp, spiky, slowly-recurring pulses - the famous self-organizing chemical clock as a modulator.

Organ [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Additive drawbar organ oscillator that sums four sine partials at the first four harmonics of Freq. Bar1, Bar2 and Bar3 set the drawbar levels of the fundamental, second and third harmonics (the fourth harmonic tracks Bar2 at half level), and Level sets the overall output. A self-running tone generator with no audio input, voiced from sine-only partials for a clean drawbar timbre.

OrganBeatOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Beating tonewheel organ: six drawbar partials (sub, fundamental, octave, twelfth, super-octave, larigot) summed with a small per-partial detune so they beat against each other like a real electromechanical organ - the Beat knob sets the chorus/age of the instrument.

OrnsteinUhlenbeckProc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ornstein-Uhlenbeck process: noise that is constantly pulled back toward a mean while being kicked randomly (dx = -theta*x dt + sigma dW), giving a smooth, band-limited wander - the canonical model of a damped particle in thermal noise and of mean-reverting signals.

Oscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Audio-rate tone generator with sine, saw, square and triangle waves (Wave) at Freq. Use it as a ring-mod or FM carrier, a drone, a sub layer, or a test tone. It takes no signal input - patch its output into RingMod, a filter, or straight to Out.

Oud [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Arabic oud: a warm, fretless plucked lute with a round, fast-decaying tone and no buzz - a softly-damped Karplus string voiced for the mellow Middle-Eastern lute. Decay sets the ring, Tone the brightness, Level the output. Trigger on in0, in1 = Pitch CV.

OutboardMotorOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Outboard motor: the sputtering buzz of a small two-stroke boat engine, an irregular high firing tone with a watery gurgle as the prop bites and cavitates, the puttering of a dinghy; the RPM knob revs it up.

OwlHootOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Owl-hoot oscillator: soft, breathy low-pitched hoots (a near-sine tone with a touch of breath noise) grouped into the rhythmic hoo-hoo-hoo phrase of an owl; the hoot rate sets the call's cadence.

PadStack [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Lush pad oscillator: the fundamental plus detuned copies an octave and two octaves up, gently detuned for slow beating. Spread sets the detune width and Octaves the upper-layer level - an instant wide synth-pad bed.

PadVoices [OSC] [FX] [SYNTH] in: — out: Audio

Detuned sine pad: seven sine partials micro-detuned around the fundamental beat slowly against each other into a warm, wide, chorused pad drone - the lush PadSynth-style stack from pure sines.

PafOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-aligned formant: a carrier centred on a formant frequency is windowed once per pitch period by a raised-cosine bell, placing a clean, movable formant peak on a harmonic series (the PAF technique) - voiced, vowel-like timbres.

PagerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pager oscillator: the two-tone sequential alert that fires a pager - one tone held, then a second different tone - followed by the alert beeps, the dispatch-radio sound that summons on-call crews; the tone pair selects the address.

Pandeiro [OSC] [FX] [SYNTH] in: Trigger out: Audio

Pandeiro: the Brazilian frame drum with jingling platinelas - a tuned membrane hit wrapped in a bright metallic jingle. Tune sets the pitch, Jingle the platinela mix, Level the output. Trigger on in0.

PanFlute [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Pan flute / shakuhachi: a breathy edge-blown voice - a sine fundamental wrapped in tuned breath noise from a resonant bandpass at the pitch, played by holding in0 as breath. Airy and hollow. Tune sets the pitch, Breath the air balance, Level the output. in0 = Breath gate, in1 = Pitch CV.

PanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pan attractor: a three-dimensional autonomous chaotic system in the Lorenz family with a single quadratic cross-term in each of two equations, tracing a compact, fast-spinning butterfly.

PaperShredderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Paper shredder: a document fed into a shredder, a steady low motor whir overlaid with a harsh ripping crunch as the blades chew the paper, the confidential-disposal sound; the feed knob sets how much paper is tearing.

PaperTearOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Paper tear: the ripping rasp of a sheet of paper being torn, a dense crackle of tiny fibre snaps whose rate and brightness rise with the tearing speed, the satisfying rrrip of paper; the speed knob sets how fast it tears.

ParabolaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parabola wave: a periodic train of parabola segments (the integral of a triangle wave), whose harmonics roll off as 1/n^2 for a tone softer and rounder than a triangle - a gentle, hollow source with very weak upper partials.

ParabolicSineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parabolic-sine oscillator: builds its cycle from the classic two-parabola sine approximation (4/pi)t - (4/pi^2)t*|t| over t in [-pi,pi] - exact at the zero crossings and peaks, with a characteristic small odd-harmonic error (~ -24 dB third) that gives it a slightly brighter, hollower tone than a true sine. The cheap-synth sine, as a distinct timbre.

ParametricPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parametric pendulum: a pendulum whose pivot is shaken vertically, so the gravity term is modulated; at the right drive it swings, period-doubles, then tumbles into full chaotic rotation - the parametric route to rotational chaos.

ParetoCdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pareto-CDF oscillator: each cycle is the Pareto cumulative 1-(1/x)^Alpha for x>=1, a power-law saturation that shoots up off the floor then asymptotes - Alpha sets how abrupt the rise, a heavy-headed ramp distinct from exponential or hyperbolic saturation. Bounded.

ParetoIVOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pareto-IV oscillator: each cycle is the Pareto type-IV PDF (A/Gamma)*$x^{(1/Gamma-1)*(1+x}$(1/Gamma))^(-A-1), a flexible heavy-tailed bump that generalizes the log-logistic (A=1) with an extra tail-shape control - Gamma sets the rise, A the tail. Peak-normalized, bounded.

ParetoNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pareto noise: a power-law source u^(-1/alpha) where a few enormous values dominate a sea of small ones (the 80/20 rule, city sizes, wealth); lowering alpha fattens the tail toward rare extreme spikes - a self-similar heavy-tailed noise.

ParetoQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pareto-quantile oscillator: each cycle is the Pareto inverse-CDF (1-p)^(-1/Alpha), a ramp that stays low then blows up toward the top as a power law - Alpha sets how violent the late acceleration is, a heavy-tailed quantile curve. Bounded.

ParticleSwarm [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Particle-swarm optimization: a swarm of candidate solutions each accelerate toward their own best-found spot and the swarm's global best while chasing a slowly-drifting optimum, so the global-best position homes in with overshoot and re-convergence - a self-organizing tracking CV.

PartyBlowerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Party blower: a party horn blown out, a reedy squawk that rises in pitch as the paper tongue unrolls and ends in a little flutter, fired at the set rate, the birthday-party toot; the rate sets the blows.

ParzenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parzen oscillator: each cycle is the Parzen window, a piecewise-cubic B-spline bump that is flatter on top and rolls off as a cubic at the edges - a smooth pulse between the parabolic Welch and the cosine windows, with a soft, woody harmonic balance.

PaulaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Paula oscillator: the Amiga's Paula chip set pitch by an integer sample-period divider off a fixed clock, the quantization that gave MOD-tracker music its particular tuning; a band-limited-ish saw stepped onto Paula's period grid.

PaulWaveletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Paul wavelet: an analytic (one-sided-spectrum) complex wavelet of integer order, more time-localized and asymmetric than the Morlet; its real part read per cycle gives a sharp leading lobe with a decaying oscillatory tail.

PCGNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

PCG noise: O'Neill's permuted congruential generator advances a linear-congruential state then applies an xorshift-and-rotate output permutation, giving statistically-excellent white noise (passes BigCrush) with a different texture than plain LCG.

PearsonIvOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pearson-IV oscillator: each cycle is the Pearson type-IV PDF (1+(x/a)$^{2)}$(-M) * e^(-Nu*atan(x/a)), a bell that Nu skews to one side and M sets the tail weight - a flexible skewed heavy-tailed shape (used for asymmetric financial returns); peak-normalized and bounded.

PehlivanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pehlivan-Uyaroglu attractor: a chaotic flow with a single quadratic term and simple linear couplings, designed for low circuit cost, tracing a smooth twin-lobed orbit - a hardware-friendly strange attractor.

PencilSharpenerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pencil sharpener: an electric pencil sharpener running, a high-pitched motor whine under a gritty grinding crunch as the blades shave the wood, the classroom-desk sound; the grit knob sets how hard it bites.

PercolationCluster [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Percolation: each ring site is independently occupied with probability p, re-rolled each step; the largest connected run of occupied sites jumps sharply once p crosses the percolation threshold - a critical-transition order parameter that flickers between fragmented and spanning.

PernarowskiBurst [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pernarowski burster: a minimal three-variable polynomial model that captures square-wave bursting through a cubic fast subsystem gated by a single slow variable - a clean, analyzable bursting oscillator.

PersistentWalk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Persistent random walk: a walker that keeps its current direction with high probability and only occasionally reverses (a correlated walk with momentum), giving long smooth runs instead of the jagged jitter of an ordinary random walk - a drifting, inertia-laden CV.

Pew [OSC] [FX] [SYNTH] in: Trigger out: Audio

Laser pew: a quick descending-pitch zap - a sine swept rapidly down with a short decay, the retro arcade/sci-fi laser shot. Tune sets the start pitch, Sweep the drop, Level the output. Trigger on in0.

PhamHidden [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hidden-attractor flow: a three-dimensional system deliberately built with no equilibrium points (a Pham-type construction), so its chaotic attractor is 'hidden' - reachable only from special initial conditions, not by tracking any fixed point.

PhaseDist [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Casio CZ phase-distortion oscillator: a phase ramp drives a cosine through a warped read so DCW morphs a sine toward the chosen shape, then scales by Level. Shape selects Saw or Pulse (true phase distortion) or three resonant modes (ResoSaw, ResoTri, ResoTrap) that are windowed hard-sync, and DCW sets the distortion amount or, in resonant modes, the resonant multiple (1 to 16). Low DCW stays near a clean sine while high DCW opens into bright sawtooth, pulse, or formant-like resonant tones.

PhaseDistort [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Casio CZ phase distortion: a sine read with a warped phase (fast then slow), synthesising a resonant, filter-swept sawtooth without a filter - the bright, vocal PD timbre. Amount sets the distortion (dark to resonant), Level the output. in0 = Gate, in1 = Pitch CV.

PhaseDistortCZ [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase distortion (Casio CZ): a linear phase ramp is bent through a movable knee before driving a cosine, so the read-out speeds up then slows within each cycle - warping a sine toward a resonant saw/pulse without any filter, the signature CZ sound.

PhaseFbOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-feedback sine: the oscillator's own output is fed back into its phase (the DX-style feedback operator), so as Feedback rises the sine warps smoothly toward a sawtooth, adding a controllable wash of harmonics from one operator.

PhaserBlastOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phaser blast: the pew of a sci-fi energy weapon, a tone that swoops sharply downward in pitch with a ringing metallic overtone and a quick decay, fired repeatedly at the set rate; the laser-gun staple.

PhaseWarp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-distortion oscillator (Casio CZ style): a pitch-locked phase ramp is bent through a movable inflection before a cosine, sweeping from sine toward a resonant, formant-like spectrum as Warp increases.

PhasingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phasing oscillator: two identical tones, one running fractionally faster, so they slide continuously in and out of phase (Steve Reich's 'Piano Phase' technique), creating ever-shifting interference patterns from a single repeated voice.

PhiPartials [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Golden-ratio additive oscillator: partials stacked at powers of the golden ratio (phi) are maximally inharmonic (the least rational spacing), producing shimmering, never-beating metallic clusters.

PhoenixFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phoenix fractal: an escape-time iteration z' = z^2 + c + p*z_prev that feeds back the previous iterate (a memory term p), producing curling phoenix-feather structures absent from the plain Mandelbrot; the orbit's real part (reseeded on escape) is the output.

PhoenixJulia [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phoenix fractal orbit: the complex map z' = z^2 + c + p*z_prev (which remembers its previous value), whose flame-shaped 'Phoenix' Julia set folds the orbit through a memoryful complex-plane chaos.

PhotocopierOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Photocopier: an office copier running a page, a steady mechanical hum under a scan-lamp carriage that whooshes across and back on each copy cycle, the document-copy sound; the cycle rate sets how fast it copies.

Piano [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Additive piano (house piano stab): sums inharmonic partials tuned by the real piano stretch formula fnsqrt(1+B*n^2) - so the overtones run slightly sharp the way struck strings do - with a bright hammer-noise attack and a low-pass that darkens as the note decays (higher partials die first). The acoustic-piano chord stab of piano house, deep house and progressive. Tune sets the pitch (in1 = Pitch CV), Inharm the string-stretch detune, Decay the note length, Hammer the attack-noise knock, Tone the initial brightness, and Level the output. A gate on in0 strikes it.

PianoHammerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Piano: a hammer-struck string modelled as a Karplus delay with a soft hammer-pulse excitation plus a slightly-sharp second partial for the inharmonic stretch of stiff piano wire; re-struck at the set rate, with a long ringing decay.

PickoverAttr [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pickover attractor: a 3-D trigonometric iterated map that folds into smooth ribbon-like surfaces, giving a flowing, less-grainy chaotic texture than the 2-D maps.

PIDChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

PID-loop chaos: a nonlinear (saturating) plant under a proportional-integral-derivative controller whose gains are pushed past the stability margin, so the closed loop oscillates and chaotically hunts around its setpoint - control-theory instability.

PierceDiode [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pierce diode: an electron beam injected across a gap whose space-charge forms an oscillating virtual cathode; past the Pierce instability threshold the current chaotically self-modulates - a plasma-physics chaos source.

PigOinkOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pig oink: the grunting snort of a pig, a low buzzy voiced burst with a nasal honk and a snuffly noise edge, repeated in quick grunts, fired at the set rate; the contented sound of the sty.

PinchersMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pinchers map: a Pickover iterated map combining a hyperbolic-tangent pinch with a sine, which squeezes the orbit into pinched, pincer-like fractal lobes - a distinctive folded chaotic texture.

PinkGen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pink noise generator: white noise shaped by the Paul Kellet 7-pole filter to roll off at 3 dB/octave, the equal-energy-per-octave noise used for room tuning and natural-sounding wind/rain textures.

PinkNoise [OSC] [FX] [SYNTH] in: — out: Audio

Pink (1/f) noise - equal energy per octave, the natural-sounding noise for wind, surf and analog hiss. Tone tilts it from full pink toward darker low-pass, Level sets the output.

Pipa [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pipa: the Chinese pear-shaped lute - a bright nylon/silk Karplus string driven by the rapid finger tremolo (lunzhi) that sustains a note as a shimmering roll; Tremolo sets the re-attack rate from single plucks to a dense buzzing roll.

PipeOrgan [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Church pipe organ (Interstellar-style): additive ranks at the true organ-stop pitches - 8’ fundamental, 4’ octave, 2 2/3’ twelfth, 2’ fifteenth, 1 3/5’ seventeenth and 1 1/3’ nineteenth (harmonics 1-6) - so the upper mutation stops reinforce the fundamental with cathedral brilliance, plus a wind ‘chiff’ on the attack and an optional tremulant. The huge sustained organ of cinematic and gothic scoring, distinct from the Hammond drawbar Organ. Tune sets the pitch (in1 = Pitch CV), Stops the registration (how many upper ranks are drawn = brightness), Chiff the attack wind-noise, Tremulant the slow amplitude vibrato, and Level the output. A gate on in0 holds the note.

PiriformOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Piriform oscillator: the pear-curve coordinate cos(t)(1 + Pinch*sin t), morphing from a pure cosine (Pinch 0) to an asymmetric pear-shaped wave (Pinch 1) that bulges on one side - a smoothly skewable timbre with shifting even-harmonic content.

PistonPumpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Piston pump: the rhythmic chuff of a reciprocating air pump or old engine, a low thump on each compression stroke followed by a breathy intake hiss, the pulsing of a piston cycling; the rate sets the strokes per second.

Plaits [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Macro-oscillator (Mutable Instruments Plaits): one voice, six synthesis Models selected by the Model param - Analog (saw/square), FM (2-op), Grain (formant), Chord (stacked intervals), Wavetable (sine-saw-square morph) and Modal (struck bar) - each shaped by Timbre and Morph. Tune sets pitch; an internal decay envelope (Decay) fires when a gate hits in0, so it plays as a drone unpatched or a pinged percussive voice when triggered. The Swiss-army oscillator covering most synthesis techniques in a single block.

PlanckTaperOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Planck-taper oscillator: each cycle is the Planck-taper window - a flat-topped pulse whose edges fall to zero through the infinitely-smooth Planck bump function - so Taper morphs from a near-rectangle (buzzy) to a gently rounded pulse with no ringing, the smoothest possible band-limited flat-top.

PlantNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Plant neuron: a model of molluscan bursting neurons where a slow calcium variable periodically gates fast spiking, producing the parabolic-bursting and chaotic spike-cluster patterns seen in real pacemaker cells.

PlasmaBallOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Plasma globe: the high crackling fizz of the dancing tendrils inside a novelty plasma ball, a high noise band modulated by a fast flicker as the filaments writhe and snap to the glass; the flicker rate sets how restless the plasma.

PlasmaLangmuirOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Langmuir oscillation: electrons in a plasma sloshing against the ion background at the plasma frequency; at large amplitude the sinusoid steepens and ‘wave-breaks’ into a sharply-peaked, cusped waveform - the cold-plasma electrostatic oscillation pushed nonlinear.

PlasmaSheath [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Plasma sheath instability: the thin charged boundary layer at a plasma-wall interface whose ion transit time and ionization feedback self-oscillate, relaxation-spiking and chaotically flickering near the instability threshold.

PlateModeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Plate-mode oscillator: sines at the dense two-dimensional bending modes of a rectangular plate (ratios near 1, 1.58, 1.94, 2.0, 2.44, 2.92, 3.16), the crowded inharmonic spectrum behind metallic plate-reverb shimmer and thunder-sheet clang.

PLLChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-locked-loop chaos: a third-order PLL whose loop filter and sinusoidal phase detector, when the input frequency steps too far, lose lock and break into chaotic phase slipping - the cycle-slip instability of real PLLs.

PlosiveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Plosive oscillator: a train of brief release bursts - a sharp noise transient with a quick spectral fall - the popping articulation of stop consonants like p, t and k; the rate sets how fast the stops repeat.

Pluck [OSC] [PHY] [FX] [SYNTH] in: Note (CV), Gate out: Audio

Karplus-Strong plucked-string physical model: a gate edge fills a delay line with a white-noise burst, then each pass averages neighbouring samples and scales by Decay so the noise settles into a pitched, decaying tone. Tune shifts the note in semitones (sets the delay length from the in0 note), Damp blends in more of the lowpass average to dull the highs, Decay sets how slowly the string rings out, and Level scales the output. Plucks on the in1 gate; short Decay and high Damp give a soft muted pluck, long Decay and low Damp a bright sustained string.

PluckBass [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Plucked bass: a tight, finger-plucked electric-bass tone over a Karplus-Strong string - warm, punchy and dynamic. Decay sets the sustain, Tone the brightness, Level the output. Trigger on in0, in1 = Pitch CV.

PluckBurst [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Gated Karplus-Strong pluck: a gate fires a one-period noise burst into a tuned, damped delay loop, so each trigger plucks a string at the set pitch - physical string synthesis you play with a gate.

PluckedBass [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Plucked bass string: a low-tuned Karplus-Strong delay loop excited by a short filtered noise burst gives a warm, rounded fingered-bass pluck with natural string decay - a self-contained bass voice.

PluckExcite [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Transient-plucked string: an envelope detector fires a one-period noise burst into a tuned, damped Karplus-Strong delay loop on every input transient, so drums or any percussive input replay as a plucked string at the set pitch - a resonant excitation effect, not a continuous resonator.

PluckPickPos [OSC] [FX] [SYNTH] in: Pitch (CV), FM out: Audio

Pick-position pluck: the noise burst exciting a Karplus-Strong string is comb-filtered by a pick-position parameter (as on a real string, plucking near the bridge thins the lows), shaping the harmonic balance of each pluck.

PMOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-modulation oscillator: a sine modulator running at Ratio times Freq offsets the carrier phase by depth Index, and the warped phase is read through the selected carrier waveform, then scaled by Level. Wave selects the carrier shape (Sine, Tri, three bent/saturated triangle variants, narrowed sine), Ratio sets the modulator-to-carrier frequency, and Index sets modulation depth. Raising Index and Ratio builds increasingly bright, inharmonic FM-style spectra.

PoinsotSpiralOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Poinsot-spiral oscillator: the projection (1/cosh(dtheta))cos(theta) of Poinsot's spiral, a cosine under a smooth hyperbolic-secant amplitude decay across each cycle - a soft plucked-like swell that fades to silence, with Decay setting how fast it dies within the period.

PointillistGrainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pointillist grain: very sparse, isolated grains scattered in time, each a short tone at a randomly-chosen pitch from a band, producing the dotted, droplet-like texture of pointillist electronic music; density and pitch-range set how busy and how wide.

PoissonKernelOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Poisson kernel: (1-r^2)/(1-2r cos theta + r^2), the harmonic-extension kernel of the disc whose harmonics roll off geometrically as r^n; the radius r continuously tunes from a flat sine (r small) to a sharp positive spike (r near 1) - a one-knob brightness pulse.

POKEY [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Atari POKEY oscillator: a square wave fed through the chip's polynomial-counter noise, sweeping from a clean tone to the dirty, buzzing distortion-bass POKEY is famous for. Dirt sets the noise blend, Width the pulse duty, Level the output. in0 = Gate, in1 = Pitch CV.

PokeyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

POKEY oscillator: the Atari 8-bit POKEY chip gated its square tone through long polynomial (LFSR) counters to make its trademark gritty, buzzy 'distortion' timbres - a square clocked by a pseudo-random divider rather than a clean divide.

PolarizationLaser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Polarization laser: a laser whose two orthogonal polarization modes compete for the same gain, switching and antiphase-flickering chaotically as the pump and anisotropy change - the polarization dynamics of VCSELs.

PolyaUrnProc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Polya urn: draw a ball, return it plus another of the same colour, repeat - so the colour proportion reinforces itself and converges to a random limit (a martingale, the model of cumulative advantage); a self-reinforcing drift that locks somewhere new each run.

PolyBlepSaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

PolyBLEP sawtooth: a naive ramp with a polynomial band-limited step correction applied at the wrap, cancelling the worst aliasing of a digital saw at low CPU cost - the standard modern virtual-analog saw.

PolyBlepSquare [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

PolyBLEP square: a naive variable-width pulse with polynomial band-limited step corrections at both edges, cancelling the dominant aliasing - the standard low-cost virtual-analog square/PWM oscillator.

PolyBlepTri [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

PolyBLEP triangle: a band-limited square (naive square plus polynomial edge corrections) is leaky-integrated into a triangle, giving a clean alias-free triangle with the soft, hollow spectrum a naive ramp triangle aliases away.

PolylogOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Polylogarithm oscillator: sums cos(k*theta)/k^s over the harmonics - the real part of the polylog Li_s(e^{i theta}) - so the Order knob (s) sets the harmonic roll-off continuously, from a bright sawtooth-like spectrum (low s) to a near-pure cosine (high s), band-limited to Nyquist.

PomeauManneville [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Intermittency map: the Pomeau-Manneville map sits near-fixed for long 'laminar' stretches then erupts into a chaotic burst before relaminarizing - quasi-periodic calm interrupted by sudden chaos, a route-to-turbulence texture.

PopcornFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Popcorn fractal (Pickover): the coupled map x' = x - hsin(y + tan(3y)), y' = y - hsin(x + tan(3x)), whose tangent terms scatter the orbit into popping clusters that outline a delicate fractal; the x-coordinate is read as a jittery CV.

PopGunOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pop gun: a toy cork gun fired, a short hollow resonant pop with a woody body thump, fired at the set rate, the playful cork-pop; the rate sets the firing.

PopulationGenetics [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Population-genetics map: an allele frequency iterated by frequency-dependent selection, where over-strong selection against the common type drives the gene pool into period-doubling and chaotic generation-to-generation swings.

PortaLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Portamento lead: a smooth, gliding monophonic saw lead that slides between notes - the expressive, singing solo voice. Glide sets the slide time, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

PortalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Portal: an open dimensional rift, a deep hum overlaid with a swirling spectral vortex (a band of noise whose pitch spirals continuously) and a faint otherworldly ring, the gateway-to-another-realm drone; the swirl rate sets the vortex speed.

PotionBrewOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Potion brew: a bubbling cauldron of magical brew, a cluster of rising bubble pops over a watery flow bed sprinkled with tiny glittering sparkle blips, the cute alchemy of mixing a potion; the density sets how vigorously it bubbles.

PourLiquidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pouring liquid: liquid glugging from a bottle or jug, a watery flow bed punctuated by rhythmic resonant glugs whose pitch rises as the vessel empties, the satisfying pour; the glug rate sets the flow.

PourOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pour: the glugging of liquid pouring from a bottle, a dense random cluster of small rising bubble pops over a soft turbulent flow-noise bed, the sound of filling a glass; density sets how fast it pours.

PowerDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Power-distribution oscillator: each cycle is the power-function PDF shape x^(Alpha-1) on [0,1] (the inverse of the Pareto), a rising ramp whose curvature Alpha controls - linear at Alpha=2, concave below, convex above - a continuously-bendable ramp distinct from a straight saw.

PowerDrillOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Power drill: the whine of an electric hand drill, a buzzy motor tone whose pitch sags and steadies as the bit bites and the load fluctuates, the rising-and-falling drone of drilling; Load sets how much the motor bogs down.

PowerLomaxOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Power-Lomax oscillator: each cycle is $x^{(Beta-1)*(1+x}$Beta)^(-Alpha-1), a power transform of the Lomax that combines an adjustable rise (Beta) with a heavy power-law tail (Alpha) - a flexible front-loaded heavy-tailed waveform. Peak-normalized, bounded.

Powerup [OSC] [FX] [SYNTH] in: Trigger out: Audio

Sci-fi power-up / energy charge: a trigger on in0 starts a tone that rises over Time - the pitch sweeps up Range octaves while bright harmonics fade in and a fast shimmer tremolo intensifies toward the peak, the 'charging up' of a weapon, shield or pickup. Tune sets the start pitch, Time the charge duration, Range the octaves it climbs, Shimmer the high-harmonic sparkle and tremolo, and Level the output. Trigger to charge; the energy peaks and holds at the top of the sweep. Distinct from the noise-based Riser - this is a tonal ascending charge.

PowerUpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Power-up: grabbing an upgrade, a fast ascending run of square-wave blips climbing a scale to a bright peak, the rewarding rising flourish of collecting a power-up; the rate sets how often it fires.

PrecessionOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Precession oscillator: a radial oscillation whose elliptical orbit slowly rotates (apsidal precession, as Mercury's orbit does), giving a rosette-tracing two-frequency wave whose slow precession rate continuously reshapes the waveform cycle to cycle.

PredatorPreyCA [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spatial predator-prey: a ring of cells that are empty, prey or predator; prey spreads into empty space, predators eat adjacent prey and reproduce, and predators starve where no prey remains - giving the chasing population waves of a spatial Lotka-Volterra system.

PredatorPreyDelay [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Delayed predator-prey: a Lotka-Volterra system whose prey self-limitation acts with a maturation delay; the delay destabilizes the coexistence equilibrium into large-amplitude, irregular boom-bust population cycles.

PredatorPreyMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Predator-prey map: a discrete-generation predator-prey system whose coupled growth and predation terms cycle and, at high growth rates, go chaotic - boom-bust ecology iterated into a jagged stepped source.

PredatorRickerMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ricker predator-prey map: a discrete-generation predator-prey system with Ricker (exponential) prey growth and saturating predation, whose coupled over-compensation drives quasi-periodic and chaotic population cycles.

PressureWasherOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pressure washer: a power washer blasting water, a high-pressure jet hiss pulsed by the throb of the pump piston, the driveway-cleaning roar; the pump rate sets the pulsation.

PrimeAdd [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Prime additive oscillator: sums sine partials only at prime-numbered harmonics (2, 3, 5, 7, 11, 13), a sparse harmonic series that sounds hollow and faintly inharmonic, distinct from a full saw/square.

PrimeAdditive [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Additive oscillator that sums only the prime-numbered harmonics (2, 3, 5, 7, 11, …) with a tunable rolloff - a hollow, woody, clarinet-like spectrum you can't get from saw/square. Partials sets how many primes and Rolloff their decay.

PrimeHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Prime-harmonic oscillator: stacks sine partials only at the prime harmonic numbers (1, 2, 3, 5, 7, 11, 13), skipping every composite overtone, for a hollow, slightly-dissonant spectrum heard nowhere in ordinary harmonic tones.

PrimeOrgan [OSC] [FX] [SYNTH] in: – out: Audio

Prime-harmonic oscillator (a first as a synth block): an additive tone built from the fundamental plus only the PRIME harmonics - the 2nd, 3rd, 5th, 7th, 11th… partials, with the composite ones (4th, 6th, 8th, 9th…) left out entirely. Because every overtone is at a prime multiple, the usual octave and fifth reinforcements are missing, giving a hollow, slightly bell-like organ timbre that no harmonic or subtractive source produces. Partials sets how many prime overtones to stack, alias-free up to Nyquist. A number-theoretic spectrum, distinct from the tilt-shaped Additive oscillator.

PrinterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dot-matrix printer: an old dot-matrix printer at work, a buzzy mid-frequency carriage drone interrupted by rhythmic print-head zips that sweep across each line, the retro-office print sound; the line rate sets the carriage speed.

ProlateSlepianOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Prolate-spheroidal (Slepian) pulse: an approximation of the discrete prolate spheroidal sequence, the window that packs the most energy into the smallest bandwidth (the optimal time-frequency-concentration window); the concentration knob sets how tightly the single lobe is focused.

Propeller [OSC] [FX] [SYNTH] in: — out: Audio

Propeller plane - a chopping blade thrum over a droning piston engine. Rpm sets the blade rate, Tone the engine brightness, Level the output.

PropellerPlaneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Propeller plane: the droning of a piston prop aircraft, a blade-pass tone with strong harmonics beating against the engine firing buzz, the steady drone of a small Cessna; the RPM knob sets the engine speed.

ProphetVoice [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Prophet-style voice: a sawtooth and a pulse oscillator detuned together through a two-pole filter, the punchy, slightly-gritty Sequential Prophet-5 poly character. Cutoff sets the filter, Reso the resonance, Level the output. in0 = Gate, in1 = Pitch CV.

PseudoVoigtOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pseudo-Voigt oscillator: each cycle is a linear blend EtaLorentzian + (1-Eta)Gaussian (the spectroscopy line-shape approximation), so Eta morphs continuously between a Gaussian core (soft, fast tails) and a Lorentzian (sharp peak, heavy tails) - one knob across the whole Voigt family. Bounded.

PsyBass [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Psytrance rolling bassline: a self-clocking 16th-note sub that retriggers a fast-attack / short-decay amp envelope and a downward filter sweep on every step, the hypnotic driving roll under a psytrance kick. A saw runs through a resonant low-pass whose cutoff starts bright and decays toward the sub each step. Tune sets the pitch (in0 = Pitch CV), Rate the step-clock frequency (tempo-sync it so the steps land on 16ths), Cutoff the filter floor, Reso the squelch, Decay the per-step length, and Level the output. Sidechain it under the kick (Pump / Ducker) for the classic off-kick pump.

Pulsar [OSC] [FX] [SYNTH] in: — out: Audio

Pulsar synthesis (Roads): a pulsaret (windowed sine) is emitted once per fundamental period, then silence. Freq sets pitch, Formant the pulsaret cycles, Width the duty.

Pulsar2 [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Pulsar-synthesis oscillator: a single-cycle pulsaret is emitted once per fundamental period then followed by silence, so Width sets a movable formant region independent of pitch - the granular/pulsar tone behind Curtis Roads' synthesis and a range of harsh-to-vocal digital timbres. Freq sets the fundamental, Width the duty (formant), Level the output. in0 = Pitch CV.

PulsarBeamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulsar-beam oscillator: a rotating neutron star sweeps its narrow radio beam past us once per spin, giving a clocklike train of very sharp pulses against silence (the lighthouse model); the duty knob sets how narrow the beam, from a click to a fat blip.

Pulsaret [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulsar synthesis oscillator: at each fundamental period a Hann-windowed sinusoidal grain (the pulsaret) is retriggered, with silence filling the rest of the cycle. Freq sets the fundamental (grain repetition rate), Formant sets the sine frequency inside the grain (formant pitch), Width sets the fraction of the period the grain occupies, and Level scales the output. Narrow widths and high formants give buzzy, formant-rich tones.

PulsaretOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulsaret: pulsar synthesis, where each period holds one short pulsaret waveform followed by a silent gap (Roads), so the period sets the audible fundamental and the pulsaret-to-gap duty sweeps a formant-like resonance through the spectrum.

PulsarSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulsar synthesis (Curtis Roads): a short pulsaret waveform of fixed duration is emitted once per (longer) pulsar period followed by silence; the period sets the pitch and the pulsaret duration sets an independent formant, bridging granular and classic synthesis.

PulseGen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zero-cross pulse generator: fires a short decaying spike on each rising zero crossing of the input, so it emits a click train locked to the input's pitch - a trigger / excitation source for resonators.

PulseStackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Super-pulse: seven pulse waves spread symmetrically around the pitch beat against each other into a thick, hollow, detuned wall - the square-wave counterpart to a supersaw, with PW setting the duty of every layer for anything from reedy to nasal.

PulseWidthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulse-width modulation: a rectangular oscillator whose duty cycle is set (and can be modulated) from a thin spike through a square to its inverse, sweeping the hollow-to-bright timbres of classic PWM string and lead patches.

PupilReflex [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pupil light reflex: the pupil area set by retinal light flux through a delayed neural loop (Longtin-Milton model); enough loop gain and delay destabilize it into the spontaneous pupil-size oscillations clinicians can induce with edge-lighting.

PuuBusinessCycle [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Puu business cycle: Tonu Puu's continuous multiplier-accelerator model where the accelerator is a cubic (van-der-Pol-like) function of income change, self-exciting a bounded economic limit cycle that goes chaotic when forced.

PWM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pulse-width-modulation oscillator (Juno-style): a single pulse wave whose duty cycle is swept by a slow LFO, so one oscillator breathes and thickens like a chorused string / pad - the classic Juno / Jupiter PWM movement behind hollow synth strings and reedy basses. Tune sets the pitch (in0 = Pitch CV), Rate the PWM LFO speed, Depth how far the width is swept, Sub an octave-down square for weight, and Level the output. Distinct from SuperPWM, which stacks many detuned pulses - this is the single classic PWM voice.

PWMLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

PWM lead: a pulse-width-modulated square lead with continuous PWM movement for a thick, animated, hollow solo voice. Width sets the duty, Move the PWM depth, Tone the brightness. in0 = Gate, in1 = Pitch CV.

PWMOsc [OSC] [FX] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Pulse-width-modulation oscillator: a variable-width square wave, from a thin nasal pulse to a full square. Width sets the duty cycle and in1 modulates it (patch an LFO for the classic fattening PWM sweep). in2 = Pitch CV.

PWMSquare [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pitch-locked PWM square: an internal square locks to the input pitch and its pulse width is set by a knob, sweeping the hollow-to-nasal PWM timbre while tracking whatever you play.

PWMStrings [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

String-machine pad: a stack of pulse-width-modulated squares detuned into a wide ensemble with a gentle highpass, the shimmering Solina/Logan string-synth wash. Cutoff sets the body, Width the PWM motion, Level the output. in0 = Gate, in1 = Pitch CV.

PythagorasTree [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Pythagoras tree: the chaos game over the two scaled-and-rotated maps that recursively sprout left and right square branches, building the classic fractal tree; the iterate's height is read out as a branching, self-similar CV.

QexponentialOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

q-exponential oscillator: each cycle is the Tsallis q-exponential (2-q)lambda(1-(1-q)lambdax)^(1/(1-q)), which is a pure exponential at q=1, has compact support for q<1, and a heavy power-law tail for q>1 - one knob spanning bounded-decay to fat-tailed decay. Peak-normalized, bounded.

QGaussOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

q-Gaussian oscillator: each cycle is a Tsallis q-Gaussian bump $(1-(1-q)x^{2)}(1/(1-q))$ with compact support, so Q morphs the pulse from a parabola-like dome to a sharp Gaussian-ish spike with hard zero edges - a tunable bell-shaped waveform that a plain Gaussian cannot reach.

QiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Qi attractor: a chaotic flow with strong quadratic cross-coupling that sweeps a wide, fast, multi-lobed orbit, a high-energy strange attractor for aggressive tones and broad modulation.

QuadMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quadratic map: the iterated map x -> a - x^2, the additive-form sibling of the logistic map with the same period-doubling cascade into chaos but a symmetric, bipolar output range.

QuadraticChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quadratic-chirp oscillator: within each cycle the phase follows t^Power, so the frequency sweeps up as a polynomial rate - gentler at the start than a log chirp and tunable from near-linear (Power~1) to a steep late ramp (Power high). A smooth accelerating glide, repeated at the fundamental.

QuadrupTwoOrbit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quadruptwo attractor: a Pickover iterated map built from signs, sines and logs that lays out a four-fold-symmetric tracery, a distinctive crystalline chaotic texture from his strange-attractor collection.

Quasar [OSC] [FX] [SYNTH] in: – out: Audio

Quasar - a pulsing stream of radio-telescope beeps and bursts over cosmic static. Pulse sets the beep rate, Tone the pitch, Level the output.

QuasarFlickerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quasar flicker: the aperiodic red-noise variability of an accreting supermassive black hole, modelled as heavily-integrated (Brownian) noise so low frequencies dominate, the slow random brightening and dimming of a quasar’s accretion disk; the cutoff sets how fast it wanders.

RabinovichOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rabinovich-Fabrikant attractor: a chaotic flow from plasma-wave physics with cubic nonlinearities, tracing an intricate, spiky orbit very sensitive to its parameters - delicate, edge-of-chaos textures.

RabiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rabi oscillation: the population of a driven two-level quantum system cycling between its states as sin^2(Omega t/2); adding detuning raises the floor and speeds the generalized Rabi frequency, morphing a full-depth raised-cosine into a shallow, faster flutter.

RadarChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Radar oscillator: a repeating short linear-FM 'chirp' pulse (the pulse-compression waveform of modern radar that sweeps frequency within each ping) followed by a listening gap; the pulse rate and sweep width set the radar's character.

RademacherOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rademacher function: the sign of sin(2^k * theta), an exact square wave at the chosen power-of-two harmonic and the orthonormal building block of the Walsh-function system; stacking several gives the binary-rate timbres of digital and Walsh synthesis.

RailHunting [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wheelset hunting: a coned railway wheelset's lateral-and-yaw oscillation driven by creep forces and flange contact; above a critical speed it self-excites into the snaking 'hunting' limit cycle and, with flange impacts, chaos.

Rain [OSC] [FX] [SYNTH] in: – out: Audio

Rain texture: a steady band-passed noise wash (the hiss of falling rain) under randomly-timed resonant droplet 'plinks' - short decaying pings at scattered pitches, the patter of drops on surfaces. Distinct from Fire's broadband crackle: rain is a wet high wash plus pitched droplets. Self-running. Wash sets the steady rain level, Drops the droplet density, Pitch the droplet resonance, Tone the wash brightness, and Level the output. Layer instances for heavier rain; add Thunder for a storm.

RainGen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rain: a Poisson scatter of little resonant water droplets, each a short pitched 'plip' at a random pitch, the density rising with the input level (silence = light drizzle, signal = downpour). A rain-bed generator played by dynamics. Density sets the drop rate, Mix the blend.

RainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rain oscillator: a dense Poisson shower of tiny filtered-noise droplet impacts whose mean rate the Intensity knob sets, from a few scattered drops to a steady downpour hiss - granular rainfall synthesis with no loops.

RainPad [OSC] [FX] [SYNTH] in: Amount out: Audio

Rain: a steady downpour - a wash of highpassed noise speckled with random droplet ticks. Density sets the droplet rate, Tone the brightness, Level the output. in0 = Amount gate.

RaisedCosineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Raised-cosine pulse: the smooth flat-topped pulse used as the inter-symbol-interference-free shaping filter in digital communications, with a roll-off knob setting how much of the period is flat versus cosine-tapered - from a wide plateau to a single cosine lobe.

RaisedCosOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Raised-cosine pulse: a flat-topped pulse with cosine-tapered edges (the Tukey window / comms pulse shape), where Width sets the duty and Taper morphs the edges from a hard rectangle (buzzy) to a soft raised-cosine bump (mellow) - band-limited-feeling shaping from one waveform.

RamaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rama oscillator: each cycle is the Rama PDF proportional to $(1+x^3)e$(-Theta*x), a lifetime distribution whose cubic factor produces the most delayed, pronounced hump of the Lindley family before the exponential decay. Peak-normalized, bounded.

RampCoreOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ramp-core morph oscillator: one phase ramp derives a saw, a triangle and a pulse, and the Shape knob crossfades continuously across all three (saw -> tri -> pulse) so a single core sweeps the classic analog waveshapes without switching; PW sets the pulse duty.

RandomFibonacci [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Random Fibonacci: the recurrence x = +/-x_prev +/- x_prev2 with random signs, whose magnitude grows like Viswanath's constant 1.1319… almost surely (a surprising result given the chaotic sign-flipping); tracked as a log-scaled, periodically-renormalized wander.

RanduNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

RANDU noise: IBM's infamous 1960s linear-congruential generator (x = 65539*x mod 2^31), notorious because its triples fall on just 15 planes in 3-D; its conspicuous low-order-bit structure makes a deliberately-flawed, lattice-correlated noise.

RanunculoidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ranunculoid oscillator: the coordinate 6sin(t) - Shapesin(6t) from the five-cusped ranunculoid epicycloid, a clean blend of the fundamental and its sixth harmonic - Shape adds a bright, bell-like upper partial without the lower odd harmonics a square or saw would bring.

RatchetOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ratchet: the rapid clicking of a spring-loaded pawl skipping over the teeth of a turning gear (a socket-wrench or noisemaker), a fast train of short broadband clicks whose rate is the tooth-pass frequency.

Rattleback [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rattleback: the curved 'celt' top whose skew-symmetric mass distribution converts spin into rocking and back, spontaneously reversing its rotation direction over and over - a real mechanical curiosity, read as the spin axis.

RattlesnakeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rattlesnake oscillator: a dry, dense buzz of noise pulses at the rattle's shake rate, the warning sound of interlocking keratin segments knocking together; the shake rate and swell set how agitated the snake.

RayleighDuffing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rayleigh-Duffing oscillator: a resonator with both a cubic stiffness (Duffing twin-well) and a velocity-cubed self-pumping (Rayleigh) term under sinusoidal forcing, combining two nonlinear-oscillator routes to chaos.

RayleighNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rayleigh noise: the magnitude of a two-component Gaussian vector (sqrt(-2 ln u)), the distribution of the envelope of narrowband noise and of wind speeds - a strictly-positive, right-skewed source, here centred for audio.

RayleighOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rayleigh self-oscillator: a nonlinear damping term pumps energy into a resonator until it settles on a stable limit cycle, the physics behind self-sustaining reed and bowed tones - a sound that starts itself.

RayleighQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rayleigh-quantile oscillator: each cycle is the Rayleigh inverse-CDF sqrt(-2*ln(1-p)), a concave ramp that rises briskly then bends toward a steep finish - the quantile counterpart of the Rayleigh distribution. Bounded.

RCLadderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

RC phase-shift oscillator: three cascaded RC stages give the 180-degree feedback for a sine oscillator, but a saturating amplifier and extra gain tip the three-pole loop into chaos - a textbook audio oscillator over the edge.

ReciprocalDistOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Reciprocal oscillator: each cycle is the reciprocal (log-uniform) distribution PDF proportional to 1/x, a smooth hyperbolic decay from a high start to a low tail - a logarithmic-falling ramp distinct from linear, exponential or power-law decays.

ReciprocalNormalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Reciprocal-normal oscillator: each cycle is the PDF of 1/X for a standard normal X, $e^{(-1/2x}$2)/x^2, which is zero at the origin, rises to an off-centre peak, then decays with a heavy tail - a distinctly hollow-then-spiked shape unlike the centre-peaked Gaussian. Peak-normalized, bounded.

Recorder [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Recorder: the simple, pure, slightly-breathy whistle of the wooden recorder - a sine with a soft second harmonic and a touch of air, played by holding in0. Breath sets the air noise, Tone the second-harmonic mix, Level the output. in0 = Breath gate, in1 = Pitch CV.

RecordScratchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Record scratch: a DJ scratching a record, a tone whose pitch swings rapidly up and down (forward-and-back hand motion) over a faint vinyl rumble, the turntablist ‘wikka-wikka’, fired at the set rate.

ReeseBass [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Reese bass: two heavily-detuned sawtooths beating against each other through a resonant lowpass, the growling, metallic drum-and-bass / neurofunk sub-bass. Detune sets the beating rate, Cutoff the filter brightness, Reso the growl. in0 = Pitch CV.

ReeseSynth [OSC] [FX] [SYNTH] in: – out: Audio

Reese bass: two saws detuned just enough to beat against each other, summed and rolled off with a low-pass, the growling, phasing drum-and-bass/neurofunk sub-bass timbre from one block.

RefrigeratorHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Refrigerator hum: the low background drone of a fridge compressor, a soft mains-derived hum with a couple of harmonics and a faint intermittent rattle, the sound you only notice when it stops; the rattle knob adds mechanical buzz.

RefWhistleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Referee whistle: the shrill blast of a sports whistle, a high near-sine tone with the fast fluttering warble of the pea rattling inside, blown in sharp bursts; the rate sets the blast pattern.

RelaxationSawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Relaxation oscillator: a capacitor charges toward a supply through a resistor, then dumps when it hits a threshold - so the ramp is the curved exponential-charge segment of real analog ramp generators, not the straight line of an ideal sawtooth.

RelayChatterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Relay chatter: an electromechanical relay rapidly pulling in and dropping out, a buzzing train of hard mechanical clicks at the chatter rate with a faint coil hum, the angry buzz of a relay fed marginal voltage; the rate sets the chatter speed.

RenyiMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Renyi beta-map: the expanding map x -> beta*x mod 1 for any real beta > 1 (the doubling/tripling maps are the integer cases), reading off the base-beta expansion of the seed - a continuously-tunable deterministic noise source.

RepressilatorGene [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Repressilator: the engineered three-gene loop where each gene’s protein represses the next in a ring (Elowitz and Leibler’s synthetic clock), producing sustained genetic-circuit oscillations from Hill-function feedback.

ReservoirEcho [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Echo-state reservoir: a pool of neurons wired with fixed random recurrent weights (derived deterministically from index) and driven by a small input, holding a fading memory of its history; a weighted readout of the reservoir gives a rich, recurrent-but-stable evolving CV.

ResoComb [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Noise-excited resonant comb: white noise fed through a tuned feedback comb rings at the pitch, giving a metallic, pitched-noise tone between a flute and a struck pipe. Reso sets the ring time, Tone the brightness.

ResonateFire [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Resonate-and-fire neuron: a damped sub-threshold oscillator that fires and resets when its rotation crosses threshold, so it responds selectively to inputs near its resonant frequency - a frequency-preferring spiking model.

ResonatorBank [OSC] [FX] [SYNTH] in: Audio out: Audio

Bank of three tuned resonators excited by the input - strike it with a click or noise and it rings like a struck object, feed it audio and it adds metallic, bell-like resonant body. Freq sets the fundamental, Spread detunes the upper resonators (harmonic to inharmonic), Decay the ring time, Mix the wet amount. A compact modal/Rings-style resonator.

ResoSawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Resonant-saw oscillator: the Casio CZ 'fake resonance' technique - a sine hard-synced to the fundamental at Reso times its frequency, windowed each cycle by a falling sawtooth ramp, so a sweepable formant peak rides on a sawtooth body. Reso sets the formant frequency in harmonics of the fundamental.

ResoTrapOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Resonant-trapezoid oscillator: a CZ-style resonant waveform whose formant sine is held at full level for the first half of each cycle then ramped down - a flat-topped resonance window giving a brighter, more sustained formant than the saw or triangle variants. Reso sets the formant in harmonics of the fundamental.

ResoTriOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Resonant-triangle oscillator: a CZ-style resonant waveform where the synced formant sine is windowed by a triangle that rises then falls across each fundamental cycle, so the formant swells in the middle of the period - a softer, more vocal resonance than the resonant-saw. Reso sets the formant in harmonics.

RespiratoryRhythm [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Respiratory rhythm: a model of the pre-Botzinger complex (the brainstem pacemaker for breathing), where a persistent-sodium current and slow inactivation generate the inspiratory burst-and-pause rhythm, irregular when stressed.

RestrictedThreeBody [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Restricted three-body: a massless test particle moving in the gravity of two orbiting masses (Sun-Earth-spacecraft), whose path through the rotating potential is chaotic and threads the Lagrange-point gateways - celestial-mechanics chaos.

ReverseBass [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Hardstyle / jumpstyle reverse bass: a trigger on in0 (the offbeat gate) restarts a sub that swells up from silence along an accelerating exponential - the 'reversed reverb tail' shape the genre is built on - then is hard-cut by the next trigger so it sits in the gap between kicks. A sine sub blended with a saw is driven through tanh for weight. Tune sets the pitch (in1 = Pitch CV), Swell the rise time, Curve the swell shape (higher = slower start, sharper rush at the end), Drive the distortion, and Level the output. Gate it on the off-beats against a kick on the downbeat.

ReverseBeepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Reverse beeper: the steady, slow square-wave beep of a reversing truck or forklift (a fixed-pitch tone gated on and off about once a second), the ubiquitous back-up warning tone.

Rhodes [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

FM electric piano (DX-style tine EP): a sine carrier phase-modulated by a sine at Ratio, where the modulator index decays fast to give the metallic 'tine' bark on the attack before settling to a mellow near-sine body - the warm electric piano of deep, soulful and lo-fi house. Tune sets the pitch (in1 = Pitch CV), Ratio the tine frequency ratio (higher = more bell/bark), Bark the attack FM depth (brightness / velocity), Tine how fast the bark fades, Decay the note length, and Level the output. A gate on in0 strikes the note.

RhodesEP [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

FM electric piano: a 1:1 carrier/modulator FM pair with a fast-decaying modulation index produces the bell-like attack and mellow body of a Rhodes/DX tine piano, struck by a gate.

RichardsCurveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Richards-curve oscillator: each cycle is the Richards generalized-logistic growth S, $1/(1+e^{(-B(t-0.5))})^{(1/Nu)}$, whose asymmetry the Nu parameter controls - the inflection slides off-centre (Nu<1 leans early, Nu>1 late), giving a lopsided S-ramp no symmetric sigmoid produces.

RickerMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ricker map: the fisheries population model x -> x e^(r(1-x)), whose over-compensation drives period-doubling into chaos as the growth rate rises - an exponential-feedback route to chaos distinct from the logistic map.

RickerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ricker / Mexican-hat wavelet: the negative second derivative of a Gaussian, a single positive bump flanked by two symmetric dips, widely used in seismic and edge-detection work; per cycle it makes a soft, DC-free thump with a controllable width.

Ride808 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 ride cymbal: a sustained metallic ping - inharmonic squares with a tonal centre and a steady, bell-like decay, more pitched and controlled than the crash. Tone sets the brightness, Decay the ring, Level the output. Trigger on in0.

RiemannOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Riemann function: the sum of sin(pi n^2 x)/n^2, Riemann's candidate for a continuous-but-nowhere-differentiable function (differentiable only at special rationals), here pitch-locked as a softly-spiky fractal waveform with quadratic-harmonic content.

RifeVincentOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rife-Vincent oscillator: each cycle is the order-2 Rife-Vincent window 1 - (4/3)cos(2pip) + (1/3)cos(4pip), a cosine-sum bell with a wider, flatter main lobe than Hann - a mellow rounded pulse with its own low-harmonic spectral balance.

RijkeTube [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rijke tube: a heated gauze in a pipe feeds energy into the air column with a time lag, so the acoustic mode self-amplifies into the loud thermoacoustic howl of a singing flame - here driven through its limit cycle into chaos.

RikitakeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rikitake dynamo: a coupled two-disk dynamo model of the Earth's magnetic field that chaotically reverses polarity, giving a signal that wanders then abruptly flips sign - chaos with dramatic sign reversals.

Rim909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-909 rimshot: a short, sharp, slightly-tonal click - a fast high-frequency ping under a bright noise transient, the 909's dry rim. Tune sets the ping pitch, Decay the click length, Level the output. Trigger on in0.

Rimshot [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog rimshot (808-style): a trigger on in0 fires a very short burst of two high resonant tones (a bright tap plus a lower ring) with an extremely fast decay for the sharp click of a rim hit. Tune sets the bright tone (the lower ring tracks it), Decay the click length (very short), and Level the output. Patch a Clock, Euclid or sequencer gate into in0.

Rimshot808 [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

TR-808 rimshot / claves: a trigger on in0 fires a very short, sharp pitched click with a tiny resonant ring - the tight 808 rim hit. Tune sets the pitch, Level the output. in1 = Pitch CV.

RimshotSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rimshot voice: a very short, sharply-decaying resonant tone with a click attack (the sound of a stick striking the rim and head together), re-triggered at the set rate; bright, dry and percussive.

RingFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ring-modulated FM: a 2-operator FM voice is multiplied by an independent third oscillator, stacking FM sidebands with ring-mod sum/difference tones into a dense, clangorous, metallic spectrum tuned by two ratio knobs - bell and gong timbres a plain FM pair can't reach.

RingInverter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ring oscillator: three logic inverters in a loop, each a slew-limited tanh stage; the odd-stage feedback self-oscillates and, with an extra cross-coupling and asymmetry, the node voltages tumble chaotically - on-chip oscillator dynamics.

RingPair [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two oscillators ring-modulated together: Ratio detunes the second against the first and Mix blends the ring product against the carrier for metallic, clangorous, bell-like inharmonics.

RingStackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ring stack: a carrier sine ring-modulated by a second sine at a tunable ratio, producing the sum-and-difference sidebands of classic ring modulation; non-integer ratios give the clangorous, bell-like inharmonic spectra heard in gongs and sci-fi tones.

RinzelNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rinzel burster: Rinzel's three-variable reduction of a bursting neuron, where a slow recovery current periodically lifts a fast spiking subsystem across its threshold - producing clean square-wave bursts of spikes.

Riser [OSC] [FX] [SYNTH] in: Trigger out: Audio

Build-up riser: a trigger on in0 restarts a sweep that rises over Time seconds. White noise is driven through a resonant band-pass whose cutoff climbs while a pitch-rising sine fades in underneath and the whole thing swells in level - the classic pre-drop tension build. Time sets the sweep length, Tone balances the noise against the rising sine, Reso sets the band-pass sharpness, and Level the output. Patch a bar clock or downbeat gate into in0; the swell accelerates toward the drop.

RiserSweepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Riser: the build-up tension sweep before a drop, a noise band and tone that both climb steadily in pitch and intensity to a peak and then reset, the staple transition of EDM; the rate sets the build length.

RissetBell [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Risset bell: five inharmonic sine partials at Risset's classic frequency ratios, each with its own exponential decay, struck by a gate - additive bell synthesis where the higher partials die first, just like a real bell.

RissetBellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Risset bell: the additive bell from Jean-Claude Risset's Introductory Catalogue of Computer-Synthesized Sounds, eleven sustained partials at his published inharmonic ratios (0.56, 0.92, 1.19, 1.7, 2, 2.74, 3, 3.76, 4.07…) - a foundational electronic-music timbre.

RissetDrumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Risset drum: the drum timbre from Risset's catalogue - a cluster of low inharmonic partials mixed with a filtered-noise bed - which fuses into a pitched-yet-percussive membrane tone; the noise knob sets how dry or rattly the skin.

RissetGliss [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shepard-Risset glissando: the continuous-frequency cousin of the Shepard tone, where octave-spaced partials slide smoothly and ceaselessly under a fixed Gaussian spectral envelope, giving the illusion of pitch gliding up (or down) forever.

RLDiodeDriven [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Driven RL-diode: a resistor-inductor-diode loop driven by a sine source, the circuit in which period-doubling to chaos was first measured in the lab; its junction-capacitance nonlinearity folds the driven current chaotically.

Robot [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Robot / computer voice (sci-fi): a pulse source ring-modulated by a low metallic carrier, shaped by a formant band-pass and stepped to a new pitch at Rate (the flat, phoneme-by-phoneme 'robot talk'), then bit-crushed for digital grain - the 1980s computer-voice and droid chatter of film and game sound design. Tune sets the base pitch (in0 = Pitch CV), Rate the speech-step speed (how fast the pitch jumps), Ring the ring-mod carrier (metallic clang), Formant the vowel centre, Crush the bit-grain, and Level the output. Distinct from the Vocoder/RingMod effects - this is a self-contained robotic voice.

∿ **RobotBeepOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Robot beeps: the chirpy data-blip language of a friendly robot, a square-wave tone whose pitch hops to a new random step on each fast beep, the expressive bleep-bloop chatter of an astromech droid; the beep rate sets the talking speed.

∿ **RocheBeatOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ellipsoidal-variation oscillator: the light curve of a close binary whose mutual tides stretch each star into an egg shape, so we see more surface twice per orbit - a double-humped brightness wave (at twice the orbital rate) with unequal minima, plus the eclipse dips.

∿ **RocketLaunchOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rocket launch: the earth-shaking roar of a rocket lifting off, a swelling wall of low broadband noise overlaid with the random crackle of supersonic combustion instabilities, building to full thrust; the build knob sets the ignition swell.

∿ **RockPaperScissors** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Replicator RPS: the evolutionary-game dynamics of a rock-paper-scissors population, which spirals around the equal-mix point in a heteroclinic cycle - the strategy fractions chase each other in slow, ever-lengthening rounds.

∿ **RoosterCrowOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rooster crow: the cock-a-doodle-doo of a cockerel at dawn, a strident voiced tone stepping through the call's rising-then-held syllables with a buzzy edge, fired at the set rate; the farmyard alarm clock.

∿ **RootRaisedCosineOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Root-raised-cosine oscillator: each cycle is the RRC pulse (the matched-filter pulse shape of digital comms), a central lobe with symmetric ringing side-lobes whose extent the roll-off Beta controls - a band-limited pulse with controllable ringing distinct from the smoothing windows. Bounded.

∿ **RoseCurveOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rose-curve oscillator: the rhodonea r = cos(k*theta) swept around the circle and projected to one axis, whose petal count (k) sets a comb of lobes per cycle - the flower-petal curve rendered as a band-limited waveform.

∿ **RosenMorseOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rosen-Morse oscillator: each cycle is the Rosen-Morse potential -Depthsech^2(x) + Tilttanh(x), combining a central well with an asymmetric step so one side ends higher than the other - a well-with-a-bias shape distinct from the symmetric Morse and sech pulses.

∿ **RosenzweigMac** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rosenzweig-MacArthur model: logistic prey with a saturating (Holling type-II) predator; a seasonal forcing of the prey's carrying capacity tips the predator-prey limit cycle into chaos - boom-bust population dynamics as a slow source.

RosslerMod [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Modified Rossler: the Rossler flow with an added cubic damping on the z equation that tames the spike and reshapes the spiral-and-fold into a smoother, rounder attractor.

RosslerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rossler attractor: integrates the three-equation Rossler flow, which spirals out then folds back, giving a smooth quasi-periodic-to-chaotic signal - a continuous strange attractor as a tone or slow mod source.

RosslerPrototype4 [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rossler prototype-4: one of Rossler's lesser-known prototype systems that produces a screw-type / toroidal chaotic attractor rather than the familiar funnel, giving a smoother, more rotational chaotic orbit.

RPSDynamics [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rock-paper-scissors dynamics: three strategies each beating the next in a cycle, evolving by replicator equations so the populations chase each other round and round in a heteroclinic cycle that slows near each corner - the canonical model of cyclic ecological coexistence.

RPSSpatial [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spatial rock-paper-scissors: a ring of cells each holding one of three species where a cell is invaded by an adjacent species that dominates it (rock crushes scissors, etc.); cyclic dominance keeps any species from winning and spins up endless rotating spiral waves of coexistence.

RubberStampOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rubber stamp: a hand stamp pressed onto a document, a soft dull thunk with a brief inky paper slap, fired at the set rate, the bureaucratic 'approved' sound; the rate sets the stamping.

RucklidgeDouble [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Double-diffusive convection: a Rucklidge-type model of fluid driven by two competing density gradients (heat and salt), whose interplay gives slow-fast oscillations and chaos distinct from single-diffusion Lorenz convection.

RucklidgeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rucklidge attractor: a model of double-diffusive convection whose orbit loops with a slow, rolling motion - a smooth, low-frequency-flavoured chaotic source good for organic modulation.

Rule110Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 110: the elementary cellular automaton proven Turing-complete, whose 32-cell row evolves complex, glider-laden patterns at the edge of chaos; the centre cell and density read out as an intricate, computation-rich control source.

Rule150Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 150: the additive (left XOR centre XOR right) elementary cellular automaton, an additive rule whose nested triangular patterns differ from Rule 90's - another deterministic, self-similar fractal bit-stream generator.

Rule184Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 184: the traffic-flow cellular automaton where 1s (cars) advance into empty cells, modelling the jam-to-free-flow transition; its conserved density gives drifting platoons of bits - a structured, density-dependent pattern.

Rule30Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 30: Wolfram's chaotic elementary cellular automaton (used as the random generator inside Mathematica), whose central column passes randomness tests; the 32-cell row evolves each step into a deterministic white-noise-like bit-stream.

Rule54Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 54: a class-IV elementary cellular automaton on the edge of chaos, whose background of blinkers hosts gliders that collide and interact like particles; the row evolution yields a structured, almost-rhythmic complex bit-stream.

Rule90Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rule 90: the XOR-of-neighbours elementary cellular automaton that draws the Sierpinski triangle; its 32-cell row evolves each clock step and the cells read out as a self-similar, additive-XOR bit pattern - deterministic fractal noise.

RulkovMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Rulkov map neuron: a discrete fast-slow iterative model that produces spiking and bursting with very few operations, iterated at a chosen rate - cheap, sharp neural dynamics from a 2-D map.

SaddleNodeFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SNIC oscillator: a system poised at a saddle-node-on-invariant-circle bifurcation, where a slow variable repeatedly drags it just across threshold, firing spikes with an arbitrarily-long, current-tunable interval - the Type-I neural-excitability rhythm.

SaitoHysteresis [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saito hysteresis circuit: a piecewise-linear oscillator whose state is steered by a hysteretic relay (a Schmitt-trigger switching element), producing chaos from a simple linear system plus memory - hysteresis-driven chaos.

SakaryaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sakarya attractor: a symmetric chaotic flow with product cross-terms tracing a twin-lobed butterfly orbit, a Lorenz-cousin strange attractor with its own folding character.

SaltzmanClimate [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saltzman ice-age model: three slow climate variables (ice, CO2, ocean) whose nonlinear feedbacks self-sustain the ~100,000-year glacial cycles and can go chaotic - geologic-timescale oscillation as a very slow modulator.

SambasFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sambas flow: a Sambas-type three-dimensional system engineered with no equilibrium point, so its chaotic attractor is hidden; its single quadratic term and constant offset trace a folded scroll reachable only from special initial states.

Sampler [OSC] [FX] [SYNTH] in: – out: Audio

Reads one of the 8 shared SampleStore slots with linear interpolation, advancing a playback position scaled by pitch and the slot/host sample-rate ratio, then scaled by Level. Slot picks the source, Pitch transposes -24 to +24 semitones, Start sets the entry point (0 to 0.99 of the buffer), and Loop chooses whether playback wraps back to Start or stops at the end. Useful for looped wavetable-style sources or one-shot sample playback depending on Loop.

SamuelsonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Samuelson model: the original linear multiplier-accelerator income oscillator made nonlinear by a saturating investment response, so it self-sustains a bounded business-cycle rhythm instead of exploding or decaying.

SandpileSOC [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Self-organized-criticality sandpile: grains drop onto a small lattice that topples whenever a site exceeds its threshold, spilling to neighbours in scale-free avalanches - the Bak-Tang-Wiesenfeld source of 1/f, power-law-distributed bursts.

Sandstorm [OSC] [FX] [SYNTH] in: – out: Audio

Desert sandstorm - a gritty mid-band noise rush with grains whipping past. Force sets the gust strength, Grit the abrasiveness, Level the output.

Sarangi [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sarangi: the bowed short-necked Indian fiddle - gut playing strings bowed in stick-slip motion over ~35 sympathetic strings (a second resonator bank) that ring in sympathy, giving the most vocal, human, halo-rich of all bowed tones. BowForce sets the grip, Sympathetic the resonant shimmer.

Sarod [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sarod: the fretless Indian plucked lute - steel strings stopped against a polished metal fingerboard give a bright, glissando-rich pluck (no frets, so no fret buzz; notes slide) with a long, ringing metallic sustain and a high sympathetic-string shimmer; the deep, resonant counterpart to the sitar. Sympathetic sets the resonant shimmer tap.

Saron [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Gamelan saron: a struck bronze-bar metallophone with a bright, clangy attack and a short metallic decay - the driving melodic bar of the gamelan. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

SaturableLaser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Passively Q-switched laser: a laser with a saturable absorber that bleaches under high intensity then recovers, dumping the cavity in giant repetitive pulses; coupling the gain, intensity and absorber gives chaotic pulse trains.

SawSwarm [OSC] [FX] [SYNTH] in: – out: Audio

Detuned saw swarm: seven sawtooths spread around a centre frequency by a detune amount sum into the huge, shimmering unison of a supersaw - a self-contained trance lead/pad voice.

SawSync [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Hard-sync sawtooth: a slave saw is reset every time the master cycle completes, so raising Ratio above 1 forces the bright, vocal, formant-rich sweep of oscillator hard-sync. Freq sets the master pitch, Ratio the slave-to-master frequency. in0 = Pitch CV.

SawtoothMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sawtooth map: an area-preserving twist map whose kick is a linear sawtooth rather than a sine (the linearized standard map), giving uniformly-hyperbolic, fully-chaotic mixing across the whole phase plane.

Sax [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Saxophone voice: a gate on in0 sounds a reedy harmonically-rich tone (a soft sawtooth) with vibrato - honking low, singing high. Freq sets the note, Tone the brightness and reediness (number of harmonics), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

SaxMultiphonic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saxophone multiphonic: a single reed feeding two simultaneously-resonant bore modes (cross-fingered) whose nonlinear flow valve makes the two pitches beat, lock and split into the gritty multiphonic chords saxophonists growl.

SaxWGOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saxophone waveguide: a single-reed bore whose conical shape gives a non-inverting reflection and therefore the full harmonic series (brighter and reedier than the odd-only clarinet); breath pressure drives it from a soft tone to a honking growl.

Saz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Saz (baglama): the Turkish/Anatolian long-neck lute - thin metal courses over a buzzing bridge give a bright, nasal, buzzing pluck, and the low drone strings ring under the melody; the driving voice of Anatolian and Alevi music. Buzz sets the bridge rattle, Drone the open-string ring.

Scanned [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Scanned synthesis (Mathews/Verplank): a 16-mass spring ring evolves slowly after a gate hammer-strike; a scan reads the mass positions at audio rate as the waveform. Tension/Damping shape the timbre’s evolution.

ScannedString [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Scanned synthesis: a ring of masses connected by springs vibrates at sub-audio rates while a scan pointer reads its evolving shape as the waveform, so the timbre morphs continuously with the string’s slow dynamics (Verplank/Mathews).

ScannedSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Scanned synthesis (Verplank-Mathews-Shaw): a ring of masses joined by springs vibrates in slow, hand-shapeable low modes while a pointer scans around the ring at audio rate to read the waveform - the wave shape itself evolves dynamically as the string sloshes.

ScatterGrainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Scatter grain: a grain stream whose onsets are randomly jittered around the nominal grain rate, smearing a steady pulse into a loose, organic spatter of grains; the jitter knob takes it from a tight stream to a chaotic scatter.

SchellingSeg [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Schelling segregation: two-type agents on a ring relocate when too few of their neighbours match them, so even a mild same-type preference cascades into fully-segregated blocks (the classic emergent-segregation result); the fraction of content agents climbs as domains coalesce.

SchmittOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Relaxation oscillator: an integrator charges toward a rail until a Schmitt trigger flips it, ramping the other way - the asymmetric ramp/square waveform of an RC relaxation oscillator core.

Schnakenberg [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Schnakenberg reaction: a minimal two-variable trimolecular autocatalysis model (a standard Turing-pattern kinetics) whose activator-substrate feedback gives smooth chemical limit-cycle oscillations.

SchoolBellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

School bell: a resonant metal bell tone rapidly re-struck by a buzzing clapper (an electric bell’s solenoid hammer), giving the loud, continuous, slightly-rattly ring that signals the end of class; the clapper rate sets the buzz.

SchumannResonanceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Schumann resonance: the standing electromagnetic waves in the cavity between the Earth’s surface and the ionosphere, whose harmonic series (7.83, 14.3, 20.8, 27.3, 33.8 Hz ratios) is summed here as the planet’s natural ‘heartbeat’ rendered as an audio tone.

SchusterMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Intermittency map: a quadratic-tangent map sitting just past a saddle-node bifurcation, so the orbit creeps slowly through the reinjection channel (laminar phase) then bursts chaotically - Schuster’s textbook type-I intermittency.

ScreamLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Screaming lead: a heavily-driven saw forced through a sweeping resonant formant for a vocal, howling, distorted solo lead. Scream sets the formant sweep, Drive the distortion, Level the output. in0 = Gate, in1 = Pitch CV.

ScrollTickOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Scroll tick: the tiny tick of a scroll-wheel detent or picker increment, a very short high resonant blip repeated at the set rate, the subtle UI feedback of moving through a list.

Seagulls [OSC] [FX] [SYNTH] in: – out: Audio

Seagulls - periodic descending squawk-cries over a faint shore wash for a coastal ambience. Rate sets the call frequency, Pitch the cry register, Level the output.

SecantFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Secant fractal: the derivative-free secant root-finding iteration on z^3-1, which uses the previous two iterates instead of a tangent; its irrational (golden-ratio) convergence order paints fractal basins distinct from Newton's, read out as a CV.

SechCubedOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sech-cubed oscillator: each cycle is sech^3(x), a smooth bump narrower and more sharply-peaked than plain sech but with the same soft exponential tails - a focused, click-free pulse with a distinct harmonic balance between the sech and Gaussian shapes.

SechPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sech pulse: each cycle is a single smooth hyperbolic-secant bump (the soliton / 1-over-cosh shape), giving a rounded, click-free pulse whose width sets the harmonic richness - from a near-sine swell to a sharp bipolar spike, all without discontinuities.

SeesawOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seesaw: a playground seesaw tipping up and down, a dry pivot squeak that rises and falls in pitch as the weight shifts across the fulcrum, the see-saw rhythm; the rate sets the tipping speed.

SEIRSeasonal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seasonal SEIR epidemic: a susceptible-exposed-infected-recovered model (adding an incubation stage to SIR) under seasonal transmission, whose extra delay deepens the chaotic, multi-peaked outbreak dynamics.

SeismicChirpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Seismic chirp: a repeating glide that sweeps from a low rumble up through a faster shake, mimicking the arrival of fast compressional P-waves followed by the larger, lower shear S-waves of an earthquake on a seismogram - a quake rendered as a looping chirp.

SelkovOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Selkov oscillator: the two-variable model of glycolytic oscillations in living cells (x'=-x+ay+x^2 y, y'=b-ay-x^2 y), a smooth biochemical limit cycle whose period and shape shift with the feed parameters.

SemiconductorLaser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Semiconductor laser: the carrier-photon rate equations with nonlinear gain saturation; under current modulation the relaxation-oscillation resonance period-doubles into the spiking chaos seen in directly-modulated laser diodes.

SemicubicalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Semicubical oscillator: each cycle is Neile's semicubical parabola sign(x)*|x|^1.5, an odd power-law curve sitting between a triangle (exponent 1) and a parabola (exponent 2) - a soft ramp with a gently flattened centre and a 1.5-power cusp, a distinct curvature no integer-power wave has.

SerpentineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Serpentine oscillator: each cycle is Newton's serpentine curve x/(x^2+1), an odd rational bump that rises to a peak then decays with long tails - a smooth, sine-adjacent waveform with a rounder centre and flatter extremes, sharpening as Width grows.

SewingMachineOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sewing machine: the steady whir of an electric sewing-machine motor under a rhythmic needle-punch thump once per stitch, the domestic chug of stitching fabric; the stitch rate sets how fast it sews.

Shaker [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shaker / maraca: a trigger on in0 fires a soft burst of band-passed noise with a quick 'tsh' envelope - the egg-shaker and maraca sound, all air and no pitch. Tone sets the noise band (low and dark to high and crisp), Decay the burst length, Level the output. Needs only a trigger; chain two at offset times for a swung shaker groove.

Shakuhachi [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Japanese shakuhachi: a breathy end-blown bamboo flute with a strong air component and expressive pitch bends - a sine wrapped in heavy breath noise with a subtle attack bend, played by holding in0. Breath sets the air, Bend the attack pitch slide, Level the output. in0 = Breath gate, in1 = Pitch CV.

Shamisen [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Japanese shamisen: a plucked three-string lute with a percussive snap and the buzzing 'sawari' resonance at the nut - a Karplus string with a nonlinear bridge that adds the signature gritty buzz. Decay sets the ring, Buzz the sawari amount, Level the output. Trigger on in0, in1 = Pitch CV.

ShankerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shanker oscillator: each cycle is the Shanker PDF proportional to (Theta+x)e^(-Theta*x), a lifetime distribution where the same Theta sets both the linear lift and the decay - giving a tightly-coupled rise/fall whose whole shape scales with one knob. Peak-normalized, bounded.

ShannonWaveletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shannon wavelet: the sinc-derived wavelet whose spectrum is a perfect band-pass brick wall (the time-domain dual of the ideal filter), giving a ringing, sidelobe-rich waveform that is the sharpest-possible band-limited packet.

ShawVdp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Forced van der Pol (Shaw): a self-oscillating van der Pol resonator driven by an external sine forces it through entrainment, quasi-periodicity and chaos as the drive grows - a textbook nonlinear-oscillator route to chaos.

SheepBaaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sheep baa: the wavering bleat of a sheep, a voiced ‘baa’ vowel with a fast nervous vibrato (the trembling flock sound) that fades, fired at the set rate; the sound of a hillside of sheep.

ShepardChordOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shepard chord: a chord whose every note is octave-doubled across the whole spectrum under a fixed bell-shaped loudness window, so the chord has no clear octave register - the harmonic counterpart of the Shepard tone, an endlessly-thick pad.

ShepardOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shepard-Risset tone: seven octave-spaced partials under a fixed bell-shaped spectral window whose centre slides continuously, creating the auditory illusion of a pitch that rises (or falls) forever without ever leaving its range.

ShepardTone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shepard tone: seven octave-spaced sines all glide upward together while a fixed bell-shaped spectral window fades in the new low octave and fades out the top, so the pitch seems to rise forever without ever getting higher - the auditory barber-pole.

ShermanBurst [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sherman beta-cell: a model of insulin-secreting pancreatic cells where a slow gating variable periodically switches fast spiking on and off, producing the square-wave bursting (and chaos) of electrical activity in islet cells.

ShieldHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shield hum: the steady protective hum of a raised energy shield, a stack of harmonics gently pulsing in amplitude with a slow breathing throb and a faint high resonance, calmer and purer than the angry force field; the throb sets the shield’s heartbeat.

Shilnikov [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shilnikov chaos: a flow with a saddle-focus equilibrium whose spiralling-out and looping-back form a homoclinic orbit; the Shilnikov theorem guarantees chaos nearby, giving bursts of growing spiral oscillation that reinject - a canonical route to chaos.

ShimizuOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shimizu-Morioka attractor: a symmetric Lorenz-like system with a simpler quadratic structure, giving a clean butterfly orbit - a tidy strange attractor for bell-like chaotic tones.

ShinrikiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Shinriki oscillator: a classic chaotic electronic circuit (a negative-resistance LC oscillator loaded by a nonlinear diode pair), one of the earliest experimentally-demonstrated chaos circuits, with a cubic-like conductance fold.

ShipHornOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ship's horn: the deep mournful blast of an ocean-liner foghorn, a very low fundamental stacked with strong harmonics blaring in long held notes with gaps, the call that carries for miles across water; the blast rate sets the signal pattern.

ShipRoll [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ship-roll capsize model: a vessel rolling in beam waves, whose restoring moment softens and vanishes past the angle of vanishing stability; near that escape it rolls chaotically before capsizing - a nonlinear naval-architecture oscillator.

Shoot [OSC] [FX] [SYNTH] in: – out: Audio

Laser shot - a fast downward pitch-zap with a bright click, the arcade blaster, auto-repeating. Rate sets the fire rate, Pitch the start frequency, Level the output.

Shriek [OSC] [FX] [SYNTH] in: – out: Audio

Shriek - a piercing high dissonant scream of two close beating tones smeared with noise. Pitch sets the height, Tone the rasp, Level the output.

SibilantOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sibilant oscillator: noise pushed into the high sibilance band (4-10 kHz) where the hiss of s and z lives, with an optional low voiced buzz for z versus s; the brightness knob slides the energy up toward a piercing 's'.

SidechainPumpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sidechain pump: the breathing 'pump' of a pad ducked by a kick-driven sidechain compressor, a tone whose volume drops sharply on each beat and swells back, the rhythmic heartbeat of house and EDM; the pump rate sets the tempo.

SIDPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SID pulse: the Commodore 64 SID chip's variable-width pulse blended with a touch of its sawtooth (the chip's 'combined waveform' trick), giving the buzzy, hollow PWM lead and bass voice that defined 8-bit demo and game music.

SIDVoice [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Commodore 64 SID oscillator: a gritty saw/pulse blend with optional ring modulation against a sub-octave - the lo-fi 8-bit chiptune-bass voice. Shape morphs saw to pulse, Ring sets the ring-mod against the sub, Level the output. in0 = Gate, in1 = Pitch CV.

SierpinskiChaosGame [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sierpinski chaos game: repeatedly jump halfway from the current point toward a randomly-chosen triangle vertex; the orbit fills the Sierpinski gasket, the simplest demonstration that randomness plus contraction yields a deterministic fractal - the x-coordinate is the output.

SigmaDeltaChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sigma-delta chaos: a second-order one-bit sigma-delta modulator fed a constant input produces a deterministic yet chaotic +/-1 bit-stream (the idle-tone/limit-cycle problem that plagues real converters) - structured digital noise from a feedback quantizer.

Sigmoid [OSC] [FX] [SYNTH] in: – out: Audio

Sigmoid sine-to-square oscillator (a first as a synth block): takes a pure sine and pushes it through a tanh sigmoid, so as Shape rises the sine's curves are progressively squashed flat - rounded at low Shape, then a soft-clipped near-square, then a hard square at the top. One knob sweeps the entire continuum from sine to square WITHOUT the abrupt harmonic jumps of switching waveforms or the aliasing of a raw square, because the edges stay smoothly rounded. A continuous waveform morph distinct from the fixed geometric shapes. Needs no input.

SilvermanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Silverman oscillator: each cycle is Silverman's kernel 0.5e^(-|x|/sqrt2)sin(|x|/sqrt2+pi/4), an exponentially-damped oscillating bump that dips into negative side-lobes before settling - a higher-order kernel with real ripple, distinct from the strictly-positive smoothing kernels. Width sets how many lobes appear.

Simmons [OSC] [FX] [SYNTH] in: Trigger out: Audio

Simmons SDS electronic tom: a sine with a steep downward pitch sweep plus a noise click - the unmistakable 80s 'pew/dooo' electronic drum. Tune sets the start pitch, Sweep the pitch-drop amount, Level the output. Trigger on in0.

SinaiMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sinai map: the Arnold cat map perturbed by a cosine coupling on the torus, a hyperbolic toral automorphism whose mixing scrambles the two coordinates into hard modular-sawtooth chaos.

SincSquaredOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sinc-squared oscillator: each cycle is (sin(pix)/(pix))^2, the Fraunhofer single-slit diffraction intensity - a bright central lobe with strictly-positive, fast-decaying side-lobes - distinct from the bipolar sinc and the periodic Laue grating function. Bounded.

SineCircleMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sine-circle map: a phase that advances by a fixed rotation plus a sine coupling, the canonical model of mode-locking, Arnold tongues and the quasiperiodic-to-chaos transition - tunable from locked to chaotic.

SineCubedOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sine-cubed oscillator: sin^3(theta), which expands to (3sin(theta) - sin(3theta))/4 - a clean fixed blend of the fundamental and a phase-inverted third harmonic, giving a slightly pinched, hollow sine with a single low overtone.

SineMapOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sine map: the iterated map x -> rsin(pix), a transcendental analogue of the logistic map with the same period-doubling road to chaos but a smoother, rounder bifurcation structure.

SinghFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Singh flow: a Singh-type three-dimensional system constructed without any equilibrium point (a hidden attractor) using a single product nonlinearity and a constant term, traced here as a folded chaotic scroll.

Siren [OSC] [FX] [SYNTH] in: – out: Audio

Rave siren oscillator: a tone whose pitch glides up and down between Low and High at Rate - the hardcore/jungle air-raid siren. Low and High set the pitch range, Rate sets how fast the pitch sweeps through the cycle, Wail skews the sweep from a smooth triangle glide toward a fast-rise / slow-fall ramp, and Level the output. A self-running source - no input needed; the glide is exponential so it tracks musically across the range.

SirenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Siren: a tone whose pitch sweeps up and down in a continuous wail (the rise-and-fall of an emergency siren), or, at fast rates, the rapid yelp mode; the sweep rate and depth set police-versus-air-raid character.

SIRSpatial [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spatial SIR: a ring of cells that are susceptible, infected or recovered; infection spreads to neighbours probabilistically, the infected recover after a step, and immunity slowly wanes back to susceptible - so epidemic waves sweep round and recur. Infected fraction is the output.

Sitar [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Plucked sitar: a Karplus-Strong string whose bridge adds the buzzing ‘jawari’ overtone shimmer through a nonlinear waveshaper in the feedback loop - the droning, sympathetic-buzz character of the Indian sitar. Tune sets the pitch, Buzz the bridge nonlinearity, Level the output. Trigger on in0, in1 = Pitch CV.

SitarDrone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sitar/tanpura resonator: a plucked-string delay loop whose feedback passes through a curved bridge that flattens the wave peaks (the jawari), generating the shimmering, buzzing, harmonic-rich drone of a sitar bridge.

SitarOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sitar: a plucked string whose vibration grazes the curved jawari bridge, adding a buzzing, shimmering nonlinearity that throws off rich high harmonics, the unmistakable twang of a sitar; re-plucked at the set rate.

Sitnikov [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sitnikov problem: a light body oscillating along the axis through two equal masses on elliptical orbits; the periodically-changing gravity makes its vertical motion chaotic - a clean celestial-mechanics chaos with bursts and dwells.

SixOpFM [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Six-operator FM: three modulator-carrier pairs summed (a DX7-style parallel algorithm), for thick, layered FM pads, basses and electric pianos. Spread detunes the three pairs, Index the FM depth, Decay the index falloff. in0 = Gate, in1 = Pitch CV.

SizzleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sizzle: food frying in a hot pan, a continuous high filtered-noise hiss sprinkled with random sharp oil-spatter crackles, the appetising sound of cooking; density sets how vigorously it fries.

SkewBaker [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Skew baker's map: the dough-kneading baker's map with unequal-width strips, so it stretches and stacks asymmetrically into a multifractal attractor with a non-uniform invariant measure - lopsided mixing chaos.

SkewLaplaceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Asymmetric-Laplace oscillator: a two-sided exponential peak whose left and right sides decay at independent rates (Left, Right), so the cycle is a lopsided tent - sharp on one flank, gradual on the other. Unlike the symmetric Laplace, the asymmetry adds even harmonics and a tilted spectral character. Bounded.

SkewNormalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Skew-normal oscillator: each cycle is the skew-normal PDF 2phi(x)Phi(Skew*x), a Gaussian bump leaned to one side by the Skew parameter (negative leans left, positive right) - a continuously asymmetrical bell, peak-normalized and bounded, distinct from the symmetric generalized-normal.

SlashOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Slash oscillator: each cycle is the slash distribution PDF (phi(0)-phi(x))/x^2 - a symmetric bump with a Gaussian-like centre but very heavy, polynomially-decaying tails - peak-normalized to a bounded waveform that sits spectrally between a Gaussian and a Lorentzian.

SleetOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sleet: a cold mix of rain and ice, soft wet droplet ticks interspersed with harder, brighter frozen-pellet clicks, the rattly in-between of a freezing rain; the mix knob shifts it from rain toward ice.

Sleighbells [OSC] [FX] [SYNTH] in: Trigger out: Audio

Sleigh bells: a bright cluster of jingling metal bells shaken together - high inharmonic metallic noise with a fast attack and a shimmering tail. Tone sets the brightness, Decay the shimmer, Level the output. Trigger on in0.

Slenthem [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Gamelan slenthem: the lowest, softest tube-resonated metallophone - a mellow, round low bar with a gentle attack and a smooth decay. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

SlideWhistleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Slide whistle: the comedy swoop of a slide whistle, a pure tone that glides smoothly up then back down in pitch (the plunger sliding), the staple of cartoon falls and rises; fired at the set rate.

SlinkyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Slinky: a metal coil spring sproinging, a shimmering cluster of metallic partials that glide downward in pitch with a fluttering coil resonance, fired at the set rate, the toy-spring boing; the rate sets the sproings.

SlitDrum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Slit/tongue drum: two woody, moderately-damped inharmonic modes ring when struck, the warm, hollow, pitched knock of a wooden tongue drum or log drum - excite with an impulse or gate.

SloshingTank [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sloshing tank: liquid in a horizontally-rocked container whose surface mode is parametrically pumped, giving the chaotic swash of fuel tanks and tuned-liquid dampers - two coupled sloshing modes exchanging energy nonlinearly.

SlurpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Slurp: the noisy gurgle of sucking the last of a drink through a straw, a bubbling tone rising in pitch as the liquid runs out, choked with air-and-liquid noise, fired at the set rate; the rude end of a milkshake.

SmaleHorseshoe [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Smale horseshoe: the canonical stretch-fold-and-reinsert map that Smale proved contains a full shift on symbols - the geometric heart of why deterministic systems are chaotic, iterated here as a fractal-mixing source.

SmokeAlarmOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Smoke alarm: a piercing ~3 kHz tone gated in the standardized Temporal-3 pattern - three half-second beeps, a pause, repeat - the internationally-mandated fire-alarm evacuation signal.

SmootherStepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Smootherstep oscillator: each cycle is Perlin's smootherstep 6t^{5-15t}4+10t^3, a quintic S-ramp with zero first AND second derivatives at both ends - even flatter shoulders and a more central slope than the cubic smoothstep, a very mellow soft sawtooth with minimal edge harmonics.

SN76489Osc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SN76489 PSG: the programmable sound generator in the SEGA Master System, BBC Micro and many arcades, whose 10-bit frequency divider quantizes pitch ever more coarsely as it rises - a clean square with the chip's characteristic high-note detuning.

Snare [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog snare drum: a trigger on in0 fires two detuned body tones plus a burst of band-passed noise (the snare wires). Tune sets the body pitch, Snap the noise level, Tone the body/noise balance, Decay the length, and Level the output.

Snare606 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-606 snare drum: a thin, buzzy snare - a short tonal body under a bright burst of highpassed noise, cheaper and more lo-fi than the 808, the gritty Drumatix backbeat. Tone sets the body-vs-noise balance, Decay the length, Level the output. Trigger on in0.

∿ **Snare707** [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-707 snare drum: a tight, bright, slightly-digital snare - a short tonal body under a crisp highpassed noise crack, the 707's punchy backbeat. Tone sets the body-vs-noise balance, Decay the length, Level the output. Trigger on in0.

∿ **Snare808** [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-808 snare drum: a trigger on in0 fires two detuned tone oscillators (the 808's shell body) plus a burst of filtered noise (the snappy wires), each with its own decay. Snappy balances the tonal body against the noise, Decay sets the length, Level the output - the dry, snappy 808 backbeat.

∿ **Snare909** [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-909 snare drum: brighter and noisier than the 808 - a short tonal thwack under a long crack of highpassed white noise, the punchy house/techno snare. Tone sets the body-vs-noise balance, Decay the noise length, Level the output. Trigger on in0.

∿ **SnareRoll** [OSC] [FX] [SYNTH] in: Gate out: Audio

Build-up snare roll: while the gate on in0 is held, fires band-passed noise hits (snare / clap) at a rate that accelerates from Start to End over the Accel ramp - the pre-drop EDM snare roll that speeds up into the drop. Start and End set the hit rate at the beginning and end of the build (End higher = it speeds up), Accel the ramp time, Tone the noise brightness, Decay the per-hit length, and Level the output. Hold the gate over the build bar and release it at the drop.

∿ **SnareSynth** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth snare drum: a gate fires a short tuned tone mixed with a noise burst (the 'snares'), each with its own decay - a self-contained drum-machine snare voice.

∿ **SnareSynthOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Snare-drum voice: two detuned tone oscillators for the drum body mixed with a burst of noise for the snare wires, re-triggered at the set rate; the snappy knob balances the dry thump against the rattly hiss.

∿ **SneezeOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sneeze: the two-part 'ah-CHOO', a rising breathy tension that builds and then releases in an explosive burst of bright turbulent noise, fired at the set rate; the involuntary nasal blast.

∿ **Snore** [OSC] [FX] [SYNTH] in: – out: Audio

Snoring - a rhythmic buzzy growl on the in-breath with a softer snort on the out-breath. Rate sets the breath pace, Tone the rasp, Level the output.

∿ **SnoreOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Snore: the rattling buzz of snoring, a low growling tone amplitude-modulated by the soft-palate flutter that swells once per breath, the comical sawing of deep sleep; the rate sets the breathing speed.

SnowflakeReiter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Reiter snowflake: a reaction-diffusion cellular automaton where water vapour diffuses across cells and freezes onto the crystal wherever it borders ice, growing the branching dendrites of a snowflake; this radial section reads out the advancing, ever-thickening growth front.

SoftSync [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Soft-sync oscillator: when the master completes a cycle it reverses the slave saw's direction of travel rather than hard-resetting its phase, so the slave folds back on itself - the gentler, more harmonically complex cousin of hard sync that adds even harmonics and a hollow, reedy formant sweep as Ratio is detuned. Freq is the master, Ratio the slave's relative pitch.

SolarRadioSweepOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Solar radio sweep: a type-III solar radio burst, where electrons flung from a flare race outward through the corona emitting at ever-lower plasma frequencies, drawing a fast high-to-low diagonal streak; repeated bursts give the crackling sweep of an active Sun.

SolarWind [OSC] [FX] [SYNTH] in: – out: Audio

Solar wind - a vast shifting cosmic-noise drone with a faint pitched whistle drifting through it. Flux sets the movement, Tone the brightness, Level the output.

Sonar [OSC] [FX] [SYNTH] in: – out: Audio

Submarine sonar - a clean periodic ping with a long ringing tail. Rate sets the ping interval, Pitch the ping frequency, Level the output.

SonarPing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sonar: emits a periodic submarine ping - a descending sine chirp followed by its own fading echo return - on a slow clock, ducking the dry signal underneath. A ready-made depth/suspense bed. Period sets the seconds between pings, Mix the blend.

SonarPingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sonar oscillator: a clean tonal ping repeated at a slow interval, each fading into a decaying echo tail as the sound returns from the deep, the unmistakable submarine ‘ping… ping…’; the interval and pitch set the vessel and depth.

SparkOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Electric spark: random sharp broadband crackle bursts (the snapping of an electric arc jumping a gap), each a very short high-passed noise transient; the density knob ramps from an occasional pop to a continuous frying arc.

SpatialPrisoners [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spatial prisoner's dilemma (Nowak-May): cooperators and defectors on a ring each play their neighbours, then copy the strategy of whichever neighbour scored highest; cooperation survives in shifting clusters instead of dying out - the cooperation fraction ebbs and flows.

SpectralTiltOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spectral-tilt oscillator: a complete harmonic series whose partial amplitudes follow a single tunable tilt, sweeping continuously from a dark sine-like tone (steep down-tilt) through sawtooth to a bright, buzzy up-tilted spectrum - one knob across the brightness axis.

Speech [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vowel/speech synthesizer (Mutable Instruments / SAM-style): a buzzing saw source is shaped by three formant band-passes tuned to vowel sounds, so it sings. Vowel morphs through A-E-I-O-U; Speak, when above zero, auto-advances the vowel at that rate for talking/chanting textures; Tune sets the pitch. A formant voice for choirs, robots and vocal pads. Distinct from the Formant filter (which colours external audio) and Vosim (a single grain pair) - this sweeps a vowel sequence over its own oscillator.

SpellCastOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spell cast: the magical flourish of casting a spell, a cluster of detuned partials sweeping upward in pitch and sparkle with a shimmering tremolo, building to a release; the rate sets how often the spell fires.

SphericalPendulum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spherical pendulum: a pendulum free to swing in two angles under a driving force, whose azimuthal and polar motions couple nonlinearly into chaotic, precessing, never-repeating swings.

SpinningTopOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spinning top: a toy top spinning on a table, a whirring hum that rises in pitch as it speeds up then wobbles and wavers as it loses momentum, the hypnotic top spin; the cycle rate sets how often it is spun.

SpinTorque [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spin-torque oscillator: a nanomagnet whose magnetization is driven into steady precession by a spin-polarized current (Slonczewski torque against Gilbert damping), and into chaos under field modulation - the heart of spintronic RF oscillators.

Spire [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spire-style multimode oscillator: sums up to 9 unison voices, each rendered by the selected Mode (Classic saw, Noise, FM, AMSync, SawPWM, HardFM, Vowel) with the sum normalized by the voice count. Mode picks the algorithm, Shape is the per-mode control (FM index, sync ratio, pulse width, or vowel position), Voices sets unison count (up to 9, matching Spire), Detune the fine +/-3% spread, and Interval fans the voices out over a musical interval (0-12 semitones, e.g. octave/fifth stacks). The Vowel mode adds three morphing formant band-passes over the summed saw for talking-synth timbres.

SpirographOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spirograph oscillator: the hypotrochoid traced by a pen held inside a small gear rolling within a ring, whose radius ratio and pen offset weave the looping rosette patterns of the toy; the x-coordinate makes an intricate multi-loop waveform.

SplashOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Splash: an object hitting water, a sharp broadband impact transient that decays into a brief scatter of high spray-droplet noise, fired at the set rate; the cannonball plunge and its spray.

SplitMixNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SplitMix noise: advances a Weyl counter by a fixed odd constant and runs it through a strong xor-shift-multiply mixing finalizer (the SplitMix algorithm), giving fast, well-distributed white noise with a long period and good statistics.

SpringSproingOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spring sproing: the twangy metallic ‘sproing’ of a released coil spring, a cluster of inharmonic partials with a fast downward pitch glide and a buzzy rattle, the cartoon spring-launch; fired at the set rate.

Sprott2DQuadratic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott 2-D quadratic map: a general two-variable map whose update is a full quadratic polynomial (the family Sprott computer-searched for elegant chaotic attractors), tuned here to one of its strange-attractor parameter sets.

SprottBOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case I flow: one of Sprott’s catalogue of algebraically-simplest chaotic systems (dx=-0.2y, dy=x+z, dz=x+y^2-z), a clean low-cost strange attractor with its own folding character.

SprottConsA [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott-style minimal flow A: a near-conservative three-term chaotic system (dx=y, dy=-x+yz, dz=1-y^2 scaled) that preserves phase-space volume almost exactly, so its chaos has the dense, non-attracting texture of a conservative system.

SprottConsB [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott-style minimal flow B: a second near-conservative chaotic system (dx=yz, dy=x-y, dz=1-xy variant with weak damping) whose orbit fills its phase space densely, giving a noisy, non-settling chaotic texture.

SprottConsC [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott-style minimal flow C: a third low-dissipation chaotic system (dx=y, dy=z, dz=-x-y^2+ small drive) that nearly conserves phase-space volume, giving the dense, space-filling texture of conservative chaos.

SprottConsD [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott-style minimal flow D: a fourth low-dissipation chaotic system (dx=-y, dy=x+z, dz=x z + 3 y^2, weakly damped) whose volume contraction is tiny, giving the dense, fat, space-filling orbit characteristic of near-conservative chaos.

SprottCOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case C flow: a minimal chaotic system (dx=yz, dy=x-y, dz=1-x^2) closely related to but distinct from case B, giving a compact, smoothly-folding attractor.

SprottD [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case D flow: a minimal chaotic system (dx=-y, dy=x+z, dz=xz+3y^2) whose orbit spirals through a tilted attractor, another distinct entry in Sprott's elegant-equations catalogue.

SprottEOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case E flow: a minimal chaotic system (dx=yz, dy=x^2-y, dz=1-4x) whose squared term gives a sharply-folded orbit - another elegant low-complexity strange attractor.

SprottGOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case G flow: a minimal chaotic system (dx=0.4x+z, dy=xz-y, dz=-x+y) with a self-driving linear term, producing a steadily-circulating chaotic orbit.

SprottHOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case H flow: a minimal chaotic system (dx=-y+z^2, dy=x+0.5y, dz=x-z) whose squared feedback folds a compact attractor, another of Sprott's algebraically-simplest chaotic systems.

SprottJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott jerk system: the simplest chaotic 'jerk' equation (a single variable with its first two derivatives and one nonlinearity), the most elementary route to chaos in one differential equation.

SprottJOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case J flow: a minimal chaotic system (dx=2z, dy=-2y+z, dz=-x+y+y^2) tracing a smoothly-circulating attractor, a distinct member of Sprott's elegant-equations catalogue.

SprottKOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case K flow: a minimal chaotic system (dx=xy-z, dy=x-y, dz=x+0.3z) whose product and self-driving terms give a steadily-circulating attractor, another entry in Sprott's elegant-equations catalogue.

SprottLinz [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott-Linz F flow: another of Sprott's algebraically minimal chaotic systems (dx=y+z, dy=-x+0.5y, dz=x^2-z), a low-cost smooth attractor with a character of its own.

SprottLOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case L flow: a minimal chaotic system (dx=y+3.9z, dy=0.9x^2-y, dz=1-x) blending a squared term with linear drives into a distinctive folded orbit.

SprottMegastable [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Megastable oscillator: a damped resonator with a spatially-periodic (sinusoidal) restoring force, so it has a countable infinity of nested coexisting limit cycles - which orbit it lands on depends on the energy, giving amplitude-dependent timbres.

SprottMOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case M flow: a minimal chaotic system (dx=-z, dy=-x^2-y, dz=1.7+1.7x+y) with a squared term that folds the orbit sharply, a distinct low-complexity strange attractor.

SprottNOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case N flow: a minimal chaotic system (dx=-2y, dy=x+z^2, dz=1+y-2z) with a squared feedback, a distinct compact strange attractor from Sprott's catalogue.

SprottOOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case O flow: a minimal chaotic system (dx=y, dy=x-z, dz=x+xz+2.7y) whose product term twists the orbit into a distinct attractor shape.

SprottOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott flow: one of Sprott's minimal three-term chaotic systems (dx=yz, dy=x-y, dz=1-xy), the simplest equations that still produce chaos - a clean, low-cost strange-attractor source.

SprottParabolicJerk [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Parabolic jerk: a third-order jerk circuit whose restoring force is a parabola in position (dz=-a z - y + x^2 - offset), folding an asymmetric single-scroll chaotic attractor - a distinct member of the elegant-jerk family.

SprottPOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case P flow: a minimal chaotic system (dx=2.7y+z, dy=-x+y^2, dz=x+y) mixing a squared term with simple linear couplings into another elegant chaotic attractor.

SprottQOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case Q flow: a minimal chaotic system (dx=-z, dy=x-y, dz=3.1x+y^2+0.5z) whose squared feedback and self-driving z term yield a distinct circulating attractor.

SprottROsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case R flow: a minimal chaotic system (dx=0.9-y, dy=0.4+z, dz=xy-z) with mostly constant drives, producing a smoothly-looping attractor distinct from the product-heavy cases.

SprottSE [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott SE flow: an entry from Sprott's 'Elegant Chaos' catalogue (a jerk system dz=-a z + y^2 - x driven slightly differently), one of the algebraically simplest dissipative chaotic systems with a single quadratic term.

SprottSOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott case S flow: a minimal chaotic system (dx=-x-4y, dy=x+z^2, dz=1+x) closing out Sprott's elegant-equations set with another compact, squared-feedback attractor.

SprottSQ [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sprott SQ flow: another 'Elegant Chaos' system whose jerk uses a signum nonlinearity (dz=-z - a y - sgn(x) + …), reducing the chaos generator to comparators and integrators - hard-switching minimal chaos.

SprottThomasSym [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Symmetric cyclic flow: a cyclically-symmetric chaotic system (each axis driven by a sine of the next plus a bias) generalizing the Thomas attractor with an adjustable symmetry-breaking bias, sweeping from a tidy lattice walk to dense chaos.

SpruceBudworm [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Spruce-budworm model: a fast insect population on a slowly-growing forest, whose fold-catastrophe outbreak-and-collapse hysteresis gives slow relaxation oscillations - the classic fast-slow ecological boom-bust.

SquareArmy [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

A wall of detuned pulse waves with movement: Voices sets how many, Detune spreads them, and Width is the pulse-width (square at 0.5) that an internal LFO slowly drifts for a hollow PWM ensemble.

SquareLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Square-wave lead: the classic hollow chiptune/synth square lead with a gentle vibrato. Vibrato sets the wobble, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

SquareNumOsc [OSC] [FX] [SYNTH] in: — out: Audio

Square-number oscillator (a first as a synth block): an additive tone (NOT a square wave) whose overtones sit on the perfect squares - the 1st, 4th, 9th, 16th, 25th… harmonics. Because the partials thin out quadratically, the few that remain are widely and unevenly spaced, giving a clear, glassy, almost-pure tone with a faint metallic ring - the spectral fingerprint of the squares. Partials sets how many square overtones to sound, alias-free up to Nyquist. A figurate spectrum distinct from the triangular and prime oscillators. Needs no input.

SqueakyToyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Squeaky toy: a rubber dog toy squeezed, a high reedy squeak that bends up sharply then sags back down on each squeeze, fired at the set rate, the comic squeak-toy sound; the rate sets the squeezes.

SquidOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SQUID oscillator: a superconducting quantum-interference loop whose trapped flux sits in a tilted double-well potential and is driven by an external field, hopping chaotically between flux states - a sensitive magnetometer pushed into chaos.

SquishOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Squish: a wet squishy splat or squelch, a burst of noise swept by a fast-closing lowpass with a gloopy pitch droop, the cartoon mud-step or slime-squish; fired at the set rate.

SSTVOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

SSTV oscillator: the warbling audio of slow-scan television, where each image line is sent as an FM sweep across the 1500-2300 Hz video band bracketed by sync tones, the burbling sound of pictures over ham radio; the line rate sets the warble speed.

Stab [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Chord stab: a trigger on in0 fires a detuned-saw triad (root, third, fifth) through a fast amp-and-filter pluck envelope - the staccato house/trance organ-stab voice. Tune sets the root pitch (in1 = Pitch CV), Chord picks the voicing (0 major / 1 minor / 2 sus4 / 3 power-octave), Detune spreads paired saws per note for width, Decay sets the stab length, Cutoff the low-pass brightness, and Level the output. Distinct from the MIDI Chord block: a self-contained audio voice, not a note processor.

StadiumBilliard [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stadium billiard: a point bouncing inside a Bunimovich stadium (a rectangle capped by semicircles), whose defocusing curved walls make every trajectory chaotic - a textbook example of hard chaos, read as the ball's position.

StadiumCheerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stadium cheer: the roar of a packed stadium erupting, a huge wall of broadband crowd noise that swells from a murmur to a deafening peak and falls back, the surge of a goal being scored; the surge rate sets the cheer waves.

StampedeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stampede: a running crowd or herd, a low rolling rumble of many overlapping footfalls thickening into a continuous thunder with random pounding accents, building as the mass approaches; the density sets the size of the rush.

StaplerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stapler: a desk stapler punched down, a sharp two-part mechanical ka-chunk (a spring-loaded press click then the metallic staple punch), fired at the set rate, the paperwork sound; the rate sets the stapling.

StarterPistolOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Starter pistol: the crack of a track-race starting gun, a very sharp broadband transient followed by a quick slap-back echo off the stadium, fired at the set rate; on your marks, get set, go.

Static [OSC] [FX] [SYNTH] in: – out: Audio

Radio static - white-noise hiss with random crackle bursts and a faint drifting carrier whine, the between-stations sound. Noise sets the hiss level, Tune the whine pitch, Level the output.

StaticShockOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Static shock: the sharp snap of an electrostatic discharge zapping your finger to a doorknob, a very short bright broadband crack with a tiny high ring, fired at the set rate; the dry winter-carpet zap.

Steam [OSC] [FX] [SYNTH] in: Trigger out: Audio

Steam / pressure release: a trigger on in0 vents a burst of band-passed noise - the hydraulic 'psssh' / hiss of escaping gas, valves and pneumatics - with a sharp attack and a decaying tail whose filter falls as the pressure drops. The mech, robot-joint and steam-pipe sound of film and game design. Tone sets the hiss brightness, Hole the resonance (how whistling / pitched the leak is), Decay the vent length, Sweep how far the filter falls as it decays, and Level the output. Trigger per vent.

SteamHissOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Steam oscillator: high-passed turbulent noise (the hiss of escaping steam) with an optional narrow whistle resonance riding on top, from a gentle radiator leak to a screaming kettle; the whistle knob mixes in the pitched peak.

SteamTrainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Steam train: the chuffing of a steam locomotive, a low thump and a burst of steam hiss on each piston exhaust repeating at the chuff rate that speeds up with the train; the rate sets how fast it rolls.

Steeldrum [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Struck steelpan (steel drum): a bright metallic note from partials near 1 : 2 : 2.8 : 4 with a medium shimmering decay - the sunny Caribbean steelpan timbre. Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

Steelpan [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal steelpan (steel drum): a trigger on in0 strikes a tuned note on an oil-drum face - a bright shimmering tone with near-harmonic partials (1 : 2 : 3 : 4) and a metallic edge, medium sustain. Tune sets the pitch, Decay the ring, Shimmer the upper-partial brightness, and Level the output.

SteelpanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Steel pan: a struck note-area of a Caribbean steel drum, a bright stack of mostly-harmonic partials (fundamental, octave, twelfth) with a fast metallic attack and ringing sustain, re-struck at the set rate; the sunny pannist tone.

StepperMotorChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stepper-motor chaos: a permanent-magnet stepper driven near its mid-frequency resonance, where the sinusoidal detent and reluctance torques make the rotor angle lose lock and oscillate chaotically - the cause of stepper stall and missed steps.

StickSlip [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stick-slip oscillator: a spring-held block dragged by a moving belt whose velocity-weakening friction makes it alternately stick and slip, the relaxation mechanism behind earthquakes, brake squeal and bowed-string sound.

∿ **Stinger** [OSC] [FX] [SYNTH] in: – out: Audio

Horror stinger - a sudden dissonant cluster stab with a sharp attack and fast decay, the jump-scare hit, auto-repeating. Rate sets the interval, Tone the harshness, Level the output.

∿ **StingerHitOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stinger hit: the sudden violent musical stab of a jump scare, a dense high dissonant cluster slammed in with a sharp attack and a fast decay (the screeching-strings shock chord), fired at the set rate; the gotcha moment.

∿ **StochasticGrainOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stochastic grain: a Xenakis-style grain field where grain onsets follow a Poisson process and pitches are drawn from a band, the granular-cloud realization of stochastic music; the density governs the statistical mean rate, not a fixed clock.

∿ **StochResonance** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stochastic-resonance switch: a particle in a double-well potential nudged by a weak periodic force plus a chaotic 'noise' term, so it hops between wells nearly in step with the drive - the counterintuitive noise-aided switching effect, deterministically rendered.

∿ **StommelBox** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stommel box model: two ocean boxes exchanging heat and salt whose density-driven flow can suddenly flip direction (the thermohaline-circulation switch behind abrupt climate change), giving slow bistable, relaxation-oscillator dynamics.

∿ **Stream** [OSC] [FX] [SYNTH] in: – out: Audio

Babbling brook - bright filtered noise with random water gurgles for a flowing-stream ambience. Flow sets the gurgle activity, Tone the brightness, Level the output.

∿ **StretchOsc** [OSC] [FX] [SYNTH] in: – out: Audio

Stretched-harmonic oscillator (a first as a synth block): models the inharmonicity of a real struck or plucked string, where stiffness makes each overtone sharper than a perfect multiple - partial k sits at ksqrt(1 + Stretchk^2) instead of exactly k. That tiny upward stretch, growing with each partial, is what gives a piano its warm, slightly detuned brilliance and a struck metal bar its character. Stretch sets the amount (0 = perfectly harmonic, up = ever more bell-like). Partials sets how many to stack. A physically-modelled inharmonic spectrum distinct from the golden and integer oscillators. Needs no input.

∿ **StringMachine** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vintage divide-down string ensemble: a full sawtooth chord run through a triple-chorus for the lush, shimmering 70s string-synth pad. Ensemble sets the chorus depth and Tone the brightness.

∿ **StringPartialOsc** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Stiff-string oscillator: harmonics stretched by the inharmonicity law f_n = nf0sqrt(1 + B*n^2), where bending stiffness pushes the upper partials progressively sharp - the slight overtone stretch that gives a piano (especially its bass) its characteristic shimmer.

Strings [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Orchestral string section: a wide ensemble of detuned saws with a slow bowed attack, section vibrato and a warm low-pass body - the sustained legato strings of cinematic and Zimmer-style scoring. Distinct from the Solina BBD ensemble effect (which colours an input) and the single Bowed string: this is the full section as a voice. Tune sets the pitch (in1 = Pitch CV), Ensemble the detune spread (section width and beating), Attack the bow swell time, Vibrato the section pitch waver, Tone the brightness, and Level the output. A gate on in0 bows the note; hold for long legato.

StruckTube [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Struck open tube / pipe: a hollow, woody-metallic pitched ‘tonk’ from striking an open pipe, ringing on its odd harmonics (1 : 3 : 5 : 7). Tune sets the pitch, Decay the ring, Level the output. Trigger on in0, in1 = Pitch CV.

StudentTNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Student-t noise: a Gaussian divided by sqrt(chi-squared/dof), whose degrees-of-freedom knob morphs the tails from heavy (Cauchy-like at dof=1) to light (Gaussian as dof grows) - a continuously tail-tunable noise.

Stylophone [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Stylophone: the buzzy, lo-fi stylus-organ - a bright square with a fast vibrato and a slightly unstable pitch, the 60s pocket-synth toy, played by holding in0. Vibrato sets the wobble, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

SuarezSchopf [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Suarez-Schopf delayed oscillator: the canonical El-Nino model T’ = T - T^3 - a T(t-tau), where a delayed negative feedback (the ocean’s Rossby-wave return) sustains an irregular interannual oscillation - delay-driven climate cyclicity.

Sub [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Dedicated sub oscillator: runs a single phase at Freq dropped by Octave, outputting either a sine or a square, then scaled by Level. Octave shifts down 0, 1, or 2 octaves, Wave selects sine or square, and Level sets the output gain. Use it to add low-end weight under a main oscillator, with sine for clean sub and square for a harder, more present low end.

Sub808 [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

808 sub-bass: a tuned sine sub with a short downward pitch-drop attack (the TR-808 bridged-T ‘damped sine’ punch), portamento glide between notes for the trap-style slide, an octave-up layer for midrange, and parallel saturation that adds upper harmonics so the sub translates on small speakers. Tune sets the pitch (in1 = Pitch CV), Glide the portamento time, Punch the pitch-drop attack, Decay the note length, Drive the parallel saturation, and Level the output. A gate on in0 strikes the note; lives in the 30-80 Hz sub range.

SubBoom [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Devastating sub-bass oscillator: a sine fundamental reinforced with two subharmonics an octave and two octaves below, then driven into saturation. Sub sets the lower-octave weight and Drive the harmonic grit so it cuts on small speakers.

SubGrowl [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Sub-growl bass: a deep sine sub blended with a driven, growling harmonic layer - clean weight at the bottom with grit on top. Growl sets the harmonic drive, Sub the sub level, Level the output. in0 = Gate, in1 = Pitch CV.

SubharmonicOsc [OSC] [FX] [SYNTH] in: — out: Audio

Undertone-series oscillator (a first as a synth block): the mirror image of an ordinary tone - instead of stacking overtones ABOVE the fundamental it stacks the UNDERTONE series below it (f, f/2, f/3, f/4…), the descending ratios that Harry Partch and the old harmoniums explored. The fundamental sits at the TOP and the partials descend, giving a deep, hollow, slightly unsettling body quite unlike any overtone-based saw or square. Partials sets how many undertones to stack. A subharmonic spectrum distinct from every overtone oscillator. Needs no input.

SubHarmonicSeriesOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Undertone-series oscillator: the mirror image of the harmonic series, stacking partials at 1, 1/2, 1/3, 1/4 … of the set frequency (the subharmonic series of Mixtur-Trautonium and just-intonation theory) - a dark, bass-heavy, minor-feeling spectrum.

SubOscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sub oscillator: the main pitch reinforced by square waves one and two octaves below it, the classic bass-thickening sub-osc found on Minimoog-style synths; the Sub1 and Sub2 knobs blend in the lower octaves for weight.

Subway [OSC] [FX] [SYNTH] in: — out: Audio

Subway train - rhythmic rail-joint clatter over a rolling rumble, with brake squeals. Speed sets the clatter tempo, Tone the metallic brightness, Level the output.

SubwayOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Subway: an electric metro train, a rising-and-falling traction-motor whine over a low rolling-stock rumble with an occasional metallic rail screech on the curves; the speed knob sets the motor pitch and rumble.

SuccessChimeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Success chime: the bright rising arpeggio of a completed task or level-up, three ascending bell tones (a major triad) ringing in quick succession, the satisfying done/achievement sound; fired at the set rate.

SujathaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sujatha oscillator: each cycle is the Sujatha PDF proportional to (1+x+x$^{2)e}$(-Theta*x), a lifetime distribution whose full quadratic numerator gives the broadest hump of the Lindley family before the exponential tail - a wide, gently-skewed bump. Peak-normalized, bounded.

SummonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Summon: calling forth a creature or boss, a deep minor chord of voices that builds from a low rumble, swelling in volume and adding upper octaves as the entity materialises, the dramatic conjuring crescendo; the rate sets the summon cycle.

SundarapandianFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sundarapandian flow: a representative three-dimensional quadratic chaotic system from the chaos-synchronization/control literature, with two product nonlinearities folding a compact double-scroll attractor.

Suona [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Suona: the Chinese double-reed shawm - a conical-bore waveguide driven by a hard double reed (two reeds beating together, a sharper, more aggressive reed table than a single-reed clarinet/sax) into a flared metal bell, giving the loud, piercing, nasal blare of Chinese wedding and folk bands. Reed sets the double-reed bite, Bright the bore.

SuperellipseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Superellipse oscillator: the Lame curve |x|^n + |y|^n = 1 whose exponent morphs the cross-section from a pinched astroid (n<1) through a circle (n=2) to a near-square 'squircle' (n large); sweeping the exponent fattens a sine toward a square.

SuperformulaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Superformula oscillator: Gielis's single polar equation that generalizes the circle into starfish, flowers, polygons and gear shapes through its symmetry and exponent parameters; the radius read around the circle becomes a hugely-variable waveform from one formula.

SuperGaussOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Super-Gaussian oscillator: each cycle is a super-Gaussian bump exp(-(x$^{2)}$Order), so Order morphs the pulse from an ordinary Gaussian (1) to an increasingly flat-topped, steep-edged near-rectangle (high) - a smoothly band-limited pulse with controllable edge sharpness.

SuperPWM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Supersaw built from detuned pulse waves: stacks several pulse oscillators spread symmetrically in tuning and averages them for a thick unison. Freq sets the center pitch, Detune spreads the voices apart in tuning, Width sets the pulse duty cycle (timbre), Voices sets how many oscillators stack, and Level scales the output. High detune and voice counts give wide, rich pad and lead tones.

SuperSawStack [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Supersaw: seven sawtooth oscillators detuned around the centre pitch and summed (the Roland JP-8000 trance-lead sound), where the Detune knob fans the partials apart into the thick, shimmering chorused saw that defines a whole genre.

SuperSquare [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Super-square: seven detuned pulse oscillators stacked into one huge unison voice - the square-wave cousin of the supersaw, giving a wide, hollow, chorused wall of sound for hardstyle leads and fat basses. Detune spreads the seven copies, Mix balances the centre against the detuned pack. in0 = Pitch CV.

SuperTri [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Detuned triangle stack: up to 9 triangle voices spread by Detune for a soft, hollow, flute-like unison with none of a saw's bite. Voices sets the thickness. A mellow alternative to the supersaw.

SupplyChain [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Supply-chain bullwhip: three echelons (retailer, wholesaler, factory) ordering to correct their inventory and pipeline with a delay, so small demand changes amplify into wild oscillating orders up the chain - the bullwhip effect as a chaotic cycle.

Surdo [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal surdo: a trigger on in0 strikes the deep Brazilian samba bass drum - a very low membrane (modes near 1 : 1.8) with a long booming decay and a slow pitch drop, the heartbeat under a samba bateria. Tune sets the low pitch, Decay the long boom, Mute tightens it toward a damped stroke, Level the output. Trigger from a Clock, Euclid or sequencer.

Susaw [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Massive X super-saw: sums 7 saws spread across a detune range, with a slow random drift added to the base frequency, then scaled by Level. Detune sets the +/-3% spread between voices, Blend lowers the 6 side voices relative to the loud centre voice, and Drift sets how much the wandering frequency offset is applied. Higher Blend gives a wider unison wash; lower Blend keeps the centre pitch dominant for a tighter lead.

SuspensionQuarterCar [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Quarter-car suspension: the sprung mass, tyre and a hardening/asymmetric damper of one wheel rolling over a periodic road, whose nonlinear damping and tyre lift-off make the body motion chaotic - vehicle ride-dynamics chaos.

SvenssonAttr [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Svensson attractor: a trigonometric iterated map (a relative of the de Jong family) tracing tight, looping fractal filaments - another distinctive strange-attractor sound source.

SwellPopOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Swell pop: a balloon or bubble inflating and bursting, a tone that swells up in pitch and volume then snaps with a sharp pop transient, the cartoon inflate-and-burst; fired at the set rate.

SwiftHohenberg [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Swift-Hohenberg mode: the amplitude equation for the convective-roll pattern-forming instability, a bistable cubic-quintic mode whose growth and saturation model the snap into and out of striped patterns.

SwingCreakOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Playground swing: a child swinging back and forth, a filtered creak of the chain that swells at each end of the arc where the swing reverses, the lazy-afternoon playground sound; the rate sets the swing tempo.

SwingEquation [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Power-grid swing: a synchronous generator's rotor angle obeying the pendulum-like swing equation against a stiff grid; a fluctuating mechanical input drives the angle through pole-slips into chaotic loss-of-synchronism - electrical-grid stability dynamics.

Sword [OSC] [FX] [SYNTH] in: Trigger out: Audio

Sword / blade SFX: a trigger on in0 fires a bright metallic swish - a band-passed noise whoosh as the blade cuts the air, layered with ringing inharmonic metal partials (the steel ‘shing’) that shimmer and decay. The unsheathe, swing and parry of film and game sound design. Tone sets the swish brightness, Ring the metallic-resonance level, Pitch the ring fundamental, Decay the ring length, and Level the output. Trigger per swing.

SymIconMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Symmetric icon: an iterated map with built-in rotational symmetry (Field-Golubitsky ‘symmetry in chaos’), whose orbit fills a kaleidoscopic n-fold-symmetric attractor - structured, crystalline chaos.

Sync [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Hard-sync oscillator: a slave saw runs at Sync times the master frequency and is reset to zero each time the master phase wraps, producing the classic sync-sweep timbre, then scaled by Level. Freq sets the master pitch and Sync sets the slave ratio (1 to 6). Higher Sync ratios add brighter, more formant-like harmonics from the forced phase resets.

SyncGrainOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sync grain: a synchronous granular stream firing one windowed grain per period at the grain rate, each grain a short burst of an internal tone, so the grain rate sets the pitch while the internal frequency colours the timbre - classic synchronous granular synthesis.

SyncLead [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Hard-sync lead: a slave saw reset by a master each cycle, swept by Sync for the bright, vocal, tearing sync sweep - the classic aggressive synth lead. Sync sets the slave ratio, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

SznajdModel [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sznajd model: a pair of agreeing neighbours convinces both of their outer neighbours of their opinion (‘united we stand, divided we fall’), the sociophysics rule whose outward persuasion drives the ring toward one of two consensus states or a stalemate.

Tabla [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal tabla: a trigger on in0 strikes a tuned Indian hand drum whose loaded membrane gives near-harmonic modes (1 : 2 : 3 : 4) plus its signature pitch-bend - the fundamental swoops as it rings, the singing ‘tun’ of the dayan. Tune sets the pitch, Decay the ring, Bend how far the pitch glides during the hit, Level the output. Trigger from a Clock, Euclid or sequencer.

TablaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tabla: the pitched ringing stroke of a North-Indian tabla, whose loaded drumhead is uniquely tuned so its membrane modes line up into a near-harmonic series (a rare pitched drum), with a sharp attack and a singing decay; re-struck at the set rate.

TacomaFlutter [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Bridge torsional flutter: a suspension-bridge deck whose cables pull only in tension (a one-sided nonlinear restoring force) couples vertical and torsional modes under wind, the Lazer-McKenna mechanism blamed for the Tacoma Narrows collapse.

Taiko [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Cinematic taiko drum: a trigger on in0 strikes a deep membrane - a sine body with a fast downward pitch bend (the skin tensioning), an inharmonic mode for the skin ‘thwack’, and a wooden stick-attack click - decaying with the round boom of a big Japanese drum. The epic ensemble percussion of trailer and Zimmer-style scoring. Tune sets the pitch (in1 = Pitch CV), Bend the attack pitch-drop depth, Decay the boom length, Stick the wooden attack-click level, Skin the inharmonic body character, and Level the output. Trigger from a Clock, Euclid or sequencer; layer several for an ensemble.

TaikoBig [OSC] [FX] [SYNTH] in: Trigger out: Audio

Big taiko drum: a deep, thunderous Japanese barrel drum - a low pitch-swept membrane boom with a hard wooden attack and a long resonant decay. Tune sets the pitch, Decay the boom, Level the output. Trigger on in0.

TaikoOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Taiko: a large Japanese drum, a deep low membrane tone built from inharmonic circular-mode ratios with a slight downward pitch bend and a noisy skin-slap transient, re-struck at the set rate; thunderous and primal.

TamaseviciusOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tamasevicius oscillator: a compact four-element chaotic circuit (the TCMNL: two capacitors, an inductor, and a nonlinear conductance) designed as a simple, robust chaos generator for secure-communication experiments.

Tambourine [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal tambourine: a trigger on in0 shakes a ring of jingles - a cluster of high, bright, inharmonic metallic partials that shimmer and ring, struck with a noise transient. Tune sets the jingle pitch, Decay the shimmer length, Jingle the metallic-partial richness, Level the output. Trigger from a Clock, Euclid or sequencer for tambourine hits.

TanClipOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Clipped-tangent oscillator: each cycle is tan(x) over a clipped argument range, so the waveform sits nearly flat then rips steeply through the centre - low Sharp gives a soft ramp, high Sharp approaches a square with a vertical zero-crossing, a transition curve no sine or saw has.

TangStatzLaser [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-mode laser model: the Tang-Statz-deMars rate equations for two competing laser modes sharing a gain medium, whose intensity cross-saturation makes the modes flicker and exchange energy chaotically - antiphase laser dynamics.

TanhWindowOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tanh-window oscillator: each cycle is a flat-topped pulse built from the difference of two tanh edges, 0.5*(tanh(k(u+R))-tanh(k(u-R))) - a plateau of width Width with smoothly-saturating skirts whose steepness Steep sets, a distinct edge profile from the cosine, Planck and Fermi flat-tops.

Tannerin [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Tannerin / electro-theremin: a controllable slide oscillator - a pure sine you glide in pitch (the ‘Good Vibrations’ sound), simpler to play than a true theremin. Glide sets the portamento, Tone the harmonic richness, Level the output. in0 = Gate, in1 = Pitch CV.

TapeHissOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tape hiss: the gentle background noise floor of analog magnetic tape, a soft band-limited hiss modulated by a very slow wow-and-flutter amplitude drift, the warm bed under a cassette recording; the tone knob sets the hiss colour.

TapeStopOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tape stop: the classic effect of a tape (or turntable) being stopped, a tone whose pitch and volume sag rapidly to a halt as the reel slows, then snaps back to full speed to loop; the rate sets how often it stops.

TargetZone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Target-zone exchange rate: Krugman’s model of a currency held within an official band, where the rate’s nonlinear S-shaped response to fundamentals plus reflection at the band edges produces bounded, irregular near-the-edge dynamics.

TausswortheNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tausworthe noise: a generalized-feedback shift-register generator (shift-and-XOR taps on a 32-bit word, the basis of combined-Tausworthe RNGs) producing fast maximal-length white noise with a distinct spectral fingerprint.

TaylorCouette [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Taylor-Couette flow: a three-mode truncation of the fluid between two rotating cylinders, modelling the cascade from smooth Couette flow through Taylor vortices to wavy, chaotic vortices as the rotation speeds up.

TeardropOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Teardrop oscillator: the coordinate sin(t)*sin^Point(t/2) traces a teardrop - rounded on one side and drawn to a sharp point on the other - so raising Point sharpens the cusp and pushes energy into higher harmonics on just one half of the cycle.

TeethChatterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Teeth chatter: the rapid clacking of teeth chattering from cold or fear, a fast train of short hard clicks with slight random timing, fired at the chatter rate; a percussive bony rattle.

Teleport [OSC] [FX] [SYNTH] in: — out: Audio

Teleport - a shimmering upward sweep that dissolves into sparkle, the sci-fi beam-out, auto-repeating. Rate sets the interval, Tone the shimmer brightness, Level the output.

TeleporterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Teleporter: the dematerialisation shimmer of a transporter beam, a cluster of detuned partials sweeping rapidly upward in pitch and sparkle until they vanish, then re-trigger; the rate sets how fast the beam cycles.

Telharmonium [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Telharmonium: the 1897 additive tonewheel organ (the first electronic instrument) - a pure sum of sine harmonics with no filter, giving a clean, glassy, drawbar-organ tone. Bright mixes in the upper harmonics, Level the output. in0 = Gate, in1 = Pitch CV.

TempleBlockOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Temple block: a hollow carved-wood block struck with a mallet, a short dry pitched knock with a quick woody decay and a faint hollow second partial, re-struck at the set rate; the muted clop of orchestral temple blocks.

Tension [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Cinematic tension drone: a slowly evolving underscore bed - a low cluster of detuned voices (root, fifth, octave, twelfth) that beat and drift against each other, swept by a slow filter LFO over a bed of filtered air noise - the sustained dread under a Zimmer-style scene. Self-running. Tune sets the root pitch (in0 = Pitch CV), Motion how fast the detuning and filter drift evolve, Width the cluster spread (dissonance), Noise the air-bed level, Tone the filter base frequency, and Level the output. Automate Tune for rising or falling tension.

TensionPulseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tension pulse: the slow throbbing pulse that builds suspense, a low tone swelling and contracting in volume with a faint dissonant overtone, the held-breath build before something happens; the pulse rate sets the heartbeat.

TentLogistic [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tent-logistic hybrid: crossfades between the angular tent map and the smooth logistic map each iteration, so one knob morphs the chaotic texture from sharp and piecewise to round and parabolic.

TentOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tent-map chaos source: the piecewise-linear tent map (x = mu * min(x, 1-x)) gives a sharp-edged chaotic stream whose character differs from the smooth logistic map - angular digital noise textures.

TentSkew [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Skew tent map: an asymmetric tent whose peak sits off-centre, so the two linear branches have different slopes; it is exactly chaotic with a tunable invariant density - a fast, sharp, biasable noise source.

TeslaCoilOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tesla coil: a musical solid-state Tesla coil, where the spark fires in tight bursts at the note pitch so the plasma arc itself becomes a buzzy, square-ish singing voice; the duty knob sets how fat and bright the discharge.

TheraminOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

A smooth, expressive lead sine with a gentle vibrato and portamento glide between pitch changes - the theremin/whistle voice. Vibrato sets the wobble depth and Glide the portamento time.

Theremin [OSC] [FX] [SYNTH] in: Pitch CV, Volume out: Audio

Theremin: a continuously-pitched sine voice with a warm vibrato, its pitch played by in0 (1V/oct) and its loudness by in1 - the hands-in-the-air theremin glide. Tune sets the base pitch, Vibrato the wobble depth, Level the output. in0 = Pitch CV, in1 = Volume CV.

ThereminOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Theremin oscillator: the pure heterodyne sine of the gesture-controlled instrument, gliding smoothly between pitches (portamento, since there are no discrete notes) with a gentle hand vibrato - the eerie, voice-like wail of the theremin.

Thermosyphon [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Thermosyphon loop: fluid in a vertical ring heated below and cooled above circulates by buoyancy, but the flow can stall and chaotically reverse direction (Welander's model) - a physical realization of Lorenz-type convection chaos.

Theta [OSC] [FX] [SYNTH] in: – out: Audio

Jacobi-theta oscillator (a first as a synth block): synthesises the theta function - a periodic version of the Gaussian bell, built as a cosine series whose harmonic k is weighted by q to the power k-squared. As Width (the q value) rises toward 1 those weights stay strong for many harmonics, so the bell sharpens into a narrow spike-train; as it falls the upper harmonics vanish and it rounds back toward a pure sine. A continuous sine-to-pulse morph rooted in one of the deepest functions in mathematics, distinct from a PWM or a filtered pulse. Needs no input.

ThetaNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Theta neuron: the canonical Type-I excitable model where the membrane state is an angle on a circle (theta'=(1-cos theta)+(1+cos theta) I); above threshold it rotates and fires a spike each lap - clean, frequency-controllable neural pulses.

ThetaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Jacobi theta-3: the function theta3(z,q) = 1 + 2 sum $q^{(n}$2) cos(2 n z), whose Gaussian-decaying harmonics make a smooth bump that sharpens into a periodic spike as the nome rises - the building block of elliptic functions, here a peaked even waveform.

ThomasConservative [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Conservative Thomas flow: the Thomas cyclically-symmetric system with its damping set to zero, so it preserves phase-space volume and the orbit performs a labyrinthine, non-attracting random-walk through a sine-driven lattice.

ThomasOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Thomas attractor: a cyclically-symmetric chaotic flow (each axis is sin of the next minus damping), giving a smooth, swirling, labyrinthine signal that drifts between order and chaos as the damping drops.

ThreeScrollOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Three-scroll attractor: a chaotic flow whose orbit threads three separate scrolls, switching between three basins - chaos with a richer multi-region structure than the usual double-scroll.

ThreeWing [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Three-wing attractor: a chaotic flow whose orbit visits three lobes ('wings'), an even-richer multi-region switching than the four-wing system, for unpredictable jumpy modulation.

Throat [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Throat singing (khoomei): a low buzzing drone with a single sharp overtone whistling above it, swept by Overtone - the Tuvan harmonic-singing voice. Overtone selects the emphasized harmonic, Drone the fundamental weight, Level the output. in0 = Gate, in1 = Pitch CV.

Thruster [OSC] [FX] [SYNTH] in: – out: Audio

Rocket thruster - a roaring filtered-noise blast with a low combustion body and pressure flutter. Thrust sets the power, Tone the brightness, Level the output.

ThruZeroFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Through-zero FM: the modulator drives the carrier frequency below zero and back, so the carrier momentarily runs in reverse - the clangorous, glitch-free deep-FM timbre analog VCOs cannot do but digital can.

ThueOsc [OSC] [FX] [SYNTH] in: – out: Audio

Thue-Morse oscillator (a first as a synth block): an additive tone that keeps harmonic k only when k is a Thue-Morse 'odious' number - one whose binary digits sum to an ODD count (1, 2, 4, 7, 8, 11…). Because the keep/drop pattern is the self-similar Thue-Morse sequence, the spectrum is fractal: the same fair-share thinning repeats at every octave, producing a hollow, evolving timbre with a built-in self-similar comb. Partials sets the highest harmonic considered, alias-free up to Nyquist. A self-similar spectrum distinct from the prime and Fibonacci oscillators.

Thunder [OSC] [FX] [SYNTH] in: Trigger out: Audio

Thunderclap: a trigger on in0 fires a sharp crack (a band of noise) followed by a deep brown-noise sub rumble that swells then rolls away over several seconds - the storm thunder of film sound design. Crack sets the initial snap level, Rumble the sub-rumble depth, Roll the length of the rolling tail, Tone the low-pass darkness of the rumble, and Level the output. Trigger over a Rain bed for a storm; short Roll reads as a near strike, long Roll as distant.

ThunderHit [OSC] [FX] [SYNTH] in: Trigger out: Audio

Thunder: a deep, rolling thunderclap - a low rumble of filtered noise and a sub sweep that booms then tumbles away. Tune sets the depth, Decay the roll, Level the output. Trigger on in0.

ThunderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Thunder oscillator: a deep, slowly-modulated brown-noise rumble punctuated by occasional sharp cracks, the long rolling boom of distant-to-near thunder; the rumble knob sets how low and heavy the body sits.

TIAOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

TIA oscillator: the Atari 2600's TIA produced pitch only by dividing a fixed clock by small integers, so its notes land on a coarse, out-of-tune scale; this models that integer-divider quantization for the console's famously rough, detuned bleeps.

TidalHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tidal oscillator: sums the principal tide-generating constituents at their real period ratios - the semidiurnal lunar M2 and solar S2 plus the diurnal K1 and O1 - so they beat into the fortnightly spring-neap cycle that tide tables predict.

TiganOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tigan (T) system: a Lorenz-like chaotic flow whose middle equation carries the cross-term, tracing a flatter, more sheet-like twin-scroll attractor than Lorenz.

Timbale [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal timbale: a trigger on in0 strikes a shallow metal-shell drum - a tuned head plus the bright metallic inharmonic ring of the steel shell (modes near 1 : 2.7 : 4.2), the cutting crack of salsa and Latin percussion. Tune sets the pitch, Decay the ring, Ring the metallic shell brightness, Level the output. Trigger from a Clock, Euclid or sequencer.

Timpani [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Tuned kettledrum (timpani): the prominent near-harmonic membrane modes (1 : 1.5 : 1.98 : 2.44 : 2.9) that give a kettledrum its definite pitch, with a soft attack thump and a downward pitch drop as the head settles. Tune sets the pitch, Decay the boom length, Level the output. Trigger on in0, in1 = Pitch CV.

TinkerbellMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tinkerbell map: a quadratic 2-D map tracing a delicate wing-shaped chaotic orbit, iterated at a set rate for a distinctive curling, fractal-edged chaos texture.

TinkerbirdMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tinkerbird map: a Tinkerbell-family quadratic 2-D map at a parameter set that opens its delicate wing into a fuller, bird-like fractal orbit - a curling, fine-edged chaotic texture distinct from the classic Tinkerbell.

TippeTop [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tippe top: a spun top that paradoxically flips to spin on its stem, its tilt angle climbing through a gyroscopic instability driven by sliding friction - a real toy whose inversion is modelled as a slow oscillatory rise-and-settle.

TireScreechOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tire screech: the squeal of skidding tires, a harsh resonant noise band that wavers in pitch as the rubber judders and grips, the cartoonish screech of a hard stop; the pitch knob sets the squeal frequency.

ToasterPopOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Toaster pop: the satisfying spring-loaded thunk of a toaster ejecting the toast, a low resonant clunk with a metallic spring ring and a faint rattle, fired at the set rate; breakfast is ready.

TodaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Toda oscillator: a resonator with the exponential restoring force of the Toda lattice (a famous integrable soliton system) which, when damped and sinusoidally driven, becomes chaotic - an asymmetric, spiky nonlinear-spring tone.

ToggleSwitchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Toggle switch: the satisfying two-blip flip of a UI toggle, a low blip then a higher blip (the switch snapping to on), fired at the set rate; the settings-toggle feedback.

Tom [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Analog tom drum: a trigger on in0 fires a sine whose pitch bends down from Tune with a gentler sweep than the Kick (no sub click), plus a short noise transient for stick attack. Tune sets the fundamental, Bend the pitch-drop amount, Decay the body length, Tone the attack-noise blend, and Level the output - the synthesized 808/909-style tom (low/mid/high by Tune). Patch a Clock, Euclid or sequencer gate into in0.

Tom808 [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

TR-808-style tom: a trigger on in0 fires a tuned sine with a downward pitch sweep and a medium decay - the round, pitched 808 tom/conga. Tune sets the pitch, Decay the length, Sweep the pitch-drop amount, Level the output. in1 = Pitch CV.

Tom909 [OSC] [FX] [SYNTH] in: Trigger out: Audio

TR-909 tom: a punchy pitch-swept tom with a touch of noise grit, brighter and tighter than the 808 tom, the house/techno fill voice. Tune sets the pitch, Decay the length, Level the output. Trigger on in0.

TomDrum [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Synth tom: a sine whose pitch drops from the strike down to the tuned note while the amplitude decays, with a touch of noise for the head attack - the rounded pitched-drum voice of a drum machine.

TomSynthOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tom-drum voice: a mid-range sine with a gentle downward pitch bend and a medium decay, the round pitched thump of a tom, re-triggered at the set rate; tune it from a high rack tom to a low floor tom.

ToppLeoneOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Topp-Leone oscillator: each cycle is the Topp-Leone PDF $2Nu(1-x)*x^{(Nu-1)*(2-x)}(Nu-1)$ on [0,1], a bounded-support shape that morphs from J-shaped (Nu<1) to a centred bump (Nu>1) - a compact waveform whose skew and peakedness Nu sets. Peak-normalized, bounded.

Tornado [OSC] [FX] [SYNTH] in: – out: Audio

Tornado - a vast roaring vortex: deep filtered-noise roar with a low rotating tone and debris. Fury sets the roar intensity, Tone the upper shriek, Level the output.

TorusOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Torus oscillator: two limit-cycle oscillators at incommensurate frequencies coupled together trace a quasiperiodic torus that, as coupling rises, wrinkles and breaks down into chaos - the Ruelle-Takens route, as an evolving two-tone source.

TotalisticCA [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Totalistic CA: a ring of three-state cells where each cell's next value depends only on the sum of itself and its neighbours through a totalistic rule code; richer than binary elementary CA, it grows nested, multi-level fractal patterns.

TractorBeamOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tractor beam: a steady humming energy beam pulsing with a slow throb, a pure low tone ring-modulated by a mid carrier and amplitude-pulsed, the sound of being slowly reeled into a mothership; the throb rate sets the beam's pulse.

TractrixOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tractrix oscillator: each cycle traces the tractrix horizontal coordinate x = t - tanh(t), which is nearly flat (cubic) through the centre then straightens to a ramp - a soft S with a long lazy middle, distinct from tanh or a linear saw.

Traffic [OSC] [FX] [SYNTH] in: — out: Audio

City traffic - a low layered road rumble with occasional passing-car doppler swooshes and distant horns. Density sets the activity, Tone the brightness, Level the output.

TrafficFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Traffic stop-and-go: three cars following each other by the optimal-velocity rule, where drivers accelerate toward a headway-dependent target speed; above a density threshold the line jams and surges in self-organized chaotic waves.

TrainletOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trainlet: a granular stream whose grains are short band-limited impulse trains - a stack of equal-amplitude harmonics (Roads's trainlet) - giving a bright, buzzy, harmonically-dense click-tone; the harmonic count sets the brightness.

TrainWhistle [OSC] [FX] [SYNTH] in: — out: Audio

Train horn - a rich chord of detuned tones blaring with a slow doppler bend, the classic locomotive whistle. Pitch sets the key, Tone the brightness, Level the output.

TransitDipOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Transit oscillator: the light curve of a star with a planet crossing it - mostly flat at full brightness, then a brief U-shaped dip with ingress and egress ramps once per orbit - the signal that the Kepler mission used to discover thousands of exoplanets.

∿ TransmutedExpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Transmuted-exponential oscillator: each cycle is lambdae^(-lambda x)(1-Alpha+2Alpha*e^(-lambda x)), a quadratic-rank-transmuted exponential where Alpha bends the pure decay - positive Alpha front-loads it, negative Alpha grows a small hump. Peak-normalized, bounded.

∿ TrapezoidalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trapezoidal oscillator: each cycle is the trapezoidal distribution - a flat-topped pulse with straight linear ramps up and down whose slope Ramp controls - sweeping from a near-rectangle (short ramp, buzzy) to a near-triangle (long ramp), a piecewise-linear flat-top distinct from the cosine and Planck tapers.

∿ Trautonium [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Trautonium: Oskar Sala's gliding subharmonic synthesizer - a sawtooth plus its strong subharmonics (f/2, f/3) for a rich, formant-heavy, organ-meets-string voice. Sub sets the subharmonic mix, Tone the brightness, Level the output. in0 = Gate, in1 = Pitch CV.

∿ TriadHarmonicOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Triad-harmonic oscillator: boosts the harmonics that spell a just major triad (the 4th, 5th and 6th partials sound a root-third-fifth chord), so a single note rings with a built-in consonant chord - the acoustic basis of the major triad fused into one tone.

∿ Triangle [OSC] [FX] [SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal triangle: a trigger on in0 strikes a metal triangle - a cluster of high inharmonic metallic partials with a long, bright, shimmering ring that sits on top of a mix. Tune sets the pitch, Decay the long sustain, Shimmer the upper-partial brightness, Level the output. A clean, sustained metallic ping, unlike the short noisy shakers. Trigger from a Clock, Euclid or sequencer.

∿ TriangleDrum [OSC] [FX] [SYNTH] in: Trigger out: Audio

Struck triangle / metallic ping: four inharmonic high partials rung together with a long shimmering decay - the bright, bell-like triangle and finger-cymbal sparkle for percussion accents. Tune sets the pitch, Decay the ring time, Level the output. Trigger on in0.

∿ TriangleModeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Triangle (idiophone) oscillator: the very inharmonic, high-and-dense partials of a struck metal triangle (ratios near 1, 2.71, 5.15, 8.43, 12.4), whose pitch is famously ambiguous - a bright, glittering metallic ping with no clear fundamental.

∿ TriangularNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Triangular noise: the sum of two independent uniform samples, giving a triangular probability density (peaked at the middle, zero at the edges) - the classic TPDF dither noise that decorrelates quantization error, softer than white.

∿ TriangularOsc [OSC] [FX] [SYNTH] in: — out: Audio

Triangular-number oscillator (a first as a synth block): an additive tone whose overtones land on the triangular numbers - the 1st, 3rd, 6th, 10th, 15th, 21st… harmonics (k(k+1)/2). The gaps between partials grow by one each step, so the spectrum starts dense and thins out smoothly as it climbs, giving a clear-but-airy body that sits between a full harmonic saw and a sparse sine stack. Partials sets how many triangular overtones to sound, alias-free up to Nyquist. A figurate-number spectrum distinct from the prime and Fibonacci oscillators. Needs no input.

TriangularQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Triangular-quantile oscillator: each cycle is the symmetric triangular inverse-CDF (sqrt(p/2) below the midpoint, 1-sqrt((1-p)/2) above), a square-root S that is steep at both ends and flat through the middle - the inverse of a triangle distribution, distinct from the logistic/erf sigmoids. Bounded.

TricornFractal [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tricorn (Mandelbar): the Mandelbrot map with the iterate conjugated before squaring (z -> conj(z)^2 + c), whose anti-holomorphic twist gives a three-cornered fractal with characteristic cusps; the reseeded-on-escape real part is the output.

TricubeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tricube oscillator: each cycle is the tricube kernel $(1-|x|^{3})3$ (used in LOESS smoothing), a compact bump with a notably flat top and a cubic-cubed edge roll-off - a distinctly plateau-like pulse different from the $(1-x^{2})n$ kernels.

TrigammaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trigamma oscillator: each cycle is the trigamma psi'(x) ~ 1/x + 1/2x^2 + 1/6x^3 (the derivative of the digamma), a convex decay that falls faster than a plain 1/x - a steep-then-gentle falling ramp distinct from exponential and logarithmic decays. Bounded.

TrillVoiceOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trill oscillator: a voiced tone rapidly gated open and shut at the flutter rate of a rolled or uvular trill (the tongue or uvula vibrating against the airflow), giving the rattling r of Spanish 'rr' or a French uvular r.

TriplingMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tripling map: the expanding circle map x -> 3x mod 1 (the base-3 shift, sibling of the doubling map), whose orbit is the ternary digits of the seed marching left - a sharp, deterministic ternary noise source.

TrisectrixOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Trisectrix oscillator: the rational coordinate t(t^2 - 3)/(t^2 + 1) from the trisectrix of Maclaurin, a pole-free curve that dips, crosses and rises with a distinctive inflected shape - a smooth asymmetric waveform with a characteristic central wiggle.

TritaveOsc [OSC] [FX] [SYNTH] in: – out: Audio

Bohlen-Pierce (tritave) oscillator (a first as a synth block): a tone built from ODD harmonics only (1, 3, 5, 7, 9, 11…), with no even partials at all - the spectrum that fits the Bohlen-Pierce scale, which divides the TRITAVE (a 3:1 ratio, an octave plus a fifth) into 13 steps instead of dividing the octave. With only odd harmonics it has a hollow, clarinet-like purity, and it consonates with the strange non-octave intervals of BP music. Partials sets how many odd harmonics to stack, alias-free up to Nyquist. A non-octave timbre distinct from the usual harmonic oscillators. Needs no input.

TriweightOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Triweight oscillator: each cycle is the triweight kernel $(1-x^2)3$, an even narrower, flatter-topped compact bump than the biweight - approaching a Gaussian shape but with true compact support (exactly zero at the edges) and an even softer harmonic spectrum.

Trombone [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Trombone: the brassy, slightly-buzzy mid-brass - a sawtooth pushed through a brass formant with a slow swell and breath, played by holding in0. Brass sets the formant bite, Breath the air, Level the output. in0 = Breath gate, in1 = Pitch CV.

Trumpet [OSC] [FX] [SYNTH] in: Gate, Pitch (CV) out: Audio

Trumpet voice: a gate on in0 sounds a bright brassy tone - harmonically rich with a brass-like formant peak that opens up as the note swells in. Freq sets the note, Bright the brassiness (upper-harmonic weight), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

TruncNormalNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Truncated-normal noise: Gaussian samples confined to a chosen number of standard deviations by re-drawing any that fall outside (rejection), giving a bell shape with hard edges and no outliers - useful where unbounded Gaussian tails would clip harshly.

TschirnhausenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tschirnhausen oscillator: each cycle traces the Tschirnhausen cubic 3t - t^3 over a span, an odd S-curve that steepens at the centre and rolls over at the extremes - a cubic waveshape between a sine and a triangle, with a gentle compression at the peaks.

Tuba [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Tuba: the deep, round, powerful low brass - a sawtooth through a low formant and lowpass with a slow swell, played by holding in0. Brass sets the bite, Breath the air, Level the output. in0 = Breath gate, in1 = Pitch CV.

TubeBell [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tubular-bell resonator: three long-decaying modes at bell-like inharmonic ratios ring when struck by the input, giving the shimmering, beating metallic tone of a chime or tubular bell.

TubeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Waveguide wind/tube oscillator: a delay-line resonator excited by breath noise models a flute or pipe. Breath sets the noise excitation and Tone the resonance damping - airy, organic pitched wind tones.

TubularBell [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Struck tubular bell / chime: five inharmonic partials (1 : 2.76 : 5.4 : 8.93 : 13.3) with a long shimmering decay and a metallic strike, the orchestral tubular-bell and clock-chime voice. Tune sets the pitch, Decay the ring time, Level the output. Trigger on in0, in1 = Pitch CV.

TubularBellOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tubular-bell oscillator: the chime partials 2, 3, 4.16, 5.43, 6.79, 8.21 whose 4:5:6-ish upper trio fuses into a clear ‘strike note’ an octave below the lowest partial (a partial that is not even present) - the orchestral-chime psychoacoustic trick.

TukeyLambdaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tukey-lambda oscillator: each cycle is the Tukey-lambda quantile $(p^{L-(1-p)}L)/L$ swept over the cycle, so Lambda morphs the waveform continuously from a near-linear ramp (L=1) through an S-curve (L=2) to a steep, edge-loaded shape - one knob spanning a whole family of symmetric waves.

TumorImmune [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tumor-immune dynamics: immune effector cells hunt a growing tumor while the tumor suppresses them, a predator-prey-with-recruitment system whose competition gives dormancy, oscillatory relapse and chaotic regimes - oncology modelling as a slow source.

TunnelDiode [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tunnel-diode oscillator: an LC loop loaded by a tunnel diode whose N-shaped (cubic) current-voltage curve has a negative-resistance region, producing relaxation oscillations and, when driven, chaos - a real microwave-era circuit.

TunnelDiodeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tunnel-diode oscillator: a negative-resistance device whose N-shaped current-voltage curve makes the circuit jump astably between two stable branches, snapping quickly across the unstable region - a fast relaxation waveform with sharp switching edges and curved dwells.

TurbineBladeMistuned [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Mistuned bladed disk: two coupled turbine blades with slightly different natural frequencies (mistuning) whose energy localizes and beats unpredictably under aerodynamic forcing - the vibration-localization that fatigues jet-engine blades.

TurkeyGobbleOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Turkey gobble: the rapid burbling gobble of a tom turkey, a voiced tone chopped by a very fast irregular warble into the unmistakable gobble-obble-obble, fired at the set rate; the strut of the barnyard.

TweekAtmosphericOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Tweek: a distant-lightning sferic that has bounced in the Earth-ionosphere waveguide, dispersing near the VLF cutoff so it ends in a brief ringing ‘tweek’ tone instead of a flat click; the staccato pinging of nighttime VLF radio, fired at the set rate.

TwinTOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Twin-T oscillator: the Twin-T notch-network sine oscillator with a saturating amplifier whose extra phase shift, past the limit cycle, breaks into chaos - a classic audio-oscillator topology pushed over the edge.

Typewriter [OSC] [FX] [SYNTH] in: – out: Audio

Typewriter - mechanical key clacks at a typing rhythm with the occasional carriage ding. Rate sets the typing speed, Tone the clack brightness, Level the output.

TypewriterOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Typewriter: the clatter of a manual typewriter, randomly-timed sharp key-strike clacks at the typing rate with the occasional high margin-bell ding, the mechanical rhythm of typed prose.

Udu [OSC] [FX] [SYNTH] in: Trigger out: Audio

Udu clay pot drum: a deep, hollow, tuned 'boop' from striking the pot's mouth, plus a finger-slap transient - a pitch-dropping sine with a noise click. Tune sets the pitch, Decay the boom, Level the output. Trigger on in0.

UedaOscillator [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Ueda oscillator: the forced cubic equation x'' + k x' + x^3 = B cos(t) with no linear restoring term, whose strange attractor ('Ueda's Japanese attractor') was one of the first chaotic attractors ever computed.

UFOHumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

UFO hum: the hovering drone of a flying saucer, a mid tone deeply vibrato'd and tremolo'd by a slow wobble plus a faster shimmer, the classic theremin-flavoured sci-fi spacecraft hum; the wobble knob sets how unstable the craft.

UnifiedChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Unified chaotic system: Lu's family that continuously interpolates between the Lorenz (alpha=0) and Chen (alpha=1) attractors as one parameter sweeps, so a single knob morphs between two classic chaotic regimes.

VacuumCleanerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vacuum cleaner: the drone of an upright vacuum, a high motor/impeller whine with harmonics riding on a broadband airflow rush, the steady household roar; the motor knob sets the suction speed and pitch.

VaidyanathanFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vaidyanathan flow: a representative three-dimensional quadratic chaotic system of the type catalogued for chaos-control and secure-communication research, with a single quadratic cross-term per equation tracing a compact twin-scroll.

VallisOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vallis ENSO model: a three-variable model of wind-driven ocean temperature that produces the irregular, multi-year El-Nino oscillation chaotically - slow, organic climate-cycle modulation.

ValueNoiseOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Value-noise oscillator: picks a new random target at the set Rate and smoothstep-interpolates toward it, producing a continuous, gently-undulating random signal - the smooth interpolated 'value noise' of procedural graphics, distinct from stepped sample-and-hold and from gradient/Perlin noise.

VAnalog [OSC] [CIRCUIT] [FX] [SYNTH] in: Pitch (CV) out: Audio

Virtual-analog oscillator: saw or pulse (Wave) with adjustable pulse width (PWM), a square sub-oscillator an octave down (Sub), and slow random Drift for analog pitch instability. Level scales the output. A warm, slightly unstable VA voice - pair with the Mini/Jup-8 analog filters for a classic synth chain.

VanDeGraaffOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Van de Graaff: a static-charge generator, a faint rising whine of the charging belt building tension until it suddenly snaps over in a sharp crackling spark to ground, then recharges; the rate sets how often it discharges.

VanDerPolHeart [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Heart-rhythm model: two coupled van-der-Pol oscillators standing in for the sinoatrial pacemaker and the ventricle; weak coupling gives a healthy beat, but detuning and forcing produce arrhythmia-like chaotic rhythms.

VarSlopeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Variable-slope oscillator: a single waveform whose rising-to-falling time ratio is continuously adjustable, morphing from an up-ramp through a symmetric triangle to a down-ramp (and every asymmetric shape between) - one knob across the linear-waveform family.

VasicekProcess [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vasicek process: dr = a(b-r)dt + sigma*dW, the Gaussian mean-reverting model of interest rates (an Ornstein-Uhlenbeck process with an explicit long-run mean); unlike CIR it can go negative, giving a symmetric colored wander around a set level.

VectorOsc [OSC] [FX] [SYNTH] in: Pitch CV out: Audio

Vector-synthesis oscillator: blends four waveforms - sine, saw, square and triangle - placed at the corners of an XY pad, with X and Y crossfading smoothly between them (the Prophet-VS / Korg Wavestation vector joystick). Sweep X/Y for evolving timbres from one oscillator. in0 = Pitch CV.

VectorSynth [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vector synthesis (Prophet-VS / SY22 style): four source waveforms sit at the corners of a square and a slowly-orbiting vector bilinearly blends between them, sweeping the timbre around the plane - a two-axis crossfade of sine, saw, square and triangle.

VelvetGen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Velvet noise generator: instead of a value every sample, it emits sparse, randomly-placed +/-1 impulses, the perceptually-smooth noise used in efficient reverbs and decorrelation - clickier and airier than white noise.

VelvetNoise [OSC][FX][SYNTH] in: – out: Audio

Velvet noise - sparse random impulses instead of dense hiss, giving a smooth, airy decorrelated texture used in reverb and diffusion. Density sets the impulse rate, Level the output.

Vibraphone [OSC][FX][SYNTH] in: Trigger, Pitch (CV) out: Audio

Modal vibraphone: a trigger on in0 strikes an aluminium bar (inharmonic partials 1 : 4 : 10, a metal cousin of the marimba) with a long shimmering sustain, plus the characteristic motor tremolo - an amplitude vibrato. Tune sets the pitch, Decay the (long) ring, Vibrato the tremolo depth, and Level the output.

VibraphoneOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Vibraphone: a struck aluminium bar with near-octave partials and a long shimmering decay, amplitude-modulated by the spinning resonator-disc tremolo that gives the vibraphone its signature wobble; re-struck at the set rate, the tremolo knob sets the vibe motor speed.

Vibraslap [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Modal vibraslap: a trigger on in0 cracks the ball against the wooden box, then the loose pins rattle and buzz as they ring out - the sharp 'crack-brrrrr' heard at the top of so many percussion breaks. Tone sets the rattle brightness, Decay how long the pins buzz, Level the output. Needs only a trigger.

VicsekModel [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Vicsek model: a swarm of self-propelled particles each steer toward the average heading of their neighbours plus angular noise; below a noise threshold they spontaneously align into collective motion (a flock), above it they scatter - the order parameter tracks that flocking transition.

VinylCrackle [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Vinyl crackle: sparse random pops with decaying tails layered over a faint filtered hiss, the lo-fi surface noise of an old record - instant nostalgia/texture under a mix.

VinylCrackleOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Vinyl crackle: the nostalgic surface noise of a vinyl record, a soft bed of filtered hiss sprinkled with random sharp dust pops and crackles, the lo-fi warmth of analog playback; density sets how worn the record.

VioletNoise [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Violet noise: white noise twice-differenced, rising +6 dB/octave so it is almost entirely high-frequency hiss - the spectrum of differentiated thermal noise, useful for sizzle and as a dither shape.

ViolinBowOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Violin: a bowed-string waveguide where a stick-slip bow-friction curve continuously pumps the delay line into a sustained sawtooth-like Helmholtz motion, the singing tone of a bowed violin; bow force and speed shape it from airy to digging.

∿ **VocalFilter** [OSC] [FX] [SYNTH] in: Audio out: Audio

Vowel / formant filter: three resonant formant peaks tuned to the vowels A-E-I-O-U, with Vowel morphing continuously between them and Reso setting the formant sharpness - turns any bright source into a talking, vocal timbre. A morphable vowel bank, distinct from the single Formant block.

∿ **VocalFoldChaos** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic vocal folds: a two-mass model of the vibrating vocal folds whose collision nonlinearity, under asymmetric tension, breaks into the subharmonic and chaotic regimes heard as vocal fry, diplophonia and creak.

∿ **VocalFry** [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Vocal fry: the low, creaky glottal rattle at the bottom of the voice - sparse, jittery pulses through a vocal formant. Rate sets the creak speed, Vowel the formant, Level the output. in0 = Gate, in1 = Pitch CV.

∿ **VonMisesNoise** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Von Mises noise: a concentration-controlled random angle on the circle (the circular analogue of the Gaussian), read out as its cosine; high concentration clusters near one phase, low concentration spreads uniformly - a wrapped, circular random source.

∿ **VortexShedding** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vortex shedding: a van-der-Pol wake oscillator coupled to an elastic cylinder (the Hartlen-Currie model of vortex-induced vibration), which locks in, beats and chaotically modulates - the aeolian-tone source behind singing wires and bridge sway.

∿ **VosimSynth** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

VOSIM (Kaegi-Tempelaars): each fundamental period holds a short burst of N decaying squared-sine pulses followed by silence; the pulse rate sets a formant and the count/decay shape its bandwidth - an efficient 1970s vocal-formant synthesis.

∿ **VosimVoice** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

VOSIM oscillator: each fundamental period fires a short burst of amplitude-decaying squared-sine pulses (the Kaegi-Tempelaars voice-simulation method), so the pulse-rate sets a formant and the period sets the pitch - vowel-like tones from a pulse train.

∿ **VoterModel** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Voter model: on a ring of two-opinion agents, each step a random agent adopts the opinion of a random neighbour; opinions drift by imitation until the whole population reaches consensus, then a perturbation reseeds it - the magnetization makes a slow random wander.

∿ **VowelFormant** [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Formant oscillator that imposes vowel resonances on a buzzy source: Vowel morphs through A-E-I-O-U formant pairs and Q sharpens the peaks, turning a saw into a singing, talking timbre.

VoxPad [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Synth-vox pad: a robotic, two-formant vocal pad on detuned pulses, morphing through vowels - the talking, choir-machine sound of vintage string synths. Vowel morphs the vowel, Width the detune, Level the output. in0 = Gate, in1 = Pitch CV.

VPS [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vector phase-shaping oscillator (Kleimola/Lazzarini/Timoney/Valimaki, DAFx 2011): the linear phase ramp is bent through a two-segment function with one movable inflection point (D, V) before a cosine - V above 1 packs extra cycles into the period for sweepable formant peaks (FM- and hard-sync-like spectra) while D shifts their position. Freq sets the pitch (in0 = Pitch CV), D the inflection time, V the cycles/formant, Level the output.

VuvuzelaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Vuvuzela: the relentless monotone drone of the South-African stadium horn, a single buzzy lip-reed tone around B-flat thick with harmonics and a faint breathy waver, the swarm-of-bees sound of the 2010 World Cup; the breath knob adds airiness.

WaldOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wald oscillator: each cycle is the inverse-Gaussian (Wald) PDF, a sharply-rising, long-right-tailed bump (a first-passage-time distribution) - Mu shifts the peak and Lambda its sharpness, peak-normalized to a bounded bipolar waveform with a distinctive asymmetric skew.

Walsh [OSC] [FX] [SYNTH] in: – out: Audio

Walsh-function oscillator (a first as a synth block): builds its waveform from the Walsh functions - the SQUARE-wave orthogonal basis that does for digital signals what sines do for analogue, used in the Hadamard transform. Each one is a product of doubling square waves, so the output is a hard +/-1 pattern with exactly Sequency sign-changes per cycle (sequency being the square-wave equivalent of frequency). The result is a buzzy, bit-exact, distinctly digital timbre - a complete non-sinusoidal basis rather than a filtered square. Sequency picks which Walsh function to sound. Needs no input.

WalshOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Walsh-function oscillator: products of square waves at octave-related rates (the Rademacher functions that build the Walsh-Hadamard basis) lock to the input pitch, giving hollow, purely-square spectra a sine basis can't.

WangBuzsaki [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wang-Buzsaki interneuron: a reduced conductance-based model of a fast-spiking cortical interneuron (the cells that pace gamma-band brain rhythms), firing narrow high-frequency spikes under injected current.

WangChenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wang-Chen system: a chaotic flow that coexists with a single stable equilibrium, so its chaotic attractor is ‘hidden’ (its basin does not touch the equilibrium) - a notable counterexample in hidden-attractor theory.

WangSunOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wang-Sun attractor: a chaotic flow with mixed linear and quadratic terms tracing a folded-ribbon orbit, another distinctive strange attractor for evolving drones and modulation.

Warp [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Phase-warping oscillator: a master phase feeds a base waveform but is reshaped by one of several Mode warps (Neutral, Wrap, Grain, Hardsync, Hard Up-Down, Bend, Mirror, Random, Jitter) before the wave is read. Wave picks the base shape (sine, saw, square, triangle), Mode selects the warp, Amount sets warp depth, and Ratio scales the slave frequency for the sync modes. Pushing Amount drives anything from subtle phase bend to per-sample jitter noise depending on Mode.

WarpDriveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Warp drive: a starship jumping to lightspeed, a deep rumble that accelerates in pitch and intensity, building to a bright whooshing peak before snapping back to start the cycle again; the rate sets the warp-charge time.

WarpOsc [OSC] [FX] [SYNTH] in: — out: Audio

Spectral-warp oscillator (a first as a synth block): a single knob that bends the whole harmonic series. Each partial sits at k raised to the power Warp, so at Warp = 1 it is an ordinary harmonic saw; below 1 the partials crowd together (a compressed, dense, organ-like spectrum) and above 1 they spread apart into an ever-more-inharmonic, bell-to-clangorous timbre. Sweeping Warp morphs continuously from harmonic to inharmonic - a one-control spectral stretch unlike any fixed waveform. Partials sets how many to stack, alias-free up to Nyquist. Needs no input.

WaspOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wasp: the angrier, higher, harsher buzz of a wasp or hornet, a bright harmonic wingbeat tone with a sharper edge and aggressive amplitude surges as it lunges, more menacing than the rounder bee; the wingbeat sets the pitch.

Water [OSC] [FX] [SYNTH] in: — out: Audio

Water / bubbles: randomly-triggered bubble ‘bloops’ - sine pings whose pitch rises sharply as they pop (the physical bubble model Asin(2pift)*e^-bt with rising f) - over an optional flowing-noise wash, the stream, drip and underwater sound of film sound design. Distinct from Rain’s falling droplet plinks: bubbles sweep up. Self-running. Flow sets the stream-noise level, Bubbles the pop density, Pitch the bubble base frequency, Spread the random pitch range, and Level the output.

WaterDrumOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Water drum: a struck clay udu/water-pot whose enclosed air cavity gives a deep boomy tone that bends quickly downward in pitch as the hand seals and opens the hole (the wet ‘bloop’ of an udu), re-struck at the set rate.

WaterHammer [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Water hammer: the pressure surge when flow in a pipe is throttled, reflecting back and forth with valve-and-column nonlinearity and cavitation cut-off; repeated throttling gives the banging, relaxation-oscillating surge of plumbing.

Waterphone [OSC] [FX] [SYNTH] in: Bow out: Audio

Waterphone: the eerie, bowed-metal horror-film instrument - several inharmonic metal-rod resonances with a slow random glissando, played by holding in0 as bow pressure. Density sets the resonance spread, Glide the pitch wander, Level the output. in0 = Bow gate.

WaterphoneRes [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Waterphone resonator: three densely-packed, slowly-decaying inharmonic modes ring and beat against each other when excited, the eerie, sliding, horror-soundtrack timbre of a bowed waterphone or singing rods.

WaterSprayOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Water spray: the steady high hiss of a hose nozzle or spray bottle, a band-passed jet of fine-mist noise with a faint flutter as droplets break up, the airy sound of misting; tone sets the nozzle fineness.

WaterTank [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-tank hydraulic oscillator: tanks filling and siphon-dumping into each other, whose threshold-triggered overflows interact into irregular, chaotic level swings - a relaxation oscillator you could build out of buckets.

WaveCrashOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wave-crash oscillator: a slow noise swell that builds, breaks into a bright broadband crash, then hisses out as foam and recedes, repeating like surf on a shore; the period knob sets the wave interval.

WaveFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Two-operator FM voice: a modulator (with its own feedback path) phase-modulates a carrier, the core DX-style FM cell - Ratio sets the harmonics, Index the brightness, Feedback adds sawtooth-like edge.

WavefolderOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wavefolder oscillator: a sine driven hard into a triangle-folding nonlinearity that reflects peaks back on themselves, the Buchla/West-Coast technique that adds shimmering high harmonics with rising drive instead of clipping them - a fold-rich timbre source.

WaveguideMesh [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Waveguide mesh: a 2-D membrane modelled as four delay-line waveguides meeting at scattering junctions, so a strike spreads, reflects off the rim and rings with the inharmonic modal spectrum of a small drum head or plate.

WaveMorph [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Morphing waveform oscillator: locks to the input pitch and crossfades a single oscillator continuously from sine through saw to square with one knob - a sweepable waveshape in one voice.

WavePacketOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wave-packet oscillator: a quantum Gaussian wave packet whose component states dephase and then periodically rephase, so its amplitude collapses to a blur and revives to a sharp peak each cycle - the quantum-revival phenomenon as a self-pulsing carrier.

WaveScan [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Additive harmonic scanner: sums a series of sine partials whose amplitudes follow a power-law roll-off, then normalizes the result. Freq sets the fundamental, Harmonics sets how many partials are stacked, Scan tilts the spectrum from bright (all harmonics equal at 0) toward dark (fundamental only as it approaches 1), and Level scales the output. Sweep Scan for a sawtooth-to-sine timbral morph.

WaveStack [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Octave saw stack: sums sawtooths at successive octaves above the fundamental with a roll-off, building the bright, organ/drawbar-like stacked-octave spectrum from one phase accumulator.

Wavetable [OSC] [FX] [SYNTH] in: – out: Audio

Wavetable oscillator: it advances a phase at Freq and reads single-cycle frames loaded from the node config, interpolating both within and between frames. Position (0 to 1) morphs across the frame stack, Warp (-1 to +1) phase-distorts the read pointer for timbral bending, Freq sets pitch (20 to 8000 Hz), and Level scales output. A pure generator (no inputs) that sweeps from one waveform to another as Position moves.

WaveTerrain [OSC] [FX] [SYNTH] in: Orbit Pos out: Audio

Wave-terrain synthesis: a Lissajous orbit traverses a 2D surface z=f(x,y) and the height is the output. Surface picks among five built-in terrains - Classic (the Mitsuhashi/Roads polynomial), Saddle, Ripple, EggCarton, Ring - or a drawn heightfield supplied in node config overrides them. Warp tilts the orbit, Ratio sets its y/x frequency, Freq the pitch, Level the output. The Orbit Pos input (in0) scrolls the orbit across the surface, so a moving CV sweeps the timbre instead of holding one static spectrum.

WaveTerrainScan [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wave-terrain synthesis: an orbit travels around a 2-D landscape z = f(x,y) and the terrain height becomes the waveform; as the orbit's radius and the terrain warp change, the timbre morphs continuously - a classic but rarely-implemented technique.

WaveTerrainXY [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

2-D wave terrain: two phasors at independent rates trace a path across a fixed two-variable surface, reading its height as the output - detuned X/Y rates sweep continuously evolving timbres a 1-D wavetable can't reach.

WebMap [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zaslavsky web map: an area-preserving kicked-oscillator map whose orbits weave an infinite crystalline 'stochastic web' of symmetric channels, a Hamiltonian chaos with a kaleidoscopic structure.

WeedWhackerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

String trimmer: a weed-whacker spinning its nylon line, a high motor whine periodically loaded by the slap of the line whipping through grass, the lawn-edging sound; the load knob sets how much grass it hits.

WeibullCdfOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weibull-CDF oscillator: each cycle is the Weibull cumulative 1-e$^{(-(x/Lambda)}$K), a stretched-exponential ramp that morphs from a concave saturating rise (K<1) through exponential (K=1) to an S-curve (K>1) - one shape knob spanning ramp families. Bounded.

WeibullNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weibull noise: samples (-ln u)^(1/k) whose shape parameter k morphs the distribution from heavy-tailed exponential (k<1) through the bell-like k~3.6 to a sharp peak (k>1) - the reliability/failure-time distribution, here a tunable-skew source.

WeibullQuantileOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weibull-quantile oscillator: each cycle is the Weibull inverse-CDF (-ln(1-p))^(1/K), a ramp whose curvature K controls - concave-then-steep for small K, near-linear and then accelerating for large K - the quantile (inverse-CDF) counterpart to the Weibull family. Bounded.

Weierstrass [OSC] [FX] [SYNTH] in: — out: Audio

Fractal-waveform oscillator (a first as a synth block): synthesises the Weierstrass function - the textbook curve that is continuous everywhere but differentiable nowhere - as a self-similar audio tone, summing eight cosine octaves whose frequencies climb by Lacunarity and amplitudes fall by Detail. The result is infinitely jagged: zoom in at any scale and the same roughness repeats. Detail sets the high-frequency weight (roughness/brightness) and Lacunarity the gap between layers. A genuinely fractal timbre, not a filtered or wavefolded one.

WeierstrassMandelbrot [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weierstrass-Mandelbrot function: the Weierstrass fractal generalized with per-octave phase offsets and a fractional-dimension exponent, a statistically-self-affine waveform used to model rough surfaces and fractional Brownian texture.

WeierstrassOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weierstrass function: a sum of cosines at geometrically-rising frequencies and shrinking amplitudes (sum a^k cos(b^k x)), the textbook continuous-everywhere yet differentiable-nowhere fractal curve, here pitch-locked as a richly-detailed waveform.

WeierstrassPOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Weierstrass P-function: the canonical doubly-periodic elliptic function, which has a double pole each period giving an even waveform with a sharp spike whose width the Sharpness knob sets - a band-limited approximation of its characteristic spiked profile.

WeiWangOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wei-Wang attractor: a chaotic flow constructed to have no equilibrium points at all (a ‘hidden’ attractor), so the orbit wanders chaotically with nowhere to settle - a modern example of hidden chaos.

WelchOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Welch oscillator: each cycle is the parabolic Welch window 1-(2p-1)^2, a single smooth quadratic arc - a rounded pulse brighter than the cosine-sum windows (a parabola has a richer even-harmonic spectrum than a raised cosine), with no discontinuities.

WestCoast [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Buchla-style west-coast oscillator: a pure sine driven into a sine wavefolder. Fold sets how hard it folds (sparse to dense harmonics) and Symmetry biases the fold for even-harmonic, brassy timbres - timbre from folding, not filtering.

WGMesh [OSC] [FX] [SYNTH] in: Strike out: Audio

2-D digital waveguide mesh (Van Duyne & Smith 1993): a strike on in0 launches a wave across an 8x8 membrane grid - the finite-difference 2-D wave equation - so it rings with the dense, inharmonic modes of a struck drumhead, plate or gong rather than a 1-D string. Not a wavetable: the timbre is the physics of a vibrating surface. Speed (the propagation/Courant coefficient) sets pitch and brightness - low for a deep tom, high for a tight gong; Damp sets the decay; Level the output. in0 = Strike.

Whale [OSC] [FX] [SYNTH] in: – out: Audio

Whale song - slow, deep, pitch-bending moans drifting through an underwater haze. Calls sets how often, Pitch the depth, Level the output.

WhaleSongOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whale-song oscillator: long, slowly-gliding low-frequency moans that rise and fall over seconds with a few harmonics, the haunting, drawn-out calls of a humpback or blue whale; the glide rate sets how mournfully the pitch bends.

WheelShimmy [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wheel shimmy: a trailing caster (shopping-cart or aircraft nose-gear) whose tyre side-force and castor geometry self-excite a violent steering wobble; the tyre’s nonlinear slip saturation bounds it into a chaotic limit cycle.

Whip [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whip crack / slapstick: a trigger on in0 fires a single sharp crack - a fast burst of bright band-passed noise with a quick downward zip, the slapstick thwack and whip-crack of cartoon and orchestral percussion. Tone sets the brightness of the crack, Decay its (very short) length, Level the output. Needs only a trigger.

WhiskOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whisking: a metal whisk beating eggs or cream in a bowl, a fast rhythmic swish of bright metallic noise modulated by the back-and-forth stroke, the steady whisking motion; the stroke rate sets the speed.

Whisper [OSC] [FX] [SYNTH] in: – out: Audio

Whispering voice - breathy mid-band noise gated into syllable-like bursts. Pace sets the syllable rate, Tone the sibilance, Level the output.

WhisperOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whisper oscillator: broadband breath noise shaped by a vowel formant resonance instead of a pitched source, producing the pitchless, airy quality of a whispered vowel; the formant knob selects which vowel colour the breath takes.

WhisperTextureOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whisper: a breathy unvoiced whisper, a band of noise shaped by slowly-shifting vocal formants with no pitch (just turbulent air forming words), the creepy disembodied whisper of horror; the formant rate sets the syllable speed.

Whistle [OSC] [FX] [SYNTH] in: Breath, Pitch CV out: Audio

Human whistle / piccolo-pure tone: an almost-sinusoidal voice with a gentle vibrato and a faint breath hiss, played by holding in0. Tune sets the pitch, Vibrato the wobble depth, Level the output. in0 = Breath gate, in1 = Pitch CV.

WhistlerWaveOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Whistler: a lightning-stroke radio impulse that travels along Earth's magnetic field lines and disperses, so the high frequencies arrive first and the pitch glides smoothly downward - the descending 'pheww' VLF whistler heard on a radio receiver; the rate sets how often a stroke fires.

Whoosh [OSC] [FX] [SYNTH] in: Trigger out: Audio

Transition whoosh: a trigger on in0 fires a band-passed noise swish whose loudness and filter both swell to a peak at the mid-point then fall away - the air pass-by used for cuts, swipes and reveals in film and trailers. A doppler-style arc, not a tonal riser. Time sets the pass duration, Tone the filter centre the sweep peaks at, Reso the band sharpness, Body a low-rumble layer for weight, and Level the output.

WienBridgeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wien-bridge oscillator: the classic op-amp RC sine generator (the topology of the first Hewlett-Packard audio oscillator), modeled as a resonant loop with a soft amplitude-stabilizing nonlinearity so it settles to a clean, low-distortion sine.

WienChaos [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Wien bridge: the textbook Wien-bridge sine oscillator with a saturating amplifier nonlinearity that, past its stable limit cycle, breaks into chaos - a familiar audio circuit driven over the edge.

WienerProcess [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wiener process: the standard Brownian motion - the integral of white noise with independent Gaussian increments - reflected at +/-1 so it stays bounded; the canonical continuous random walk underlying every diffusion model, as a slow drifting CV.

WienRobinson [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Chaotic Wien-Robinson oscillator: the balanced Wien-Robinson bridge sine source with a cubic amplifier nonlinearity, whose deep notch and extra feedback path destabilize the limit cycle into chaos - a bench-oscillator route to chaos.

WillamowskiRossler [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Willamowski-Rossler reaction: a mass-action chemical model with autocatalytic and cross-inhibitory steps that produces genuine chaos in concentration space - one of the few polynomial chemical systems proven to be chaotic.

WilsonCowanOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wilson-Cowan neural mass: coupled excitatory and inhibitory neuron populations with sigmoid activation oscillate and, when driven, break into chaos - the cortical-rhythm model behind brain-wave dynamics, as a sound source.

WilsonNeuron [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wilson neuron: Hugh Wilson's polynomial reduction of Hodgkin-Huxley to two variables (voltage + a recovery current), reproducing regular and fast spiking with simple cubic terms - an efficient biophysical spiker.

WimolBanlue [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wimol-Banlue attractor: a chaotic flow built around a hyperbolic-tangent and absolute-value nonlinearity, tracing a symmetric butterfly orbit with smoothly-saturating dynamics.

Wind [OSC] [FX] [SYNTH] in: – out: Audio

Wind texture from filtered noise: white noise is fed through a resonant TPT band-pass whose center frequency is modulated by a slow 0.3 Hz gust LFO. Cutoff sets the band-pass center frequency, Reso sets the resonance (how whistling/pitched the noise becomes), Gust scales the LFO cutoff sweep depth, and Level scales the output. Produces ambient wind, breath and atmospheric beds.

WindChimes [OSC] [FX] [SYNTH] in: – out: Audio

Wind chimes - randomly struck tuned metal tubes (FM bells on a pentatonic scale) drifting in a breeze. Activity sets how often they ring, Tone the metallic brightness, Level the output.

WindGen [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wind: band-passed noise swept into a howling wind, with the input signal's level driving the gusts - feed it silence for a steady breeze, or a signal to make the wind surge and moan with the music. A weather-bed generator that responds to what you play. Speed sets the base gustiness, Tone the wind colour, Mix the blend.

WindGustOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wind gust: broadband turbulent noise whose loudness is slowly and randomly modulated to swell into gusts and fall to lulls (a von-Karman-like gust envelope on a noise carrier), the restless rise-and-fade of real wind for atmospheres and whoosh textures.

WindmiOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

WINDMI attractor: a low-order model of the solar-wind/magnetosphere/ionosphere energy coupling whose exponential nonlinearity drives a chaotic geomagnetic-storm-like flow - space-weather chaos as a modulation source.

WindNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Resonant wind noise: white noise driven through a tunable resonant band-pass whistles and howls like wind through a gap - a self-contained breath/surf/wind texture generator (deterministic RNG).

WindowSlideOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sliding window: a sash window or patio door sliding in its track, a low rolling rumble of filtered noise with a faint periodic judder as it rides the rollers, fired at the set rate; the rumble of opening a heavy slider.

WindPad [OSC] [FX] [SYNTH] in: Amount out: Audio

Wind: a howling, gusting wind texture - white noise through a resonant bandpass whose pitch and level wander randomly, from a gentle breeze to a storm. Gust sets the wander, Pitch the centre, Level the output. in0 = Amount gate.

WindupSpringOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wind-up spring: the runaway-then-slowing whir of a wind-up toy's spring-driven escapement, a click train that starts fast and gradually slows as the mainspring unwinds, looping back to full wind; the run length sets how long before it rewinds.

WinfreeOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Winfree oscillators: a population of phase oscillators that influence each other through a pulse-shaped phase-response curve (the original biological-rhythm sync model, predating Kuramoto); the mean field locks, desynchronizes or beats irregularly.

WingRock [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wing rock: a slender delta-wing aircraft at high angle of attack whose asymmetric vortex lift gives negative roll damping, self-exciting the rolling ‘wing-rock’ limit cycle that, when forced, becomes chaotic - a flight-dynamics instability.

WireWorldCA [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

WireWorld: a four-state automaton (empty, conductor, electron head, electron tail) on which electrons flow down wires - a head becomes a tail, a tail becomes wire, and wire turns to head when one or two neighbours are heads; logic gates emerge. Head density is the output.

WobbleBass [OSC] [FX] [SYNTH] in: Gate, Pitch CV out: Audio

Dubstep wobble bass: a sawtooth through a resonant lowpass swept by a built-in LFO - the talking, wobbling bass of dubstep. Wobble sets the LFO rate, Cutoff the centre, Drive the grit. in0 = Gate, in1 = Pitch CV.

WobbleBoardOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wobble board: the flexing-sheet-of-metal warble (the Rolf Harris wobble board / thunder-sheet wobble), a low tone deeply modulated in both pitch and amplitude by a slow wobble, the boing-oing-oing drone; the wobble rate sets the flex speed.

WolfHowlOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wolf-howl oscillator: a voiced tone that swells up to a sustained pitch and then slowly falls, shaped by a moving vocal formant, the long mournful arc of a wolf's howl; the howl rate sets how drawn-out each cry is.

WolfTone [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wolf tone: when a bowed-string pitch coincides with a strong body resonance, the two modes fight and the note stutters and beats uncontrollably; this models that coupled string-plus-body oscillator in its warbling, near-chaotic regime.

Wolves [OSC] [FX] [SYNTH] in: – out: Audio

Wolf pack - mournful howls that glide up to a sustained pitch and fall away, layered for a pack. Howls sets how often they call, Pitch the howl register, Level the output.

WompWompOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Sad trombone: the womp-womp-womp-wommmp of comic failure, a brassy buzzy tone stepping down through three short notes and a long final slide, the price-is-wrong loss sting; fired at the set rate.

Woodblock [OSC] [FX] [SYNTH] in: Trigger, Pitch out: Audio

Woodblock / clave: a trigger on in0 fires a short, bright resonant ping - the dry, percussive wood knock. Tune sets the pitch, Decay the ring length, Level the output. in1 = Pitch CV.

WoodblockHit [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Woodblock: two short, high, dry resonant modes ring when struck by the input, the hard hollow knock of a woodblock or temple block - feed it an impulse or gate to play.

WoodChipperOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wood chipper: branches fed into a chipper, a heavy low motor grind punctuated by violent splintering chunk impacts as the blades shatter the wood, the brutal yard-waste sound; the feed knob sets how often it chews.

WoodpeckerOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Woodpecker drill: a fast train of sharp knocks, each a quickly-damped resonant ping at the wood pitch, fired at the Rate - the hollow, machine-gun drumming of a woodpecker on a trunk; Rate sets the drilling speed, Pitch the resonance of the wood.

WoodsSaxonOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Woods-Saxon oscillator: each cycle is the nuclear Woods-Saxon profile 1/(1+e^((|x|-R)/a)), a flat-topped pulse with Fermi-shaped (logistic) edges - a flat plateau falling through a sigmoid skirt, a distinct edge profile from the cosine, Planck and linear flat-tops. Soft sets the edge width.

Wormhole [OSC] [FX] [SYNTH] in: – out: Audio

Wormhole - a swirling vortex of FM-swept tones spiralling through space-time. Swirl sets the sweep rate, Tone the brightness, Level the output.

WrappedCauchyOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wrapped-Cauchy oscillator: each cycle is the wrapped-Cauchy circular density (1-r$^{2)/(1+r}$2-2r*cos(theta)), a periodic Lorentzian peak whose sharpness Rho sets - from a gentle cosine bump (Rho~0) to a needle-like spike (Rho near 1). Naturally band-limited and periodic.

WrappedExpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wrapped-exponential oscillator: each cycle is e^(-Lambda*theta) over theta in [0,2pi), a sawtooth whose ramp is exponential rather than linear - it starts high and decays to a floor before snapping back, Lambda setting the decay curvature. A curved saw distinct from the linear ramp.

WrappedNormalOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wrapped-normal oscillator: each cycle is a Gaussian wrapped around the circle (summed over neighbouring turns), a periodic bell whose width Sigma sets - from a sharp central peak to nearly flat as it wraps fully. The Gaussian-peak counterpart of the wrapped-Cauchy/von Mises. Peak-normalized, bounded.

WrapStackOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Wrap stack: the phase is multiplied by each integer 1..N and wrapped to a bipolar ramp, then the wrapped ramps are summed with a 1/k weighting - a sawtooth-like spectrum built purely from modulo phase-wrapping, brightening as you add stacks.

Wurlitzer [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Wurlitzer electric piano: a reedy tine voice - a sine fundamental with an odd-harmonic bark on the attack and a warm, woody decay, the bluesy 200A timbre. Tone sets the bark brightness, Decay the note length, Level the output. Trigger on in0, in1 = Pitch CV.

XgammaOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Xgamma oscillator: each cycle is the xgamma PDF proportional to (1 + (Theta/2)x$^{2)e}$(-Theta*x), a mixture of an exponential and a gamma component - the quadratic weight scales with Theta, so the hump grows and shifts as the decay sharpens. Peak-normalized, bounded.

XorShiftNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Xorshift noise: Marsaglia's xorshift generator (three shift-and-XOR operations on a 32-bit word) produces fast, long-period high-quality white noise with a different spectral signature than a linear-congruential source.

Xylophone [OSC] [FX] [SYNTH] in: Trigger, Pitch CV out: Audio

Struck wooden bar (xylophone): a bright modal voice from three inharmonic partials tuned 1 : 3 : 6 like a rosewood bar, with the high partials decaying fastest for the characteristic hard, woody 'tock'. Tune sets the pitch, Decay the ring (short = staccato), Level the output. Trigger on in0, in1 = Pitch CV.

YangChenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Yang-Chen attractor: a Lorenz-family chaotic system with two coexisting attractors depending on initial conditions, tracing an energetic twin-scroll orbit.

YangMills [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Yang-Mills mechanics: two coupled particles in the x^2 y^2 potential of classical gauge-field mechanics, an energy-conserving system that is chaotic at essentially all energies - a famously fully-chaotic Hamiltonian.

YangSystem [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Yang system: a chaotic flow lying between the Lorenz and Chen attractors in parameter space, where a small change in one coefficient switches the topology of the twin-scroll - a bridge system in the generalized Lorenz family.

YawnOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Yawn: the long drawn-out ‘aaah’ of a yawn, a breathy voiced vowel whose pitch and brightness arch up and then sink as the jaw opens and closes, fired at the slow yawn rate; the sound of sleepiness.

YukawaOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Yukawa oscillator: each cycle is the screened-Coulomb potential e^(-Screen*r)/r, a sharp spike at the origin that decays faster than a bare 1/r tail - Screen sets how quickly the tail is cut off, from a long Coulomb-like decay to a tight pulse. Peak-normalized, bounded.

YuleProcess [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Yule process: a pure-birth Markov chain where each individual independently spawns at a constant rate, so the population grows exponentially with super-exponential bursts (the model of preferential-attachment and species radiation) - a log-scaled accelerating ramp, reseeded.

YuWangOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Yu-Wang attractor: a chaotic flow with an exponential nonlinearity that produces a sharply-folded, asymmetric orbit - an unusual strange attractor with a distinctive crackling character.

Zap [OSC][FX][SYNTH] in: Trigger out: Audio

Psytrance zap: a trigger on in0 fires a fast descending pitch sweep - a bright saw that drops from Top times the pitch down to Tune in a few milliseconds with a short decay, the classic laser/zap blip and FX accent. Tune sets the landing pitch, Top the start-pitch multiplier (sweep height), Fall the sweep speed, Decay the blip length, and Level the output. Distinct from Impact: a bright high descending laser, not a deep sub boom.

ZaslavskyMap [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Zaslavsky map: a dissipative kicked-rotor map whose attractor shows a stretched, striped fractal structure - a richly-textured 2-D chaos different from the smooth-flow attractors.

ZeemanMachine [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Zeeman catastrophe machine: a bistable elastic system whose state sits in a double-well potential and snaps abruptly between wells (a cusp catastrophe) as a control parameter is swept - sudden hysteretic jumps, not smooth chaos.

ZernikeOsc [OSC][FX][SYNTH] in: Pitch (CV) out: Audio

Zernike oscillator: each cycle is a Zernike radial polynomial R_n^0 (the optical-aberration basis) - 2r^2-1, 6r^{4-6r}2+1 or 20r^{6-30r}4+12r^2-1 by Order - a bipolar polynomial that oscillates with increasing lobes across the cycle, distinct from the Chebyshev/Legendre bases.

ZeroCrossFM [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zero-cross-tracked FM: an internal sine carrier locks to the input's pitch (from its zero crossings) and is phase-modulated by the input itself, generating FM sidebands that follow whatever you feed it.

ZhouChenOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zhou-Chen attractor: a chaotic flow with two quadratic cross-terms and a constant drive that traces a compact, fast-spinning twin-lobed orbit, a distinct entry in the Lorenz-family zoo.

ZhuChenFlow [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zhu-Chen flow: a Lorenz-family chaotic system with a constant drive in its third equation, shifting the attractor off-centre into an asymmetric twin-scroll - another systematically-derived member of the generalized Lorenz zoo.

ZipfNoise [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zipf noise: picks discrete levels with probability proportional to 1/rank^s (the Zipf/zeta law of word frequencies and city sizes), so a few low levels dominate and high ones are rare - a heavy-headed quantized random source distinct from uniform noise.

ZipperOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zipper: the brrrip of a zip fastener, a very fast dense train of tiny tooth clicks whose rate (and so pitch) rises as the slider accelerates; the speed knob sets how fast it is yanked.

ZipUpOsc [OSC] [FX] [SYNTH] in: Pitch (CV) out: Audio

Zip: a fast zipper or whoosh-up, a buzzy tone that races rapidly upward in pitch with a granular zipping texture, the cartoon quick-exit or zip-up; fired at the set rate.

Physical (20)

AcousticGuitar [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Acoustic guitar (physical model): six digital-waveguide strings (Extended Karplus-Strong -

Banjo [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Banjo (physical model): five digital-waveguide steel strings (Extended Karplus-Strong - tuned

Celesta [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Celesta (physical model): steel bars struck by FELT hammers (soft, so few high partials) over

Cello [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Cello (physical model): a self-oscillating bowed string (STK two-delay waveguide - neck + bridge

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☐ **ChurchBell** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Church bell (physical model): a cast bronze bell tuned to the true-harmonic partials - hum (0.5),

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☐ **Clavichord** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Clavichord (physical model): a brass tangent strikes the string and stays in contact, so key

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☐ **DoubleBass** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Double bass (physical model): the bowed STK two-delay waveguide in the double bass's deep range,

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☐ **Dulcimer** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Appalachian dulcimer (physical model): a strummed melody string over two open drone strings

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☐ **GrandPiano** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Grand piano (physical model, modal synthesis): the 16 string partials are placed DIRECTLY at

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☐ **Harp** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Harp (physical model): seven diatonic digital-waveguide strings (Extended Karplus-Strong -

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☐ **HurdyGurdy** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Hurdy-gurdy (physical model): a rosined wheel, turned by the crank, continuously bows ALL the

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☐ **Kantele** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Kantele (physical model): the Finnish box zither - open metal strings over a shallow wooden

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☐ **Nyckelharpa** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Nyckelharpa (physical model): the Swedish keyed fiddle - a self-oscillating bowed string (STK

☐ **SingingBowl** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Tibetan singing bowl (physical model): a struck standing bell - four inharmonic shell modes

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☐ **SteelTongue** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Steel tongue drum (physical model): cut steel tongues of a tank/Hapi drum, each TUNED so its

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☐ **TubularBells** [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Tubular bells / orchestral chimes (physical model): a struck metal tube whose prominent

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TuningFork [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Tuning fork (physical model): a struck fork rings in the modes of a clamped-free bar (ratios

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Viola [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Viola (physical model): the bowed STK two-delay waveguide in the viola's alto range, coupled to

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ViolaDaGamba [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Viola da gamba (physical model): the bowed STK two-delay waveguide voiced as a bass viol -

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Violin [FX] [PHY] [FX] [SYNTH] in: Audio out: Audio

Violin (physical model): a self-oscillating bowed string - an STK two-delay waveguide

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Pitch (28)

Doppler [PIT] [FX] [SYNTH] in: Audio, Trigger out: Audio

Doppler pass-by: makes the input sound like a source moving past the listener - on a trigger the pitch bends up as it approaches then down as it recedes (a varispeed delay line), the level swells to a peak at the closest point, and a low-pass opens at the pass then closes with distance. The car / jet / projectile fly-by of film sound design. Speed sets the pass duration, Amount the pitch-bend depth (how fast the source moves), Distance the closest approach (level and tone contrast), and Mix blends against dry. in0 = Audio, in1 = Trigger.

DriftPitch [PIT] [FX] [SYNTH] in: Audio out: Audio

Analog pitch drift: a quasi-random slow modulator (two incommensurate LFOs) sweeps a fractional delay so pitch wanders like a detuning analog oscillator - organic, never-repeating instability.

FreqDouble [PIT] [FX] [SYNTH] in: Audio out: Audio

Frequency doubler: squaring the signal generates a component at twice the frequency; a DC blocker removes the rectification offset, leaving an octave-up tone that tracks the input.

FreqShift [PIT] [FX] [SYNTH] in: Audio out: Audio

Bode-style single-sideband frequency shifter: splits the input at in0 into in-phase and 90-degree branches via a polyphase IIR Hilbert allpass pair, then heterodynes against an internal carrier to shift every partial by a fixed Hz amount. Shift sets the offset (-1000 to +1000 Hz) and Mix blends the shifted signal against the dry input. Because the shift is in Hz rather than ratio, harmonic relationships break, giving inharmonic, metallic, clangorous tones.

GranPitch [PIT] [FX] [SYNTH] in: Audio out: Audio

Granular pitch shifter: two delay read-heads sweep at the pitch-shift ratio and crossfade with overlapping windows, transposing the input up or down in real time without changing its length - classic time-domain pitch shifting.

H910 [PIT][FX][SYNTH] in: Audio out: Audio

Harmonizer: a pitch shifter feeding a feedback delay, so each repeat is pitched again into a rising or falling cascade (the classic H910 fused pitch+delay).

Harmonizer [PIT][FX][SYNTH] in: Audio out: Audio

Single-voice harmonizer that pitch-shifts the input by a chosen musical interval using overlapping windowed delay-line reads, then adds it back to the dry signal. Interval sets the transposition (+/-24 semitones), Detune adds up to 30 cents of offset for a less rigid blend, and Level sets the harmony voice gain. Mix scales the added voice into the output; useful for parallel harmonies and octave doubling.

HarmonizerFixed [PIT][FX][SYNTH] in: Audio out: Audio

Diatonic harmonizer: granular-pitch-shifts the input by a fixed musical interval (third, fourth, fifth, octave) and mixes it under the dry signal, adding a parallel harmony voice without a separate sequencer.

InstFreq [PIT][FX][SYNTH] in: Audio out: Audio

Instantaneous frequency: differentiates the unwrapped phase of the analytic signal (input plus its quadrature) to estimate the signal's frequency continuously - a fast pitch/movement CV that responds within a cycle.

MicroShift [PIT][FX][SYNTH] in: Audio out: Audio

Micro-pitch doubler that generates two delay-line voices detuned slightly up and slightly down, then sums them with the dry signal to widen the source. Detune sets the symmetric pitch offset (0-50 cents) and Delay sets the base tap time (0-40 ms) before the shifted voices. Mix sets dry/wet; small Detune thickens, larger Detune broadens into a chorus-like spread.

Octaver [PIT][FX][SYNTH] in: Audio out: Audio

Sub-octave generator: a flip-flop toggled at each rising zero crossing produces a half-rate square tracked by the input envelope, low-passed by Tone for warmth and mixed under the Dry signal at Sub. Monophonic, analog-pedal style - best on clean single notes and bass. It glitches on chords, as expected of the technique.

Octavox [PIT][FX][SYNTH] in: Audio out: Audio

A cloud of four pitch-shifted voices at diatonic intervals, each on its own delay tap, summed into a shimmering harmonized pitch-delay (Octavox/Quadravox class).

OctUp [PIT][FX][SYNTH] in: Audio out: Audio

Octave-up doubler: full-wave rectification (which doubles the fundamental) followed by a DC blocker yields an octave-up tone tracking the input, blended back with the dry signal.

PhaseMult [PIT][FX][SYNTH] in: Audio out: Audio

Harmonic phase multiplier: tracks the input period and emits a clean saw at an exact integer multiple of its pitch, generating a precise upper harmonic locked to the source (octaves, fifths-as-3x, etc.).

Pitch2Midi [PIT] [FX] [SYNTH] in: Audio out: Audio

Monophonic pitch tracker that measures the period between rising zero-crossings, converts it to a MIDI note number, and emits it as a control value. Glide smooths the tracked pitch over time for portamento, while Out switches the output between the normalized pitch CV and a simple gate that goes high whenever the input is active. Intended to drive other modules from a played signal rather than to process audio.

PitchCorrect [PIT] [FX] [SYNTH] in: Audio out: Audio

Rough auto-tune that detects the input period via zero-crossings, finds the nearest semitone, and applies a crossfaded two-tap windowed pitch shift to pull the note onto pitch. Amount scales how far the signal is moved toward the target semitone, from no correction to full snapping, and Mix blends the corrected voice against the dry input. Best on clean monophonic sources; artifacts increase with stronger correction.

PitchCV [PIT] [FX] [SYNTH] in: Audio out: Audio

Pitch-to-CV converter: tracks the input's fundamental from its zero crossings and outputs the frequency as a 0..1 control signal between two limits - a monophonic pitch follower for driving filters or other oscillators.

PitchQuant [PIT] [FX] [SYNTH] in: Audio out: Audio

Pitch quantizer: tracks the input pitch from zero crossings, snaps it to the nearest equal-tempered semitone, and emits a clean saw at the quantized pitch - a monophonic auto-tune-to-grid oscillator.

PitchShift [PIT] [FX] [SYNTH] in: Audio out: Audio

Granular pitch shifter using two overlapping, cross-faded grains (Window 20-120 ms) to transpose +/-24 semitones without changing length. Feedback re-injects the shifted output for cascading and shimmer effects, and Mix sets dry/wet. Shorter windows tighten transients but add warble; longer windows smooth the tone at the cost of latency.

PitchShiftFB [PIT] [FX] [SYNTH] in: Audio out: Audio

Feedback pitch shifter: the granular-shifted output is fed back into the buffer, so each pass is transposed again, stacking an ever-climbing (or descending) cascade of shifted copies - the harmonic-staircase / barber-pole pitch effect.

ScaleSnap [PIT] [FX] [SYNTH] in: Audio out: Audio

Scale quantizer: snaps a 0..1 control input to the nearest note of a chromatic, major or minor scale across a chosen octave span - the pitch-quantizer that turns a smooth or random CV into in-key steps.

Shifter [PIT] [FX] [SYNTH] in: Audio out: Audio

Pitch / frequency / ring shifter in one (Ableton Shifter-style). Mode switches between granular Pitch shift (transpose by Coarse semitones + Fine cents, length preserved), Bode Frequency shift (Coarse/Fine reinterpreted as Hz via a polyphase-IIR Hilbert SSB shifter, breaking harmonics into inharmonic/metallic tones), and Ring modulation (multiply by an internal sine carrier at the Coarse/Fine frequency). Window sets the pitch-mode grain size; Mix blends dry/wet.

Spinback [PIT] [FX] [SYNTH] in: Audio, Gate out: Audio

Turntable brake: a gate on in1 starts reading the captured buffer at steadily falling speed, so playback drops in pitch and slows to a dead stop like braking a record. Time sets how long the brake takes to reach a halt (50..3000 ms) and Mix blends the braked signal against the dry input. Releasing the gate stops braking and returns to live audio.

SubHarm [PIT] [FX] [SYNTH] in: Audio out: Audio

Sub-harmonic sine synthesizer: tracks the input period from zero crossings and synthesizes a clean sine one octave below it - a smooth tonal sub, distinct from the square-wave flip-flop SubOctave.

SubOctave [PIT] [FX] [SYNTH] in: Audio out: Audio

Sub-octave generator: a flip-flop toggled on each rising zero crossing produces a square one octave below the input pitch (analog octave-divider style), smoothed and mixed with the dry signal.

TimeStretch [PIT] [FX] [SYNTH] in: Audio out: Audio

Granular time-smear that writes the input into a delay line and reads it back with two overlapping Hann-windowed grains advancing slower than real time. Stretch sets how slowly the read position moves (lower values smear the sound out further) and Grain sets the grain length in milliseconds, trading smoothness against fluttering artifacts. Mix blends the stretched texture against the dry signal for frozen, drifting timbres.

Unison [PIT] [FX] [SYNTH] in: Audio out: Audio

Unison thickener: turns a mono input into up to 16 pitch-detuned voices (the supersaw sound) using overlapping ramped-delay pitch shifters at symmetric cent offsets, so the centre stays at pitch and the outer voices spread apart. Detune sets the spread, Blend the outer-voice level and Mix dry/wet. Wide and lush on synths, pads and backing vocals.

ZeroLockRamp [PIT] [FX] [SYNTH] in: Audio out: Audio

Zero-cross-locked ramp: measures the input period from its rising zero crossings and emits a clean phase ramp (saw) at a multiple of that pitch - a band-limited ‘reconstruct the oscillator’ from any monophonic source.

Reverb (43)

Aetherizer [REV] [FX] [SYNTH] in: Audio out: Audio

Grain cloud (Absynth Aetherizer-style): continuously records the input and sprays a swarm of overlapping Hann-windowed grains read back from random recent positions, each at a randomized pitch, so a held sound dissolves into an evolving shimmering cloud. Size sets grain length, Density the grains per second, Pitch the centre transpose, Spread the random pitch scatter, Scatter how far back grains are picked, and Mix the wet amount. A texture/sound-design effect distinct from the granular delay and granular reverb.

AllpassChain [REV] [FX] [SYNTH] in: Audio out: Audio

All-pass diffusion chain: four first-order all-pass stages in series smear phase without touching magnitude, thickening and diffusing transients - a building block for reverb pre-diffusion or a standalone phasey wash.

Ascend [REV] [FX] [SYNTH] in: Audio out: Audio

Pitch-rising shimmer reverb: a long delay whose feedback is pitch-shifted up an octave (granular two-tap) and damped, so the tail ascends and blooms into ethereal pads. Size sets the delay length, Shimmer the octave-up feedback amount and Damp the tail brightness. Lush on pads, vocals and ambient guitar - see Crystals for reverse+pitch and Gravity for huge inverted tails.

CombAllpass [REV] [FX] [SYNTH] in: Audio out: Audio

Schroeder allpass: a tuned delay-line allpass (feed-forward and feedback comb combined) that passes all frequencies flat but smears phase into a short coloured tail - the building block of classic reverb diffusion, here exposed standalone and tuned in Hz. Distinct from the first-order AllpassChain.

CombAllpassDiffuse [REV] [FX] [SYNTH] in: Audio out: Audio

Comb-into-allpass diffuser: a tuned feedback comb followed by an all-pass smears its periodic echoes into a denser, less-metallic tail - the comb-plus-allpass building block of Schroeder/Moorer reverbs in one cell.

CombHaze [REV] [FX] [SYNTH] in: Audio out: Audio

Comb haze: four short feedback comb taps at incommensurate delays share one buffer and sum into a dense, metallic, slightly detuned resonance with a damping low-pass on the tail - a small bright box between a resonator and a reverb.

CombReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Two-comb reverb: a pair of damped feedback combs at incommensurate delays sum into a short, coloured reverb tail - the core of a Schroeder reverb in one block.

CombSwarm [REV] [FX] [SYNTH] in: Audio out: Audio

Comb swarm: four damped feedback combs at closely-detuned incommensurate delays beat against each other into a shimmering, chorused metallic resonance - denser and more animated than a single comb, between a resonator and a small bright reverb.

Convolution [REV] [FX] [SYNTH] in: Audio out: Audio

Convolution reverb: input is buffered into blocks and convolved with an impulse response via partitioned FFT, then streamed back with block latency. The IR comes from the node config (an audio file path loads that real IR, otherwise a decay time in seconds generates a decaying-noise tail); PreDelay offsets the wet by 0 to 200 ms, Tone is a one-pole low-pass on the wet, and Mix blends dry/wet. Use it to place sounds in real spaces or to apply any captured IR.

CrushVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Lo-fi crush reverb: a pair of damped feedback combs whose recirculating signal is bit-crushed each pass, so the tail degrades into a grainy, quantized, ever-coarser decay - a digital ruin between a reverb and a sample-rate reducer.

Crystals [REV] [SYNTH] in: Audio out: Audio

Shimmer reverb: a 4-line FDN whose output is pitch-shifted an octave up and re-injected into the network, building rising crystalline overtones. Size scales the base delay times up to 6x, Shimmer sets how much octave-up signal feeds back in, and Damp sets the per-line low-pass that darkens repeats. Mix blends the wet sum against the dry; higher Shimmer ascends further, higher Damp tames the brightness.

Diffuser [REV] [FX] [SYNTH] in: Audio out: Audio

Allpass diffuser: a cascade of four Schroeder all-pass sections with prime-spaced delays smears transients into a short, dense burst without adding a reverb tail or colouring the magnitude spectrum - the diffusion stage that lives inside reverbs, exposed on its own to thicken percussion, blur clicks, or pre-soften a signal before further processing. Amount sets the all-pass coefficient (smear density), Size scales the delay times, and Mix blends against the dry signal.

DiffusorComb [REV] [FX] [SYNTH] in: Audio out: Audio

Diffused comb: a pair of allpass diffusers smear the input before it enters a tuned damped comb, so the comb's metallic resonance is softened into a smooth, dense, plate-like ring rather than a sharp peak - between a resonator and a small reverb.

DuckVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb whose wet signal is attenuated by an envelope follower tracking the dry input, so the tail dips while you play and swells back in the gaps. Size sets the room length and Damp the high-frequency decay, Duck sets how strongly the dry level suppresses the wet, and Mix blends the ducked reverb against the dry signal. Keeps the reverb from washing out a busy performance while preserving the tail between phrases.

EarlyReflections [REV] [FX] [SYNTH] in: Audio out: Audio

Early reflections: sums eight delayed taps at prime-spaced room-like times with decaying gains - the discrete first echoes a room produces before the diffuse tail, with no feedback. Adds spatial depth and 'air' without the wash of a full reverb. Size scales the room dimensions (tap times); Mix blends against the dry signal.

ErbeVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Modeless morphing reverb (Make Noise / Soundhack Erbe-Verb): a continuously-variable feedback-delay-network reverb with no presets - dial Size and Decay anywhere from a tight room through a cavernous hall, and Tilt from dark to bright. Mod animates the delay network for chorused, evolving and unreal spaces, and Mix blends wet against dry. One reverb that morphs smoothly across the whole space rather than switching between fixed types.

FDNReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Feedback-delay-network reverb: four delay lines recirculate through a lossless Hadamard mixing matrix, the modern algorithmic-reverb core that builds a dense, smooth, hall-like tail from just four taps.

Freezer [REV] [FX] [SYNTH] in: Audio out: Audio

Audio freezer that continuously buffers the input and, when Freeze is engaged, loops the most recently captured window indefinitely. Size sets the length of the looped window in milliseconds, trading a short stuttering grain against a longer evolving pad, and Mix blends the frozen loop against the dry signal. With Freeze off it passes the input through while keeping the buffer primed for an instant capture.

FreezeReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Freeze reverb: a damped comb-feedback reverberator whose Freeze control pushes the feedback toward unity while muting the input, capturing the current sound and sustaining it forever as a smooth pad - the 'infinite reverb / hold' effect.

At Freeze 0 it is an ordinary decaying reverb (set by Feedback); turn it up to lock the tail. Damping rolls off the highs in the loop.

FreezeVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb that can latch into an infinite-hold freeze mode, sustaining the current tail indefinitely instead of decaying. Size sets the room length and Damp the high-frequency decay, Freeze above its midpoint engages the infinite hold, and Mix blends the reverb against the dry signal. Use the freeze to build sustained drones and pads from a single moment of input.

GatedReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Gated reverb: a dense reverb whose tail is cut off abruptly by a gate that closes when the dry input falls quiet, the explosive, truncated drum-reverb sound of 1980s records.

GatedVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb whose wet tail is muted by a fast envelope-driven gate, so the reverb only passes while the dry input stays above a threshold. Size sets the room length and Damp the high-frequency decay, Threshold sets the level below which the tail is cut, and Mix blends the gated reverb against the dry signal. Reproduces the abrupt, chopped-off reverb of 80s drum production.

GrainVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Granular reverb that windows the input with a moving Hann grain and injects it into a 4-line feedback delay network for diffusion. Size scales the four delay lengths up to 6x for longer tails, Grain sets the grain rate in milliseconds, and Damp applies a low-pass in each feedback line to darken successive reflections. Mix blends the diffused tail against the dry input for a smeared, granular ambience.

Gravity [REV] [FX] [SYNTH] in: Audio out: Audio

Oversized 4-line feedback delay network (FDN) reverb, each line damped by a low-pass and length-modulated by a shared LFO. Size stretches the base delay times up to 9x, Gravity raises feedback toward unity for near-infinite tails, Mod sets the LFO modulation depth, and Freeze pins feedback to 1.0 while muting new input. Mix blends the Hadamard-mixed wet sum against the dry; pushed hard it yields huge, slowly-evolving, freezable spaces.

Iceverb [REV] [FX] [SYNTH] in: Audio out: Audio

Four lowpass-damped comb filters form the reverb body, then a self-oscillating resonant state-variable filter is applied to the tail for a cold, ringing sheen. Size sets the comb feedback/decay (up to 0.95), Color sets the resonant filter frequency (~100 Hz..14 kHz), Icing sets its resonance toward self-oscillation, and Mix blends wet against dry. High Icing makes the tail sing into a metallic, icy ring.

MangledVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb whose fully wet output is fed through a tanh saturation stage for gritty, broken ambience. Size sets the room size and tail length, Damp rolls off high frequencies in the decay, and Drive pushes the reverb output harder into the nonlinearity for increasing crush. Mix blends the distorted tail against the dry signal, ranging from a lightly dirtied space to a heavily mangled wash.

Massive [REV] [FX] [SYNTH] in: Audio out: Audio

Feedback-delay-network reverb (Valhalla Supermassive style): four modulated delay lines mixed through a unitary Hadamard matrix give huge, smooth, near-infinite tails. Size scales delay length up to multi-second, Decay sets feedback, Damp absorbs highs in the loop, and Mod choruses the lines for shimmer. Mode picks Lush / Huge / Cosmos voicings, from tight to cathedral to cosmic.

ModEchoVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Fused reverb and modulated delay: the input feeds a wow-modulated feedback delay and a short comb reverb tank at once, blending echo and space in one block.

ModVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb whose room size is continuously modulated by a sine LFO, detuning the tail for a chorused, shimmering character. Size sets the base room length and Damp the high-frequency decay, Rate sets the LFO speed (0.05-5 Hz) and Depth how far it sweeps the size, and Mix blends the moving reverb against the dry signal. Adds motion and width to otherwise static reverb tails.

NonlinVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Nonlinear 4-line FDN reverb whose wet tail is gated by an envelope follower: a transient above the running level retriggers a hold, then the gain holds flat before cutting off. Size scales the short base delay times up to 5x, while Hold sets how long (10-500 ms) the tail sustains at full level before it decays away. Mix sets dry/wet; the held-then-cut behaviour gives gated, non-natural reverb bursts rather than a smooth decay.

Plate [REV] [FX] [SYNTH] in: Audio out: Audio

Plate reverb: the input runs through four series allpass diffusers and then into four parallel damped feedback combs that are summed for the wet tail. Decay sets the comb feedback (0 to 0.95), Damp is a one-pole low-pass inside the feedback that darkens later reflections, Size scales the delay-line lengths, and Mix blends dry/wet. Dense, smooth metallic decay suited to plates and tight ambiences.

PlateReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Plate reverb: a pair of all-pass input diffusers feed a damped recirculating delay tank (the Dattorro topology), producing the dense, bright, metallic-smooth decay of a studio plate rather than a room.

Plonk [REV] [FX] [SYNTH] in: Audio out: Audio

Modal resonator (Rings/Plonk-style): excites a bank of five tuned band-pass resonators with the input, imposing the ringing modes of a struck object - bar, plate, bell, drumhead - onto any signal. Tune sets the fundamental, Structure morphs the partials from a harmonic stack toward inharmonic metallic ratios, Decay sets the ring time (resonator Q), and Mix the wet blend. Feed it drums, noise or a synth for plucked, struck and resonated tones. Distinct from the single-mode Resonator and the self-struck Modal oscillator.

Reverb [REV] [FX] [SYNTH] in: Audio out: Audio

Algorithmic room reverb (Freeverb-style comb/all-pass network) with Size scaling the decay time and Damp rolling off the highs in the tail. Width sets the stereo spread and Mix the dry/wet blend. Large Size with low Damp gives long bright halls; high Damp yields short dark rooms - see Massive, Plate, Spring and Shimmer for character-specific spaces.

ReverseVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Reverb whose wet level swells up from silence after each detected onset, fading the tail in rather than out for a backward, blooming effect. Size sets the room length and Damp the high-frequency decay, Swell sets the fade-in time (1-500 ms) restarted on every new transient, and Mix blends the swelling reverb against the dry signal. Produces the rising, sucked-in quality of reversed reverb.

SchroederAP [REV] [FX] [SYNTH] in: Audio out: Audio

Schroeder all-pass diffuser: the classic flat-magnitude, phase-smearing all-pass that thickens transients into a diffuse tail - one reverb-grade diffusion stage you can chain or use alone.

Shimmer [REV] [FX] [SYNTH] in: Audio out: Audio

Shimmer reverb: a delay line whose feedback path is pitch-shifted by a two-window granular shifter so repeats climb in pitch each pass. Shift sets the transposition in semitones (-12 to +24), Feedback (0 to 0.95) controls how long the tail sustains, Size sets the delay time in ms, Damp low-passes the feedback, and Mix blends dry/wet. With Shift at +12 it builds the classic ethereal rising-octave wash.

ShimmerVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Shimmer reverb: the reverb's feedback path is granular-pitch-shifted up an octave each pass, so the tail blooms into an ethereal, ever-rising angelic wash - the signature ambient/post-rock reverb.

SlewVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Slew-limiter wash: six slew-rate limiters in series, fed back on the last stage, smear every edge into a soft triangular blur that pools into a short ambient tail - a reverb-like space built entirely from slew limiting, no delay lines.

Spring [REV] [FX] [SYNTH] in: Audio out: Audio

Spring-reverb approximation: a short (~30 ms) delay loop diffused through an all-pass cascade for the characteristic dispersive boing, with a Tone low-pass in the feedback. Decay sets the tail length and Mix the blend. The classic guitar-amp and dub reverb - splashy on transients, drippy at high Decay.

SpringReverb [REV] [FX] [SYNTH] in: Audio out: Audio

Spring-tank reverb: a feedback delay whose loop runs through a chain of first-order all-pass sections, giving the frequency-dependent (dispersive) delay that produces a real spring's characteristic 'boing' and metallic chirp - distinct from the dense diffusion of plate or hall reverbs. Time sets the tank length, Decay the tail, Dispersion the all-pass chirp amount, Damp the high-frequency loss, and Mix the wet/dry blend.

SpringVerb [REV] [FX] [SYNTH] in: Audio out: Audio

Spring reverb: an all-pass-dispersed feedback delay recreates the chirpy, boingy, frequency-smeared bounce of a reverb spring tank, the surf-guitar/amp reverb sound rather than a smooth hall.

Sympathetic [REV] [FX] [SYNTH] in: Audio out: Audio

Sympathetic-string resonator bank: four high-Q tuned resonators (like the drone strings of a sitar or the undamped strings of a piano) ring in sympathy with the input, voiced as a chord above the Tune root. Tune sets the root pitch, Decay how long the strings ring (resonator Q), Spread the chord voicing, and Mix the resonated blend. Feed it a plucked or percussive source for shimmering resonant tails - distinct from reverb and from a single modal resonator.

Routing (1)

SeqSwitch [RTE] [FX] [SYNTH] in: Clock, In A, In B, In C, In D out: Audio

Sequential switch: a clock on in0 steps the output through the In A..D signal inputs in turn, wrapping after Steps - route four sources into one in a repeating cycle, or scan CVs into a sequence. Like a rotary switch advanced by a clock instead of by hand.

Sequencing (186)

Abundant [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Abundant-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is an abundant number - one whose proper divisors sum to MORE than itself (12: 1+2+3+4+6 = 16 > 12). These are the 'over-rich' integers, the opposite of the deficient majority, with the perfect numbers sitting exactly on the boundary. Abundant numbers cluster around the highly-composite values, so the gate fires in factor-rich bursts - a number-theoretic rhythm tied to divisor structure, distinct from the digit-based Harshad. A rising edge on in1 resets.

Achilles [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Achilles-number sequencer (a first as a synth block): steps through the Achilles numbers 72, 108, 200, 288, 392, 432, 500, 648… - integers that are POWERFUL (every prime factor appears at least squared) yet are NOT themselves a perfect power. Like the hero, they are strong but imperfect - 72 = 2^3 * 3^2 cannot be written as a single base to a single exponent. Each clock advances one term, folded modulo Base. A strong-but-imperfect class distinct from the plain powerful sequencer. A rising edge on in1 resets.

Aliquot [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Aliquot-sequence sequencer (a first as a synth block): each in0 clock advances the step n and outputs how many steps its aliquot sequence runs before it terminates - the sequence where each term is the sum of the PROPER divisors of the last (12 -> 16 -> 15 -> 9 -> 4 -> 3 -> 1). Most chains fall to 1, but the PERFECT numbers (6, 28…) are fixed points that never move, the AMICABLE pairs cycle forever, and some chains climb without any known end - one of number theory's great open questions. The output is the chain length (capped), spiking at the perfect and untouchable numbers. A divisor-iteration melody distinct from the single-step Sigma. A rising edge on in1 resets.

Amicable [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Amicable-number sequencer (a first as a synth block): steps through the members of amicable pairs 220, 284, 1184, 1210, 2620, 2924… - two numbers each equal to the sum of the OTHER'S proper divisors (the divisors of 220 sum to 284 and vice versa). Known to the Pythagoreans as a symbol of friendship, these pairs are rare and were hunted for centuries. Each clock advances one amicable number, folded modulo Base. A mutual-divisor-sum class distinct from the self-referential perfect numbers. A rising edge on in1 resets.

Apery [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Apery-number sequencer (a first as a synth block): steps through the Apery numbers 1, 5, 73, 1445, 33001, 819005… - the extraordinary integers Roger Apery used in 1979 to prove that zeta(3) is irrational, a result that had eluded everyone for centuries. They satisfy a deep three-term recurrence with cubic coefficients and are sums of products of squared binomial coefficients. The exact value is folded modulo Base, giving a fast, wide-leaping climb. A zeta-irrationality witness distinct from the polynomial figurate sequencers. A rising edge on in1 resets.

▪▫▪▫ **Arrangement** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Arrangement sequencer (a first as a synth block): steps through the total number of arrangements 1, 2, 5, 16, 65, 326, 1957… - the count of ALL ordered selections (of any length) from n items, equal to the sum of n!/k! over k. It obeys the tidy rule a(n) = na(n-1) + 1, and the values are the nearest integers to n!e. Folded modulo Base. A partial-permutation total distinct from the full-factorial and derangement sequencers. A rising edge on in1 resets.

▪▫▪▫ **Automaton** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Elementary cellular-automaton sequencer (a first as a synth block): a 16-cell row of one-bit cells evolves one generation on every rising edge of the in0 clock, each cell's next state set by the 8-bit Rule applied to its left/centre/right neighbours (Wolfram's elementary CA, wrapped). Rule 90 paints a fractal Sierpinski pattern, 30 is chaotic, 110 is Turing-complete. The output is the live cell density (0..1 stepped CV) - a deterministic yet endlessly evolving pattern, unlike the random shift-register of Turing or Marbles. A rising edge on in1 reseeds a single centre cell.

▪▫▪▫ **Automorphic** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Automorphic-number gate (a first as a synth block): the output goes high when the step n is automorphic - its SQUARE ends in n itself (5 squared is 25, 6 squared is 36, 25 squared is 625, 76 squared is 5776…). These 'curious' self-reproducing numbers are extraordinarily rare - only a handful below each power of ten - so the gate fires on a very sparse, irregular pulse. A self-squaring digit rhythm distinct from the figurate and prime gates. A rising edge on in1 resets.

▪▫▪▫ **BakSneppen** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Bak-Sneppen evolution source (a first as a synth block): sixteen species sit in a ring, each with a random fitness; every clock the least-fit species - and its two neighbours - are wiped out and replaced with new random fitnesses. Driving out the weakest drags the whole ecosystem upward in fits and starts: long calm stretches of slow improvement shattered by sudden cascades of extinction, the punctuated equilibrium of self-organised criticality. The output is the current minimum fitness, climbing toward a critical threshold then collapsing. A rising edge on in1 reseeds.

▪▫▪▫ **Ballot** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Ballot-triangle sequencer (a first as a synth block): reads Catalan's ballot triangle row by row - the number of ways a count can stay ahead in an election where each entry is the sum of the one to its left and the one above (1; 1,1; 1,2,2; 1,3,5,5; 1,4,9,14,14…) - modulo Base. Built by C(n,k) = C(n,k-1) + C(n-1,k), so mod Base is exact, and its right edge IS the Catalan numbers. The combinatorics of never-falling-behind paths as a melody, distinct from the Pascal and Stirling triangles. A rising edge on in1 resets.

▪▫▪▫ **BaumSweet** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Baum-Sweet gate (a first as a synth block): the output goes high when the step n is a Baum-Sweet number - one whose binary representation contains NO block of consecutive 0s of odd length (n = 0 counts as high by convention). It is one of the classic automatic sequences (computable by a tiny finite automaton reading n's bits) yet its pattern of 1s and 0s looks irregular and unpredictable - 1,1,0,1,1,0,0,1,0,1… A deterministic, self-similar gate built purely from the shape of the zero-runs in binary, distinct from the bit-COUNT sequences. A rising edge on in1 resets.

▪▫▪▫ **Beatty** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Golden-ratio (Beatty) rhythm gate (a first as a synth block): fires high when the clock step lands on floor(n * Ratio) for n = 1, 2, 3… With Ratio at the golden mean (~1.618) the hits fall on 1, 3, 4, 6, 8, 9… - a quasi-periodic pattern that never settles into a loop, the irrational cousin of the Euclidean rhythm. By Beatty's theorem two such sequences with complementary ratios partition every beat exactly once. Use it for organic, non-repeating grooves. A rising edge on in1 resets.

■□■□ **Bell** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Bell-number sequencer (a first as a synth block): outputs the Bell numbers - 1,1,2,5,15,52,203,877… - the count of ways to partition a set of n elements, taken modulo Base, one per in0 clock. They are built here by Aitken's array (pure additions, the analogue of Pascal's triangle for set partitions), so taking them mod Base is exact. The values explode combinatorially, so mod Base they scramble into a busy, evenly-spread pattern - a melody from the deep counting numbers of combinatorics, distinct from the additive recurrences. A rising edge on in1 resets to the top.

■□■□ **BinaryPalindrome** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Binary-palindrome sequencer (a first as a synth block): steps through the numbers that read the same forwards and backwards in BINARY 0, 1, 3, 5, 7, 9, 15, 17, 21, 27… (5 = 101, 9 = 1001, 21 = 10101). Mirror symmetry in base two rather than base ten, so the pattern is quite different from the decimal palindromes. Each clock advances one term, folded modulo Base. A base-two mirror class distinct from the decimal Palindrome sequencer. A rising edge on in1 resets.

■□■□ **BitReversal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Bit-reversal sequencer (a first as a synth block): emits the bit-reversal permutation - count up 0, 1, 2, 3… but flip the binary digits end-for-end before reading the value (with 4 bits: 0, 8, 4, 12, 2, 10, 6, 14…). This is the scrambled order in which the fast Fourier transform visits its data, a perfect shuffle that maximally spreads consecutive indices apart. Bits sets the word length (and so the loop length, 2^Bits). A deterministic maximal-spread permutation distinct from the random and quasi-random sources. A rising edge on in1 resets.

■□■□ **Boustrophedon** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Boustrophedon-triangle sequencer (a first as a synth block): reads out the Seidel-Entringer-Arnold triangle one cell at a time, row by row. The triangle is built ox-plough style - each row is filled by snaking left-to-right then right-to-left, accumulating the row above - and its diagonal is the Euler zigzag sequence. Walking the interior cells instead of just the diagonal gives a rippling, terraced contour folded modulo Base, distinct from the single-line Euler, Tangent and Secant sequencers that only sample its edges. A rising edge on in1 resets.

■□■□ **BrianBrain** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Brian's Brain cellular automaton (a first as a synth block): a 4x4 torus grid of three-state cells - ready, firing, dying - evolves on every rising edge of the in0 clock. A ready cell ignites only when exactly two neighbours are firing; a firing cell always passes to dying; a dying cell falls back to ready. Where Conway's Game of Life has two states, this refractory third state keeps structures restless - they almost never settle, throwing off gliders and sparks. Output is the firing-cell density as a 0..1 stepped CV. A rising edge on in1 reseeds; Seed sets the random fill.

■□■□ **Cantor** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Cantor-set rhythm gate (a first as a synth block): the step counter is read in base 3 and the output goes high only when NO ternary digit is 1 - exactly the membership rule of the Cantor middle-thirds set. The result is a self-similar fractal on/off pattern: dense clusters of hits separated by silences of every scale, the same gappy structure repeating inside itself at 1/3, 1/9, 1/27… A deeply uneven yet deterministic rhythm with built-in fractal phrasing, unlike the even fair-share of Thue-Morse. A rising edge on in1 resets the counter.

■□■□ **Catalan** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Catalan-number sequencer (a first as a synth block): outputs the Catalan numbers - 1,1,2,5,14,42,132,429,1430… - taken modulo Base, one per in0 clock. They count an astonishing range of things (the ways to balance n pairs of brackets, to triangulate a polygon, the paths that never cross the diagonal) and are built here by the convolution C(n) = sum

C(i)C(n-1-i), so taking them mod Base is exact. They explode faster than 2^n, so mod Base they scramble into a busy pattern - a melody from the most ubiquitous numbers in combinatorics, distinct from the Bell and Motzkin counts. A rising edge on in1 resets.

■□■□ **CenteredHexagonal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Centered-hexagonal-number sequencer (a first as a synth block): steps through the centered hexagonal numbers 1, 7, 19, 37, 61, 91, 127, 169, 217… = 3n(n+1) + 1 - a central dot ringed by hexagons (the hex or honeycomb numbers). Their partial sums are the perfect cubes, and the gaps between them are the multiples of six. Each clock advances one term, folded modulo Base. A honeycomb figurate distinct from the star and pyramidal sequencers. A rising edge on in1 resets.

■□■□ **CenteredSquare** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Centered-square-number sequencer (a first as a synth block): steps through the centered square numbers 1, 5, 13, 25, 41, 61, 85, 113, 145… = 2n^2 + 2n + 1 - a central dot ringed by successive square frames (the pattern of a diamond growing outward). They are the sums of two consecutive squares, and the gaps between them run through the multiples of four. Each clock advances one term, folded modulo Base. A centered figurate distinct from the ordinary square sequencer. A rising edge on in1 resets.

■□■□ **CenteredTriangular** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Centered-triangular-number sequencer (a first as a synth block): steps through the centered triangular numbers 1, 4, 10, 19, 31, 46, 64, 85, 109… = (3n^2 + 3n + 2)/2 - a central dot surrounded by growing triangular rings. The gaps between terms run through the multiples of three. Each clock advances one term, folded modulo Base. A centered figurate distinct from the ordinary triangular sequencer. A rising edge on in1 resets.

■□■□ **CentralBinomial** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Central-binomial sequencer (a first as a synth block): outputs the central binomial coefficients C(2n, n) - 1,2,6,20,70,252,924… - the biggest number in each EVEN row of Pascal's triangle (the count of equal-length up/down paths, the heart of the random walk). They are built here straight from Pascal's additive triangle, so modulo Base is exact, and they grow like 4^n so mod Base they scramble busily. The combinatorial spine of Pascal's triangle as a melody, distinct from the Catalan numbers that are derived from them. A rising edge on in1 resets.

■□■□ **Champernowne** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Champernowne-constant sequencer (a first as a synth block): walks the decimal digits of Champernowne's constant 0.123456789101112131415… - simply the counting numbers written out and concatenated - emitting one digit per in0 clock as a 0..1 stepped melody (digit / 9). Because the constant is provably NORMAL, the digit stream eventually contains every finite pattern, every motif, infinitely often: it climbs 1..9 then folds into the rolling two- and three-digit runs of 10,11,12…, a melody that is rigidly deterministic yet endlessly inventive. Distinct from the integer sequences - this plays the digits themselves. A rising edge on in1 resets.

■□■□ **ChenPrime** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Chen-prime sequencer (a first as a synth block): steps through the Chen primes - primes p for which p+2 is either prime OR a product of two primes (a semiprime): 2, 3, 5, 7, 11, 13, 17, 19, 23, 29… Chen Jingrun proved there are infinitely many, a celebrated partial result toward the twin-prime and Goldbach conjectures. They are denser than the twin primes but still a proper prime subset. Each clock advances one term, folded modulo Base. A near-twin prime class distinct from the strict twin-prime sequencer. A rising edge on in1 resets.

■□■□ **CircularPrime** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Circular-prime sequencer (a first as a synth block): steps through the circular primes 2, 3, 5, 7, 11, 13, 17, 31, 37, 71, 73, 79, 97, 113… - primes that stay prime under EVERY cyclic rotation of their digits (197 -> 971 -> 719 are all prime). A rare and elegant prime property combining primality with digit rotation. Each clock advances one term, folded modulo Base. A rotation-stable prime class distinct from the twin and Sophie-Germain sequencers. A rising edge on in1 resets.

■□■□ **Collatz** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Collatz-conjecture sequencer (a first as a synth block): walks the famous 3n+1 trajectory - on each rising edge of the in0 clock the number halves if even or becomes 3n+1 if odd, and the output is its altitude (log2 of the value, 0..1) so you hear the trajectory's jagged climbs and plunges. When it reaches 1 the tour restarts from Start. A rising edge on in1 also restarts. Different Start values give wildly different melodic contours from one deterministic rule.

■□■□ **CollatzPeak** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Collatz-peak sequencer (a first as a synth block): each in0 clock advances the step n and outputs the HIGHEST value the Collatz trajectory of n ever climbs to before falling to 1, folded into Steps. Some starting numbers shoot up to enormous heights before collapsing (27 soars past 9000 on its way down), and the peak heights leap about unpredictably from one n to the next. Where CollatzSteps counts the journey's LENGTH, this measures its greatest ALTITUDE - a different chaotic contour from the same famous iteration. A rising edge on in1 resets.

■□■□ **CollatzSteps** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Collatz-stopping-time sequencer (a first as a synth block): each in0 clock advances the step n and outputs how many steps n takes to reach 1 under the Collatz rule (halve it if even, triple-plus-one if odd) - the total stopping time, folded into Steps. Whether EVERY number reaches 1 is the famous unsolved Collatz conjecture; the step counts jump around wildly with no pattern (n=27 takes 111 steps). Where the Collatz block plays the trajectory itself, this plays its LENGTH - a chaotic-looking melody from an unproven theorem. A rising edge on in1 resets.

■□■□ **ContinuedFraction** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Continued-fraction sequencer (a first as a synth block): emits the partial quotients of the square root of N - the integers a0; a1, a2… in sqrt(N) = a0 + 1/(a1 + 1/(a2 + …)). For any non-square N this expansion is eventually PERIODIC (sqrt(7) = 2; 1,1,1,4 repeating; sqrt(23) = 4; 1,3,1,8…), and the periodic block is palindromic, so the sequencer produces a clean repeating melodic loop whose shape is fixed by N. A number-theoretic period distinct from the digit-orbit and Farey sequencers. A rising edge on in1 resets.

■□■□ **Conway** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Conway-Hofstadter sequencer (a first as a synth block): the recurrence a(n) = a(a(n-1)) + a(n - a(n-1)) with a(1)=a(2)=1 - Conway's '$10000 sequence', so named for the prize he offered to anyone who could tame its wild early behaviour. The output is a(n)/n, which lurches chaotically near the start, swinging widely above and below 0.5, then (as Mallows proved) gradually settles toward exactly 0.5 - a turbulence that calms as it goes. Each in0 clock advances one term. A famous self-referential drift distinct from the Hofstadter Q-sequence. A rising edge on in1 resets; the first 512 terms then loop.

■□■□ **CopelandErdos** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Copeland-Erdos sequencer (a first as a synth block): reads out the digits of the Copeland-Erdos constant 0.235711131719232931… - formed by writing the prime numbers one after another. Copeland and Erdos proved this constant is normal in base ten (its digits are perfectly evenly distributed), so the melody is a statistically uniform digit stream that nonetheless encodes the primes in order. Each clock advances one decimal digit, output as a 0..1 pitch step. A prime-digit stream distinct from the Champernowne and prime-gap sequencers. A rising edge on in1 resets.

▪▫▪▫ **CyclicCA** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Cyclic cellular automaton (a first as a synth block): every cell holds one of N colours arranged in a cycle, and a cell advances to the next colour if at least Threshold neighbours already wear it - rock beats scissors beats paper beats rock. From noise it self-organises into rotating spiral waves that endlessly consume one another. Each rising edge of the in0 clock steps one generation; the output is the grid's mean colour as a smoothly cycling CV. A demolition-derby of colours quite unlike the Game-of-Life family. A rising edge on in1 reseeds.

▪▫▪▫ **DeBruijn** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

De Bruijn sequence gate (a first as a synth block): plays a cyclic binary sequence in which every possible 5-bit pattern appears exactly once before the 32-step cycle repeats - the shortest sequence that tours all sub-patterns. Advancing on the in0 clock it fires a dense, structured, maximally-varied rhythm that never presents the same five-step window twice within a bar. Where Euclid spreads pulses evenly and ThueMorse is self-similar, the de Bruijn cycle is exhaustive. A rising edge on in1 resets to the start.

▪▫▪▫ **Decagonal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Decagonal-number sequencer (a first as a synth block): steps through the ten-sided polygonal numbers 1, 10, 27, 52, 85, 126, 175, 232… = n(4n-3) - dots arranged in nested ten-gons, the gaps between terms growing by 8 each step. Each clock advances one term, folded modulo Base. A ten-gon figurate climb distinct from the octagonal and nonagonal sequencers. A rising edge on in1 resets.

▪▫▪▫ **Dedekind** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Dedekind-psi sequencer (a first as a synth block): each in0 clock advances the step n and outputs psi(n)/n - 1, where psi is the Dedekind psi function, n times the product of (1 + 1/p) over the primes p dividing n (the count of points on a certain modular curve). It is the mirror image of the totient: where the totient REMOVES a fraction for each prime, psi ADDS one, so the output sits near zero at the primes and rises at the factor-rich numbers - the inverse contour of the Jordan and Totient curves. A multiplicative-growth melody distinct from the totient-family dips. A rising edge on in1 resets.

▪▫▪▫ **Delannoy** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Delannoy-number sequencer (a first as a synth block): outputs the central Delannoy numbers - 1,3,13,63,321,1683… - which count the paths a KING takes across a chessboard from one corner to the other, moving only right, up or diagonally. They are built here by the lattice recurrence D(i,j) = D(i-1,j) + D(i,j-1) + D(i-1,j-1) read down the diagonal, so taking them modulo Base is exact (pure additions). A king-path melody from two-dimensional lattice combinatorics, distinct from the one-dimensional recurrences. A rising edge on in1 resets.

▪▫▪▫ **Derangement** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Derangement sequencer (a first as a synth block): steps through the subfactorials 1, 0, 1, 2, 9, 44, 265, 1854… - the number of ways to shuffle n items so that NONE ends up in its own place (the hat-check problem). They satisfy D(n) = (n-1)(D(n-1) + D(n-2)) and the ratio D(n)/n! homes in on 1/e, so almost exactly 37 percent of all shufflings are derangements. Folded modulo Base. A fixed-point-free permutation count distinct from the factorial and Bell sequencers. A rising edge on in1 resets.

▪▫▪▫ **DigitCube** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Digit-cube orbit sequencer (a first as a synth block): iterates the map that replaces a number by the sum of the CUBES of its digits, whose orbits settle on the fixed points 1, 153, 370, 371, 407 (the three-digit narcissistic numbers) or short cycles such as 55 -> 250 -> 133 -> 55 and 136 -> 244 -> 136. Each clock advances one step and emits the value folded into a pitch, reseeding to tour fresh orbits. A cube-of-digits dynamic distinct from the factorial and reverse-add orbits. A rising edge on in1 resets.

▪▫▪▫ **DigitFactorial** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Digit-factorial orbit sequencer (a first as a synth block): iterates the factorion map - replace a number by the sum of the factorials of its digits - whose orbits famously fall into the fixed points 1, 2, 145, 40585 or the cycle 169 -> 363601 -> 1454 -> 169. Each clock advances one step and emits the value folded into a pitch, reseeding every so often to tour fresh orbits. A factorial-of-digits dynamic distinct from the reverse-add and cube orbits. A rising edge on in1 resets.

▪▫▪▫ **DigitProduct** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Digit-product sequencer (a first as a synth block): each in0 clock advances the step n and outputs the product of n's decimal digits (24 -> 2*4 = 8, 39 -> 27), folded into Steps. Unlike the steadily-climbing digit SUM, the product swings wildly and collapses to ZERO whenever any digit is 0 (every tenth number, all the hundreds…), punching rhythmic holes into the melody. A multiplicative digit contour with its own jagged, gap-ridden shape distinct from the digit sum and reversal. A rising edge on in1 resets.

▪▫▪▫ **DigitReverse** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Digit-reversal sequencer (a first as a synth block): each in0 clock advances the step n and outputs n with its decimal digits written backwards (123 -> 321, 250 -> 52), folded into Steps. The reversal scrambles the steady count into a jagged, self-similar permutation - the ones digit becomes the loudest place, so the melody leaps by big jumps within each block of ten and resets at each new power of ten. A digit-permutation contour distinct from the digit-SUM and the value-based sequences. A rising edge on in1 resets.

▪▫▪▫ **DigitSum** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Digit-sum sequencer (a first as a synth block): each in0 clock advances the step n and outputs the sum of n's digits in base Base, folded into Steps as a 0..1 melody. As n counts up, the digit sum climbs steadily then drops sharply at every carry (9 -> 10, 99 -> 100), tracing a self-similar sawtooth whose teeth get longer in higher place values - the fingerprint of positional notation itself. Base reshapes the pattern (base 2 gives the popcount/binary-weight contour, base 10 the familiar decimal roll). A digit-structure melody distinct from the value-based recurrences. A rising edge on in1 resets.

▪▫▪▫ **DistinctParts** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Distinct-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of DISTINCT positive integers 1, 1, 1, 2, 2, 3, 4, 5, 6, 8, 10, 12, 15… (4 = 4 = 3+1, so two ways). Euler proved this equals the count of partitions into ODD parts, a famous identity. It grows smoothly and sub-exponentially, so folded modulo Base it gives a gentle, slowly-rising organic contour distinct from the explosive zigzag sequencers and the unrestricted-partition family. A rising edge on in1 resets.

▪▫▪▫ **Divisors** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Divisor-count sequencer (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs d(n), the number of divisors of n, folded into Steps as a 0..1 melody. Primes drop to the floor (d = 2) while the highly-composite numbers (12, 24, 36, 48, 60…) leap up with many divisors, so the line is a spiky staircase that spotlights the most factor-rich integers - the arithmetic opposite of the Totient dip. A melody driven by raw divisibility, distinct from the additive and word sequences. A rising edge on in1 resets.

▪▫▪▫ **Dragon** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Dragon-curve (paperfolding) sequencer (a first as a synth block): outputs the regular paperfolding sequence - the fold directions of the famous dragon-curve fractal - as a 0/1 gate that advances on the in0 clock (1,1,0,1,1,0,0,1…). Strip the trailing zeros from the step counter and the next bit decides the turn, giving a sequence that is perfectly self-similar (it contains scaled copies of itself) yet never periodic. Between Euclid's even spread and ThueMorse's parity, the dragon fold is the fractal one. A rising edge on in1 resets.

▪▫▪▫ **EKG** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

EKG-sequence sequencer (a first as a synth block): walks the EKG (or electrocardiogram) sequence 1, 2, 4, 6, 3, 9, 12, 8, 10, 5, 15… where each term is the smallest number not yet used that shares a common factor with the previous one. Plotted, it traces a jagged heartbeat-like trace, and it is a proved permutation of the positive integers with a deep three-line structure. Each clock advances one term, folded modulo Base into a pitch. A gcd-chained permutation distinct from the digit and prime sequencers. A rising edge on in1 resets.

▪▫▪▫ **EuclidGate** [SEQ] [CV] [FX] [SYNTH] in: Clock out: CV

Euclidean gate generator: distributes Pulses as evenly as possible across Steps and emits that rhythm, advancing one step on each rising edge of the in0 clock - the maximally-even Euclidean patterns (e.g. 3-in-8 = the tresillo) behind so much generative rhythm. Rotate shifts the pattern's starting point.

▪▫▪▫ **Euler** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Euler zigzag sequencer (a first as a synth block): steps through the up-down (alternating-permutation) numbers 1, 1, 1, 2, 5, 16, 61, 272, 1385… that count the ways to arrange 1..n so the values go up, down, up, down. Built from the boustrophedon triangle that snakes left-then-right across each row, it is the combinatorial heart shared by the tangent and secant functions. The term is folded modulo Base into a stepped pitch, giving a fast-accelerating melodic climb distinct from the factorial Pisano and Catalan sequencers. A rising edge on in1 resets.

▪▫▪▫ **Eulerian** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Eulerian-number sequencer (a first as a synth block): reads the triangle of Eulerian numbers row by row - A(n,k), the number of permutations of n items with exactly k ascents (1; 1,1; 1,4,1; 1,11,11,1; 1,26,66,26,1…) - modulo Base. Built by A(n,k) = (k+1)A(n-1,k) + (n-k)A(n-1,k-1), so mod Base is exact. The rows are symmetric and sum to a factorial, and they are the coefficients that connect powers to binomial sums. A permutation-ascent triangle melody distinct from the Stirling and Pascal triangles. A rising edge on in1 resets.

▪▫▪▫ **Farey** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Farey-sequence sequencer (a first as a synth block): walks the Farey sequence of order N - every fraction in 0..1 whose denominator is at most N, listed in increasing size (for N=5: 0, 1/5, 1/4, 1/3, 2/5, 1/2, 3/5, 2/3, 3/4, 4/5, 1). It steps with the exact integer mediant recurrence, so each clock outputs the next fraction directly as a control voltage and loops back at 1. A rising, unevenly-spaced rational staircase whose gaps mirror the deep structure of the rationals, distinct from the digit-orbit sequencers. A rising edge on in1 resets.

▪▫▪▫ **FemaleMale** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hofstadter Female-Male sequencer (a first as a synth block): outputs the 'Female' half of Hofstadter's famous pair of INTERTWINED sequences, F(n) = n - M(F(n-1)) and M(n) = n - F(M(n-1)) - two sequences each defined in terms of the OTHER, an elegant mutual recursion. F grows smoothly (close to the golden-ratio staircase) while leaning on M at every step. Each in0 clock advances one term, folded into Steps. A mutually-recursive melody distinct from the single self-referential Hofstadter sequences. A rising edge on in1 resets.

■□■□ **Fibbinary** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fibbinary gate (a first as a synth block): the output goes high when the step n is a fibbinary number - one whose binary representation has NO two adjacent 1-bits (0,1,2,4,5,8,9,10,16…). These are exactly the numbers expressible as a sum of non-consecutive powers of two, the base-2 mirror of Zeckendorf's non-consecutive Fibonacci sums. The gate fires in a sparse, self-similar pattern whose density follows the golden ratio, distinct from the prime and happy-number gates. A rising edge on in1 resets.

■□■□ **Fibonacci** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fibonacci-number gate (a first as a synth block): the output goes high when the step n is itself a Fibonacci number - 0, 1, 2, 3, 5, 8, 13, 21, 34, 55… Because each Fibonacci number is roughly the golden ratio times the last, the gaps grow GEOMETRICALLY, so the gate fires fast and densely at first then thins out exponentially - a self-similar, golden-spaced rhythm quite unlike the polynomial slow-down of the square and triangular gates. The test is Gessel's: n is Fibonacci exactly when 5n^2+4 or 5n^2-4 is a perfect square. A rising edge on in1 resets.

■□■□ **FibonacciWord** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fibonacci-word sequencer (a first as a synth block): the infinite binary word grown by the substitution 0 -> 01, 1 -> 0, giving 0100101001001010010100100… - the most-ordered aperiodic sequence there is, the symbolic shadow of the golden ratio. The 1s land exactly where the golden-ratio Beatty sequence says, so it is a self-similar two-note rhythm that never repeats yet is utterly deterministic. Each clock advances one symbol, output as a 0 or 1 gate. A golden-ratio word distinct from the Thue-Morse and Rudin-Shapiro sequencers. A rising edge on in1 resets.

■□■□ **FibWord** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fibonacci-word sequencer (a first as a synth block): the infinite Fibonacci word over {0,1} grown by the substitution 0 -> 0 1, 1 -> 0 (its own fixed point) - 0,1,0,0,1,0,1,0,0,1,0,0,1… - emitted as a 0/1 gate, one symbol per in0 clock. It is the canonical Sturmian sequence: the 1s fall at the golden-ratio spacing, the quasicrystal of one dimension, balanced so any two windows of equal length hold the same count of 1s to within one. A structured aperiodic gate built on phi, distinct from the self-describing Kolakoski. A rising edge on in1 resets; the first 256 symbols then loop.

■□■□ **Fubini** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fubini-number sequencer (a first as a synth block): steps through the ordered Bell (Fubini) numbers 1, 1, 3, 13, 75, 541, 4683… - the number of ways to rank n items allowing ties (the outcomes of a race with photo finishes), equivalently the number of ordered set partitions. They grow faster than the plain Bell numbers because the blocks are ordered. The exact value is folded modulo Base. A weak-ordering count distinct from the unordered Bell sequencer. A rising edge on in1 resets.

■□■□ **Galton** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Galton-board (bean machine) source (a first as a synth block): each rising edge of the in0 clock drops a bean through Rows rows of pins, bouncing left or right on a fair coin at every pin; the output is the bin it settles in, 0..1. Because that bin is the SUM of Rows coin flips it is binomially distributed - clustered around the centre and rare at the edges, the discrete bell curve - unlike the flat, uniform spread of the Random source. Rows sets the curve's sharpness (more rows = tighter central bias). A Gaussian-weighted random step for natural-feeling movement. A rising edge on in1 reseeds.

■□■□ **GameOfLife** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Conway's Game of Life sequencer (a first as a synth block): a 32-cell torus grid (8 wide x 4 tall) evolves one Conway generation on every rising edge of the in0 clock - a live cell with 2 or 3 neighbours survives, a dead cell with exactly

3 is born (the classic B3/S23) - and the output is the live population density as a 0..1 stepped CV. Where the 1-D Automaton runs a row rule, this is full 2-D Life with its gliders, blinkers and still-lifes: the population breathes, oscillates or settles into stable colonies. A rising edge on in1 reseeds the grid; Seed sets the random fill. Genuinely alive modulation.

▪▫▪▫ GoldenAngle [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Golden-angle (phyllotaxis) sequencer (a first as a synth block): each in0 clock adds the golden ratio's fractional part (~0.618) to a running position wrapped into 0..1 - the additive recurrence that places sunflower seeds and pinecone scales at the 137.5-degree golden angle, the most irrational rotation there is. The result is a maximally even, low-discrepancy melody: successive notes land as far from all previous notes as possible, filling the range without ever repeating or clustering. The optimal spread, distinct from the bit-reversal Van der Corput. A rising edge on in1 resets to the start.

▪▫▪▫ Golomb [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Golomb self-describing sequencer (a first as a synth block): outputs Golomb's sequence - 1,2,2,3,3,4,4,4,5,5,5… - the unique non-decreasing sequence that DESCRIBES ITSELF: the number n appears in it exactly a(n) times (1 appears once, 2 appears twice, 3 appears three times, and so on). It climbs in ever-longer plateaus whose lengths are given by the sequence's own earlier values. Each in0 clock advances one term, folded into Steps, giving a gently-rising staircase of stretching steps. A self-describing melody distinct from the self-referential Hofstadter recurrences. A rising edge on in1 resets.

▪▫▪▫ Gould [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Gould-sequence sequencer (a first as a synth block): outputs Gould's sequence - 2 raised to the number of 1-bits of the step n - folded into Steps, one per in0 clock. By Kummer's theorem this is exactly the count of ODD entries in row n of Pascal's triangle, so plotting it draws the Sierpinski-triangle skyline: it leaps to a new power of two on the dense numbers and collapses to 1 at the powers of two. A doubling, self-similar contour distinct from the plain popcount it is built on. A rising edge on in1 resets.

▪▫▪▫ GrayCode [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Gray-code sequencer (a first as a synth block): advances a counter on each rising edge of the in0 clock and outputs its reflected binary Gray code (n XOR n»1), folded into 16 steps - so successive values always differ by exactly one bit. The result is a stepped CV that walks 0, 1, 3, 2, 6, 7, 5, 4… visiting levels in an order where every move is a single minimal change, giving smooth one-step contours that still tour the whole range without repeating until the cycle completes. A rising edge on in1 resets.

▪▫▪▫ Halton [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Halton-sequence sequencer (a first as a synth block): a quasi-random low-discrepancy sequence built from the radical inverse - take the index, write it in base b, then flip the digits across the decimal point (1,2,3 in base 2 give .1, .01, .11 = 0.5, 0.25, 0.75). Unlike true randomness it fills the interval evenly, never clumping or leaving gaps, the workhorse of quasi-Monte-Carlo sampling. As a melody it sounds scattered yet always lands in the holes it left before. Base picks the digit base. A deterministic gap-filling scatter distinct from the pseudo-random generators. A rising edge on in1 resets.

▪▫▪▫ Hamming [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hamming-number sequencer (a first as a synth block): steps through the regular or 5-smooth numbers 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 20, 24, 25, 27… - the integers whose only prime factors are 2, 3 and 5. Famous from Dijkstra's programming exercise, they are exactly the just-intonation frequency ratios, so the sequence is a ladder of harmonically pure intervals. Each clock advances one term, folded modulo Base. A smooth-number ladder distinct from the prime and figurate sequencers. A rising edge on in1 resets.

▪▫▪▫ **Happy** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Happy-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is a happy number - one that reaches 1 when you repeatedly replace it by the sum of the squares of its digits (7 -> 49 -> 97 -> 130 -> 10 -> 1). The unhappy numbers instead fall forever into the cycle 4,16,37,58,89,145,42,20. The happy numbers are scattered with no pattern (1,7,10,13,19,23,28,31,32,44…), so the gate fires on a sparse, irregular, deterministic rhythm distinct from the primes. A rising edge on in1 resets.

▪▫▪▫ **Harshad** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Harshad-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is a Harshad (Niven) number - one divisible by the sum of its own digits (18 is, since 1+8=9 divides 18; 19 is not). Harshad numbers are common among small values then thin out gradually, so the gate fires densely at first and slowly sparsens - a denser, more regular cousin of the prime gate, with little runs and gaps tied to the decimal digits. A rising edge on in1 resets.

▪▫▪▫ **Heptagonal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Heptagonal-number sequencer (a first as a synth block): steps through the seven-sided polygonal numbers 1, 7, 18, 34, 55, 81, 112, 148… = n(5n-3)/2 - dots arranged in nested heptagons. Like all polygonal numbers they are second differences constant (here the gaps grow by 5 each time). Each clock advances one term, folded modulo Base. A seven-gon figurate climb distinct from the triangular, square and pentagonal sequencers. A rising edge on in1 resets.

▪▫▪▫ **Hexagonal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hexagonal-number sequencer (a first as a synth block): steps through the six-sided polygonal numbers 1, 6, 15, 28, 45, 66, 91, 120… = n(2n-1) - dots in nested hexagons. They are exactly the odd-indexed triangular numbers, and every perfect number is hexagonal. Each clock advances one term, folded modulo Base. A six-gon figurate climb distinct from the triangular and pentagonal sequencers. A rising edge on in1 resets.

▪▫▪▫ **HighlyComposite** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Highly-composite-number sequencer (a first as a synth block): steps through the highly composite numbers 1, 2, 4, 6, 12, 24, 36, 48, 60, 120… - each having MORE divisors than any smaller number, the record-breakers of factor-richness studied by Ramanujan. They are the anti-primes (12 and 60 and 360 are why clocks and circles are divided as they are). Each clock advances one record-setter, folded modulo Base. A divisor-maximising class distinct from the prime and figurate sequencers. A rising edge on in1 resets.

▪▫▪▫ **Hipparchus** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hipparchus sequencer (a first as a synth block): steps through the little Schroeder (super-Catalan) numbers 1, 1, 3, 11, 45, 197, 903, 4279… which count the ways to insert non-crossing brackets into a row of n items, or to subdivide a polygon by non-crossing diagonals. Plutarch records that Hipparchus counted the tenth term (103049) two thousand years before they were rediscovered. The exact value is folded modulo Base. A bracketing count distinct from the Catalan and Motzkin sequencers. A rising edge on in1 resets.

▪▫▪▫ **Hofstadter** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hofstadter Q-sequence source (a first as a synth block): the chaotic meta-Fibonacci recurrence Q(n) = Q(n - Q(n-1)) + Q(n - Q(n-2)) with Q(1)=Q(2)=1 - but where Fibonacci looks a fixed distance back, this looks back by its OWN recent values, so it never settles and wanders erratically around n/2. Each rising edge of the in0 clock advances one term; the output is Q(n)/n, a turbulent drift hovering near 0.5 with unpredictable excursions - deterministic chaos from pure integer recursion, distinct from the smooth Fibonacci-period Pisano. A rising edge on in1 resets; the first 512 terms then loop.

▪▫▪▫ **HofstadterG** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hofstadter G-sequence sequencer (a first as a synth block): the self-referential recurrence G(n) = n - G(G(n-1)) with G(0)=0, which folds the integers down by feeding the sequence's own output back into itself TWICE. Despite the tangled recursion it grows smoothly at the golden-ratio rate (G(n) is almost exactly floor(n/phi)), so it makes a gently-climbing staircase with an occasional flat step where the recursion catches up. Each in0 clock advances one term, folded into Steps. A self-referential melody distinct from the chaotic Q-sequence. A rising edge on in1 resets.

▪▫▪▫ **HofstadterH** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hofstadter H-sequence sequencer (a first as a synth block): the recurrence H(n) = n - H(H(H(n-1))) with H(0)=0 - the same self-referential idea as the G-sequence but nesting the sequence inside itself THREE deep, so it grows at the 'plastic number' rate (~n times 0.6823) rather than the golden ratio. The result is a smoothly-climbing staircase that rises a touch slower than G, with its own pattern of flat treads. Each in0 clock advances one term, folded into Steps. A deeper self-recursion distinct from the G-sequence. A rising edge on in1 resets.

▪▫▪▫ **HofstadterQ** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Hofstadter Q-sequencer (a first as a synth block): the chaotic meta-Fibonacci sequence Q(n) = Q(n - Q(n-1)) + Q(n - Q(n-2)), seeded Q(1) = Q(2) = 1, giving 1, 1, 2, 3, 3, 4, 5, 5, 6, 6, 6, 8… Unlike Fibonacci it looks BACK by amounts that the sequence itself decides, so it wobbles unpredictably and it is unknown whether it stays defined forever. A self-indexing, semi-chaotic climb folded modulo Base. A strange-loop recurrence distinct from the smooth linear sequencers. A rising edge on in1 resets.

▪▫▪▫ **Ikeda** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Ikeda-map chaos sequencer (a first as a synth block): iterates the Ikeda map - a model of light circulating in a nonlinear optical cavity - whose state rotates by an angle that itself depends on the state, folding the plane into an intricate strange attractor. On each rising edge of the in0 clock it advances one step and outputs the x coordinate as a stepped CV that hops chaotically around the attractor's hooked arms. U sets how deep into chaos it runs. A different flavour again from the angular Henon, smooth Rossler or tumbling pendulum. A rising edge on in1 reseeds.

▪▫▪▫ **Jacobsthal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Jacobsthal-sequence sequencer (a first as a synth block): the recurrence J(n) = J(n-1) + 2*J(n-2) - giving 0,1,1,3,5,11,21,43,85,171… - a Fibonacci cousin whose doubled second term makes successive ratios tend to 2 instead of the golden ratio. Each rising edge of the in0 clock advances one term, taken modulo Steps. Its faster, near-doubling growth gives a melody that leaps up the scale more aggressively than Fibonacci, a distinct integer-recurrence line from the Pisano, Padovan and Pell sequencers. A rising edge on in1 resets.

▪▫▪▫ **JacobsthalLucas** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Jacobsthal-Lucas sequencer (a first as a synth block): steps through the Jacobsthal-Lucas numbers 2, 1, 5, 7, 17, 31, 65, 127, 257… defined by a(n) = a(n-1) + 2*a(n-2), the companion to the Jacobsthal sequence. Their values hug the powers of two (each is 2^n plus or minus one), so the sequence is a wobble around a clean octave doubling. Folded modulo Base. A near-power-of-two companion distinct from the Jacobsthal and Mersenne sequencers. A rising edge on in1 resets.

▪▫▪▫ **Jordan** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Jordan-totient sequencer (a first as a synth block): each in0 clock advances the step n and outputs J2(n)/n^2, where J2 is Jordan's totient - a generalisation of Euler's totient that counts pairs coprime to n rather than single numbers, given by n^2 times the product of (1 - 1/p^2) over the primes p dividing n. The output rides high (near 1) at the primes and dips at the factor-rich numbers, a SHARPER version of the totient curve because each prime cuts by 1/p^2. A second-order coprimality melody distinct from the ordinary Totient. A rising edge on in1 resets.

▪▫▪▫ **Kaprekar** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Kaprekar-routine sequencer (a first as a synth block): each in0 clock advances the step n and counts how many rounds of Kaprekar's routine its four-digit form takes to reach 6174 - the magic Kaprekar constant. One round sorts the digits up and down and subtracts the smaller from the larger (e.g. 3524 -> 5432 - 2345 = 3087 -> … -> 6174); EVERY four-digit number with at least two different digits funnels to 6174 in at most seven rounds, while repeated-digit numbers fall to zero. The output is that round count over 7, a strange staircase rising and falling with no obvious pattern. A rising edge on in1 resets.

▪▫▪▫ **Keith** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Keith-number gate (a first as a synth block): the output goes high when the step n is a Keith number (14, 19, 28, 47, 61, 75, 197…). To test n, seed a Fibonacci-like sequence with its digits - then each new term is the sum of the previous (digit-count) terms; n is a Keith number if that sequence eventually lands exactly on n. They are far rarer than the primes (only a hundred or so below a trillion), so the gate fires on a very sparse, irregular pulse - a deep, hard-to-find numerical rhythm distinct from the prime, happy and lucky gates. A rising edge on in1 resets.

▪▫▪▫ **Kolakoski** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Kolakoski self-describing sequencer (a first as a synth block): the unique sequence over {1,2} that is its OWN run-length encoding - 1,2,2,1,1,2,1,2,2,1,2,2… - so reading off the lengths of its own runs reproduces it exactly. Each rising edge of the in0 clock emits the next symbol as a two-level CV (1 low, 2 high): a quasi-periodic gate pattern that never settles into a loop and has no underlying period, aperiodic yet fully deterministic. A rhythmic counterpart to the number-theoretic pitch sequences. A rising edge on in1 resets; the first 256 symbols are generated, then the phrase repeats.

▪▫▪▫ **LaggedFibonacci** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Lagged-Fibonacci sequencer (a first as a synth block): generalises the Fibonacci rule by adding two FAR-apart earlier terms - x(n) = x(n-7) + x(n-10) modulo a power of two - keeping a short ring of past values. This classic pseudo-random generator has an enormously long period and the additive, memory-of-the-past structure gives it a subtly different texture from the bit-twiddling generators. It plays a deterministic scrambled melody that drifts and never quite repeats. Bits sets the pitch range. An additive-recurrence noise distinct from the shift-register ones. A rising edge on in1 resets.

▪▫▪▫ **Lah** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Lah-number sequencer (a first as a synth block): reads the triangle of unsigned Lah numbers row by row - L(n,k), the number of ways to partition n items into k non-empty ORDERED lists (1; 2,1; 6,6,1; 24,36,12,1…) - modulo Base. Built by L(n,k) = (n+k-1)*L(n-1,k) + L(n-1,k-1), so mod Base is exact. The Lah numbers are the 'Stirling numbers of the third kind' that convert rising factorials into falling ones. An ordered-partition triangle melody distinct from the two Stirling triangles. A rising edge on in1 resets.

▪▫▪▫ **Langton** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Langton's Ant turmite sequencer (a first as a synth block): a single 'ant' walks an 8x4 torus of cells - turning right on a white cell, left on a black one, flipping each cell it leaves, then stepping forward. From a blank grid it scribbles chaotically for a while, then abruptly locks into a periodic 'highway' - emergent order from one trivial rule. Each rising edge of the in0 clock advances the ant Steps cells; the output is the ant's column position as a 0..1 stepped CV. A rising edge on in1 resets the ant to the centre of a clear grid. Unlike the cellular automata this is a single moving agent, not a population.

▪▫▪▫ **Leonardo** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Leonardo-number sequencer (a first as a synth block): steps through the Leonardo numbers 1, 1, 3, 5, 9, 15, 25, 41, 67, 109… defined by L(n) = L(n-1) + L(n-2) + 1 - the Fibonacci rule with a plus-one twist. They are the sizes of the balanced binary trees used in Dijkstra's smoothsort, and each is one less than twice a Fibonacci number. A nearly-Fibonacci climb, slightly steeper, folded modulo Base. A rising edge on in1 resets.

▪▫▪▫ **LFSR** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Linear-feedback shift-register sequencer (a first as a synth block): clocks a 16-bit Galois LFSR with maximal-length feedback taps, the classic hardware pseudo-random generator behind scramblers, CRCs and chip-tune noise. It marches through all 65535 non-zero states in a fixed, repeatable order before looping, so it sounds random yet plays the exact same long pattern every time - a deterministic random melody. Bits sets how many low bits are read out as the pitch CV. A shift-register noise distinct from the word-mixing Xorshift. A rising edge on in1 resets the register.

▪▫▪▫ **Liouville** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Liouville-function gate (a first as a synth block): each rising edge of the in0 clock advances the step n and the output goes high when lambda(n) = +1, low when -1 - lambda being (-1) raised to the TOTAL number of prime factors of n counted with multiplicity. It is the multiplicative cousin of Thue-Morse (which counts bits, this counts prime factors): a deeply structured, conjecturally-balanced two-level pattern tied to the Riemann hypothesis through its running sum. An aperiodic gate built from the deepest texture of the primes. A rising edge on in1 resets.

▪▫▪▫ **LookSay** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Look-and-say sequencer (a first as a synth block): each in0 clock advances one term of Conway's look-and-say sequence - 1, 11, 21, 1211, 111221, 312211… - where each term is read aloud to make the next ('one 1' -> 11, 'two 1s' -> 21, 'one 2 one 1' -> 1211). The output is the LENGTH of each term, folded into Steps; those lengths grow by Conway's constant (~1.303685) each step, so the melody climbs in a fixed geometric ratio - an 'audioactive' sequence whose only digits are ever 1, 2 and 3. A self-describing melody distinct from the Kolakoski self-description. A rising edge on in1 resets; the first 16 terms then loop.

▪▫▪▫ **Lucas** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Lucas-number sequencer (a first as a synth block): the companion of the Fibonacci sequence - same add-the-last-two rule but seeded 2, 1 instead of 0, 1, giving 2,1,3,4,7,11,18,29,47,76… Its ratios converge on the very same golden ratio, so it shares Fibonacci's growth while landing on different values (the Lucas numbers turn up in primality testing and the closed form for Fibonacci itself). Each in0 clock advances one term, taken modulo Steps - a golden-growth melody with its own distinct contour. A rising edge on in1 resets.

▪▫▪▫ **Lucky** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Lucky-number gate (a first as a synth block): the output goes high when the step count is a 'lucky number' (1,3,7,9,13,15,21,25,31…) - survivors of a sieve like the prime sieve but which crosses out by POSITION instead of by value: keep every number, delete every 2nd, then from the survivors delete every 3rd, then every 7th, and so on. Astonishingly the luckies share many statistical traits with the primes (twin luckies, a similar density) despite being built from counting alone, not divisibility. A different deterministic sparse rhythm from the prime gate. A rising edge on in1 resets; the first 256-step phrase then loops.

■□■□ **LyndonWords** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Lyndon-word sequencer (a first as a synth block): steps through the number of binary Lyndon words of length n 2, 1, 2, 3, 6, 9, 18, 30, 56, 99… - the aperiodic necklaces, strings that are strictly smallest among all their rotations. Every string factors uniquely into a descending sequence of Lyndon words (the Chen-Fox-Lyndon theorem), making them a basis for free Lie algebras. Counted by Moebius inversion of the powers of two, the exact value is folded modulo Base. A primitive-string count distinct from the full necklace sequencer. A rising edge on in1 resets.

■□■□ **Markov** [SEQ] [CV] [FX] [SYNTH] in: In, Clock out: CV

Style-learning sequencer (a first as a synth block): quantises the input at in0 into 4 states and continuously learns the transition probabilities between them (a Markov chain; Memory sets how fast old transitions fade). On each rising edge of the in1 clock it samples the learned transitions from its current state to pick the next, improvising a NEW sequence in the style of what it heard - feed it a melody or CV and it riffs endless variations that obey the same move-to-move statistics. Output is the generated state as a 0..1 stepped CV.

■□■□ **MephistoWaltz** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Mephisto-waltz sequencer (a first as a synth block): the binary word grown by the substitution 0 -> 001, 1 -> 110, giving 001001110001001110110… - named after Liszt's waltz for its restless triple-time feel. It is an automatic sequence with curious correlation properties, closely related to the ternary expansion of position. Each clock advances one symbol, output as a 0 or 1 gate, giving a lurching three-against-two rhythm. A triple-time word distinct from the Fibonacci and Thue-Morse sequencers. A rising edge on in1 resets.

■□■□ **Mersenne** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Mersenne-number sequencer (a first as a synth block): steps through the Mersenne numbers 2^n - 1 = 1, 3, 7, 15, 31, 63, 127, 255… - the numbers that are all-ones in binary. When the exponent is prime these are the candidates for Mersenne PRIMES, the form that has held the record for the largest known prime for decades, hunted by the GIMPS distributed search. The exact value 2^n - 1 is folded modulo Base, giving a doubling-then-wrapping staircase. A power-of-two-minus-one climb distinct from the additive sequencers. A rising edge on in1 resets.

■□■□ **Mertens** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Mertens-function source (a first as a synth block): each in0 clock advances the step n and outputs the running sum of the Mobius function, M(n) = mu(1) + mu(2) + … + mu(n) - a famous slowly-staggering walk that creeps up and down by +/-1 and 0 as squarefree numbers of even and odd prime-count alternate. Whether it stays bounded by sqrt(n) (the disproved Mertens conjecture) is tied to the Riemann hypothesis. The output is centred on 0.5, drifting either side like a number-theoretic Brownian path that is in fact fully deterministic. Distinct from the per-step Mobius block. A rising edge on in1 resets.

■□■□ **MianChowla** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Mian-Chowla sequencer (a first as a synth block): the greedy Sidon sequence 1, 2, 4, 8, 13, 21, 31, 45, 66, 81… - built by always taking the smallest next integer such that ALL the pairwise sums stay distinct (no two pairs add to the same value). Sidon sets like this are prized in signal design and radio astronomy for their flat autocorrelation. The terms spread apart ever wider as collisions become harder to avoid. Folded modulo Base. A distinct-sums greedy set distinct from the prime and figurate sequencers. A rising edge on in1 resets.

▪▫▪▫ **MiddleSquare** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Middle-square-Weyl sequencer (a first as a synth block): a modern repair of von Neumann's 1949 middle-square method - square the number and keep the middle digits - which on its own quickly collapses to zero. Adding a Weyl sequence (a steadily advancing counter) each step cures the collapse, yielding a fast, high-quality pseudo-random stream that passes stringent statistical tests. It plays a deterministic but thoroughly scrambled melody. Bits sets the read-out pitch range. A squared-digit generator distinct from the shift-register and congruential ones. A rising edge on in1 resets.

▪▫▪▫ **Mobius** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Mobius-function source (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs the Mobius function mu(n) as a three-level CV - high for a squarefree number with an even count of prime factors (+1), centre for a non-squarefree number (0, it has a repeated prime), low for squarefree with an odd count (-1). The squarefree/squareful flicker and the parity swings produce a jittery three-state pattern that encodes the prime-factor structure of each integer. A number-theoretic trit stream found nowhere else in modular gear. A rising edge on in1 resets.

▪▫▪▫ **Motzkin** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Motzkin-number sequencer (a first as a synth block): outputs the Motzkin numbers - 1,1,2,4,9,21,51,127,323… - taken modulo Base, one per in0 clock. They count the ways to draw non-crossing chords between points on a circle (the chords-and-non-edges cousin of the bracket-counting Catalan), and are built here by the convolution M(n) = M(n-1) + sum M(i)M(n-2-i), so mod Base is exact. A combinatorial melody with its own growth rate (~3^n), distinct from the Catalan and Bell counts. A rising edge on in1 resets.

▪▫▪▫ **MultRoot** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Multiplicative-digital-root sequencer (a first as a synth block): each in0 clock advances the step n and repeatedly replaces it by the product of its digits until a single digit is left - the multiplicative digital root (39 -> 27 -> 14 -> 4). The output is that final digit over 9. Because any number containing a 0 (or that ever reaches one) collapses to 0, the melody is dotted with zeros among the 1..9 values, a sparse fingerprint of the multiplicative structure of each integer. Distinct from the additive digital root, which simply cycles 1..9. A rising edge on in1 resets.

▪▫▪▫ **Narayana** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Narayana's-cows sequencer (a first as a synth block): the 14th-century recurrence N(n) = N(n-1) + N(n-3) - giving 1,1,1,2,3,4,6,9,13,19,28… - from the Indian problem of how a herd grows if each cow bears a calf every year from its fourth year on. Its ratios converge on the SUPERGOLDEN ratio (~1.4656). Each rising edge of the in0 clock advances one term, taken modulo Steps. Its three-step memory gives a gentler, more loping climb than Fibonacci, distinct from the Padovan (N(n-2)+N(n-3)) it is often confused with. A rising edge on in1 resets.

▪▫▪▫ **Necklace** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Necklace sequencer (a first as a synth block): steps through the number of distinct two-colour necklaces of n beads 1, 2, 3, 4, 6, 8, 14, 20, 36, 60… - binary strings of length n counted as the same when rotated into one another. Burnside's counting lemma gives the count as the average over rotations, mixing Euler's totient with powers of two. The exact value is folded modulo Base. A cyclic-symmetry count distinct from the linear word sequencers. A rising edge on in1 resets.

▪▫▪▫ **Nonagonal** [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Nonagonal-number sequencer (a first as a synth block): steps through the nine-sided polygonal numbers 1, 9, 24, 46, 75, 111, 154, 204… = n(7n-5)/2 - dots arranged in nested nine-gons, the gaps between terms growing by 7 each step. Each clock advances one term, folded modulo Base. A nine-gon figurate climb distinct from the lower polygonal sequencers. A rising edge on in1 resets.

▪▫▪▫ Octagonal [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Octagonal-number sequencer (a first as a synth block): steps through the eight-sided polygonal numbers 1, 8, 21, 40, 65, 96, 133, 176… = n(3n-2) - dots arranged in nested octagons, the gaps between terms growing by 6 each step. Each clock advances one term, folded modulo Base. An eight-gon figurate climb distinct from the lower polygonal sequencers. A rising edge on in1 resets.

▪▫▪▫ Octahedral [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Octahedral-number sequencer (a first as a synth block): steps through the octahedral numbers 1, 6, 19, 44, 85, 146… = n(2nn + 1)/3 - the count of spheres stacked into a regular octahedron (two square pyramids base to base). Each is the sum of two consecutive square-pyramidal numbers. Each clock advances one term, folded modulo Base. A different polyhedral packing distinct from the tetrahedral and pyramidal sequencers. A rising edge on in1 resets.

▪▫▪▫ Padovan [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Padovan-sequence sequencer (a first as a synth block): the plastic-number analogue of Fibonacci - P(n) = P(n-2) + P(n-3), giving 1,1,1,2,2,3,4,5,7,9,12,16,21… - whose ratio of successive terms tends to the plastic number ~1.3247 instead of the golden ratio. Each rising edge of the in0 clock advances one term, taken modulo Steps so it stays in range. Its slower, gentler growth makes a melody that climbs more lazily than Fibonacci, distinct from the Pisano (Fibonacci mod m) and Perrin recurrences. A rising edge on in1 resets.

▪▫▪▫ Palindrome [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Palindrome gate (a first as a synth block): the output goes high when the step n reads the same backwards as forwards in decimal (1, 2, …, 9, 11, 22, …, 99, 101, 111, 121…). The palindromes thin out in a self-similar way - dense among the small numbers, then clustering near each new power of ten - so the gate fires in rhythmic bursts that stretch apart as the count climbs. A mirror-symmetry digit rhythm distinct from the divisibility and prime gates. A rising edge on in1 resets.

▪▫▪▫ Partition [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Partition-number sequencer (a first as a synth block): outputs p(n), the number of ways to write n as a sum of positive integers (4 = 4, 3+1, 2+2, 2+1+1, 1+1+1+1 so p(4)=5), taken modulo Base, one per in0 clock. The partition numbers - 1,1,2,3,5,7,11,15,22,30,42… - grow almost as fast as the primes thin out (Hardy-Ramanujan), and are built here by Euler's pentagonal-number recurrence (additions and subtractions only), so mod Base is exact. A melody from the deep additive combinatorics of the integers, distinct from the multiplicative divisor functions. A rising edge on in1 resets.

▪▫▪▫ PartitionsIntoPrimes [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Prime-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of PRIME numbers 1, 0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 6, 7… (7 = 7 = 5+2 = 3+2+2, so three ways). It restricts the famous partition function to prime parts only, tying the additive world of partitions to the multiplicative primes. Computed by the exact prime-restricted partition recurrence, folded modulo Base. A prime-restricted partition count distinct from the full partition and distinct-parts sequencers. A rising edge on in1 resets.

▪▫▪▫ PartitionsIntoSquares [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Square-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of perfect squares 1, 1, 1, 1, 2, 2, 2, 2, 3, 4… (allowing repeats, order ignored: 4 = 4 = 1+1+1+1, so two ways). It connects the additive partition world to the squares, and is bounded above by the four-squares theorem that says four are always enough. Folded modulo Base. A square-part partition count distinct from the prime-part and ordinary partition sequencers. A rising edge on in1 resets.

▪▫▪▫ PartitionsIntoTriangular [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Triangular-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of triangular numbers 1, 1, 1, 2, 2, 2, 4, 4, 4, 6… (the parts drawn from 1, 3, 6, 10, 15…). Gauss proved every number is a sum of at most three triangular numbers (his EUREKA theorem), and this sequence counts all the ways. Folded modulo Base. A triangular-part partition count distinct from the square-part and prime-part sequencers. A rising edge on in1 resets.

▪▫▪▫ Pascal [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pascal's-triangle sequencer (a first as a synth block): reads Pascal's triangle row by row - 1; 1,1; 1,2,1; 1,3,3,1… - taking each binomial coefficient modulo Base and emitting it as a 0..1 stepped value, one per in0 clock. Modulo 2 the pattern IS the Sierpinski triangle (Lucas' theorem): a self-similar fractal of 1s and 0s; higher Base reveals the richer nested structure of binomials mod a number. A fractal melody/gate from the most elementary combinatorics, distinct from the cellular-automaton fractals. A rising edge on in1 resets to the top of the triangle.

▪▫▪▫ PCG [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Permuted-congruential sequencer (a first as a synth block): the modern PCG generator - a plain linear-congruential engine (the oldest pseudo-random method) whose weak low bits are fixed by an output permutation that xor-shifts and then bit-rotates the state. The result is small, fast and statistically excellent, far better than the bare congruential generator it is built on. It plays a deterministic, very thoroughly scrambled melody. Bits sets the pitch range. A permuted-LCG noise distinct from the shift-register and squared-digit generators. A rising edge on in1 resets.

▪▫▪▫ Pell [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pell-sequence sequencer (a first as a synth block): the recurrence P(n) = 2*P(n-1) + P(n-2) - giving 0,1,2,5,12,29,70,169,408… - whose ratios converge on the SILVER ratio (1 + sqrt2 ~ 2.414), the silver-mean analogue of Fibonacci's golden. The Pell numbers also give the best rational approximations to sqrt2 (the side-and-diagonal numbers). Each rising edge of the in0 clock advances one term, taken modulo Steps - a faster-climbing, silver-ratio melody distinct from the golden Fibonacci/Pisano and the plastic Padovan. A rising edge on in1 resets.

▪▫▪▫ PellLucas [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pell-Lucas sequencer (a first as a synth block): steps through the companion Pell numbers 2, 2, 6, 14, 34, 82, 198, 478… defined by Q(n) = 2*Q(n-1) + Q(n-2) - the Pell recurrence (doubling the previous term) seeded to give the numerators of the rational approximations to the square root of 2. They grow by the silver ratio 1 + sqrt(2), a faster climb than the golden Fibonacci. Folded modulo Base. A silver-ratio companion sequence distinct from the Pell and Fibonacci sequencers. A rising edge on in1 resets.

▪▫▪▫ Pentagonal [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pentagonal-number gate (a first as a synth block): the output goes high when the step n is a pentagonal number - 1, 5, 12, 22, 35, 51, 70… (k(3k-1)/2), the dots that tile a growing pentagon. These are the very numbers in Euler's pentagonal-number theorem that drive the partition function, and n is one exactly when 24n+1 is a perfect square whose root is one less than a multiple of 6. The gate fires on a decelerating but sparser rhythm than the squares or triangulars. A rising edge on in1 resets.

▪▫▪▫ Pentanacci [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pentanacci sequencer (a first as a synth block): the five-step Fibonacci - each term is the sum of the previous FIVE: 0, 0, 0, 0, 1, 1, 2, 4, 8, 16, 31, 61, 120… For the first several terms it doubles exactly (powers of two) before the deeper memory pulls it just below, its growth ratio approaching 2 as the number of summed terms rises. A deep-memory recurrence distinct from the two-term Fibonacci and three-term Tribonacci. Folded modulo Base. A rising edge on in1 resets.

▪▫▪▫ Pentatope [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pentatope-number sequencer (a first as a synth block): steps through the pentatope (4-simplex) numbers 1, 5, 15, 35, 70, 126… = n(n+1)(n+2)(n+3)/24 - the four-dimensional analogue of the tetrahedral numbers, the running sums of the tetrahedral numbers and the fourth diagonal of Pascal's triangle (so every fifth binomial coefficient C(n,4)). Each clock advances one term, folded modulo Base. A 4-D figurate climb distinct from the 3-D tetrahedral sequencer. A rising edge on in1 resets.

▪▫▪▫ Pernicious [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Pernicious-number sequencer (a first as a synth block): steps through the pernicious numbers 3, 5, 6, 7, 9, 10, 11, 12, 13, 14… - integers whose binary representation contains a PRIME number of 1-bits (3 = 11 has two ones, 7 = 111 has three). A property living entirely in the binary digits, bridging primality and bit patterns. Each clock advances one term, folded modulo Base. A binary-popcount-prime class distinct from the decimal-digit sequencers. A rising edge on in1 resets.

▪▫▪▫ Perrin [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Perrin-sequence sequencer (a first as a synth block): the recurrence P(n) = P(n-2) + P(n-3) seeded 3,0,2 - giving 3,0,2,3,2,5,5,7,10,12,17,22,29… - famous for the Perrin pseudoprime test (n divides P(n) almost exactly when n is prime). Each rising edge of the in0 clock advances one term, taken modulo Steps. It shares Padovan's plastic-number growth but a different opening, so its melody starts on a falling figure (3,0,2) before climbing - a distinct number-theoretic line from Padovan or the Fibonacci-based Pisano. A rising edge on in1 resets.

▪▫▪▫ Persistence [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Multiplicative-persistence sequencer (a first as a synth block): each in0 clock advances the step n and outputs its multiplicative persistence - how many times you must replace the number by the product of its digits before reaching a single digit (39 -> 27 -> 14 -> 4 takes three steps, so its persistence is 3). Astonishingly, no number below 10^233 is known to need more than 11 steps, an unexplained ceiling. Most numbers collapse in one or two steps, so the output is a low, spiky pattern that occasionally jumps. The DEPTH of digit-product collapse, distinct from the MultRoot block that plays its endpoint. A rising edge on in1 resets.

▪▫▪▫ Pisano [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Fibonacci-modulo sequencer (a first as a synth block): runs the Fibonacci recurrence modulo M, so the output is (..0,1,1,2,3,5,8,13..) wrapped into M levels - a melodic loop whose length is the Pisano period of M (often surprisingly long and lopsided). Advance on the in0 clock; Modulus sets the wrap and therefore the loop length and contour. Deterministic and exactly repeating yet rarely an obvious cycle - the additive cousin of the multiplicative sequencers. A rising edge on in1 resets to 0,1.

▪▫▪▫ PlanePartition [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Plane-partition sequencer (a first as a synth block): steps through the number of plane partitions of n 1, 1, 3, 6, 13, 24, 48, 86, 160… - stacks of unit cubes pushed into a corner so the heights never increase going away from the walls, the three-dimensional generalisation of ordinary integer partitions. MacMahon's beautiful product formula counts them. The exact value is folded modulo Base. A 3-D partition count distinct from the flat partition and figurate sequencers. A rising edge on in1 resets.

▪▫▪▫ **Polydivisible** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Polydivisible-number sequencer (a first as a synth block): steps through the polydivisible numbers 1, 2, …, 9, 10, 12, 14, 16, 18, 20… - numbers where the first digit is divisible by 1, the first two digits form a number divisible by 2, the first three by 3, and so on all the way down. A cascade of divisibility through the digits; there are only finitely many (the longest has 25 digits). Each clock advances one term, folded modulo Base. A prefix-divisibility class distinct from the whole-number divisibility sequencers. A rising edge on in1 resets.

▪▫▪▫ **Popcount** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Popcount sequencer (a first as a synth block): outputs the Hamming weight of the step n - the number of 1-bits in its binary representation - folded into Steps, one per in0 clock. As n counts up the bit-count rises and falls in a self-similar fractal pattern (0,1,1,2,1,2,2,3…), climbing on dense numbers and dropping to 1 at every power of two. Where Thue-Morse keeps only the PARITY of this count, Popcount plays the full value - a binary-weight melody distinct from the digit-sum and ruler contours. A rising edge on in1 resets.

▪▫▪▫ **Power2** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Power-of-two gate (a first as a synth block): the output goes high only when the step n is an exact power of two - 1, 2, 4, 8, 16, 32, 64… Each gap is double the last, so the gate fires on the sparsest possible self-similar rhythm: it hits, waits twice as long, hits, waits twice as long again, forever halving in density. The test is the classic bit trick (n AND n-1) == 0. The most extreme decelerando of the number gates, ideal for octave-spaced accents and bar-doubling structure. A rising edge on in1 resets.

▪▫▪▫ **Powerful** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Powerful-number sequencer (a first as a synth block): steps through the powerful numbers 1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, 72, 81, 100… - integers in which every prime factor appears at least squared (so each is a product of a perfect square and a perfect cube). They are sparse and clump near the squares and cubes, giving a sequence that climbs in widening, uneven leaps. Each clock advances one term, folded modulo Base. A factor-exponent class distinct from the squarefree and smooth sequencers. A rising edge on in1 resets.

▪▫▪▫ **Practical** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Practical-number sequencer (a first as a synth block): steps through the practical numbers 1, 2, 4, 6, 8, 12, 16, 18, 20, 24… - integers n such that EVERY smaller number can be written as a sum of distinct divisors of n. They behave like little change-making systems (12 is why a dozen is handy), and they are about as dense as the primes yet defined multiplicatively. Each clock advances one term, folded modulo Base. A subset-sum-complete class distinct from the prime and abundant sequencers. A rising edge on in1 resets.

▪▫▪▫ **Prime** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Prime-step gate (a first as a synth block): a step counter advances on each rising edge of the in0 clock and the output goes high only when the count is a prime number (2,3,5,7,11,13…). Because primes thin out as they grow, the gate fires densely at first then ever more sparsely and irregularly - a deterministic yet non-repeating rhythm with no built-in period. Use it to trigger events, mask a sequence, or gate an effect on the primes. A rising edge on in1 resets the counter.

■□■□ PrimeOmega [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Prime-factor-count sequencer (a first as a synth block): each in0 clock advances the step n and outputs big-Omega(n) - the number of prime factors of n counted WITH multiplicity (so 12 = 223 gives 3) - folded into Steps. It rises to a local peak at the powers and products of small primes and drops to 1 at every prime, climbing on average like log-log n. The total weight of a number's prime factorisation as a melody, distinct from the parity-only Liouville and the divisor-counting Divisors. A rising edge on in1 resets.

■□■□ PrimePi [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Prime-counting sequencer (a first as a synth block): each in0 clock advances the step n and outputs pi(n), the number of primes less than or equal to n, folded into Steps. Unlike the prime GATE (which only fires on the primes themselves) this plays the running TALLY - a monotone staircase that steps up by one at every prime and holds flat across the composite runs, the function at the heart of the Prime Number Theorem (pi(n) ~ n / ln n). The widening flat steps trace how the primes thin out. A counting melody distinct from the prime gate. A rising edge on in1 resets.

■□■□ Primorial [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Primorial sequencer (a first as a synth block): steps through the primorials 2, 6, 30, 210, 2310, 30030, 510510… - the running products of the prime numbers (2, then 23, then 23*5, and so on), the prime analogue of the factorial. They explode even faster than factorials and are central to proofs about prime gaps and to Euclid's proof that the primes never end. The exact product is folded modulo Base, giving a fast-cycling pattern as each new prime multiplies in. A prime-product climb distinct from the additive prime sequencers. A rising edge on in1 resets.

■□■□ Pronic [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Pronic-number gate (a first as a synth block): the output goes high when the step n is a pronic (oblong) number - the product of two consecutive integers, 0, 2, 6, 12, 20, 30, 42… (k(k+1)), the count of dots in a rectangle one taller than it is wide, and exactly twice each triangular number. n is pronic when 4n+1 is a perfect square. The gate fires on a decelerating rhythm interleaved exactly halfway between the squares. A rising edge on in1 resets.

■□■□ Queens [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

N-queens sequencer (a first as a synth block): steps through the number of ways to place n non-attacking queens on an n-by-n chessboard 1, 1, 0, 0, 2, 10, 4, 40, 92, 352, 724… - one of the most famous combinatorial search problems. The count is zero for the impossible 2 and 3 boards, then climbs irregularly; the eight-queens puzzle has its celebrated 92 solutions. Each clock recomputes the count by backtracking and folds it modulo Base. A board-packing count distinct from every closed-form sequencer. A rising edge on in1 resets.

■□■□ Radical [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Radical sequencer (a first as a synth block): each in0 clock advances the step n and outputs the RADICAL of n - the product of its distinct prime factors, with all repeats stripped out (so 12 = 223 has radical 23 = 6, and 8 = 22*2 has radical just 2) - folded into Steps. The radical equals n itself for the squarefree numbers and drops far below it for the prime-power-heavy ones, so the melody dips sharply at every perfect power. The squarefree kernel of a number as a melody, central to the abc conjecture, distinct from the additive divisor functions. A rising edge on in1 resets.

■□■□ Recaman [SEQ][CV][FX][SYNTH] in: Clock, Reset out: CV

Recaman's-sequence sequencer (a first as a synth block): walks the famous OEIS A005132 path a(n) = a(n-1) - n when that step lands on a positive integer not already visited, else a(n-1) + n - the jump-back-or-leap-forward self-avoiding walk that is renowned for sounding musical (the Numberphile melody). Each rising edge of the in0 clock advances one term, folded into Steps to keep it in range; the contour darts down and springs up with none of the symmetry of a periodic pattern. Distinct from the monotone hailstone of Collatz. A rising edge on in1 resets to the 256-term phrase start.

▪▫▪▫ **Refactorable** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Refactorable-number sequencer (a first as a synth block): steps through the refactorable (or tau) numbers 1, 2, 8, 9, 12, 18, 24, 36, 40, 56… - integers that are divisible by their OWN number of divisors (12 has six divisors and 12/6 = 2). A self-referential factor property: the count of divisors itself divides the number. Each clock advances one term, folded modulo Base. A self-divisor class distinct from the perfect and abundant sequencers. A rising edge on in1 resets.

▪▫▪▫ **Repdigit** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Repdigit gate (a first as a synth block): the output goes high when every digit of the step n is the SAME (1, 2, …, 9, 11, 22, …, 99, 111, 222, …, 1111…). These monodigit numbers (the repunits 1, 11, 111… among them) are extremely sparse - only nine in each block of digit-length - so the gate fires in tight little clusters of nine right after each power of ten, then falls silent for a long stretch. The most regular of the rare digit gates, distinct from the palindrome and prime gates. A rising edge on in1 resets.

▪▫▪▫ **ReverseAdd** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Reverse-and-add orbit sequencer (a first as a synth block): walks the famous 196-algorithm - take a number, add it to its own digit-reversal, repeat - which almost always tumbles into a palindrome but for a few stubborn seeds (196, 879…) may climb forever. Each clock takes one reverse-add step and emits the running value folded into a pitch; when it grows large it reseeds to the next small number, so you hear orbit after orbit of this digit-mixing climb. A self-referential digit dynamic distinct from the power-sum orbits. A rising edge on in1 resets.

▪▫▪▫ **Riordan** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Riordan-number sequencer (a first as a synth block): steps through the Riordan numbers 1, 0, 1, 1, 3, 6, 15, 36, 91, 232… - the Motzkin paths with no flat steps at ground level, equivalently certain trees and the number of ways to colour cyclic arrangements. They obey $(n+1)a(n) = (n-1)(2a(n-1) + 3a(n-2))$ and are a close relative of the Motzkin and Catalan families. The exact value is folded modulo Base. A flat-step-free path count distinct from the Motzkin sequencer. A rising edge on in1 resets.

▪▫▪▫ **RogersRamanujan1** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Rogers-Ramanujan (first) sequencer (a first as a synth block): steps through the partitions of n into parts that leave remainder 1 or 4 when divided by 5: 1, 1, 1, 1, 2, 2, 3, 3, 4, 5… The celebrated first Rogers-Ramanujan identity proves this exactly equals the partitions whose parts differ by at least 2 - a deep and surprising equality between a congruence condition and a gap condition. Folded modulo Base. A gap-condition partition count distinct from the ordinary partition sequencer. A rising edge on in1 resets.

▪▫▪▫ **RogersRamanujan2** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Rogers-Ramanujan (second) sequencer (a first as a synth block): steps through the partitions of n into parts that leave remainder 2 or 3 when divided by 5: 1, 0, 1, 1, 1, 1, 2, 2, 3, 3… The second Rogers-Ramanujan identity proves this equals the partitions whose parts are at least 2 and differ by at least 2. The companion to the first identity, shifted by the smallest allowed part. Folded modulo Base. A shifted gap-condition count distinct from the first Rogers-Ramanujan sequencer. A rising edge on in1 resets.

▪▫▪▫ **RudinShapiro** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Rudin-Shapiro sequencer (a first as a synth block): outputs the Rudin-Shapiro sequence as a 0/1 gate - high when the number of overlapping ‘11’ pairs in the step counter’s binary is even (the +1 term), low when odd. Its defining property is remarkable: it is fully deterministic yet its power spectrum is FLAT, like white noise - the +/-1 sequence whose partial sums stay bounded by sqrt-of-n, the closest a structured pattern gets to acoustically random. A rhythm that sounds scattered and unpredictable but never actually repeats, distinct from the nested symmetry of Thue-Morse. A rising edge on in1 resets the counter.

▪▫▪▫ **Ruler** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Ruler-sequence sequencer (a first as a synth block): outputs the 2-adic valuation of the step n - the number of times 2 divides it, equivalently the count of trailing zeros in its binary - folded into Steps, one per in0 clock. The result is exactly the tick pattern on a ruler: 0,1,0,2,0,1,0,3,0,1,0,2… where every other mark is short, every fourth is taller, every eighth taller still, forever self-similar. A perfectly nested fractal staircase that drives rhythmic accents and melodic emphasis, distinct from the digit-sum and popcount contours. A rising edge on in1 resets.

▪▫▪▫ **Sandpile** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Abelian sandpile (a first as a synth block): drops a grain on the centre of a 4x4 pile each clock; when any cell reaches four grains it topples, sending one to each neighbour and possibly starting a chain reaction off the open edges. The output is the avalanche size - how many topplings that grain triggered - which is silent most of the time then spikes, following the power-law statistics of self-organised criticality. Most hits do nothing; occasionally one grain unleashes a cascade. A model of avalanches, forest fires and earthquakes as a rhythmic CV. A rising edge on in1 clears the pile.

▪▫▪▫ **Schroeder** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Schroeder-number sequencer (a first as a synth block): outputs the large Schroeder numbers - 1,2,6,22,90,394,1806… - which count the lattice paths from one corner of a grid to the opposite one using east, north and diagonal steps that never cross the diagonal (and a dozen other things in combinatorics). They are built here by the convolution S(n) = S(n-1) + sum S(k)S(n-1-k), so taking them modulo Base is exact. Growing faster than 5^n, mod Base they scramble into a busy pattern. A lattice-path melody distinct from the Catalan and Motzkin counts. A rising edge on in1 resets.

▪▫▪▫ **Secant** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Secant-number sequencer (a first as a synth block): steps through the even-indexed zigzag numbers 1, 1, 5, 61, 1385, 50521… that appear as the coefficients in the Taylor series of sec(x) (the unsigned Euler numbers). These count the alternating permutations of EVEN length and grow a touch slower than the tangent family, so the melody climbs steeply but a step behind Tangent once folded modulo Base. A trigonometric-combinatorial line distinct from the Euler and Tangent sequencers. A rising edge on in1 resets.

▪▫▪▫ **SecondOrder** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Second-order (reversible) cellular-automaton sequencer (a first as a synth block): runs an elementary CA with memory - the next row is the Rule applied to the current row XOR-ed with the PREVIOUS row. That backward coupling makes the automaton time-reversible (you can run it backwards to recover any past state), so it conserves information and never settles into a fixed pattern, churning endlessly. Each clock advances one generation on a 31-cell ring, reading a 12-cell window out as a stepped CV. A reversible CA distinct from the dissipative Wolfram and Totalistic ones. A rising edge on in1 resets.

▪▫▪▫ **SelfConjugate** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Self-conjugate-partition sequencer (a first as a synth block): steps through the number of self-conjugate partitions of n 1, 1, 0, 1, 1, 1, 1, 1, 2, 2… - the partitions whose Young diagram is symmetric across its main diagonal (unchanged when rows and columns are swapped). Euler showed these are exactly the partitions into DISTINCT ODD parts. Computed by the distinct-odd-parts recurrence, folded modulo Base. A symmetric-diagram count distinct from the full and distinct partition sequencers. A rising edge on in1 resets.

▪▫▪▫ **SelfNumber** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Self-number sequencer (a first as a synth block): steps through the self (or Colombian) numbers 1, 3, 5, 7, 9, 20, 31, 42, 53, 64, 75, 86, 97, 108… - integers that CANNOT be written as some smaller number plus its own digit sum, so they have no 'generator' and stand alone. Discovered by the Indian mathematician Kaprekar, they thin out in a curious near-periodic pattern tied to base ten. Each clock advances one term, folded modulo Base. A digit-generator gap sequence distinct from the other digit sequencers. A rising edge on in1 resets.

▪▫▪▫ **Sigma** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Divisor-sum sequencer (a first as a synth block): each in0 clock advances the step n and outputs sigma(n), the sum of ALL of n's divisors (sigma(6) = 1+2+3+6 = 12), folded into Steps as a 0..1 melody. Where the Totient counts what is coprime to n and Divisors counts how many divisors it has, Sigma adds them all up - so it climbs highest at the abundant, factor-rich numbers and sits low (sigma(p) = p+1) at the primes. The perfect numbers are exactly where sigma(n) = 2n. A melody from the total weight of divisibility, distinct from the other arithmetic functions. A rising edge on in1 resets.

▪▫▪▫ **SmallOmega** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Distinct-prime-factor sequencer (a first as a synth block): each in0 clock advances the step n and outputs little-omega(n) - the number of DISTINCT prime factors of n (so 12 = 223 gives just 2, since only 2 and 3 are distinct) - folded into Steps. It stays at 1 across all the prime powers and rises only when a number draws in a NEW prime, so it climbs very slowly and spotlights the squarefree, many-prime numbers. The count of distinct primes as a melody, distinct from the with-multiplicity PrimeOmega. A rising edge on in1 resets.

▪▫▪▫ **Smith** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Smith-number gate (a first as a synth block): the output goes high when the step n is a Smith number - a composite whose digits sum to the SAME total as the digits of all its prime factors added up (4 = 22 has digit sum 4 and factor-digit sum 2+2 = 4; 22 = 211 gives 4 and 2+1+1 = 4). Named after a phone number that turned out to have this property, they are scattered irregularly among the integers, so the gate fires on a sparse, unpredictable rhythm tied to the deep link between a number's digits and its factorisation. A rising edge on in1 resets.

▪▫▪▫ **SophieGermain** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Sophie-Germain-prime sequencer (a first as a synth block): steps through the primes p for which 2p+1 is ALSO prime - 2, 3, 5, 11, 23, 29, 41, 53, 83, 89… Named after the mathematician who used them to attack Fermat's Last Theorem, they are central to cryptography (the safe primes 2p+1 resist certain attacks). They form a sparse, irregular subset of the primes. Each clock advances one term, folded modulo Base. A cryptographic prime class distinct from the twin-prime and full prime sequencers. A rising edge on in1 resets.

▪▫▪▫ **Square** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Perfect-square gate (a first as a synth block): the output goes high when the step n is a perfect square - 0, 1, 4, 9, 16, 25, 36… The gaps between squares grow by the odd numbers (1, 3, 5, 7…), so the gate fires on a smoothly and quadratically decelerating rhythm: a quick flurry at the start that stretches into ever-longer silences faster than the triangular gate. A perfect-square pulse distinct from the triangular and prime gates. A rising edge on in1 resets.

▪▫▪▫ **Squarefree** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Squarefree-number sequencer (a first as a synth block): steps through the squarefree numbers 1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 17, 19, 21, 22, 23… - integers divisible by no perfect square (no repeated prime factor). They have density 6/pi-squared, about 61 percent of all integers, so the sequence skips only the few that carry a squared factor (4, 8, 9, 12, 16…), giving a near-chromatic run with occasional gaps. Each clock advances one term, folded modulo Base. The exact complement of the powerful numbers. A rising edge on in1 resets.

▪▫▪▫ **SquarePyramidal** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Square-pyramidal-number sequencer (a first as a synth block): steps through the square-pyramidal numbers 1, 5, 14, 30, 55, 91… = n(n+1)(2n+1)/6 - the count of spheres stacked in a pyramid with a square base, the running sums of the perfect squares. They appear in the classic cannonball problem (only 4900 is both square and square-pyramidal). Each clock advances one term, folded modulo Base. A square-based packing distinct from the triangular tetrahedral sequencer. A rising edge on in1 resets.

▪▫▪▫ **Stanley** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Stanley sequencer (a first as a synth block): steps through the Stanley sequence 0, 1, 3, 4, 9, 10, 12, 13, 27, 28… - built greedily by always adding the smallest integer that creates NO three-term arithmetic progression with any two earlier terms. Remarkably the result is exactly the numbers whose base-three representation contains no digit 2, giving it a clean self-similar Cantor-like structure. Folded modulo Base. A progression-free greedy set distinct from the Sidon (Mian-Chowla) set. A rising edge on in1 resets.

▪▫▪▫ **StarNumber** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Star-number sequencer (a first as a synth block): steps through the star (or centred-hexagram) numbers 1, 13, 37, 73, 121, 181… = 6n(n-1) + 1 - dots arranged in a six-pointed Star of David, a centred hexagonal number with six triangular points added. They climb in even, widely-spaced jumps. Each clock advances one term, folded modulo Base. A centred star figurate distinct from the solid polyhedral sequencers. A rising edge on in1 resets.

▪▫▪▫ **Stern** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Stern diatomic sequencer (a first as a synth block): outputs Stern's diatomic sequence (the fusc function) - 0,1,1,2,1,3,2,3,1,4,3,5… - in which consecutive pairs enumerate every positive rational number exactly once, the arithmetic behind the Stern-Brocot tree. Each rising edge of the in0 clock advances one term, folded into 16 steps; the contour rises and falls in a self-similar fractal staircase that never repeats. A deterministic, number-theoretic melody distinct from the Fibonacci-based Pisano. A rising edge on in1 resets.

▪▫▪▫ **Stirling1** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Stirling-first-kind sequencer (a first as a synth block): reads the UNSIGNED Stirling triangle of the FIRST kind row by row - the number of permutations of n items having exactly k disjoint cycles (1; 1,1; 2,3,1; 6,11,6,1…) - modulo Base. Built by S(n,k) = S(n-1,k-1) + (n-1)*S(n-1,k), so mod Base is exact. Where the second kind counts set partitions, the first kind counts permutation cycles, and each row sums to a factorial. A cycle-counting triangle melody distinct from the second-kind and Pascal triangles. A rising edge on in1 resets.

▪▫▪▫ **Stirling2** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Stirling-second-kind sequencer (a first as a synth block): reads Stirling's triangle of the SECOND kind row by row - S(n,k), the number of ways to split n objects into exactly k non-empty groups (1; 1,1; 1,3,1; 1,7,6,1…) - taken modulo Base. Built by the recurrence S(n,k) = S(n-1,k-1) + k*S(n-1,k), so mod Base is exact. Each row sums to a Bell number, so this is the partition-counting triangle behind the Bell sequence. A set-partition triangle melody distinct from the Pascal and Catalan triangles. A rising edge on in1 resets.

▪▫▪▫ **Tangent** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tangent-number sequencer (a first as a synth block): steps through the odd-indexed zigzag numbers 1, 2, 16, 272, 7936, 353792… that appear as the coefficients in the Taylor series of tan(x). These are the alternating permutations of ODD length, and they explode far faster than the secant family, so the melody rockets upward in big leaps once folded modulo Base. A trigonometric-combinatorial climb distinct from the Euler and Catalan sequencers. A rising edge on in1 resets.

▪▫▪▫ **TentMap** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tent-map chaos sequencer (a first as a synth block): iterates x' = R * min(x, 1-x), the simplest piecewise-linear chaotic map - a symmetric tent that stretches and folds the unit interval. At R near 2 it is fully chaotic, producing a stepped CV that never repeats yet stays evenly spread across the range (unlike the lopsided logistic Chaos). Each rising edge of the in0 clock takes one iteration; R below 2 tames it toward order. A rising edge on in1 reseeds. The angular, uniform cousin of the smooth attractors.

▪▫▪▫ **TernaryTree** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Ternary-tree sequencer (a first as a synth block): steps through the Fuss-Catalan numbers 1, 1, 3, 12, 55, 273, 1428, 7752… = C(3n, n)/(2n+1) - the count of rooted trees in which every node has exactly zero or three children (the ternary analogue of the Catalan binary trees), and of the ways to triangulate a polygon with non-crossing diagonals into quadrilaterals. The exact value is folded modulo Base. A three-way branching count distinct from the binary Catalan sequencer. A rising edge on in1 resets.

▪▫▪▫ **Tetrahedral** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tetrahedral-number sequencer (a first as a synth block): steps through the tetrahedral numbers 1, 4, 10, 20, 35, 56, 84, 120… = n(n+1)(n+2)/6 - the count of spheres stacked in a triangular pyramid (cannonball-pile shape), the running sums of the triangular numbers. They are the third diagonal of Pascal's triangle. Each clock advances one term, folded modulo Base. A 3-D figurate climb distinct from the flat polygonal and the 4-D pentatope sequencers. A rising edge on in1 resets.

▪▫▪▫ **Tetranacci** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tetranacci sequencer (a first as a synth block): the four-step Fibonacci, T(n) = T(n-1) + T(n-2) + T(n-3) + T(n-4) - 0,0,0,1,1,2,4,8,15,29,56… - each term the sum of the FOUR before it instead of two, so its ratio of successive terms climbs toward ~1.9276 rather than the golden 1.618. Each in0 clock advances one term, taken modulo Steps. Its faster four-term growth gives a melody that accelerates up the scale more steeply than Fibonacci or Tribonacci. A higher-order recurrence distinct from the Pisano and Padovan sequencers. A rising edge on in1 resets.

▪▫▪▫ **ThueMorse** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Thue-Morse sequencer (a first as a synth block): outputs the Thue-Morse sequence - the parity of the number of 1-bits in a step counter - as a 0/1 gate that advances on every rising edge of the in0 clock (0,1,1,0,1,0,0,1…). It is the canonical fair-share, cube-free, self-similar binary sequence: never periodic yet never random, full of nested symmetry. Use it as an aperiodic rhythm, a generative on/off mask, or a melodic gate that feels structured but never repeats. A rising edge on in1 resets the counter.

▪▫▪▫ **Totalistic** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Totalistic cellular-automaton sequencer (a first as a synth block): runs a one-dimensional CA whose next cell depends only on the SUM of its three-cell neighbourhood (0 to 3), not the exact pattern - so the whole rule is a 4-bit Code. This is the symmetric, count-only cousin of the Wolfram CA, the same rule space Wolfram used to study totalistic complexity. Each clock advances one generation on a 31-cell ring, reading a 12-cell window out as a stepped CV. A sum-driven automaton distinct from the pattern-driven Wolfram and second-order CAs. A rising edge on in1 resets to a single live cell.

▪▫▪▫ **Totient** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Euler-totient sequencer (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs phi(n)/n - the fraction of integers up to n that share no factor with it. The contour is jagged and number-aware: it sits high near primes (phi(p)/p = 1 - 1/p, almost 1) and plunges at the highly-composite, factor-rich numbers (6, 30, 210…), tracing the multiplicative texture of the integers as a melody. A deterministic line whose ups and downs encode coprimality itself, unlike the additive Stern or the word sequences. A rising edge on in1 resets.

▪▫▪▫ **Triangular** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Triangular-number gate (a first as a synth block): the output goes high when the step n is a triangular number - one that counts the dots in a filled triangle, 0,1,3,6,10,15,21,28,36,45… (n(n+1)/2). The test is the classic one: n is triangular exactly when 8n+1 is a perfect square. Because the gaps between successive triangular numbers grow by one each time, the gate fires on a steadily-rarefying rhythm - dense at first, then ever-longer silences - a smoothly-decelerating pulse distinct from the irregular prime and happy-number gates. A rising edge on in1 resets.

▪▫▪▫ **Tribonacci** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tribonacci-word sequencer (a first as a synth block): the three-symbol cousin of the Fibonacci word, grown by the substitution a -> a b, b -> a c, c -> a (its own fixed point) and emitted as a three-level CV (0, 0.5, 1), one symbol per in0 clock. Where the Fibonacci word is built on the golden ratio, this is built on the tribonacci constant ~1.839 - a balanced, aperiodic THREE-state pattern that never repeats and tiles the line like a 1-D quasicrystal. A richer alphabet than the binary words, distinct from the Fibonacci word and Kolakoski. A rising edge on in1 resets; the first 256 symbols then loop.

▪▫▪▫ **TribonacciWord** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Tribonacci-word sequencer (a first as a synth block): the infinite three-symbol word grown by the substitution 0 -> 01, 1 -> 02, 2 -> 0, giving 0102010010201010201… - the ternary cousin of the Fibonacci word, the symbolic shadow of the Tribonacci constant. Its three symbols appear in the Tribonacci proportions and tile the line self-similarly, a three-note aperiodic pattern. Each clock advances one symbol, output as 0, 0.5 or 1. A Tribonacci-ratio word distinct from the two-symbol Fibonacci word. A rising edge on in1 resets.

▪▫▪▫ **TrigSeq** [SEQ] [CV] [FX] [SYNTH] in: Clock out: CV

Trigger sequencer: steps through an on/off pattern on each rising edge of the in0 clock, emitting a trigger on the steps you switch on - a compact drum/gate sequencer. Steps sets the loop length, Pattern is the on/off bitmask (0-255 = 8 steps). in0 = Clock.

▪▫▪▫ **TwinPrime** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Twin-prime sequencer (a first as a synth block): steps through the lesser members of the twin-prime pairs 3, 5, 11, 17, 29, 41, 59, 71, 101, 107… - primes p for which p+2 is ALSO prime (3 and 5, 11 and 13, 17 and 19). Whether there are infinitely many is one of the oldest open problems in mathematics. They thin out faster than the primes, giving a sparse, irregular climb. Each clock advances one term, folded modulo Base. A prime-pair subset distinct from the full prime sequencer. A rising edge on in1 resets.

▪▫▪▫ **Ulam** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Ulam-sequence sequencer (a first as a synth block): the additive sequence 1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28… where each new term is the smallest integer larger than the last that is the sum of two distinct earlier terms in EXACTLY one way. Despite that purely additive rule it develops a deep, still-unexplained near-periodicity (a hidden spacing near 21.6) that number theory cannot yet account for. Each rising edge of the in0 clock advances one term, folded into Steps; a quasi-periodic melody distinct from the Fibonacci-based Pisano and the rational-enumerating Stern. A rising edge on in1 resets.

▪▫▪▫ **Undulating** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Undulating-number gate (a first as a synth block): the output goes high when the step n has digits that strictly ALTERNATE between two different values in an a-b-a-b-a pattern (101, 121, 131…, 212, 232…, 12121…). These wavy numbers ripple back and forth between two digits, and they are sparse and clustered, so the gate fires in little flurries near each band of three-, then four-, then five-digit numbers. A zig-zag digit rhythm distinct from the palindrome and repdigit gates. A rising edge on in1 resets.

▪▫▪▫ **VanDerCorput** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Van der Corput low-discrepancy sequencer (a first as a synth block): outputs the radical-inverse sequence - reverse the bits of the step counter and read them as a fraction - so each new value lands in the largest remaining gap (0.5, 0.25, 0.75, 0.125, 0.625…). The result spreads evenly across the range yet never repeats a spacing, the quasi-random-but-balanced pattern used in quasi-Monte-Carlo sampling. Advance on the in0 clock; a rising edge on in1 resets. A more even, less clumpy alternative to a random source.

▪▫▪▫ **Wedderburn** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Wedderburn-Etherington sequencer (a first as a synth block): steps through the Wedderburn-Etherington numbers 1, 1, 1, 2, 3, 6, 11, 23, 46, 98, 207, 451… which count the number of distinct ways to bracket a product of n identical items when the operation is commutative but NOT associative - equivalently the unordered binary trees with n leaves. They arise in counting hydrocarbon isomers and phylogenetic trees, and grow by a deep nonlinear self-convolution. Folded modulo Base. A combinatorial tree count distinct from the linear recurrences. A rising edge on in1 resets.

▪▫▪▫ **Weyl** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Weyl-sequence sequencer (a first as a synth block): the additive recurrence x <- frac(x + alpha) with an IRRATIONAL step, whose iterates are equidistributed by Weyl's theorem - they spread out perfectly evenly and never repeat. With the golden ratio as the step it becomes the optimal one-dimensional low-discrepancy sequence, each new note falling in the largest remaining gap. Ratio selects the irrational (golden, root 2, root 3, plastic), each giving a different even scatter. A number-theoretic equidistribution distinct from the chaotic and pseudo-random sources. A rising edge on in1 resets.

▪▫▪▫ **Wireworld** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Wireworld cellular automaton (a first as a synth block): a 4x4 grid of cells in four states - empty, electron head, electron tail, conductor - evolves on each rising edge of the in0 clock. A head becomes a tail, a tail becomes a conductor, and a conductor becomes a head only if exactly one or two neighbours are heads, so electrons stream along the conductor wires like current in a circuit. The output is the electron-head density as a stepped CV - sparks chasing each other around closed loops. Seed sets the initial spark density; a rising edge on in1 reseeds.

▪▫▪▫ **Wolfram** [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Elementary cellular-automaton sequencer (a first as a synth block): runs Stephen Wolfram's one-dimensional 2-state CA on a 31-cell ring - every cell updates from itself and its two neighbours by the 8-bit Rule code (Rule 30 is chaotic and was used as a random generator, Rule 90 draws the Sierpinski triangle, Rule 110 is Turing-complete, Rule 184 models traffic). Each clock advances one generation, reading a 12-cell window of the row out as a stepped CV. The whole zoo of CA behaviour - order, chaos and the complex edge between them - from one integer. A rising edge on in1 resets to a single live cell.

▪▫▪▫ Xorshift [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Xorshift-PRNG sequencer (a first as a synth block): clocks George Marsaglia's xorshift generator - the running 32-bit word is repeatedly shifted and XOR-ed into itself (left 13, right 17, left 5) - one of the fastest pseudo-random generators known. It visits all 2^32 - 1 non-zero states before repeating, so like the LFSR it plays a deterministic but seemingly random melody, with a denser, more thoroughly scrambled feel from mixing the whole word at once. Bits sets the read-out pitch range. A word-mixing noise distinct from the bit-shifting LFSR. A rising edge on in1 resets the seed.

▪▫▪▫ Zeckendorf [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Zeckendorf-weight sequencer (a first as a synth block): every positive integer has a UNIQUE representation as a sum of non-consecutive Fibonacci numbers (Zeckendorf's theorem); each rising edge of the in0 clock advances the step n and outputs how many Fibonacci terms that representation needs, folded into Steps. The weight rises and falls as the greedy Fibonacci-base digits shuffle - the 'Fibonacci-base digit sum', a self-similar contour related to the golden ratio and the Fibonacci word. A melody from base-phi numeration, distinct from the decimal Champernowne. A rising edge on in1 resets.

▪▫▪▫ Zuckerman [SEQ] [CV] [FX] [SYNTH] in: Clock, Reset out: CV

Zuckerman-number sequencer (a first as a synth block): steps through the Zuckerman numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 15… - integers divisible by the PRODUCT of their own digits (12 has digit product 2 and 12/2 = 6; numbers containing a zero are excluded). A self-referential multiplicative digit property. Each clock advances one term, folded modulo Base. A digit-product-divisible class distinct from the digit-sum Harshad sequencers. A rising edge on in1 resets.

Spectral (13)

FormantShift [SPC] [FX] [SYNTH] in: Audio out: Audio

Formant shift: per STFT frame it captures the magnitude envelope, resamples it by a ratio of 2^(Shift/12), then re-weights each bin to impose the shifted envelope while leaving the underlying partials in place. Shift (+/-12 semitones) moves the formants up or down, so pitch stays fixed while timbre changes; Mix blends against the dry input. Produces vocal gender/size effects without affecting the played note.

LinearEQ [SPC] [FX] [SYNTH] in: Audio out: Audio

Linear-phase 3-band EQ: per STFT frame it applies a single magnitude gain to each FFT bin grouped into low (below half/8), mid, and high (from half/2) bands, so the output has zero phase shift. Low, Mid, and High set each band's gain in dB (converted via 10^(dB/20)); Mix blends against the dry input. The linear-phase response avoids the phase smearing of analog EQ, useful on transient or stereo material.

MatchEQ [SPC] [FX] [SYNTH] in: Audio out: Audio

Matching EQ: with Capture on it learns each bin's magnitude as a reference curve; with Capture off it scales every bin toward that stored reference per frame. Capture latches learn vs apply, Amount (0..1) blends between no change and a full match (bin gain clamped to 8x); Mix blends against the dry input. Reshapes the input's tonal balance to resemble a previously captured reference.

Soothe [SPC] [FX] [SYNTH] in: Audio out: Audio

Dynamic resonance suppressor: per STFT frame it tracks a smoothed local average up the bins and pulls any bin above that average down toward it, scaled by Amount. Amount (0..1) sets how aggressively over-average peaks are tamed (0 leaves them, 1 flattens them to the average); Mix blends against the dry input. Reduces harsh resonant spikes for a smoother, more even spectrum.

SpecBlur [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral blur: per STFT frame each bin's magnitude is a Blur-weighted mix of its previous value and the current one, smearing magnitudes over time while keeping live phase. Blur (0..1) sets the smoothing amount - higher values hold past energy longer; Mix blends against the dry input. Softens transients and stretches the sound into a sustained pad.

SpecComp [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral compressor: per STFT frame it finds the loudest bin, then attenuates every bin above Threshold (relative to that peak) back toward the threshold. Threshold sets the relative knee and Amount scales how hard over-threshold bins are pulled down; Mix blends against the dry input. Evens out spectral dynamics, taming dominant partials for a flatter, denser spectrum.

SpecDelay [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral echo: per STFT frame each bin's magnitude is summed with its previous-frame value scaled by Feedback, and that running magnitude is fed back (live phase retained). Feedback (0..0.95) sets how much each bin's energy persists and decays across frames; Mix blends against the dry input. Produces a smeared, frequency-domain tail closer to a freeze/reverb than a discrete time echo.

SpecFilter [SPC] [FX] [SYNTH] in: Audio out: Audio

Per-bin spectral filter: runs a streaming STFT and zeroes whole FFT bins on one side of a cutoff bin for a brick-wall response in the frequency domain. Cutoff is the bin index of the edge and Mode selects low-pass (zero bins above) or high-pass (zero bins below); Mix blends against the dry input. The hard bin cut gives a far steeper, ringing-free roll-off than an analog filter.

SpecFreeze [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral freeze: while Freeze is off it tracks each FFT bin's magnitude per frame; when Freeze is on it holds those captured magnitudes (keeping the live phase) to sustain the last spectrum indefinitely. Freeze is the on/off latch and Mix blends the held spectrum against the dry input. Turns a transient moment into a steady drone or pad.

SpecGate [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral gate: runs a streaming STFT and, per frame, zeroes any FFT bin whose magnitude falls below Threshold relative to the loudest bin. Threshold sets that cutoff fraction (0 passes everything, 1 keeps only the peak); Mix blends the gated spectrum against the dry input. Keeps only the strongest partials, thinning noise and weak harmonics into a sparser, cleaner tone.

SpecMorph [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral smoothing: per STFT frame it runs a one-pole average up the bins, replacing each magnitude with a Smooth-weighted blend of the running value and the current bin (live phase retained). Smooth (0..1) sets how strongly the envelope is smeared across frequency - higher values flatten spectral detail; Mix blends against the dry input. Blurs sharp formants into a softer, more uniform tone.

Spectral [SPC] [FX] [SYNTH] in: Audio out: Audio

FFT spectral-resynthesis oscillator: a streaming 1024-pt STFT phase vocoder (75% overlap) that FFTs each frame, reshapes the spectrum, then iFFTs. Stretch scales the spectral envelope by resampling bins, Shift moves bins up/down for a frequency shift, and Blur smears each bin's magnitude over time. Mix crossfades the resynthesis against the dry signal - pushes toward metallic, harmonically reshaped tones.

SpectralReso [SPC] [FX] [SYNTH] in: Audio out: Audio

Spectral resonator: per STFT frame each bin is compared with its two neighbors; local peaks are boosted by up to 1 + Amount x 2 while non-peak bins are cut by up to Amount x 0.5. Amount (0..1) sets both the peak boost and the valley attenuation; Mix blends against the dry input. Sharpens the spectrum into a more resonant, harmonically pronounced, vowel-like tone.

Stereo (18)

AutoPan [STR] [FX] [SYNTH] in: Audio out: Audio

LFO auto-panner for a mono source: a sine LFO driven from the shared transport time sweeps a constant-power pan automatically. Rate sets the sweep speed from 0.05 to 10 Hz and Depth scales how far the source travels. Out picks L or R, and because the LFO is phase-coherent across instances the two outputs track in opposition for stable motion.

AutoWidth [STR] [FX] [SYNTH] in: L, R out: Audio

Auto-widening LFO: a sine LFO swings the mid/side width of the L/R pair so the stereo image rhythmically breathes between narrow and wide. Rate sets the sweep speed, Depth how far the width travels around a centred 1.0, and the LFO is driven from the shared transport time so it is phase-coherent across instances. Where AutoPan moves a mono source side to side and StereoWidth sets a fixed spread, AutoWidth animates the width of an existing stereo signal. in0 is L, in1 is R; Out picks the channel.

Balance [STR] [FX] [SYNTH] in: L, R out: Audio

Stereo balance for an existing L/R pair: tilts the level between the two input channels without collapsing them to mono - Balance at -1 mutes the right and passes the left, +1 does the reverse, 0 leaves both untouched. Where Pan places a single mono source and StereoWidth changes the mid/side spread, Balance simply re-weights a stereo signal's sides (the classic balance knob). in0 is L, in1 is R; Out picks which channel this instance emits.

Binaural [STR] [FX] [SYNTH] in: L, R out: Audio

Binaural 3D panner: positions the (summed) input at an Azimuth around the head using an interaural time delay, an interaural level difference and a head-shadow low-pass on the far ear - a simplified HRTF that places a sound left/right with real depth on headphones, beyond a flat constant-power pan. Azimuth sweeps -90..90 degrees, Distance dims and dulls far sources, and Out picks L or R.

Crossfeed [STR] [FX] [SYNTH] in: L, R out: Audio

Headphone crossfeed: low-passes the opposite channel and bleeds it into this one by Amount, emulating how each ear hears both speakers in a room. Amount sets the bleed level from 0 to 1. Out picks L or R, choosing which channel is the destination and which is the filtered source; eases the hard-panned separation that fatigues on headphones.

Doubler [STR] [FX] [SYNTH] in: Audio out: Audio

Vocal doubler: writes the input to a delay line and reads back a single detuned, delayed copy whose time drifts via a slow c.time-phased LFO for a natural double-tracked feel. Detune scales the drift depth, Delay sets the base delay time, and Out shifts the copy slightly shorter or longer (0.9x or 1.1x of the base) to place it as the left or right voice. Layer two instances on opposite Out settings for a widened stereo double.

FreqPanner [STR] [FX] [SYNTH] in: L, R out: Audio

Frequency-split panner: sums the input to mono, low-passes it at Freq, and applies a constant-power pan to only the high band while the low band stays centred. Freq sets the split from 100 to 4000 Hz and Pan positions the highs from -1 to +1. Out picks L or R, applying the matching pan-law leg to the highs over the shared centred lows.

Haas [STR] [FX] [SYNTH] in: Audio out: Audio

Haas widener for a mono source: feeds the input into a delay line and outputs either the dry signal or a copy delayed by a few milliseconds. Delay sets the offset from 0 to 40 ms, exploiting the precedence effect to throw the image to one side. Out picks L (dry) or R (delayed) so the pair forms a wide stereo spread from one mono input.

Imager [STR] [FX] [SYNTH] in: L, R out: Audio

Multiband stereo imager (Ozone-Imager-style): mono the low end and widen the highs - low frequencies (below Freq) are summed to the centre so the bass stays tight and phase-coherent, while the side content above Freq is spread by Width for an airy top. Low directional bass widened only weakens and phase-cancels the low end, so this keeps the ‘stereo rose’ shape: narrow at the bottom, widest at the top. Freq sets the mono crossover (80-120 Hz typical), Width the high-band spread (1 = unchanged, >1 wider, 0 fully mono), and Out picks the channel. in0 = L, in1 = R; run two instances on opposite Out for the stereo pair.

MidSide [STR] [FX] [SYNTH] in: L, R out: Audio

Mid/side processor with independent gain on each component: splits L/R into mid and side, applies MidGain and SideGain in dB, then recombines. Lowering MidGain or raising SideGain widens the image while the reverse focuses the centre. Out picks L or R for the recombined channel; handy for level-matching centre versus stereo content.

MonoMaker [STR] [FX] [SYNTH] in: L, R out: Audio

Bass mono-maker: a one-pole low-pass at Freq splits the spectrum so content below the crossover is summed to mono while the highs stay in their own channel. Freq sets the crossover from 20 to 500 Hz. Out picks L or R, supplying that channel’s highs over the shared mono low end; tightens low-frequency phase for vinyl and club playback.

Pan [STR] [FX] [SYNTH] in: Audio out: Audio

Static constant-power panner for a mono source: places the input at a fixed position from -1 (hard left) to +1 (hard right) using an equal-power (sin/cos) law so perceived loudness stays constant across the sweep. The plain manual counterpart to the LFO-driven AutoPan and the frequency-split FreqPanner - set a source’s place in the image once. Out picks L or R, supplying that channel’s pan-law leg; centred (Pan 0) sends equal power to both.

Panner [STR] [FX] [SYNTH] in: Audio out: Audio

Constant-power panner for a mono source: maps Pan (-1 to +1) onto a sine/cosine gain pair so total power stays even across the field. Pan at 0 centres the source; the +/-1 extremes hard-pan it. Out picks L or R, each instance applying the matching sine or cosine leg of the pan law.

Rotate [STR] [FX] [SYNTH] in: L, R out: Audio

Stereo field rotation: rotates the L/R signal vector by Angle, tilting the whole stereo image (a hard-left source swings toward centre and on toward the right) - a true rotation rather than a width change. Angle sets the rotation in degrees and Out picks L or R. Subtle angles re-balance a lopsided image; +/-90 swaps the channels.

Shuffle [STR] [FX] [SYNTH] in: L, R out: Audio

Blumlein shuffler (frequency-dependent stereo width): splits the side (L-R) signal at a crossover and applies Bass width below and Treble width above, so you can collapse the low end toward mono for a tight, phase-safe centre while widening the highs - the classic mastering width control. Freq sets the crossover, Bass/Treble the width on each side of it, and Out picks L or R.

StereoImager [STR] [FX] [SYNTH] in: L, R out: Audio

Frequency-dependent imager: splits the side signal at Freq with a one-pole low-pass and scales only the high side band by Width, leaving the low side untouched. Width sets the high-band spread and Freq sets the split point from 100 to 2000 Hz. Out picks L or R for the recombined channel; widens air and detail while keeping the low end mono-safe.

Stereoize [STR] [FX] [SYNTH] in: Audio out: Audio

Mono-to-stereo decorrelator: runs the input through a four-stage all-pass chain to phase-shift it without changing tone, then blends that toward the dry signal by Amount. Amount controls how much decorrelated signal is mixed in. Out picks L (dry) or R (decorrelated) so the stereo pair widens a mono source while keeping a centred reference.

StereoWidth [STR] [FX] [SYNTH] in: L, R out: Audio

Mid/side stereo widener: decomposes the L/R pair into mid (sum) and side (difference), scales the side by Width, then recombines. Width below 1 narrows toward mono and above 1 (up to 2) exaggerates the stereo spread. Out picks L or R, so one instance emits each recombined channel; useful for tightening or opening a stereo bus.

Utility (147)

Abs [UTL] [FX] [SYNTH] in: Audio out: Audio

Absolute value / full-wave rectifier: outputs |in0| * Gain plus Offset, so

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AbsDiff [UTL] [FX] [SYNTH] in: Audio out: Audio

Unsigned difference: outputs |in0 - in1| * Scale, the distance between the

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AbsDiffSq [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Squared difference: (a-b)^2 scaled, which is zero when the two inputs match and grows quadratically as they diverge - a sharp similarity/error detector for sidechaining, gating on signal mismatch, or driving modulation from how far apart two sources are.

AbsMax [UTL] [FX] [SYNTH] in: Audio out: Audio

Largest magnitude: outputs the greater of |in0| and |in1| times Gain - the

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AbsSum [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Sum of magnitudes: |a|+|b| scaled, the combined rectified level of two sources - a quick total-energy/activity signal for driving envelopes, gates or meters from a pair of inputs, distinct from a signed mix or a max-of-magnitudes.

Acos [UTL] [FX] [SYNTH] in: Audio out: Audio

Arccosine shaper: clamps in0 to [-1, 1] then outputs acos(in0) / pi, a smooth

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Analyzer [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Signal analyzer: passes in 1 straight through (out = in 1, transparent in the chain) while its detail panel shows a live 3D FFT waterfall + oscilloscope of in 1, and a Lissajous (X-Y) phase scope of in 1 against in 2. A measurement tool, not an effect - drop it inline to see the spectrum feeding a filter, the harmonics a distortion adds, what an envelope follower tracks, or the stereo/phase relationship of two signals.

ArgMax [UTL] [FX] [SYNTH] in: Audio out: Audio

Winner-take-all index: outputs which of in0..in3 is currently the largest, as

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Asin [UTL] [FX] [SYNTH] in: Audio out: Audio

Arcsine shaper: clamps in0 to [-1, 1] then outputs asin(in0) * 2/pi, an odd

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Asinh [UTL] [FX] [SYNTH] in: Audio out: Audio

Inverse-hyperbolic-sine compander: outputs asinh(Drive * in0) / asinh(Drive), a

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Atan [UTL] [FX] [SYNTH] in: Audio out: Audio

Arctangent shaper: outputs (2/pi) * atan(Drive * in0), a smooth odd S-curve that

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Atan2 [UTL] [FX] [SYNTH] in: Audio out: Audio

Two-argument arctangent: outputs the angle of the (in0, in1) vector mapped to

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Atanh [UTL] [FX] [SYNTH] in: Audio out: Audio

Inverse-hyperbolic-tangent expander: clamps in0 to +/-Edge then outputs

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Attenuvert [UTL] [FX] [SYNTH] in: Audio out: Audio

Attenuverter + offset: scales the input by a bipolar gain (can invert) and adds a DC offset - the fundamental CV-shaping utility for inverting, attenuating and biasing control or audio signals.

Average [UTL] [FX] [SYNTH] in: Audio out: Audio

Mean of the four signal inputs, weighted only by how many are non-zero so

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BeatRepeat [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Beat repeat: on each trigger it grabs the most recent slice and loops it for the slice length, freezing the groove into a glitchy stutter until the next trigger - the classic beat-repeat/roll effect.

Bernoulli [UTL] [FX] [SYNTH] in: Audio out: Audio

Bernoulli gate (coin-flip router): on each rising edge of the in0 trigger it flips a weighted coin and either passes that gate through or blocks it, with Probability setting the odds. A stream of clocks becomes a randomly thinned-out pattern - the building block for generative rhythms and stochastic sequencing. Unlike the deterministic ClockDiv, every pulse is an independent chance event; the decision latches until the next trigger.

BitRepeat [UTL] [FX] [SYNTH] in: Audio out: Audio

Fixed-rate sample-hold decimator: re-samples the input at a chosen rate and holds between samples (the deterministic, non-random cousin of a glitch hold) - classic downsampling aliasing and lo-fi character.

BrightCV [UTL] [FX] [SYNTH] in: Audio out: Audio

Brightness follower: compares the energy above a split frequency to the total energy and outputs the ratio, a cheap spectral-centroid-style ‘how bright is it right now’ control voltage for driving timbre-reactive patches.

Burst [UTL] [FX] [SYNTH] in: Audio out: Audio

Burst generator: a rising edge on in0 fires a burst of Count evenly-spaced trigger pulses at Rate, then falls silent until the next input edge - turn one hit into a ratchet, a drum-roll trigger or a stutter clock. Count sets how many pulses and Rate their spacing; re-triggering mid-burst restarts the count. Patch its output into envelopes, sample-and-holds or sequencer clocks.

CentroidalMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Centroidal mean: outputs (2/3)($|a|^{2+|a||b|+|b|}$2)/(|a|+|b|), the x-coordinate of the centroid of the region under a line between the two values - another above-arithmetic mean with its own bias toward the larger input, distinct from the contraharmonic. Gain scales the result.

ChordQuant [UTL] [FX] [SYNTH] in: Audio out: Audio

Chord quantizer: snaps in0 - a pitch in semitones (the Note source, a

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Clamp [UTL] [FX] [SYNTH] in: Audio out: Audio

Hard range limiter: constrains in0 to the [Lo, Hi] window with no

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ClockDiv [UTL] [FX] [SYNTH] in: Audio out: Audio

Clock divider: passes only every Div-th rising edge of the in0 trigger, turning a fast gate or clock into a slower one for polyrhythms and rhythmic gating. The output mirrors the input pulse on the pulses it lets through and is silent otherwise, and the count stays phase-locked to the input.

ClockMult [UTL] [FX] [SYNTH] in: Clock out: Audio

Clock multiplier: measures the period of the in0 clock and emits Mult evenly-spaced pulses per input beat - the complement to ClockDiv, for faster subdivisions and ratchets. Locks to the input tempo on every edge so it tracks tempo changes.

Comparator [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Comparator: outputs a gate that goes high when input A rises above input B and low when it falls below by a hysteresis margin - turns any signal into a clean trigger with no chatter near the threshold.

Compare [UTL] [FX] [SYNTH] in: Audio out: Audio

Schmitt comparator: outputs +1 (high) when in0 rises above

Complement [UTL] [FX] [SYNTH] in: Audio out: Audio

Unit-range inverter: outputs Center - in0 * Scale, defaulting to 1 - in0 so

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Constant [UTL] [FX] [SYNTH] in: Audio out: Audio

DC source: emits a fixed Value every sample with no input. Patch it

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ContraharmonicMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Contraharmonic mean: outputs $(|a|^{2+|b|}2)/(|a|+|b|)$ of the two inputs, a mean that always lies above the arithmetic mean and is pulled toward whichever input is larger - useful for level-combining that favours the louder source. Gain scales the result.

CopySign [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Copy-sign: outputs the magnitude of A carrying the sign (polarity) of B - so B’s zero-crossings/phase drive A’s, useful for re-polarizing a signal, building ring-mod-like polarity gating, or forcing one source to follow another’s sign pattern.

Correlate [UTL] [FX] [SYNTH] in: A, B out: Audio

Two-signal relatedness meter (a first as a synth block): outputs a running normalised cross-correlation of in0 against in1 - how strongly the two signals move together - as a control voltage. About +1 when they rise and fall in lock-step, 0 when independent, -1 when they mirror each other. Window sets the averaging time. Detect when two modulators line up, gate an effect on the agreement of two sources, or measure stereo/phase coherence. Both inputs must vary (a constant has no correlation).

Cosh [UTL] [FX] [SYNTH] in: Audio out: Audio

Hyperbolic-cosine bowl: warps in0 through a normalised (cosh(in0 * Drive) - 1) /

Counter [UTL] [FX] [SYNTH] in: Trigger, Reset out: Audio

Pulse counter / staircase: each rising edge of the in0 trigger advances an internal count, and the output is that count scaled to 0..1 over Steps before wrapping to zero - a stepped CV ramp for sequences and rhythmic builds. A rising edge on in1 resets the count so it can be re-synced to a bar.

CrestCV [UTL] [FX] [SYNTH] in: Audio out: Audio

Crest-factor follower: outputs the ratio of peak level to RMS level, which is high for spiky percussive material and low for sustained tones - a 'how punchy is it' control voltage for transient-aware patching.

Crossfade [UTL] [FX] [SYNTH] in: Audio out: Audio

Blends from in0 to in1 by a fade position of Mix plus the in2 CV

CrossfadeLoop [UTL] [FX] [SYNTH] in: Audio out: Audio

Crossfade looper: two overlapping read-heads replay the most recent window forward on a loop with triangular crossfades, so the captured texture repeats seamlessly with no click at the loop point - an instant stutter/loop layer.

CrossTrig [UTL] [FX] [SYNTH] in: Audio out: Audio

Crossing detector: emits a one-sample High pulse each time in0 crosses the Level (zero by default), on the Rising, Falling or Both edge as selected. Turns an oscillator or LFO into a clock (a zero-crossing tick), derives a trigger from a gate's edge, or fires an event when a signal passes a threshold. Pairs with TrigToGate to stretch the pulse into a sustained gate.

CVBridge [UTL] [FX] [SYNTH] in: Audio out: Audio

Control-voltage utility that conditions a single input value in one of three Mode settings: attenuvert (scale by Scale plus Offset), note-to-1V/oct pitch conversion, or bipolar-to-unipolar remap. Scale provides positive or negative gain including inversion, and Offset shifts the result up or down. Useful for routing and rescaling modulation or pitch control between modules.

DCBlock [UTL] [FX] [SYNTH] in: Audio out: Audio

Removes DC offset and sub-sonic bias with a one-pole high-pass that subtracts the previous input and feeds back a 0.9995 coefficient, passing audio while draining any constant component. It takes no parameters and always runs at the same fixed corner. Place it after asymmetric distortion or rectification stages to keep the signal centred.

Deadband [UTL] [FX] [SYNTH] in: Audio out: Audio

Centre dead-zone: passes in0 straight through once it moves more than Width

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Decibels [UTL] [FX] [SYNTH] in: Audio out: Audio

Decibel <-> linear-gain converter. In dB->Gain mode it outputs 10^(in0 / 20),

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Dither [UTL] [FX] [SYNTH] in: Audio out: Audio

Bit-depth reducer with triangular (TPDF) dither: builds a quantization step from Bits, adds triangular noise from two xorshift draws scaled by Amount, then rounds to the step. Bits sets the target word length from 8 to 24 and Amount scales the dither level. The dither decorrelates quantization error so low-level detail survives reduction instead of distorting.

DitherCrush [UTL] [FX] [SYNTH] in: Audio out: Audio

Dithered bit reducer: requantizes to a chosen bit depth after adding triangular (TPDF) dither, so the bit-crush decorrelates into a smooth noise floor instead of correlated distortion - lo-fi done the right way.

Divide [UTL] [FX] [SYNTH] in: Audio out: Audio

Safe divider: in0 divided by (in1 + Denom), with the result limited to

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Entropy [UTL] [FX] [SYNTH] in: Audio out: Audio

Information-theoretic modulation source (a first as a synth block): outputs a running Shannon-entropy estimate of the input - near 0 for silence or a steady tone, rising toward 1 for noisy, unpredictable signals. It folds recent samples into a decaying 16-bin amplitude histogram and reports the normalised entropy of that distribution, answering ‘how disordered is this signal right now?’ as a control voltage. Window sets the memory (higher = slower, smoother). Patch it so an effect opens up only when the input turns chaotic.

Equal [UTL] [FX] [SYNTH] in: Audio out: Audio

Tolerance comparator: outputs High when in0 sits within +/-Tol of (in1 plus

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Erf [UTL] [FX] [SYNTH] in: Audio out: Audio

Error-function sigmoid: outputs erf(Drive * in0), the Gaussian-integral S-curve

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Exp [UTL] [FX] [SYNTH] in: Audio out: Audio

Linear-to-exponential converter: outputs 2^(in0 * Range), turning a linear

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Expr [UTL] [FX] [SYNTH] in: In A, In B, In C out: Audio

User-written per-sample expression compiled by an RT-safe VM

FlipFlop [UTL] [FX] [SYNTH] in: Trigger, Reset out: Audio

T (toggle) flip-flop: each rising edge of the in0 trigger flips the output between 0 and 1, halving the clock and turning a stream of triggers into a square gate. A rising edge on in1 resets it to 0. The bistable building block for clock division and on/off latching.

Fold [UTL] [FX] [SYNTH] in: Audio out: Audio

Triangle wavefolder: when |in0 * Drive| exceeds Threshold the excess is

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Frac [UTL] [FX] [SYNTH] in: Audio out: Audio

Fractional part: outputs in0 minus its floor, the [0, 1) remainder after the

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Gain [UTL] [FX] [SYNTH] in: Audio out: Audio

Gain trim (dB) plus a DC bias offset.

GateDelay [UTL] [FX] [SYNTH] in: Trigger out: Audio

Trigger delay: a rising edge on in0 is held for Time, then re-emitted as a short pulse - shift a trigger later in time to stagger hits, build echoes of a clock or offset a sequence. Each input edge arms an independent delay.

Gaussian [UTL] [FX] [SYNTH] in: Audio out: Audio

Bell-curve shaper: outputs exp(-((in0 - Centre)^2) / (2 * Width^2)), a smooth

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GeoMean [UTL] [FX] [SYNTH] in: Audio out: Audio

Geometric mean of two inputs: outputs sign(in0in1) sqrt(|in0 * in1|) *

GiniMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Gini mean: outputs ((a^{R+b}R)/(a^{S+b}S))^(1/(R-S)) of the two input magnitudes, a two-exponent generalization of the power means - R and S together place the result anywhere from harmonic-like to contraharmonic-like, the most flexible of the two-source mean combiners. Bounded.

GlitchHold [UTL] [FX] [SYNTH] in: Audio out: Audio

Sample-level glitch hold: with a set probability it freezes the current sample for a random run length, producing buffer-underrun-style digital stutter and aliasing artifacts (RT-safe deterministic RNG).

⚙ **GrainCloud** [UTL] [FX] [SYNTH] in: Audio out: Audio

Granular cloud: continuously sprays short windowed grains plucked from random positions in the recent input buffer, smearing the source into an evolving textural cloud - the core of granular texture synthesis.

⚙ **GranScrub** [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Granular scrubber: a position CV moves a windowed grain freely through the recent input buffer, so you can freeze, slow-scrub or jump around the recent audio in real time - tape-scrub/timestretch by hand.

⚙ **GranTimeStretch** [UTL] [FX] [SYNTH] in: Audio out: Audio

Granular time stretch: overlapping grains advance their read position slower (or faster) than real time, stretching or compressing the recent audio's duration while leaving its pitch unchanged - time-domain timestretching.

⚙ **HeronianMean** [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Heronian mean: outputs (|a| + sqrt(|a||b|) + |b|)/3, the mean used for the volume of a frustum - it sits between the geometric and arithmetic means, a gentle blend of two sources that leans slightly toward their average. Gain scales the result.

⚙ **HilbertEnv** [UTL] [FX] [SYNTH] in: Audio out: Audio

Instantaneous amplitude: forms an approximate quadrature (90-degree) partner of the signal via an all-pass and takes the magnitude of the analytic signal, tracking the true envelope even within a single cycle - smoother than a rectified follower.

⚙ **Hypot** [UTL] [FX] [SYNTH] in: Audio out: Audio

Vector magnitude: outputs sqrt(in0^2 + in1^2) scaled by Gain - the Euclidean

⚙ **HypotDiff** [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Hypotenuse difference: outputs sqrt(max(0, a^2 - b^2)), the remaining leg of a right triangle given the hypotenuse A and one side B (and zero when B exceeds A) - a complement to the usual hypot, useful for de-correlation, sideband math or gating when one signal overtakes another. Gain scales the result.

⚙ **Hysteresis** [UTL] [FX] [SYNTH] in: Audio out: Audio

Schmitt trigger with memory: the output latches High once in0 rises above the upper threshold (Center + Width) and stays High until in0 falls below the lower threshold (Center - Width), then latches Low. Because it remembers its last state, a noisy or slowly-drifting signal around one level produces a clean debounced gate instead of chattering - unlike the stateless Compare, which has no memory.

⚙ **Integrator** [UTL] [FX] [SYNTH] in: Signal, Reset out: Audio

Leaky integrator / accumulator (CV): sums in0 over time so a constant input ramps and a periodic input is smoothed and phase-shifted a quarter cycle. Rate scales the input before accumulation, Leak bleeds the accumulator toward zero (0 =

a near-perfect integrator that holds, higher = a shorter-memory smoother), and a rising edge on in1 resets the sum to zero. The running output is safety-limited. Distinct from Slew (which rate-limits a signal) and Counter (which counts edges).

Latch [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Bistable latch / flip-flop: remembers a High/Low state across time. In Toggle mode each rising edge on in0 flips the state (a clock divide-by-two, or tap-to-toggle). In Set/Reset mode a rising edge on in0 sets the output High and a rising edge on in1 sets it Low (a one-bit memory cell driven by two triggers). Outputs High when set and Low when clear - the missing fundamental logic block alongside Compare, Logic and Hysteresis.

LehmerMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Lehmer mean: outputs $(|a|^{p+|b|}p)/(|a|^{(p-1)+|b|}(p-1))$, a one-parameter family that gives the harmonic mean at Power=0, the arithmetic at 1 and the contraharmonic at 2 - so Power continuously biases the combination from the smaller toward the larger input.

LmsCanceller [UTL] [FX] [SYNTH] in: In A, In B out: Audio

LMS adaptive canceller: an 8-tap adaptive FIR (least-mean-squares) that filters the reference input (in 2) to best match the correlated component of the primary input (in 1), then subtracts it - the classic adaptive setup for stripping hum, a tone, or any reference-correlated noise out of a signal. StepSize sets adaptation speed vs stability; the output is the residual error.

Log [UTL] [FX] [SYNTH] in: Audio out: Audio

Exponential-to-linear converter: outputs log2(in0) / Range, the inverse of Exp -

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Logic [UTL] [FX] [SYNTH] in: Audio out: Audio

Boolean gate combiner: treats in0 and in1 as logic highs when they

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LogicGate [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Gate logic: treats two inputs as boolean gates and outputs their AND, OR, XOR or NAND - the combinational-logic building block for deriving new triggers and rhythms from two existing gate streams.

Logistic [UTL] [FX] [SYNTH] in: Audio out: Audio

Logistic sigmoid: outputs 1 / (1 + exp(-Gain * (in0 - Centre))), the classic

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Logit [UTL] [FX] [SYNTH] in: Audio out: Audio

Logit (inverse-logistic) expander: clamps in0 to [1-Edge, Edge] then outputs

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LogMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Logarithmic mean: outputs (|a|-|b|)/(ln|a|-ln|b|) (and |a| when they are equal), the mean from heat-transfer analysis that lies strictly between the geometric and arithmetic means - a smooth combiner biased toward the smaller value. Gain scales the result.

LogSumExp [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Smooth maximum: $\log(e^{(k*a)+e}(k*b))/k$, a differentiable soft-max of two signals that rounds the corner where a plain max would kink. Sharp sets how tightly it hugs the true maximum (high) versus averaging the two inputs (low) - a smooth selector/combiner.

Manhattan [UTL] [FX] [SYNTH] in: Audio out: Audio

Taxicab (L1) magnitude: outputs (|in0| + |in1|) * Gain, the sum of the two

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Map [UTL] [FX] [SYNTH] in: Audio out: Audio

Affine range mapper: rescales in0 from its [In Lo, In Hi] range onto a

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Max [UTL] [FX] [SYNTH] in: Audio out: Audio

Outputs the larger of in0 and in1 (after adding the In B offset to in1).

Median3 [UTL] [FX] [SYNTH] in: Audio out: Audio

Median of three inputs: outputs the middle value of in0, in1, in2, discarding

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Meter [UTL] [FX] [SYNTH] in: Audio out: Audio

Analysis meter that tracks the running mean-square of the input and outputs its RMS level as a scalar control value. Smooth sets the one-pole averaging coefficient, with higher values giving slower, steadier readings and lower values reacting faster to transients. Used to derive a level signal for metering or to drive other modules from program loudness.

Min [UTL] [FX] [SYNTH] in: Audio out: Audio

Outputs the smaller of in0 and in1 (after adding the In B offset to in1).

MinkowskiMix [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Minkowski (p-norm) mixer: combines two inputs by their L-p norm $(|a|^{Norm+|b|}Norm)$^(1/Norm), carrying the sign of their sum - at Norm=1 it sums magnitudes, at Norm=2 it is the Euclidean/RMS blend, and as Norm grows it approaches a max() of the two. A tunable interpolation between additive and maximum mixing. Bounded.

MinMax [UTL] [FX] [SYNTH] in: A, B out: Audio

Minimum / maximum selector: outputs either the smaller or the larger of the two input voltages each sample - an analog comparator-style utility for combining envelopes, CVs or gates (max = OR-like, min = AND-like). Mode picks min or max. in0, in1 = inputs.

MinMaxGate [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Two-input logic combiner: outputs the minimum, maximum or absolute difference of two signals - the analog-logic building block for combining CVs, rectifying differences or wave-multiplexing two sources.

MinMaxMorph [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Order-statistic morph: blends the sample-by-sample minimum and maximum of two inputs by Morph - at 0 it outputs min(a,b) (an AM-like duck to the quieter signal), at 1 it outputs max(a,b) (an OR-style combine to the louder), and at 0.5 their average. A continuous slider across the order statistics of two signals, distinct from additive or ring mixing.

Mixer [UTL] [FX] [SYNTH] in: In A, In B, In C out: Audio

Sums three scaled inputs plus a DC offset.

Modulo [UTL] [FX] [SYNTH] in: Audio out: Audio

Euclidean remainder: outputs in0 wrapped into [0, Mod), the non-

Multiply [UTL] [FX] [SYNTH] in: Audio out: Audio

Two-quadrant multiplier / VCA: in0 times in1 times Scale. With an

Mux [UTL] [FX] [SYNTH] in: Audio out: Audio

Four-way input selector: routes one of in0..in3 to the output based on the

Negate [UTL] [FX] [SYNTH] in: Audio out: Audio

Sign inverter: outputs -in0 scaled by Scale (default 1) plus an Offset.

NlmsCanceller [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Normalized-LMS canceller: an 8-tap adaptive FIR like the plain LMS canceller, but the step is divided by the reference signal's instantaneous power - so convergence speed no longer depends on input level and it adapts faster and more stably across quiet and loud passages. Filters reference (in 2) to cancel its correlated part from primary (in 1); outputs the residual.

NoiseRate [UTL] [FX] [SYNTH] in: Audio out: Audio

Noisiness follower: measures the zero-crossing rate of the signal (low for tones, high for noise) and outputs it as a control voltage, a tone-vs-noise discriminator for driving gates, filters or morphs.

NullTest [UTL] [FX] [SYNTH] in: A, B out: Audio

Null tester that subtracts the second input from the first and outputs the residual difference, scaled by Gain. When two signals are identical the output is silence, and any audible residual reveals the difference between them. Gain amplifies that residual so small mismatches become easy to hear, making this a tool for verifying bit-for-bit equivalence between processing chains.

PeakFollow [UTL] [FX] [SYNTH] in: Audio out: Audio

Peak follower: tracks the instantaneous peak of the signal and decays back down at a set rate, a fast-attack/slow-release detector that produces a clean envelope CV for triggering or amplitude-following.

PeakHold [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Peak / hold detector: latches the extreme value in0 reaches and holds it, optionally bleeding back toward zero at Decay (Decay 0 = hold forever). Mode picks the maximum (Peak), the minimum (Trough), or the largest magnitude (Abs Peak). A rising edge on the in1 reset clears the held value to the current input. Where EnvFollow smooths the amplitude with attack/release, PeakHold captures the strict extreme - peak-metering CV, grabbing the loudest transient, or a max-tracking sample-and-hold.

Periodicity [UTL] [FX] [SYNTH] in: Audio out: Audio

Self-similarity detector (a first as a synth block): writes the input to a delay line and outputs the running normalised correlation between the signal now and the signal one Period ago - how strongly it repeats itself at that lag. About 1 for a steady tone or loop whose cycle matches Period, near 0 for noise or a one-shot, so it reports the STRENGTH of repetition rather than the pitch (set Period to the cycle you are listening for). Window sets the averaging time. Drive an effect by how locked-in a groove is.

PhaseDiff [UTL] [FX] [SYNTH] in: Audio out: Audio

Wrapped phase distance: treats in0 and in1 as phases on a [-1, 1] cycle and

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PhaseInvert [UTL] [FX] [SYNTH] in: Audio out: Audio

Flips the polarity of the signal, negating every sample when engaged and passing it untouched otherwise. Invert toggles the inversion on or off. Use it to correct out-of-phase tracks or to null one source against another.

Power [UTL] [FX] [SYNTH] in: Audio out: Audio

Sign-preserving exponent: outputs sign(in0) * |in0|^Exp, so it reshapes

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PowerMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Power mean: the generalized Holder mean of the two input magnitudes ((|a|$^{p+|b|}$p)/2)^(1/p), where Power sweeps continuously through harmonic (-1), geometric (0), arithmetic (1) and RMS (2) means - one knob spanning every classical average for combining envelopes or control signals.

Quadratic [UTL] [FX] [SYNTH] in: Audio out: Audio

Second-order polynomial shaper: outputs Ain0^2 + Bin0 + C, a full parabola

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Quantize [UTL] [FX] [SYNTH] in: Audio out: Audio

Rounds in0 to the nearest of Steps evenly-spaced levels across the

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Reciprocal [UTL] [FX] [SYNTH] in: Audio out: Audio

One-over-x: 1 / (in0 + Bias), limited to +/-Limit and floored away

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RectifyCV [UTL] [FX] [SYNTH] in: Audio out: Audio

Rectifier: passes only the positive half, only the negative half, or the absolute value of the signal, then offsets it - the analog full/half-wave rectifier for deriving envelopes and folding control voltages.

RevGrain [UTL] [FX] [SYNTH] in: Audio out: Audio

Reverse-grain looper: continuously plays the most recent window of input backwards on a loop, layering a reversed echo of what just happened - granular reverse texture with no transport.

RMS3 [UTL] [FX] [SYNTH] in: Audio out: Audio

Root-mean-square of three inputs: outputs sqrt((in0^2 + in1^2 + in2^2) / 3) *

RmsFollow [UTL] [FX] [SYNTH] in: Audio out: Audio

RMS follower: tracks the running root-mean-square of the signal rather than its peak, giving a smooth loudness-correlated envelope (the detector loudness meters and RMS compressors use) - steadier than a rectified peak follower.

Round [UTL] [FX] [SYNTH] in: Audio out: Audio

Integer rounding with a selectable Mode (nearest, floor, ceil, truncate)

SampHold [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Sample and hold: captures the value of the signal input at the instant the trigger input rises and holds it until the next trigger - the classic stepped/random voltage source when fed noise and a clock.

SampleSkip [UTL] [FX] [SYNTH] in: Audio out: Audio

Random sample dropout: with a set probability each sample is discarded and the previous one held, simulating a glitching digital link - bit-rot crackle and stutter, deterministic (RT-safe) RNG.

ScaleQuant [UTL] [FX] [SYNTH] in: Audio out: Audio

Musical scale / pitch quantizer: snaps in0 - read as a pitch in

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Select [UTL] [FX] [SYNTH] in: Audio out: Audio

Two-way signal switch: routes in0 when the in2 selector (plus the

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Semitones [UTL] [FX] [SYNTH] in: Audio out: Audio

Semitone-to-ratio pitch converter: outputs 2^((in0 * Range) / 12), turning a

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ShiftReg [UTL] [FX] [SYNTH] in: Clock, Signal out: Audio

Analog shift register: each rising edge of the in0 clock shifts the in1 voltage one stage down a chain, and the output is the value Stages steps ago - a tapped delay for CV that turns a melody or random source into canonic, echoing pitch patterns. The classic Buchla/Triadex-style sequencing trick.

Sign [UTL] [FX] [SYNTH] in: Audio out: Audio

Sign extractor with a dead zone: outputs -1, 0, or +1 depending on

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SignGate [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Polarity transplant: outputs the magnitude of input A carrying the sign of input B - a one-bit ring modulator that imposes B's zero-crossing structure on A's amplitude.

Sinc [UTL] [FX] [SYNTH] in: Audio out: Audio

Normalised sinc: outputs sin(pi * in0 * Scale) / (pi * in0 * Scale), the cardinal

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Sinh [UTL] [FX] [SYNTH] in: Audio out: Audio

Hyperbolic-sine shaper: warps in0 through a normalised sinh(in0 * Drive) /

Skew [UTL] [FX] [SYNTH] in: Audio out: Audio

Asymmetric curve bender: warps in0 across the [0,1] range so its midpoint slides

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SlewExp [UTL] [FX] [SYNTH] in: Audio out: Audio

Exponential slew limiter: limits how fast the output can follow, but with an exponential (RC-style) approach whose speed grows with the distance to the target - smooth glide / portamento with a natural curve, separate rise and fall.

SlewLimAsym [UTL] [FX] [SYNTH] in: Audio out: Audio

Asymmetric slew limiter: caps how fast the output may rise and fall independently with a hard (linear, constant-slope) limit, for lag/portamento or attack-vs-release shaping with a straight-line edge rather than an exponential curve.

Slope [UTL] [FX] [SYNTH] in: Signal out: Audio

Differentiator / rate-of-change detector (CV): outputs how fast in0 is moving - positive while it rises, negative while it falls, zero when steady - so it reads the slope of an envelope, LFO or any control signal. Scale sets the sensitivity and Smooth low-passes the result to tame noise on fast signals. Pair it with Compare to fire a trigger on a rising or falling edge. The inverse complement of the Integrator.

SlopeDetect [UTL] [FX] [SYNTH] in: Audio out: Audio

Slope detector: outputs a smoothed derivative of the input, positive while it rises and negative while it falls - a movement/transient CV for triggering on direction changes or driving slew-dependent effects.

SmoothMin [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Polynomial smooth minimum: blends toward min(a,b) but rounds the crossover with a quadratic over a window K (the computer-graphics smin) - a compact-support soft-min distinct from the exponential log-sum-exp, for gentle kink-free combining or ducking of two signals.

Smoothstep [UTL] [FX] [SYNTH] in: Audio out: Audio

Hermite smoothstep ramp: maps in0 across the [Edge0, Edge1] range through

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Spread [UTL] [FX] [SYNTH] in: Audio out: Audio

Dispersion of three inputs: outputs (max - min) of in0, in1, in2 times Gain -

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Stochastic [UTL] [FX] [SYNTH] in: Audio out: Audio

Stochastic-resonance detector (a first as a synth block): the real phenomenon where adding the RIGHT amount of noise lets a weak, sub-threshold signal cross a detector it otherwise never could - too little and the signal stays buried, too much and it drowns. Noise sets the injected level, Threshold the detector, Smooth the output ballistic, and the output is the recovered crossing envelope. Sweep Noise to find the resonance sweet spot where a rhythm hidden below the threshold suddenly emerges.

StolarskyMean [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Stolarsky mean: outputs ((a^{p-b}p)/(p(a-b)))^(1/(p-1)) of the two input magnitudes, a difference-based family that interpolates the logarithmic, geometric, identric and power means as Power varies - a richly-tunable two-source combiner. Bounded.

Subtract [UTL] [FX] [SYNTH] in: Audio out: Audio

Difference node: in0 minus in1 minus in2, plus a DC Offset. Feed two

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Sum [UTL] [FX] [SYNTH] in: A, B, C, D out: Audio

Summing node: adds inputs in 1..4 into a single output with no per-input gain - the quick way to merge oscillators, parallel chains or control signals without wiring up a full Mixer.

Surprise [UTL] [FX] [SYNTH] in: Audio out: Audio

Adaptive novelty detector (a first as a synth block): keeps a self-tuning one-tap linear predictor of the input and outputs the magnitude of its prediction error - a spike whenever the signal does something the recent past did not predict, near silence while it stays predictable. Unlike Slope (the raw derivative, which stays high through any steady ramp) Surprise LEARNS the trend and fires only on genuine novelty: a new note, a sudden jump, a broken pattern. Rate sets how fast it adapts, Smooth the output ballistic.

Switch [UTL] [FX] [SYNTH] in: In A, In B, In C, In D out: Audio

Clocked sequential switch: routes one of the signal inputs in0..in2 to the output and advances to the next on every rising edge of the in3 clock - an addressed router (Doepfer A-151 style) that steps through your sources in time. Steps sets how many inputs are in the rotation (2 or 3). Where Mux selects an input by a manual index and Select chooses between two by level, Switch cycles automatically on a clock.

TapeBrake [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Tape stop: when the gate goes high the playback speed of a short buffer ramps down to a standstill (pitch and tempo falling together), then spins back up when released - the DJ/turntable brake effect.

Threshold [UTL] [FX] [SYNTH] in: Audio out: Audio

Unipolar threshold gate: outputs High when in0 is at or above Threshold and

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TrackAndHold [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Track and hold: follows the signal input while the gate is open and freezes the last value when the gate closes - a glitch-free way to freeze a moving signal or build sample-and-hold with a variable aperture.

TrackHold [UTL] [FX] [SYNTH] in: Signal, Gate out: Audio

Track-and-hold (Doepfer A-148): while the gate at in1 is high the output follows the in0 signal; when the gate goes low it freezes and holds the last value until the gate returns. Unlike Sample-and-hold (which grabs one instant on a trigger), it tracks continuously through the whole gate - use it to freeze a slow LFO or envelope, glide-and-hold pitch, or window a control signal. A core CV utility.

TrigToGate [UTL] [FX] [SYNTH] in: Audio out: Audio

Trigger-to-gate: each rising edge of the in0 trigger opens a gate that stays high for a fixed Length, then closes - turning a momentary pulse (a clock tick, a Compare edge) into a sustained gate of predictable duration. Retriggering restarts the timer, so a stream of pulses keeps the gate open. Drive envelopes, VCAs or effects from short triggers.

TriShape [UTL] [FX] [SYNTH] in: Audio out: Audio

Phasor-to-triangle shaper: treats in0 as a phase and outputs a triangle that

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WaveSetRepeat [UTL] [FX] [SYNTH] in: Audio out: Audio

Waveset repeat: detects each cycle between rising zero crossings (a 'waveset') and replays it a number of times before moving on, the Trevor-Wishart waveset-distortion technique that drops pitch and adds buzzy sub-harmonics.

Window [UTL] [FX] [SYNTH] in: Audio out: Audio

Window comparator: outputs High when in0 sits inside the [Lo, Hi] band and

WindowComp [UTL] [FX] [SYNTH] in: Signal out: Audio

Window comparator: outputs a high gate whenever the in0 voltage sits inside the Low..High window, and low when it strays outside - a band detector for triggering events when a CV enters a range, or for extracting a slice of a modulation curve.

WindowGate [UTL] [FX] [SYNTH] in: Audio out: Audio

Window comparator: outputs a high gate only while the input sits between a lower and upper bound, the analog window-detector used to fire triggers when a control voltage passes through a specific range.

Wrap [UTL] [FX] [SYNTH] in: Audio out: Audio

Phase-style wrap: folds in0 back into the [-Range, Range] window by

XFade [UTL] [FX] [SYNTH] in: In A, In B out: Audio

Equal-power crossfade: blends two inputs with a constant-power (sine/cosine) law so the perceived loudness stays even across the whole sweep - the correct way to morph or A/B between two signals.

Synth (zpwr-synth) (309 blocks)

Effect (6)

Crusher [FX] [SYNTH] in: Audio out: Audio

Bit-depth reduction plus downsampling (bitcrush).

Folder [FX] [SYNTH] in: Audio out: Audio

Wavefolder: west-coast sine folding with bias.

SynthDelay [FX] [SYNTH] in: Audio, Feedback Mod, Time Mod out: Audio

Delay line with feedback and a modulation input.

SynthDrive [FX] [SYNTH] in: Audio out: Audio

Saturation/overdrive with output trim.

SynthRingMod [FX] [SYNTH] in: Audio, Carrier out: Audio

Ring and amplitude modulation of two inputs.

SynthWaveshaper [FX] [SYNTH] in: Audio out: Audio

Waveshaper: tanh/arctan/polynomial/sine transfer curves.

Envelope (1)

Env [ENV] [CV] [SYNTH] in: Gate out: CV

DAHDSR envelope generator triggered by the gate input. Delay waits before the attack; Hold sustains the peak before the decay (both 0 = classic ADSR).

Filter (2)

DiodeLadder [FLT] [CIRCUIT] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

Diode-ladder low-pass filter (TB-303 lineage) with resonance.

SynthFilter [FLT] [SYNTH] in: Audio, Reso Mod, Cutoff Mod out: Audio

State-variable filter (low/high/band) with resonance and pitch mod.

Modulation (4)

NoiseLFO [MOD] [SYNTH] in: – out: Audio

Noise modulation source (white/pink/brown) with optional sample-and-hold.

StepLFO [MOD] [SYNTH] in: – out: Audio

Step LFO: descending stair or per-step random sample-and-hold, smoothed.

SynthLFO [MOD] [SYNTH] in: Audio out: Audio

Low-frequency oscillator (sine/tri/saw/square) modulation source. A trigger into in0 (e.g. the note Gate) resets the phase (retrigger); leave it unpatched to free-run. Phase offsets the start point; Smooth slews the output for a softer shape.

SynthSampleHold [MOD] [SYNTH] in: Signal out: Audio

Sample-and-hold of the input at a clocked rate, with glide.

Oscillator (291)

AccordionVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Accordion voice: two or three free reeds sounding the same note slightly detuned (musette tuning) under steady bellows, the warm wavering chorus of an accordion/bandoneon as a note-driven polyphonic voice. in0 = note, in1 = pitch mod. Octave/Fine tune it, Musette the detune wetness, Level the output.

AcidVCO [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

TB-303-style acid voice: a single oscillator switchable between the 303's bright sawtooth and square, with the characteristic sub-heavy low end, built to be slammed through a resonant lowpass and envelope for squelchy acid lines. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Wave selects saw or square, Sub the sub-octave level, Level the output.

AcousticGuitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Acoustic-guitar voice: a steel-string Karplus-Strong pluck with a bright attack and a resonant wooden-body colour, the warm ringing tone of a strummed acoustic. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Damping the decay, Level the output.

AgogoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Agogo voice: the gate strikes one of two pitched metal agogo bells (alternating high/low on each strike) with a bright clangy attack and short ring, the Brazilian samba bell. in1 = gate. HiPitch/LoPitch set the two bells, Decay the ring, Level the output.

AltoFluteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Alto-flute voice: an alto flute, a mellow sine a fourth below the concert flute with a warm breathy air wash and a smooth attack, the haunting middle-register woodwind. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Breath the air noise, Level the output.

AltoSaxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Alto-sax voice: an alto saxophone, a reedy sawtooth pushed through a resonant bandpass formant with a breathy edge, a quick attack and an expressive vibrato, the soulful jazz lead. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Honk the formant bite, Level the output.

AnvilVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Anvil voice: a struck iron anvil, a bright clanging cluster of dense inharmonic metal partials with a sharp attack and a noisy ring, the blacksmith's hammer-on-anvil clang. in0 = note, in1 = gate (strike). Octave tunes it, Decay the ring, Level the output.

ArpVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Arpeggiator voice: from one held note it generates a built-in arpeggio, cycling through a chord shape at the set rate in up, down, up-down or random order - an instant single-finger arp. in0 = note, in1 = pitch mod. Rate the arp speed, Chord the triad shape, Mode the direction, Octaves the range, Level the output.

BagpipeVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bagpipe voice: a fixed low drone reed sounding continuously beneath a melody chanter reed at the note pitch, both buzzy and slightly detuned, the relentless skirl of Highland pipes. in0 = note, in1 = pitch mod. Octave tunes the chanter, Drone the drone level, Level the output.

BalafonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Balafon voice: a West-African gourd-resonated xylophone, a struck wooden bar whose gourd resonators add a buzzy nasal rattle, the warm buzzing tone of the balafon. in0 = note, in1 = gate (strike). Octave tunes it, Buzz the gourd rattle, Level the output.

BalalaikaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Balalaika voice: a thin bright triangular-bodied Russian lute, a lightly-damped Karplus pluck with a sharp trebly attack and quick decay, often strummed fast; the plinky tone of the balalaika. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

BandoneonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bandoneon voice: the concertina of tango, two or three free reeds slightly detuned under bellows for a wheezy, expressive, vibrato-laden chord tone, more reedy and tremulous than an accordion. in0 = note, in1 = gate (bellows). Octave/Fine tune it, Detune the reed spread, Level the output.

BanjoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Banjo voice: a bright lightly-damped Karplus string coupled into a quick drum-head resonance, the twangy plinky fast-decaying attack of a 5-string banjo. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

BaritoneSaxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Baritone-sax voice: the big low saxophone, a fat reedy sawtooth through a deep bandpass formant with a heavy breathy growl, a soft attack and vibrato, the honking bass of the sax section. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Honk the formant bite, Level the output.

BassClarinetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bass-clarinet voice: a low single-reed cylindrical-bore waveguide whose phase-inverting reflection gives the hollow odd-harmonic woody growl an octave below the clarinet. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Dark the bore filter, Level the output.

BassFluteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bass-flute voice: a bass flute, a soft sine tone two octaves below the concert flute with a thick breathy air column and a slow attack, the dark velvety low woodwind. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Breath the air noise, Level the output.

BassoonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bassoon voice: the low double-reed, a buzzy sawtooth through a dark lowpass that leaves a reedy growl in the bass, with a soft attack and slow vibrato, the woody foundation of the woodwind section. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Reed the growl, Level the output.

BassoonVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bassoon voice: a double-reed low conical bore, woody and round in the bass with all harmonics, self-oscillating on breath, the comic baritone of the orchestra. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Dark the bore filter, Level the output.

BellTowerVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Church-bell voice: the gate strikes a large cast bell, voiced from its named partials - hum, prime, tierce (minor third), quint and nominal - with the long swelling-and-fading sustain of a tower bell. in0 = note, in1 = gate. Octave tunes it, Decay the toll length, Level the output.

BerimbauVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Berimbau voice: a Brazilian musical bow (the capoeira instrument), a single buzzy plucked string whose pitch toggles with the coin pressure, over a gourd resonance and a faint caxixi rattle. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Buzz the gourd buzz, Level the output.

BlownBottleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Blown-bottle voice: noise blown across the mouth of a Helmholtz resonator (a bottle) excites a breathy near-sine tone at the cavity resonance tracking the note, the hooting whistle of blowing over a bottle. in0 = note, in1 = gate (breath). Octave/Fine tune it, Noise the breath amount, Resonance the cavity Q, Level the output.

∿ BongoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bongo voice: the gate strikes a small high-pitched hand drum, a bright tight membrane tone with a fast pitch snap and a quick decay, the chattering high partner of the conga. in0 = note (transposes), in1 = gate. Tune sets the head pitch, Decay the ring, Level the output.

∿ BoobamVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Boobam voice: a tuned tubular drum, a struck membrane over a pitched tube giving a clean focused mid-pitched 'boo' with a quick pitch drop, the melodic boobam. in0 = note, in1 = gate (strike). Octave tunes it, Decay the body, Level the output.

∿ BouzoukiVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bouzouki voice: a Greek long-neck lute, a bright trichordo Karplus pluck layered with an octave course for the metallic ringing sustain of rebetiko. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Octave-course level, Level the output.

∿ BowedPadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bowed-pad voice: a sustained string-section pad built from several bowed-saw layers detuned and gently chorused, with a slow bow-pressure swell on the gate, the lush legato bed of a string ensemble. in0 = note, in1 = pitch mod. Octave/Fine tune it, Spread the section width, Swell the attack, Level the output.

∿ BowVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Bowed-string voice: a pitch-tuned delay-line string driven by a stick-slip bow-friction curve while the gate holds, so it self-oscillates into a sustained, singing Helmholtz tone (the bowed cousin of PluckString). in0 = note, in1 = gate (bow on). Octave/Fine tune it, BowForce the grip, Bright the string filter, Level the output.

∿ BrassFallVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Brass-fall voice: a brass note that ends with a doit-fall, a sawtooth that sounds steady then bends its pitch downward across the note like the falling tail of a big-band brass hit; the fall depth sets how far it drops. in0 = note, in1 = gate (breath). Octave tunes it, Fall the drop in semitones, Bright the blare, Level the output.

∿ BrassLeadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Brass-section lead voice: a stack of detuned saws through a lowpass whose cutoff swells open on the gate (the slow brass attack) then settles, the punchy synth-brass of 80s pop. in0 = note, in1 = gate. Octave/Fine tune it, Swell the attack-opening time, Tone the cutoff ceiling, Level the output.

∿ BrassSectionVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Brass-section voice: a stacked horn section, several detuned saws through a lowpass whose cutoff swells open on the gate (the brass attack) for a punchy ensemble fanfare. in0 = note, in1 = gate. Octave tunes it, Swell the attack-opening, Tone the brightness, Level the output.

∿ BrassSwellVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Brass-swell voice: a whole brass section in a slow crescendo, several detuned sawtooth voices through a lowpass that opens as the section swells from nothing up to a full blare, the cinematic brass hit-build. in0 = note, in1 = gate (breath). Octave tunes it, Swell the crescendo speed, Spread the section width, Level the output.

Buzz [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Buzzer / band-limited impulse train: sums Harmonics equal-amplitude harmonics of the note in closed form (the Dirichlet kernel sin(Npiphase)/(Nsin(piphase))), giving a bright brassy pulse with a flat harmonic series and no aliasing - the classic Csound buzz generator. Harmonics sets the partial count (brightness), Octave and Fine tune it, and Level scales the output. A rich raw source to feed subtractive filters or formant shaping. The harmonic count is auto-capped so the top partial stays below Nyquist.

Bytebeat [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Bytebeat oscillator: an algorithmic 8-bit generator that runs a bitwise integer formula over a pitch-driven sample counter, the low byte becoming the waveform - the chiptune/demoscene 'bytebeat' technique that turns shift-and-mask arithmetic into gnarly, ever-evolving digital timbres no analog or wavetable model can make. Formula selects the expression, Octave and Fine drive the counter rate (pitch), and Level scales the output. A world of aliased, glitchy, harmonically rich tones from one line of integer maths.

CajonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cajon voice: the gate strikes a box drum, switching by note register between a deep bass thud (low notes) and a snare-wire buzz against the front face (high notes), the flamenco/acoustic box drum. in0 = note (selects bass vs slap), in1 = gate. Tune sets the bass pitch, Snare the wire buzz, Level the output.

CastanetsVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Castanets voice: the gate clicks a pair of hardwood shells, a very short sharp wooden double-click with a high resonant tick, the flamenco castanet snap. in1 = gate. Tone sets the pitch, Level the output.

CelestaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Celesta voice: a felt-hammer-struck metal bar over a wooden resonator, a soft sweet bell-like tone with gentle inharmonic overtones and a medium decay, the celesta of orchestral magic. in0 = note, in1 = gate (strike). Octave tunes it, Decay the ring, Level the output.

CelesteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Celeste voice: a felt hammer striking a small steel plate over a wooden resonator, the gate triggering a soft sweet bell-like tone with a gentle attack and pure near-octave partials, the celesta. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

CelloVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cello voice: the tenor/bass bowed string, a rich sawtooth Helmholtz motion through a deep resonant body with a slow bow attack and a warm vibrato, the soulful cello. in0 = note, in1 = gate (bow). Octave/Fine tune it, Vibrato the waver, Body the resonance, Level the output.

CelloVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cello voice: the bowed-string waveguide tuned to the cello's low range with a darker bow filter, giving a deep woody resonant sustain. in0 = note, in1 = gate (bow). Octave/Fine tune it, BowForce the grip, Dark the bow filter, Level the output.

ChamberOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Chamber-organ voice: a small soft pipe organ, gentle stopped-flute sine ranks with a slow breathy speak and an instant release, the intimate baroque continuo organ. in0 = note, in1 = gate. Octave tunes it, Ranks the added octave pipes, Tone the cutoff, Level the output.

ChaosVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Chaotic-oscillator voice: a saw whose pitch and amplitude are continuously perturbed by a logistic-map chaotic generator, so the tone wobbles and shimmers unpredictably (deterministic chaos as modulation) around the played note. in0 = note, in1 = pitch mod. Octave/Fine tune it, Chaos the logistic r (order->chaos), Depth the modulation amount, Level the output.

CharangoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Charango voice: a small bright Andean lute with five paired courses, a high lightly-damped Karplus pluck with a fast shimmering double-course beating, plinky and trebly. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

ChiptuneLead [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

NES-style chiptune lead: a band-limited pulse with a stepped duty cycle (12.5/25/50/75%) and a fast internal arpeggio that retriggers a chord shape, the 8-bit square-lead sound of an NES/Famicom. in0 = note, in1 = pitch mod. Octave/Fine tune it, Duty the pulse width step, Arp the arpeggio speed, Chord the arpeggiated interval shape, Level the output.

ChoirAahVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Choir 'aah' voice: a bright open-vowel choir, a sawtooth shaped by the bright 'aah' vocal formants with a slow ensemble vibrato across a few detuned voices, the lush massed sound of a choir on aah. in0 = note, in1 = pitch mod. Octave/Fine tune it, Spread the section width, Level the output.

ChoirOohVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Choir 'ooh' voice: a dark rounded-vowel choir, the same massed-voice approach but shaped by the low 'ooh' formants for a warm hooting pad, the soft backing of a choir on ooh. in0 = note, in1 = pitch mod. Octave/Fine tune it, Spread the section width, Level the output.

ChoirVox [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Choir voice: a sawtooth shaped by three fixed vocal formants into a sustained 'aah'/'ooh' choir vowel, with a gentle ensemble vibrato for a sung, breathy pad. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Vowel morphs aah->ooh, Vibrato the pitch waver, Level the output.

ChordOsc [OSC] [SYNTH] in: Pitch (CV) out: Audio

Chord oscillator: one note drives a detuned chord of saws.

ChordStab [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Paraphonic chord-stab voice: from a single note it generates a whole triad (major, minor or sus) of detuned saws in one voice, the instant rave/house stab you play with one finger. in0 = note, in1 = pitch mod. Octave/Fine tune it, Chord selects the triad quality, Detune the stack width, Level the output.

CimbalomVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cimbalom voice: a Hungarian concert hammered dulcimer, a metal string struck hard (bright quick-attack Karplus) with a resonant box body and long ringing sustain, the dramatic cimbalom of folk and orchestral music. in0 = note, in1 = gate (strike). Octave/Fine tune it, Level the output.

ClarinetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Clarinet voice: a single-reed clarinet, a square wave (odd harmonics only) through a soft lowpass that gives the hollow woody tone of the cylindrical bore, with a smooth attack and light vibrato, the agile mellow woodwind. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Tone the brightness, Level the output.

ClavesVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Claves voice: the gate strikes two resonant hardwood sticks, a clean high-pitched woody ‘tok’ with a quick decay, the crisp clave click that carries a Latin rhythm. in0 = note (transposes), in1 = gate. Tone sets the pitch, Level the output.

ClaveVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Clave voice: a struck wooden clave or woodblock, a high resonant near-sine ping with a fast woody decay, the bright Latin-percussion click. in1 = gate (trigger). Tune the pitch, Level the output.

ClavichordVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Clavichord voice: a string struck and stopped by a metal tangent, a quiet bright Karplus pluck with a hard metallic attack and a tiny pitch-bend wobble (Bebung) from key pressure, the intimate medieval keyboard. in0 = note, in1 = gate (strike). Octave/Fine tune it, Bebung the pressure vibrato, Level the output.

ClavVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Clavinet voice: a bright struck-string Karplus pluck with a hard attack click and a tight, percussive decay run through a comb-like pickup colour, the funky electric clavinet of 70s funk. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Damping the decay, Level the output.

ColLegnoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Col-legno voice: striking the string with the wood of the bow, a short percussive woody tick with a faint pitched component and a dry rattle, the eerie col-legno effect. in0 = note, in1 = gate (strike). Octave tunes it, Level the output.

CombOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Combo-organ voice (Vox/Farfisa): a bright reedy additive organ summing the first few harmonics into the buzzy, slightly-hollow combo-organ tone of 60s garage rock, distinct from the smoother tonewheel organ. in0 = note, in1 = pitch mod. Octave tunes it, Bright the upper-harmonic level, Level the output.

CombResonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Comb-resonator voice: a pitch-rate impulse train fed into a tuned feedback comb filter, whose evenly-spaced resonant peaks ring out a hollow, metallic, plucked-flanger tone distinct from the noise-burst Karplus pluck. in0 = note, in1 = pitch mod. Octave/Fine tune it, Feedback the resonance/decay, Bright the damping, Level the output.

CongaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Conga voice: the gate strikes a hand-drum, a clear pitched membrane tone with a slight upward pitch snap and a short skin decay, the warm open-hand hit of a conga. in0 = note (transposes), in1 = gate. Tune sets the head pitch, Decay the ring, Level the output.

ContrabassoonVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Contrabassoon voice: the deepest double-reed, a low conical-bore waveguide an octave below the bassoon, a dark rumbling foundation reed. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Dark the bore filter, Level the output.

ContrabassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Contrabass voice: the lowest bowed/plucked string, a deep heavily-damped Karplus pluck (pizzicato) with a round woody body and a long fundamental, the orchestral double bass. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Damping the decay, Level the output.

CornetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cornet voice: a conical-bore cousin of the trumpet, the same buzzy sawtooth but rounder and sweeter through a softer lowpass, the mellow lyrical lead of the brass band. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Bright the blare, Level the output.

CowbellVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cowbell voice: the gate strikes two square oscillators a non-harmonic interval apart (the 808 cowbell ratio) gated by a sharp decay, the clanky percussion staple. in0 = note (transposes), in1 = gate. Tune sets the pitch, Decay the length, Level the output.

CrossModVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cross-modulation voice: two oscillators each frequency-modulate the other (mutual cross-FM), a feedback timbre generator that breaks into rich, chaotic, metallic spectra a one-way FM operator cannot reach. in0 = note, in1 = pitch mod. Octave tunes it, Ratio the second oscillator's pitch, Amount the cross-FM depth, Level the output.

CrotaleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Crotale voice: a small tuned bronze disc (antique cymbal), a very high bright ping with widely-stretched inharmonic partials and a long shimmering ring, the crystalline crotale. in0 = note, in1 = gate (strike). Octave tunes it, Decay the ring, Level the output.

CS80Voice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

CS-80 voice: the Yamaha CS-80 brass-lead character (Vangelis's Blade Runner sound), two detuned oscillators with a ring-modulated edge through a resonant lowpass, fat and expressive. in0 = note, in1 = pitch mod. Octave tunes it, Ring the ring-mod amount, Tone the cutoff, Level the output.

DCO [OSC] [CIRCUIT] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Analog-modeled DCO (Roland Juno digitally-controlled oscillator character): rock-stable pitch (no drift) with a band-limited (PolyBLEP) saw and variable-width pulse blended by Pulse, plus a square Sub oscillator one octave down - the clean, punchy Juno foundation that sits perfectly under a chorus. Octave/Fine tune it, PWM sets the pulse width, Pulse blends saw->pulse, Sub the sub-oscillator level, and Level the output. in1 = FM, in2 = pitch mod.

DidgeridooVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Didgeridoo voice: a low lip-buzzed drone under a slowly-moving vocal-tract formant carving the wah overtones, plus a breath rhythm, the meditative drone of a didgeridoo as a playable voice. in0 = note, in1 = gate (breath). Octave tunes the drone, WahRate the formant motion, Wah the depth, Level the output.

DistortedGuitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Distorted-guitar voice: an overdriven electric, a high-sustain Karplus string slammed through heavy tanh distortion for a singing, harmonically-saturated rock lead with long feedback-like sustain. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Drive the distortion, Level the output.

DjembeVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Djembe voice: the gate strikes a West-African goblet drum, a deep bass membrane tone for an open hit blended with a sharp skin-slap noise transient, the rich tone-and-slap of a djembe. in0 = note (transposes), in1 = gate. Tune sets the bass pitch, Slap the slap-noise amount, Decay the ring, Level the output.

DoubleBassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Double-bass voice: the lowest bowed string, a dark sawtooth Helmholtz motion through a deep heavy body with a slow bow attack, the foundation of the string section. in0 = note, in1 = gate (bow). Octave/Fine tune it, Vibrato the waver, Dark the body, Level the output.

DownLifterVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Downlifter FX voice: while the gate is held a tone sweeps steadily downward in pitch over the fall time, the tension-release 'downlifter' that follows a drop; releasing resets it. in1 = gate. Start/End set the sweep range, Time the fall duration, Level the output.

DrawbarOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Drawbar-organ voice: a generic additive drawbar organ, eight sine footages you blend from mellow flutes to bright reeds with an instant gate, the all-purpose organ palette. in0 = note, in1 = gate. Octave tunes it, Lower the 16'/8' fundamentals, Upper the bright harmonics, Level the output.

DroneVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Evolving-drone voice: a wide stack of detuned saws whose detune slowly breathes under a triple-rate LFO, building a thick, ever-shifting sustained drone for ambient beds. in0 = note, in1 = pitch mod. Octave/Fine tune it, Spread the base detune, Motion the LFO depth, Level the output.

DualFMVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Dual-modulator FM voice: a sine carrier phase-modulated by two stacked modulator operators at independent ratios, building richer, more evolving FM spectra than a single 2-op pair (a compact 3-op stack). in0 = note, in1 = FM in, in2 = pitch mod. Octave tunes it, RatioA/RatioB the two modulator pitches, Index the FM depth, Level the output.

DualVCO [OSC] [CIRCUIT] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Analog-modeled dual VCO (the fat two-oscillator lead/bass, Moog/Prophet character): two band-limited (PolyBLEP) oscillators - osc2 offset by Interval semitones and Detune cents - summed for the classic thick two-VCO tone, with slow analog drift. Distinct from the 7-voice Susaw: this is the tuned 2-oscillator voice. Octave tunes both, Interval sets osc2's pitch offset, Detune the fine beating, Shape sweeps triangle->saw->pulse, Mix the osc1/osc2 balance, and Level the output. in1 = FM, in2 = pitch mod.

DustVoice [OSC] [SYNTH] in: Pitch (CV) out: Audio

Dust voice: a stochastic generator firing random unit impulses at a mean Density (the granular 'dust' source of Xenakis/Roads), a crackling sparse-to-dense impulse field for percussion textures and excitation. Density sets the mean impulses per second, Tone a gentle lowpass colour, Level the output.

ElectricGrandVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Electric-grand voice (CP-70 style): a Karplus struck-string piano fed through a magnetic pickup, giving an acoustic body with a focused, slightly-electric comb colour, the stage-piano of 70s ballads. in0 = note, in1 = gate (strike). Octave/Fine tune it, Pickup the comb depth, Level the output.

ElectricGuitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Electric-guitar voice: a clean electric pluck, a Karplus string with a magnetic-pickup comb colour and a tight medium sustain, the bell-clean Strat tone before any amp drive. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Tone the pickup brightness, Level the output.

EnglishHornVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

English-horn voice: the alto oboe, a double-reed conical-bore waveguide warmer and rounder than the oboe, the plaintive cor anglais. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Warmth the bore filter, Level the output.

ErhuVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Erhu voice: a bowed two-string Chinese fiddle, a bowed-string waveguide with a narrow nasal bandpass body and a singing vibrato, the expressive crying tone of the erhu. in0 = note, in1 = gate (bow). Octave/Fine tune it, BowForce the grip, Vibrato the waver, Level the output.

EuphoniumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Euphonium voice: a tenor-tuba, a broad warm sawtooth through a rounded lowpass with a smooth legato attack, the singing baritone of the brass band sitting between trombone and tuba. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Warmth the body, Level the output.

FairlightVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Fairlight voice: the gritty low-bit-rate digital sampler character of the 1980s CMI, a sawtooth deliberately bit-crushed and sample-rate-reduced for the famous lo-fi digital sheen of early sampling. in0 = note, in1 = pitch mod. Octave/Fine tune it, Bits the quantization, Crush the downsample, Level the output.

FarfisaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Farfisa voice: a 1960s transistor combo organ, a bright thin reedy tone from a fixed stack of square-ish harmonics with a fast gate, the buzzy garage/ska organ. in0 = note, in1 = gate. Octave tunes it, Bright the upper-harmonic edge, Vibrato the waver, Level the output.

FifeVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Fife voice: a high marching fife, a shrill near-sine tone an octave above the flute with a strong air hiss and a fast attack, the piercing whistle of the fife-and-drum corps. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Breath the air noise, Level the output.

FlugelhornVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Flugelhorn voice: a mellow conical-bore lip-reed brass, darker and rounder than a trumpet, the warm flugelhorn of jazz ballads. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Mellow the bell filter, Level the output.

FluteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Flute voice: a concert flute, a nearly pure sine tone coloured by a wash of breath noise across the embouchure with a quick attack and a light vibrato, the airy lyrical woodwind. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Breath the air noise, Level the output.

FluteVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Flute waveguide voice: an open-bore delay line excited by an air-jet cubic nonlinearity while the gate (breath) is held, giving the breathy, pure tone of a flute or whistle - the jet-driven counterpart to the reed-driven ReedVoice. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Noise the air hiss, Level the output.

FMBassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

FM-bass voice: a punchy two-operator FM bass with a tight pitch-tracked modulator and a fast index decay for a clicky attack over a deep body, the snappy digital bass of FM synths. in0 = note, in1 = gate. Octave tunes it, Ratio the modulator pitch, Click the attack index, Decay the body, Level the output.

FMBellVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Two-operator FM bell voice: the gate strikes a sine carrier phase-modulated by a non-integer-ratio sine operator whose index decays, so the spectrum starts bright and clangorous and settles to a pure tone - the classic DX bell/marimba.

in0 = note, in1 = gate. Octave tunes it, Ratio the modulator pitch, Index the initial FM depth, Decay the ring/index time, Level the output.

FMStackVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Six-operator FM voice: six sine operators at harmonic ratios wired by a selectable algorithm (a serial modulator chain, three 2-op pairs, or all-parallel additive), the full DX-style FM engine in one block. in0 = note, in1 = FM in, in2 = pitch mod. Octave tunes it, Algorithm picks the wiring, Index the FM depth, Level the output.

FormantGrowl [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Growl-bass voice: a sawtooth driven hard through tanh distortion then a resonant bandpass formant, giving the snarling, vocal, talkbox-like growl of a dubstep/neuro bass. in0 = note, in1 = formant sweep (added to Formant), in2 = pitch mod. Octave/Fine tune it, Drive the distortion, Formant the vowel peak, Reso its sharpness, Level the output.

FormantOsc [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Vowel oscillator: a sawtooth driven through two resonant formant band-passes whose centre frequencies morph across the A-E-I-O-U vowels (Vowel), giving a singing/talking tone. Unlike FOF granular synthesis this is a buzz-through-resonators model; Res sets formant sharpness and Level the output.

FormantSeqVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Vowel-sequencer voice: a sawtooth driven through a formant bandpass that steps through a vowel sequence (A-E-I-O-U) at the set rate, so a held note sings a changing vowel pattern like a chanting voice. in0 = note, in1 = pitch mod. Octave/Fine tune it, Rate the vowel-step speed, Reso the formant sharpness, Level the output.

FractalNoiseVoice [OSC] [SYNTH] in: Pitch (CV) out: Audio

Fractal-noise voice: a 1/f^beta colored noise generator built by summing octave-spaced random layers (fractional Brownian motion), tunable continuously from white through pink to deep red/brown noise. in0 = unused, in1 = unused. Color sets the spectral slope (0 white .. 1 brown), Level the output.

FrenchHornVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

French-horn voice: a mellow orchestral horn, a sawtooth rounded by a soft lowpass with a slow legato attack and a warm vibrato, the noble round brass that sits behind the section. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Mellow the roundness, Level the output.

FrenchHornVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

French-horn voice: a lip-reed bore with a mellow flared bell, self-oscillating on breath into the warm rounded brass of a horn, softer than trumpet. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Mellow the bell filter, Level the output.

GamelanVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Gamelan voice: a struck bronze metallophone bar (saron/gender), the gate ringing a cluster of stretched inharmonic partials with a shimmering beating decay, the gong-bar tone of an Indonesian gamelan. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

GeoBlend [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Drawable geometric oscillator (u-he Zebra GeoBlend-style): trace two hand-drawn single-cycle geometries and Morph continuously between them. Draw Geo A and Geo B in the editor (drag points, double-click to add/remove); Morph crossfades A->B for evolving timbres no fixed wave can make. Octave/Fine tune it, Level scales the output.

GeoMorph [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Geometric oscillator (u-he Zebra GeoMorph-style): synthesises the waveform of a point tracing a regular polygon, so Order - the number of sides, continuously morphable - sets the harmonic richness, from a near-sine at low orders to bright buzzy formant-like spectra at high orders, while Teeth skews the geometry for asymmetric reedy timbres. A vertex/geometry-based oscillator distinct from wavetable and additive synthesis. Octave and Fine tune it, Level scales the output.

GlassArmonicaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Glass-armonica voice: the pure, ethereal tone of Franklin's rubbed-glass instrument, a near-sine fundamental with faint even partials that swells in slowly under the finger and shimmers with a gentle beating, the ghostly sound of wet glass bowls. in0 = note, in1 = pitch mod. Octave/Fine tune it, Swell the attack rise, Level the output.

GlassPad [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Glassy FM pad: a two-operator FM voice with a high inharmonic modulator ratio and a low index, giving the clear, shimmering, bell-glass pad timbre of classic DX electric-piano/pad patches. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Ratio the modulator pitch, Index the FM depth, Level the output.

GlideLeadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Glide lead voice: a single detuned-saw mono lead with portamento that slides smoothly from the previous note to the new one, the expressive sliding synth-lead of leads and basslines. in0 = note, in1 = pitch mod. Octave/Fine tune it, Glide the portamento time, Detune the unison width, Level the output.

GlitchVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Glitch/lo-fi voice: a saw oscillator deliberately bit-crushed and sample-rate-reduced so the note dissolves into aliased, quantized digital grit, the broken-CD/circuit-bent texture. in0 = note, in1 = pitch mod. Octave/Fine tune it, Bits the quantization depth, Crush the downsample factor, Level the output.

GlockenspielVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Glockenspiel voice: the gate strikes a small steel bar, a high bright bell tone with widely-stretched metallic partials and a long shimmering decay, the sparkle of the glockenspiel/orchestra bells. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

GongVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Gong voice: the gate strikes a large gong, a dense cluster of detuned inharmonic partials that builds in shimmer after the strike (the gong's signature swell-up) before slowly fading, the crashing wash of a tam-tam. in0 = note, in1 = gate. Octave tunes it, Decay the wash length, Level the output.

∿ **GrandPianoVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Grand-piano voice: a struck-string Karplus model with a bright hammer attack, slightly stretched inharmonic partials and a resonant soundboard body, the rich acoustic grand. in0 = note, in1 = gate (strike). Octave/Fine tune it, Hammer the attack brightness, Level the output.

∿ **GuiroVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Guiro voice: while the gate is held a notched gourd is scraped, a rasping ratchet of fast clicks rising slightly in rate across the stroke, the Latin guiro scrape. in1 = gate. Rate sets the scrape speed, Tone the rasp brightness, Level the output.

∿ **GuzhengVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Guzheng voice: a Chinese zither, a warm Karplus pluck followed by an expressive downward pitch-bend tail (the left-hand vibrato/slide pressing the string), the lyrical ornament of the guzheng. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Bend the slide depth, Level the output.

∿ **HammondVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hammond voice: a tonewheel drawbar organ, nine sine partials at the classic organ footages summed into the full B3 tone with a percussive key-click attack and an instant gate, the soul/gospel organ. in0 = note, in1 = gate. Octave tunes it, Drawbars the upper-partial brightness, Click the key percussion, Level the output.

∿ **HandClapVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hand-clap voice: a clap, a series of three rapid noise bursts (many hands not-quite-together) followed by a short reverberant noise tail through a bandpass, the classic drum-machine clap. in1 = gate (trigger). Tone the brightness, Level the output.

∿ **HandpanVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Handpan voice: the gate strikes a note-dimple of a hang/handpan, a soft fundamental locked with its octave and compound-fifth overtones for a warm meditative bell-like ding. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

∿ **HardKick** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hardstyle/rawstyle kick generator: a gate-struck kick with a fast downward pitch sweep from the click transient to the body fundamental, an exponential amp decay and built-in tanh distortion for the signature hard punch. in0 = note (optional, transposes the body), in1 = gate (strike). Tune sets the body pitch, Punch the pitch-sweep depth, PSweep its time, Decay the body length, Click the attack transient, Drive the distortion. Patch a reverse-bass tail after it for the full hardstyle kick.

∿ **HarmonicaVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Harmonica voice: a buzzy free-reed tone through its own resonance with a draw-bend, the bluesy reedy timbre of a blues harp as a note-driven voice. in0 = note, in1 = pitch mod. Octave/Fine tune it, Bend the draw-bend pitch droop, Reed the resonance, Level the output.

∿ **HarmonicsStringVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Harmonics-string voice: natural string harmonics (flageolet), a near-pure sine touched with a single octave overtone and a light bow attack, the glassy whistling flageolet tone. in0 = note, in1 = gate (bow). Octave/Fine tune it, Octave2 the overtone level, Level the output.

HarmoniumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Harmonium voice: a hand-pumped reed harmonium, a buzzy free-reed sawtooth with a breathy bellows shimmer and a slow swell, the Indian/folk drone organ. in0 = note, in1 = gate. Octave/Fine tune it, Reed the buzz, Bellows the swell shimmer, Level the output.

HarpVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Harp voice: a concert-harp string, a soft-attack Karplus pluck with very light damping for a long shimmering decay, gentle and bright, ideal for arpeggiated runs. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

HonkyTonkVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Honky-tonk-piano voice: a saloon upright with two strings per note slightly out of tune, a detuned dual-Karplus pluck that beats and jangles, the boozy bar-room piano. in0 = note, in1 = gate (strike). Octave tunes it, Detune the string spread, Level the output.

Hoover [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Hoover / Mentasm lead (Alpha Juno ‘What The…’ rave stab): faithful to the original single-DCO patch - a sawtooth combed by a pulse (Roland ‘sawtooth PWM’, the signature hoover vowel), a square sub two octaves down, a light detuned ensemble standing in for the Juno chorus, and the note-on pitch sweep - the staple of rave/hardcore/hardstyle. Movement comes from the PWM comb, not a detuned saw stack. in0 = note, in1 = gate (arms the sweep), in2 = pitch mod. Octave/Fine tune it, PWM the comb width, Chorus the ensemble width, Sweep the note-on pitch-drop depth, SweepT its time, Level the output.

HurdyGurdyVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hurdy-gurdy voice: a rosined wheel bows a melody string and a constant drone string together while the buzzing-bridge ‘dog’ rattles rhythmically, the medieval drone-fiddle. in0 = note, in1 = pitch mod. Octave tunes the melody, Buzz the dog-bridge rattle rate, Level the output.

JewsHarpVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Jew’s-harp voice: a fixed-pitch twanging lamella whose buzz is filtered by a sweepable mouth-cavity formant that the player shapes, the boing-wow of a jaw harp; the note sets the cavity formant. in0 = note (formant), in1 = gate (twang). Twang sets the lamella pitch, FormantRate the wah motion, Level the output.

JunoChorusVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Juno-chorus voice (Roland Juno character): a single DCO sawtooth thickened by a lush stereo-style chorus (a slow detuned doubling) through a gentle lowpass, the iconic warm Juno pad. in0 = note, in1 = pitch mod. Octave tunes it, Chorus the doubling depth, Tone the cutoff, Level the output.

JupiterPadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Jupiter-pad voice (Roland JP-8 character): a lush six-saw super-detuned pad through a gentle resonant lowpass with a slow filter drift, the wide warm analog-poly pad of the early 80s. in0 = note, in1 = pitch mod. Octave tunes it, Detune the width, Tone the cutoff, Level the output.

KalimbaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Kalimba/mbira voice: the gate plucks a short metal tine, a bright fundamental with a quick buzzy attack and a fast decay plus a faint octave partial, the warm plinking thumb-piano tone. in0 = note, in1 = gate (pluck). Octave tunes it, Buzz the attack-noise amount, Decay the ring time, Level the output.

Karplus [OSC] [SYNTH] in: Pitch (CV) out: Audio

Karplus-Strong plucked string: noise burst in a tuned damped delay.

KazooVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Kazoo voice: humming into a kazoo, a buzzy nasal tone where the voiced pitch drives a vibrating membrane that adds a raspy comb-buzz to every note, the comic party-favour timbre. in0 = note, in1 = pitch mod. Octave/Fine tune it, Buzz the membrane rasp, Level the output.

KickVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Synth kick voice: the gate strikes a sine whose pitch drops sharply from a click to a low body, with an exponential amp decay and optional drive, the 808/909 bass-drum as a note-driven voice. in0 = note (transposes the body), in1 = gate. Tune sets the body pitch, Punch the pitch-drop depth, Decay the length, Drive the saturation, Level the output.

KoraVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Kora voice: a West-African 21-string harp-lute, a bright soft-attack Karplus pluck with a long shimmering decay and a faint gourd-body resonance, the cascading harp tone of the kora. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

KotoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Koto voice: a plucked silk/nylon string over a movable bridge, a Karplus pluck with a bright attack and a woody body resonance and a faint bridge buzz, the shimmering tone of the Japanese koto. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Damping the decay, Level the output.

KSDrumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Karplus drum voice: a very short, heavily-damped Karplus-Strong delay excited by a noise burst on the gate, so instead of a pitched pluck it produces a tuned membrane/tom thud - physical-model percussion. in0 = note, in1 = gate. Octave tunes it, Damping the skin tightness, Decay the membrane ring, Level the output.

LogDrumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Log-drum voice: a wooden slit drum, a struck low hollow-wood thunk with two pitched tongues and a medium woody decay, the mellow tongue-drum tone. in0 = note, in1 = gate (strike). Octave tunes it, Decay the ring, Level the output.

LuteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Lute voice: a Renaissance gut-string lute, a soft warm Karplus pluck with gentle attack and a rounded mellow body, the intimate plucked tone of early music. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

MalletVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Mallet voice: a percussive two-operator FM tone with a fast index decay and a short amp decay, giving the woody/glassy struck-mallet attack of a marimba or vibraphone patch. in0 = note, in1 = gate. Octave tunes it, Ratio the modulator pitch, Decay the length, Level the output.

MandolinVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Mandolin voice: a bright paired-course plucked string with a fast tremolo re-pluck while held (the mandolin's signature shimmer), two slightly-detuned Karplus strings beating together. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Tremolo the re-pluck rate, Level the output.

MarimbaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Marimba voice: the gate strikes a tuned rosewood bar voiced from its deep-cut 1:4:10 partials over a resonator-tube hum, the warm woody mallet tone of a marimba. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

MellotronVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Mellotron voice: the tape-replay string machine of prog rock, a bowed-saw string tone with a slow attack, a wavering tape wow/flutter and occasional dropout, the haunting wobble of looping tape. in0 = note, in1 = gate. Octave/Fine tune it, Flutter the tape wobble, Level the output.

MelodicaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Melodica voice: a breath-blown free reed with a bright buzzy sawtooth-through-reed-resonance tone, the cheerful schoolroom keyboard-harmonica. in0 = note, in1 = gate (breath). Octave/Fine tune it, Reed the resonance brightness, Level the output.

ModalResonator [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Modal-resonator voice: the gate strikes a bank of seven inharmonic partials each with its own exponential decay, ringing out a struck metal-bar/bell tone; Inharm morphs from a harmonic stack toward stretched bell ratios. in0 = note, in1 = gate (strike). Octave tunes it, Inharm the partial stretch, Decay the ring time, Level the output.

MoogBassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Moog-bass voice: the fat foundation bass of a Minimoog, a stack of detuned saws driven through a resonant 4-pole-style lowpass with overdrive, the warm growling synth bass. in0 = note, in1 = pitch mod. Octave tunes it, Cutoff the filter, Reso the resonance, Drive the saturation, Level the output.

MS20LeadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

MS-20-lead voice (Korg MS-20 character): two detuned oscillators slammed through an aggressive, screaming resonant lowpass with overdrive, the gritty acid-scream mono lead. in0 = note, in1 = pitch mod. Octave tunes it, Cutoff the filter, Reso the screech, Drive the grit, Level the output.

MusicalSawVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Musical-saw voice: a bowed flexible saw blade, an almost-pure sine (the flat sweet spot vibrating) with a wide slow expressive vibrato and a faint metallic shimmer, the ghostly vocal wail of the singing saw. in0 = note, in1 = gate (bow). Octave/Fine tune it, VibDepth the wide vibrato, Level the output.

MusicBoxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Music-box voice: the gate plucks a tuned steel comb tooth, a soft pure bell-like tone with a fast attack, a faint inharmonic shimmer and a medium decay, the delicate plink of a wind-up music box. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

MutedTrumpetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Muted-trumpet voice: a trumpet with a straight mute, a sawtooth squeezed through a narrow resonant bandpass that gives the pinched nasal honk of a muted horn, the jazzy wah-brass colour. in0 = note, in1 = gate (breath). Octave/Fine tune it, Mute the bandpass pinch, Vibrato the waver, Level the output.

NoiseCymbalVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Cymbal/crash voice: the gate fires a dense bank of inharmonic squares high-passed into a shimmering metallic wash with a long decay, the synth crash/ride. in1 = gate. Tone sets the metallic region, Decay the ring (crash vs ride), Level the output.

NoiseHat [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hi-hat percussion voice: the gate fires a burst of metallic noise (six detuned square oscillators) high-passed and gated by a fast decay, a closed/open hi-hat depending on the decay time. in1 = gate (strike). Tone sets the metallic pitch region, Decay the length (closed vs open), Level the output.

NoiseSweep [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Riser/sweep voice: while the gate is held, white noise is pushed through a bandpass whose centre frequency rises steadily, building the upward whoosh/riser used as a transition effect; releasing the gate resets the sweep. in0 = unused, in1 = gate. Start/End set the sweep range, Time the rise duration, Reso the band sharpness, Level the output.

NoiseTuned [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Resonant tuned-noise voice: white noise driven through a steep state-variable bandpass tracking the note pitch, so noise acquires a definite pitch (the whistling, breathy, wind-instrument-edge tone). Resonance sets how tonal versus airy. in0 = note, in1 = pitch mod. Octave/Fine tune it, Resonance the bandpass Q, Level the output.

NylonGuitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Nylon-guitar voice: a classical nylon-string pluck, a softer-attack Karplus with heavier high-frequency damping for a warm, mellow, woody fingerstyle tone. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Warmth the damping, Level the output.

OboeVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Oboe voice: a double-reed oboe, a bright narrow-pulse waveform through a lowpass that keeps its reedy nasal upper harmonics, with a focused attack and gentle vibrato, the plaintive penetrating woodwind. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Reed the nasal bite, Level the output.

OboeVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Oboe voice: a double-reed narrow-conical bore waveguide, bright and nasal with the full harmonic series, self-oscillating on breath. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Bright the bore filter, Level the output.

OcarinaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Ocarina voice: a Helmholtz vessel flute, breath exciting a single resonant cavity mode so it produces an almost-pure tone with little harmonic content and a breathy edge, the round sweet whistle of an ocarina. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air noise, Level the output.

OndesMartenotVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Ondes Martenot voice: the ethereal early-electronic instrument, a pure sine that glides smoothly between notes (ribbon portamento) with a gentle expressive vibrato and a faint added harmonic, the ghostly voice of Messiaen. in0 = note, in1 = pitch mod. Octave/Fine tune it, Glide the ribbon portamento, Vibrato the waver, Level the output.

Operator [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Single FM operator: a sine tuned to Ratio times the note, phase-modulated by the in1 input (scaled by FM) plus its own Feedback - the patchable DX-style building block you chain to build any FM algorithm.

OrchHitVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Orchestra-hit voice: the gate fires a dense cluster of harmonics plus a short noise transient that decays fast, the iconic sampled ‘orch hit’ stab of 80s/90s pop and hip-hop, re-synthesised. in0 = note, in1 = gate. Octave tunes it, Spread the cluster width, Decay the length, Level the output.

Osc [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Virtual-analog oscillator: saw/square/tri with PWM, sub and unison detune.

OudVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Oud voice: a fretless Middle-Eastern lute, a warm round-bodied Karplus pluck with heavier high-frequency damping for a mellow woody tone and no fret buzz, the deep voice of the oud. in0 = note, in1 = gate (pluck). Octave/Fine tune it (fretless, so any microtone), Warmth the damping, Level the output.

PadEnsemble [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

String-ensemble pad voice (Solina/string-machine character): a wide stack of detuned saws fed through a triple-LFO ensemble chorus, giving the lush, swirling, BBD-chorused string-synth pad of the 1970s. in0 = note, in1 = pitch mod. Octave/Fine tune it, Ensemble the chorus depth/width, Tone a gentle high-cut, Level the output.

PADsynthVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

PADsynth pad voice (Nasca Octavian Paul): each harmonic is spread into a small band of closely-detuned partials instead of a single line, so the spectrum is smeared into a lush, animated, ever-evolving choir/string pad that a plain additive stack can never sound. in0 = note, in1 = pitch mod. Octave tunes it, Bandwidth the per-harmonic spread, Bright the harmonic rolloff, Level the output.

PAF [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Phase-Aligned Formant oscillator (IRCAM/Csound paf-style): a single formant built from a harmonic carrier centred on the Formant frequency, multiplied by a phase-aligned bell window whose width Bandwidth sets, all locked to the note's fundamental so the spectrum stays clean and clickless. Distinct from FOF (grain bursts), Vosim (sin-squared pulses) and FormantOsc (resonator-filtered saw): here one formant is a single phase-aligned carrier times window. Formant places the peak, Bandwidth its spectral width, Level the output.

PalmMuteGuitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Palm-mute guitar voice: the chugging palm-muted electric, a heavily-damped Karplus string with a short percussive thunk and a touch of drive, the tight rhythm-chug of metal and punk. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Drive the grit, Level the output.

PanFluteVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Pan-flute voice: a stopped-pipe edge tone, breath noise exciting a quarter-wave resonant pipe (odd harmonics, hollow and breathy) tracking the note, the haunting hoot of pan pipes. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air noise, Level the output.

PercClap [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Hand-clap percussion voice: the gate fires three quick stuttered bandpassed-noise bursts followed by a diffuse tail (the classic clap recipe), a one-shot percussion voice. in0 = unused, in1 = gate (strike). Tone sets the bandpass centre, Spread the burst spacing, Decay the tail length, Level the output.

PhaseDistVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Casio-CZ phase-distortion voice: a sine read with a warped phase index so the waveform morphs from sine toward a resonant saw/pulse as Amount rises (the phase-distortion synthesis of the CZ-101), giving sweepable formant-like resonance without a filter. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Amount the distortion depth, Reso the pseudo-resonance harmonic, Level the output.

PianetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Pianet voice: a Hohner Pianet electric piano, a Karplus reed plucked by a sticky pad giving a bright buzzy attack and a short woody decay, the punchy 60s EP heard on early pop records. in0 = note, in1 = gate (strike). Octave/Fine tune it, Level the output.

PiccoloTrumpetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Piccolo-trumpet voice: a very high bright lip-reed brass (the Baroque clarino trumpet), a piercing brilliant fanfare voice an octave above the trumpet. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Bright the bell filter, Level the output.

PiccoloVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Piccolo voice: the highest orchestral woodwind, a small open-bore air-jet waveguide voiced very high and bright with a breathy edge, the piercing whistle of a piccolo. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air, Level the output.

PipaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Pipa voice: a Chinese pear-shaped lute famous for its tremolo (lunzhi), a bright Karplus pluck rapidly re-plucked while held for a continuous shimmering roll. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Tremolo the roll rate, Level the output.

PipeOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Pipe-organ voice: a stack of flue-pipe ranks (octave-spaced near-sine pipes) with a breathy chiff transient on each attack, the majestic additive tone of a church pipe organ, distinct from the buzzy tonewheel organ. in0 = note, in1 = gate. Octave tunes it, Ranks the number of pipe ranks, Chiff the attack breath, Level the output.

PitchRiserVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tonal-riser FX voice: while the gate is held a tone sweeps steadily upward in pitch over the rise time, building the tension-riser/uplifter used before a drop; releasing the gate resets it. in1 = gate. Start/End set the sweep pitch range, Time the rise duration, Level the output.

PizzicatoStringVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Pizzicato-string voice: a plucked (not bowed) orchestral string, a short bright Karplus pluck with a quick decay and a woody body, the staccato pizz of the string section. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Level the output.

PluckChordVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Plucked-chord voice: from one note it sounds a whole triad as three detuned decaying tones with a shared noise-burst attack, the strummed-chord stab of a plucked guitar or harp in a single paraphonic voice. in0 = note, in1 = gate (strum). Octave tunes it, Chord the triad quality, Decay the ring, Level the output.

PluckedHarmonicVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Flageolet pluck voice: a Karplus-Strong string lightly touched at a node so the fundamental is damped and only the bright harmonic (the flageolet/natural-harmonic whistle of a guitar or harp) rings. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Node selects which harmonic sounds, Level the output.

PluckString [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Karplus-Strong plucked-string voice: the gate fires a noise burst into a pitch-tuned delay line damped by a lowpass in the feedback, the classic physical-modelled pluck. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Damping sets how fast the high end dies (nylon to muted), Bright the excitation colour, Level the output.

PolysixPadVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Polysix-pad voice (Korg Polysix character): warm detuned saws through a soft lowpass with a built-in chorus shimmer, the lush, slightly-fuzzy analog poly pad. in0 = note, in1 = pitch mod. Octave tunes it, Detune the width, Tone the cutoff, Level the output.

PowerChordVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Power-chord voice: an overdriven guitar power chord from one note, a Karplus string plus an added fifth tap slammed through heavy distortion for the root-fifth wall of rock rhythm guitar. in0 = note, in1 = gate (strum). Octave/Fine tune it, Drive the distortion, Level the output.

PPGWaveVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

PPG-wave voice: the cold digital wavetable character of the PPG Wave, a stepped harmonic spectrum scanned by a slow sweep so the timbre morphs through gritty digital waveforms, the metallic-pad sound of 80s wavetable synths. in0 = note, in1 = pitch mod. Octave tunes it, Scan the wavetable position, Level the output.

ProphetSyncVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Prophet-sync voice (Prophet-5 character): a hard-sync lead where a slave saw is reset by a master oscillator, with a sweepable sync ratio for the searing, vowel-shifting sync-lead sweep. in0 = note, in1 = pitch mod. Octave tunes it, Sync the slave ratio, Level the output.

PWMPad [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

PWM pad voice: a band-limited pulse oscillator whose width is swept by an LFO so the harmonic content breathes in and out (the classic pulse-width-modulation string/pad movement), thickened by a detuned unison. in0 = note, in1 = pitch mod. Octave/Fine tune it, Rate the PWM LFO speed, Depth the width sweep, Level the output.

QanunVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Qanun voice: an Arabic plucked box-zither, a bright trichordo Karplus pluck with fast double-course beating and a quick decay, the rippling tremolo runs of the qanun. in0 = note, in1 = gate (pluck). Octave/Fine tune it (microtones supported), Level the output.

RecorderVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Recorder voice: a fipple flute, an open-bore air-jet waveguide with a pure, slightly-hollow tone and a soft breath edge, the simple sweet sound of a recorder. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air, Level the output.

ReedOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Reed-organ voice: a pump/parlour reed organ, a mellow set of reedy harmonics softened by a lowpass with a gentle swell, the church-hall harmonium tone. in0 = note, in1 = gate. Octave tunes it, Reed the harmonic bite, Tone the cutoff, Level the output.

ReedVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Reed-instrument voice: a cylindrical-bore delay-line waveguide driven by a clipped single-reed nonlinearity while the gate (breath) is held, self-oscillating into the hollow odd-harmonic tone of a clarinet-family reed. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Bright the bore filter, Level the output.

Reese [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

DnB / neurofunk Reese bass: a stack of heavily detuned saws beating against each other (the classic Kevin Saunderson 'Reese' tone) through a low-pass and tanh drive, for the growling, metallic, ever-moving low end of drum & bass. in0 = note, in1 = pitch mod. Octave/Fine tune it, Detune sets the beating spread (the Reese character), Voices the saw count, Tone the low-pass, Drive the distortion, Level the output. Modulate Detune or Tone for the neuro morph.

ResoBass [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Resonant squelch-bass voice: a saw run through a self-enveloped resonant lowpass whose cutoff snaps open on the gate and decays, the acid/squelch bass where the filter envelope is the sound. in0 = note, in1 = gate. Octave tunes it, Cutoff the base frequency, EnvAmt the sweep depth, Decay the filter-envelope time, Level the output.

ResonBodyVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Resonant-body voice: a short noise/impulse excitation on the gate is rung through a bank of fixed inharmonic modal resonances (an instrument-body resonator), giving a struck or scraped wooden/metal-box tone shaped by the body, not the exciter. in0 = note, in1 = gate. Octave tunes the body, Decay the ring, Bright the high-mode mix, Level the output.

ResoPluckVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Resonant-pluck voice: the gate fires a sharp impulse into a steep resonant bandpass tuned to the note, which rings down into a pinged pitched tone (a filter-rung pluck, distinct from the comb-based pluck). in0 = note, in1 = gate. Octave/Fine tune it, Resonance the ring length/Q, Level the output.

ReverseCymbalVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Reverse-cymbal voice: while the gate is held a metallic noise wash swells from silence to a peak (the reversed-cymbal riser that announces a downbeat), resetting when the gate releases. in1 = gate. Time sets the swell duration, Tone the brightness, Level the output.

RhodesVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Electric-piano voice (DX/Rhodes tine): a two-operator FM bell-tine where a high-ratio modulator with a fast-decaying index gives the struck-tine attack settling to a pure tone, the classic FM e-piano. in0 = note, in1 = gate. Octave tunes it, Ratio the tine modulator pitch, Bark the attack brightness, Decay the note length, Level the output.

RideCymbalVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Ride-cymbal voice: the gate strikes a ride cymbal, a pingy struck bell tone riding on a sustained metallic noise wash with a long decay, the steady ping-and-wash of a jazz ride. in1 = gate. Tone sets the ping pitch, Decay the wash, Level the output.

RimshotVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Rimshot voice: a drum rimshot / sidestick, a very short sharp pitched click with a quick woody resonance, the tight tick of striking the drum rim. in1 = gate (trigger). Tune the pitch, Level the output.

RimVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Rimshot voice: the gate fires a very short sharply-decaying resonant click (stick on the rim), bright and dry. in0 = note (transposes), in1 = gate. Tune sets the pitch, Level the output.

∿ RingVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Ring-modulation voice: two note-tracking sine oscillators (a carrier and a Ratio-tuned modulator) multiplied to produce the clangorous metallic sum-and-difference spectrum of ring modulation, the inharmonic bell/robot tone subtractive synths cannot reach. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Ratio the modulator pitch, Mix dry-carrier vs ring, Level the output.

∿ Risset [OSC] [SYNTH] in: Pitch (CV), Gate out: Audio

Risset bell: a gate-struck additive bell from seven partials at the classic Jean-Claude Risset frequency ratios, each with its own amplitude and exponential decay (higher partials ring shorter), giving the shimmering slightly-detuned metallic bell tone that single-FM or harmonic-additive voices cannot. in0 is the note, in1 the strike gate. Decay sets the ring time, Inharm morphs from a harmonic stack (0) to the full inharmonic Risset ratios (1), and Level scales the output.

∿ RotaryOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Rotary-organ voice: a drawbar organ through a Leslie rotary speaker, the additive tone swept by a rotating Doppler vibrato and amplitude throb, the swirling rock/jazz organ. in0 = note, in1 = gate. Octave tunes it, Speed the rotor rate, Depth the swirl, Level the output.

∿ RotoTomVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Roto-tom voice: a tunable single-head tom, a struck membrane with a clean tuned body and a moderate pitch drop, the melodic roto-tom heard in fills. in0 = note, in1 = gate (strike). Octave tunes it, Decay the body, Level the output.

∿ Sample [OSC] [SYNTH] in: – out: Audio

Ableton-Simpler-style sample playback: pitch-tracked from the loaded slot with selectable interpolation (Linear/Cubic/None), playback Mode (Forward/Reverse/Ping-Pong), an independent loop region (Loop Start..End) with click-free loop Crossfade. Start/End set the played window; Loop wraps back to Loop Start.

∿ SampleHoldSynthVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sample-and-hold voice: a tone whose pitch is re-randomised to a new quantised scale step on each clock tick (the stepped random-pitch bleeps of a sample-and-hold sequencer), driven from the note as the pitch centre. in0 = note, in1 = pitch mod. Octave tunes it, Rate the clock speed, Range the random pitch spread in semitones, Level the output.

∿ SanturVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Santur voice: a Persian hammered dulcimer, a bright metal string struck by a light mallet (a quick-attack Karplus pluck) with a silvery sustain and faint detuned-course shimmer, the cascading santur. in0 = note, in1 = gate (strike). Octave/Fine tune it, Level the output.

∿ SaxSectionVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sax-section voice: a horn section of reeds, several detuned reed-shaped tones (sawtooth softened by a reed waveshape) through a bright bandpass formant, the honking blend of a sax section. in0 = note, in1 = pitch mod. Octave tunes it, Detune the section width, Reed the formant bite, Level the output.

SaxVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Saxophone waveguide voice: a single-reed bore whose conical shape gives the full harmonic series (brighter and reedier than the odd-only clarinet ReedVoice), self-oscillating while the gate breath is held. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Bright the bore filter, Level the output.

ScannedVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Scanned-synthesis voice (Verplank/Mathews): a ring of masses connected by springs is plucked and slowly oscillates, and its instantaneous shape is read out as a dynamic wavetable at the note pitch, so the timbre morphs organically as the physical surface wobbles. in0 = note, in1 = pitch mod. Octave tunes it, Tension the spring stiffness, Damping the settle rate, Level the output.

Screech [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Hardstyle screech lead: a stack of detuned saws driven through heavy tanh distortion and a resonant formant band-pass, giving the screaming metallic gabber/hardstyle lead that a clean saw or single distortion cannot. in0 = note, in1 = formant sweep (added to Formant), in2 = pitch mod. Octave/Fine tune it, Detune spreads the saws, Drive the distortion, Formant the screech vowel peak, Reso its sharpness, Level the output.

SH101BassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

SH-101-bass voice (Roland SH-101 character): a punchy mono saw blended with a square and a sub-oscillator an octave down, through a snappy lowpass, the classic acid/electro bass. in0 = note, in1 = pitch mod. Octave tunes it, Sub the sub-osc level, Cutoff the filter, Level the output.

ShakerVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Shaker voice: the gate fires a short burst of band-passed noise with a soft attack and quick decay, the maraca/shaker/cabasa rattle. in1 = gate. Tone sets the brightness, Decay the length, Level the output.

ShakuhachiVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Shakuhachi voice: a Japanese bamboo flute, an open-bore air-jet waveguide voiced with a heavy breathy mura-iki edge and a gentle pitch waver, the meditative haunting tone of the shakuhachi. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air, MuraIki the breath-noise depth, Level the output.

ShamisenVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Shamisen voice: a plucked three-string lute struck with a large plectrum, a punchy Karplus pluck with the buzzing sawari timbre (a deliberate sympathetic buzz at the nut) and a percussive attack thwack, the snap of the Japanese shamisen. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Sawari the buzz, Level the output.

ShengVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sheng voice: a Chinese mouth-organ, several free reeds sounding a note plus its octave and fifth together as a bright reedy cluster under steady breath, the shimmering chord-drone of the sheng. in0 = note, in1 = gate (breath). Octave/Fine tune it, Cluster the added fifth/octave level, Level the output.

Shepard [OSC] [SYNTH] in: – out: Audio

Shepard-Risset tone: seven octave-spaced sine partials sweep continuously through the spectrum while a bell-shaped amplitude window fades each one in at the bottom and out at the top, producing the endless-glissando illusion of a pitch that rises (or falls) forever without ever changing register. Speed sets the glide rate, Direction up or down, Range the octave span the partials cover, and Level the output. Needs no note input - it is a self-running tone generator.

SitarVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sitar voice: a plucked Karplus string whose vibration grazes the curved jawari bridge, adding a buzzing shimmer of high harmonics, the twang of an Indian sitar as a playable voice. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Buzz the jawari amount, Level the output.

SlapBassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Slap-bass voice: a funk slap-and-pop electric bass, a low Karplus string with a sharp percussive slap transient at the attack and a round body, the thumb-slap groove of funk. in0 = note, in1 = gate (slap). Octave/Fine tune it, Slap the transient amount, Level the output.

SleighBellsVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sleigh-bells voice: the gate shakes a strand of sleigh bells, a cluster of small bright bell tones at slightly-random pitches ringing together with a quick shimmering decay, the festive jingle of bells. in1 = gate. Pitch sets the bell size, Decay the ring, Level the output.

SnapVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Finger-snap voice: the gate fires a short band-passed noise click with a fast snap envelope, the dry percussive pop of a finger snap. in1 = gate. Tone sets the click pitch, Level the output.

SnareVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Synth snare voice: the gate strikes a short tone body (two detuned sines) mixed with a bandpassed noise burst for the wires, the analog drum-machine snare. in0 = note (transposes the body), in1 = gate. Tune sets the body pitch, Snappy the noise-vs-tone balance, Decay the length, Level the output.

SopranoSaxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Soprano-sax voice: a high single-reed conical-bore waveguide with the full bright harmonic series, the reedy soaring tone of a soprano saxophone. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Bright the bore filter, Level the output.

SousaphoneVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sousaphone voice: the marching-band tuba, a very low rounded sawtooth through a dark lowpass with a soft breathy attack, the big oom-pah bass wrapped around the player. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Dark the body, Level the output.

∿ **SquareStack** [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Detuned-square super-voice: seven band-limited pulse oscillators detuned around the note (a square-wave cousin of the supersaw) for a hollow, buzzy, hugely-wide stack that a single pulse cannot make. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Detune spreads the stack, PW the pulse width, Level the output.

∿ **SteelGuitarVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Steel-guitar voice: a plucked string that glides smoothly to each new note (the lap/pedal-steel slide), a Karplus pluck whose pitch portamentos from the previous note, with a warm sustain. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Glide the slide time, Level the output.

∿ **SteelpanVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Steel-pan voice: the gate strikes a note-area of a Caribbean steel drum, a bright clustered stack of mostly-harmonic partials with a fast metallic attack and ringing sustain, the sunny tone of the pan. in0 = note, in1 = gate. Octave tunes it, Decay the ring, Level the output.

∿ **StringEnsembleVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

String-ensemble voice: a whole bowed section, several detuned sawtooth voices through a body lowpass with a slow swell and a built-in ensemble shimmer, the lush ‘strings’ pad. in0 = note, in1 = gate (bow). Octave tunes it, Spread the section width, Tone the cutoff, Level the output.

∿ **StutterVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Trance-gate voice: a detuned-saw tone chopped by a hard rhythmic on/off gate at the stutter rate with adjustable duty, the pumping gated-synth stab of trance and EDM. in0 = note, in1 = pitch mod. Octave/Fine tune it, Rate the gate speed, Duty the on-fraction, Level the output.

∿ **SubDrive** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sub-bass voice: a pure sine sub with a fast downward pitch envelope on the gate and a tanh drive for harmonic weight, the 808-style tuned sub-bass that sits under a mix. in0 = note, in1 = gate. Octave tunes it, PSweep the pitch-drop depth, Drive the saturation, Decay the amp tail, Level the output.

∿ **SubDropVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sub-drop FX voice: the gate fires a deep sine whose pitch sinks slowly over the drop time from the note down into sub-bass, the cinematic ‘sub drop’ impact tail under a downbeat. in0 = note, in1 = gate. DropTime sets the fall length, Decay the amp tail, Drive the saturation, Level the output.

∿ **SulPonticelloVoice** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sul-ponticello voice: bowing very close to the bridge, a thin glassy bowed tone unusually rich in high inharmonic-feeling overtones with a metallic edge, the scratchy ethereal ponticello colour. in0 = note, in1 = gate (bow). Octave tunes it, Vibrato the waver, Edge the harmonic bite, Level the output.

∿ **SuperPad** [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Lush super-pad voice: a wide seven-saw unison plus a sub octave through a gentle resonant lowpass, the big, warm, slightly-detuned pad/chord bed. in0 = note, in1 = pitch mod. Octave/Fine tune it, Detune the unison width, Tone the lowpass brightness, Sub the sub-octave level, Level the output.

Supersaw [OSC] [SYNTH] in: Pitch (CV) out: Audio

JP-8000 supersaw: seven detuned saws blended from centre to sides.

SyncStabVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Sync-stab voice: a hard-sync lead where a slave oscillator is reset by a master at the note pitch, struck as a short gated stab with a fast filter drop, the aggressive sync-stab of rave and big-room. in0 = note, in1 = gate. Octave tunes it, Sync the slave/master ratio, Decay the stab length, Level the output.

SyncVCO [OSC] [CIRCUIT] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Analog-modeled hard-sync oscillator (the classic sync-sweep lead): a master oscillator at the note pitch resets a slave whose frequency is Sync times higher, so sweeping Sync (from an envelope or LFO into in1) produces the screaming formant sweep of oscillator hard sync. Octave tunes the master, Sync sets the slave/master ratio, Shape the slave waveform (saw->pulse), and Level the output. in1 = sync-sweep modulation (added to Sync), in2 = pitch mod. Naive reset (bright/aliased by design - that is the sync sound).

SynthAdditive [OSC] [SYNTH] in: Pitch (CV) out: Audio

Additive oscillator: harmonic partials with rolloff and odd/even balance.

SynthFM [OSC] [SYNTH] in: Carrier, Modulator out: Audio

Two-operator FM with a selectable modulator waveform (Wave: sine / triangle / saw / square / half-sine / abs-sine / squared / pulse) for richer-than-sine spectra, and an audio input on in 2 that acts as a third FM operator. Ratio tunes the modulator, Index sets FM depth, Octave the pitch, and Voices/Detune stack a detuned unison.

SynthGranular [OSC] [SYNTH] in: – out: Audio

Granular oscillator: sprays overlapping grains from the sample slot. Pitch transposes each grain, Scan auto-advances the read position through the sample, and Dir plays grains in reverse.

SynthNoise [OSC] [SYNTH] in: – out: Audio

Colored noise source (white/pink/brown).

SynthSub [OSC] [SYNTH] in: Pitch (CV) out: Audio

Sub oscillator: sine or square, octaves down, detuned.

SynthSync [OSC] [SYNTH] in: Pitch (CV) out: Audio

Hard-sync oscillator: a master phasor resets a detuned slave saw.

SynthVector [OSC] [SYNTH] in: Pitch (CV) out: Audio

Vector oscillator: XY morph across sine/saw/square/triangle.

SynthVoxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Synth-vox voice: a synthetic choir pad, a bright saw stack pushed through a sweeping vowel bandpass that slowly morphs vowels, the wordless ‘aaah-ooh’ synth-choir of soundtrack pads. in0 = note, in1 = pitch mod. Octave tunes it, VowelRate the morph speed, Level the output.

TablaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tabla voice: the gate strikes a North-Indian tabla, the loaded head tuned so its membrane modes line up near-harmonically into a pitched ringing ‘tin’, a rare pitched drum. in0 = note (transposes), in1 = gate. Tune sets the pitch, Decay the sing, Level the output.

TaikoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Taiko voice: the gate strikes a large Japanese drum, a deep low membrane tone from inharmonic circular modes with a downward pitch bend and a noisy skin slap, thunderous and primal. in0 = note (transposes), in1 = gate. Tune sets the pitch, Decay the boom, Level the output.

TalkingDrumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Talking-drum voice: a West-African hourglass pressure drum, a struck membrane whose pitch bends upward through the note (the squeezed tension-cords) for the speech-like glide of the talking drum. in0 = note, in1 = gate (strike). Octave tunes it, Bend the pitch glide, Level the output.

TambourineVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tambourine voice: the gate shakes a tambourine, a burst of bright jingling metal zils (a cluster of high inharmonic squares) with a fast decay, the shimmery jingle of a hand tambourine. in1 = gate. Tone sets the jingle brightness, Level the output.

TempleBlockVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Temple-block voice: a hollow wooden temple block (mokugyo), a struck dull woody resonance with a quick muffled decay and a faint hollow knock, the rounded percussive woodblock. in0 = note, in1 = gate (strike). Octave tunes it, Level the output.

TenorSaxVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tenor-sax voice: a tenor saxophone, a fuller reedy sawtooth through a lower bandpass formant than the alto, with a breathy growl, a quick attack and vibrato, the warm tenor lead. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Honk the formant bite, Level the output.

TheatreOrganVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Theatre-organ voice: a Wurlitzer cinema organ, lush tibia sine ranks thickened by a celeste detune and a wide tremulant, the silent-movie/skating-rink organ. in0 = note, in1 = gate. Octave tunes it, Celeste the rank detune, Tremulant the depth, Level the output.

ThereminVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Theremin voice: the gesture-controlled classic, a pure heterodyne sine gliding between pitches with portamento and a gentle hand vibrato, the eerie sci-fi wail played without touch. in0 = note, in1 = pitch mod. Octave/Fine tune it, Glide the portamento, Vibrato the waver, Level the output.

ThunderSheetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Thunder-sheet voice: the gate shakes a large metal sheet, a low rolling rumble of filtered noise under a shimmering metallic resonance that swells and slowly fades, the orchestral thunder-sheet effect. in1 = gate. Decay sets the roll length, Shimmer the metallic content, Level the output.

TimbaleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Timbale voice: the gate strikes a shallow metal-shell drum, a bright tight membrane tone with a hard metallic rim ring, the cutting crack of Latin timbales. in0 = note (transposes), in1 = gate. Tune sets the head pitch, Rim the rim-ring amount, Decay the ring, Level the output.

TinWhistleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tin-whistle voice: an Irish penny whistle, a bright fipple-flute air-jet waveguide with a strong breathy chiff, the lively folk whistle of Celtic music. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the air, Chiff the attack noise, Level the output.

TomDrumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tom-drum voice: a pitched tom (the only fully note-tracked drum), a sine whose pitch drops from the played note with a resonant skin body and medium decay, the melodic tom fill. in0 = note, in1 = gate (trigger). Octave tunes it, Decay the body, Level the output.

TomVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tom-drum voice: the gate strikes a mid-range sine with a gentle downward pitch bend and a medium decay, the round pitched thump of a tom. in0 = note (transposes), in1 = gate. Tune sets the pitch, Bend the drop depth, Decay the length, Level the output.

TonewheelOrgan [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tonewheel-organ voice: a Hammond-style additive organ summing nine drawbar harmonics (16′,5 1/3′,8′,4′,2 2/3′,2′,1 3/5′,1 1/3′,1′) at preset registrations, with a percussion click on the gate and a slow rotary chorus. in0 = note, in1 = gate (percussion strike). Octave tunes it, Drawbars selects the registration, Percussion the click level, Rotary the chorus depth, Level the output.

TonewheelVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tonewheel voice: a vintage tonewheel organ with gentle harmonic leakage and a touch of motor hum between the wheels, the warm imperfect electromechanical organ. in0 = note, in1 = gate. Octave tunes it, Drawbars the brightness, Leakage the wheel cross-talk, Level the output.

ToyPianoVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Toy-piano voice: a child's toy piano with small struck metal rods, a very short bright Karplus pluck with strongly inharmonic, bell-like overtones, the tinkly plinky toy sound. in0 = note, in1 = gate (strike). Octave/Fine tune it, Level the output.

TR606HatVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

606-hi-hat voice: the Roland TR-606 hi-hat, a thinner brighter metallic noise through a high bandpass with a tight decay, the trashy lo-fi hat of the 606. in1 = gate (trigger). Tone the brightness, Decay the length, Level the output.

TR808HatVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

808-hi-hat voice: the TR-808 hi-hat, a cluster of six detuned square oscillators through a high bandpass with a fast decay, the metallic tick of the 808. in1 = gate (trigger). Tone the brightness, Decay the open/closed length, Level the output.

TR808KickVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

808-kick voice: the booming Roland TR-808 bass drum, a sine whose pitch drops fast from a tuned start with a long amplitude decay and a click transient, the hip-hop sub-kick. in1 = gate (trigger). Tune the start pitch, Decay the boom length, Click the attack, Level the output.

TR808SnareVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

808-snare voice: the TR-808 snare, two detuned tonal bodies plus a band of noise, decaying quickly, the snappy electro snare. in1 = gate (trigger). Tune the body pitch, Snappy the noise level, Decay the length, Level the output.

TR909KickVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

909-kick voice: the punchy Roland TR-909 bass drum, a sine with a faster, shorter pitch sweep and a prominent noisy click attack, the harder techno/house kick. in1 = gate (trigger). Tune the pitch, Decay the body, Punch the click, Level the output.

TR909SnareVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

909-snare voice: the TR-909 snare, a tonal body under a heavier, brighter band of noise than the 808, the crisp sampled-flavoured house snare. in1 = gate (trigger). Tune the body, Snappy the noise, Decay the length, Level the output.

TrautoniumVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Trautonium voice: the subharmonic synthesizer of the 1930s (used in Hitchcock's The Birds), a buzzy tone built from the undertone series - the fundamental plus pitches at 1/2, 1/3, 1/4 of it - for a hollow, organ-of-the-deep timbre. in0 = note, in1 = pitch mod. Octave/Fine tune it, Subs the undertone mix, Level the output.

∿ TremoloStringVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tremolo-string voice: a bowed string played with rapid back-and-forth bow strokes, a sawtooth body tone chopped by a fast amplitude tremolo, the shimmering suspense tremolo of film scores. in0 = note, in1 = gate (bow). Octave tunes it, TremRate the bow speed, Tone the cutoff, Level the output.

∿ TriangleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Triangle (percussion) voice: the gate strikes a small steel triangle, a high bright cluster of inharmonic partials ringing with a long shimmering decay. in0 = note, in1 = gate. Pitch sets the bar size, Decay the ring, Level the output.

∿ TromboneVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Trombone voice: a tenor slide trombone, a broad sawtooth through a warmer lowpass than the trumpet with a slightly slower attack and gentle vibrato, the rich mid-brass. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Bright the blare, Level the output.

∿ TromboneVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Trombone voice: a lip-reed bore with a long cylindrical-then-flared bore that glides smoothly between notes (the slide portamento), the bold brass of a trombone. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Glide the slide time, Level the output.

∿ TrumpetVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Trumpet voice: a Bb trumpet, a bright buzzy sawtooth driven through a resonant lowpass whose cutoff opens with the breath so the tone blares brighter the harder it is blown, a quick lip attack and a light vibrato, the lead brass. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Bright the blare, Level the output.

∿ TubaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tuba voice: the lowest brass, a deep sawtooth through a dark heavy lowpass with a slow breathy attack, the oom-pah foundation of the brass band. in0 = note, in1 = gate (breath). Octave/Fine tune it, Vibrato the waver, Dark the body, Level the output.

∿ TubaVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tuba voice: a large lip-reed conical bore in the deep bass, self-oscillating on breath into the round fat foundation brass of a tuba. in0 = note, in1 = gate (breath). Octave/Fine tune it, Breath the pressure, Level the output.

∿ TubularBellVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Tubular-bell voice: the gate strikes an orchestral chime tube, voiced from the tube's partials with the perceived strike pitch arising from the 4:5:6 cluster, ringing with a long majestic decay. in0 = note, in1 = gate. Octave tunes it, Decay the toll, Level the output.

∿ TwelveStringVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Twelve-string guitar voice: a Karplus pluck layered with a shimmering octave-course tap, giving the rich chorused jangle of a 12-string acoustic. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Shimmer the octave-course level, Level the output.

VCO [OSC] [CIRCUIT] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Analog-modeled VCO (Moog/classic voltage-controlled oscillator character): a band-limited (PolyBLEP, anti-aliased) triangle->saw->pulse morph with slow analog Drift detuning the pitch and a gentle waveshape Warmth, for the fat, slightly-unstable vintage tone the clean base Osc does not model. Octave/Fine tune it, Shape sweeps triangle->saw->pulse, PWM sets the pulse width, Drift the analog pitch wander, Warmth a soft saturation, and Level the output. in1 = FM, in2 = pitch mod.

VeenaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Veena voice: a South-Indian plucked lute, a Karplus pluck over a sustained sympathetic drone string, with a slight buzz from the wide bridge, the resonant raga voice of the veena. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Drone the sympathetic-string level, Level the output.

VibraphoneVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Vibraphone voice: the gate strikes an aluminium bar with near-octave partials and a long decay, amplitude-modulated by the spinning resonator-disc tremolo for the vibe's signature wobble. in0 = note, in1 = gate. Octave tunes it, Tremolo the motor speed, Decay the ring, Level the output.

VibraslapVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Vibraslap voice: the gate strikes a vibraslap, a sharp crack followed by the long rattling buzz of its loose metal pins chattering against the wooden box, the rattlesnake-buzz of the Latin percussion staple. in1 = gate. Decay sets the rattle length, Level the output.

ViolaVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Viola voice: the alto bowed string, a sawtooth Helmholtz motion through a warmer mid body than the violin with a slow bow attack and vibrato, the mellow viola. in0 = note, in1 = gate (bow). Octave/Fine tune it, Vibrato the waver, Warmth the body, Level the output.

ViolinVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Violin voice: a bowed string, a sawtooth approximating the Helmholtz motion through a bright body resonance with a slow bow attack and an expressive vibrato, the singing high violin. in0 = note, in1 = gate (bow). Octave/Fine tune it, Vibrato the waver, Bright the body, Level the output.

ViolinVoiceWG [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Violin voice: a bowed-string waveguide self-oscillating into Helmholtz motion while the gate (bow) is held, with a singing vibrato, the orchestral violin as a playable voice. in0 = note, in1 = gate (bow). Octave/Fine tune it, BowForce the grip, Vibrato the waver, Level the output.

VocoderVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Vocoder voice: a 3-band channel vocoder that imposes the spectral envelope of the modulator signal (in2) onto a sawtooth carrier at the played note, so the carrier 'talks' with the dynamics of the modulator - classic robot-voice cross-synthesis. in0 = note, in1 = pitch mod, in2 = modulator audio. Octave/Fine tune it, Carrier the saw/pulse mix, Level the output.

Vosim [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

VOSIM (VOcal SIMulation) oscillator: each fundamental period emits a burst of Pulses sin-squared pulses at FRatio times the pitch, each one Decay-scaled, then silence until the next period - the 1970s Kaegi-Tempelaars vocal/formant technique. FRatio places the formant region (vowel colour), Pulses sets the burst length and Decay the spectral tilt. Distinct from FOF granular and resonator formant synthesis; output is a unipolar pulse train.

VowelLeadVoice [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Vowel-morph lead voice: a sawtooth pushed through a sweepable formant bandpass that morphs across the A-E-I-O-U vowels, a mono talk-box lead with portamento glide. in0 = note, in1 = vowel sweep (added to Vowel), in2 = pitch mod. Octave/Fine tune it, Vowel the formant position, Glide the portamento, Level the output.

VoxContinentalVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Vox Continental voice: the bright British combo organ, a hollow drawbar blend leaning on the fundamental and a strong octave with a reedy upper sparkle and fast gate, the 60s-beat organ. in0 = note, in1 = gate. Octave tunes it, Sparkle the upper drawbars, Vibrato the waver, Level the output.

WalshVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Walsh-function voice: the timbre is built by summing Walsh functions (the +/-1 square-wave orthogonal basis) instead of sines, giving a hard-edged blocky digital spectrum impossible with sine additive synthesis. in0 = note, in1 = pitch mod. Octave/Fine tune it, Order the number of Walsh terms (brightness), Level the output.

Wavefold [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

West-coast wavefolder voice: a sine pushed through a multi-stage triangle wavefolder (the Buchla/Serge timbre source) so raising Fold generates ever-denser harmonics by reflecting the wave off the rails, with Symmetry adding even harmonics. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Fold the folding gain, Symmetry the DC offset, Level the output.

WaveshapeLead [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Waveshaper lead: a sine driven through a Chebyshev-polynomial transfer curve so the Harmonics knob adds a precise number of harmonics (timbre by direct spectral control), the bright synthetic lead of waveshaping synthesis. in0 = note, in1 = FM, in2 = pitch mod. Octave/Fine tune it, Harmonics the order, Drive the input gain, Level the output.

WaveTerrainVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Wave-terrain voice: the output is the height of a fixed 2-D surface z=f(x,y) read along an elliptical orbit whose angle advances at the note pitch, so changing the orbit radius sweeps the terrain and the timbre - wave-terrain synthesis. in0 = note, in1 = pitch mod. Octave tunes it, Radius the orbit size, Warp the terrain shape, Level the output.

WhipCrackVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Whip-crack voice: the gate fires a fast downward pitch zip ending in a sharp noise crack, the supersonic snap of a cracking whip. in1 = gate. TopHz sets the zip start, Crack the snap brightness, Level the output.

WhistleVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Human-whistle voice: a person whistling, a nearly-pure sine with a soft breath-noise edge and an expressive vibrato that deepens on held notes, the airy melodic whistle. in0 = note, in1 = pitch mod. Octave/Fine tune it, Breath the air noise, Vibrato the waver, Level the output.

WindChimeVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Wind-chime voice: random gusts strike a set of tuned metal tubes, sounding soft pentatonic bell tones at random times with long shimmering decays, the gentle tinkle of a wind chime in a breeze. in0 = note (sets the chime key), in1 = unused. Activity sets how often the wind strikes, Decay the ring, Level the output.

WindVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Breath/wind voice: white noise shaped by a resonant bandpass tracking the note pitch plus a slow gust LFO on amplitude, the airy whistling of wind through a tube or an overblown breath tone. in0 = note, in1 = pitch mod. Octave/Fine tune it, Resonance the bandpass Q (airy vs tonal), Gust the amplitude wobble, Level the output.

WobbleBassVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Wobble-bass voice (dubstep): a detuned-saw bass driven through a resonant lowpass whose cutoff is swept by an LFO at a musical rate, the rhythmic ‘wub-wub’ of dubstep. in0 = note, in1 = pitch mod. Octave tunes it, Wobble the LFO rate, Depth the sweep, Reso the filter bite, Level the output.

WoodblockVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Woodblock voice: the gate strikes a hollow wooden block, a dry pitched knock with a quick woody decay and a faint hollow second partial, the orchestral woodblock clop. in0 = note (transposes), in1 = gate. Tone sets the pitch, Level the output.

Wt [OSC] [SYNTH] in: Pitch (CV), FM, Pitch Mod out: Audio

Wavetable oscillator: morphs a table bank with unison voices and Serum-style WARP (Bend/PWM/Sync/Mirror).

WurlitzerVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Wurlitzer voice: the reedy electric-piano tone of a Wurlitzer, a two-operator FM with a softer bark and a slight asymmetric grit distinct from the glassier Rhodes, struck on the gate with a medium decay. in0 = note, in1 = gate. Octave tunes it, Ratio the reed modulator, Bark the attack bite, Decay the length, Level the output.

XylophoneVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Xylophone voice: a hard-mallet-struck wooden bar, a bright additive tone with stretched inharmonic overtones and a short snappy decay, the percussive wooden xylophone. in0 = note, in1 = gate (strike). Octave tunes it, Decay the ring, Level the output.

ZapLaser [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Zap/laser FX voice: the gate fires a sine whose pitch plummets from a high start in a fast exponential drop, the retro sci-fi laser/zap blip. in0 = note (sets the floor), in1 = gate. Top sets the start pitch, Drop the sweep speed, Decay the amp tail, Level the output.

ZitherVoice [OSC] [SYNTH] in: Pitch (CV), FM out: Audio

Zither voice: a multi-string plucked board that layers the plucked melody note with a sustained open drone string a fifth below, the autoharp/zither bed; the gate plucks the melody. in0 = note, in1 = gate (pluck). Octave/Fine tune it, Drone the open-string level, Level the output.

Pitch (1)

SynthGlide [PIT] [SYNTH] in: Note (CV) out: Audio

Portamento: exponential glide of the note pitch.

Utility (4)

Scaler [UTL] [SYNTH] in: Signal out: Audio

CV utility: scale, offset and bipolar curve-bend a control signal.

SynthGain [UTL] [SYNTH] in: Audio out: Audio

Gain trim in decibels.

SynthMixer [UTL] [SYNTH] in: In A, In B out: Audio

Sums two inputs with independent levels.

VCA [UTL] [SYNTH] in: Audio, Env out: Audio

Voltage-controlled amplifier: multiplies the signal by the control input.

MIDI (zpwr-midi-fx) (563 blocks)

Control (5)

ATProc [CTL] [CV] [MIDI] in: Audio out: MIDI

Scales aftertouch, optionally converting it to a CC.

BendProc [CTL] [CV] [MIDI] in: Audio out: MIDI

Scales and/or inverts incoming pitch-bend.

CC2Note [CTL] [CV] [MIDI] in: Audio out: MIDI

A watched CC crossing a threshold triggers a note.

CCMap [CTL] [CV] [MIDI] in: Audio out: MIDI

Remaps a CC number and scales/offsets its value.

Note2CC [CTL] [CV] [MIDI] in: Audio out: MIDI

Emits a CC from each note (pitch or velocity).

Dynamics (162)

AbundanceVelocity [DYN] [MIDI] in: Audio out: MIDI

Abundance velocity: accents each note by the signed abundance sigma(n)-2n, so deficient counts read soft, perfect counts sit at centre and abundant counts read loud; Depth blends with the played velocity.

AbundancyVelocity [DYN] [MIDI] in: Audio out: MIDI

Abundancy velocity: drives velocity from the abundancy index sigma(n)/n of the count - near 1 for primes, climbing past 2 for perfect and abundant numbers - so divisor-rich counts hit hardest; Depth blends with the played velocity.

Accent [DYN] [MIDI] in: Audio out: MIDI

Boosts velocity on every Nth note.

AccentEveryN [DYN] [MIDI] in: Audio out: MIDI

Accent every N: boosts the velocity of every Nth note and slightly ducks the rest, stamping a steady metric downbeat onto an even stream - the simplest way to imply a time signature in the dynamics.

AccentOnHigh [DYN] [MIDI] in: Audio out: MIDI

Accent on high: boosts the velocity of note-ons at or above a pitch threshold, brightening the top of the keyboard.

AccentOnLow [DYN] [MIDI] in: Audio out: MIDI

Accent on low: boosts the velocity of note-ons at or below a pitch threshold, reinforcing the bass register.

AccentPattern [DYN] [MIDI] in: Audio out: MIDI

Accent pattern: boosts velocity on the on-steps of a 16-step accent bitmask and trims it elsewhere, stamping a repeating dynamic groove onto an even stream of notes.

AliquotStepsVelocity [DYN] [MIDI] in: Audio out: MIDI

Aliquot-steps velocity: accents each note by how many steps the count's aliquot sequence (repeatedly summing proper divisors) takes before terminating or settling, an unpredictable number-theory accent; Depth blends with the played velocity.

AliquotVelocity [DYN] [MIDI] in: Audio out: MIDI

Aliquot velocity: drives velocity from the aliquot sum (the sum of a number's proper divisors) of the note count, the quantity behind perfect, abundant and deficient numbers; Depth blends with the played velocity.

AlternatingSumVelocity [DYN] [MIDI] in: Audio out: MIDI

Alternating-sum velocity: drives velocity from the alternating digit sum of the count (the signed digit total behind the divisible-by-11 test), an oscillating accent contour; Depth blends with the played velocity.

BackbeatAccent [DYN] [MIDI] in: Audio out: MIDI

Backbeat accent: boosts the velocity of notes on beats 2 and 4 (the backbeat) and slightly softens the others, stamping the rock/pop snare-accent feel onto a flat stream without dropping notes.

BeattyVelocity [DYN] [MIDI] in: Audio out: MIDI

Beatty velocity: drives velocity from the Beatty sequence floor(n*sqrt2) (an irrational-rotation staircase that, with its complement, partitions the integers); Depth blends with the played velocity.

BernoulliSignVelocity [DYN] [MIDI] in: Audio out: MIDI

Bernoulli-sign velocity: alternates two velocity levels by the sign of the Bernoulli numbers (the alternating-sign even-index sequence from number theory and calculus), with a neutral level on the zero-valued odd indices.

BinaryRunVelocity [DYN] [MIDI] in: Audio out: MIDI

Binary-run velocity: accents each note by the longest run of identical bits in its binary count, spiking on counts like 7, 15, 31 and 56; Depth blends with the played velocity.

BinaryWeightVelocity [DYN] [MIDI] in: Audio out: MIDI

Binary-weight velocity: drives velocity from the number of 1-bits (Hamming weight) in the note count, a self-similar accent pattern that pulses with the binary structure of the ordinal; Depth blends it with the played velocity.

CalkinWilfVelocity [DYN] [MIDI] in: Audio out: MIDI

Calkin-Wilf velocity: drives velocity from the Calkin-Wilf rational at the count (fusc(n)/fusc(n+1)), the breadth-first enumeration that lists every positive fraction exactly once; Depth blends with the played velocity.

CatalanVelocity [DYN] [MIDI] in: Audio out: MIDI

Catalan velocity: steps velocity through the Catalan numbers (1,1,2,5,14,42,…) taken modulo a range, the combinatorial sequence counting balanced brackets and binary trees; Depth blends with the played velocity.

CollatzOddStepsVelocity [DYN] [MIDI] in: Audio out: MIDI

Collatz-odd-steps velocity: accents each note by how many odd (3n+1) rises its count takes on the way down the Collatz trajectory, an erratic hailstone-flavoured accent; Depth blends with the played velocity.

CollatzPeakVelocity [DYN] [MIDI] in: Audio out: MIDI

Collatz-peak velocity: accents each note by the log-height of the highest value its count reaches in the Collatz (3n+1) hailstone trajectory, so counts that soar before falling hit hardest; Depth blends with the played velocity.

CollatzVelocity [DYN] [MIDI] in: Audio out: MIDI

Collatz velocity: maps each note's velocity to the number of 3n+1 steps its count takes to reach 1, an erratic deterministic accent sequence from the unsolved conjecture; Depth blends against the played velocity.

CompositeRunVelocity [DYN] [MIDI] in: Audio out: MIDI

Composite-run velocity: accents each note by how many composite counts have passed since the last prime, ramping up through prime gaps and snapping back to zero on each prime; Depth blends with the played velocity.

CototientVelocity [DYN] [MIDI] in: Audio out: MIDI

Cototient velocity: accents each note by the cototient n minus Euler's totient (the count of integers up to n that share a factor with it), a divisor-flavoured accent complementary to the totient; Depth blends with the played velocity.

CrescendoLoop [DYN] [MIDI] in: Audio out: MIDI

Crescendo loop: velocity climbs from Min to Max across Length notes then snaps back and repeats, an automatic sawtooth dynamic swell driven by the note count rather than the clock.

CrescendoRamp [DYN] [MIDI] in: Audio out: MIDI

Crescendo ramp: ramps note-on velocity up linearly across a phrase of N notes then resets, an automatic swell from soft to loud; Depth blends with the played velocity.

DigitalRootVelocity [DYN] [MIDI] in: Audio out: MIDI

Digital-root velocity: accents each note by the digital root (the repeated digit-sum, 1..9) of its count, a perfectly periodic nine-step accent staircase from elementary number theory; Depth blends with the played velocity.

DigitEntropyVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-entropy velocity: drives velocity from how many distinct decimal digits the count uses, so repdigits read soft and varied counts read loud; Depth blends with the played velocity.

DigitMaxVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-max velocity: drives velocity from the largest decimal digit of the note count, a coarse stepping accent that climbs and resets with the leading digits; Depth blends with the played velocity.

DigitMinVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-min velocity: drives velocity from the smallest decimal digit of the note count, the mirror of digit-max; Depth blends with the played velocity.

DigitProductVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-product velocity: accents each note by the product of its count's decimal digits, so counts with big digits hit hard and those with small digits stay soft; Depth blends with the played velocity.

DigitReverseVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-reverse velocity: drives velocity from the count read backwards (123 becomes 321), an erratic accent that scrambles the steady climb of the counter into a jumpy pattern; Depth blends with the played velocity.

DigitRotateVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-rotate velocity: rotates the count's decimal digits one place (the last digit jumps to the front) and reads velocity from the result, scrambling the steady counter into a jumpy pattern; Depth blends with the played velocity.

DigitSpreadVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-spread velocity: drives velocity from the spread between the largest and smallest decimal digit of the note count, so repdigit counts read soft and mixed-digit counts read loud; Depth blends with the played velocity.

DigitSumVelocity [DYN] [MIDI] in: Audio out: MIDI

Digit-sum velocity: derives velocity from the decimal digit-sum of the note count, a self-similar sawtooth-of-sawtooths accent that resets every power of ten; Depth blends it with the incoming velocity.

DiminuendoRamp [DYN] [MIDI] in: Audio out: MIDI

Diminuendo ramp: ramps note-on velocity down linearly across a phrase of N notes then resets, an automatic fade from loud to soft; Depth blends with the played velocity.

DistinctPrimeVelocity [DYN] [MIDI] in: Audio out: MIDI

Distinct-prime velocity: accents each note by little-omega, the number of distinct prime factors of its count (primes softest, highly-composite counts loudest); Depth blends with the played velocity.

DivisorCountVelocity [DYN] [MIDI] in: Audio out: MIDI

Divisor-count velocity: makes notes louder on counts with many divisors, so highly-composite ordinals (12, 24, 36…) land as accents and primes stay quiet; Depth blends against the played velocity.

DivisorProductVelocity [DYN] [MIDI] in: Audio out: MIDI

Divisor-product velocity: drives velocity from the product of the note count's divisors (taken modulo a range), a number-theoretic accent that swings high for highly-composite counts; Depth blends with the played velocity.

DivisorSumVelocity [DYN] [MIDI] in: Audio out: MIDI

Divisor-sum velocity: drives velocity from sigma(n), the sum of all divisors of the note count, so primes read low and highly-composite counts read loud; Depth blends with the played velocity.

EuclideanAccentVelocity [DYN] [MIDI] in: Audio out: MIDI

Euclidean accent: boosts the velocity of notes landing on a Euclidean (Bjorklund) rhythm and softens the rest, stamping an evenly-distributed accent groove onto a steady stream of notes without dropping any.

EuclideanVelocity [DYN] [MIDI] in: Audio out: MIDI

Euclidean velocity: accents the notes that fall on the onsets of a Euclidean rhythm of Pulses-in-Steps and softens the rest, stamping a maximally-even world-rhythm groove onto the dynamics.

ExpDecayVelocity [DYN] [MIDI] in: Audio out: MIDI

Exp-decay velocity: velocity starts loud and decays exponentially across a phrase of N notes then resets, an automatic decrescendo per phrase; Depth blends with the played velocity.

FactorialBaseVelocity [DYN] [MIDI] in: Audio out: MIDI

Factorial-base velocity: accents each note by the digit-sum of its count written in the factorial number system (where place values are 1!,2!,3!,…), an exotic mixed-radix accent contour; Depth blends with the played velocity.

FactorialModVelocity [DYN] [MIDI] in: Audio out: MIDI

Factorial-mod velocity: drives velocity from the count's factorial taken modulo 127, which collapses to zero past a small index (Wilson's theorem territory), giving a sharp early flurry then silence-velocity; Depth blends with the played velocity.

FibVelocity [DYN] [MIDI] in: Audio out: MIDI

Fibonacci velocity: drives note velocity from the Fibonacci sequence taken modulo 128, climbing then wrapping in the golden cadence; Depth blends the pattern against the incoming velocity.

GcdVelocity [DYN] [MIDI] in: Audio out: MIDI

GCD velocity: sets velocity from the greatest common divisor of the note count and a modulus, so counts sharing big factors with the modulus hit harder - a quietly-periodic, factor-driven accent; Depth blends with the played velocity.

GhostNoteEveryN [DYN] [MIDI] in: Audio out: MIDI

Ghost-note every N: softens every Nth note-on down to a quiet ghost-note level, dropping a recurring accent into the background for a syncopated groove.

GhostNoteInject [DYN] [MIDI] in: Audio out: MIDI

Ghost-note inject: randomly demotes a fraction of notes to a soft ‘ghost’ level, scattering the quiet in-between hits that give drum and bass parts their human, shuffling feel.

GoldbachCountVelocity [DYN] [MIDI] in: Audio out: MIDI

Goldbach-count velocity: accents each even count by the number of distinct ways it splits into a sum of two primes (its Goldbach partitions), a rising and jagged accent; Depth blends with the played velocity.

GoldenAngleVelocity [DYN] [MIDI] in: Audio out: MIDI

Golden-angle velocity: rotates each note count by the 137.5-degree golden angle (the phyllotaxis spiral that arranges sunflower seeds) and reads velocity from the resulting angle, an evenly-scattering accent; Depth blends with the played velocity.

GoldenSectionAccent [DYN] [MIDI] in: Audio out: MIDI

Golden-section accent: places a strong accent at the start and at the golden-ratio point (~0.618) of a repeating cycle, an aesthetically-balanced asymmetric stress pattern instead of an even backbeat.

GoldenVel [DYN] [MIDI] in: Audio out: MIDI

Golden velocity: replaces each note’s velocity with a value from the golden-ratio low-discrepancy sequence, which fills the dynamic range far more evenly than random, so accents never clump; Depth blends it against the played velocity.

GrayCodeVelocity [DYN] [MIDI] in: Audio out: MIDI

Gray-code velocity: drives velocity from the reflected-binary Gray code of the note count (successive values differ by one bit), giving a smoothly-rotating, single-step-change accent pattern; Depth blends with the played velocity.

HammingDistanceVelocity [DYN] [MIDI] in: Audio out: MIDI

Hamming-distance velocity: drives velocity from how many bits flip between successive note counts (the Hamming distance of consecutive integers), a small spiky accent that jumps at binary carries; Depth blends with the played velocity.

┬ **HappyVelocity** [DYN] [MIDI] in: Audio out: MIDI

Happy velocity: accents each note by how its count behaves under the happy-number process (sum of squared digits) - loud and bright if it reaches 1, darker if it falls into the sad cycle; Depth blends with the played velocity.

┬ **HarmonicNumberVelocity** [DYN] [MIDI] in: Audio out: MIDI

Harmonic-number velocity: drives velocity from the harmonic number H(n)=1+1/2+…+1/n, a smooth slowly-saturating logarithmic climb across the note count; Depth blends with the played velocity.

┬ **HarmonicVelocity** [DYN] [MIDI] in: Audio out: MIDI

Harmonic velocity: shapes velocity along a 1/(k+1) harmonic-decay curve over a repeating Length-note cycle, so each phrase opens strong and tapers - an automatic accent-and-decay groove.

┬ **Humanize** [DYN] [MIDI] in: Audio out: MIDI

Adds random timing and velocity jitter for a human feel.

┬ **HumanizeVelocity** [DYN] [MIDI] in: Audio out: MIDI

Humanize velocity: adds a small random variation to each note-on velocity so repeated hits no longer read as identical, a subtle natural-feel dither.

┬ **IntervalVelocity** [DYN] [MIDI] in: Audio out: MIDI

Interval velocity: sets each note’s velocity from the size of its leap from the previous note, so wide jumps hit hard and stepwise motion stays gentle - dynamics that follow the melodic contour; Depth blends with the played velocity.

┬ **JacobiVelocity** [DYN] [MIDI] in: Audio out: MIDI

Jacobi-symbol velocity: picks one of three velocity levels from the Jacobi symbol of the count over an odd Modulus (+1, -1 or 0), a quadratic-residue signature from number theory; the modulus sets the pattern’s period.

┬ **JacobsthalVelocity** [DYN] [MIDI] in: Audio out: MIDI

Jacobsthal velocity: drives velocity from the Jacobsthal numbers (J(n)=J(n-1)+2J(n-2): 1,1,3,5,11,21,…), a Fibonacci cousin tied to powers of two; Depth blends with the played velocity.

┬ **KaprekarVelocity** [DYN] [MIDI] in: Audio out: MIDI

Kaprekar velocity: drives velocity from how many digit-sort-and-subtract steps the count takes to reach Kaprekar’s constant 6174, an erratic 0-7 accent from a famous digit routine; Depth blends with the played velocity.

┬ **KolakoskiVelocity** [DYN] [MIDI] in: Audio out: MIDI

Kolakoski velocity: alternates two velocity levels following the self-describing Kolakoski sequence (whose run-lengths are the sequence itself), giving a hypnotically self-similar yet non-repeating accent pattern.

LeonardoVelocity [DYN] [MIDI] in: Audio out: MIDI

Leonardo velocity: drives velocity from the Leonardo numbers (L(n)=L(n-1)+L(n-2)+1: 1,1,3,5,9,15,25,…), the smoothsort sequence kin to Fibonacci; Depth blends with the played velocity.

LogVelocity [DYN] [MIDI] in: Audio out: MIDI

Log velocity: drives velocity from the base-2 logarithm of the note count, a gentle ever-slowing ramp that doubles its reach each octave of counts; Depth blends with the played velocity.

LookSayVelocity [DYN] [MIDI] in: Audio out: MIDI

Look-and-say velocity: drives velocity from the growing length of the look-and-say sequence (1, 11, 21, 1211, …) at the note count, a self-describing sequence that lengthens by Conway's constant; Depth blends with the played velocity.

LucasVelocity [DYN] [MIDI] in: Audio out: MIDI

Lucas velocity: steps velocity through the Lucas numbers (the Fibonacci companion 2,1,3,4,7,11,…) taken modulo a range, a golden-cadence accent contour; Depth blends with the played velocity.

MarkovVelocity [DYN] [MIDI] in: Audio out: MIDI

Markov velocity: walks the velocity through a near-neighbour Markov chain of levels, so dynamics drift coherently (mostly small changes, occasional leaps) rather than jumping randomly - a stochastic but musical accent contour.

MelodicAccent [DYN] [MIDI] in: Audio out: MIDI

Melodic accent: accents notes that step up from the previous one and softens those that step down, so the dynamics follow the rise and fall of the melodic contour - phrasing that breathes with the line.

MobiusVelocity [DYN] [MIDI] in: Audio out: MIDI

Moebius velocity: picks one of three velocity levels from the Moebius function mu of the count - high for squarefree-even-factor counts, low for squarefree-odd, neutral for counts with a squared factor - a number-theoretic three-state accent.

ModInverseVelocity [DYN] [MIDI] in: Audio out: MIDI

Mod-inverse velocity: drives velocity from the modular inverse of the note count under a chosen modulus (zero when the count shares a factor with the modulus), a number-theoretic scramble; Depth blends with the played velocity.

ModRangeVelocity [DYN] [MIDI] in: Audio out: MIDI

Mod-range velocity: ramps velocity in a repeating sawtooth across a chosen period of notes, a steady rising-then-resetting accent staircase; Depth blends with the played velocity.

MoebiusVelocity [DYN] [MIDI] in: Audio out: MIDI

Moebius velocity: accents notes by the Moebius function of the count - loud on squarefree numbers with an even number of prime factors, soft on odd, and mid-level on square-divisible ones; Depth blends with the played velocity.

NarayanaVelocity [DYN] [MIDI] in: Audio out: MIDI

Narayana velocity: drives velocity from Narayana's-cows sequence (a(n)=a(n-1)+a(n-3): 1,2,3,4,6,9,13,19,…), whose ratio tends to the supergolden ratio; Depth blends with the played velocity.

NibbleSwapVelocity [DYN] [MIDI] in: Audio out: MIDI

Nibble-swap velocity: scrambles velocity by swapping the high and low 4-bit nibbles of the note count, a byte-twiddling accent that jumps in a fixed but jagged pattern; Depth blends with the played velocity.

OmegaVelocity [DYN] [MIDI] in: Audio out: MIDI

Omega velocity: sets velocity from the number of prime factors (with multiplicity) of the count, so smooth primes play soft and factor-heavy numbers play hard; Depth blends it with the incoming velocity.

PadovanVelocity [DYN] [MIDI] in: Audio out: MIDI

Padovan velocity: steps velocity through the Padovan sequence (P(n)=P(n-2)+P(n-3), the plastic-number recurrence) taken modulo a range, a slow, gently-rolling accent contour; Depth blends with the incoming velocity.

PartitionVelocity [DYN] [MIDI] in: Audio out: MIDI

Partition velocity: drives velocity from p(n), the number of ways to write the count as a sum of positive integers, a fast-growing combinatorial accent; Depth blends with the played velocity.

PellVelocity [DYN] [MIDI] in: Audio out: MIDI

Pell velocity: drives velocity from the Pell numbers (P(n)=2P(n-1)+P(n-2): 1,2,5,12,29,70,…), the silver-ratio cousin of Fibonacci; Depth blends with the played velocity.

PerrinVelocity [DYN] [MIDI] in: Audio out: MIDI

Perrin velocity: drives velocity from the Perrin sequence (P(n)=P(n-2)+P(n-3): 3,2,3,2,5,5,7,10,…), famous for its primality-test conjecture; Depth blends with the played velocity.

PersistenceVelocity [DYN] [MIDI] in: Audio out: MIDI

Persistence velocity: accents each note by the multiplicative-persistence depth of its count (how many digit-product steps reach a single digit), a sparse, spiky accent rooted in a famous open digit problem; Depth blends with the played velocity.

PitchClassVelocity [DYN] [MIDI] in: Audio out: MIDI

Pitch-class velocity: accents every note of a chosen pitch class and softens the rest, so a tonic or any scale degree rings out across the whole keyboard - a harmonic spotlight on one note name.

PopcountVelocity [DYN] [MIDI] in: Audio out: MIDI

Popcount velocity: drives velocity from the number of 1-bits in the note count (its binary Hamming weight), a jagged bit-pattern accent; Depth blends with the played velocity.

PowerOfTwoFactorVelocity [DYN] [MIDI] in: Audio out: MIDI

Power-of-two-factor velocity: drives velocity from the largest power of two that divides the count, so odd counts read soft and highly-even counts read loud (a binary-divisibility accent); Depth blends with the played velocity.

PrimeCountVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime-count velocity: drives velocity from the prime-counting function pi(count) - how many primes are at or below the note count - taken modulo a span, a slowly-climbing-then-wrapping accent rooted in the distribution of primes.

PrimeFactorVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime-factor velocity: sets velocity from the largest prime factor of the count (log-scaled), so prime counts ring loud and smooth, small-factored counts stay soft; Depth blends with the played velocity.

PrimeGapVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime-gap velocity: drives velocity from the size of the prime gap bracketing the note count (the distance between the primes just below and just above it), so notes near large prime deserts hit harder; Depth blends with the played velocity.

PrimeIndexAccent [DYN] [MIDI] in: Audio out: MIDI

Prime-index accent: accents the prime-numbered notes in the stream (the 2nd, 3rd, 5th, 7th, 11th, … to pass) and softens the rest, an irregular number-theoretic accent pattern.

PrimeIndexVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime-index velocity: drives velocity from the n-th prime number (the prime sitting at the running count's index), an irregularly-climbing accent straight from the prime sequence; Depth blends with the played velocity.

PrimePiVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime-pi velocity: drives velocity from the prime-counting function pi(count) - how many primes are at or below the count - a slowly-climbing staircase accent from analytic number theory; Depth blends with the played velocity.

PrimeVelocity [DYN] [MIDI] in: Audio out: MIDI

Prime velocity: drives velocity from the gaps between successive prime numbers, an irregular yet deterministic accent sequence that never settles into an obvious loop; Depth blends it with the incoming velocity.

PrimorialModVelocity [DYN] [MIDI] in: Audio out: MIDI

Primorial-mod velocity: drives velocity from the primorial (product of the first k primes) taken modulo 127, a fast-scrambling accent that jumps as each new prime multiplies in; Depth blends with the played velocity.

RadicalVelocity [DYN] [MIDI] in: Audio out: MIDI

Radical velocity: accents each note by the radical of its count (the product of its distinct prime factors), so squarefree counts read high and prime-power counts read low; Depth blends with the played velocity.

Ramp [DYN] [MIDI] in: Audio out: MIDI

Velocity ramp over a run of notes, ascending or descending.

RandomHoldVelocity [DYN] [MIDI] in: Audio out: MIDI

Random-hold velocity: picks a random velocity and holds it for Hold notes before re-rolling, a sample-and-hold dynamic that gives blocky terraced changes rather than per-note jumps; Depth blends with the played velocity.

RegisterAccent [DYN] [MIDI] in: Audio out: MIDI

Register accent: boosts the velocity of notes inside a Low..High pitch window and softens those outside, spotlighting one register's dynamics without dropping any notes.

RepeatDecay [DYN] [MIDI] in: Audio out: MIDI

Repeat decay: each immediate repetition of the same note plays quieter by a decay factor, taming machine-gun retriggers into a natural fade and resetting the moment the pitch changes.

ReverseBitsVelocity [DYN] [MIDI] in: Audio out: MIDI

Bit-reverse velocity: drives velocity from the 8-bit bit-reversal of the note count, scrambling the steady counter into the scattered order used by FFT bit-reversal; Depth blends with the played velocity.

ReverseSubtractVelocity [DYN] [MIDI] in: Audio out: MIDI

Reverse-subtract velocity: drives velocity from the absolute difference between the count and its digit-reversal (the 196-algorithm / Kaprekar step), zero on palindromes and large on lopsided counts; Depth blends with the played velocity.

RoughnessVelocity [DYN] [MIDI] in: Audio out: MIDI

Roughness velocity: drives velocity from the smallest prime factor of the note count, the complement of smoothness, so even counts read low and prime-like counts read high; Depth blends with the played velocity.

RudinShapiroAccent [DYN] [MIDI] in: Audio out: MIDI

Rudin-Shapiro accent: pushes each note's velocity up or down by the +/-1 Rudin-Shapiro sequence of its count, an aperiodic accent pattern with a famously flat (white-like) spectrum - structured stress that never repeats predictably.

RulerVelocity [DYN] [MIDI] in: Audio out: MIDI

Ruler velocity: accents notes by the ruler sequence (the power of two dividing the count: 0,1,0,2,0,1,0,3,…), so every other note is a light tick and the downbeats of each binary level land harder - a self-similar metric accent.

┬ RunningSumVelocity [DYN] [MIDI] in: Audio out: MIDI

Running-sum velocity: accumulates a small per-note increment into a velocity that drifts upward and wraps, a slowly-cycling ramp untied to pitch; Depth blends with the played velocity.

┬ ScaleDegreeAccent [DYN] [MIDI] in: Audio out: MIDI

Scale-degree accent: accents notes by their function in the key - tonic loudest, then fifth and third, with weak and chromatic degrees softened - carving a dynamic shape that follows the harmony.

┬ ScaleHighlight [DYN] [MIDI] in: Audio out: MIDI

Scale highlight: accents notes that belong to the chosen Root/Scale and softens the chromatic outsiders, spotlighting the key without removing any notes - a scale-aware dynamic emphasis.

┬ SigmaVelocity [DYN] [MIDI] in: Audio out: MIDI

Sigma velocity: drives velocity from sigma(count), the sum of all divisors of the note count, so highly-divisible counts hit harder - an accent contour straight out of number theory; Depth blends with the played velocity.

┬ SineWaveVelocity [DYN] [MIDI] in: Audio out: MIDI

Sine-wave velocity: shapes velocity as a smooth sine across a chosen period of notes, a gentle rise-and-fall dynamic swell repeating every Period notes; Depth blends with the played velocity.

┬ SmoothnessVelocity [DYN] [MIDI] in: Audio out: MIDI

Smoothness velocity: drives velocity from the largest prime factor of the note count, so smooth (small-factor) counts read soft and prime counts read loud; Depth blends with the played velocity.

┬ SquareDistanceVelocity [DYN] [MIDI] in: Audio out: MIDI

Square-distance velocity: accents notes by how close the count sits to a perfect square - loud right on the squares, fading in the gaps - a pulsing accent that widens as the squares spread apart; Depth blends with the played velocity.

┬ SquareRootVelocity [DYN] [MIDI] in: Audio out: MIDI

Square-root velocity: drives velocity from the integer square root of the note count, a staircase whose steps grow ever wider; Depth blends with the played velocity.

┬ SquareWaveVelocity [DYN] [MIDI] in: Audio out: MIDI

Square-wave velocity: alternates every note between two fixed velocity levels, a hard 2-step on/off accent that drives a mechanical pumping groove.

┬ StaircaseVelocity [DYN] [MIDI] in: Audio out: MIDI

Staircase velocity: velocity climbs in quantized steps up a staircase across a chosen number of steps then drops back, a terraced crescendo accent; Depth blends with the played velocity.

┬ **SternBrocotVelocity** [DYN] [MIDI] in: Audio out: MIDI

Stern-Brocot velocity: shapes velocity from Stern's diatomic (fusc) sequence, the fractal numerators of the Stern-Brocot tree of rationals, giving a self-similar sawtooth-of-sawtooths accent; Depth blends with the incoming velocity.

┬ **SternDiatomicVelocity** [DYN] [MIDI] in: Audio out: MIDI

Stern-diatomic velocity: accents each note by Stern's diatomic (fusc) sequence 1,1,2,1,3,2,3,1,4,… whose consecutive ratios enumerate every rational once (the Stern-Brocot tree); Depth blends with the played velocity.

┬ **SwingVelocity** [DYN] [MIDI] in: Audio out: MIDI

Swing velocity: accents on-beat notes and softens off-beat ones, applying the dynamic side of a swing/shuffle feel (the loud-soft alternation) independent of timing - instant groove from even input.

┬ **TernaryDigitVelocity** [DYN] [MIDI] in: Audio out: MIDI

Ternary-digit velocity: drives velocity from the base-3 digit sum of the note count, a self-similar accent that climbs and resets on ternary carries; Depth blends with the played velocity.

┬ **ThueMorseTernaryVel** [DYN] [MIDI] in: Audio out: MIDI

Ternary Thue-Morse velocity: picks one of three velocity levels from the base-3 Thue-Morse sequence (digit-sum mod 3) of the count, a self-similar three-level accent that avoids short repeats; Depth blends with the played velocity.

┬ **ThueMorseVelocity** [DYN] [MIDI] in: Audio out: MIDI

Thue-Morse velocity: alternates two velocity levels by the cube-free Thue-Morse parity of the note count, accenting in a self-similar, never-quite-repeating pattern instead of a simple every-other.

┬ **TotientRatioVelocity** [DYN] [MIDI] in: Audio out: MIDI

Totient-ratio velocity: drives velocity from Euler's totient ratio phi(n)/n, so primes read near-full and highly-composite counts read soft, an arithmetic density accent; Depth blends with the played velocity.

┬ **TotientStepsVelocity** [DYN] [MIDI] in: Audio out: MIDI

Totient-depth velocity: accents each note by how many times Euler's totient must be applied to its count to reach 1 (the iterated-totient depth, roughly log-scaled); Depth blends with the played velocity.

┬ **TotientVelocity** [DYN] [MIDI] in: Audio out: MIDI

Totient velocity: drives velocity from Euler's totient ratio phi(n)/n of the count, dipping on numbers rich in small prime factors and peaking on primes - a number-theoretic accent contour; Depth blends with the played velocity.

┬ **TrailingOnesVelocity** [DYN] [MIDI] in: Audio out: MIDI

Trailing-ones velocity: accents each note by how many 1-bits trail at the bottom of its binary count (0,1,0,2,0,1,0,3,…), a carry-driven spiky pattern; Depth blends with the played velocity.

┬ **TriangularModVelocity** [DYN] [MIDI] in: Audio out: MIDI

Triangular-mod velocity: shapes velocity as a triangle wave rising and falling across a chosen period of notes, a smooth swell-and-fade accent contour; Depth blends with the played velocity.

┬ **TribonacciVelocity** [DYN] [MIDI] in: Audio out: MIDI

Tribonacci velocity: drives velocity from the Tribonacci numbers where each term sums the previous three (0,1,1,2,4,7,13,24,44,…); Depth blends with the played velocity.

┬ **TurnaroundAccent** [DYN] [MIDI] in: Audio out: MIDI

Turnaround accent: boosts the velocity of notes that are local melodic turning points - the peaks and valleys where the line changes direction - emphasising the contour's corners.

┬ **UnitaryDivisorVelocity** [DYN] [MIDI] in: Audio out: MIDI

Unitary-divisor velocity: accents each note by the sum of its count's unitary divisors (divisors that share no factor with their cofactor), a coprime-divisor variant of the sigma accent; Depth blends with the played velocity.

┬ **VelByPitch** [DYN] [MIDI] in: Audio out: MIDI

Velocity by pitch: sets each note's velocity from its keyboard position (a Slope per semitone around middle C plus a Base), so high notes can ring louder or softer - keyboard-position dynamics.

┬ **VelClip** [DYN] [MIDI] in: Audio out: MIDI

Clamps velocity between Min and Max.

┬ **VelCompress** [DYN] [MIDI] in: Audio out: MIDI

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

┬ **VelCurve** [DYN] [MIDI] in: Audio out: MIDI

Reshapes velocity through a power curve.

┬ **VelDrift** [DYN] [MIDI] in: Audio out: MIDI

Velocity drift: each note's velocity takes one step of a bounded random walk, so dynamics wander smoothly and correlatedly over time instead of jumping independently like Humanize - the slow swell and ebb of a tiring player.

┬ **VelGate** [DYN] [MIDI] in: Audio out: MIDI

Velocity gate: drops any note whose velocity falls outside the Min..Max window, so only soft (or only hard) hits pass - a dynamics-dependent filter for splitting ghosted from accented playing.

┬ VelGateHysteresis [DYN] [MIDI] in: Audio out: MIDI

Velocity Schmitt gate: opens once a note exceeds the High velocity and stays open until one drops below Low, so soft passages between accents either all pass or all mute - hysteretic dynamic gating that avoids chattering at the threshold.

┬ VelInvert [DYN] [MIDI] in: Audio out: MIDI

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

┬ VelKey [DYN] [MIDI] in: Audio out: MIDI

Keyboard velocity tracking: scales each note's velocity by how far its pitch sits from a Center note, Tilt setting how strongly and which way - higher notes louder (or softer), the dynamic key-tracking of real instruments.

┬ VelLength [DYN] [MIDI] in: Audio out: MIDI

Note duration scales with velocity: hard notes ring for Max ms, soft notes for Min (set Max < Min to make accents staccato) - the played articulation where dynamics shape note length.

┬ Velocity [DYN] [MIDI] in: Audio out: MIDI

Scales and offsets note velocity, with a modulation amount.

┬ VelocityAlternate [DYN] [MIDI] in: Audio out: MIDI

Velocity alternate: alternates note-on velocities between two fixed levels (loud, soft, loud, soft), an automatic backbeat-style dynamic groove.

┬ VelocityBoost [DYN] [MIDI] in: Audio out: MIDI

Velocity boost: adds a fixed amount (positive or negative) to every note-on velocity, clamped to the legal range, a quick dynamics offset.

┬ VelocityClip [DYN] [MIDI] in: Audio out: MIDI

Velocity clip: hard-limits note-on velocities to a floor and ceiling, clamping anything outside the window to the edges (a dynamic gate/limiter).

┬ VelocityCompress [DYN] [MIDI] in: Audio out: MIDI

Velocity compress: pulls every note-on velocity toward the centre by the set ratio, narrowing the dynamic range so soft and loud notes sit closer together.

┬ VelocityCurve [DYN] [MIDI] in: Audio out: MIDI

Velocity curve: reshapes note-on velocity through a gamma curve - values above 1 soften the response (more playing in the quiet range), below 1 harden it.

VelocityDrift [DYN] [MIDI] in: Audio out: MIDI

Velocity drift: applies a slow bounded random walk to note-on velocities so the dynamics wander gently up and down over time rather than jumping; Step sets the walk speed.

VelocityExpand [DYN] [MIDI] in: Audio out: MIDI

Velocity expand: pushes velocities away from the mid-level (the opposite of compression), so soft notes get softer and loud notes louder - widening the dynamic range of a flatly-played or over-quantized part.

VelocityFloor [DYN] [MIDI] in: Audio out: MIDI

Velocity floor: lifts any note played softer than the floor up to it, guaranteeing every note speaks at a minimum level - a reverse noise gate for taming ghost notes and uneven controllers.

VelocityFold [DYN] [MIDI] in: Audio out: MIDI

Velocity fold: folds note-on velocities that exceed a ceiling back downward (a wavefolder for dynamics), so very hard hits read softer in a non-monotonic way.

VelocityFromInterval [DYN] [MIDI] in: Audio out: MIDI

Velocity from interval: accents each note by how far it leaps from the previous note, so large melodic jumps hit harder than stepwise motion; Depth blends with the played velocity.

VelocityFromPitch [DYN] [MIDI] in: Audio out: MIDI

Velocity from pitch: derives velocity from the note's pitch so higher notes play louder (or, inverted, lower notes louder), a keyboard-tilt dynamic; Depth blends with the played velocity.

VelocityGammaCurve [DYN] [MIDI] in: Audio out: MIDI

Velocity gamma curve: reshapes velocity through a gamma transfer curve - values below 1 lift soft notes and compress dynamics, above 1 push them down and expand - a smooth nonlinear dynamics bend.

VelocityGate [DYN] [MIDI] in: Audio out: MIDI

Velocity gate: passes only notes whose velocity falls inside a Low..High window, masking soft or loud notes (a dynamics filter / range splitter); Invert mutes the window instead of keeping it.

VelocityLFO [DYN] [MIDI] in: Audio out: MIDI

Velocity LFO: sweeps velocity up and down with a slow sine over a set number of notes, an automatic dynamic swell-and-fade that breathes life into a flat sequence; Depth sets how deep the swell.

VelocityMultiply [DYN] [MIDI] in: Audio out: MIDI

Velocity multiply: scales every note-on velocity by a fixed gain factor, a simple dynamics trim that boosts or attenuates the whole part.

┬ VelocityQuantize [DYN] [MIDI] in: Audio out: MIDI

Velocity quantize: snaps note-on velocities to a small number of evenly-spaced levels, a terraced/stepped dynamic reminiscent of early samplers and harpsichords.

┬ VelocityRandomScale [DYN] [MIDI] in: Audio out: MIDI

Velocity random scale: multiplies each note's velocity by an independent random gain, scattering the dynamics for a more human, less machine-perfect feel (timing untouched, unlike Humanize); Amount sets the scatter width.

┬ VelocityRandomWalk [DYN] [MIDI] in: Audio out: MIDI

Velocity random walk: nudges velocity by a smoothed, bounded random walk so the dynamics drift and breathe over time rather than jumping independently each note, a gradual humanizing of touch.

┬ VelocitySmooth [DYN] [MIDI] in: Audio out: MIDI

Velocity smooth: glides each note's velocity toward a running average of the recent ones, ironing out wild dynamic jumps into an even, controlled performance; Amount sets how strongly it pulls toward the average.

┬ VelocitySwap [DYN] [MIDI] in: Audio out: MIDI

Velocity swap: inverts each note-on velocity (128 minus the value) so the softest hits become the loudest and vice versa; Depth blends with the played velocity.

┬ VelStep [DYN] [MIDI] in: Audio out: MIDI

Velocity step: quantizes velocity to a small number of levels (a dynamics bit-crush), flattening expressive playing into stepped terraces from gentle 8-level to brutal 2-level.

┬ WythoffVelocity [DYN] [MIDI] in: Audio out: MIDI

Wythoff velocity: drives velocity from the lower Wythoff sequence floor(n*phi) (the golden Beatty sequence, the winning positions of Wythoff's game: 1,3,4,6,8,9,11,…); Depth blends with the played velocity.

┬ ZeckendorfVelocity [DYN] [MIDI] in: Audio out: MIDI

Zeckendorf velocity: accents each note by how many Fibonacci numbers it takes to sum to the note count (its Zeckendorf representation length), a number-theoretic accent that grows in golden steps; Depth blends with the played velocity.

┬ ZipfVelocity [DYN] [MIDI] in: Audio out: MIDI

Zipf velocity: draws each note's velocity from a Zipf (1/rank) distribution, so a few dynamic levels dominate and loud accents are rare - the power-law dynamics of natural performance; Depth blends it against the played velocity.

┬ ZoneVelocityScale [DYN] [MIDI] in: Audio out: MIDI

Zone velocity scale: scales the velocity of notes inside [Low,High] by Gain, a per-zone dynamics trim that lets a split layer sit louder or softer than the rest.

Expression (3)

Fall [FX] [MIDI] in: Audio out: MIDI

Jazz fall-off: on release the pitch sweeps down by Depth over Time while the note rings, then the note releases and the bend snaps back to centre. The note-off is delayed to the end of the fall.

MidiVibrato [FX] [MIDI] in: Audio out: MIDI

A pitch-bend LFO that fades in after an Onset delay while any note is held - the delayed, played-in vibrato of a string or voice rather than an instant wobble. Rate in Hz, Depth as bend amount.

Scoop [FX] [MIDI] in: Audio out: MIDI

Bends up into each struck note from Depth below, resolving to pitch over Time - the vocal/brass scoop, a pitch-bend ramp that ends at centre so no bend is left hanging.

Harmony (51)

AutoInvert [HRM] [MIDI] in: Audio out: MIDI

Cycles the inversion of each struck chord: every trigger rotates which of the lowest notes jump an octave (by Rotate), so a repeated chord keeps climbing through its inversions - movement ChordVoicing's static shapes don't give.

Bassline [HRM] [MIDI] in: Audio out: MIDI

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

Chord [HRM] [MIDI] in: Audio out: MIDI

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

ChordLock [HRM] [MIDI] in: Audio out: MIDI

Quantizes every note to the nearest tone of a fixed chord (Root + Type), tighter than Scale's scale-quantize - locks an improvisation to a chord's notes.

ChordMem [HRM] [MIDI] in: Audio out: MIDI

Each key plays a stored chord shape (semitone intervals from node config).

ChordProg [HRM] [MIDI] in: Audio out: MIDI

Each note triggers the next chord in a stored root progression (node config).

ChordVoicing [HRM] [MIDI] in: Audio out: MIDI

Re-voices each note into a jazz shape (open / shell / quartal / spread).

ConsonanceGate [HRM] [MIDI] in: Audio out: MIDI

Consonance gate: passes a note only when its interval from the previous one is consonant (unison, third, fourth, fifth, sixth or octave), filtering a chromatic line down to smooth, dissonance-free melodic motion.

ContourGate [HRM] [MIDI] in: Audio out: MIDI

Contour gate: passes only the rising moves (or only the falling moves) of a melody relative to the previous kept note, isolating the ascending or descending gestures of a line.

ContourReverseGate [HRM] [MIDI] in: Audio out: MIDI

Contour-reverse gate: passes a note only when the melody changes direction (a peak or valley), keeping just the turning points of a line and dropping the runs between them; Invert keeps the runs.

Counterpoint [HRM] [MIDI] in: Audio out: MIDI

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

Diatonic [HRM] [MIDI] in: Audio out: MIDI

Transposes by scale DEGREES within Root/Scale (a third, a fifth…) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

DiatonicChord [HRM] [MIDI] in: Audio out: MIDI

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th…) along Root/Scale, so single-finger melodies become correct chords in the key.

DiatonicHarmonize [HRM] [MIDI] in: Audio out: MIDI

Diatonic harmonize: adds a second voice a chosen number of scale steps above each note within a major key, so the harmony note always stays in key (true diatonic thirds/sixths, unlike a fixed chromatic interval).

DiatonicTranspose [HRM] [MIDI] in: Audio out: MIDI

Diatonic transpose: shifts every note by a number of whole scale degrees within a major key, so the melody moves up or down the staff while always landing on in-key notes - true in-key transposition, unlike the chromatic kind.

Drone [HRM] [MIDI] in: Audio out: MIDI

Doubles every note with a fixed pedal root.

GhostNote [HRM] [MIDI] in: Audio out: MIDI

Ghost note: doubles every note with a quiet copy a set interval below (default an octave), at a fraction of the original velocity - the subliminal low reinforcement drummers and bassists use to thicken a line.

Glissando [HRM] [MIDI] in: Audio out: MIDI

Fills a stepwise run between consecutive notes.

GravityQuantize [HRM] [MIDI] in: Audio out: MIDI

Gravity quantize: pulls every note toward a center pitch by an adjustable strength, as if the keyboard had a gravitational well - gentle settings nudge stray notes inward, strong settings collapse a whole part toward one tone.

HarmonicQuantize [HRM] [MIDI] in: Audio out: MIDI

Harmonic quantize: re-pitches each note to the nearest member of the natural harmonic series above a root, snapping a chromatic part onto the pure, stretched-then-compressed intervals of the overtone series.

Harmonize [HRM] [MIDI] in: Audio out: MIDI

Adds up to three fixed-interval harmony voices to each note.

IntervalClassGate [HRM] [MIDI] in: Audio out: MIDI

Interval-class gate: passes a note only when its leap from the previous kept note belongs to a chosen interval class (0 unison/octave .. 6 tritone), filtering a line down to one harmonic colour of motion; Invert keeps the rest.

Invert [HRM] [MIDI] in: Audio out: MIDI

Mirrors pitch around a pivot note.

LeapGate [HRM] [MIDI] in: Audio out: MIDI

Leap gate: passes only stepwise motion (intervals below the threshold) or only leaps (intervals at or above it) relative to the previous kept note, splitting a melody into its smooth and its jagged motion.

MidiCluster [HRM] [MIDI] in: Audio out: MIDI

Adds symmetric tone-cluster voices above and below each note.

MidiFold [HRM] [MIDI] in: Audio out: MIDI

Folds out-of-range notes back inside a Low..High window.

MirrorPivot [HRM] [MIDI] in: Audio out: MIDI

Mirror pivot: reflects every note about a fixed pivot pitch (out = 2*Pivot - note), turning rising lines into falling ones and inverting every interval around the chosen axis - melodic inversion with a movable mirror.

ModalShift [HRM] [MIDI] in: Audio out: MIDI

Modal shift: brightens or darkens the key by raising the fourth toward Lydian or lowering the third, sixth and seventh toward natural minor, recolouring a part's mode without retransposing it.

NegativeHarmony [HRM] [MIDI] in: Audio out: MIDI

Negative harmony: reflects every note about the axis halfway between the tonic and dominant (Ernst Levy's mirror), swapping major for minor and turning a progression into its tonal shadow while keeping the same harmonic function.

NeoRiemann [HRM] [MIDI] in: Audio out: MIDI

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

Octave [HRM] [MIDI] in: Audio out: MIDI

Doubles each note up and/or down by whole octaves.

OctaveClamp [HRM] [MIDI] in: Audio out: MIDI

Octave clamp: folds every incoming note into the single octave starting at Center by collapsing its pitch class, flattening wide-ranging input into one register - a pitch-class monitor or tight-range re-voicer.

OctaveFoldGate [HRM] [MIDI] in: Audio out: MIDI

Octave fold: collapses every note into a single octave above a base note (keeping only its pitch class), squashing a wide-ranging part into one register - useful for bass-monitoring or extreme register clamping.

OctaveStackAdd [HRM] [MIDI] in: Audio out: MIDI

Octave stack: layers each note with a copy an octave above and/or below, fattening a thin line into a multi-octave stack (organ-style); choose which octaves to add.

OvertoneStack [HRM] [MIDI] in: Audio out: MIDI

Overtone stack: layers quieter harmonic-series voices (octave, octave+fifth, two octaves) above each note with a geometric Rolloff, thickening a single note toward an organ-like harmonic chord.

PitchClassGate [HRM] [MIDI] in: Audio out: MIDI

Pitch-class gate: passes only notes whose pitch class belongs to the chosen Root and Mode set (chromatic, major, minor, pentatonic or whole-tone), filtering a stream down to a key without transposing anything.

PitchMirror [HRM] [MIDI] in: Audio out: MIDI

Pitch mirror: reflects every note around a fixed axis pitch, so ascending lines become descending and the melody is turned upside-down about a chosen centre (a literal pitch inversion, key-independent).

PowerChordAdd [HRM] [MIDI] in: Audio out: MIDI

Power chord: doubles every note with a perfect fifth above (and optionally the octave on top), instantly turning a single line into the root-fifth power chords of rock guitar.

RangeCompress [HRM] [MIDI] in: Audio out: MIDI

Range compress: linearly squeezes the played pitch range toward a center note, narrowing a wide-ranging part into a tighter register (or, near ratio 1, leaving it alone) - a melodic-ambitus compressor.

RangeExpand [HRM] [MIDI] in: Audio out: MIDI

Range expand: stretches the played pitch range away from a center note (the inverse of range-compress), exaggerating a cramped melody into a wider, more dramatic ambitus; near ratio 1 it leaves the part alone.

RangeGate [HRM] [MIDI] in: Audio out: MIDI

Range gate: passes only notes whose pitch lies inside a Low..High window, masking a register in or out of the part; Invert mutes the window instead of keeping it.

ReflectOctave [HRM] [MIDI] in: Audio out: MIDI

Reflect octave: mirrors each note's pitch class about a pivot while keeping it in its own octave, inverting the melodic contour within every register at once - octave-preserving interval inversion.

RepeatedNoteGate [HRM] [MIDI] in: Audio out: MIDI

Repeated-note gate: drops a note-on when it repeats the previous kept pitch, passing only when the melody actually changes note - a de-stutter that removes hammered repeats; Invert keeps only the repeats.

Scale [HRM] [MIDI] in: Audio out: MIDI

Quantizes incoming notes to the nearest degree of the chosen root and scale.

SubHarmonicAdd [HRM] [MIDI] in: Audio out: MIDI

Sub-harmonic add: reinforces every note with undertone voices an octave (Sub1) and an octave-plus-fifth (Sub2) below at adjustable levels - the subharmonic weight of a Trautonium or organ bourdon.

SubharmonicShift [HRM] [MIDI] in: Audio out: MIDI

Subharmonic shift: drops each note to the nearest undertone (subharmonic) of a ceiling pitch, the mirror of the harmonic series, building dark, minor-tilted lines beneath an upper anchor note.

Transpose [HRM] [MIDI] in: Audio out: MIDI

Shifts every note by a fixed semitone amount plus a modulatable offset.

TwelveToneRow [HRM] [MIDI] in: Audio out: MIDI

Twelve-tone row: enforces Schoenberg's serial rule - no pitch class may sound again until all twelve have been played - dropping early repeats so the stream is forced into complete chromatic rows.

UnisonDetuneAdd [HRM] [MIDI] in: Audio out: MIDI

Unison detune: layers a second copy of each note slightly detuned (via per-note microtuning), thickening a mono line into a wide chorused unison without a separate effect.

VoiceLead [HRM] [MIDI] in: Audio out: MIDI

Smooth voice leading: places each note's pitch class in the octave nearest the previous output, so a line or chord progression moves by the smallest interval. Range caps the leap; past it the original octave is kept.

WholeToneSnap [HRM] [MIDI] in: Audio out: MIDI

Whole-tone snap: quantizes every note to the nearest degree of the six-note whole-tone scale from Root, dissolving tonality into the dreamy, rootless ambiguity that scale is famous for.

Modulation (60)

AvgPitchCC [MOD] [MIDI] in: Audio out: MIDI

Average-pitch CC: emits a controller that follows the mean pitch of all held notes (the chord's centroid), a smooth register-position modulation source.

BendFromVelocity [MOD] [MIDI] in: Audio out: MIDI

Bend from velocity: emits a pitch-bend on each note-on scaled by how hard you played, so harder hits bend up further (an expressive velocity-to-pitch gesture); Depth sets the range.

BluesInflect [MOD] [MIDI] in: Audio out: MIDI

Blues inflect: bends the third, fifth and seventh of a major key downward toward the blue notes (the cracks between the piano keys that a blues singer or guitarist bends into), adding microtonal blues feel without changing the notes played.

BohlenPierceTune [MOD] [MIDI] in: Audio out: MIDI

Bohlen-Pierce tune: retunes the keyboard to the Bohlen-Pierce scale, which divides not the octave but the 3:1 'tritave' into 13 equal steps, giving an alien but surprisingly consonant tuning built on odd-harmonic ratios.

ByzantineTune [MOD] [MIDI] in: Audio out: MIDI

Byzantine tune: retunes to a Byzantine chant scale, whose soft-chromatic genus uses neutral intervals absent from Western tuning, the sound of Eastern Orthodox liturgical melody.

CarlosAlphaTune [MOD] [MIDI] in: Audio out: MIDI

Carlos Alpha tune: retunes to Wendy Carlos's Alpha scale of 78-cent equal steps, a non-octave temperament engineered to nail pure major thirds and fifths at the cost of ever repeating at the octave - a shimmering, otherworldly intonation.

CarlosBetaTune [MOD] [MIDI] in: Audio out: MIDI

Carlos Beta tune: retunes to Wendy Carlos's Beta scale, built from a ~63.8-cent step that deliberately does not repeat at the octave, giving the uncanny non-octave harmony of her Beauty in the Beast.

CarlosGammaTune [MOD] [MIDI] in: Audio out: MIDI

Carlos Gamma tune: retunes to Wendy Carlos's Gamma scale, an ultra-fine ~35.1-cent micro-step tuning that renders both perfect fifths and major thirds nearly pure, the smoothest of her invented scales.

CCGlide [MOD] [MIDI] in: Audio out: MIDI

Portamento for a CC: smooths the watched CC toward each newly received value over Time, emitting interpolated CC each block - de-zippers a stepped controller, where MidiSlew only glides the internal scalar.

CCLFO [MOD] [MIDI] in: Audio out: MIDI

Continuous CC LFO generator (sine/triangle/saw/square) emitted on a chosen CC at a free Rate - automates a synth parameter over MIDI, where MidiLFO drives only the internal mod-matrix scalar.

CCSeq [MOD] [MIDI] in: Audio out: MIDI

Tempo-synced step CC sequencer: one CC value per synced step cycling an 8-step shape (overridable via the node's config list) - a modulation sequencer in the CC domain, where StepSeqMidi sequences notes.

ChordSpanCC [MOD] [MIDI] in: Audio out: MIDI

Chord-span CC: emits a controller that reflects the interval span (highest minus lowest) of the held notes, full at Range semitones - wide voicings push it up.

ExpressionSwell [MOD] [MIDI] in: Audio out: MIDI

Expression swell: emits a CC11 expression value that ramps up from zero after each note-on over the swell time, an automatic volume fade-in / crescendo per note for strings and pads.

GamelanDetune [MOD] [MIDI] in: Audio out: MIDI

Gamelan detune: tunes alternate keys slightly apart so paired notes shimmer with the slow acoustic beating of a gamelan's deliberately-mistuned instrument pairs (ombak) - an ensemble-wide chorused waver.

Generative [MOD] [MIDI] in: Audio out: MIDI

Probabilistic melodic random walk synced to a rate.

GoldenAngleDetune [MOD] [MIDI] in: Audio out: MIDI

Golden-angle detune: scatters each successive note's microtuning by the golden angle, the maximally-even irrational spacing, so repeated notes never land on the same detune - an organic, never-repeating pitch shimmer.

HarmonicSeriesSnap [MOD] [MIDI] in: Audio out: MIDI

Harmonic-series snap: detunes each note to the nearest member of the natural harmonic series above a root, replacing tempered intervals with the pure, beatless ratios of just intonation as a brass or didgeridoo would sound them.

HarmonicSeriesTune [MOD] [MIDI] in: Audio out: MIDI

Harmonic-series tune: retunes every note to the nearest member of the natural overtone series of a root (the just-intonation pitches a vibrating string actually produces), so chords lock into pure, beatless ratios that no equal scale can reach.

HighNoteCC [MOD] [MIDI] in: Audio out: MIDI

High-note CC: emits a controller that tracks the pitch of the highest held note, so the top voice of your playing steers a synth parameter (filter, level, …).

HirajoshiTune [MOD] [MIDI] in: Audio out: MIDI

Hirajoshi tune: retunes to the Japanese hirajoshi pentatonic, the dark, koto-flavoured five-note scale of traditional Japanese music, snapped from the played notes.

IntervalCC [MOD] [MIDI] in: Audio out: MIDI

Interval CC: emits a controller on each note-on from how far it leaps from the previous note, so stepwise lines read low and big jumps read high, full at Range semitones.

JustMinorTune [MOD] [MIDI] in: Audio out: MIDI

Just-minor tune: retunes to a 5-limit just-intonation natural minor (pure 6/5 third and 3/2 fifth), giving beatless minor chords impossible in equal temperament.

LowNoteCC [MOD] [MIDI] in: Audio out: MIDI

Low-note CC: emits a controller that tracks the pitch of the lowest held note, so the bass voice of your playing steers a synth parameter.

MaqamRastTune [MOD] [MIDI] in: Audio out: MIDI

Maqam Rast tune: retunes to the Arabic maqam Rast, whose neutral third and sixth sit a quarter-tone below the Western major, giving the characteristic in-between intervals of Middle-Eastern melody.

MeantoneTune [MOD] [MIDI] in: Audio out: MIDI

Meantone tune: retunes to quarter-comma meantone, the Renaissance/Baroque temperament with pure (beatless) major thirds bought at the price of a howling 'wolf' fifth, the sound of early-music keyboards.

MicrotuneSpread [MOD] [MIDI] in: Audio out: MIDI

Microtune spread: gives every note a small random pitch detune up to +/-Cents via its microtuning offset, loosening rigid tuning into a chorused, slightly-out ensemble shimmer.

MidiEnv [MOD] [MIDI] in: Audio out: MIDI

ADSR envelope triggered by the note gate, on the scalar control output.

MidiLFO [MOD] [MIDI] in: Audio out: MIDI

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

MidiRandom [MOD] [MIDI] in: Audio out: MIDI

Generates random notes within a pitch range at a synced rate.

MidiSampleHold [MOD] [MIDI] in: Audio out: MIDI

Samples and holds the incoming control value on each note.

MidiSlew [MOD] [MIDI] in: Audio out: MIDI

Smooths (glides) the control value over a set time.

NEdoRetune [MOD] [MIDI] in: Audio out: MIDI

N-EDO retune: detunes every note from 12-tone equal temperament to the nearest step of an N-equal-divisions-of-the-octave scale, opening up 19, 24, 31 or any microtonal grid without changing the played note numbers.

NoteOnTrigCC [MOD] [MIDI] in: Audio out: MIDI

Note-on trigger CC: fires a CC pulse that jumps to full on every note-on and decays away, a built-in trigger envelope you can route to a synth parameter; Decay sets the fall time.

NoteToPressure [MOD] [MIDI] in: Audio out: MIDI

Note-to-pressure: emits a channel-aftertouch value scaled from each note's velocity, so how hard you strike also drives any pressure-mapped destination (vibrato, filter, swell) - turning a non-aftertouch keyboard into an expressive one.

PartchTune [MOD] [MIDI] in: Audio out: MIDI

Partch tune: retunes to an 11-limit just-intonation subset in the spirit of Harry Partch's 43-tone tonality, the ratio-based microtonality of his hand-built instruments.

PelogTune [MOD] [MIDI] in: Audio out: MIDI

Pelog tune: retunes to a Javanese pelog scale - seven unequal steps with characteristic narrow and wide intervals - the more tense, expressive counterpart to slendro in gamelan music.

PersianTune [MOD] [MIDI] in: Audio out: MIDI

Persian tune: retunes to a Persian dastgah scale with its characteristic koron (quarter-flat) degrees, the microtonal intervals of classical Persian music played on the tar and santur.

PitchBendLFO [MOD] [MIDI] in: Audio out: MIDI

Pitch-bend LFO: emits a free-running pitch-bend sine that wobbles the tuning continuously whether or not notes are playing - a global vibrato/auto-bend source independent of any single note; Rate and Depth shape the sweep.

PitchClassCC [MOD] [MIDI] in: Audio out: MIDI

Pitch-class CC: emits a controller on each note-on from the note's pitch class (its position 0-11 within the octave, octave-independent), a harmonic-colour modulation source.

PitchDrift [MOD] [MIDI] in: Audio out: MIDI

Nudges each note's pitch by a small random detune up to +/-Cents (per-note, polyphonic) - the gentle oscillator instability of analog gear, applied to pitch where Humanize only scatters timing and velocity.

PitchScoop [MOD] [MIDI] in: Audio out: MIDI

Pitch scoop: emits a pitch-bend that starts bent below each new note and swoops up to true pitch over the scoop time, the expressive slide-into-note of a fretless or vocal attack.

PolyCountCC [MOD] [MIDI] in: Audio out: MIDI

Poly-count CC: emits a controller that reflects how many notes are held (chord density), reaching full value at Max voices - play thicker chords to open a parameter.

PressSwell [MOD] [MIDI] in: Audio out: MIDI

Generates a channel-pressure (aftertouch) swell that rises to Max over Rise while any note is held, snapping back to zero on release - an automatic crescendo for pads and strings, where ATProc only transforms incoming aftertouch.

PythagoreanTune [MOD] [MIDI] in: Audio out: MIDI

Pythagorean tune: retunes to 3-limit Pythagorean intonation built from a spiral of pure fifths, giving brilliant, wide major thirds and the bright, slightly-sharp leading tones of medieval and string-ensemble tuning.

QuarterToneShift [MOD] [MIDI] in: Audio out: MIDI

Quarter-tone shift: detunes every note by up to a quarter-tone via its microtuning offset, sliding the whole performance off the 12-tone grid into 24-tone territory without changing the note numbers.

QuarterToneTune [MOD] [MIDI] in: Audio out: MIDI

Quarter-tone tune: retunes to 24-tone equal temperament, snapping every note to the nearest 50-cent quarter-tone, the standard grid of Arabic maqam notation and much 20th-century microtonal music.

∿ RagaBhairavTune [MOD] [MIDI] in: Audio out: MIDI

Raga Bhairav tune: retunes to the just-intonation shrutis of the North-Indian morning raga Bhairav (with its flat second and flat sixth), bringing the pure, beatless intervals of Indian classical tuning.

∿ RandOctave [MOD] [MIDI] in: Audio out: MIDI

Randomly shifts notes by up to +/- Range octaves with a chance.

∿ RandomCC [MOD] [MIDI] in: Audio out: MIDI

Random CC: fires a random value on a chosen controller number with every note-on, an instant source of evolving per-note modulation (filter, pan, anything) that never repeats; Amount sets the spread around center.

∿ SeptimalTune [MOD] [MIDI] in: Audio out: MIDI

Septimal tune: retunes to a 7-limit just scale including the harmonic seventh (7/4), the deep, locked-in dominant-seventh sound prized in barbershop and blues harmony.

∿ SlendroTune [MOD] [MIDI] in: Audio out: MIDI

Slendro tune: retunes the keyboard to a Javanese gamelan slendro scale - five near-equal steps spanning the octave - giving the floating, anchorless quality of Indonesian slendro tuning, snapped from the nearest played note.

∿ StretchTuning [MOD] [MIDI] in: Audio out: MIDI

Stretch tuning: applies a Railsback-style octave stretch, tuning notes progressively sharp as they rise and flat as they fall, the psychoacoustic tuning real pianos use so octaves sound in tune to the ear.

∿ SubharmonicTune [MOD] [MIDI] in: Audio out: MIDI

Subharmonic tune: retunes every note to the nearest member of the undertone (subharmonic) series descending from a top pitch, the mirror image of the overtone series that gives eerie, hollow, just-intoned minor sonorities.

∿ SwellPressure [MOD] [MIDI] in: Audio out: MIDI

Swell pressure: emits a channel-pressure that rises the longer notes are held and falls once they release, an automatic aftertouch swell for sustained pads; Rate sets the speed.

∿ ThaiTune [MOD] [MIDI] in: Audio out: MIDI

Thai tune: retunes to the Thai 7-tone equal temperament (seven equal steps to the octave), the tuning of the Thai ranat and pi phat ensemble that sits between the Western diatonic pitches.

∿ TurkishMakamTune [MOD] [MIDI] in: Audio out: MIDI

Turkish makam tune: retunes to a Turkish makam built on Holdrian (53-comma) steps, whose microtonal neutral seconds and sevenths give the characteristic intervals of Ottoman classical melody.

VelToCC [MOD] [MIDI] in: Audio out: MIDI

Turns each note's velocity into a CC value emitted just before the note, so a synth's filter or level opens with how hard you played - dynamics routed to expression, where Note2CC sends pitch.

VelToDetune [MOD] [MIDI] in: Audio out: MIDI

Velocity-to-detune: maps how hard each note is played to a microtuning detune, so dynamics bend pitch slightly sharp or flat - the pitch-pulls-with-force behaviour of acoustic instruments and analog drift.

WerckmeisterTune [MOD] [MIDI] in: Audio out: MIDI

Werckmeister tune: retunes to Werckmeister III, the famous 1691 well-temperament whose unequal fifths give each key its own colour (the 'well-tempered' tuning of Bach's era) - subtle, key-dependent detuning from equal temperament.

YoungTune [MOD] [MIDI] in: Audio out: MIDI

Young tune: retunes to Thomas Young's 1799 well-temperament, where each key has its own subtly different colour - near-pure common keys, spicier remote ones - the way Classical-era keyboards actually sounded.

Probability (180)

AbundantGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Abundant gate: passes note-ons only on abundant counts - numbers whose proper divisors sum to more than themselves (12, 18, 20, 24…) - thinning the stream onto the divisor-rich integers; Invert keeps the deficient ones.

AchillesGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Achilles gate: passes note-ons on Achilles counts - powerful numbers (every prime factor squared) that are not themselves a perfect power (72,108,200,288,…); Invert keeps the rest.

AdditivePrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Additive-prime gate: passes note-ons on additive-prime counts that are prime and whose digit sum is also prime (2,3,5,7,11,23,29,41,43,…); Invert keeps the rest.

AlmostPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Almost-prime gate: passes note-ons on k-almost-prime counts having exactly K prime factors counted with multiplicity (K=1 primes, K=2 semiprimes, K=3 …), one knob spanning the whole almost-prime family; Invert keeps the rest.

AntiprimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Antiprime gate: passes note-ons on highly-composite (antiprime) counts that have more divisors than every smaller number (1,2,4,6,12,24,36,48,60,…), the most-divisible records; Invert keeps the rest.

AntiRepeatGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Anti-repeat gate: drops a note when it repeats the immediately-previous pitch, forcing the line to keep moving instead of hammering one note - an automatic de-stutter for melodic variety.

AscendingDigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Ascending-digit gate: passes note-ons whose decimal digits strictly increase from left to right (1..9,12,13,…,123,124,…); Invert keeps the rest.

AutomorphicGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Automorphic gate: passes note-ons on automorphic counts whose square ends in the number itself (5,6,25,76,376,625,…), a curious self-reproducing-under-squaring set; Invert keeps the rest.

BellNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Bell-number gate: passes note-ons on Bell-number counts (1,2,5,15,52,203,877,…), which count the ways to partition a set into groups; their fast growth thins the stream quickly. Invert keeps the rest.

BinaryPalindromeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Binary-palindrome gate: passes note-ons whose count reads the same in binary forwards and backwards (1,3,5,7,9,15,17,21,…), a self-mirroring bit pattern distinct from decimal palindromes; Invert keeps the rest.

BinaryWeightGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Binary-weight gate: passes note-ons whose count has exactly the chosen number of 1-bits in binary (its Hamming weight), a bit-pattern sieve that thins to sparser, more clustered hits as the weight rises; Invert keeps the rest.

BlumIntegerGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Blum-integer gate: passes note-ons on Blum counts that are the product of two distinct primes both congruent to 3 mod 4 (21,33,57,69,77,…), the moduli behind Blum-Blum-Shub cryptography; Invert keeps the rest.

CakeNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Cake-number gate: passes note-ons on cake counts (the most pieces a cake splits into with k planar cuts: 1,2,4,8,15,26,42,…), the 3-D analogue of the lazy-caterer numbers; Invert keeps the rest.

CarmichaelGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Carmichael gate: passes note-ons on Carmichael counts - squarefree composites that fool the Fermat primality test (561,1105,1729,…) - via Korselt's criterion; an extremely sparse set. Invert keeps the rest.

CatalanGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Catalan gate: passes note-ons only on Catalan-number counts (1,2,5,14,42,…), the combinatorial sequence counting trees and bracketings, which thins out fast as the values explode; Invert keeps the others.

CatalanParityGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Catalan-parity gate: passes note-ons when the Catalan number C(count) is odd - which happens only when the count-plus-one is a power of two - giving a sparse, exponentially-spaced pattern; Invert keeps the rest.

CenteredCubeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-cube gate: passes note-ons on centered-cube counts (1,9,35,91,189,…), the dots in a cube grown shell by shell around a center; Invert keeps the rest.

CenteredDecagonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-decagonal gate: passes note-ons on centered-decagonal counts (1,11,31,61,101,…), a ten-sided figure grown ring by ring around a center; Invert keeps the rest.

CenteredHexGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-hexagonal gate: passes note-ons on centered-hexagonal counts (1,7,19,37,61,…), the numbers of dots in a growing hexagonal grid (and the rows of Pascal-triangle hex packing); Invert keeps the rest.

CenteredNonagonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-nonagonal gate: passes note-ons on centered-nonagonal counts (1,10,28,55,91,…), a nine-sided figure grown ring by ring around a center, which are also every third triangular number; Invert keeps the rest.

CenteredPentagonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-pentagonal gate: passes note-ons on centered-pentagonal counts (1,6,16,31,51,76,…), a pentagon grown ring by ring around a center dot; Invert keeps the rest.

CenteredSquareGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-square gate: passes note-ons on centered-square counts (1,5,13,25,41,…), the dots in a square diamond grown ring by ring; Invert keeps the rest.

CenteredTriangularGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Centered-triangular gate: passes note-ons on centered-triangular counts (1,4,10,19,31,…), a triangular grid grown outward from a central dot; Invert keeps the rest.

Chance [PRB] [CV] [MIDI] in: Audio out: MIDI

Probabilistic gate: lets each note through with the set probability.

CircularPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Circular-prime gate: passes note-ons on circular-prime counts where every cyclic rotation of the decimal digits is itself prime (2,3,5,7,11,13,17,37,79,…); Invert keeps the rest.

CollatzGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Collatz gate: runs the 3n+1 process on each note's count and passes it when the number of steps to reach 1 is even - an erratic but deterministic gate straight from the unsolved Collatz conjecture; Invert takes the odd-length ones.

CollatzPeakGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Collatz-peak gate: passes note-ons whose count sends its Collatz (3n+1) trajectory soaring above Mult times the starting value, spotlighting the counts that take the wildest hailstone excursions; Invert keeps the rest.

ContraryGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Contrary gate: passes a note only if it moves in the opposite direction to the previous melodic step, forcing the line into constant zig-zag contrary motion and filtering out runs that keep climbing or falling.

CoprimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Coprime gate: passes a note only when its count shares no common factor with the modulus (gcd = 1), thinning the stream onto the totatives of the modulus - a number-theoretic sieve; Invert keeps the non-coprime counts.

CoprimeStepGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Coprime-step gate: passes a note only when its running count shares no common factor with the previous kept count (their gcd is 1), a number-theoretic thinning that favours relatively-prime spacings; Invert keeps the rest.

CousinPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Cousin-prime gate: passes note-ons on counts in a cousin-prime pair (two primes four apart: 3,7,13,17,19,23,…); Invert keeps the rest.

CubeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Cube gate: passes note-ons on perfect-cube counts (1,8,27,64,125,…), an exponentially-widening figurate spacing; Invert keeps the rest.

CullenGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Cullen gate: passes note-ons on Cullen-number counts (n*2^n+1: 3,9,25,65,161,385,…), the +1 companions of the Woodall numbers; Invert keeps the rest.

DeficientGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Deficient gate: passes note-ons on deficient counts whose proper divisors sum to less than the number (1,2,3,4,5,7,8,9,10,11,…; the majority of integers); Invert keeps the abundant/perfect ones.

DelannoyGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Delannoy gate: passes note-ons on central-Delannoy counts (1,3,13,63,321,1683,…), the king-move lattice paths from corner to corner of a grid; Invert keeps the rest.

DescendingDigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Descending-digit gate: passes note-ons of 2+ digits whose decimal digits strictly decrease from left to right (10,20,21,30,31,32,…); Invert keeps the rest.

DigitalRootGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digital-root gate: passes note-ons whose count has a chosen digital root (the single digit reached by repeatedly summing digits), giving a clean period-9 repeating pattern; Invert keeps the rest.

DigitalRootPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digital-root-prime gate: passes note-ons whose digital root is a prime digit (2, 3, 5 or 7), a clean period-9 repeating pattern; Invert keeps the rest.

DigitCountGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-count gate: passes note-ons whose running count has exactly the chosen number of decimal digits, opening for a block of counts then closing - a coarse decimal-magnitude window; Invert keeps the rest.

DigitFactorialGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Factorion gate: passes note-ons on the rare factorion counts equal to the sum of the factorials of their digits (1,2,145,40585); mostly silent, firing only on those four. Invert keeps the rest.

DigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-sum gate: passes a note unless the decimal digit-sum of its count is divisible by Modulo, carving a self-similar arithmetic rhythm out of a steady stream; Invert keeps only the divisible ones.

DigitProductGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-product gate: passes a note only when the product of its count's decimal digits exceeds a threshold, so counts containing a zero or small digits drop out - a digit-driven thinning with a tunable density. Invert keeps the rest.

DigitProductPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-product-prime gate: passes note-ons where the product of the nonzero decimal digits is prime - exactly one prime digit (2,3,5,7) among ones (2,3,5,7,12,13,15,…); Invert keeps the rest.

DigitSumGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-sum gate: passes note-ons whose decimal digit sum equals the chosen target, a repeating digit-driven pattern whose density you set with the target; Invert keeps the rest.

DigitSumSquareGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Digit-sum-square gate: passes note-ons whose decimal digit sum is a perfect square (1,4,9,16,…), so counts summing to 1, 4, 9 or 16 pass; Invert keeps the rest.

DivisorCountGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Divisor-count gate: passes note-ons whose count has exactly the chosen number of divisors (tau=2 selects primes, tau=3 prime squares, and so on), a divisor-structure sieve; Invert keeps the rest.

DivisorParityGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Divisor-parity gate: passes note-ons on counts with an odd number of divisors, which are exactly the perfect squares (1,4,9,16,25,…) - a gate that fires on an ever-widening square spacing; Invert keeps the rest.

DoubleFactorialGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Double-factorial gate: passes note-ons on double-factorial counts n!! - the product of integers of the same parity down to 1 (1,2,3,8,15,48,105,384,…); Invert keeps the rest.

DudeneyGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Dudeney gate: passes note-ons on Dudeney counts that are perfect cubes whose digit sum equals the cube root (1,512,4913,5832,17576,19683); an extremely rare set. Invert keeps the rest.

EmirpGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Emirp gate: passes note-ons on emirp counts - primes that turn into a different prime when their digits are reversed (13,17,31,37,71,73,79,…); Invert keeps the rest.

EquidigitalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Equidigital gate: passes note-ons on equidigital counts that take the same number of digits to write as their prime factorization does (1,2,3,5,7,10,…); Invert keeps the rest.

EvenDigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Even-digit gate: passes note-ons whose count is written with only even decimal digits (2,4,6,8,20,22,24,…), a simple digit-pattern thinning; Invert passes counts containing an odd digit.

EvenDigitSumGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Even-digit-sum gate: passes note-ons whose decimal digit sum is even, a near-even-split digit pattern; Invert keeps the odd-digit-sum counts.

ExtravagantGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Extravagant gate: passes note-ons on extravagant counts whose prime factorization takes MORE digits to write than the number itself (4,6,8,9,12,18,…); Invert keeps the rest.

FermatPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Fermat-prime gate: passes note-ons on the five known Fermat primes (2$^{(2}$k)+1: 3,5,17,257,65537), the primes behind constructible polygons; an exceedingly rare set. Invert keeps the rest.

FibGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Fibonacci gate: passes a note unless the Fibonacci number of its ordinal is divisible by Modulo, carving a golden, self-similar rhythm out of a steady stream; Invert keeps the divisible ones.

FibonacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Fibonacci gate: passes note-ons on Fibonacci-number counts (1,2,3,5,8,13,21,34,…), the golden-ratio recurrence whose gaps grow geometrically; Invert keeps the rest.

FibonacciPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Fibonacci-prime gate: passes note-ons on Fibonacci counts that are also prime (2,3,5,13,89,233,1597,…), a doubly-special sparse set; Invert keeps the rest.

FrugalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Frugal gate: passes note-ons on frugal counts that take fewer digits to write than their prime factorization does (125=5^3, 128=2^7, …), an economical-number sieve; Invert keeps the rest.

GapGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Gap gate: passes a note only after at least Gap notes have gone by since the last one it let through, a refractory thinner that guarantees a minimum spacing between surviving notes regardless of input density.

GeneralizedPentagonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Generalized-pentagonal gate: passes note-ons on generalized-pentagonal counts (1,2,5,7,12,15,22,26,…), the exponents in Euler's pentagonal-number-theorem partition product; Invert keeps the rest.

GoldenGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Golden gate: passes notes by a golden-ratio low-discrepancy sequence against Density, so the kept notes spread far more evenly through time than random chance ever would - a deterministic, clump-free thinning.

GolombGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Golomb gate: passes note-ons on Golomb-sequence counts (1,2,3,4,5,…; the non-decreasing self-describing sequence where term n says how often n appears), a gently-climbing staircase set; Invert keeps the rest.

HammingNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Hamming-number gate: passes note-ons on regular (5-smooth) counts whose only prime factors are 2, 3 and 5 (1,2,3,4,5,6,8,9,10,12,15,16,…), the Hamming/Humble numbers of computing lore; Invert keeps the rest.

HappyGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Happy gate: passes notes on happy-number counts - those whose repeated sum-of-squared-digits reaches 1 - and drops the sad ones that fall into the 4-16-37 cycle; Invert flips which set plays.

HeptanacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Heptanacci gate: passes note-ons on heptanacci counts where each term is the sum of the previous seven (1,2,4,8,16,32,64,127,…), approaching a doubling sequence; Invert keeps the rest.

HexagonalPyramidalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Hexagonal-pyramidal gate: passes note-ons on hexagonal-pyramidal counts (1,7,25,63,129,…), the cannonballs in a hexagon-based pyramid; Invert keeps the rest.

HexanacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Hexanacci gate: passes note-ons on hexanacci counts where each term is the sum of the previous six (1,2,4,8,16,32,63,125,…); Invert keeps the rest.

HighlyAbundantGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Highly-abundant gate: passes note-ons on highly-abundant counts whose divisor sum beats that of every smaller number (1,2,3,4,6,8,10,12,16,18,20,24,…); Invert keeps the rest.

HighlyCompositeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Highly-composite gate: passes note-ons only on counts that set a new record for number of divisors (1,2,4,6,12,24,36,48,60,…), the maximally-factorable integers; a rare, widening-gap accent pulse. Invert keeps the rest.

HoaxNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Hoax-number gate: passes note-ons on hoax counts whose digit sum equals the summed digit sums of their DISTINCT prime factors (22,58,84,…), the squarefree-distinct cousin of Smith numbers; Invert keeps the rest.

HofstadterQGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Hofstadter-Q gate: passes note-ons on values that occur in the chaotic Hofstadter Q meta-sequence (Q(n)=Q(n-Q(n-1))+Q(n-Q(n-2))), an erratic self-referential set from Godel, Escher, Bach; Invert keeps the rest.

IntervalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Interval gate: passes a note only if its leap from the previous kept note lies between Min and Max semitones, filtering a line down to small steps or to wide jumps - a melodic-contour sieve.

JacobsthalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Jacobsthal gate: passes note-ons on Jacobsthal-number counts (1,3,5,11,21,43,…; J(n)=J(n-1)+2J(n-2)), a Fibonacci-like sequence whose ratio tends to 2; Invert keeps the rest.

KaprekarGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Kaprekar gate: passes note-ons on Kaprekar counts whose square can be split into two parts that add back to the number (1,9,45,55,99,297,703,999,…); Invert keeps the rest.

KeithNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Keith-number gate: passes note-ons on Keith (repfigit) counts - numbers that reappear in the Fibonacci-like sequence seeded by their own digits (14,19,28,47,…) - an extremely rare self-referential set; Invert keeps the rest.

LazyCatererGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Lazy-caterer gate: passes note-ons on central-polygonal (lazy-caterer) counts (1,2,4,7,11,16,22,…), the maximum pieces a pancake splits into with n straight cuts; Invert keeps the rest.

LeonardoGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Leonardo gate: passes note-ons on Leonardo-number counts (L(n)=L(n-1)+L(n-2)+1: 1,3,5,9,15,25,41,…), the sizes behind Dijkstra's smoothsort heaps; Invert keeps the rest.

LeylandGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Leyland gate: passes note-ons on Leyland-number counts (x^y+y^x for x,y>=2: 8,17,32,54,57,100,145,…), a sparse two-exponent set; Invert keeps the rest.

LiouvilleGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Liouville gate: passes note-ons when the Liouville function is +1 (the count has an even number of prime factors with multiplicity), an almost-balanced parity stream from analytic number theory; Invert keeps the odd-factor counts.

LucasGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Lucas gate: passes note-ons only on Lucas-number counts (2,1,3,4,7,11,18,…), the Fibonacci companion sequence, thinning the stream onto golden-ratio-spaced ordinals; Invert keeps the rest.

LucasPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Lucas-prime gate: passes note-ons on Lucas counts that are also prime (2,3,7,11,29,47,199,521,…); Invert keeps the rest.

LuckyNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Lucky-number gate: passes note-ons on Ulam's lucky counts, the survivors of a Josephus-style sieve (1,3,7,9,13,15,21,25,…) that share many properties with the primes; Invert keeps the rest.

MagicConstantGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Magic-constant gate: passes note-ons on magic-square constant counts n(n^2+1)/2 - the common row/column/diagonal sum of an n x n magic square (1,5,15,34,65,111,…); Invert keeps the rest.

MersenneGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Mersenne gate: passes note-ons on Mersenne-number counts (one less than a power of two: 1,3,7,15,31,63,127), the all-ones binary numbers; their exponential spacing thins the stream fast. Invert keeps the rest.

MersennePrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Mersenne-prime gate: passes note-ons on Mersenne-prime counts (2^p-1 that are prime: 3,7,31,127,…), an extremely sparse, exponentially-spaced set tied to the perfect numbers; Invert keeps the rest.

MertensGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Mertens gate: passes note-ons while the Mertens function (the running sum of the Moebius function up to the count) is positive, a slow random-walk-like sign whose growth is tied to the Riemann hypothesis; Invert keeps the negative stretches.

MoebiusGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Moebius gate: passes a note only when its count is squarefree (the Moebius function is non-zero), dropping any note whose ordinal is divisible by a perfect square - a sieve-flavoured aperiodic gate; Invert keeps the square-divisible ones.

MoranGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Moran gate: passes note-ons on Moran counts - Harshad numbers whose quotient (number over its digit sum) is prime (18,21,27,42,45,…); Invert keeps the rest.

MotzkinGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Motzkin gate: passes note-ons on Motzkin-number counts (1,2,4,9,21,51,127,…), the combinatorial sequence counting non-crossing chords and lattice paths; Invert keeps the rest.

MultiplicativePersistenceGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Multiplicative-persistence gate: passes note-ons whose multiplicative persistence - the number of times you replace the count by the product of its digits before hitting a single digit - equals the chosen target; Invert keeps the rest.

NarayanaGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Narayana gate: passes note-ons on Narayana's-cows counts (a(n)=a(n-1)+a(n-3): 1,2,3,4,6,9,13,19,28,…), a slow recurrence whose ratio tends to the supergolden ratio; Invert keeps the rest.

NarcissisticGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Narcissistic gate: passes note-ons on Armstrong (narcissistic) counts that equal the sum of their own digits each raised to the number-of-digits power (1,153,370,371,407,…), a rare self-referential set; Invert keeps the rest.

NivenGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Niven (Harshad) gate: passes note-ons on counts divisible by the sum of their own digits (1..10,12,18,20,21,24,…), a moderately-dense digit-driven set; Invert keeps the rest.

NontotientGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Nontotient gate: passes note-ons on nontotient counts that are never the output of Euler's totient for any number (14,26,34,38,50,…; all even), an unreachable set; Invert keeps the rest.

NudeNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Nude-number gate: passes note-ons on nude counts that are divisible by each of their own nonzero digits (1..9,11,12,15,22,24,…), a polydivisible-by-digit set; Invert keeps the rest.

OctahedralGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Octahedral gate: passes note-ons on octahedral counts (k(2k^2+1)/3: 1,6,19,44,85,…), the 3-D figurate numbers counting points in an octahedron; Invert keeps the rest.

OddDigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Odd-digit gate: passes note-ons whose count is written with only odd decimal digits (1,3,5,7,9,11,13,15,…), the digit-pattern complement of the even-digit gate; Invert passes counts containing an even digit.

OdiousGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Odious gate: passes note-ons on odious counts with an odd number of 1-bits in binary (1,2,4,7,8,11,13,14,…), the Thue-Morse complement of the evil numbers; Invert keeps the evil counts.

OreNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Ore-number gate: passes note-ons on Ore (harmonic-divisor) counts whose divisors have an integer harmonic mean (1,6,28,140,496,…), a divisor-rich set overlapping the perfect numbers; Invert keeps the rest.

PadovanGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Padovan gate: passes note-ons on Padovan-number counts (1,2,3,4,5,7,9,12,16,…; P(n)=P(n-2)+P(n-3)), the plastic-number sequence; Invert keeps the rest.

PalindromeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Palindrome gate: passes note-ons only on counts that read the same forwards and backwards (1,2,…,9,11,22,…,101,…), an irregular, self-mirroring thinning; Invert keeps the non-palindromes.

PalindromicPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Palindromic-prime gate: passes note-ons on counts that are both prime and a decimal palindrome (2,3,5,7,11,101,131,151,…), the intersection of two famous sparse sets; Invert keeps the rest.

PalindromicSquareGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Palindromic-square gate: passes note-ons whose square reads the same forwards and backwards (1,2,3,11,22,26,101,111,…), the roots of palindromic squares; Invert keeps the rest.

PandigitalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pandigital gate: passes note-ons on 1-to-k pandigital counts whose digits are exactly a permutation of 1..(digit count), using each once with no zeros (1,12,21,123,132,…); Invert keeps the rest.

PartitionGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Partition gate: passes note-ons on integer-partition counts p(k) (1,2,3,5,7,11,15,22,30,…), the number of ways to write k as a sum, which thins out as it climbs; Invert keeps the others.

PartitionParityGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Partition-parity gate: passes note-ons when the integer-partition number p(count) is odd, a famously-irregular parity pattern (the subject of Ramanujan's congruences); Invert keeps the even-p counts.

PellGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pell gate: passes note-ons on Pell-number counts (1,2,5,12,29,70,…; P(n)=2P(n-1)+P(n-2)), the silver-ratio sequence behind the best rational approximations to root-2; Invert keeps the others.

PellLucasGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pell-Lucas gate: passes note-ons on companion-Pell counts (Q(n)=2Q(n-1)+Q(n-2): 2,6,14,34,82,198,…), the Lucas-style partner of the Pell numbers; Invert keeps the rest.

PentagonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pentagonal gate: passes note-ons on pentagonal-number counts (1,5,12,22,35,…; k(3k-1)/2), the figurate numbers from Euler's pentagonal-number theorem; Invert keeps the rest.

PentanacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pentanacci gate: passes note-ons on pentanacci counts where each term is the sum of the previous five (1,2,4,8,16,31,61,120,…), a fast-growing Fibonacci generalization; Invert keeps the rest.

PerfectNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Perfect-number gate: passes note-ons on perfect counts equal to the sum of their proper divisors (6,28,496,8128), an exceedingly rare and ancient set; mostly silent, firing only on those landmarks. Invert keeps the rest.

PerfectPowerGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Perfect-power gate: passes note-ons on counts that are an exact power a^b with b>=2 (1,4,8,9,16,25,27,32,…), merging the squares, cubes and higher powers into one set; Invert keeps the rest.

PerrinGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Perrin gate: passes note-ons on Perrin-sequence counts (3,2,5,5,7,10,12,…; P(n)=P(n-2)+P(n-3)), famous because n almost always divides P(n) exactly when n is prime; Invert keeps the rest.

PoliteGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Polite gate: passes note-ons on polite counts that can be written as a sum of two or more consecutive integers (everything except the powers of two); Invert keeps just the powers of two.

PolydivisibleGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Polydivisible gate: passes note-ons on polydivisible counts where the first k digits form a number divisible by k for every prefix (12,15,24,…,120,…); Invert keeps the rest.

PolygonalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Polygonal gate: passes note-ons on s-gonal figurate-number counts for any chosen number of Sides - triangular at 3, square at 4, pentagonal at 5, and so on - one knob covering the whole figurate family; Invert keeps the rest.

PowerfulNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Powerful-number gate: passes note-ons on powerful counts where every prime factor occurs at least squared (1,4,8,9,16,25,27,32,36,…), the dense-factor opposite of squarefree; Invert keeps the rest.

PowerOfTwoGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Power-of-two gate: passes note-ons only on counts that are exact powers of two (1,2,4,8,16,32,…), an octave-spaced exponentially-thinning pattern; Invert keeps the rest.

PracticalNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Practical-number gate: passes note-ons on practical counts, where every smaller integer is a sum of distinct divisors of the count (1,2,4,6,8,12,…; all powers of two and more), a divisor-rich set close to the abundant numbers; Invert keeps the rest.

PrimeDigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Prime-digit gate: passes note-ons whose count is written with only the prime digits 2, 3, 5 and 7 (2,3,5,7,22,23,25,…), a digit-pattern sieve; Invert passes counts containing a non-prime digit.

PrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Number-theory gate: counts incoming notes and lets one through only when its ordinal is a prime number (2,3,5,7,11,…), so the texture thins out unpredictably as primes spread; Invert keeps the composites instead.

PrimeIntervalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Prime-interval gate: passes a note only when its leap from the previous one spans a prime number of semitones (2,3,5,7,11,…), favouring the seconds, thirds, fourths and fifths while blocking octaves and tritones - a number-theoretic melodic filter.

PrimePowerGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Prime-power gate: passes note-ons only when the count is a power of a single prime (2,3,4,5,7,8,9,…), a sparse self-similar pattern from analytic number theory; Invert keeps everything else.

PrimeTripletGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Prime-triplet gate: passes note-ons on counts belonging to a prime triplet - a prime with two more primes within a span of six (5,7,11,13,17,19,…); Invert keeps the rest.

PrimorialGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Primorial gate: passes note-ons on primorial counts - products of the first k primes (2,6,30,210,2310,…) - an extremely sparse, fast-growing set; Invert keeps the rest.

ProbabilityByPitch [PRB] [CV] [MIDI] in: Audio out: MIDI

Probability by pitch: the chance a note passes rises or falls with its keyboard position, so you can thin out the bass while keeping the top (or the reverse) - register-weighted probabilistic gating.

ProbabilityRamp [PRB] [CV] [MIDI] in: Audio out: MIDI

Probability ramp: the pass chance sweeps from Start to End across a repeating cycle of notes, so phrases can fade in (sparse to dense) or thin out (dense to sparse) on a loop - an evolving probabilistic gate.

PronicGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Pronic gate: passes note-ons on pronic (oblong) counts n(n+1): 2,6,12,20,30,…, the products of consecutive integers; Invert keeps the rest.

ProthGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Proth gate: passes note-ons on Proth-number counts (k*2^n+1 with k odd and k below 2^n: 3,5,9,13,17,25,…), the numbers with a fast dedicated primality test; Invert keeps the rest.

RandomGateWalk [PRB] [CV] [MIDI] in: Audio out: MIDI

Random-gate walk: the pass probability itself drifts on a bounded random walk, so the music breathes between dense and sparse passages instead of thinning uniformly like a fixed-chance gate.

RecamanGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Recaman gate: passes note-ons on counts that appear in Recaman's sequence (jump back by n if new and positive, else forward), the famously erratic jumping sequence; its darting coverage thins the stream unpredictably. Invert keeps the rest.

RefactorableGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Refactorable gate: passes note-ons on refactorable (tau) counts whose own number of divisors divides the number (1,2,8,9,12,18,24,36,…); Invert keeps the rest.

RepdigitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Repdigit gate: passes note-ons on repdigit counts whose decimal digits are all identical (1..9,11,22,…,99,111,…), a sparse uniform-digit pattern distinct from the all-ones repunits; Invert keeps the rest.

RepunitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Repunit gate: passes note-ons on repunit counts whose decimal digits are all ones (1,11,111,1111,…), an extremely sparse, widely-spaced pattern; Invert keeps the rest.

RhombicDodecahedralGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Rhombic-dodecahedral gate: passes note-ons on rhombic-dodecahedral counts (1,15,65,175,369,…), the 3-D figurate numbers of that crystal shape; Invert keeps the rest.

RoughNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Rough-number gate: passes note-ons whose count's smallest prime factor is at least Bound (a B-rough number with no small factors), the complement of smoothness; raising Bound makes the gate fire only on prime-like counts. Invert keeps the rest.

SafePrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Safe-prime gate: passes note-ons on safe-prime counts - primes p for which (p-1)/2 is also prime (5,7,11,23,47,59,…), the partners of the Sophie-Germain primes used in cryptography; Invert keeps the rest.

SchroederGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Schroeder gate: passes note-ons on large-Schroeder counts (1,2,6,22,90,394,1806,…), the lattice-path numbers counting subdivisions and super-Catalan structures; Invert keeps the rest.

SelfNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Self-number gate: passes note-ons on self (Colombian) counts that cannot be written as some smaller number plus that number's digit-sum (1,3,5,7,9,20,31,…), a self-referential sieve; Invert keeps the rest.

SemiperfectGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Semiperfect gate: passes note-ons on semiperfect (pseudoperfect) counts where some subset of the proper divisors sums to the number (6,12,18,20,24,28,…); Invert keeps the rest.

SemiprimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Semiprime gate: passes note-ons on semiprime counts - numbers that are the product of exactly two primes (4,6,9,10,14,15,21,…) - a moderately-sparse, multiplicatively-defined pattern; Invert keeps the rest.

SexyPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Sexy-prime gate: passes note-ons on counts in a sexy-prime pair (two primes six apart: 5,7,11,13,17,23,…), so named from the Latin sex for six; Invert keeps the rest.

SmithNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Smith-number gate: passes note-ons on Smith counts whose digit sum equals the digit sum of their prime factorization (4,22,27,58,85,…), a quirky composite set; Invert keeps the rest.

SmoothNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Smooth-number gate: passes note-ons whose count's largest prime factor stays at or below Bound (a B-smooth number, the easy-to-factor integers behind sieve algorithms); raising Bound opens the gate on more counts. Invert keeps the rest.

SophieGermainGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Sophie-Germain gate: passes note-ons on Sophie-Germain-prime counts - primes p where 2p+1 is also prime (2,3,5,11,23,…), the primes behind safe-prime cryptography; Invert keeps the rest.

SphenicGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Sphenic gate: passes note-ons on sphenic counts - squarefree products of exactly three distinct primes (30,42,66,70,78,…) - a multiplicatively-defined, moderately-sparse set; Invert keeps the rest.

SquarefreeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Squarefree gate: passes note-ons on squarefree counts with no repeated prime factor (1,2,3,5,6,7,10,11,13,…), about 61 percent of integers; Invert keeps the square-divisible counts.

SquareGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Square gate: passes note-ons only on perfect-square counts (1,4,9,16,25,…), so notes get steadily sparser as the gaps between squares widen; Invert plays everything except the squares.

SquarePyramidalGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Square-pyramidal gate: passes note-ons on square-pyramidal counts (1,5,14,30,55,…), the number of cannonballs in a square-based pyramid; Invert keeps the rest.

SquareTriangularGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Square-triangular gate: passes note-ons on the rare counts that are simultaneously a perfect square and a triangular number (1,36,1225,41616,…); Invert keeps the rest.

StarNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Star-number gate: passes note-ons on centered star (hexagram) counts (1,13,37,73,121,…), the dots in a growing six-pointed star; Invert keeps the rest.

StellaOctangulaGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Stella-octangula gate: passes note-ons on stella-octangula counts (k(2k^2-1): 1,14,51,124,245,…), the points of a growing star-tetrahedron; Invert keeps the rest.

StrobogrammaticGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Strobogrammatic gate: passes note-ons on counts that read the same when rotated 180 degrees (0,1,8,11,69,88,96,…), a visual digit-symmetry set; Invert keeps the rest.

SunnyNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Sunny-number gate: passes note-ons on sunny counts where the number plus one is a perfect square (3,8,15,24,35,48,…), one less than each square; Invert keeps the rest.

SuperabundantGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Superabundant gate: passes note-ons on superabundant counts whose ratio sigma(n)/n beats every smaller number (1,2,4,6,12,24,36,48,60,120,…), the densest-divisor records; Invert keeps the rest.

SylvesterGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Sylvester gate: passes note-ons on Sylvester's-sequence counts (2,3,7,43,1807,…; each one more than the product of all previous), a doubly-exponential set so sparse it fires only a handful of times; Invert keeps the rest.

TetrahedralGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Tetrahedral gate: passes note-ons on tetrahedral-number counts n(n+1)(n+2)/6 (1,4,10,20,35,56,…), the 3-D figurate numbers counting stacked spheres; their cubic spacing thins fast. Invert keeps the rest.

TetranacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Tetranacci gate: passes note-ons on tetranacci counts where each term is the sum of the previous four (1,2,4,8,15,29,56,108,…); Invert keeps the rest.

ThabitGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Thabit gate: passes note-ons on Thabit (321-) number counts (3*2^n-1: 2,5,11,23,47,95,191,…), the medieval numbers behind amicable-pair constructions; Invert keeps the rest.

ThueMorseGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Thue-Morse gate: passes notes where the cube-free Thue-Morse sequence (the bit-parity of the note count) is 0, an aperiodic-yet-structured pattern that never settles into a short loop; Invert flips which half plays.

TrailingZeroGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Trailing-zero gate: passes note-ons whose count has exactly the chosen number of trailing binary zeros (its 2-adic valuation), selecting one rung of the ruler sequence - odd counts at 0, doubly-even at higher; Invert keeps the rest.

TriangularGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Triangular gate: passes note-ons on triangular-number counts (1,3,6,10,15,…), thinning the stream on the handshake-number sequence; Invert keeps the non-triangular notes.

TribonacciGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Tribonacci gate: passes note-ons on Tribonacci-sequence counts (1,2,4,7,13,24,44,…; each the sum of the previous three), spreading wider than Fibonacci; Invert keeps the rest.

TrimorphicGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Trimorphic gate: passes note-ons on trimorphic counts whose cube ends in the number itself (1,4,5,6,9,24,25,49,…), the cube-analogue of automorphic numbers; Invert keeps the rest.

TruncatablePrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Right-truncatable-prime gate: passes note-ons on primes that stay prime as each trailing digit is chopped off (2,3,5,7,23,29,31,37,53,…), a famously finite set; Invert keeps the rest.

TwinPrimeGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Twin-prime gate: passes note-ons on counts belonging to a twin-prime pair (a prime with another prime two away: 3,5,7,11,13,17,19,…), thinning onto the famously-clustered twin primes; Invert keeps the rest.

UlamGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Ulam gate: passes note-ons on Ulam-number counts (each the smallest integer that is a sum of two earlier terms in exactly one way: 1,2,3,4,6,8,11,…), a mysteriously near-periodic set; Invert keeps the rest.

UndulatingGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Undulating gate: passes note-ons on undulating counts whose digits zig-zag high-low-high (121,132,231,1212,…), a wave-like digit pattern; Invert keeps the rest.

UndulatingNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Undulating-number gate: passes note-ons on counts of 3+ digits that undulate in an alternating ababab pattern with two different digits (121,131,212,232,…); Invert keeps the rest.

UntouchableGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Untouchable gate: passes note-ons on untouchable counts that can never be the sum of the proper divisors of any number (2,5,52,88,96,…), an unreachable set in the aliquot map; Invert keeps the rest.

VelocityFloorGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Velocity floor gate: drops note-ons whose velocity falls below a floor, filtering out the quietest notes (off events for dropped notes are removed too to avoid stuck notes); Invert keeps only the quiet ones.

VelocityProbGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Velocity-probability gate: passes each note with a probability that rises with its velocity, so soft notes thin out and accents survive - a dynamics-driven random thinning; Bias lifts the baseline pass chance.

VelRandomGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Velocity-random gate: passes each note with a probability that rises with its velocity, so accented hits survive while ghost notes thin out - dynamics-weighted probabilistic gating.

WedderburnEthringtonGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Wedderburn-Etherington gate: passes note-ons on the counts of unordered binary trees (1,2,3,6,11,23,46,98,…), a combinatorial sequence from chemistry and phylogenetics; Invert keeps the rest.

WeirdNumberGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Weird-number gate: passes note-ons on weird counts that are abundant yet have no subset of proper divisors summing to the number (70,836,4030,…), a rare paradoxical set; Invert keeps the rest.

WoodallGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Woodall gate: passes note-ons on Woodall-number counts (n*2^n-1: 1,7,23,63,159,383,…), an exponentially-sparse set partnered with the Cullen numbers; Invert keeps the rest.

WythoffGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Wythoff gate: passes note-ons on the lower-Wythoff counts floor(n*phi) (1,3,4,6,8,9,11,…), the golden-ratio Beatty sequence that, with its complement, partitions the integers; Invert keeps the upper-Wythoff notes.

ZuckermanGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Zuckerman gate: passes note-ons on Zuckerman counts divisible by the product of their own (nonzero) decimal digits (1..9,11,12,15,24,36,…); Invert keeps the rest.

ZumkellerGate [PRB] [CV] [MIDI] in: Audio out: MIDI

Zumkeller gate: passes note-ons on Zumkeller counts whose divisors can be split into two sets of equal sum (6,12,20,24,28,30,…), a balanced-divisor set; Invert keeps the rest.

Routing (29)

Channel [RTE] [MIDI] in: Audio out: MIDI

Reassigns notes to a fixed MIDI channel.

ChannelByPitch [RTE] [MIDI] in: Audio out: MIDI

Channel-by-pitch: routes notes to different MIDI channels by register, splitting the keyboard into Zones equal bands so a single part can drive several multitimbral instruments stacked across the range.

ChannelByZone [RTE] [MIDI] in: Audio out: MIDI

Channel by zone: routes notes below the Split point to channel ChanLo and notes at or above it to ChanHi, splitting one keyboard across two MIDI channels (multitimbral split).

ClampToRange [RTE] [MIDI] in: Audio out: MIDI

Clamp to range: pulls every out-of-range note onto the nearest boundary of [Low,High], folding a wide part flat against the edges of a fixed register.

FixedNote [RTE] [MIDI] in: Audio out: MIDI

Forces every note to a single fixed pitch (drum / trigger).

FoldToRange [RTE] [MIDI] in: Audio out: MIDI

Fold to range: shifts every note up or down by whole octaves until it sits inside the [Low,High] window, packing a wide-range part into a fixed register while preserving each note’s pitch class.

KeyboardSplit3 [RTE] [MIDI] in: Audio out: MIDI

Three-zone split: divides the keyboard at two split points into low, mid and high zones, each transposed by its own amount - a quick way to layer a bassline, a pad and a lead from one part across three regions.

KeySwitch [RTE] [MIDI] in: Audio out: MIDI

Splits the keyboard at a note; one zone passes, the other is filtered.

KeyZoneGate [RTE] [MIDI] in: Audio out: MIDI

Key-zone gate: passes only notes whose pitch sits inside [Low,High] and drops the rest (their note-offs are removed too to avoid stuck notes); Invert keeps only the notes outside the zone.

MaxPolyphony [RTE] [MIDI] in: Audio out: MIDI

Max polyphony: caps how many notes may sound at once, dropping any note-on past the limit until a voice frees up - a voice-stealing-free polyphony clamp for thinning dense chords.

Merge [RTE] [MIDI] in: Audio out: MIDI

Passes both note-stream inputs through, merged into one stream.

MidiLatch [RTE] [MIDI] in: Audio out: MIDI

Holds notes after release until the next note arrives.

MirrorZone [RTE] [MIDI] in: Audio out: MIDI

Mirror zone: reflects the notes inside [Low,High] around the zone centre (a melodic inversion confined to one key zone), leaving notes outside the zone alone.

MuteZone [RTE] [MIDI] in: Audio out: MIDI

Mute zone: drops every note whose pitch sits inside [Low,High] while letting the rest of the keyboard through, punching a silent notch into the range.

NoteFilter [RTE] [MIDI] in: Audio out: MIDI

Passes only notes inside a key and velocity window.

NoteLimit [RTE] [MIDI] in: Audio out: MIDI

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

NoteRangeTranspose [RTE] [MIDI] in: Audio out: MIDI

Note-range transpose: transposes by Semitones only the notes whose pitch sits inside [Low,High], leaving everything outside the zone at its original pitch.

NoteToGateCC [RTE] [MIDI] in: Audio out: MIDI

Note-to-gate CC: emits a chosen controller high while any note is on and low when it releases, turning note events into a gate/trigger control signal for envelopes, side-chains or external gear.

NoteToPitchBend [RTE] [MIDI] in: Audio out: MIDI

Note-to-pitch-bend: emits a pitch-bend proportional to each note's distance from a center pitch, so notes far from the center are pre-bent - a way to map keyboard position onto continuous detune or to drive a synth's bend input from played pitch.

PanByPitch [RTE] [MIDI] in: Audio out: MIDI

Pan by pitch: emits a pan CC (controller 10) that places low notes to the left and high notes to the right, an automatic keyboard-position stereo spread as a synth or sampler plays.

PolyChan [RTE] [MIDI] in: Audio out: MIDI

Round-robins each new note to the next of Voices MIDI channels from Base, spreading a polyphonic part across a multitimbral rig - one note per channel, the classic voice-allocation split (distinct from MPEConvert's per-note MPE channels).

RandomPanCC [RTE] [MIDI] in: Audio out: MIDI

Random-pan CC: fires a random pan (controller 10) value on every note-on, scattering successive notes across the stereo field for an automatic, ever-shifting spatial spread; Spread sets how wide the scatter.

Split [RTE] [MIDI] in: Audio out: MIDI

Keyboard split: notes below the split point transpose by Low, the rest by High.

SplitTranspose [RTE] [MIDI] in: Audio out: MIDI

Split transpose: a keyboard split where notes at or above the Split point are transposed by Semitones and notes below pass untouched, so two hands play in different registers.

Thin [RTE] [MIDI] in: Audio out: MIDI

Note-density limiter: drops any note-on arriving within MinGap ms of the last one passed, decluttering a busy stream or fast repeats - distinct from NoteLimit, which caps simultaneous polyphony. A dropped note's off is dropped too.

Transformer [RTE] [MIDI] in: Audio out: MIDI

Rule: notes within a range are transposed and velocity-scaled.

VelocityZoneGate [RTE] [MIDI] in: Audio out: MIDI

Velocity-zone gate: passes only notes whose velocity falls inside [Min,Max] and drops the rest; Invert keeps only the notes outside the velocity window.

WrapAround [RTE] [MIDI] in: Audio out: MIDI

Wrap around: notes that rise above the High edge wrap back to the Low end of the range (and vice versa), a modular pitch fold that keeps a part inside a register while letting runs cycle around.

⟜ **ZoneOctaveShift** [RTE] [MIDI] in: Audio out: MIDI

Zone octave shift: shifts the notes inside [Low,High] by whole Octaves, leaving notes outside the zone where they are, a register jump confined to one key range.

Sequencing (63)

Appoggiatura [SEQ] [CV] [MIDI] in: Audio out: MIDI

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

Arp [SEQ] [CV] [MIDI] in: Audio out: MIDI

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

BeattyGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Beatty gate: passes note-ons on a Beatty sequence floor(n*Ratio) for any irrational ratio, a Sturmian, maximally-even thinning whose density is set by the ratio; Invert keeps the complement.

BurstGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Burst gate: lets OnLen consecutive notes through, mutes the next OffLen, and repeats, chopping a steady run into rhythmic bursts and rests - a note-count gate for stutter and ratchet feels.

Cascade [SEQ] [CV] [MIDI] in: Audio out: MIDI

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

ChordArp [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu-style chord arpeggiator: each key builds a full chord (Type) and the result is arpeggiated upward across Octaves at a synced Rate - one finger plays a moving arpeggio of the whole chord. Releasing the key stops its run.

ClaveGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Clave gate: passes notes on the pulses of the Afro-Cuban son clave, the five-stroke rhythmic key of salsa and Latin music, selectable as the 3-2 or 2-3 orientation; Invert keeps the off-clave notes.

ClockDivideGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Clock divider: passes one note-on out of every Divide that arrive and drops the rest, a simple note-rate divider for thinning fast passages or extracting a downbeat pulse.

CongruentGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Congruent gate: passes a note only when its running count is congruent to Remainder modulo Modulus, a fully-parameterized periodic gate that covers every-Nth-note and offbeat patterns from two numbers; Invert keeps the rest.

▪▫▪▫ CrossPulseGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cross-pulse gate: lays two pulse grids of different periods over the note stream and passes notes where they coincide (AND) or where either fires (OR), generating polyrhythmic and cross-rhythmic gating from two simple counts.

▪▫▪▫ Drunk [SEQ] [CV] [MIDI] in: Audio out: MIDI

Random-walk melodic sequencer: each synced Rate step nudges the pitch by a random +/-Step semitones, bounded within +/-Range of the lowest held note - Brownian melody anchored to the chord, unlike MidiRandom's memoryless picks.

▪▫▪▫ DurationFromPitch [SEQ] [CV] [MIDI] in: Audio out: MIDI

Duration from pitch: maps each note's pitch to its length so high notes play short and low notes long (or, inverted, the reverse), a register-tilt articulation.

▪▫▪▫ Echo [SEQ] [CV] [MIDI] in: Audio out: MIDI

MIDI delay: repeats notes at a synced time with velocity feedback decay.

▪▫▪▫ EuclidComplementGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean complement: passes notes that fall on the rests of a Euclidean rhythm of Pulses-in-Steps, the photo-negative of a Euclid sequencer - an instant interlocking counter-rhythm.

▪▫▪▫ EuclideanGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean gate: passes notes on the pulses of a Euclidean (Bjorklund) rhythm that spreads Pulses as evenly as possible over Steps - the algorithm that reproduces clave, tresillo and most world rhythms from two numbers; Rotation shifts the pattern. Invert keeps the rest.

▪▫▪▫ EuclidMel [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean melody: spreads Pulses evenly across Steps (Bjorklund) and, on each hit, advances a cursor through the held chord - the rhythm is Euclidean and the pitch walks the chord, unlike SeqEuclid which stabs all held notes per pulse.

▪▫▪▫ Flam [SEQ] [CV] [MIDI] in: Audio out: MIDI

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

▪▫▪▫ GateArp [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp gate lane: walks the held chord upward across Octaves while the note length cycles a per-step staccato/legato pattern, so the rhythm breathes - short stabs then sustained notes. Gate scales the pattern (>1 overlaps into the next step).

▪▫▪▫ **GateLengthAccent** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length accent: holds every Nth note (the downbeat) for the Long duration and clips the rest to Short, a rhythmic gate accent.

▪▫▪▫ **GateLengthAlternate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length alternate: alternates note durations between two lengths (long, short, long, short), a bouncing gate groove.

▪▫▪▫ **GateLengthFromInterval** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length from interval: grows each note's duration with how far it leaps from the previous note, so big melodic jumps ring longer than stepwise motion.

▪▫▪▫ **GateLengthInvVelocity** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length inverse velocity: louder notes are clipped short and soft notes ring long (the inverse of VelLength), a punch-then-decay articulation.

▪▫▪▫ **GateLengthLFO** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length LFO: sweeps note duration between Min and Max with a sine that completes one cycle every Period notes, a breathing gate that swells and shrinks.

▪▫▪▫ **GateLengthRamp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length ramp: ramps note duration linearly from Min to Max across Steps notes then resets, an automatic gate crescendo.

▪▫▪▫ **GateLengthRandom** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gate-length random: gives each note a random duration between Min and Max, humanising the gate so no two notes ring exactly the same length.

▪▫▪▫ **GoldenWordGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Golden-word gate: gates notes by the infinite Fibonacci word (the Sturmian 0/1 sequence at golden-ratio spacing), the most-ordered aperiodic rhythm there is; Invert flips which letter plays.

▪▫▪▫ **KolakoskiGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Kolakoski gate: gates notes by the self-describing Kolakoski 1,2 sequence (whose run-lengths reproduce the sequence itself), a hypnotically self-similar aperiodic pattern; Invert flips which symbol plays.

▪▫▪▫ **MidiBrianBrain** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

▪▫▪▫ **MidiGameOfLife** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lifes become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

▪▫▪▫ **MidiLangton** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

▪▫▪▫ **MidiMarkov** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Learns the note-to-note transition statistics of what you play (a 12x12 pitch-class chain) and, on each synced step, walks the chain to improvise a new line in the same style. Memory fades old transitions.

▪▫▪▫ **MidiQuantize** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Snaps note starts to a synced grid by a strength amount.

▪▫▪▫ **MidiTranceGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

▪▫▪▫ **MidiTuring** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

▪▫▪▫ **Mordent** [SEQ] [CV] [MIDI] in: Audio out: MIDI

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

▪▫▪▫ **NoteLength** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Forces a fixed note duration (gate length).

▪▫▪▫ **NoteOffDelay** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Extends each note's gate by delaying its note-off.

▪▫▪▫ **OctaveArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp octave lane: arpeggiates the held chord while a per-step octave-offset Pattern leaps the line up and down by whole octaves (alternating, staircase, bounce…) instead of climbing monotonically like StepArp.

▪▫▪▫ **PaperfoldGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Paperfold gate: gates notes by the regular paperfolding (dragon-curve) sequence of the count, a 2-automatic pattern that folds in on itself at every scale; Invert flips the kept and dropped halves.

▪▫▪▫ **PatternGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gates note-ons through a 16-step on/off pattern (bitmask).

▪▫▪▫ **Polymeter** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA,LenB) steps - instant polymetric movement from one chord.

▪▫▪▫ **PolyrhythmGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Polyrhythm gate: passes notes that land on a multiple of either count A or count B, overlaying two pulse grids (e.g. 3 against 4) into an interlocking cross-rhythm of accents.

▪▫▪▫ **RatchetArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp repeat lane: walks the held chord upward across Octaves while a per-step pattern retriggers selected steps multiple times (capped by Max), packing rolls into the line. Unlike SeqRatchet's fixed count, the repeat count varies per step.

▪▫▪▫ **Ricochet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Accelerating bouncing retriggers with decaying velocity.

▪▫▪▫ **Roll** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

▪▫▪▫ **SeqEuclid** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

▪▫▪▫ **SeqRatchet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Retriggers each note Count times within a synced step.

▪▫▪▫ **Shift** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Delays every note by Delay ms (groove nudge / lay-back), pushing note-ons and offs back by the same amount so durations are preserved - the per-track timing offset of a groove sequencer.

▪▫▪▫ **SkipArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp skip lane: arpeggiates the held chord upward across Octaves, but each step has a Skip probability of being dropped to a rest, thinning the pattern differently every cycle for generative movement.

▪▫▪▫ **SkipNote** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Skip note: drops every Nth note-on and lets the rest through, the inverse of a clock divider - a quick way to punch regular holes in a run of notes.

▪▫▪▫ **Spiral** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Transposing MIDI delay: each repeat is delayed by Time and shifted by Interval semitones with velocity decaying by Feedback, so one note spins off a climbing or falling staircase of echoes - unlike Echo, which repeats the same pitch.

▪▫▪▫ **StaccatoCut** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Staccato cut: clips every note to a short fixed length for a crisp detached articulation, no matter how long the key was held.

▪▫▪▫ **StepArp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu-style step arpeggiator: walks the held chord strictly upward across Octaves octaves at a synced Rate, accenting the first step of each cycle - the rolling, octave-jumping arp of Xfer Cthulhu's sequencer. Distinct from Arp's mode-cycling: this climbs the chord in order.

▪▫▪▫ **StepSeqMidi** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Melodic step sequencer - semitone offsets (node config) clock the held root.

▪▫▪▫ **Strum** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Spreads a chord's notes over time, up or down, like a guitar strum.

▪▫▪▫ **Swing** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Pushes off-beat eighth-notes later for a swing/shuffle feel.

▪▫▪▫ **TenutoHold** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Tenuto hold: extends every note to a long fixed length so notes sustain and run into each other, a held legato-style gate.

TransArp [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp transpose lane: arpeggiates the held chord upward across Octaves while a per-step semitone Pattern (fifth pedals, triad arps, scalar runs) is added on top, so the line outlines a melodic contour, not just the chord tones.

TresilloGate [SEQ] [CV] [MIDI] in: Audio out: MIDI

Tresillo gate: passes notes on the tresillo 3-3-2 pattern, the eight-step rhythmic cell underlying reggaeton, habanera and countless Latin grooves; Invert keeps the rest.

Trill [SEQ] [CV] [MIDI] in: Audio out: MIDI

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

Tuplet [SEQ] [CV] [MIDI] in: Audio out: MIDI

Snaps note starts to a Tuplets-per-beat grid (3 = triplets, 5 = quintuplets) by Strength, for a triplet or odd-tuplet feel where MidiQuantize only snaps to the straight grid.

Turn [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

VelArp [SEQ] [CV] [MIDI] in: Audio out: MIDI

Cthulhu's arp velocity lane as a ramp: arpeggiates the held chord while velocity sweeps from From to To across Steps then resets, so every pass swells or fades without a per-step editor.

Timing (2)

HumanizeTiming [FX] [MIDI] in: Audio out: MIDI

Humanize timing: nudges each event's timing by a small random amount (up to the set sample range) so a rigid sequenced part feels played by hand.

StrumChord [FX] [MIDI] in: Audio out: MIDI

Strum chord: staggers the start of simultaneously-played notes by a small delay so a chord is strummed across the strings rather than struck as a block, with selectable up- or down-stroke direction; Spread sets the strum speed.

Tuning (3)

JustTune [TUN] [MIDI] in: Audio out: MIDI

Adaptive just intonation: retunes each note toward its 5-limit JI ratio relative to Key via per-note detune (polyphonic), Amount blending from equal temperament to pure - the beatless thirds and fifths of real harmony.

Microtuner [TUN] [MIDI] in: Audio out: MIDI

Stretched-octave microtuning via per-note pitch detune.

TuneTable [TUN] [MIDI] in: Audio out: MIDI

Custom microtuning: detunes each note by a per-pitch-class cents offset from the node's 12-value config list (relative to Key) - meantone, Pythagorean, maqam or gamelan, any temperament. Amount scales the table.

Voicing (5)

MidiUnison [VOI] [MIDI] in: Audio out: MIDI

Doubles each note into multiple voices, optionally spread across channels.

Mono [VOI] [MIDI] in: Audio out: MIDI

Monophonic note priority (last / lowest / highest).

MPEConvert [VOI] [MIDI] in: Audio out: MIDI

Assigns each note its own rotating channel for per-note MPE expression.

Slide [VOI] [MIDI] in: Audio out: MIDI

Cthulhu's slide: mono legato glue. Picks one held note by Priority (last/low/high) and emits the new note-on before releasing the old one, so a portamento synth slurs between notes instead of retriggering.

Sustain [VOI] [MIDI] in: Audio out: MIDI

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

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